SIEMENS

MAGNETOM

MR

Functional Description

Harmony *syngo* MR
Symphony *syngo* MR
Sonata *syngo* MR

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Section

1 Introduction

This Introduction is intended to familiarize you with the structure of the MAGNETOM Harmony, Symphony, Sonata documentation. Here will also be explained the print format as well as the on-line documentation and how to use it.

Service Documentation

Paper Documentation

The official service documents consist of the Functional Description (**FUN**), the Trouble Shooting Guide (**TSG**) and the Replacement of Parts (**ROP**) and are available in paper form as DIN A5 ring books and on CD/DVD. These are handed out in the training course and will be the ONLY hardcopy version you will get since all future revisions will be available in electronic form ONLY. Remember this when you make your notes ;-)

In addition, there are 3 DIN A4 binders delivered with every system which include:

- Diagrams
- Installation and Start Up instructions
- Preventative Maintenance and Safety Related Tests
- Protocols and records of service activities and modifications

Electronic Documentation

This document has a built-in navigation system to facilitate a quick navigation to the information you are looking for. **Text links** are recognizable by their BLUE color. Links within the **graphics** are usually denoted with a colored triangle placed in the upper left corner of box within the graphic that is linked. There are several cases where this could not be realized so an attempt has been made to indicate a link using some other means. In any case, almost all graphics will have at least one link, so move the mouse over any graphic. If the cursor changes to a hand anywhere, it's a link.

Files in electronic format:

- Functional Description (FUN)
- Trouble Shooting Guide (TSG)
- Replacement of Parts (ROP)
- Planning Guide (PG)
- Diagrams (DIA)
- Maintenance Instruction (MI)
- Safety Information (SAFE)

In addition to these fundamental documents there are also several supplementary documents published on an irregular basis and instrumental in keeping the CSE informed on service issues. These include:

- Speed infos (SI)
- Update or Modification instructions (UI)
- Knowledge Database entries
- Quality Database
- Report Database
- Spare Parts List

This information is found under the CS INTRAnet portal and should be consulted regularly. The CSE can be informed automatically via e-Mail on the release of Speed Infos if he or she has an account on the Knowledge Database. This functionality must be enabled. Please refer to the information pages within the Knowledge Database on how to do this.

Document Structure

The three major service documents FUN, TSG and ROP have a common sub-division including 13 sections, or parts:

- Introduction
- System - general info regarding the system
- Software
- Host / Imager
- Control
- RF System
- Patient Handling
- Gradients
- Magnet System
- Cooling System
- Power Distribution
- Tune Up
- Changes to last Version - a list changes made from the last version

As you can see, these sections cover the major hardware sub-systems, software and Tune Up.

FUN

The Functional Description contains information on HOW the MR system components and sub-components work. The FUN is usually referred to when wanting to get detailed information on a particular component and not necessarily used consistently in the service workflow. The FUN serves as the basis for the training and is gone into detail during the course. Hopefully it is also read at that time. We doubt it, though. The Block Diagrams within the FUN are especially useful to gain an overview of the functional blocks (usually field replacable units, **FRUs**) and their inter connectivity.

TSG

The TSG is the primary document referred to when dealing with service-related problems. Each of the hardware sections contain two sub-sections:

- **Strategy** : This section gives information, usually in the form of error lists or table of symptoms, to help determine WHERE the problem is and WHICH test procedures can or should be performed to help isolate the defective or mal-functioning component. Often times you will find useful information on the LEDs, switch functions, etc. that are not found elsewhere.
- **Procedures** : This section provides the instructions on performing both the automated test loops and manual testing procedures.

ROP

The ROP gives instructions on replacing FRUs as well as the removal of covers, the configuration of components and software when necessary and a list of tune up or calibration procedures which must be performed after the replacement of any component.

Other Service Documents

The structure of all other documents is specific to the type of document and is deemed unnecessary to explain in detail here.

Sources of Documentation

There are several sources available where the various documents can be found.

CD-ROM

All of the service documentation EXCEPT the Speed infos, Update Instructions and Knowledge Database entries have been placed onto a CD or DVD ROM. For CSEs who are entering the service for a particular modality will be given this CD/DVD during their first modality training course. Updates are issued automatically to CSEs that have been entered into the distribution database. The responsibility for entering CSEs into this database lies with the countries and regional service offices within the countries, i.e., YOUR boss. If you are not getting CD/DVD updates, inform your supervisor.

INTRANET

The best source for the latest documentation and information is the Siemens Intranet. All Siemens service employees will have access to the intranet. A list of the links to this information cannot be given here, since they do change from time to time. It is best if you familiarize yourself with the CS web pages and where this information can be found. Bookmarking these pages is recommended, with the reminder that they can and will change with time.

Training Course

The service CD/DVD (s) for the MR product family being trained for will be distributed during the training course. There may be additional informational material distributed specifically for the training course. These materials will differ for each course.

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Section
2 System

The MAGNETOM Harmony, Symphony and Sonata represent a new generation of MR imaging systems designed to meet the specific needs of radiologists and brings the new and improved technologies over previous MR product lines. Customer surveys provided many new ideas and their wishes led to the appearance and features of these machines.

Overview

The product line described in this document include:
- MAGNETOM Harmony, a 1 Tesla system
- MAGNETOM Symphony, a 1.5 Tesla system
- MAGNETOM Sonata, a 1.5 Tesla system with stronger gradients

These systems were not only designed to improve the performances of the major sub-systems - RF, gradients, magnet and digital processing - with the latest technological advancements, but to optimize both hardware and software for better image quality, higher resolutions, optimized radiological workflows and improved patient comfort.

Since the initial introduction of the Harmony and Symphony systems in 1997, they have since gone through two metamorphosis that have not only changed their outward looks, but also their performance. New or improved hardware components (gradient amplifier and coil, RF receive chain and RF coils) and software application improvements have accompanied these systems over the years:
- Harmony / Symphony
- Symphony Quantum
- Harmony / Symphony / Sonata syngo
- Harmony / Symphony / Sonata Maestro
- Symphony, a Tim system (not described in this document)

Figure 1 *System Block Diagram*

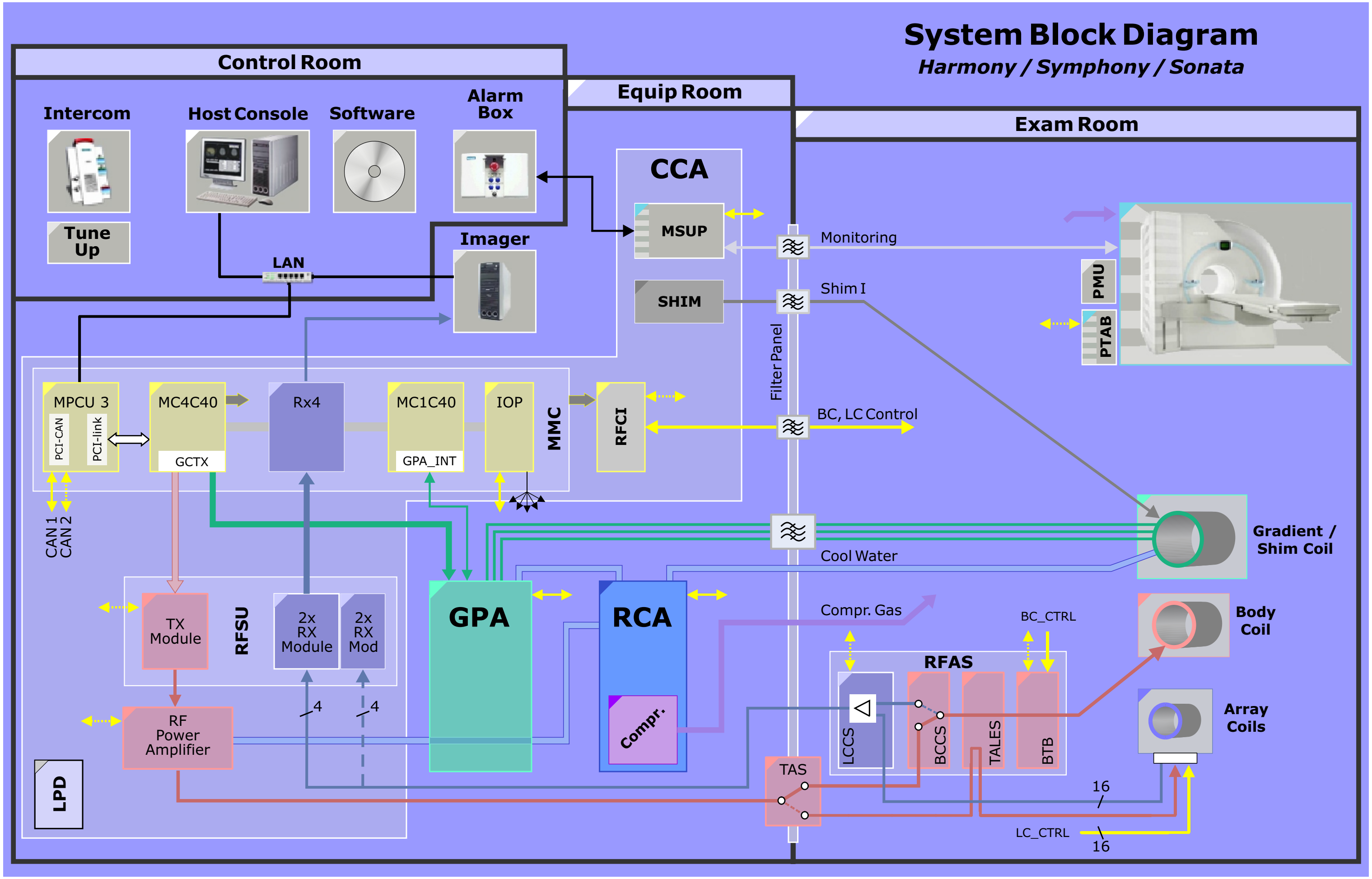


Figure 2 Equipment Room Components

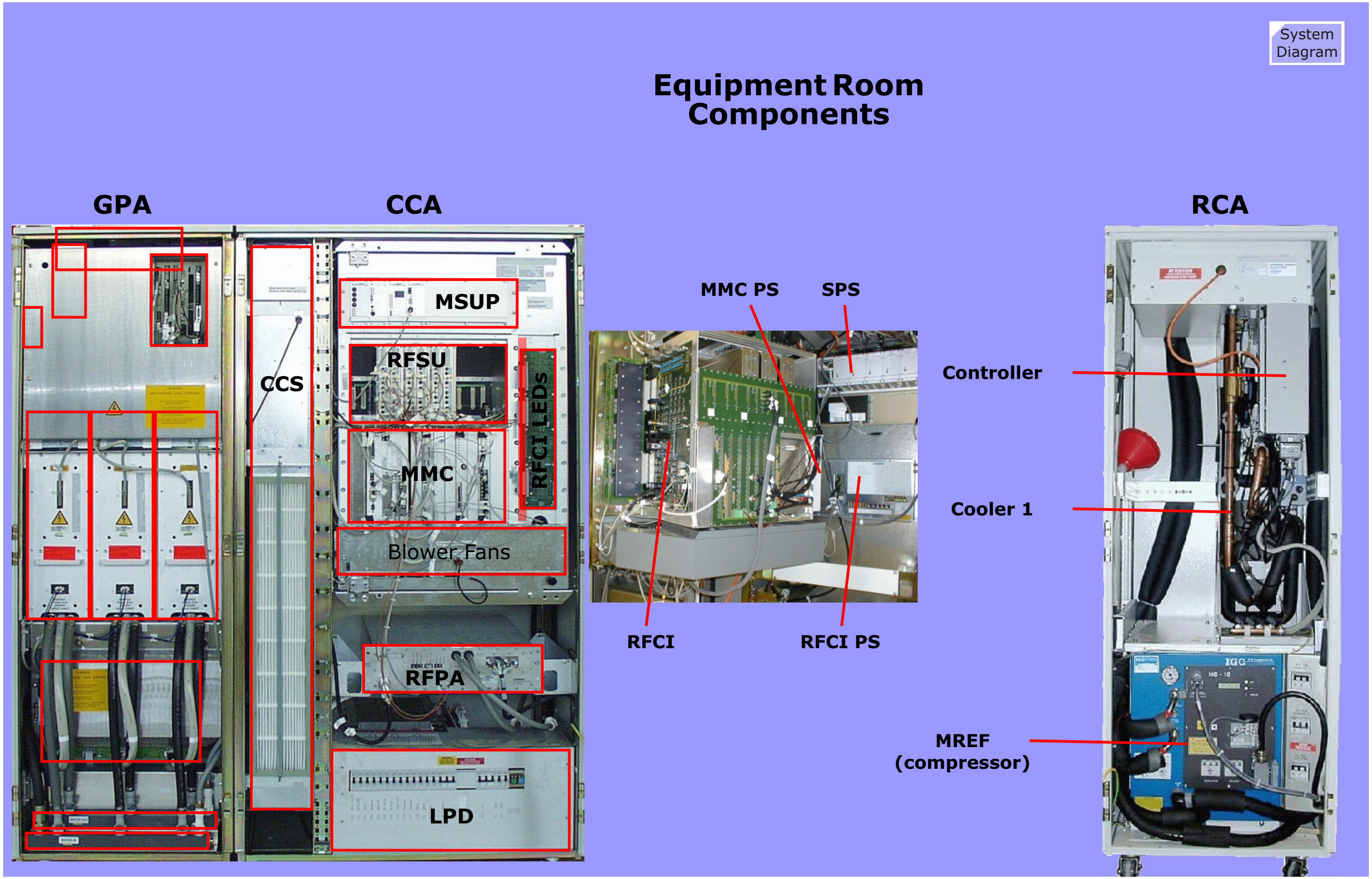
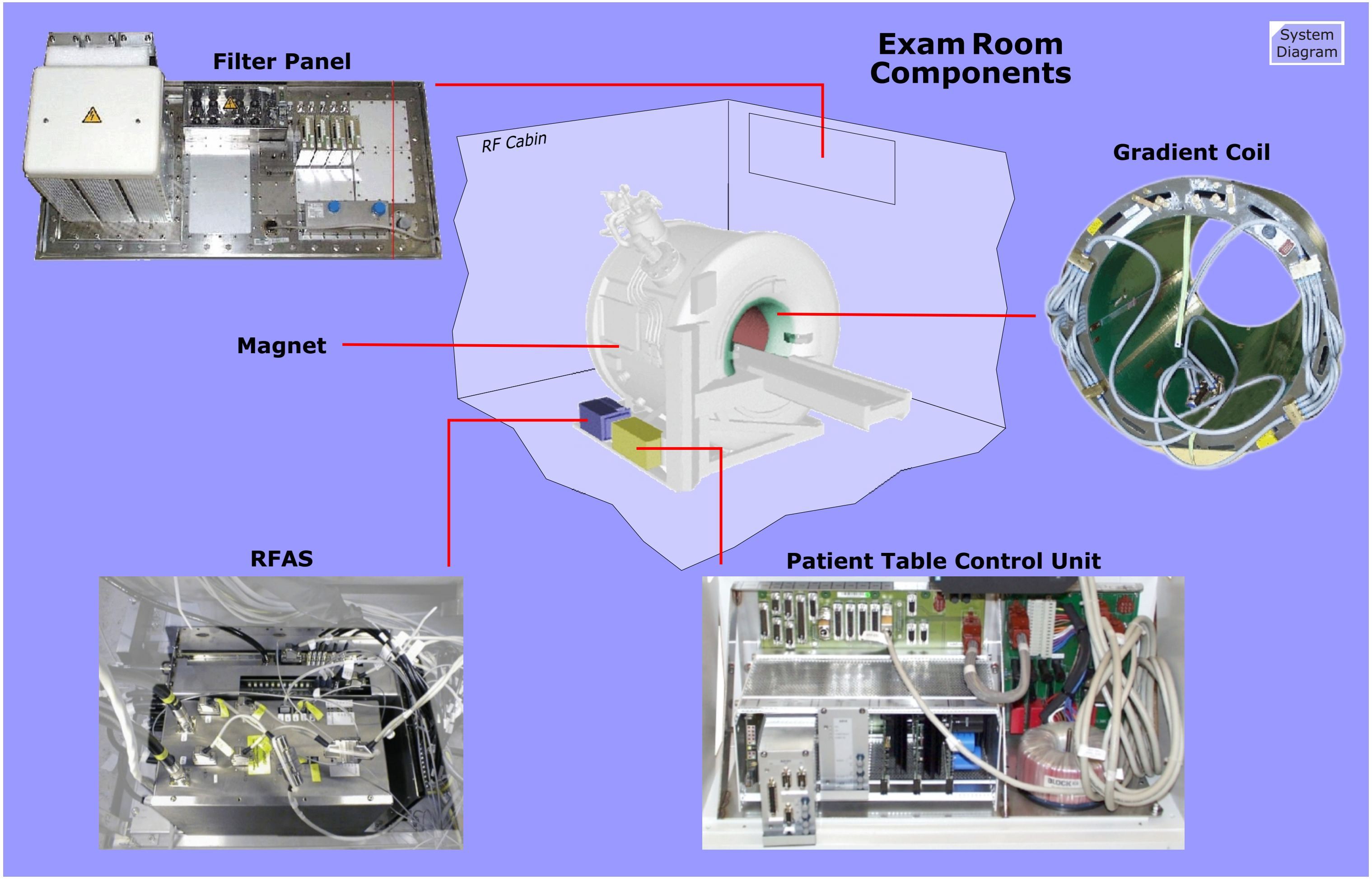


Figure 3 *Exam Room Components*



Magnet System

The magnet system generates the basic magnetic field. The components required are described below. The electrical shim coils and trays for the passive shim plates are integrated into the gradient coil.

Magnet

The magnet comprises the superconducting magnet including the systems for cooling (interface for helium fill/refill, cold head) and monitoring the magnet during operation.

Refrigeration System

The magnet has two cryo-shields surrounding the helium vessel containing the superconducting magnet for the purpose of reducing the influx of radiated energy and thus reducing the helium boil off. These shields are cooled by a conventional refrigeration system consisting of a compressor and a 2 stage cold head which maintains their temperatures at 80K and 20K respectively.

Magnet Supervision

Due to the potential risks involved with a cryogen-filled magnet a supervision unit is indispensable. The Magnet Supervision unit supervises and controls the magnet during operation, ramps it down in emergency situations (ERDU), and it provides user-accessible operating controls for the magnet system.

Magnet Covers

The magnet covers consist of the front, back and side covers, the turret covers, the apertures including all mechanisms required for installing the covers, as well as the components located directly underneath the covers. The equipment required for the tunnel lighting, patient ventilation, and light localizer are integrated into the magnet covers (fan, lamps, cables, light localizer).

Shim Power Supply - SPS

An electrical shim system consisting of 5 shim coils, integrated into every gradient coil, and a Shim Power Supply allow for interactive fine-tuning of the B0-field to compensate local field inhomogeneities caused by the patient and the surrounding environment.

Passive Shim - PSHIM

The passive shim consists of shim trays strategically placed within the Gradient Coil, wherein iron shim plates can be placed to "smooth out" the inherent static magnet inhomogeneities. Shimming the magnet passively requires a special Array Shim measuring device and shim hardware (shim and isolation plates).

Magnet Power Supply - MPS

The Magnet Power Supply consists of the unit which energizes/de-energizes the magnet, including the charge cable to the magnet, the cabling for the voltage supply and Magnet Supervision.

As a service tool, the MPS is not part of the delivery volume of the MR system and is brought on-site only as required. In contrast to the previous MPS K2200 of Harmony/Symphony, cooling water is no longer necessary. The power supply is also more robust and easier to transport and handle.

Array Shim Device - SD

The Array Shim Device is a service and installation tool for magnet shimming. It consists of mechanical supports, MR probes, preamplifier, multiplexer, etc., required to measure the magnetic field via the shim plot. As a service tool, the Array Shim Device is not part of the delivery volume of the MR system and is brought on-site only as required.

Gradient System

In addition to the basic magnetic field, the gradient system generates linear dynamic magnetic field gradients in the three room orientations for spatial encoding. The gradient system includes the gradient amplifier and the gradient coil.

Gradient Amplifier - GPA

The gradient amplifier consists of the components necessary for generating and regulating the gradient currents. This includes the electronics cabinet, the power-line filter, the gradient power-supply lines and the electronics for monitoring the temperature of the gradient coils.

Gradient Coil - GC

The gradient coil comprises the coil systems for generating the gradient fields, the system for cooling the coil, as well as the sensors for temperature monitoring plus the hardware for installing the coil in the magnet. The gradient coil also includes the compartments for passive shimming of the magnet's main field and the electrical shim coils for active shimming.

RF System

The RF system generates the RF fields and transmits them to the patient (transmit function). It also receives the MR signals emitted by the patient and provides the signals in digital format for image reconstruction (receive system).

RF Small Signal Unit - RFSU

The RFSU converts the digital transmit signal from the control unit to an analog signal and mixes it up to the system frequency. Parallel to this, the RFSU converts the analog MR signals picked up by the local coils into an intermediate frequency and digitizes them into digital format for demodulation and filtering by the receive hardware in the Measurement Control section.

RF Power Amplifier - RFPA

The RFPA amplifies the RF signal processed in the RF small signal unit to the amplitude necessary for exciting the hydrogen nuclei (1H). For diagnostic applications, the defining parameters are peak and continuous output as well as the reproducibility of pulse patterns, i.e., temporal stability, since these parameters represent the prerequisite technical conditions for realizable examination protocols and the reproducibility of examination results.

RF Application System - RFAS

The RFAS is a group of components whose function is to interface the common RF front-end to the various transmit and receive coils. The TAS is a switch unit responsible for directing the transmit signal to either the Body Coil or a transmit-capable Local coil. The BCCS provides the interface components required for the circular polarized Body coil system and includes preamplifiers when the Body coil is used to receive. The BTB provides the impedance matching components required to compensate patient influences for optimized RF transmit energy efficiency. The TALES is an RF RMS voltmeter responsible for measuring the forward and reflected RF transmit energies. These values are used as the basis for the SAR monitoring. The LCCS acts as a switchboard for the various RF receive channels coming from the Local or Body coils and includes preamplifiers for the MR signals as well as a signal combining circuit to allow combination of adjacent coil elements to a single receive channel for increased FoVs for a limited number of receive channels. The RF cables routed to the resonators as well as the transmit lines from the filter plate to the BCCS are also part of the RF application system.

RF Coil Interface - RFCI

The RFCI provides the coil control signals for dynamic de-tuning and the necessary operating voltages for the RF components in the examination room.

RF Filter - RFFIL

MR examinations are performed in an RF-shielded room or cabin which protects against external interferences and unacceptable RF exposure. This room as well as the RF-sealed feed-throughs for the cables are part of the RF filter panel.

Body Coil - BC

The system includes a whole body transmit/receive coil for emitting RF to the patient for the slice excitation and also as a receive coil for producing full FOV localizer scans used normally for slice positioning. The body coil is impedance tuned to the patient before the start of the sequence. Manual tuning elements are found within the coils allowing an adjustment of the coil elements for service purposes. The Body Coil also serves as the inner cover lining of the magnet bore.

Local Coils

Anatomically optimized receive coils pick up the sensitively small MR echoes and preamplify them out of the noise floor. These signals are transferred to the RFAS over a connector-cable system within the Patient Table. Most Local Coils are circular polarized for optimized signal reception and all Local coils do not require patient-related tuning.

Spectroscopy - MKO

The MKO spectroscopy option allows the use of different nuclei (standard imaging uses 1H) to perform Single Voxel Spectroscopy (SVS) and Chemical Shift Imaging (CSI). The option requires additional hardware (broadband RFPA, SAMI) and a software package.

Patient Handling

The Patient Handling system provides components for interaction with the patient such as the patient table, lighting, ventilation and patient monitoring. Prior to the MR examination the patient is positioned on the table using the positioning accessories for additional comfort. The patient is then moved into the magnet bore. During the examination the patient is monitored accordingly, via a video camera and or physiological monitoring equipment.

Patient Table - PTAB

The patient table is used to position as well as move the patient into the magnet bore. The motor-assisted table movement is controlled via the front side control units located at the magnet. The standard positioning support (excluding the specific positioning supports for the local coils) as well as the patient trolley are components of the patient table. The standard patient table is designed for partial body applications up to 140 cm. An optional Panoramic patient table functionality is available for automated table movements.

Patient Trolley - PTROL

To facilitate changing patients, a removable tabletop carrier and corresponding trolley are also provided as an option. The patient trolley together with the removable tabletop of the patient table allows for patient positioning outside the RF room. In the examination room, the patient lying on the removable tabletop is lifted onto the patient table with the help of the trolley. No additional patient positioning is required.

Intercom - COM

Since the patient is usually alone in the examination room during the measurements, patient and physician communicate via the intercom. The intercom consists of the configurations necessary in the examination room and the operating console.

Patient Monitoring - VID

Whenever required (no direct face-to-face interaction between physician and patient, e.g. through a window), the patient is monitored during the examination via a video camera. The set-up consists of the camera, the monitor as well as the cabling.

System Control

System control comprises all of the system's computer functions:

- Host computer
- image reconstruction system
- measurement control
- physiological signal acquisition.

The system control is the main communications interface for the overall system. It converts examination protocols or user input to the corresponding control signals and transmits these signals to the affected components via a CAN (Controller Area Network) communications bus. It also monitors the functional status of the various subcomponents and processes these accordingly for system monitoring.

Host Computer - HOST

The hardware comprising the host computer includes all processor components required for patient data management, image handling (image display, post-processing, evaluation, hardcopy documentation, archiving), and measurement sequence programming. This includes all elements of the processor's user interface, e.g. monitors, satellite console, console table, power distribution MRC, keyboards, mouse, and data carriers, etc.

Image Reconstruction System - MRIR

The image reconstruction system computes the final images from the digitized receive signals of the RF small signal unit. Optional software applications also allow automated image post-processing functions to be performed by the Imager in real-time.

Measurement Control - MMC

The Modular Measurement Control has its own processor system which performs the RF, gradient, timing and acquisition events described within the sequence. It also provides the hardware for signal generation and generation of the hardware specific dynamic control signals as well as a communications network providing the channels for controlling the peripheral components and providing status and error feedback. The cables and power supplies necessary for controlling, communicating with and monitoring the peripheral components are part of the measurement controller package.

Physiological Signal Acquisition - PMU

The PMU acquires the ECG signals, the pulse and respiration of the patient and forwards it to the measurement controller. The PMU electronics contains all sensors, voltage supplies as well as the cables for communication with the measurement controller.

In-Room MRC

With the optional In-Room MRC it is possible to start measurements inside the examination room and check the images immediately for supporting interventional tasks.

System Environmentals

Line Power Distribution - LPD

The line power distribution (aka. Power Distribution System - PDS) supplies the system components with electrical power. The power distributor ensures that the system is connected at a central point as specified to the power-supply side of the system. The power distributor provides the different components with individually fuse-protected supply voltages. The main power distributor is located in the main electronics cabinet which includes space for installing additional components.

Cabinet Cooling System - CCS

The CCS provides cooling of the electronic components with the CCA and GPA cabinets. The CCS comes in two versions:

- water cooled - primary water is used to provide heat exchange of warmed air. Air conditioning of the technical room is not necessary.
- air cooled - uses cooled air from an air conditioned environment for the internal cabinet cooling

Cooling System

The main power-consuming components of the system are water-cooled. Cooling is performed with water at a temperature of 20°C. This keeps condensation from building up on the cooled electronics components and eliminates the need for complicated insulation measures, as long as basic values for temperature and humidity are maintained.

The water can be tempered using a water cooler (chiller = external cooler) or chilled in-house water. A regulated water-water heat exchanger (RCA) discharges lost heat via an existing cold water pipe network (on-site primary cold water).

System Cabling - CABLE

The system cabling includes all control, signal, data, communications and supply cables that have to be attached to the system during on-site installation.

Software

The software used in the MAGNETOM Harmony, Symphony and Sonata is NUMARIS 4. This is a completely new design, with adoptions to the hardware, especially the MMC.

Options

To meet the customer´s needs in an optimal way, some additional components will be available as options.

UPS - for Host and imager only
Patient Monitor - video monitoring of patient
Advanced Shim - option is available for better magnetic field homogeneity and subsequently better image quality. Five shim coils are already part of the gradient coil.

List of Abbreviations:

System Documentation Manuals:

DIA	Diagram Manual
FUN	Functional Description
INS	Installation Manual (2 volumes)
LOG	Logbook
PGD	Planning Guide Deutsch
PGE	Planning Guide English
TSG	Trouble Shooting Guide

General:

ABT	Advanced Bipolar CMOS Technology
ACU	Air Condition Unit
ACT	Advanced CMOS Technology
ADC	Analog Digital Converter
AGP	Accelerated Graphics Port
ANSI	American National Standard Institute
AS	Active shielded coil
AS39s	Active shielded coil for project 039
ASIC	Application specific IC
AVC	Active Vibration Control
BC	Body Coil
BCCS	Body Coil Channel Selector
BLOB	binary large object
BNC	Bayonet connector named after Neil-Concelman
BPL	Backplane
BR	Bildrechner (image reconstruction)
BTB	Body Tune Box
CAN	Computer Area Network
CCA	Control Cabinet
CCS	Cabinet Cooling System
Codec	Compressor and Decompressor
CORA	Cost Optimized RF Amplifier
COV	(Magnet)-cover
CORE	Clinically Optimized Regional Exams
CP	circular polarized
CPU	Central Processing Unit
CSE	Customer Service Engineer (replaces FSE)
CV	Coil Voltage (in RF small signal)
CW	Control Word
DaLi	Data Link
DAC	Digital Analog Converter
DB	Database
DCL	Demountable Current Lead (Magnet)
DC	Direct Current Input (RF small signal)
DDR SDRAM	Double Data Rate Synchronous Dynamic RAM
DIMM	Dual In-Line Memory Modules
DORA	Double Resonant RF Amplifier (RFPA)
DICO	Directional Coupler
DSL	digital subscriber line
DSP	Digital Signal Processor
DSV	Diameter Spherical Volume

DVD	Digital Versatile Disk
DYSCON	Dynamic Switch Control (Pin-diodes)
ECC	eddy current compensation
ECG	electro-cardiogram
ED	DSP to calculate the ECC
EFI	External Field Interference
EIS	External Interference Shielding
EPR	electronic patient record
ERDU	emergency run-down unit
ETL	ERDU test load
FOC	fiber optic cable
FoV	Field of View
FBG, pcb	Printed circuit board (germ: Flachbaugruppe)
FRU	Field Replaceable Unit
FSC	Full Scale
FWHM	Full Width at Half Maximum
GAL	Generic Array Logic
GC	Gradient Coil
GC-DSP	Gradient and Control-DSP
GCTX	Board for gradient and TX-control (also RF!)
GPA	Gradient Power Amplifier
GPS	Gradient Power Supply
GSSU	Gradient Small Signal Unit
HDCL	hardwired de-mountable current lead
HF	High Frequency
HN	connector type for high voltages
ICE	Image Calculation Environment

IDE	Integrated Device Equipment
IDEA	integrated development environment for Application
IF	Intermediate Frequency
Imager	Image Processor (also MRIR)
IMD	Intermodulation Distortion
IOP	Input/Output and Power-Board
IPA	Integrated Panoramic Array
IQ	Image Quality
ISA	Industrial Standard Architecture
ISO	International Standard Organization
LAN	Local Area Network
LC	Local Coil
LCCS	Local Coil Connector Switch
LO	Local Oscillator
LP	linear polarized
LPD	Line Power Distribution
LNA	Low Noise Amplifier
LVD	Low Voltage Differential
LWL	Lichtwellenleiter (fiber optic cable)
MC	Measurement Control
MC4C40	p.c.b. used in the MMC with 4 DSPs TMS320C40
MDH	Measurement Data Header
MHSEL	measurement hybrid select (germ: Meßhybrid-Select)
MMC	Modular Measurement Control
MOFI	ASIC for Modulation/Demodulation and Filtering
MPS	Magnet Power Supply

MPCU	Measurement, Physiological and Communication Unit
MRIR	MR Image Reconstruction
MRSIG	Magnetic Resonance Signal
MRSC	MR Satellite Console
MSC	Measurement System Control
MSU	Magnet Supervision Unit
MSUP	Magnet Supervision
MWS	Mantelwelle Sperre (RF Trap)
N	connector type named after Paul Neill
NCO	Numeric Controlled Oscillator
NF	Line Filter (germ: Netz Filter)
NOE	Nuclear Overhauser Effect
NUMARIS	Nuclear Medicine Acquisition and Reconstruction Imaging Software
OCXO	Oven Controlled Crystal Oscillator
OEM	Original Equipment Manufacturer
OMT	Oxford Magnet Technology
OVC	Outer Vacuum Chamber
OR24	1.0 T magnet type for MAGNETOM Harmony
OR70	1.5 T magnet type for MAGNETOM Symphony
PAT	Parallel Acquisition Technique
PCB	Printed circuit board
PCI	Peripheral Component Interconnect-Bus
PCIx	extended PCI bus
PCU	Physiological Communication Unit
PD	Phase Detector
PDS	Power Distribution System
PF	Power Forward
PIO	Port In Out
PLL	Phase Locked Loop
PMS	Patient Monitoring System
PMU F	Physiological Monitoring Unit Frontend
PPC	Power PC, used in MPCU
PS	Power Stage (gradient system)
PS	Power Supply
PSHIM	Passive Shim
PR	Power Reflected, Power Reverse
PT	Patient Table
PTAB	Patient Table
PTFE	Polytetraflourethelene (teflon)
psia	pounds per square inch absolute
psig	pounds per square inch gauge
QLA	quick latch - subminiatur coax RF connector
RC	Resonance Converter
RCA	Refrigerator Cabinet
RFAS	RF Application System
RFCI	RF Cabin Interface
RFPA	RF Power Amplifier for 42MHz, 63MHz
RFSU	RF Signal Unit
RFSWD	Radio Frequency Safety Watchdog
ROI	Region of Interest
ROPE	Respiratory Ordered Phase Encoding
RT51	Operating system of the intelligent CAN nodes
RX	Receiver

RX4	Receive board for 4 RF channels
SAMI	Spectroscopy Amplifier Interface
SCSI	Small Computer System Interface
SCRU	Screening Unit (see MSUP, Magnet Supervision)
SCT	Sequence Control Task
SE	Spin Echo...
S/E	Transmit/receive... (germ: Sende/Empfang)
SeSo	Service Software^
SGA	switched gain amplifier
SIPMOS	Siemens Power MOSFET
SLIO	Serial Linked Input Output
SPS	Shim Power Supply
SSB	Single Side Band
STIMO	stimulation monitor
TALES	Transmit Antenna Level Sensor
TAS_C	Transmit Antenna Selector Circular
TiP	Table plate (germ: Tischplatte)
TP	Carrying plate (germ: Trägerplatte)
TRA	Transmitter
TX	Transmit
UPS	Uninterrupted Power Supply
USB	Universal Serial Bus
VCCO	Voltage Controlled Ceramic Oscillator
VCXO	Voltage Controlled Crystal Oscillator
VM	Virtual Machine
VPN	virtual private network
VxWorks	Operating system of MPCU
WAN	Wide Area Network

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Section

3 Software

This section covers the most expensive part of the system: software. You can't see, smell, taste or feel it, yet it is a major cause of eye, heart, lower back and stomach diseases as well as hair-loss and impotency. Scientific studies have found, however, that approx. 0.2% of the population is immune to the side-effects of software. Through observation of this minority group it could be proved that these individuals exhibit typical characteristics. It is not allowed to reveal these details to the general public, but we can say this much: they like to have all their socks in one draw and feel an affiliation to the letter "C++". If you belong to this group, we know you'll enjoy this section. The rest of you, wear the appropriate safety gear, exercise due caution and visit your doctor regularly.

Introduction

This description will give a *general* overview of the *syngo* MR software and its components. The main emphasis will be the service relevant aspects of the software. In most cases, repairing software problems will entail either fixing a bad configuration or re-installing software entirely. But, to satisfy curiosities, some background knowledge of the software and what is it SUPPOSED to do will also be presented.

Overview

Software has now become, by far, the single most expensive component of our MR systems. It has and is being developed by more than 300 programmers across four continents and is continually growing in functionality, size and number of unexpected features.

To gain our bearings, lets first define the software components running on the system. In general, the **software** components running in the system can be classified into one of three groups:

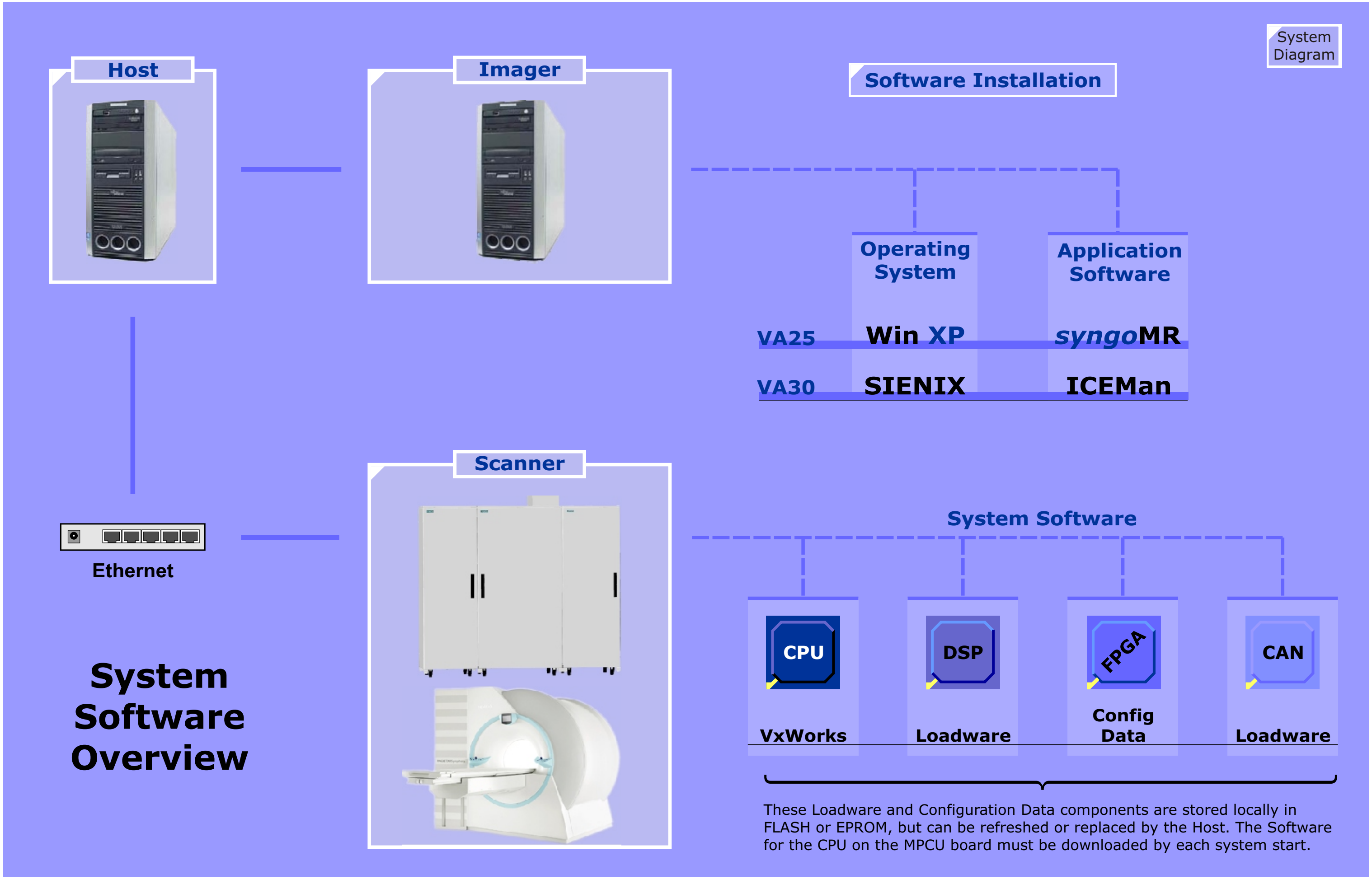
- Operating System (OS)
- Applications (*syngo*MR)
- Scanner Peripheral Control (CPU and FPGA loadware)

Of course, there can be no software without hardware. The **hardware** components running one or more of these software groups :

- MRC and MRSC **Host**
- MR Image Reconstruction MRIR (**Imager**)
- **Scanner** hardware peripheral units such as the Measurement, Physiological & Communication Unit (MPCU) and various low-level hardware and peripheral controllers

The software components mentioned above which require interaction from and with the user or service engineer will be looked at more closely, that is primarily the software running on the Host. The Imager software is invisible to the user and generally does not require configuration or handling and therefore will not be covered, although some tips will be given which may help determine whether it is running properly. The scanner software is discussed in the Control System section.

Figure 4 *System Software Overview*



Imager Software

The software for the Imager as such will not be covered since there are no service actions involving direct interaction with the Imager software. There are some issues, however, that will be mentioned here that could be helpful when dealing with the Imager.

Remote Connection

One of the problems faced when dealing with imager problems is the fact that it has no monitor or keyboard which would allow a look "inside". A method of establishing a virtual remote connection to the Imager is possible by using the **Remote Desktop** program of Host's OS Windows XP.

Using this remote connection requires that the LAN connection to the Imager is working and the Imager is in a condition to accept the connection. If the Imager's software is out to lunch, then this procedure will not be very helpful.

Direct Connection

If the remote connection over the Host is not possible, then the only other alternative to see what is going on inside the Imager is to connect a monitor and terminal to it.

A monitor is also necessary to determine whether all the raw drives are being recognized. The raw drives can only be seen during the boot of the Imager.

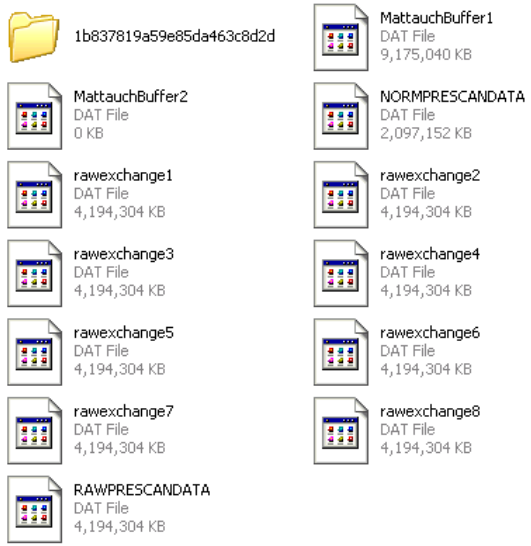
The Q Disk

The presence of the Q Disk MED_SYSTEM Q:\"Imager IP number" on the Host is an indication that the communication link to the Imager was successfully started.

Grafik: Help > Info Disks... highlight Q

Raw Data Disk F:

Below is the contents of drive F:. You can see that each of the four raw data drives contains two 4GB raw exchange buffer files. The other files are also spread out over the four disks. This is what you'll find on it:



Next

Imager syngo General License

The Imager uses components from the general syngo package and therefore requires a license. The license file is located on the Host under the directory C:/MedCom/tftp and on the Imager under C:\MedCom\config\licensing.
The Host copies this file to the Imager. If the Imager is changed, a new license file will need to be installed on the Host. Follow the replacement procedures correctly!

Host Software

The Host software includes a Microsoft OS and the *syngo* MR® software applications required for setting up scan protocols, image post-processing, image archiving functions and other organizational tasks such as providing the imager and hardware controllers with software during the scanner boot up procedure.

The software differences between MRC and MRSC are determined by the **configuration** after a software installation.

syngo®

syngo is a software platform developed especially for medical systems and applications and provides the basis for modality specific control and applications as well as applications for a wide range of post-processing and other radiological workflow-related functions. *syngo* has a modular structure and can be divided into these major functional blocks (refer to Figure 5):

- the *syngo* Platform
- the *syngo* Applications
- the *syngo* User Interface

syngo Platform

The *syngo* platform utilizes a Common Software Architecture (CSA) which provides fundamental functionalities such as basic image processing tools, a DICOM data model, an image database and networking capabilities.

A **Patient Browser** enables access to and allows one to navigate through the patient Database. Images in the database can be sent, exported and imported to and from remote network nodes or local mass storage media. The Patient Browser is **configurable** to adapt to individual workflow preferences.

syngo Applications

The common *syngo* applications are the same for all syngo implementations accross all modalities and include the Viewing, Filming and 3D processing functionalities.

MR Specific *syngo* Applications

The application software developed for MR systems includes:

- Exam - image producing application
- Mean Curve
- ARGUS Flow
- Perf MR - Neuro Perfusion
- BOLD
- Spectroscopy Evaluation
- Nero 3D - 3D Offline fMRI

syngo User Interface

syngo provides a user friendly graphical interface providing a common interface across all modalities, workstations, radiological and clinical information systems implementing the *syngo* software.

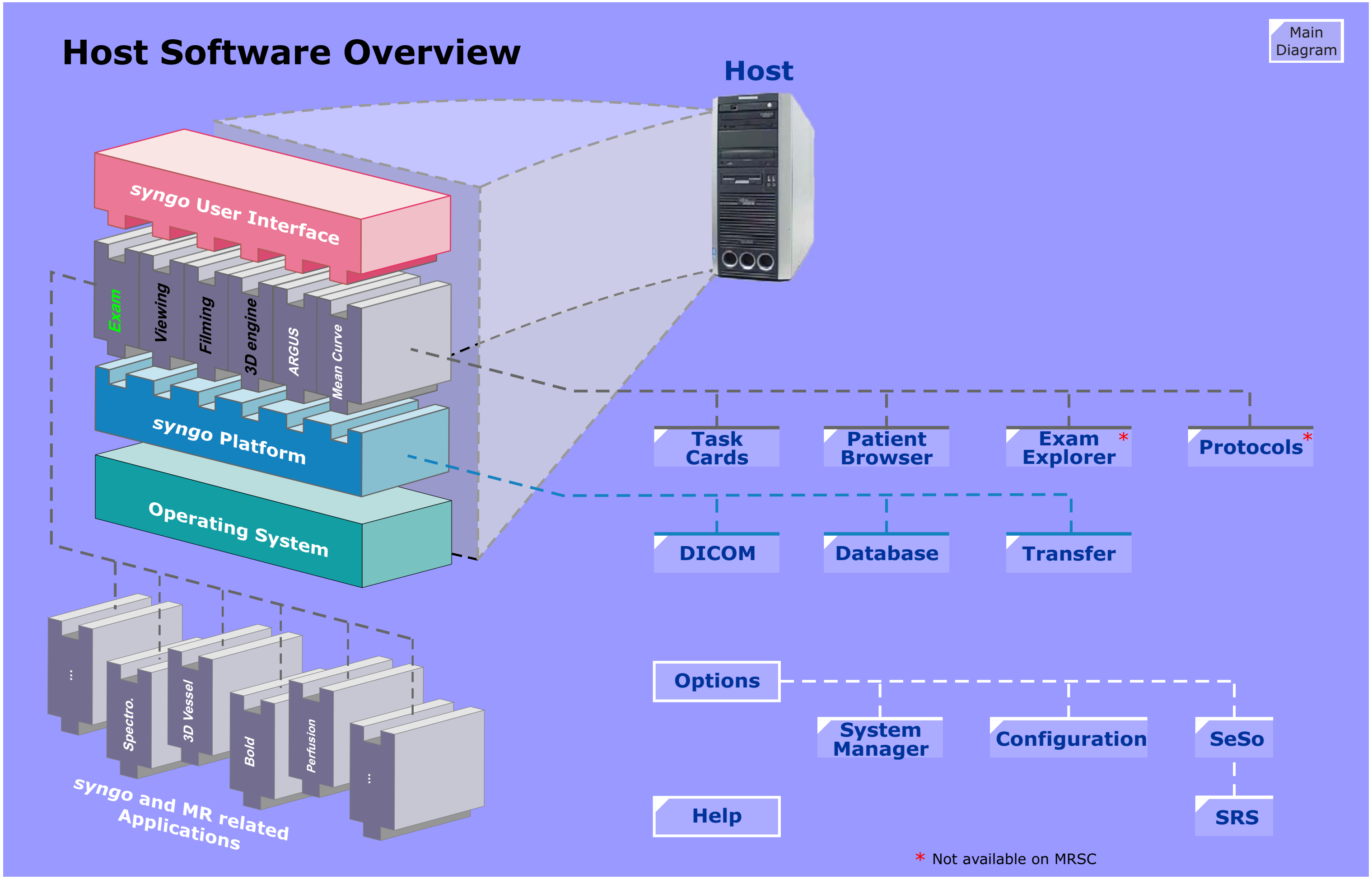
The *syngo* software together with the MR *specific application packages* is called *syngo* MR software

Much of the software just mentioned can be configured and adapted to the workflow requirements and personal preferences. This will be discussed throughout this section.

Scanner Software

The downloading of the scanner software is described in the CONTROL section.

Figure 5 *Host Software Overview*



syngo MR

Task Cards

The *syngo* and *syngo*MR Applications are loaded onto Task Cards (see Figure 6) which are automatically started when the Host is booted or can be started manually.

The arrangement of the task cards reflects routine workflow in the hospital or practice and their layout supports the examination procedure.

Task Cards which are produced by the syngo software factory and available on all *syngo*-based modalities are :

- Viewing
- Filming
- 3D (option)

The **Filming** Task Card will only be loaded after a DICOM camera has been configured. If a DICOM camera is not physically available, it is possible to configure a dummy camera in order to get the Filming Task Card. This task card is an integral part of syngo and does not require additional licensing.

The **3D** Task Card is an optional package requiring additional licensing.

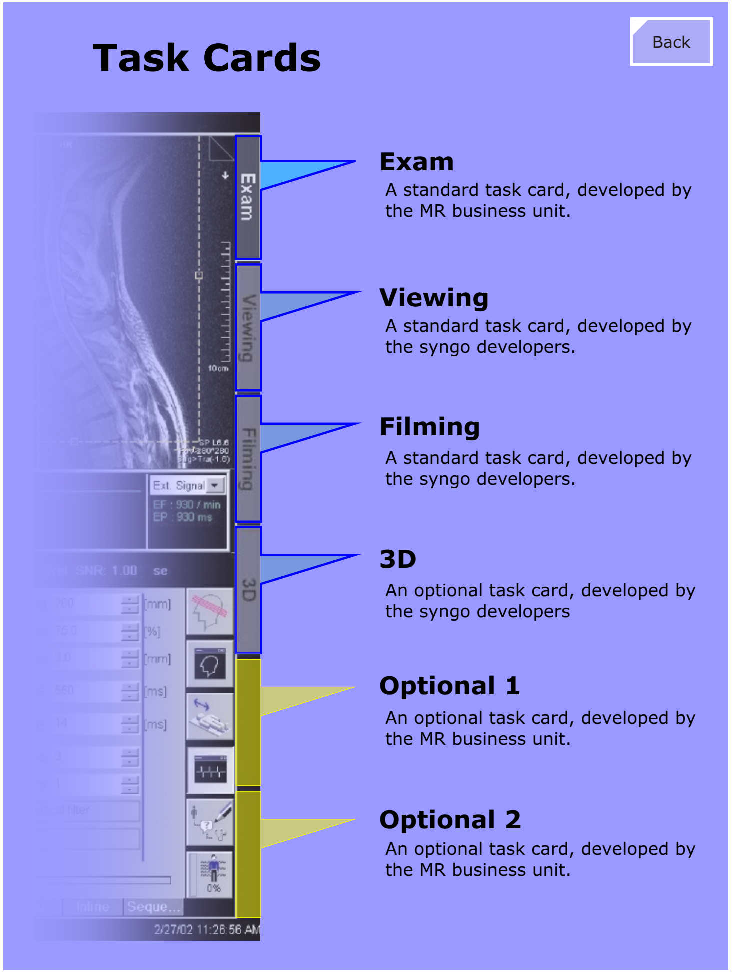
syngo MR specific Task Cards

- **Exam**
- **Argus**
- **Bold**
- **Mean Curve**
- **Spectroscopy**
- **3D Vessel View**
- **Perfusion**

Task Cards in **orange** are optional

The **Exam** Task Card is used to define and run MR examinations and is only available on the MRC. All other MR specific task cards are optional and will require additional licenses to be enabled.

Figure 6 *syngo MR Task Cards*



Licensing

The 3D optional Task Card and optional software application packages require licenses before they can be used. Licenses must be ordered explicitly for the MRC or MRSC. A license issued for the MRC will not run on the MRSC or vice versa.

Licenses can only be ordered from either the Licensing Center in Forchheim, Germany or from your Uptime Service Center or Regional Service Center in your country or time zone.

Task Card Menus

Patient Browser menu

Patient Applications Transfer Edit View Filter Evaluation Sort Private Applications Options Help

Viewing Task Card menu bar

Patient Applications Transfer Edit View Image Tools Evaluation Scroll System Options Help

Filming Task Card menu bar

Patient Applications Edit Film Image System Options Help

Although all the functions and commands found under these menus are described in very minute detail in the Operators manual (print number MR-05000.621.01) there are several functions that are useful to know for Service engineers and will be discussed here :

- The Patient Browser
- The Transfer Menu
- The Exam Explorer
- The System Menu
- The Configuration Menu

The Patient Browser

Overview

The Patient Browser is a user interface for displaying patient images stored on either the local patient database or stored on external mass media (CD, DVD). The Patient Browser can also be used to import and export images in the local database to and from CD/DVD as well as initiating image transfers over a DICOM network connection. These functions are described in the Image Transfer section below.

Configuring the Patient Browser

The Patient Browser can be configured for numerous display options, such as selecting to display the images in either a "tree" mode graphically or "list" form as a text list. It can also be configured to display various patient and/or scanner information and parameters as well as the Work Status flags, which is described below.

Opening the Configuration Mask

The configuration mask for the Patient Browser is started under the Options > Configuration menu. The window appearing will look much like the Windows Control Panel, having several icons for the various configurable entities. Under them you will find the Patient Browser. Clicking it will call up the Browser configuration mask.

Setting Work Status Flags

Figure 7 shows the top half of the *General* configuration mask with the various Work Status flags that can be configured. The Work Status flags serve to support the work flow of the imaging process. Once images have been created, the will not stay indefinitely on the modality, but will be printed, sent to an archive device or archived locally to mass storage media. Flags can be set according to the user’s work flow. Once images have been printed or sent to archive, the flags are used to indicate that the images have reached their final destination and can now be deleted from the modalities database to free up room for further images. The local database should only be considered a t e m p o r a r y storage.

If the "**Confirm Deletion**" switch has been enabled, the software will display a window when the user attempts to delete images whose appropriate flags as set in the "**Permit Delete**" section have not been set.

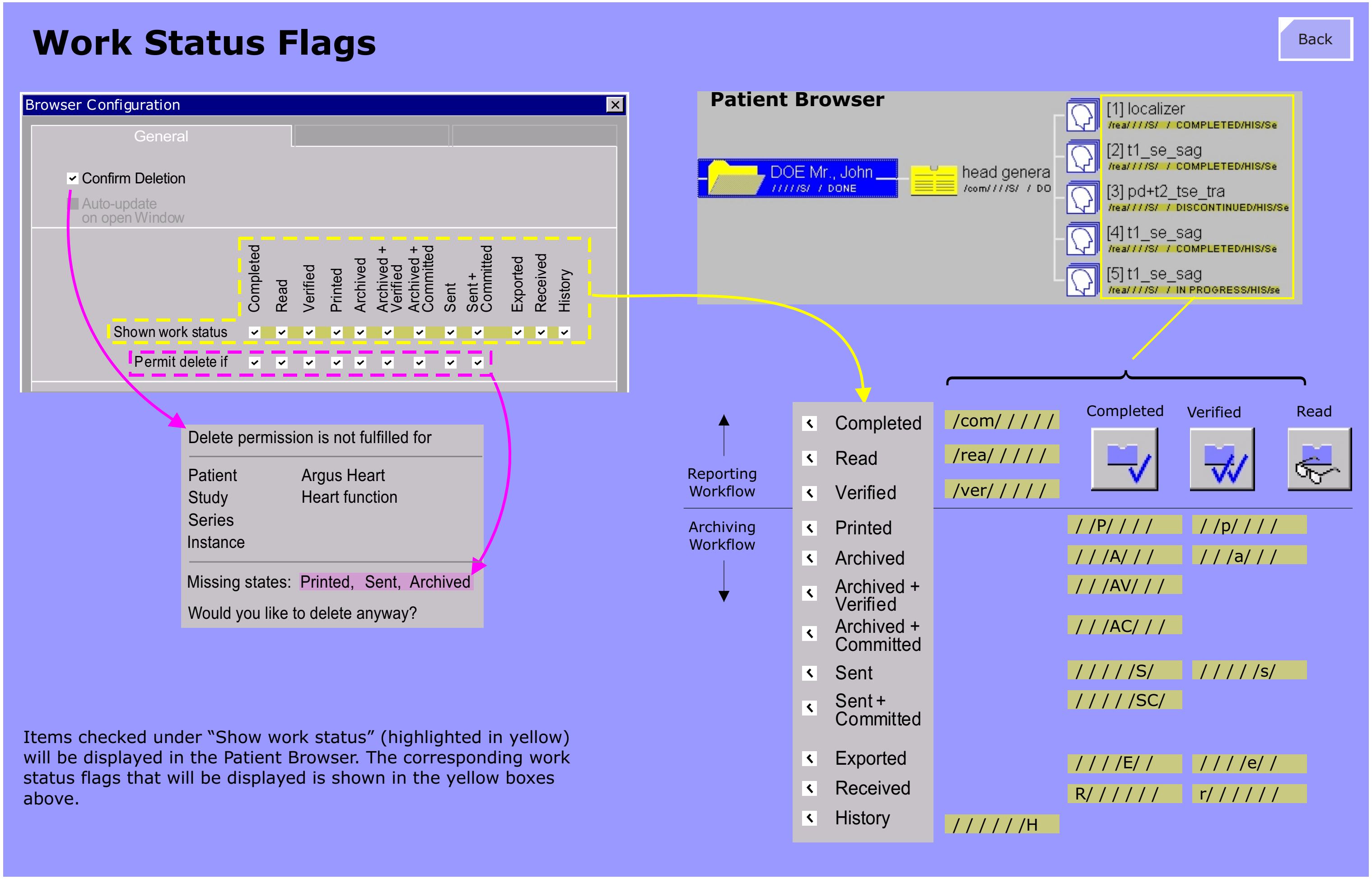
The "**Show Work Status**" section selects which work status flags will be displayed in the Patient Browser, as shown in Figure 7. THese flags are primarily grouped into "Reporting Workflow" and Archiving Workflow". The Reporting workflow flags are set with buttons on the Patient Browser. The Archiving flags are set automatically by the software. More on this can be found in the Image Transfer section below.

Image Protection Flags

In order to prevent unintentional deletion of patient images there is a built-in protection mechanism using work status flags. *Printed*, *Archived*, *Sent* and *Received* are set by the software automatically.

The work status flags are shown in the tree view of the patient browser automatically. Additionally it can be configured, which of the flags are displayed in the tree view and the list view of the patient browser.

Figure 7 *Work Status Flags*



Items checked under “Show work status” (highlighted in yellow) will be displayed in the Patient Browser. The corresponding work status flags that will be displayed is shown in the yellow boxes above.

Image Transfer

Transferring images to mass media or remote network nodes is performed via commands located under the Transfer menu found on the Viewing Task card or the Patient Browser.

Figure 8 *Transfer Menu*

Back

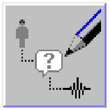
There are several commands found under the Transfer menu to allow transfer of images to other network nodes or mass storage media. These are described in the following table.

Work Status Flags

Command	Description	Flag set
Archive to CD-R	Used to archive images to the local CD ROM. The CD burner is configured as an archive device during the software installation.	A
Archive to	Used to archive data to computer systems over the network which have been configured as a DICOM archive node. When selected, a dialog box opens with a list of the available archive nodes.	A
Send to	Allows sending images over the network.	S
Export to	Used to burn images in DICOM format to a CD-ROM, even if the DC ROM is not configured as an archive device.	E
Import from Off-line	If an import / export directory is configured in the service software configuration the **Import from Off-line** can be used to import patient data in DICOM format from this directory into the local database. The import / export directory can be defined as an internal directory at the own host or as a remote directory shared at any other network computer.	R
Export to Off-line	If an import / export directory has been configured, this command can be used to export patient data in DICOM format to this directory. The import / export directory can be defined as an local directory or as a remote directory shared at any other network computer.	none

The Flags that are set will only be displayed within the Patient Browser if they have been set to be displayed which is described under the Setting Work Status flags section on page 29.

Exam Explorer



The Exam Explorer is a program to view and manage the examination protocols using a graphical interface, a la Microsoft's Windows Explorer. With the Exam Explorer it is possible copy or move protocols between **Regions**, **Exams** and **Programs** using the same drag and drop techniques know from the Microsoft Windows operating system. Right-clicking the protocol brings up a context menu providing commands to modify, rename, copy or delete protocols.

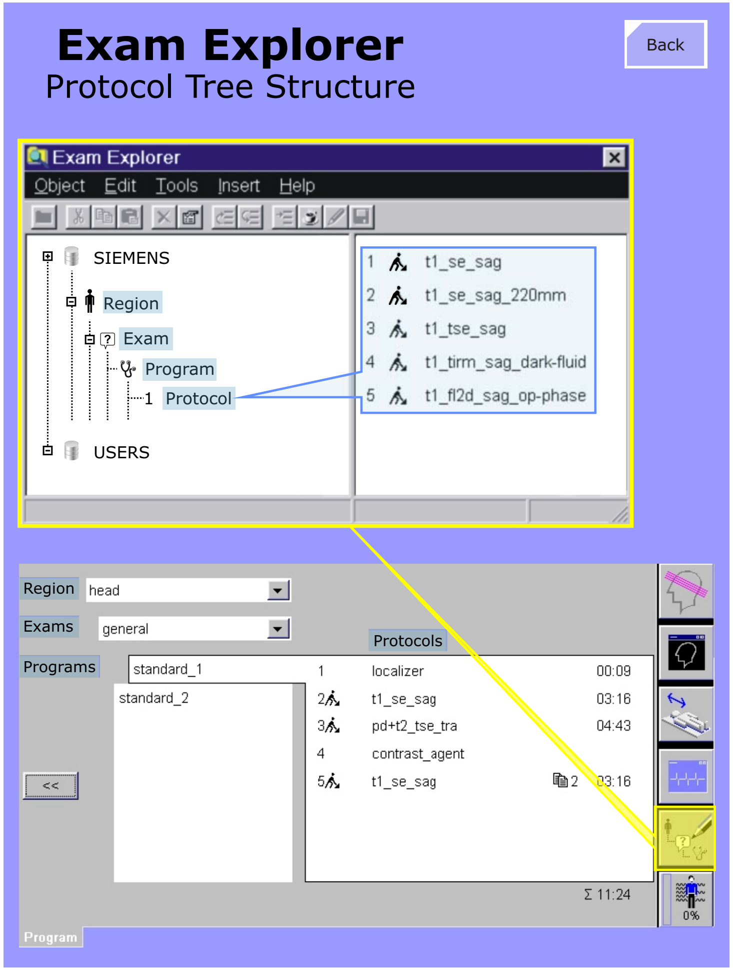
The Exam explorer can be opened from the Exam task card by either select **View** > **Exam Explorer** from the main menu or via the icon show above.

The Protocol Trees

Protocols are found under two main groups: **SIEMENS** and **USER**. The protocols located under the SIEMENS tree are factory defaults and read only, they cannot be modified. The USER programs can be modified and are used by the customers for their examinations.

Protocols are organized within a strict hierarchy as shown in Figure 9. The top level consists of **Regions.** Regions correspond to the anatomical region to be examined. Regions consist of **Exams,** which in turn contain **Programs** which are a grouping of **Protocols**.

Figure 9 *Exam Explorer*



Protocol Management

A New Begin

After installing and configuring software during a new system installation all the default SIEMENS protocols are copied to the USER tree (refer to Figure 10). At the moment, this is accomplished by starting the executable file C:\medcom\bin\MRExamDbTool.exe and performing the "Copy SIEMENS to USER tree" function. This simply copies the delivered Siemens protocols to a customer directory. The Siemens default protocols cannot be modified.

Backup of Protocols

After a new installation, application specialists will visit the customer for training and to help the customer optimize the protocols for the applications they will be using. Also, the customers themselves will probably modify and save protocols in the User tree as time goes on. Although it is part of the yearly preventative maintenance to make a Backup of the customer protocols it would, however, be a good idea to do this immediately after the application specialists have made their visit and another one or two months later since the customer will probably make many protocol modifications themselves.

The **N4Customer** package in the Backup and Restore tool in SeSo is used to make a backup of the customer (User) protocols. This Backup routine performs a COMPLETE backup of ALL protocols under the USER tree. That is, all standard (default) protocols that were originally copied from the SIEMENS tree as well as protocols that were copied or modified by application specialist and/or the customer afterwards. The archive flag (see below) plays no role in this Backup routine.

The Archive Flag

When the standard (default) Siemens protocols are modified in any way and saved, an archive flag will be set. The purpose of the archive flag is somewhat confusing and so will be explained. Unlike the archive flag known under Windows and other operating systems, the protocol archive flag is not used to make a delta copy of modified protocols since the last backup. This flag is only evaluated when protocols backed up from an earlier software version are to be restored to a system that has been updated with a newer software version (see below).

Restoring Protocols

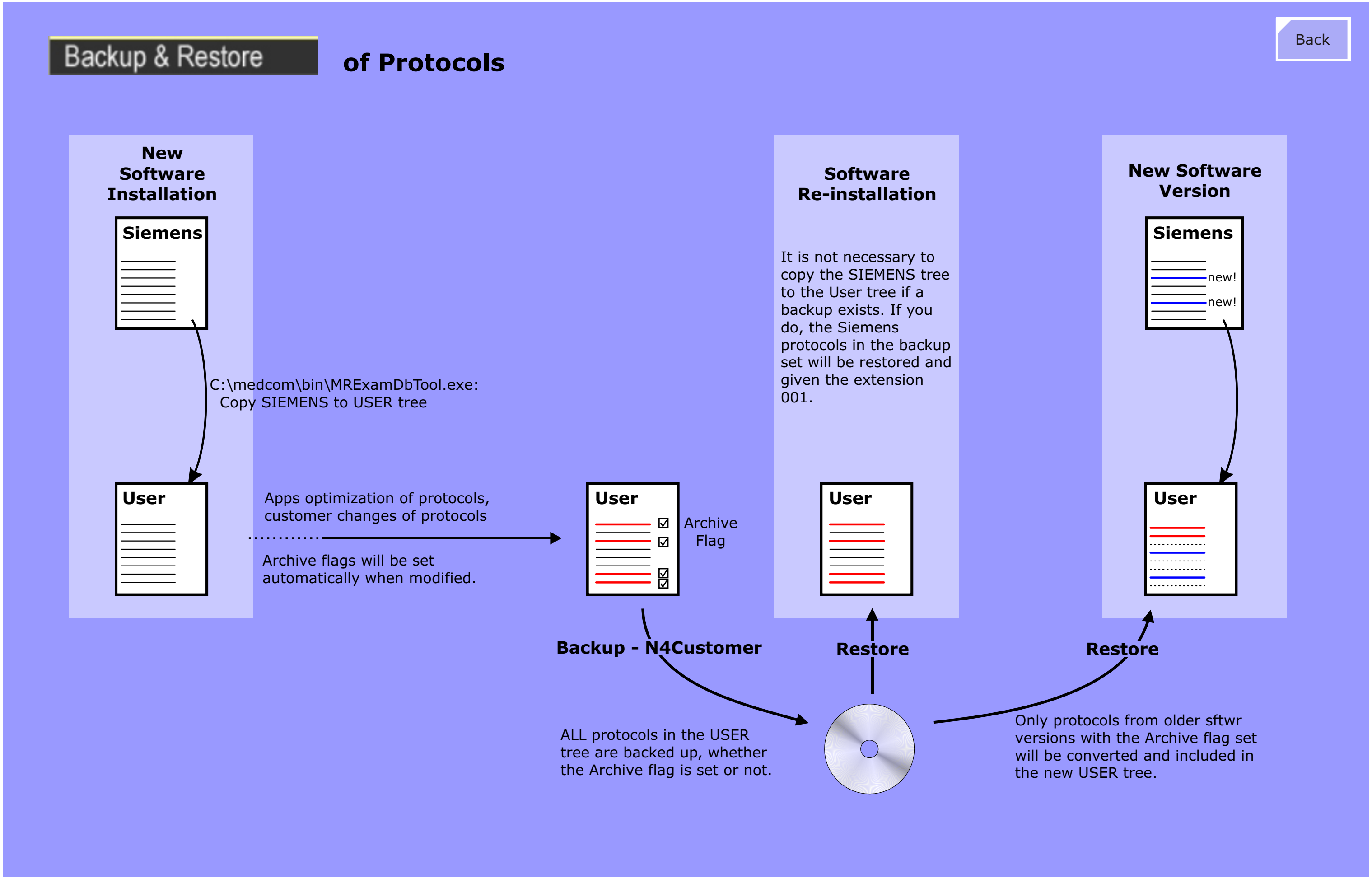
When restoring protocols after a software re-installation or software update (new software version) the User tree will have all the protocols it had when the Backup was made. So far, so good.

If, however, a Restore is to be made because the customer accidently deleted one or more protocols, keep in mind, that a Restore simply restores EVERYTHING, including those protocols that were not deleted or lost. In this case, the protocols in the backup having the same names as those in the User tree will be restored but given the extension 001. All of these protocols are now doubled! Inform your customer about this.

When the time comes to install a new software version or update, normally new and improved protocols will be included. Of course the customer will want to keep the optimized protocols from the previous version so a Restore of the protocols from the previous version will be made to the new software version. As mentioned above, only the protocols with the Archive flag set will be converted and restored. Standard protocols not having the archive flag set are not restored if there is a newer version.

If the customer decides it wants to keep the old protocols, then someone (?) will have to MANUALLY set the archive flag for EVERY protocol. We now have a similar situation as mentioned above: many of the protocols in the old and new versions will have the same name and when a Restore of the old protocols is made to the new version you will have many protocols doubled. The old ones will have the extension 001 as mentioned above.

Figure 10 *Protocol Backup Flowchart*



Options > System Manager

The System Manager provides status information of the Host, Imager and MR scanner. It is invoked over the menu: **System** > **Control**.

Host Card

The System Manager / Host card displays the status of the loaded applications. The Exam-Task, Viewing-Task, Filming-Task and the 3D-Task are normally running (green arrow up). Optional tasks which are not loaded are marked with a red arrow down.

Applications which are hanging and not responding can be closed and restarted without having to shut down *syngo* MR or the operating system. It may work. But then again, it may not.

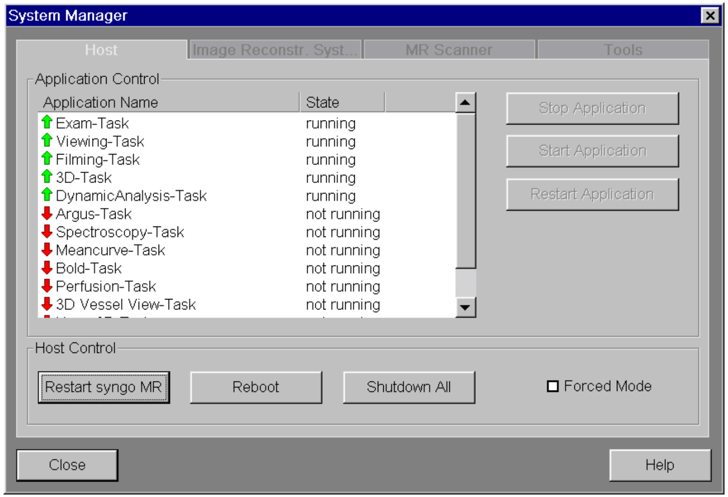
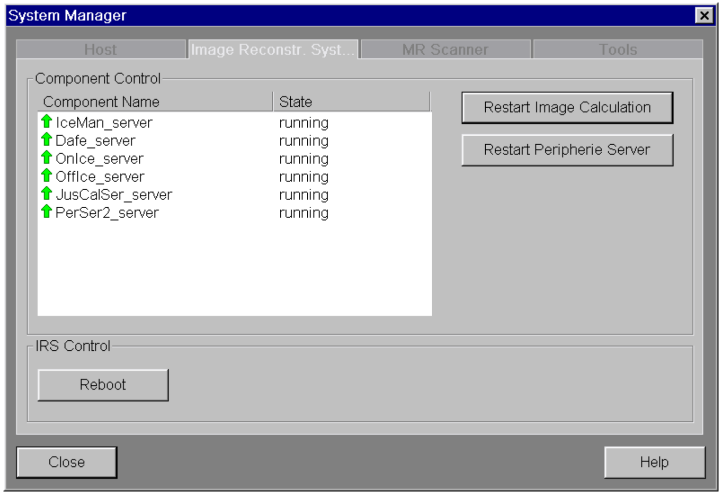


Image Reconstruction System Card

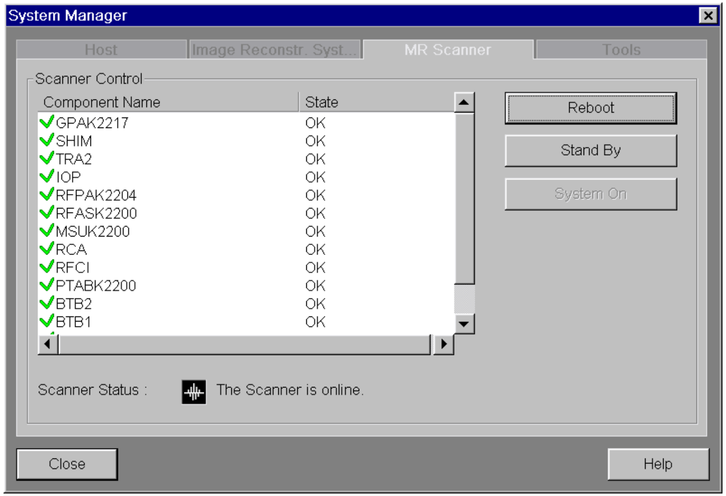
The Image Reconstruction System card displays the applications running on the Imager. From here you can reboot the imager hardware or attempt to restart the application servers. Use this only if you are feeling lucky.



MR Scanner Card

The state information is displayed for all monitored hardware components (i.e., have CAN components (SLIOs or Modules) or CPUs) within the MR scanner. These are listed under "Component Name". The current status of the components (OK, Not OK) is shown in the State column. The State flag here indicates that all conditions required for an "OK" state have been met, i.e., no errors or faults occurred or is in an undefined state. The list is updated automatically. During the reboot phase, the message "The Scanner is not online" will be displayed. When all scanner components are up and running "MR scanner is ready" is displayed.

The **System On** and **Stand By** buttons can be used to turn the scanner components on and off. The **Reboot** function is necessary to reset the NOT OK state if an error occurred.

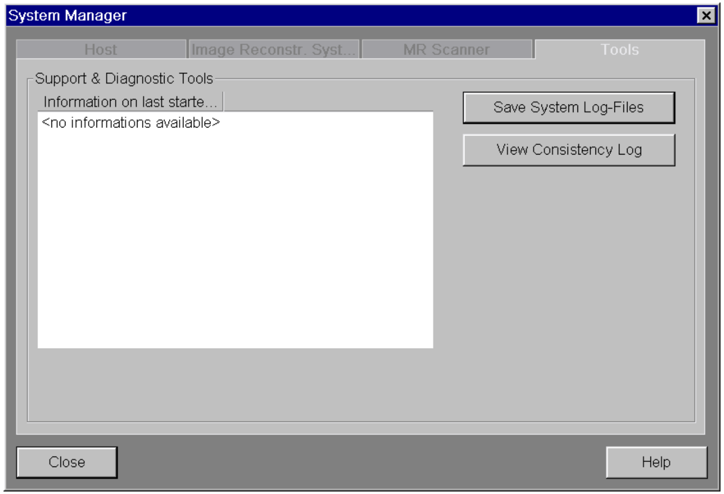


Tools Card

The Tools card is used to acquire diagnostic information or check important system files. If a software error occurs during scanning, it is recommended to create a MrSaveLog file immediately. It contains valuable diagnostic information valid at the time the error occurred. It is stored in the C:\MedCom\MriDiagnostic folder as a ZIP file.

The MR Consistency-Checker checks the system files for changes every time the system boots. The result is saved in a log file. You can view the content of this file in the Support & Diagnostic Tools window by clicking the View Consistency Log button.

CAUTION Never install additional software on the system. Doing so could replace existing system files and corrupt the scanner software.



Options > Configuration

Regional Settings

The user interface language and the keyboard settings can be configured under *Options > Configuration > Regional Settings*.

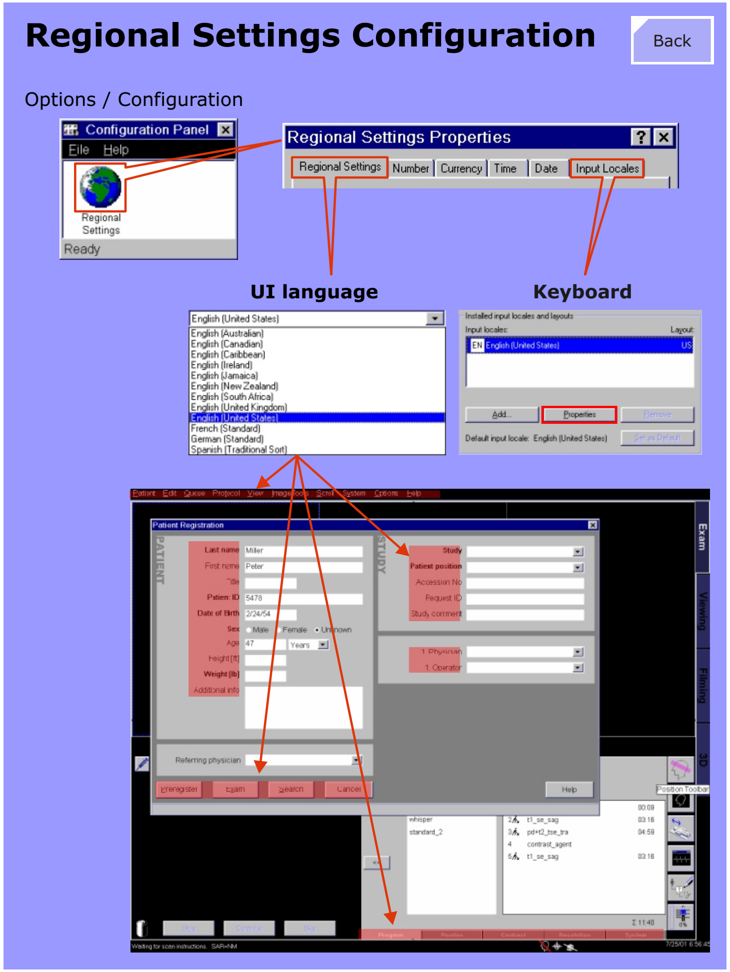
The language selection made here in the Regional Settings menu applies only for the Task Card menus, user interfaces and all fields (depicted by the areas in red in Figure 11) within pop-up windows as well as for the application online help.

NOTE An exception is the Service Software interfaces which are available only in the english language.

The language of the data displayed within the white fields in the Patient Browser and Patient Registration UIs is determined by the **DICOM Character Set**.

Setting up the time, including the daylight saving time, is discussed in Service Software.

Figure 11 *Regional Settings*

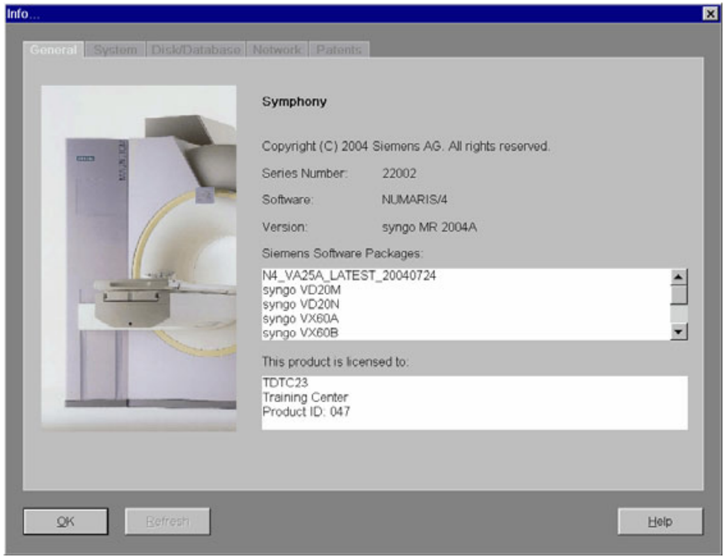


Help > Info

syngo MR also displays information about your scanner system. This includes information regarding memory capacity and availability. Select the menu **Help** > **Info** menu item.

General Card

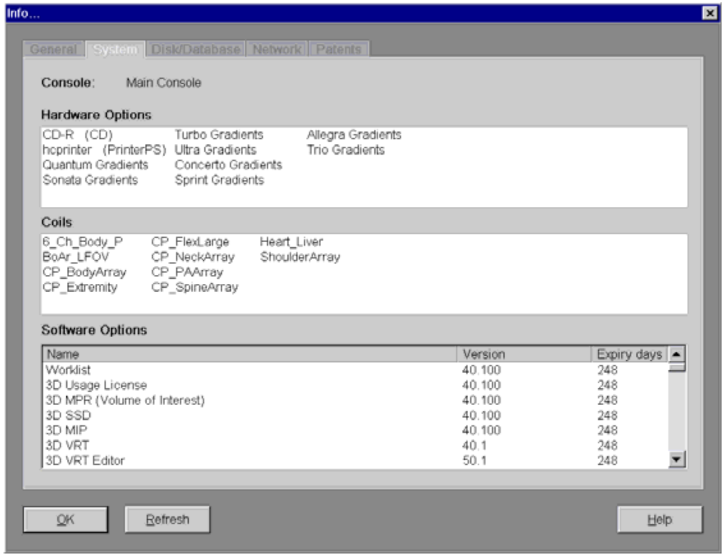
General system information is required, e.g. when a customer contacts Siemens Service regarding problems. Problems can be solved more quickly if we know the serial number of the system and the software version.



System Card

The System card provides further information about the scanner system. Here you will find information about hardware and software options installed as well as the available coils.

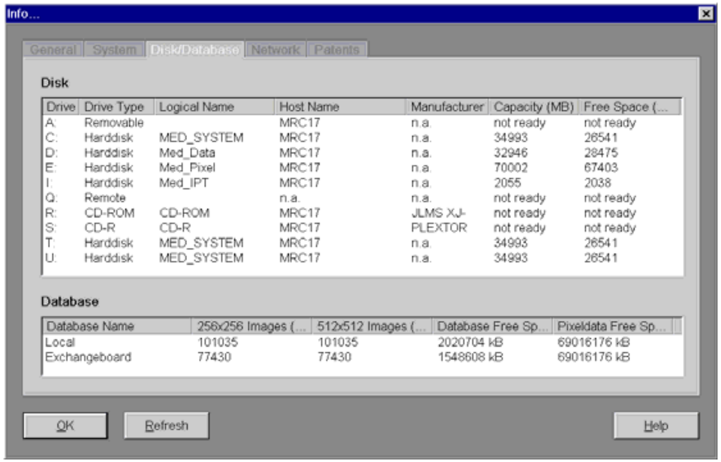
Also, the expiration date (if applicable) of software option licenses are shown.



Disk/Database Card

The **Disk** list displays all physical and logical (network) drives as well as their capacities and available space, expressed in MB and number of images.

All local databases are listed under Database. You are also shown how many images can still be stored in each matrix.

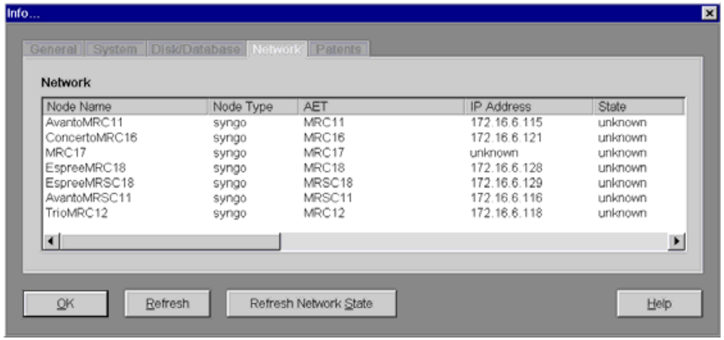


The following information is displayed:

- Name of the database
- Number of images 256 x 256 matrix images that can still be stored
- Number of images 512 x 512 matrix images that can still be stored
- Number of raw data sets that can still be stored

Network Card

If your console or satellite console is connected to a network, you may view network-specific information on the Network card such as the node names (computer names) and IP addresses.



DICOM

Digital Imaging and Communication in Medicine

Overview

This section provides some general DICOM information as it pertains to *syngo* MR. Detailed information about DICOM can be found any and everywhere else. The most comprehensive source of DICOM materials within the organization will be found at the Connectivity Competence Center (CCC) intrAnet page.

DICOM Services

DICOM provides several services which allows communication with other remote network devices for the purpose of sending, receiving images or other DICOM objects, reporting actual examination progress status and performed procedures (examinations, etc.), archiving images or other objects to mass storage media (CD, DVD, MOD) and also for printing hardcopy images to film. All of these services are described and standardized by the DICOM protocol.

Implemented DICOM Services

The following DICOM services have been implemented in the current version of *syngo* MR:

- Storage
- Storage Commitment
- Query/Retrieve
- Basic Print
- Basic Worklist
- Modality Performed Procedure Step
- Media Storage (CD, MOD and DVD)

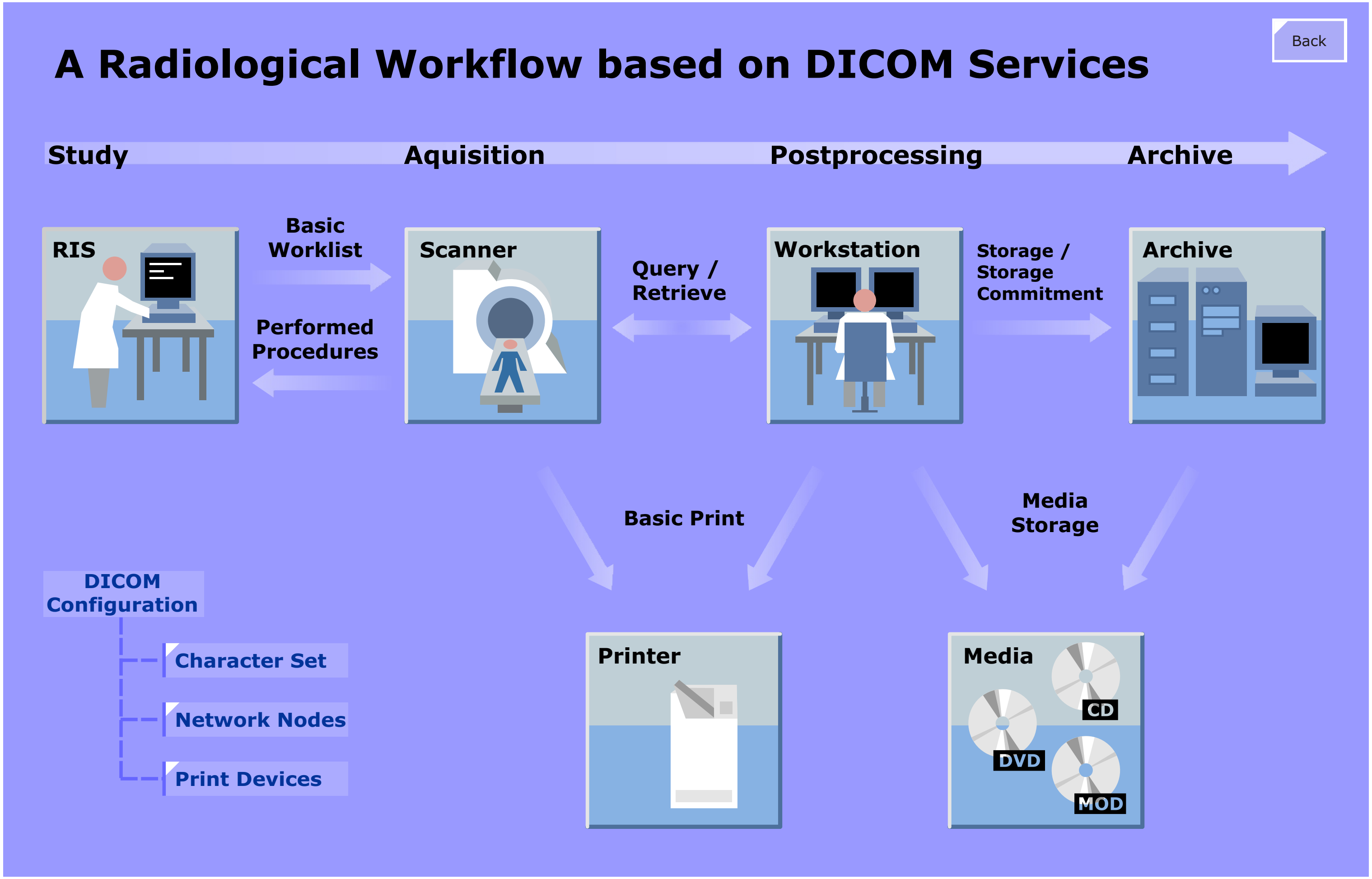
These services allow for a full integration of this Magnetom system into any radiological workflow scheme.

DICOM at MRC and MRSC

If the MRSC has been configured to share the database with the MRC, the transfer of images between the two units is NOT in the DICOM format but the native format of the Versant database. If the MRSC has been configured to have its own Local database, communication of images between the two units is performed using the DICOM protocol. In this case, both units must be properly configured for DICOM.

The DICOM Conformance Statements should be referred to for detailed information on the DICOM implementation i.e., which DICOM services and objects are supported as well as useful configuration information. The DICOM Conformance Statements can be found on the Siemens Intra- and Internet pages.

Figure 12 *Overview of DICOM Services*



DICOM Configuration

Overview

The DICOM services of both the local and remote DICOM nodes are configured under the Service Software (SeSo) configuration tool.

There are currently seven DICOM configuration pages :

- General
- Character Set (see Figure 13)
- Offline Devices
- Network Nodes (see Figure 14)
- Print Devices (see Figure 15)
- HC Overview LUT Files
- HIS/RIS Nodes

General

This mask provides several DICOM communication parameter settings such as local AE titles, maximum number of associations (simultaneous connections), protocol data unit sizes and time-outs. These parameters are usually set to default values, but have been provided to allow for "fine tuning" of DICOM communications to equipment with older DICOM implementations.

Character Set

The DICOM Character Set mask allows you to determine the character set which will be used for locally created objects, such as patient records created using the local patient registration and images created by the scanner (see Figure 13). Syngo, however, is capable of displaying all character sets for objects created by equipment using other character sets as those used in countries with latin characters sets. So, for example, if images created by a system in Japan using the Japanese character set where to be imported onto a system in EUrope, the Japanese characters would be properly displayed (not that it would do you any good...).

The DICOM character set is not to be confused with the fonts usage by the Windows XP operating system. syngo is a program running under the Windows OS, so the text in the menus and program fields will use the character set and language as defined in the "Regional Settings".

Currently three DICOM character sets are available.

- ISO-IR6: 7Bit ASCII e.g. for the US
- ISO-IR100: 8Bit ASCII e.g. Latin1
  includes special characters such as ä,ö,ü,ß, œ, é, ê and other strange characters used in many foreign languages.
- ISO-IR13: Japanese character set

Look at the online help for the selection of "multiple character sets". The selected character set will be written in the image header as DICOM attribute 0008, 0005.

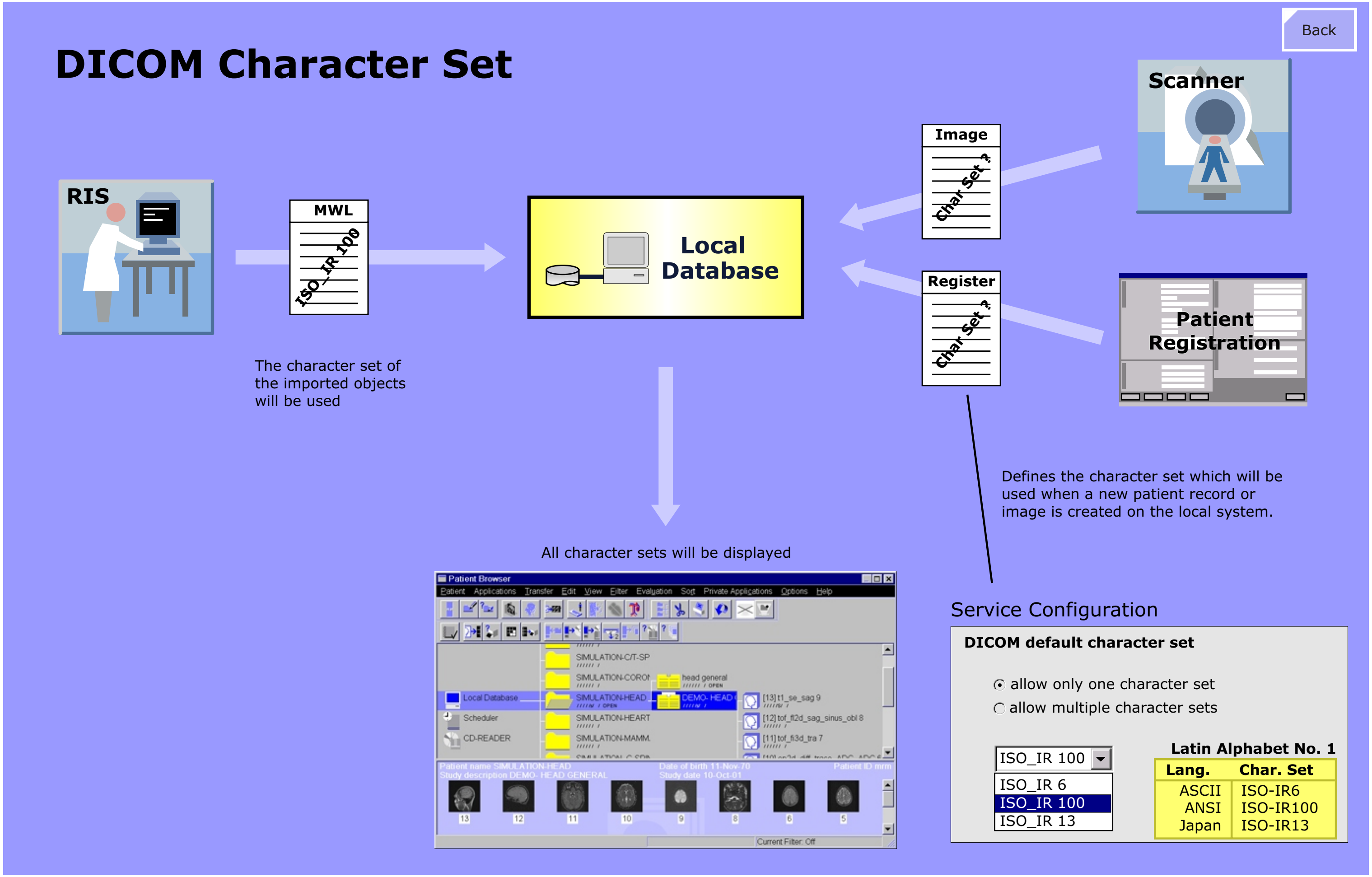
If a Radiological Information System (RIS) or Hospital Information System (HIS) uses a different character set for patient registration, the character set of the worklists will be taken over so that the patient records and images created by the modality will use that character set. The local character set setting will be over-ridden.

This behavior is true for patient registration only.

Offline Devices

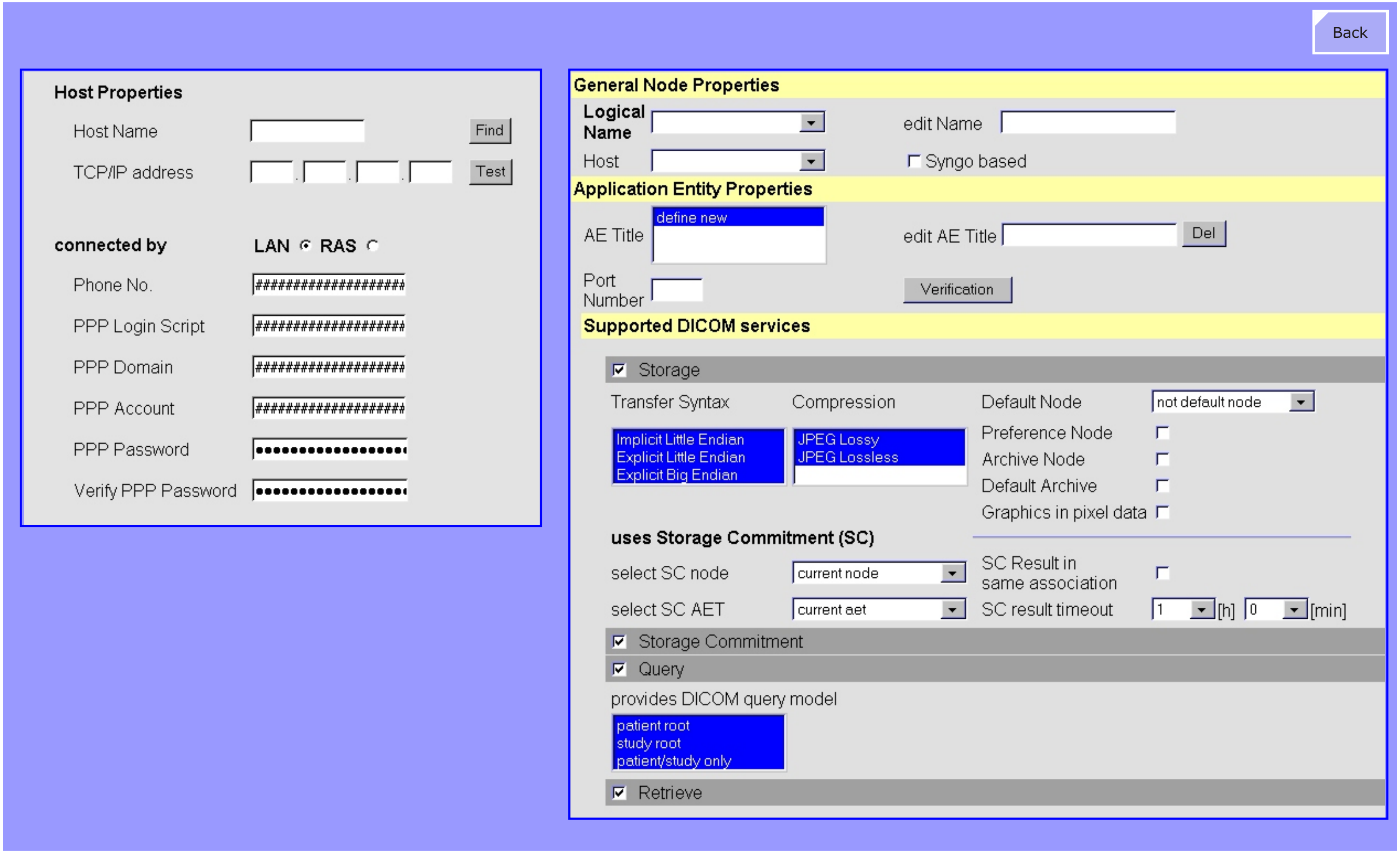
This mask provides the configuration of the local mass storage devices such as CD, floppies (!!!) and the older MOD optical drives. DVDs will be supported soon. Real soon.

Figure 13 *DICOM Character Set*



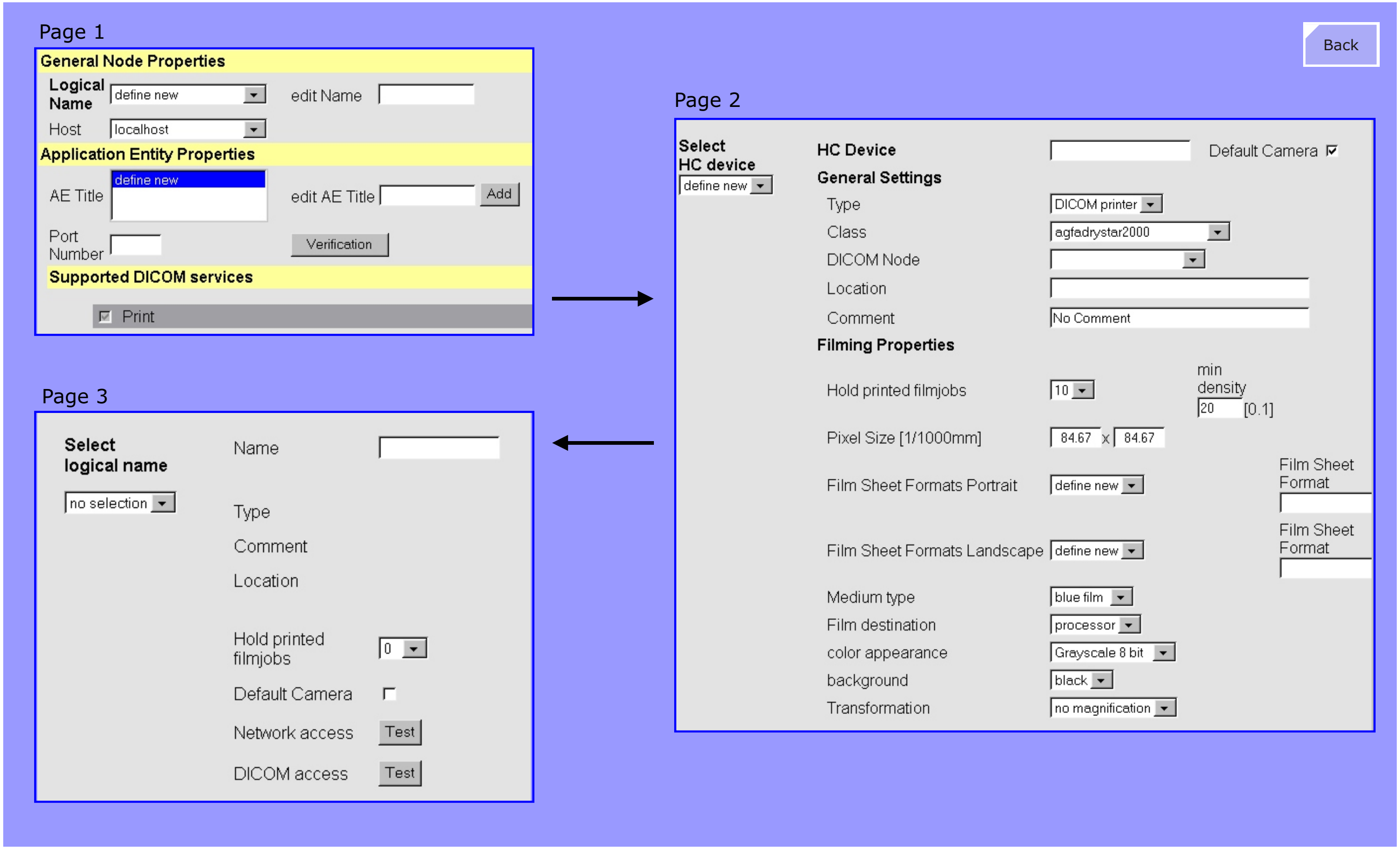
Network Nodes

Figure 14 *Configuration masks for DICOM nodes*



Print Devices

Figure 15 *Configuration masks for DICOM cameras*



Database

The database used by *syngo* has been structured to accommodate both local storage of patient images and the transferring of patient images and records to remote DICOM nodes. The database has built-in translators for translating native database records into the DICOM format and vice versa.

Overview

The database structure is seen in Figure 16. It consists of four individual databases:

- the local database (indigenous - native - inherent)
- the scheduler
- the archive
- the exchange board

Database Structure

The **Local** database (D:\Database\Local) is the main database containing the image header records as well as pointers to the actual pixel files stored on drive "E".

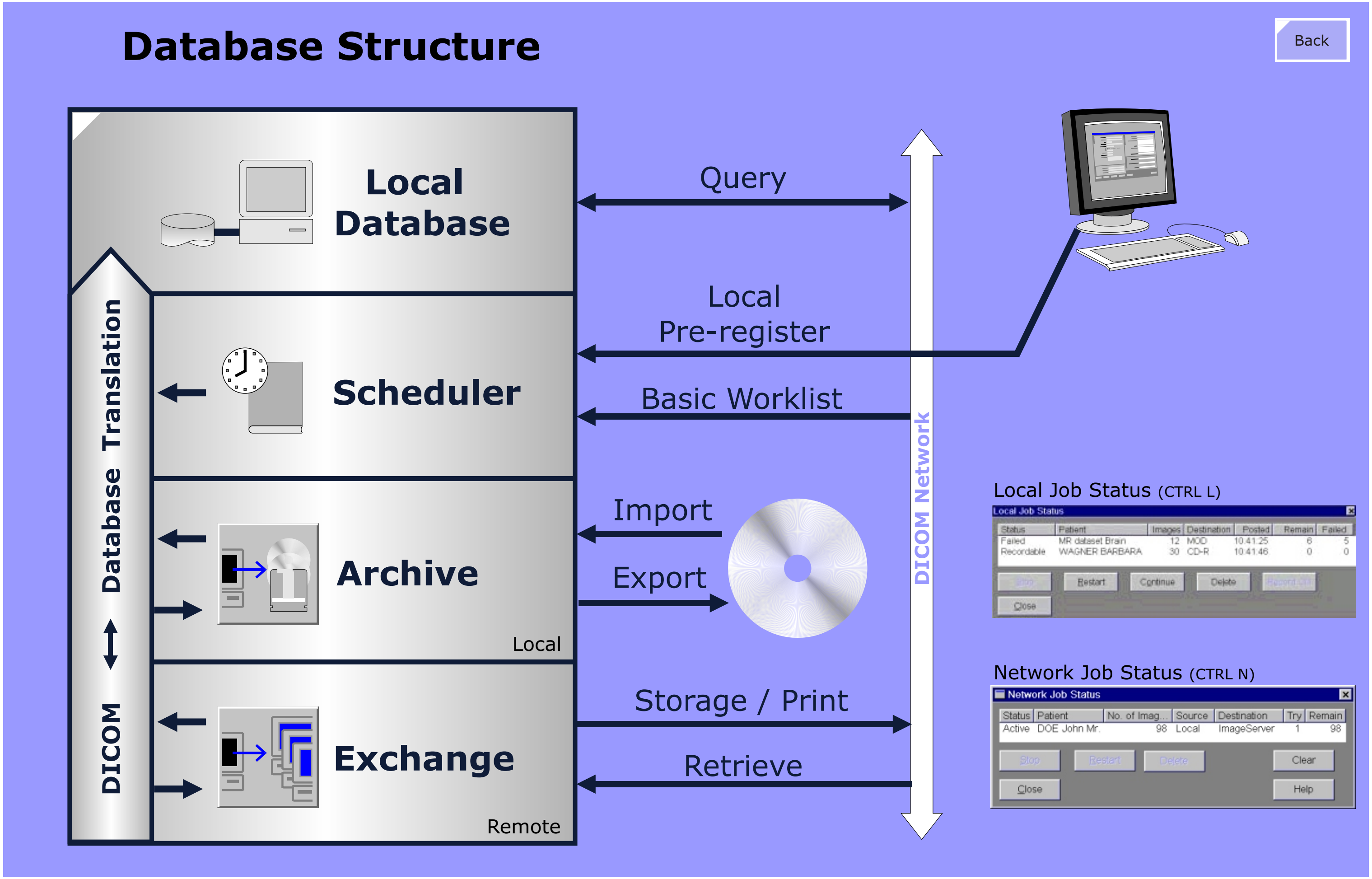
Entries in the **Scheduler** database (D:\Database\SchedulerStore) can be created by the pre-register dialog in the Patient Registration window or are retrieved from a HIS/RIS system via the DICOM Worklist service. These records are only temporary and will be transferred to the Local database as soon as the patient examination is begun. Therefore, the records in this database have no images and no pointers to drive "E".

The **Archive** database (D:\Database\ArchiveStore) is used as an Input/Output interface of the Local database to external mass media storage devices (CD, DVD, etc.).
When images in the Local database are to be written to mass media, a list of the images to be written is created which contains only pointers to the images in the Local database. The archive procedure takes the image records from the Local database, combines it with the pixel data to create DICOM files and writes them to the chosen mass storage device (which, by the way, must be p r o p e r l y configured in the Service Software). When the procedure is finished, the entries in the archive store database are deleted.
The archive store database is also used to import images from a CD to the Local database. If the CD has a DICOM format, it will be automatically mounted and the contents (DICOMDIR) displayed. With the "Transfer > Import" command the image headers can be imported into the local database and the pixel data are imported to the pixel drive "E".

The **Exchange Board** database (D:\Database\Exchangeboard) is an Input/Output interface to a network connection. When transferring images over a network connection, the header information and the pixel data is combined to produce a DICOM file and placed in the exchange board database from where it can be sent over the network. Also film jobs are placed in the exchange board database. After a film job is finished the last X number of film jobs are kept in the job list, where X is set in the "Print Devices" section of the configuration . This allows to restart a job in case of film jams or any other printer problems occur. When images are displayed for evaluation the relevant information of manipulations is stored in the exchange board database as well.

NOTE The host name is part of the image information in the local database. Therefore all images have to be archived and deleted on the local database before the host name is changed!

Figure 16 *Database Structure*



Physical Arrangement of Database

Figure 17 shows the physical arrangement of the databases in respect to hardware devices, namely disk drives and CD recorders or DVD drives.

MRC Database

On the MRC the database is distributed over two hard drives

- drive "D" (disk 1) is used for the databases. Local database, scheduler store, archive store and exchange board,
- drive "E" (disk 2) is used to store pixel data only,

MRSC Database

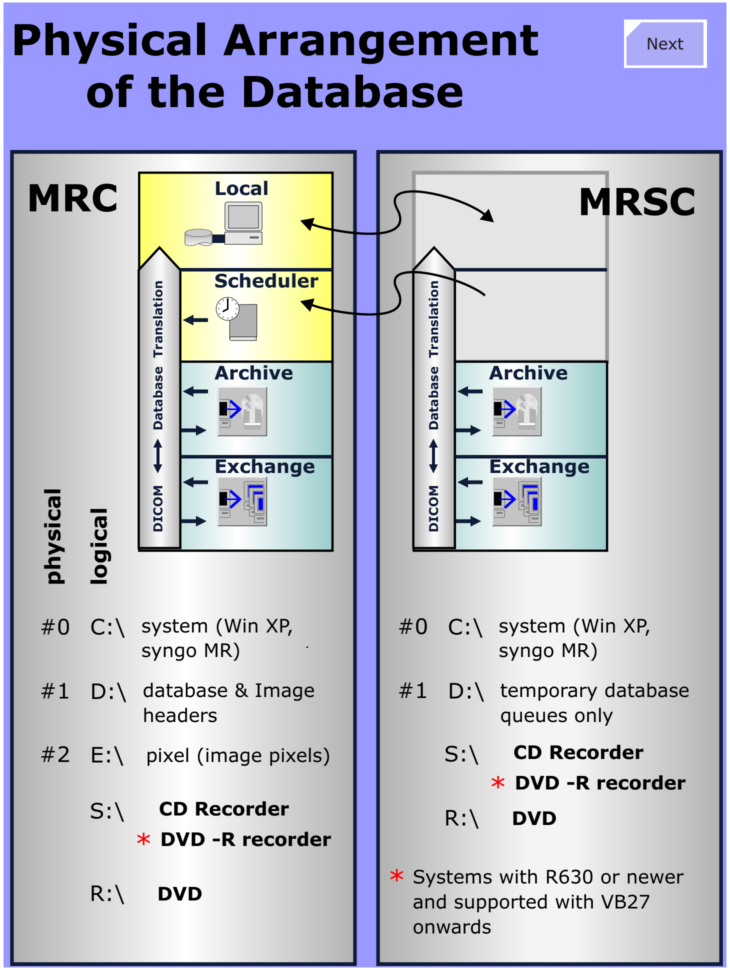
The MRSC uses two hard disks only:

- drive "C" (disk 0) is used as system software disk,
- drive "D" (disk 1) is used for the databases. Here just the archive store and exchange board databases are available,
- drive "S" (CD-R) is used to store patient data on CD-ROMs and
- drive "R" (DVD-ROM) is used to import patient data into the database.

There are two different database configurations possible for the MRSC. It can be configured to operate as a satellite of the database running on the MRC Host or set up to operate as a stand-alone database.

These two configurations are described on the following pages.

Figure 17 *Database Overview*



Shared (Remote) Database

If the MRSC is configured to share the MRC database, there will be only two databases present on the MRSC:

- archive store database (D:\Database\ArchiveStore)
- exchange board database (D:\Database\Exchangeboard)

In this configuration, the MRSC shares the Local database and the Scheduler database of the MRC Host database. Data transferred between the MRC database and the MRSC are NOT in DICOM format, but in the native Versant format. This has the benefit that images pulled from the MRC database do not have to be converted into DICOM first before being transferred. Another benefit in this configuration is, there is only one copy of images. The results of post-process operations performed on images are stored directly to the MRC database. There is also only ONE set of images and thus no extra care has to be taken that images are actual and up-to-date.

Setup of the MRC

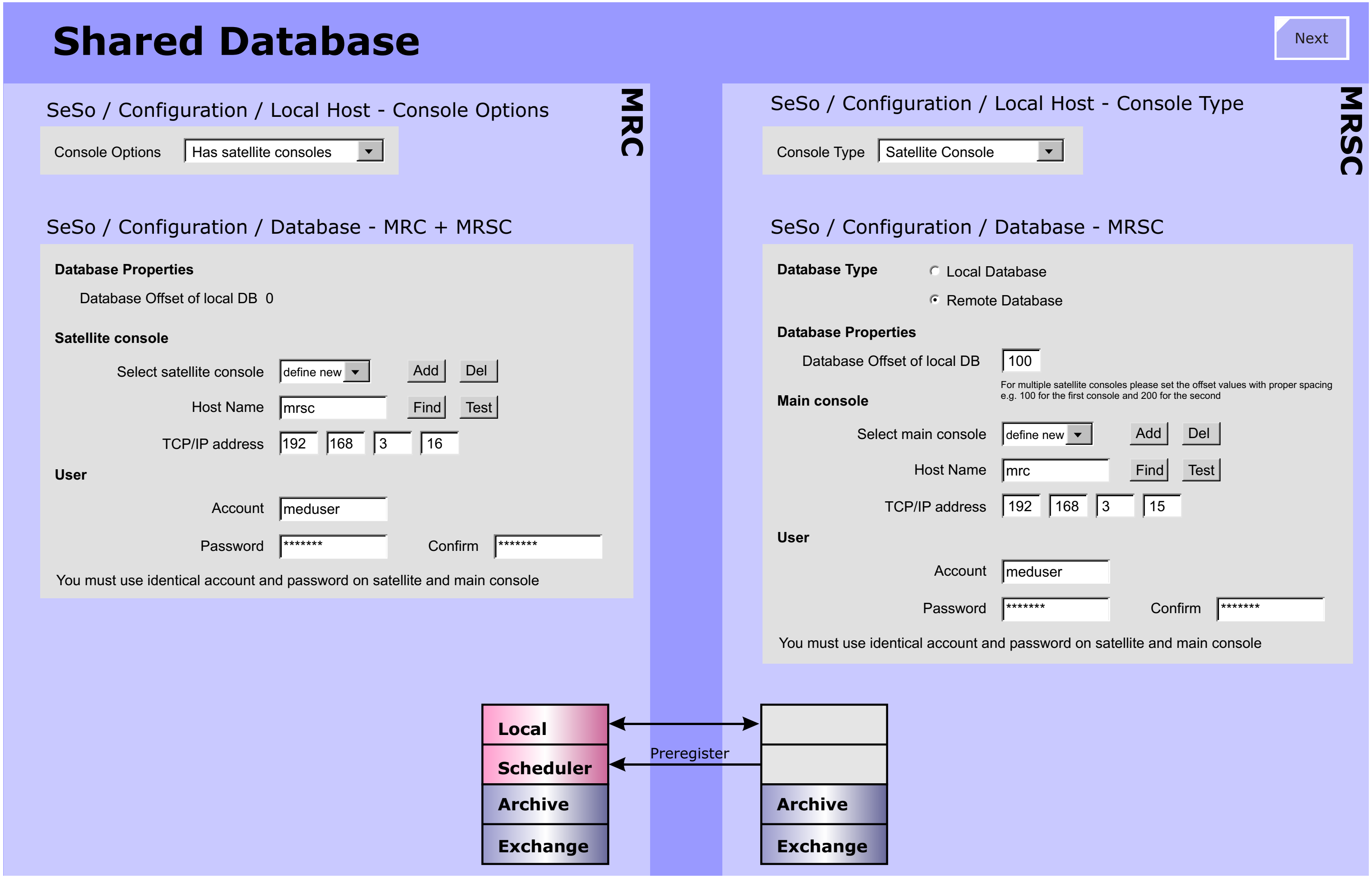
The *Console Options* under the SeSo / Configuration / Local Host to be set to "Has satellite consoles" to indicate that a satellite will be accessing its database. The database index is set to zero automatically

Setup of the MRSC:

- the "remote database" option must be set
- the database offset value set to 100

It is absolutely important to set up different database index values for the MRC and MRSC. It has to do with the way the databases communicate between themselves.

Figure 18 *MRSC Database Configurations*



Stand Alone (Local) Database

The second possibility is to configure the MRSC so that it has its own database. This configuration practically turns the MRSC into a workstation, much like a LEONARDO viewing station. In this case there will be four databases at the MRSC:

- local database (D:\Database\Local)
- scheduler store database (D:\Database\SchedulerStore)
- archive store database (D:\Database\ArchiveStore)
- exchange board database (D:\Database\Exchangeboard)

Setup of the MRC

If the MRSC is being set up with a stand-alone database, then the MRC must be configured to "Has no satellite consoles" under SeSo / Configuration / Local Host / Console Options.

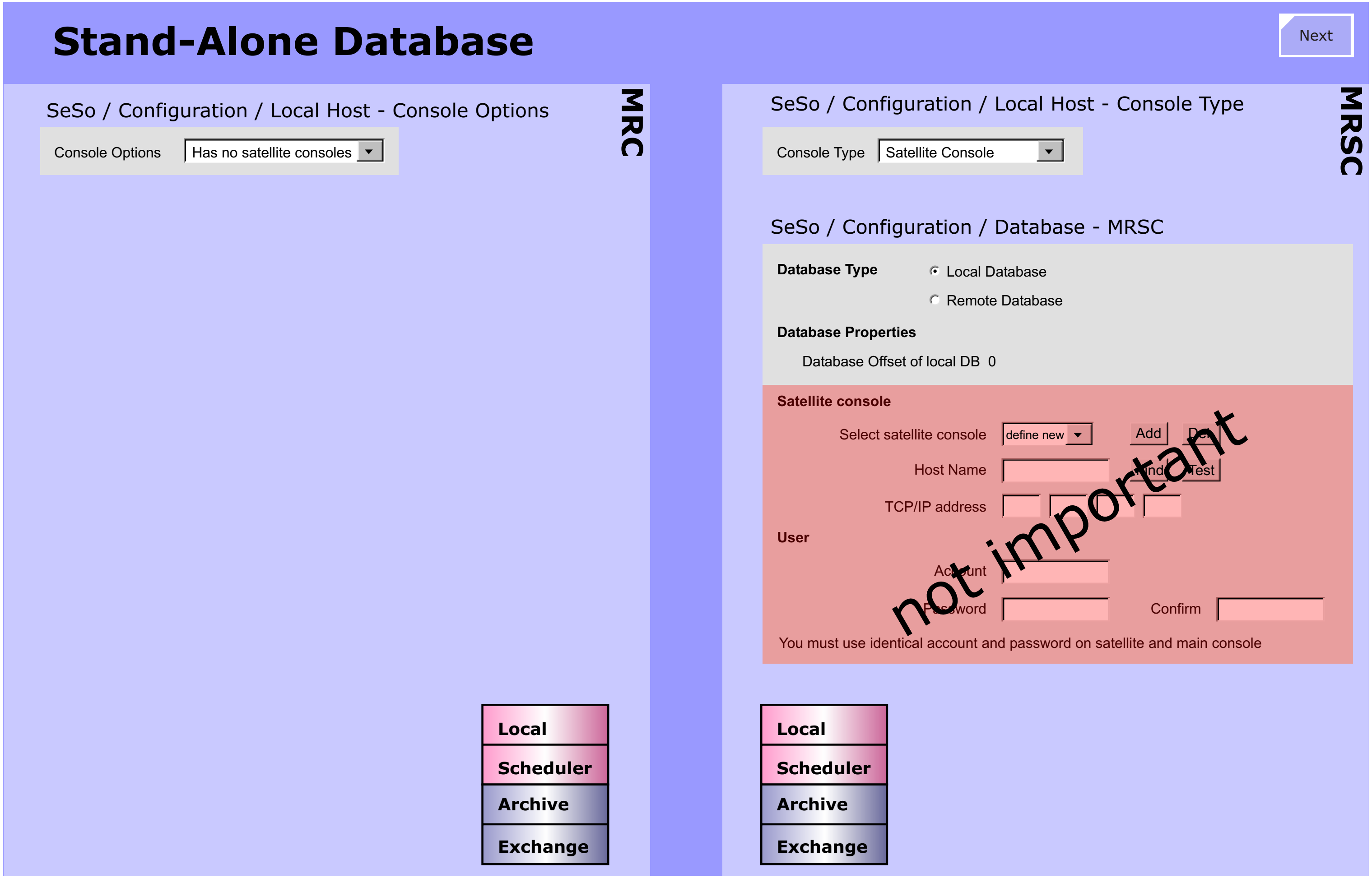
Setup of the MRSC

The MRSC will be configured as a "Satellite Console" in the configuration and the database index will be set to zero automatically.

Of course, now that the MRSC is effect just another DICOM node, it has to be configured as a DICOM node on the MRC and vice versa.

NOTE If the satellite console is configured as a workstation, images transferred between the MRC and MRSC will be made using DICOM. In this scenario the use of work and status flags needs to be considered.

Figure 19 MRSC Database as Stand-Alone



Database Repair

The image header and image pixel data are stored on separate disks, thus if one of the disks goes bad the whole database is damaged. For this reason, there is **NO database repair tool**.

Due to the structure of the database, there are dependencies between the Local and Scheduler databases so that if one of them becomes corrupt it will be necessary to recreate them both.

Therefore, repairing the database involves deleting the defective one and creating a new one. This procedure is described in the TSG.

Exchange Database Repair

Due to the problematic of transferring DICOM data and print jobs over a network connection it can happen that the Exchange Board gets "stuck". It is now possible to delete this database part separately. This is performed by starting the batch file: C:medcom\utils\ExchbRemove_MR.bat. After deleting this database you must reboot. During the reboot, the system will recreate the Exchange Board automatically.

Database Access

The possible database access functions are shown in Figure 16. Here you see two groups: internal accesses and external accesses.

Internal access means the own Host software accesses to the database to import DICOM images from a CD-ROM or MOD, to record or archive DICOM images on a CD-ROM using the CD-R drive, to prepare the virtual film sheet of the Filming task card, to register a patient or preregister a patient locally.

External access means to send and receive images via DICOM services storage and query / retrieve, to print images via DICOM basic print or to preregister patients at a HIS or RIS and get that patient data via DICOM service worklist.

Image Protection

In order to prevent unintentional deletion of patient images there is a built-in protection mechanism using work status flags. *Printed*, *Archived*, *Sent* and *Received* are set by the software automatically. The flags Completed, Read and Verified can be set by the operator himself. Icons for doing this can be configured to be available in the tool bar of the Patient Browser.

The work status flags are shown in the tree view of the patient browser automatically. Additionally it can be configured, which of the flags are displayed in the tree view and the list view of the patient browser.

The patient browser configuration tools also allows to get a warning message in case of operations not to be performed without confirmation.

Database Status Request

There are different possibilities to check the size and status of the databases.

via User Interface

Use the main menu Help > Info and select the *Disk/Database* card.

This command is available on the MRC and MRSC.

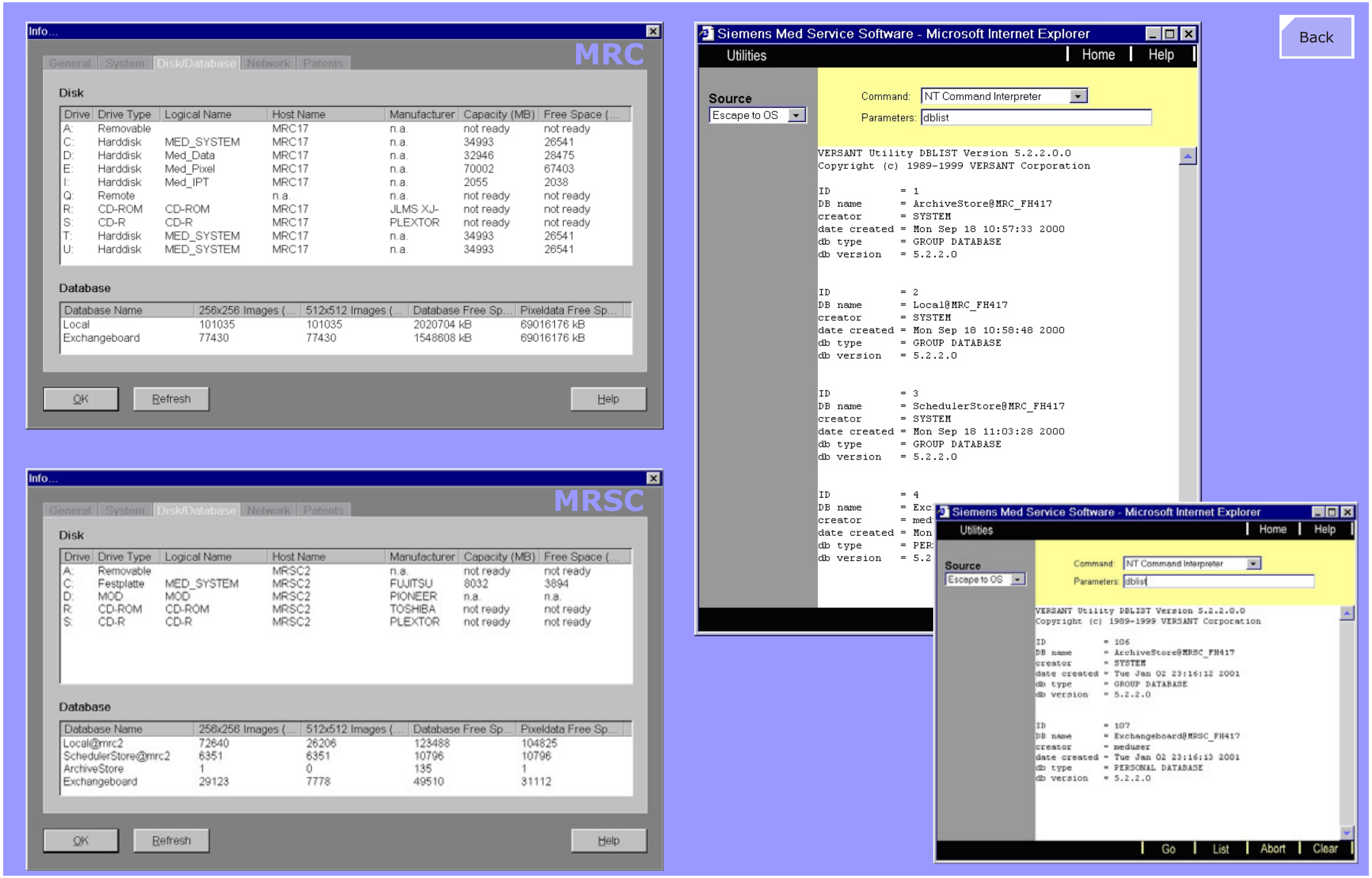
MRC: the

via Service Software

Another way is to start the service software, select Utilities Source Escape to OS and type in the command **dblist** and click Go.

These functionalities are available at the MRC and MRSC.

Figure 20 *Database Status*

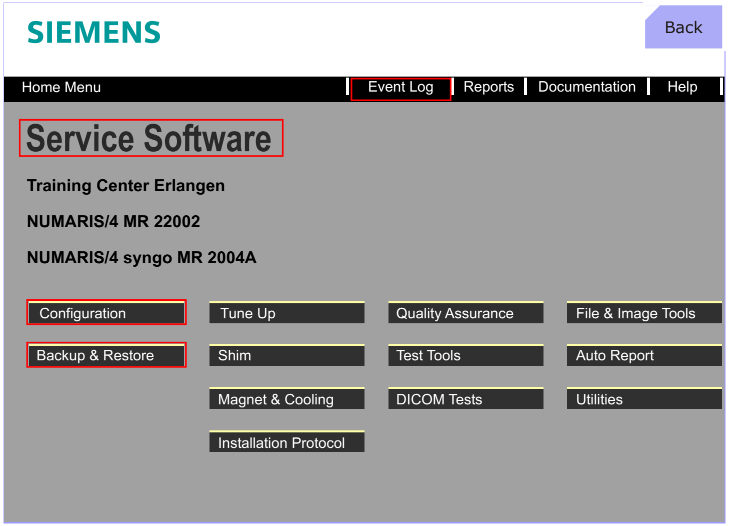


Service Software (SeSo)

Overview

After selecting Local Service in the menu Options Service and typing in the required password, the SeSo platform (shown below) will be displayed.

Figure 21 *Service Software User Interface*



SAM - Service Access Manager

SAM prevents or limits access to the various service functions via a password protection. Figure 23 gives a list of the service levels and the available service functions for each level which has its own unique password. Services levels can be purchased from either the customer or even third arty service organisations according to the level of their expertise or competence. Service passwards are limited in time and must be renewed after a time period which is determined by Siemens.

Local Access

SAM password protects the different service levels which allows the access of the available features to certain persons only.

Customer: The first level is accessible without password by the operator. Here only configuration parameters can be displayed.

In addition this level is used automatically if the operator selects the Event Log in the menu Options Service Eventlog.

SAM also distinguishes between local access and remote access.

There is the service password only for local access which allows to access the service software user interface without further limitations, since the CSE is at the system and has full control over the system anyway.

Remote Access

If wanting to log onto a system remotely there are three modes the local operator can select:

No Access - remote system access is denied and any pre-established connection will be immediately discontinued.

Limited Access - In this mode the remote connection is granted access privileges that allow only read / look type functions e.g. reading the Event Log, transfer images files etc.

Full Access - In this mode operator operator relinquishes control and the remote connection is given full control over the system. The mouse and keyboard will be locked and the screen will blocked. In this mode the remote session can perform service tests and system checks (QA). One limitation is no automated patient table movements will be allowed, for obvious safety reasons.

Figure 22 *Service Access Manager*

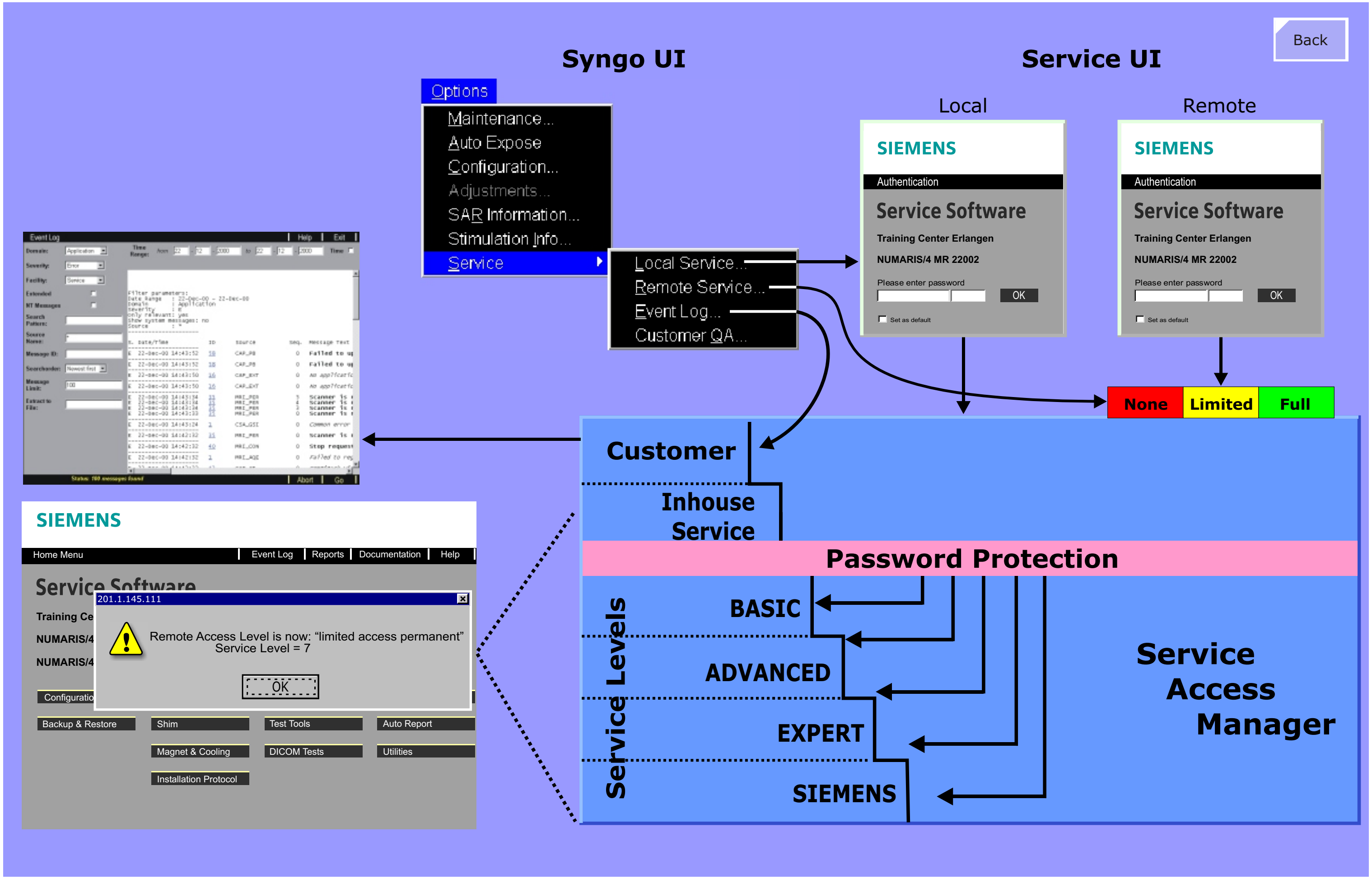


Figure 23 *Service Levels*

Back

Service Levels and their Functions					
Service Level	**Remote**				**Local *1**
	No Access	**Limited**	**Limited Permanent**	**Full**	
Customer					QA (Customer) Configuration (System ID)
Inhouse Service			Configuration (View) Backup (Auto)	Configuration (Edit) Restore (Manual)	Backup & Restore (Manual)
BASIC	No remote access possible	Same as Limited Permanent, except access is limited in time.	Service QA (View) Utilities: Send Message Process (List) Component Mgr. (List)	Service QA (Scan) TuneUp (no Plug&Play) Utilities: Escape to OS System (Start/Stop) Process (Start/Stop) Component Mgr. (Start/Stop)	TuneUp (with Plug&Play)
ADVANCED			EventLog Reports	TestTools DICOM Test Tools	
EXPERT			File & Image Tools		
SIEMENS			Documentation (System/HQ) Installation Protocol (View) Remote Application Support		Documentation (System) Install. Protocol (View&Edit)

*1 All Full-Mode functions are available to the user in the local service mode.

Event Log

In order to understand the different possibilities which can be selected in the Event Log, the following overview should be helpful. Note there are three sources of event messages:

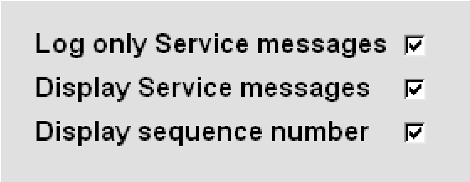
- the Windows OS
- Remote service (SAM)
- *syngo* MR

Regardless of the source, all messages are written into the same Event Log.

Pre-Filters

There is an additional pre-filter of the *syngo* MR messages that can be configured in the service software (see figure below). Enabling the "Log only Service messages" filter will block messages that are not service relevant (this should ALWAYS be enabled). The "Display service messages", when enabled, allows service messages to be entered (should also ALWAYS be enabled). The "display sequence number" will sort out the sequence of the error messages when more than one message is created at the same time. This is very useful since the first message is usually the one cased by the actual error,all following are often only the result of the first.

Figure 24 *syngo MR message filters*



These switches are set per default in the SeSo Configuration. If they are not set... well, we haven’t noticed any difference.

Post Filters

Several

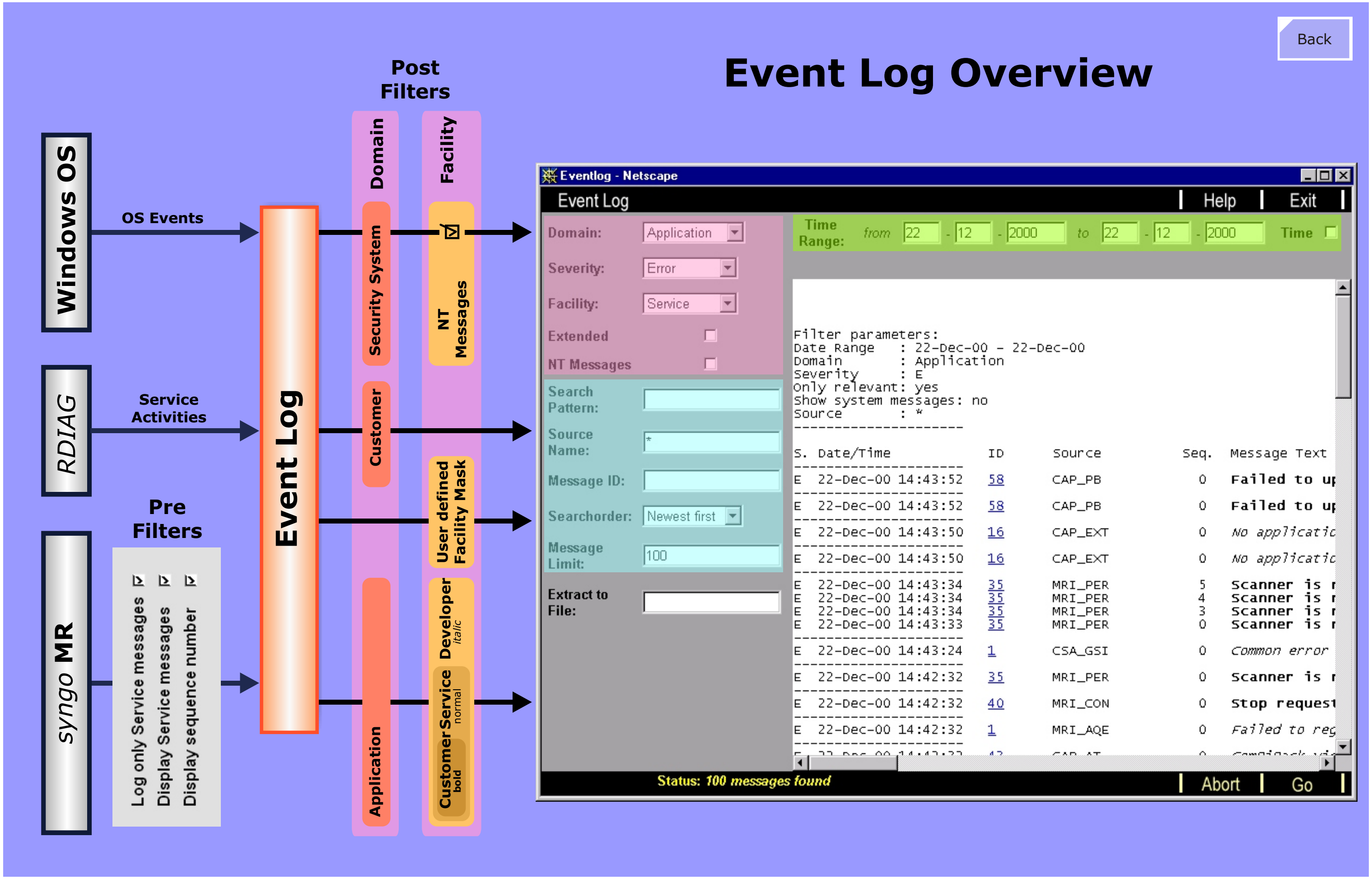
Domain

With the Domain filter only messages from the selected message source will be displayed.

Facility

For the display facility Service it depends on the service configuration parameter "Display Service Messages" whether service relevant Developer messages are displayed or not.

Figure 25 *Overview of Event Log*



SeSo Configuration

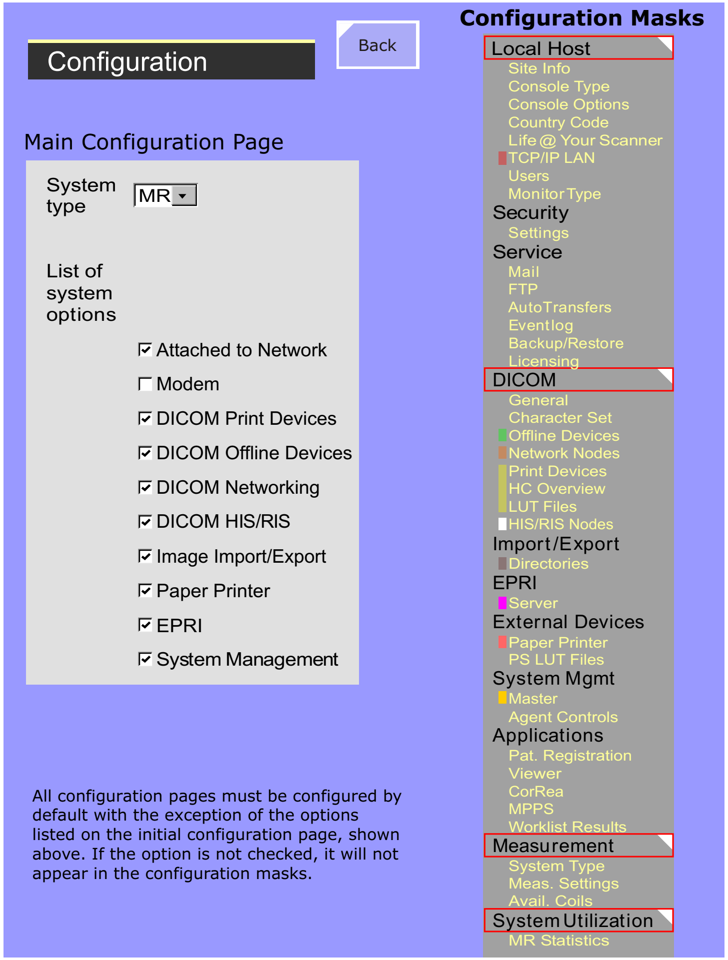
When entering the SeSo / Configuration page a List of system options is displayed with corresponding check boxes that can set if that option is to be configured (see Figure 26). The list of options is shown on the config mask on the left side and the list of the corresponding configuration pages on the right, with colored boxes showing the options to the corresponding configuration page(s).

Even if no options would be checked all the default configuration pages (pages in the list on without colored boxes in Figure 26) will still be available and must be configured.

The following configuration masks will be discussed on the following pages:

- Local Host - TCP/IP LAN (see Figure 27)
- Measurement Settings (see Figure 29)
- System Utilization (see Figure 28)

Figure 26 *SeSo Configuration Tool*



Local Hosts - TCP/IP LAN

There are two networks implemented on the MR system:

- **internal**: connects the MRS Host to the MR scanner Control Unit and Imager.
- **External**: connects the MRC Host to the MRSC Host and customer LAN

The IP addresses of the internal *syngo* MR network should be 192.168.2.x (as they are default). 192.168.2.1 is used by the Host, 192.168.2.2 is used by the MPCU and 192.168.2.3 is used by the Imager. Only if these network IP addresses are used in the hospital LAN the internal syngo MR IP addresses have to be changed.

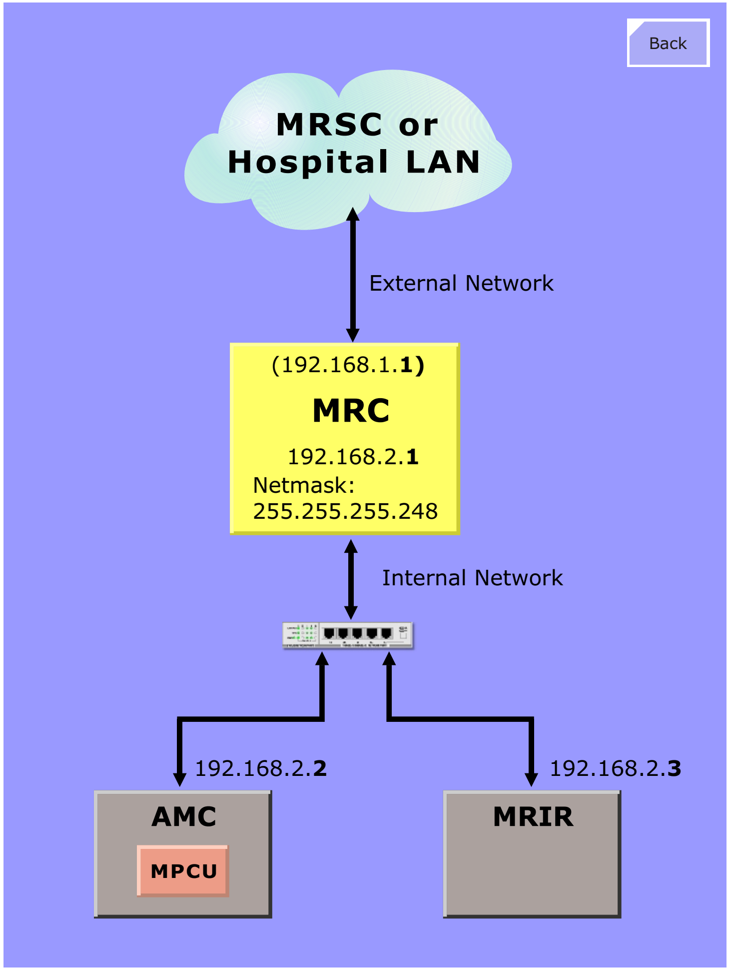
Figure 27 shows the default network addresses and net mask for the internal *syngo* MR network to communicate between the Host, the AMC and Imager (MRIR).

At the Host you have to configure an IP address and a net mask for **NIC1** (network interface controller 1, located on the Host motherboard) and for **NIC2** (PCI card in slot 1). NIC1 is used to communicate to the MRSC and hospital LAN, NIC2 is used to communicate to the AMC and the Imager. The IP addresses and net masks for the AMC (i.e. the MPCU) and for the Imager are loaded automatically from the Host during boot of the MR scanner.

IP Conflicts

If there should be a conflict in the case the customer lAN is using the same subnet, then you must change the IP address of the MPCU. You just change the IP address for NIC2 at the Host and the software automatically defines a new IP address for the MPCU and for the Imager. One possibility would be to change the NIC2 IP at the Host to 10.1.1.1 The MPCU IP will automatically be set to 10.1.1.2 (next IP after 10.1.1.1) and the IP for the Imager set up to 10.1.1.3 (second IP after 10.1.1.1).

Figure 27 *Host Network Topology*



Measurement Settings

Here you set up the MR specific measurement configuration.

In the first mask the name of the MAGNETOM scanner has to be selected.

The second mask is shown in Figure 29. Here you configure the number of receiver channels, the gradient coil, the gradient power amplifier, etc.

The third mask configures the coils available at your system to show these coils in the QA platform afterwards.

System Utilization

In the mask MrStatistic you set up the region, the type of institution and some more criteria and you activate the utilization with the check box System Utilization Activated.

Figure 28 *System Utilization*

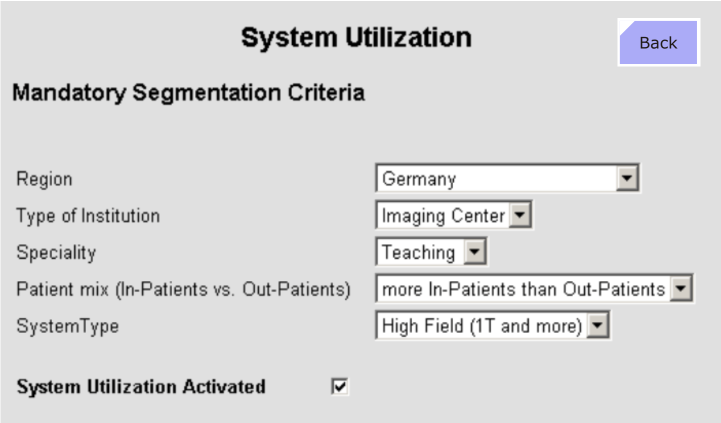
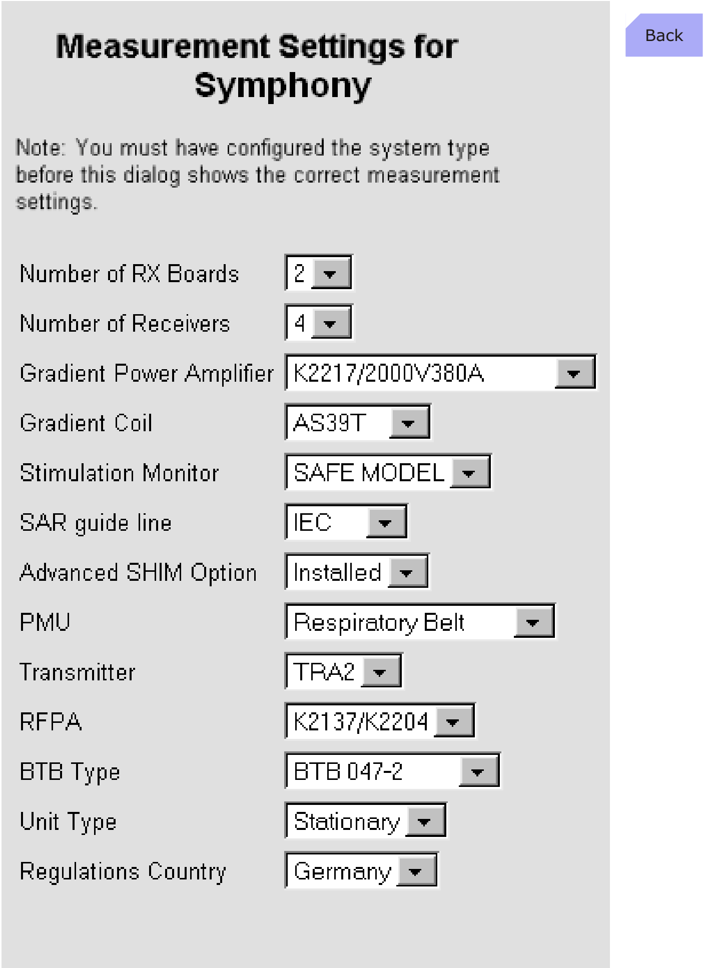


Figure 29 *Measurement Settings*



Backup & Restore

Using the Backup Tool

The backup tool can be started in the service software Home menu with the button Backup & Restore.

There are several predefined backup packages, some of the MR specific packages are discussed here, see Using the Restore Tool.

To make a backup put in a CD-ROM into the CD-R drive, start the service software and open the menu Backup & Restore. Then select the Command Backup, select Drive [S] CD-R, select the right backup package under Packages e.g., Numaris4 and click Go. In the footer you see Backup/Restore Running ..., now the backup file for the selected package is prepared and after a while it is burned to the CD-ROM. While the backup is running you get some text output and also time stamps like:

Start Backup: <day>, <data>, <time> and

End Backup: <day>, <data>, <time> Duration: <min:sec>

After the backup has finished you see the message Ready and you find the corresponding backup file:

<name of package>-<date>-<time>.ar

Example:

Numaris4-10-02-2004-10-11-37.ar.

Using the Restore Tool

The restore tool can be started in the service software Home menu with the button Backup & Restore.

To restore, put in a backup CD-ROM into the DVD-ROM drive, start the service software and open the menu Backup & Restore. Then select the Command Restore, select Drive [R] DVD-ROM, select the right backup file under Archive (e.g., Numaris4-10-02-2004-10-11-37.ar) and under Groups select all groups or several groups or one group only and click Go. In the footer you see Backup/Restore Running ..., now the backup file is restored at your system. While the backup is running you get some text output and also time stamps.

NUMARIS 4

A restore of Numaris4 should be used after software re-installation or software update only. All protocols from the user tree with and without archive flag are backed up. Restoring Numaris4 after software re-installation means all protocols are available again. After a software update the protocols without archive flag are deleted. In addition to the protocols other software parts will be restored e.g. registry entries.

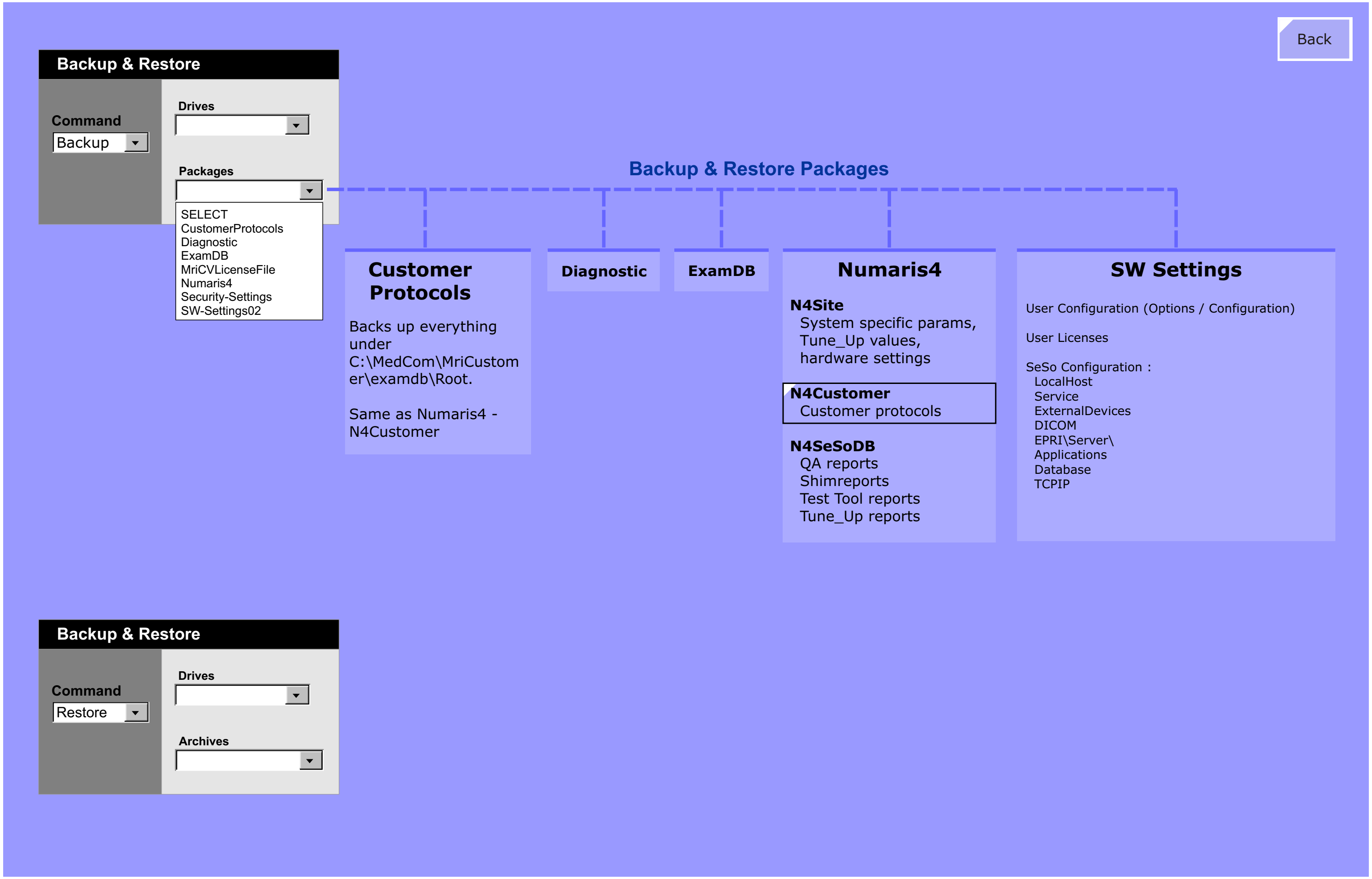
Customer Protocols

A restore of a CustomerProtocols archive file should only be used if all user protocols included in an older backup should be restored. Note: All previous user protocols are deleted during restore!

ExamDB

The customer can import and export protocols trough the Exam Explorer and therefore this Backup package is no longer needed.

Figure 30 *Backup and Restore*



Siemens Remote Services

Overview

Siemens Remote Services, or SRS, provides several remote services such as remote repair, diagnostics, phone support. event monitoring and utilization management that allow our service organization to attain higher efficiency in service delivery through avoidance of on-site service calls as well as attaining better planning, instruction and pre-clarification through SRS. These benefits also translate into clear advantages to the customer. With SRS we can help the customer to avoid system failures and resulting down-time and even reduce repair times. The remote technology employed by the SRS also makes future-oriented value-added services such as online application support or optimization of system usage through Utilization Management possible.

It is important to customers connected to SRS from the very beginning, especially during the warranty period. This gives the customer the opportunity to become acquainted with the advantages of SRS services right from the start. The SRS connection also allows us to perform more cost-effective services internally during the warranty period.

The individual services and customer benefits of Siemens Remote Services are being marketed through various features of our Performance Plans service agreements. Guaranteed system uptime, shorter response times, and application support are currently being marketed. New modules such as Utilization Management (regular evaluation of usage data, etc.), are already demonstrating great potential in the pilot phase and will be available soon. To be able to actually offer these services, and to guarantee cost-effective service, the remote connection is therefore a basic prerequisite for all service agreements.

One of the advantages of SRS is optimized system availability. For example, we can realize an increase of system availability from 95% to 98% solely with SRS. Just what 3% more availability can mean for the customer is illustrated in the following example.

Agreed service times:

250 work days with 9 hours (8:00 – 17:00) = 2250 hours

3% UPTIME = 67.5 hours (corresponds to 7.5 work days)

If one assumes an average number of exams per day at 20, then the customer will be able to perform approximately 150 exams with 3 percent more uptime.

The SRS solution has implemented the necessary technical and organizational mechanisms to assure the protection of patient and system data. This includes preventing unauthorized access (fire walls, passwords), protecting data during transmission (encryption) and the logging of all remote activities (who, what, where and when).

SRS not only helps on technical problems, usage-related problems can be addressed as well. Transmitting images, for example, ensures improved analysis and application advice. Future plans include a remote desktop facility to allow real-time applications help or usage-related trouble-shooting to support our customers even faster and more effectively than before. An additional service is Utilization Management which supports optimal of system operation.

The routers and the cost for installing a dedicated telephone line is assumed by Siemens. The SRS connection uses toll-free telephone numbers so there are also no telephone costs incurred to the customer.

SRS is mainly used by the support centers USC, TSC, HSC and finally by the business units (BU) such as MR. The responsible engineers will use the available SRS connections to access medical systems via pre configured tools. Depending on the country specific situation also CSEs can access medical systems via SRS.

Connections to SRS

Connections between the medical modalities and the SRS servers are made using routers. Routers are intelligent devices providing

Level 3 (IP address) network communications and can be programmed to allow or block certain IP addresses or ranges from passing through the device. For this reason, they offer much more security than MODEMS or other simple point-to-point communication types. Routers also often include a built-in fire wall so even Level 4 protection (TCP ports) is possible.

Routers, like switches, have the added advantage in that several systems can be connected to a single router, the number only being limited by the size of the router. Routers used for remote connections come in different versions for use over the various communications infrastructures available today. They are:

- ISDN a fully digital comms format found chiefly in Europe, but also in other countries. The maximum transmit speed is 64k bits/s.
- POTS (plain old telephone service) - a format that is available throughout the world. Its bandwidth, however is very limited from 36.6 to 56 k bits/s.
- **DSL** (Digital subscribe Line) - a relatively (2002) new comms format that is fully digital and has high bandwidth. Max transmit rates are in excess of 1M bit/s.
- **VPN** (virtual private network) - a comms format that is used over internet connections. It enjoys the highest security and its speed is only limited to the connection speed to the provider.

The routers have to be configured by the Siemens Medical router specialists, who work together with the network administrators in the hospitals. The router must also be configured on the modality, a procedure described in detail in the Software Installation, under Remote Connectivity. This is a responsibility of the CSEs.

Authentication and Authorization

The central maintenance and dial-in platform (SRS portal) used by the UPTIME Service Center is located on the company intranet and cannot be accessed externally. Access requires a valid Siemens Remote Service user ID and password.

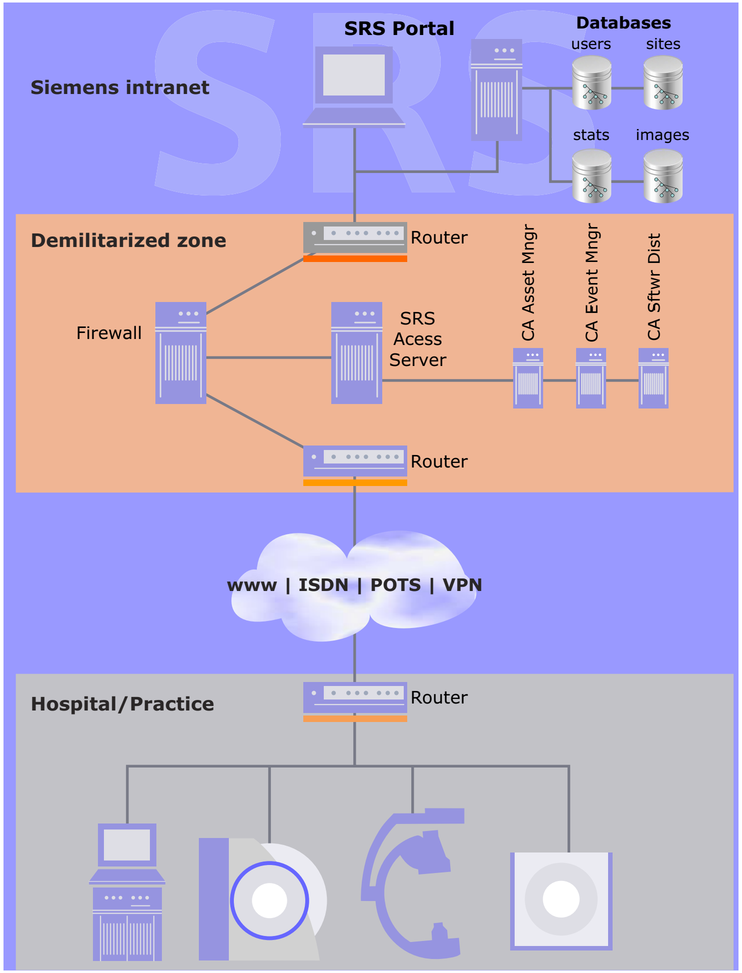
A multi-level service domain concept defines which users are permitted to access which systems. This means that Siemens service engineers only access those customer systems for which they are expressly authorized. Additionally, only those SRS functions for which the engineer is explicitly authorized are released. Other systems in the customer network not maintained by Siemens Medical Solutions cannot be accessed via this platform.

Demilitarized Zone

To protect the Siemens intranet and that of the customer from reciprocal problems and attacks the SRS access server, a Linux server, is secured through a demilitarized zone (DMZ). Connections from the Siemens service engineer to the customer system, and vice versa, are not “put through directly.” They terminate in the SRS access server using a reverse proxy function. This means that a connection established from the Siemens intranet is terminated in the SRS access server. This server then establishes the connection to the customer’s system and mirrors the communication coming from the customer back to the intranet. The possibility of a communication between the Siemens intranet and the customer’s network over not explicitly authorized protocols is thereby prevented. Mirroring occurs for predefined protocols only. This architecture is designed to prevent:

- Unauthorized access from one network to the other (e.g., hackers)
- Access from a third-party network (e.g., the Internet)
- Transmission of viruses or similar harmful programs from one network to the other. In addition, we do not store any critical data in the DMZ, in particular, customer access data.

Figure 31 *SRS Access Server Structure*



SRS <> Modality Port Connects

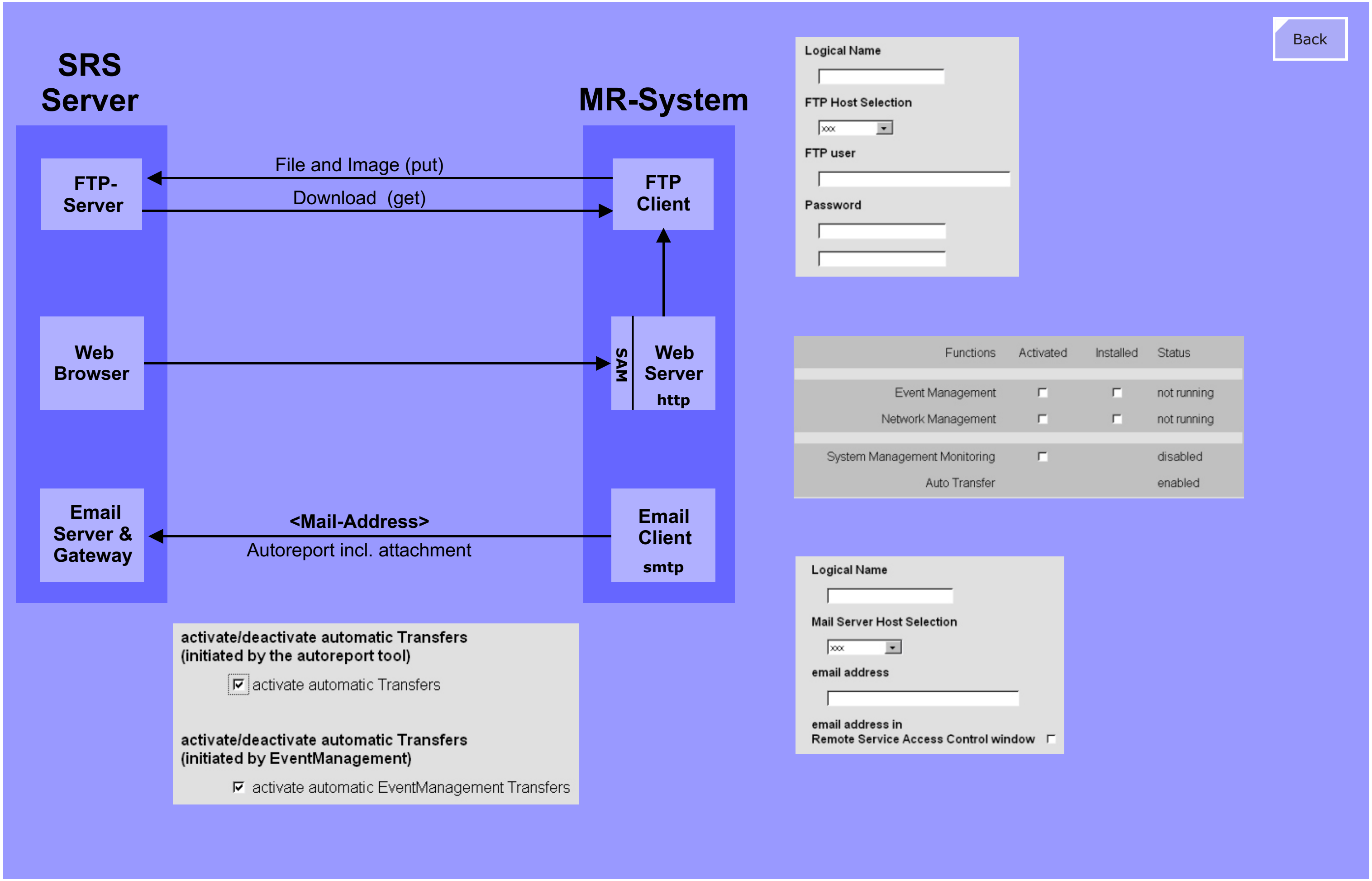
Figure 32 shows an overview of the protocols used for the various SRS functionalities such as File & Image Tools, Auto reporting, System Utilization and other Value Added services.

How to test the ftp connection to the SRS Server is shown in the document Installation Software, section Remote Connectivity: Test FTP login to the Remote server.

The figure below shows an overview of the SRS mail connection.

How to test the mail connection to the SRS Server is shown in the document Installation Software, section Remote Connectivity.: Testing connectivity from system to RDIAG server.

Figure 32 SRS Connection Protocols



Event Monitoring

Proactive Service

What does "pro-active" actually mean and how does this differ from remote diagnostics?

In contrast to traditional remote diagnostics where the source of errors must be analyzed reactively, Event-Management, enables connected systems to be monitored continually and preventively. Should a defined system parameter deviate from a pre-set value, a message is automatically generated for the System Management Center. The deviation is communicated to the responsible USC, corrected online there, or a planned service visit is scheduled.

Preventive online monitoring is currently employed using intelligent software of the newest generation of syngo-based and non-syngo-based systems and in PACS networks (CT, MR, AX, PACS, NM-PET). The pro-active service approach and its implementation is constantly being developed.

Magnet Monitoring

Magnet Monitoring via SRS provides several reports for:

- Helium/Shield Monitoring
- Helium Level Low
- Shield Temperature High
- No Data for 3 Weeks
- QDB Connectivity
- Country-Specific Connectivity

Value Added Services

Utilization Management

An additional service is Utilization Management which supports optimal of system operation. Utilization Management reports are created monthly and sent to the customer, the customer can use these utilization reports to improve the usage of his MR scanner.

To be used, the Utilization Management has to be activated in software under Configuration / System Utilization. Additionally an Autoreport template for Utilization Management has to be set up as well.

Virus Protection

We'll sell it only if the customer connects to SRS.

Quality Database

An extremely useful source of information concerning system performance values of systems world-wide. Check it out!

Software Installation

Overview

Reloading software is described in the Installation Software syngo MR 2004V document.

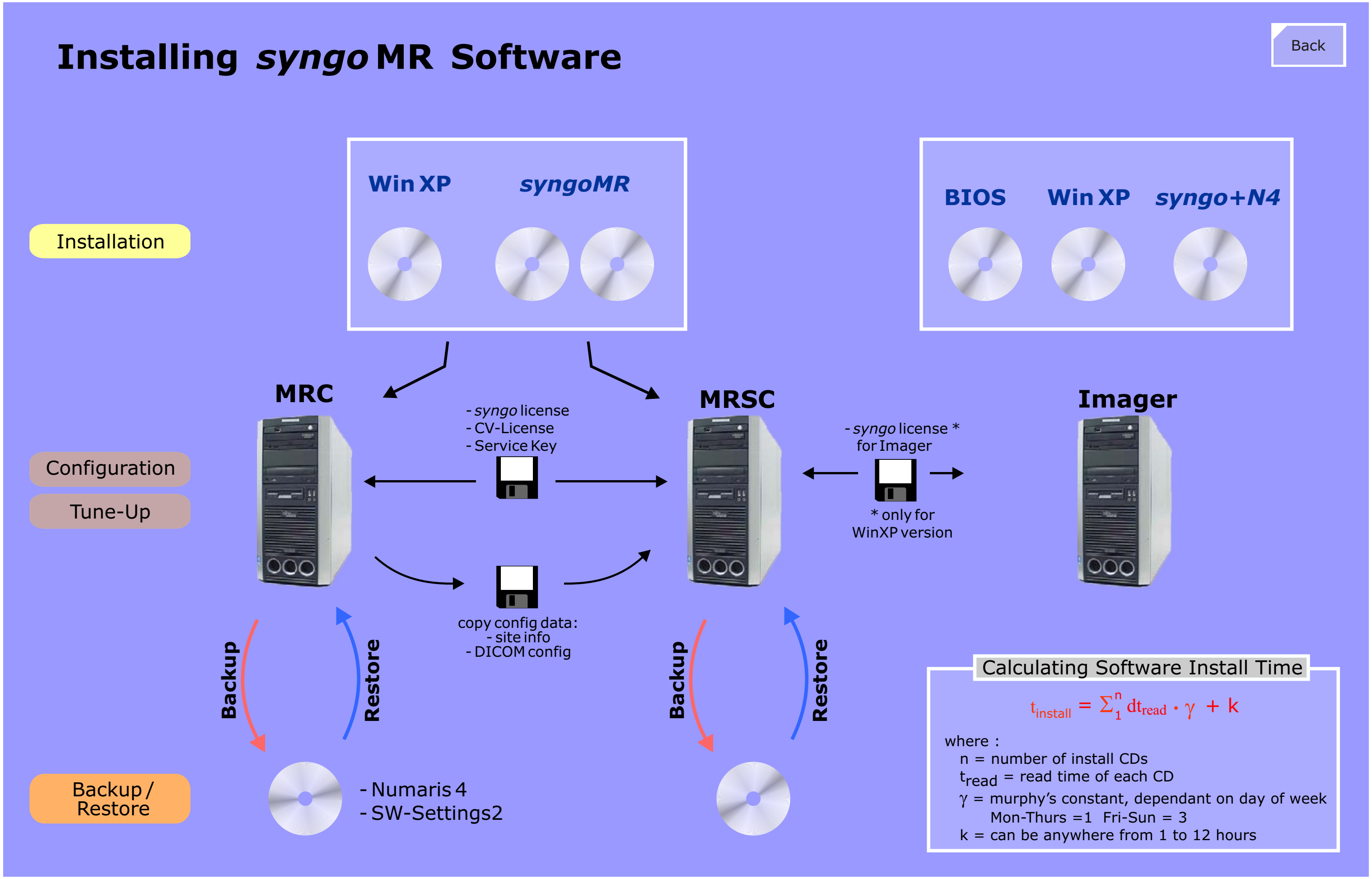
syngo MR is the name given to the complete software bundle for the MRC Host and MRSC Hot consisting of the operating system (Windows XP,) the general syngo software components and the MR specific software components (NUMARIS/4).

The software bundle for the Imager consists of an operating system (Windows or LINUX) and the MR imager specific software.

Software Reload

The software reload for the Imager is started during boot from CD-ROM, everything else runs *a u t o m a t i c a l l y*. Any questions?

Figure 33 *Software Installation*



Section

4 Host / Imager

Introduction

The Host interfaces to the MR scanner via the MPCU which also acts as the main controlling unit for the scanner hardware. The MPCU is located in the Advanced Measurement Control (AMC)

Although the Host and Imager components are individual components they have been combined here in Part 4 since they work very close together.

In addition to the MRC, there is also an optional satellite MR console, or **MRSC**. Although practically the same as the MRC, it is only intended for post-processing of images and thus may not contain some hardware components found in the MRC. Also, there will be some differences in the installed software since the MRSC doesn't require all that system control software! A description of the main differences are given on the next page.

Host (MRC)

The major tasks of the Host of the MRC are:

- User Interface (syngo)
- take sequence measurement parameters from user and pass these on to the Measurement Control (AMC)
- image post-processing and image display functionalities
- archiving (mass storage, filming)

The MRC is connected to the MR system via an ethernet LAN connection. This network connection will be referred to as the "internal network".

Host (MRSC)

The MR satellite console (MRSC) is an option for

- image viewing
- post processing
- archiving images

The MRSC host connects to the Host MRC over an additional ethernet switch, delivered with the MRSC option, and is required to decouple the large volume of image data transfers from the customer's LAN. This network connection will be referred to as the "outside network", or "Customer network".

The configuration of the MRSC computer is much the same as the MRC Host with the following exceptions:

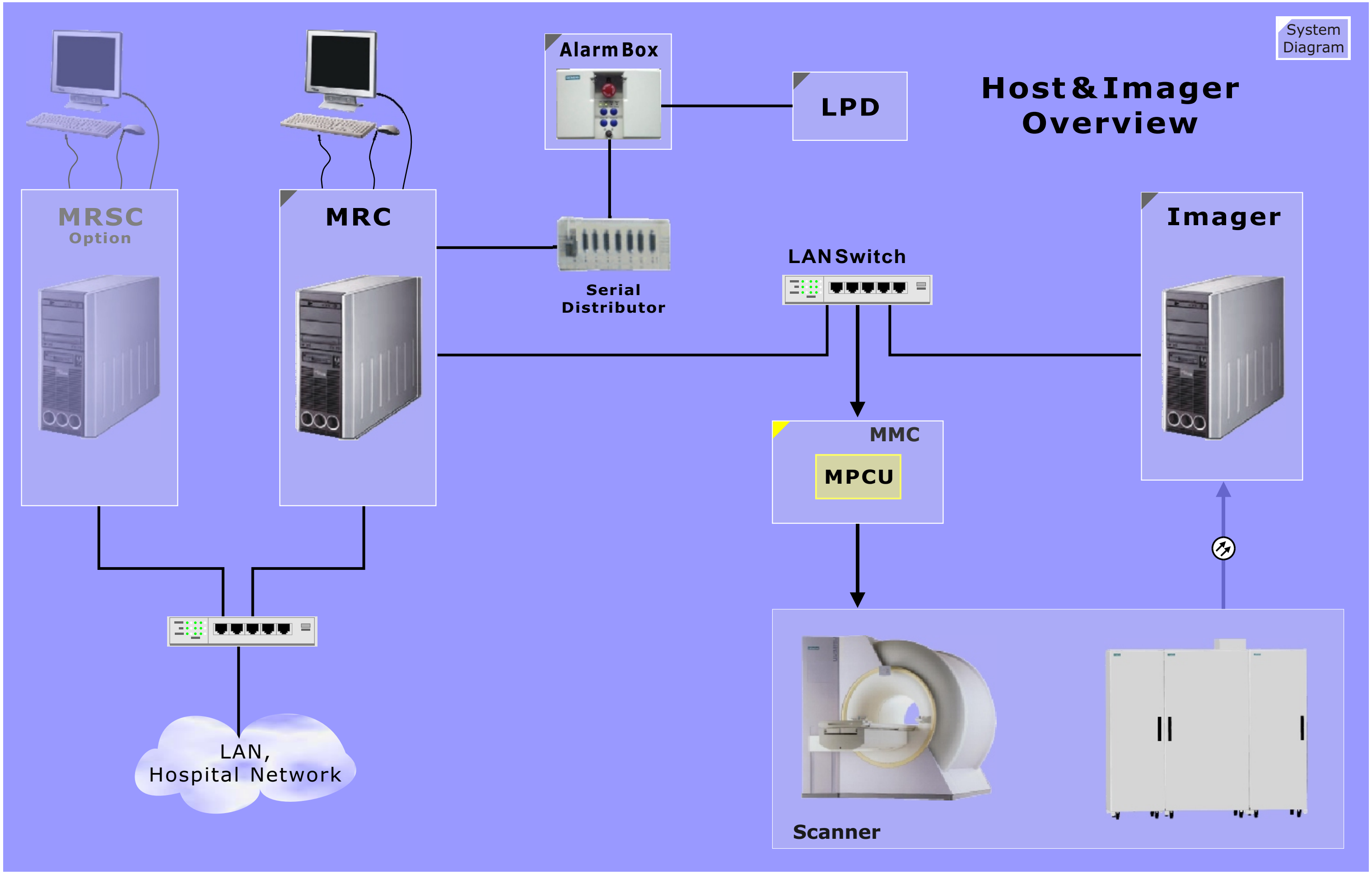
- NO hard drive pixel data (database and pixel data on one hard disk)
- Only 1 network interface - to communicate to the MRC
- NO Moxa interface card.

Imager (MRIR - MR Image Reconstruction)

The raw data coming from the MMC will be stored temporarily on one or more pixel disks.

The Ethernet connection provides the link from/to the MRC host for parameter downloading and uploading the image data for storage on the MRC pixel disk.

Figure 34 Host - Imager Overview



Host

MRC

The MR Console (MRC) and MR Satellite Console (MRSC) consist of the following components:

- a high resolution **LCD Display** : output device
- **keyboard** and **mouse**: input devices
- **Host** (Intel based PC) - basic model identical for both MRC and MRSC, however, configurations are different.

Developments are rapid in the computer industry and therefore this description applies only for the current configuration of the model implemented at the time of creation or modification of this document. Future versions of course will have faster processors, more memory, larger disk capacity and more software.

MRSC

The configuration of the satellite console is similar to the MRC. Components that are not part of the MRSC

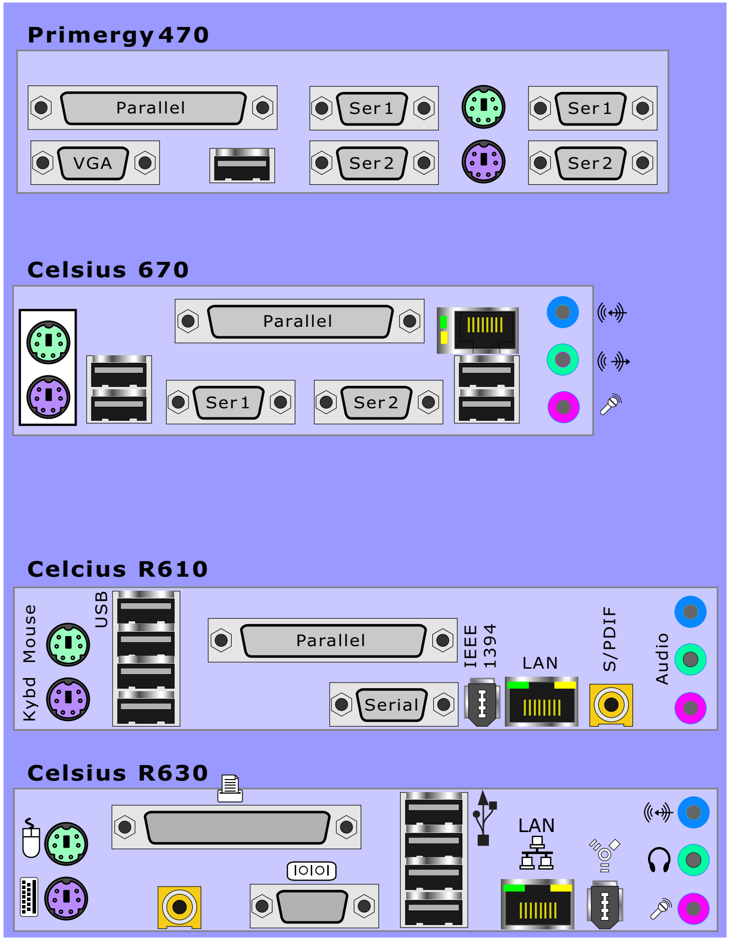
- no MOXA 8 x Serial Interface
- no Ethernet PCI card
- only two hard disks

Host Configurations

NOTE Please refer to the Spare Parts Catalog for ordering and configuration information of the various Host models.

I/O Connection Plate

Figure 35 *Host Motherboard Rear Panel*



Imager (MRIR)

Overview

The task of the Imager (MRIR) is the collection of raw data and to perform the image reconstruction.

Function

The digital raw data enters the PCI-Receiver via a fibre optic cable. Via DMA the raw data will be stored in the main RAM. The preprocessed data will be stored onto the Raw Data Disk(s) if the amount of data exceeds a certain amount (e.g. 600 MB).

After the measurement has finished, the image reconstruction will be performed. Depending on the sequence, this can be partially done on the fly during the measurement.

The resulting images will be transferred via Ethernet to the pixel disk of the MRC-Host.

Raw data will be only available for display when storage is enabled before the measurement. The second possibility to display raw data is a re-transformation from the images. This is always possible.

A serial communication line from the MRC to the Imager was shipped with the older systems, but has never been used.

NOTE Please refer to the Spare Parts Catalog for ordering and configuration information.

Keyboard / Mouse

Keyboard and mouse have to be connected temporarily, only during sw-installation and configuration.

NOTE The keyboard- and mouse adapter must be connected always. It simulates keyboard and mouse, so that Windows can boot properly!

Monitor

The Monitor (CRT or LCD) is required only for sw-installation and configuration.

The display is set for VGA (640x480). A number of LCDs SCD-1897-M (type 2) can not be used.

Storage Devices

Besides the system hard drive, there are four hard drives for the raw data buffering.

The CD-ROM - and floppy drives are used for software installation and -updates (Win NT and NUMARIS).

In-Room MRC

The "In-Room MRC" option extends the **monitor** and **mouse** of the MRC inside the exam room to allow the user to operate the system in an interventional mode. A **keyboard** is not part of this option since in all likely hood the Doctor's hands are full of blood and fleshy bits and pieces anyway.

There are two versions of the IN Room option which will be called OLD and NEW. The diagram in Figure 36 shows the new version. The old version can be Besides power supplies, there are only cables for the LCD-display and the mouse connected to the corresponding distribution box.

Function

Monitor

The video signal to the MRC and In Room MRC monitors are identical. The video signal from the Host is simply split and amplified by either the Interconnect Box (new version) or a video distributor (old version).

Mouse

Control of the two mouses of the MRC and In Room MRC is somewhat more complicated. As soon as one of the mouses is used, the other is disabled. After a delay of appr. 5 seconds the other mouse will be enabled again. THis function is also part of the Interconnect Box of the new version or by an extra Mouse Distributor box of the old version.

An additional (optional) display can be connected at the video distributor. See diagrams for details

Figure 36 *In-Room MRC (old)*

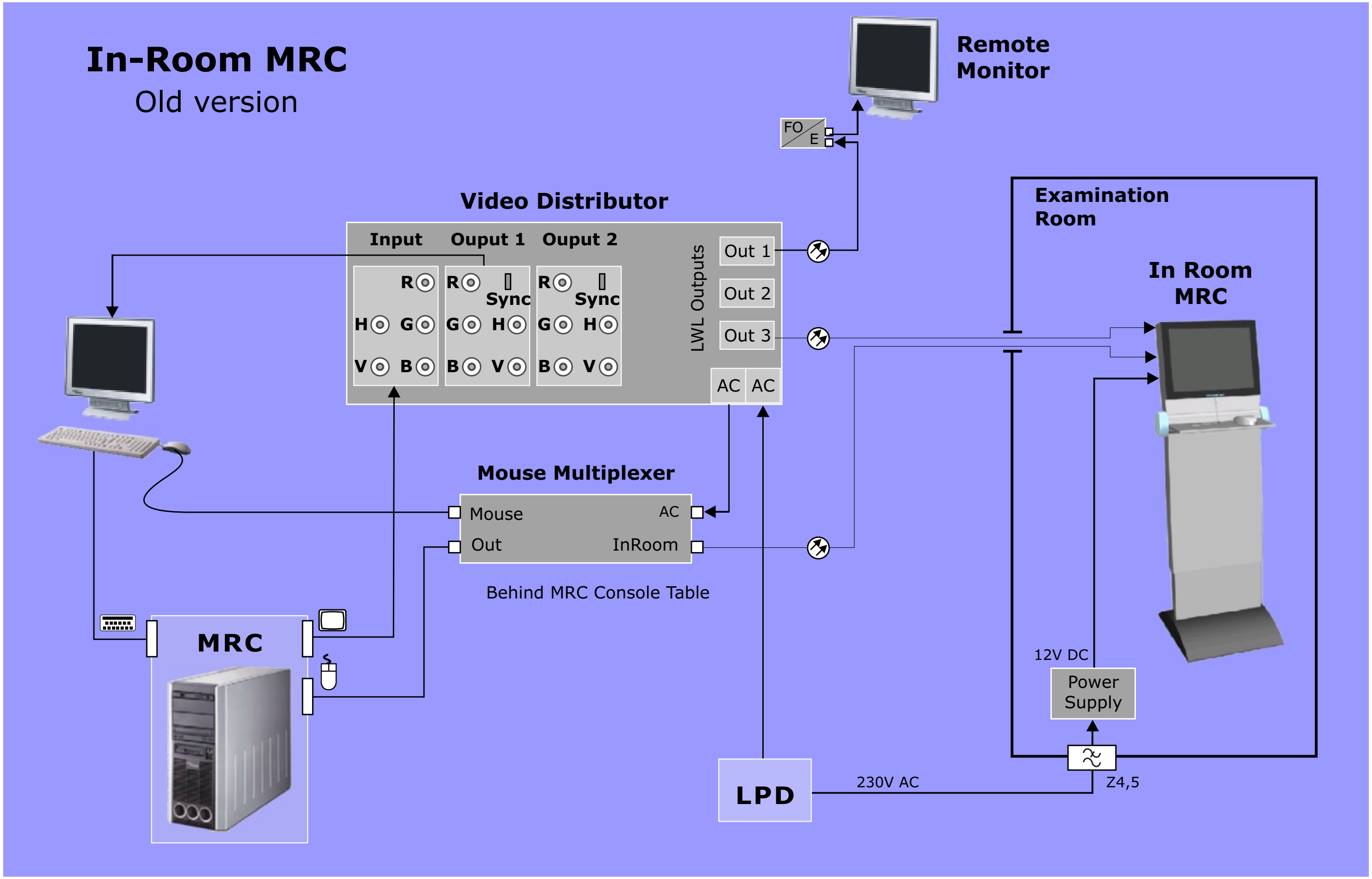
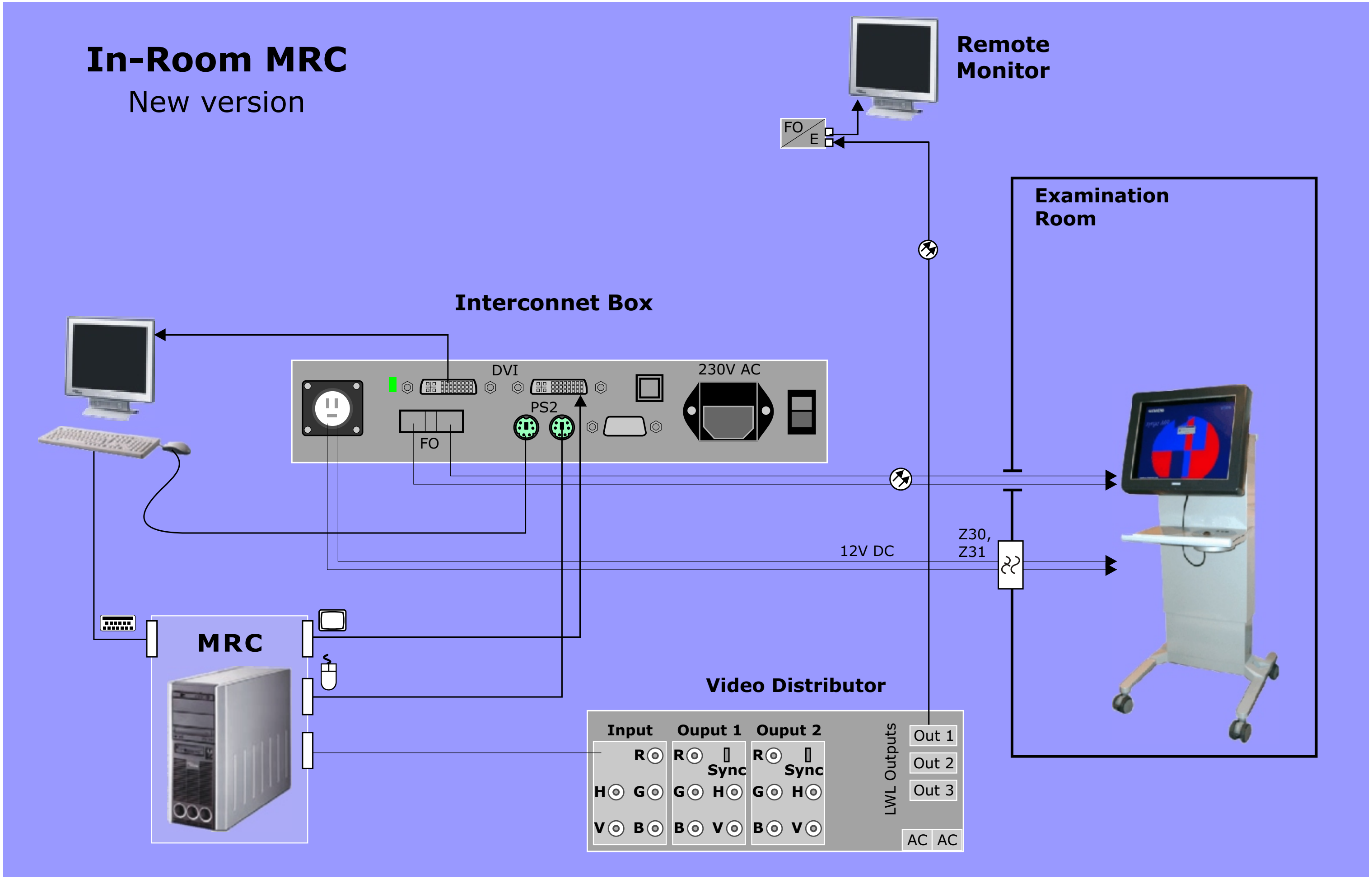


Figure 37 *In-Room MRC (new)*



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Section

5 Control

You are now in the Control section of the Functional Description. It begins with an introduction to the MMC - Modular Measurement Control. This is followed by a functional description of the software and then a description of the hardware components. You will find both text and graphics in the descriptions. The pictures can be looked at in color (they will be shown only in black and white if printed, however). The text can be read. Experience has shown that the maximum efficiency is achieved when combining the two.

Introduction

The Modular Measurement Control (MMC) is the name given the component group responsible for exercising system and sequence control. The MMC has two major tasks:

Sequence Control

Responsible for generating the gradient, RF and receive data readout events (including pulse compensations necessitated by system and site specific variables) as well as all sequence timing signals required to generate and acquire image data in a slice or 3D image volume. The resultant MR echoes are sampled under control of the MMC and provided to the image processor for the final image reconstruction.

Communications

As the system’s central intelligence unit, it has the further responsibility to provide a communication interface to the hardware periphery for supplying software, parameters, static control for those requiring them and a status/error feedback path. The industry standard CAN bus format is used for this.

The Concept

The MMC hardware platform provides the computational power for running the sequences (C++ programs) in real-time and the signal processing components in the form of DSPs to create the RF pulses, gradient pulses (including signal compensations of hardware and environmental variables), sequence timing signals as well as RF SAR and gradient stimulation monitoring. Since the advent of high-power DSP components it has been possible to integrate the MMC components to just 5 boards allowing for a relatively compact design. The MMC functional description has been divided into two parts:

- **Functional** - describes in details the functions performed by the MMC components.
- **Hardware** - describes the individual hardware components and their tasks.

Functionality

In this section a description of the MMC's functionality will be presented. The Modular Measurement Control responsibilities can be grouped into the following tasks:

- Scanner Software Download
- Sequence Control
- Supervision and Error Handling
- System Standby

Scanner Software Download (boot)

As seen in the Software section, there are several components which require software. These components can be basically grouped into two groups:

- CPUs, DSPs and FPGAs
- CAN Modules

CPUs, DSPs and FPGAs

The DSP and FPGA components on the MC4C40, MC1C40, and RX4 boards do not have, for various reasons, their software stored in non-volatile memories. The MPCU has a boot PROM containing boot protocols for establishing communications to the Host for downloading its own software and the software for the other MMC components. They have to be downloaded at each power-on or scanner reboot.

CAN Modules

Many of the peripheral hardware units and sub-systems of the scanner, (Gradient amplifier, RF System components, Patient Table, etc.) have control components, either CAN Modules or CAN SLIOs, that provide hardware control functions and a feedback path for local monitoring. The programs for the CAN Modules are stored in non-volatile memories - EEPROMs or Flash memory - and programmed in the factory with the most current version. CAN software updates delivered with newer software packages will automatically be downloaded to the CAN Modules.

At power on, the CAN Modules software versions are checked and if necessary (i.e., new software or parts replacement) will be updated with actual versions. This task is performed by the MPCU. The software modules for the scanner hardware are physically located on the Host system disk. This process is described below and shown in Figure 38.

MPCU Boot Procedure

POST

At power on, the MPCU performs an initialization and self test of its on-board circuitry. During this test routine, the BIOS writes progress codes, also known as POST codes, to LEDS connected on the parallel port (X14 on the MPCU). The LEDs display these codes in hex format.

Load Operating System

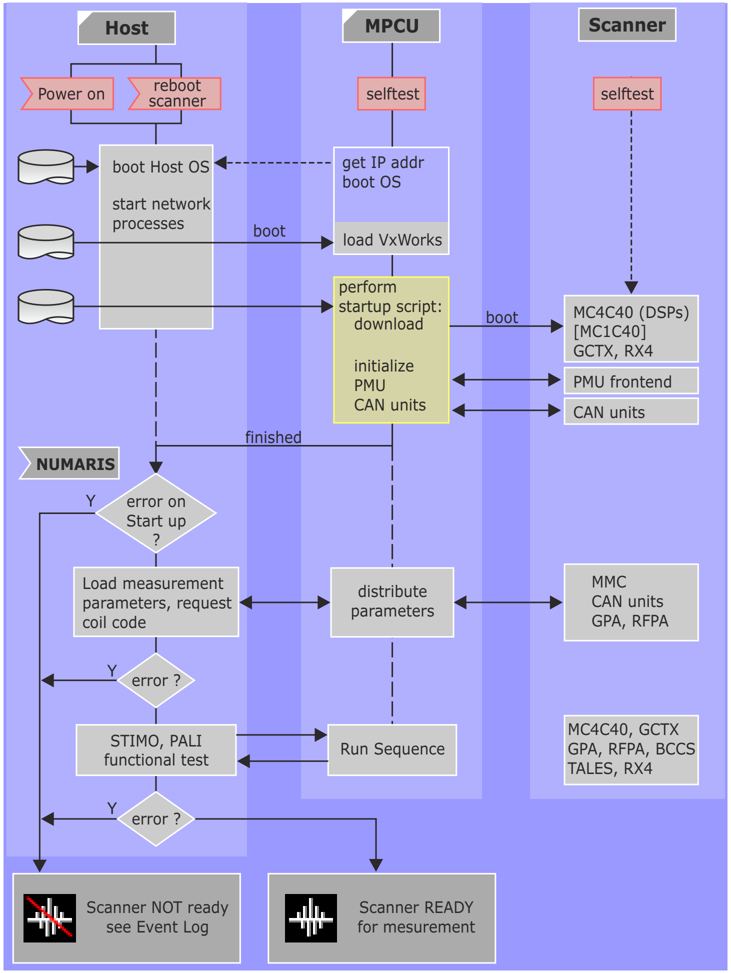
After the power on self-tests, the MPCU resets the other MMC hardware components and starts the boot loader program in EPROM. The boot loader contacts the Host and requests an IP address and its OS. The Host responds, assuming its on and running, and provides the MPCU with the name and location of the requested files.

Startup Script

After the MPCU has loaded its OS, the next step is to perform a start-up script. In this script all further steps are defined:

- Load and start MPCU tasks
- Boot the DSPs in MC4C40 (+MC1C40)
- Boot GCTX and RX4
- Check CAN units and load loadware if necessary
- Load parameters into CAN SLIOs

Figure 38 *MMC Software Download*



The "start-up script" executed by the MPCU contains the instructions for downloading the software into the MMC units. Status and error messages will be logged into the file C:\Medcom\log\mpcu_startup.log. Interpretation of this file needs detailed knowledge, but it could be helpful for support. This file will be overwritten by the start of the start-up scrip (e.g. with "System / Control, Reboot Scanner").

For detailed messages of the MPCU activities another trace can be started with System/Run -> StartMPCUTrace. Messages will be logged into C:\Medcom\log\MPCUTrace.log.

Loadware

PROMs are only used in the MPCU and in the intelligent CAN units. Loadware will be loaded into MPCU and the other units inside the MMC (e.g. MC4C40, [MC1C40], GCTX, RX4).

Loadware for the PMU and CAN units is initially loaded into EEPROMs at the factory or during a software update. A unit will only be downloaded if the version is not correct.

Parameters

For most units parameters required for measurements will be supplied.

PALI Function Test

A special measurement sequence will be started that should exceed immediately some predefined low SAR limits for that measurement. If is tested if the PALI can detect this problem.

Indications

Unfortunately there are just a few indications for a loaded system:

- CLK_20 LED of the RX4 is on
- All STAT LEDs on the MC4C40 are on
- STAT LED on the MC1C40 is on
- RFPA_ON signal active (pull cable to check!)

Sequence Control

The sequence control tasks can be summarized as follows:

- RF Control
- RF Safety WatchDog (RFSWD)
- Gradient Control
- Gradient Safety WatchDog (GSWD)
- System Compensations
- Dynamic Control Signals

The sequence file is a compiled C-program, which is loaded into the MPCU memory. In preparation for the measurement, the MPCU will check for system readiness. If everything is ok, the SCT (Sequence Control Task - the sequence generator program running on the MPCU) will initiate the start of the sequence and begins feeding the sequence information to the DSPs. During the measurement the MPCU surveys for any error or status change from the MMC or periphery hardware components, and in case of any errors the sequence can be stopped and the operator informed.

RF Control

As the sequence progresses, the MPCU feeds the TX_DSP on the MC4C40 board responsible for the generation of RF-pulses with the corresponding requests and the RF pulse amplitudes are generated (calculated) in real-time. The pulse generation must also include system specific corrections and compensations such as correction of RFPA non-linearities, B0 field correction.

Dynamic timing signals must also be provided for RF components RFPA (unblanking), RFIS (coil detuning), RCCS (receive matrix switching), BCCS (T/R switch) to name a few. These sequence timing signals are generated by the PCI_TX DSPs on requests from the MPCU.

RF Safety Watchdog (RFSWD)

The RF Safety WatchDog (RFSWD) - also called PALI: power absorption limiter - provides patient safety monitoring by measuring the specific absorption rate (SAR) of the RF energy being irradiated into the patient. This energy causes warming of the patient both locally and globally, that is, warming in the direct vicinity of the transmitting coil and via blood circulatory a global warming of the body results. The RFSWD must quantify the amount of heat-causing energy being applied to the patient, calculate the distribution of these energies throughout the body and check irradiation limits which may have been exceeded. If any limit is reached the RFSWD can disable the RF transmission.

The RFSWD is implemented as two independent processes:

Look-ahead Monitor

The first is a software look-ahead monitor running on the **Host** which evaluates the sequence and parameters (i.e. protocol) as they are selected and entered by the user. If the look-ahead monitor detects the protocol will cause excessive RF it issues warnings to the effect and suggests new parameters to bring the SAR under the limits. A sequence can only be loaded and started if it will not exceed the prescribed SAR limits.

There are two levels of SAR limits: normal mode (NM) and first level (FL). First level limits are higher than those of the normal mode and may only be selected under the explicit allowance of the user.

Online Monitor

The second process is realized in hardware and monitors the RF in real-time. The actual RF transmit pulses are sampled by directional couplers located in the RF Power Amplifier and the TALES, quantified by an ADC on the RX_Module and read in by the RX-DSP on the MC4C40 via the RX4 board. The RFSWD program running on the RX_DSP keeps a running tabulation of the applied RF pulses and if limits are exceeded before the conclusion of the measurement, the sequence will be immediately terminated. An error message will inform the operator that the RF limit was exceeded.

In case there is still outgoing RF measured in the next 1ms, the RFPA_ON signal, will be disabled. In addition to the error message, another pop-up window is shown instructing the operator to get the patient out of the magnet.

Plausibility and Consistency Checks

After each sequence, the RFSWD makes a plausibility check of the measured RF. If the measured RF values are outside some range, the values are deemed non-realistic and an error will be generated. Also, the calculated SAR values performed by the On-line monitor are compared to the actual RF values measured during the sequence, if these two values deviate by some value an "inconsistency" error will occur. Both of these error are assuming there is anything wrong with the components used in measuring the RF.

Gradient Control

The gradient pulses are generated in much the same way as those for the RF. As the sequence progresses the MPCU supplies the GC_DSP on the MC4C40 board with requests for pulse generation of the logical slice selection, read-out and phase encoding gradients. A rotation matrix is then sent so the GC_DSP can calculate the logical amplitudes (SS, RO, PE) into the physical of X, Y or Z. The resulting amplitude values are passed onto the ECC_DSP which makes corrections for the eddy-current, cross-term and B0 compensations before sending the data to the gradient amplifier. The amplitude data is for each axis 18 bits wide and some additional timing and control signals are sent as well. There are more signals than pins in the cable to the gradient so the data has to be multiplexed.

Gradient Safety Watchdog (GSWD) -Stimulation Monitor

The GSWD is only implemented on all Sonata, Trio systems and Symphony systems with the Quantum gradient option.

When applying strong gradient pulses during an MR examination muscular stimulations of the patient can occur. The cause of these stimulations is the development of electrical fields within the patient’s nerve fibers which are induced by the dynamic magnetic field generated by the gradient coil. The magnitude of these induced electrical fields is, for any given sequence type, proportional to the change of the magnet field in time, expressed otherwise as dB/dt.

A stimulation occurs when a characteristic threshold of the electrical field is exceeded. The corresponding dB/dt value needed to exceed this limit depends on the patient’s anatomy and physiology, the geometrical and physical attributes of the gradient coil and the position of the patient within this coil.

dB/dt is determined by the amplitude and rise time of the gradient pulses. In actual imaging conditions, dB/dt is never constant, but is dependant on the sequence type and sequence parameters (e.g. slice thickness, FOV, Matrix size, TR, TE, number of slices, etc.). The stimulation thresholds are further influenced by the timely organization of the individual gradient pulses, the total number of pulses their repetition time and the coincidence of any or all three gradients at any one time.

The Gradient Safety WatchDog (GSWD), also called stimulation Monitor (STIMO), provides the safety against patient stimulation. The stimulations monitor (STIMO) is realized in three parts:

- a Look-ahead monitoring with a "SAFE Model"
- a Look-ahead monitoring using the legal dB/dt-Limits
- a run-time monitoring with a "SAFE Model"

Look-ahead Monitor

The look-ahead monitor runs simultaneously with the parameterization of a sequence. If sequence parameters are chosen that will cause a stimulation, a warning message is issued to the effect and eventually changes are suggested which will prevent stimulations.

The "Safe Model"

In the SAFE-Model, the physiological stimulation is approximated by filtering the differentiated gradient pulses. Each gradient pulse is differentiated and then filtered by applying at least two exponential functions using two time constants. The filtered signals are then added together using a weighting function and then compared to an established limit. The established limits have been derived empirically through studies done on real people (they are all still living). If these limits are exceeded, stimulations are to be expected.

Online Monitor

The actual value gradient currents are sent to the first RX_Module in the RFSU, digitized, and sent to the MC1C40 over the MMC backplane.

The STIMO_DSP on the MC1C40 calculates the actual gradient values and compares them to stimulation limits. If a limit is exceeded, the GPA will be disabled via a signal.

System Compensations

Coil related Adjustments

Before a sequence measurement can begin there are some system and patient dependent adjustments which must be performed. These include:

- Body Coil tune (Adj/Tune)
- adjust frequency (Adj/Fre)
- adjust transmitter (Adj/Tra)
- adjust shim

All of these adjustments are in themselves sequences. The sequence is loaded into the MPCU and performed as any other sequence. The measured raw data will be transferred to the Imager (MRIR) where the evaluation is performed.

3D Shim

The 3D Shim procedure is to compensate the inhomogeneities of the magnetic field. A short sequence is used to determine the 1st-order and 2nd-order distortions. In order to get the field homogeneous, the 1st order terms are compensated by driving an offset current to the gradient coils. For compensating the 2nd order terms, all systems are already equipped with 5 shim coils in the gradient coil assembly. To drive a current also through these coils, you need to have the shim option installed. If no shim option is present, the 3D Shim will only calculate the 3 linear gradient offsets.

Eddy Current Compensation

The eddy current compensation is performed digitally in the ECC_DSP on the MC4C40 board. The correction values are determined in the tune-up and are calculated online. The correction also includes a B_0-component, which is implemented by changing the synthesizer frequency dynamically during the sequence.

Gradient Delay

The gradient delay compensation for the three axis is performed by the TICO section of the GCTX. The necessary time parameters are determined in the tune-up procedure.

Dynamic Control Signals

Control signals necessary for the dynamic detuning, are set by the GC-DSP, respectively the ED-DSP via the backplane to the RFCI. Remember, you need tune and detune signals for the body and local coils.

Various dynamic control signals required by several RF components and the gradient amplifier are generated by the TICO section of the GCTX, converted to fiber optic signals by the IOP and distributed to the components. The dynamic detuning signal for the local coils is sent to the RFCI electrically over a cable from the MMC backplane.

NOTE This cable is NOT tested by SeSo!

Supervision, Error Handling

During the sequence the MPCU also supervises the scanner hardware for errors. There are two main events that can cause an abort of the sequence:

- input of an error message from an unit
- the system state changes to ’not normal’

Error Messages can be generated by any of the units. If an error is detected, it will be classified and sent to the MPCU. Here, dependent on the context the MPCU will take appropriate actions. The following **error classes** are established:

- Warning
- Error and
- Alarm

In all cases the operator will be informed and there will be a new entry in the **NUMARIS Eventlog**. A running sequence will be stopped only in case there is an ’Error’ or an ’Alarm’ classification. The SCT will stop feeding the DSPs.

In addition to the error messages, there is also the System State. For each of the CAN units there can be up to 32 bit of status information available, giving detailed unit status.

If a status bit changes, e.g. due to an over temperature in the gradient coil or a malfunction in the chiller, the MPCU will be informed automatically via the CAN bus. This new state will be compared with the predefined patterns for each of the units, representing the normal working state. This definition is stored in the status file of NUMARIS and can not be changed by the operator. If the comparison shows a discrepancy an error will be created.

There are some cases, where the peripheral unit already knows, that the status change will lead to an error condition. In such a case, the unit itself already will create an error message. So it can happen, that an error condition results in two error messages sent to NUMARIS!

System Standby Mode

System Power ON_OFF Control

The Host is able to put the system into standby by sending a signal to the appropriate relay in the Line Power Distribution via the serial connection to the Alarm Box. In the Standby-mode all system components are switched off, with the exception of the MRC and MRSC Hosts.

Hardware

Figure 39 *MMC Layout*

Block Diagram

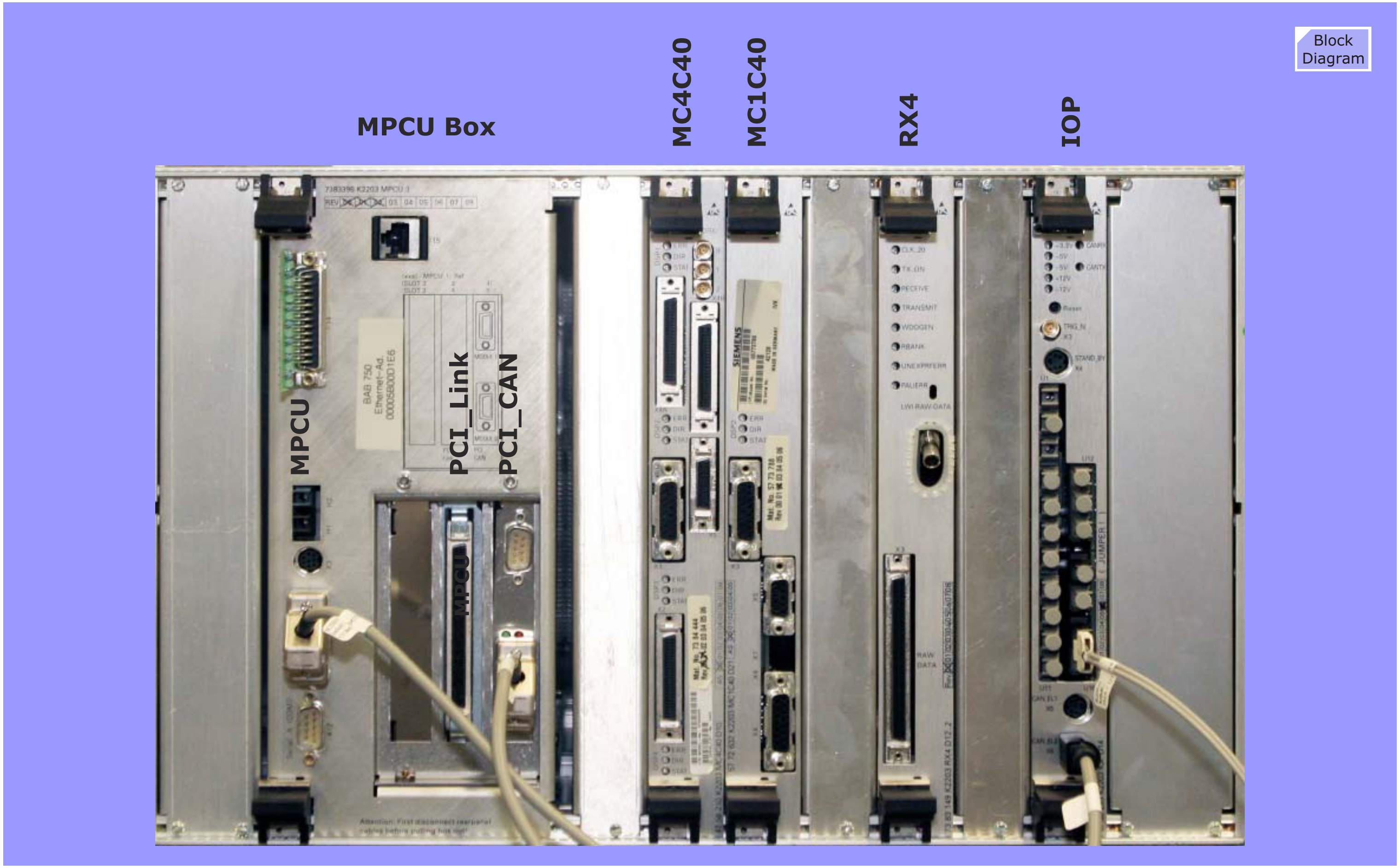
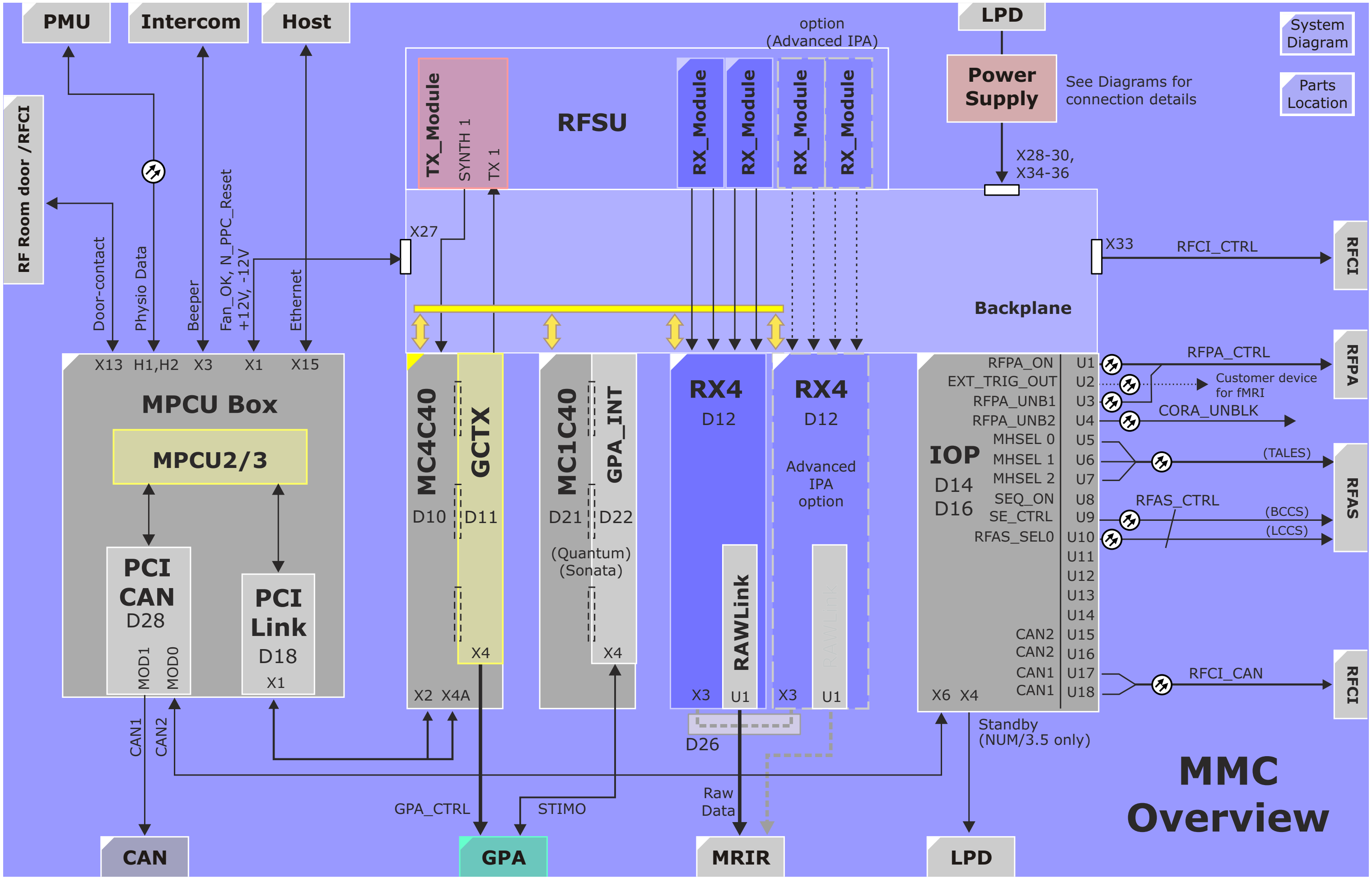


Figure 40 *MMC Block Diagram*



MPCU Box

The MPCU Box is a housing supplying mechanical and electrical connections of the MPCU, PCI_CAN and PCI_Link boards as well as providing connectors for external cabling to the MMC and periphery.

The **MPCU** is a single-board industrial controller using the PowerPC CPU with **VxWorks** as operating system. Standard interfaces include an Ethernet adapter, a parallel port, and both electrical and FO serial interfaces. Systems running *syngo* MR software (systems produced after July 2000) are equipped with an MPCU2 and systems after July 2001 with the MPCU3 board, while older systems running NUMARIS 3,5 software will have a MPCU 1.

The PCI_Link and PCI_CAN interfaces are installed in slot 4 and slot 5 respectively.

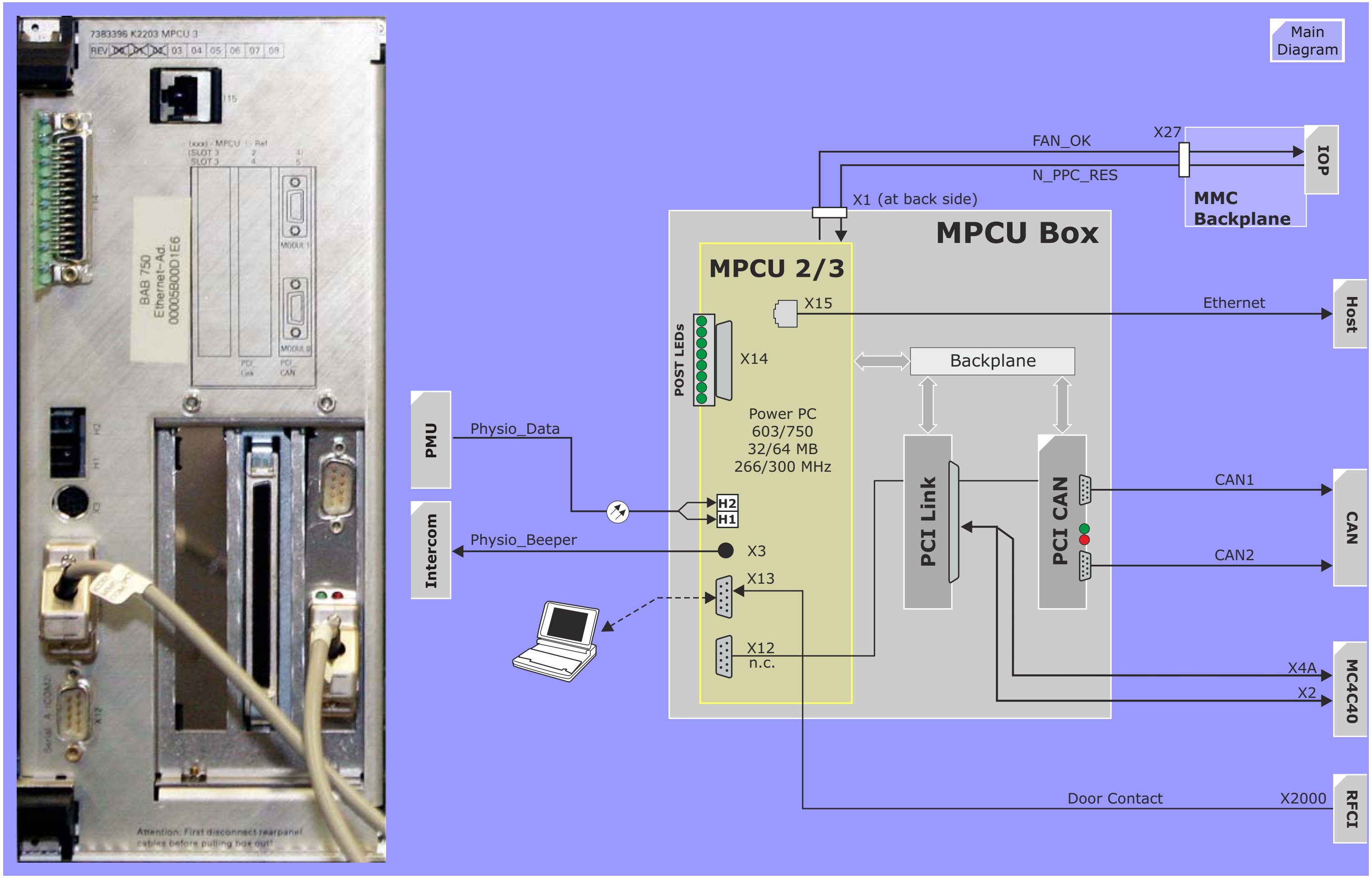
X12 is currently not used.

A service terminal can be used at x13 for setup of boot parameters in the BIOS.

PCI-Link Interface

This board provides a digital data interface between the MPCU and the DSPs on the MC4C40. The bus drivers and monitoring of this bus are found on the MC4C40 (see MC4C40 LED description).

Figure 41 *MPCU Box Block Diagram*



PCI_CAN Interface

The PCI-CAN interface provides two electrical CAN busses and one RS232 interface, which is not used.

CAN Controllers

The CAN controllers provide two separate busses:

- **Active** - used to connect periphery that must remain active during the sequence
- **Mute** - used to connect peripheral devices that must be put into standby during the sequence, in order to prevent noise

The ACTIVE bus is labelled as CAN Bus 1.

The MUTE bus as CAN Bus 2

Outputs

Opto-couplers are used to provide galvanic isolation between the MMC and the periphery. The output connectors are DB-9.

The RS232 port is internally connected to X12 of the MPCU-Box and is currently not used.

LEDs

Only the CAN Bus 2 controller has LEDs:

- GREEN: BUS RUNNING, ON indicates Bus initialized
- RED : ERROR, is on during boot until the CAN controller has been initialized

Figure 42 *PCI_CAN Block Diagram*

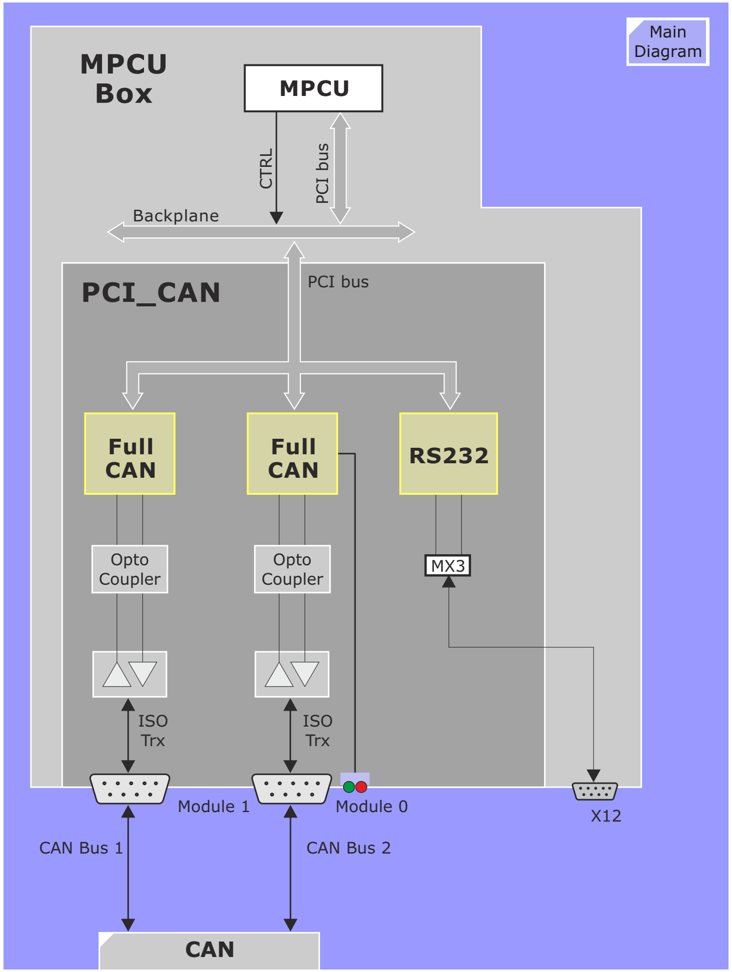
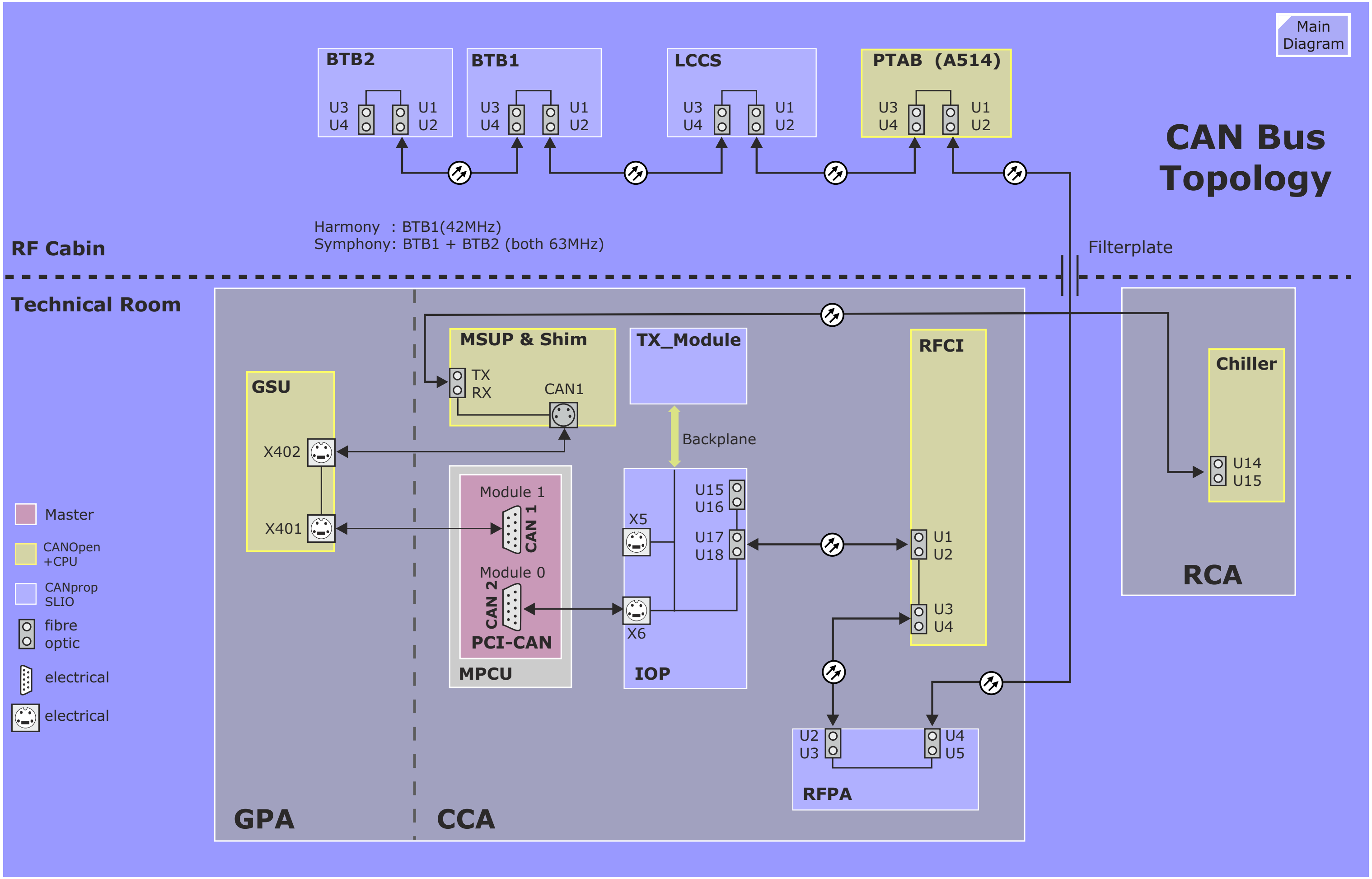


Figure 43 *CAN Bus Topology*



CAN Bus

The CAN Bus is a serial bus used for smaller computer networks. It operates in our implementation with a speed of 100kbit/s.

Figure 43 gives an overview of the components connected to the CAN bus, and the connection type. There are two connection types: an optical CAN bus, using a pair of fibre optic cables, and an electrical connection, using a twisted pair cable.

There are two CAN busses leaving the CAN master.

- CAN1 is always alive, it connects MPCU, GSU (GPA), MSU and RCA (Chiller).
- CAN2 connects MPCU, IOP, TX-module, RFCI, RFPA, PTAB, LCCS, BTB1 and, in case of Symphony, BTB2. CAN2 will be set into a ’sleep mode’ during a measurement, not to create any image artifacts. The units can be wakened up via an interrupt of the connected hardware, thus enabling error messages to MPCU.

NOTE In the diagrams and on some hardware components you will find the names ´CAN1´ and ´CAN2´ for the connectors at one single unit. This is misleading. It just shows the difference between "incoming" and "outgoing" bus.

There are LEDs indicating bus activity on the CAN2 bus at the IOP, RFPA and BTB. At the CAN1 bus there is no such an indication.

In the idle state there is no activity on the bus.

In the diagram you find that some CAN units are called SLIO and some are called CAN Module.

- **CAN** SLIO is a serial-linked-IO, this means it just has input and output ports, no intelligence.
- CAN Module has a CPU and a loadable software program.

SLIOs have to be polled by the MPCU, while the CAN+CPU units can actively access the bus.

NOTE Troubleshooting hint:
Since the CAN bus is a bus, the cables can be connected to different units.
E.g. at the IOP board there are the spare connectors x5 (mini DIN) and J15, J16 (FOC) available.
Just keep in mind, that the CAN bus tests will abort if two subsequent units (as the test expects) are not responding.

CAN Modules (CAN & CPU)

CAN modules are relatively small piggy-back devices which contain a CPU. There are five such modules used in the scanner, most of which are identical. A CAN module is identified by a 4-bit unit ID which is provided on the board into which the Can Module is plugged into.

As with all CPU circuits there is also some RAM and a PROM for Firmware. The firmware is the same for all CAN modules and contains only a boot, initialization and loader program.

The application specific software for the CAN Modules (called loadware - LW) is contained in an EEPROM and will be downloaded during installation or software update.

LED

After power ON, the CPU will boot and initialize its environment with the Firmware contained in PROM. At this time the FW LED will blink. After completing the boot process it will load the Loadware from the EEPROM, if available, and turn the LW LED on and the FW LED off.

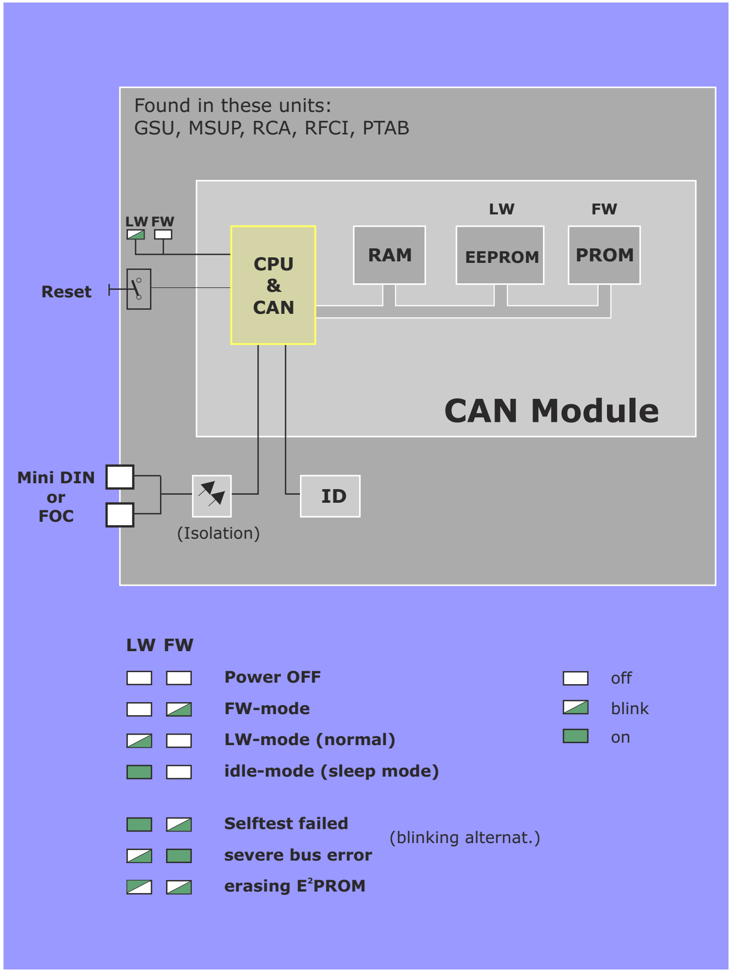
The FW / LW LEDs are used to indicated other conditions according to the diagram at right.

With a reset button the CPU will be forced to restart like power ON.

After a reset a "Reload All" has to be performed, since unit specific NUMARIS parameters may be required.

During a measurement all units at CAN bus 2 will be set into "sleep mode" in order to avoid noise. This is indicated by a permanently lit LW LED. Any activity on the bus will wake up the SLIO units. The CAN+CPU units will wake up when they are addressed by the MPCU. This is normally the case for the PTAB after the measurement but not for the RFCI.

Figure 44 *CAN Modules*



Initialization of CAN Units

After power on (or reset) the FW on the CAN+CPU units perform a power up test.

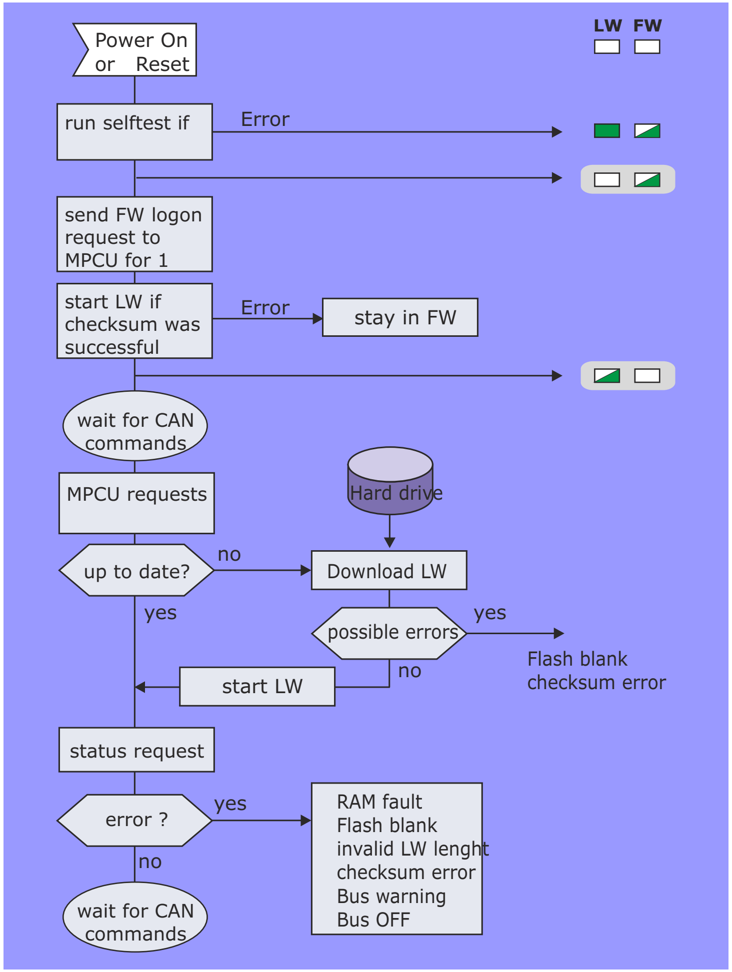
Then the LW will be started if the checksum is ok.

The MPCU will check the LW version and compare it with the corresponding file on the system disk. In case of a difference the valid version will be downloaded. Normally this procedure will be skipped.

After a status request the unit is waiting for commands from the MPCU e.g. NUMARIS parameters for a measurement.

NOTE The hardware of the piggy modules is identical but the loadware is not! During power on, the CPU switches from FW to LW. This LW is designed for a specific unit.
It is not predictable what this software will do in a different environment. Therefore don't swap the piggies!

Figure 45 *CAN Initialization*



MC4C40/GCTX

MC4C40

The MC4C40 is a generic processor board with four C40 DSPs from Texas Instruments. Each DSP is supplied with 128kByte of memory and provided with links to all other DSPs and a RS485 interface for out-going signals to the TX_Module. The gradient pulse amplitudes are supplied over the GCTX board described below.

The DSPs are named according to their functions as follows:

- DSP1 for transmit amplitudes (TX)
- DSP2 for gradient amplitudes (GC)
- DSP3 for SAR calculation and image header (RX)
- DSP4 for eddy current compensation (ECC)

GCTX

The GCTX is a specially designed board realized as a piggy back (it plugs on top of the MC4C40 board) containing all the functional circuitry for the sequence RF pulse generation, the data delay circuits and synchronization pipeline for the gradient pulses and the TICO section which is responsible for providing the timing for the sequence events. The main tasks are:

- The TICO generates the sequence timing for Grad, RF-TX (incl. PFPA_UNBL) and RF-RX. Also the control signals for the RF coil dynamic detuning are generated here. They are sent to the RFCI via the backplane.
- The Grad unit synchronizes the prepared Grad amplitude (from DSP2/4) with the delay generator.
- The RF-TX performs the digital modulation with the MOFI chip. The NCO provides the carrier (Local Oscillator) for the
- TX_IF signal (slice position) and demodulation on the RX4 (RXLO_xx)

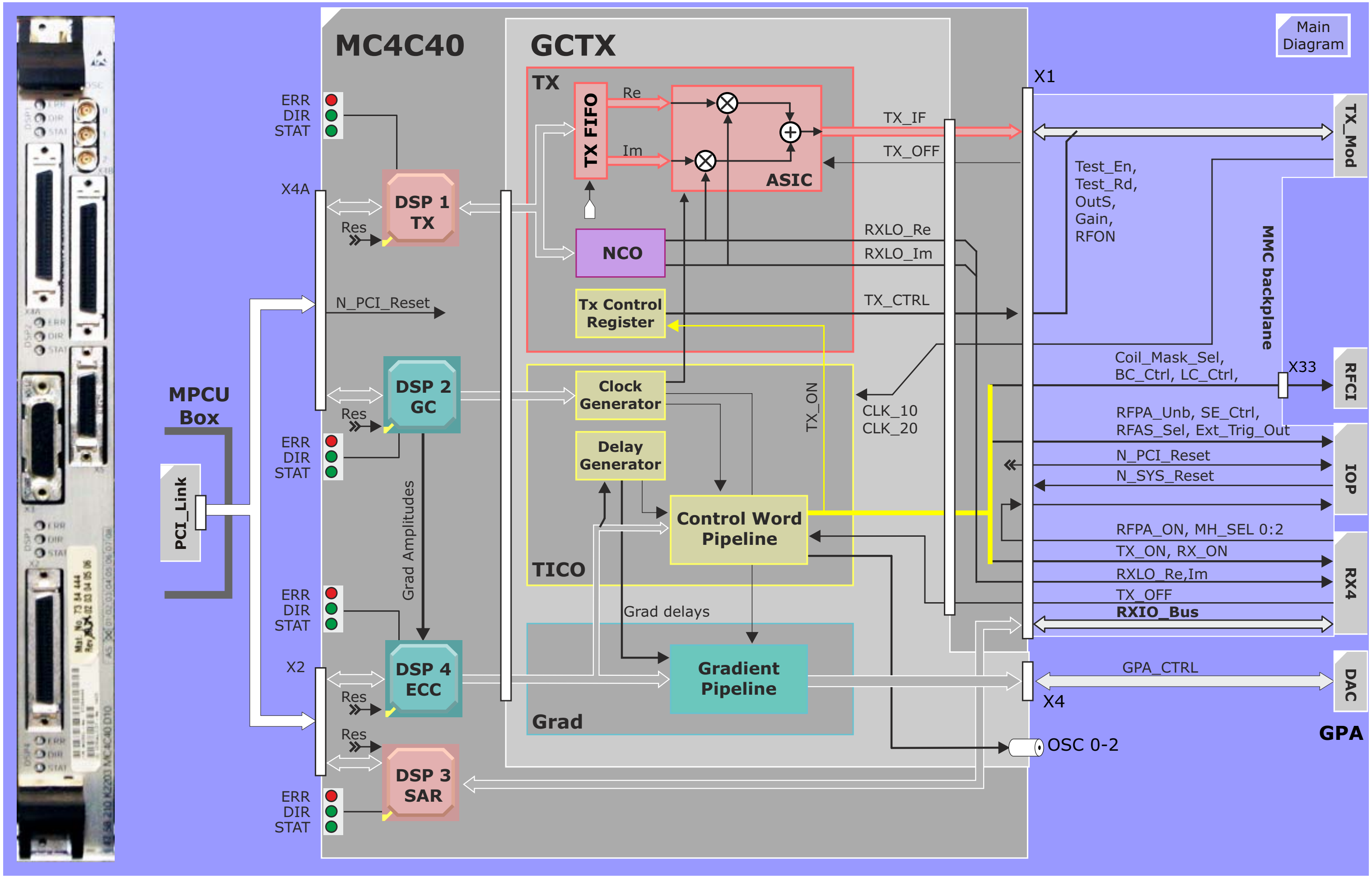
The gradient amplitudes are delivered to the Gradient Amplifier over connector X4B.

LEDs

LED	Description
ERR	Link error. THe data link between the MPCU/PCI_Link and the MC4C40 DSP is not ok.
DIR	direction of data flow to/from PCI_Link
STAT	status ok when on, must be on if software is running

For the detailed description of the functions performed by the AMC components, refer to the Functional section above.

Figure 46 *MC4C40 / GCTX Block Diagram*



MC1C40 / GPA_INT

Overview

The MC1C40 is in principle identical to the MC4C40 with the exception that it contains only 1 C40 DSP. It is responsible for the Gradient Safety Watchdog functionality (stimulation monitoring) for Quantum gradient on the Symphony and Sonata systems.

MC1C40

The stimulation values are calculated on the MC1C40 using the actual gradient values coming from the GPA. If limits are exceeded, the GPA will be disabled.

GPA_INT

The GPA INT is connected to the MC1C40 as a piggy back. The only used front connection is X4 which receives the actual current values of the three gradient axes from the GPA.

Outputs

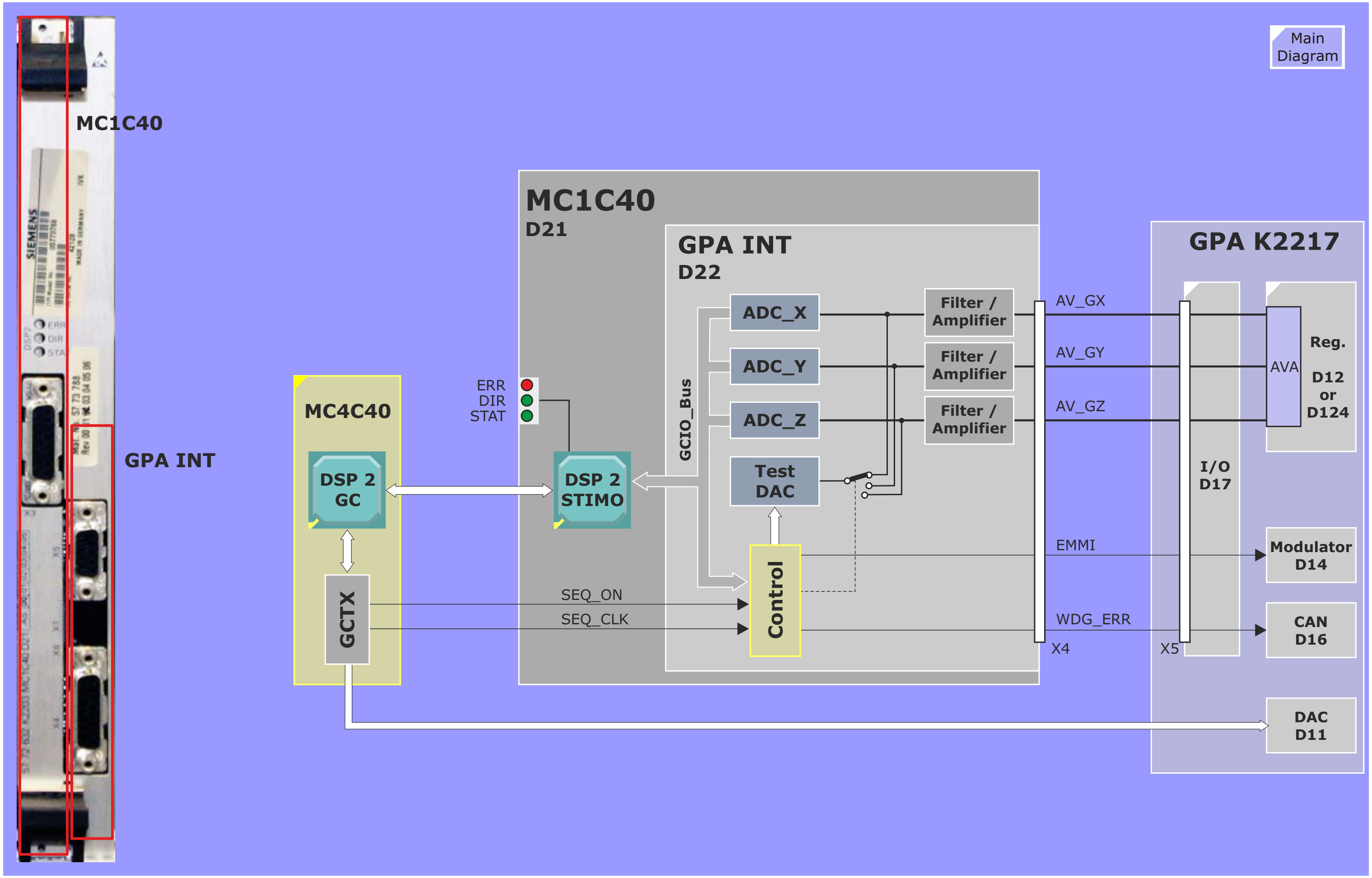
In addition two signals are control and status for the GPA:

- GPA_ON -> Enables gradients, but disables if stimulation limits are exceeded
- WDG_ERR -> Watchdog error

LEDs

LED	Description
ERR	Link error. THe data link between the MPCU/PCI_Link and the MC4C40 DSP is not ok.
DIR	direction of data flow to/from PCI_Link
STAT	status ok when on, must be on if software is running

Figure 47 *MC1C40 / GPA INT Block Diagram*



RX4

The RX4 board supplies four demodulator and filter circuits for up to four RF channels.

The Advanced IPA option includes 4 additional RF channels requiring a second RX4 board.

Inputs

As the RX4 block diagram shows, the main task of the RX4 is to demodulate the receive signals. As input signals there are the 4 digitized MR signals RX1_IF...RX4_IF coming from the *RX_Module* in the RFSU and the digital demodulation signals RXLO_RE and RXLO_IM generated from the 10MHz clock by the GCTX. And all signals are in 16 bit format. This explains the 435 pin (!!!) backplane connector. Don't mess any up, with your luck they will all be used...

Demodulation

Via the MOFI interface the data is sent into two ASICs called MOFI1 and MOFI2 (MOFI stands for modulator and filter). Here the digitized MR signals will be demodulated, or better said, the demodulation will be calculated. Digital demodulation has the advantage that expensive analog components (LP filter, demodulator, ADCs) are not required and there is absolutely no distortion of the signal - one more step towards excellent image quality.

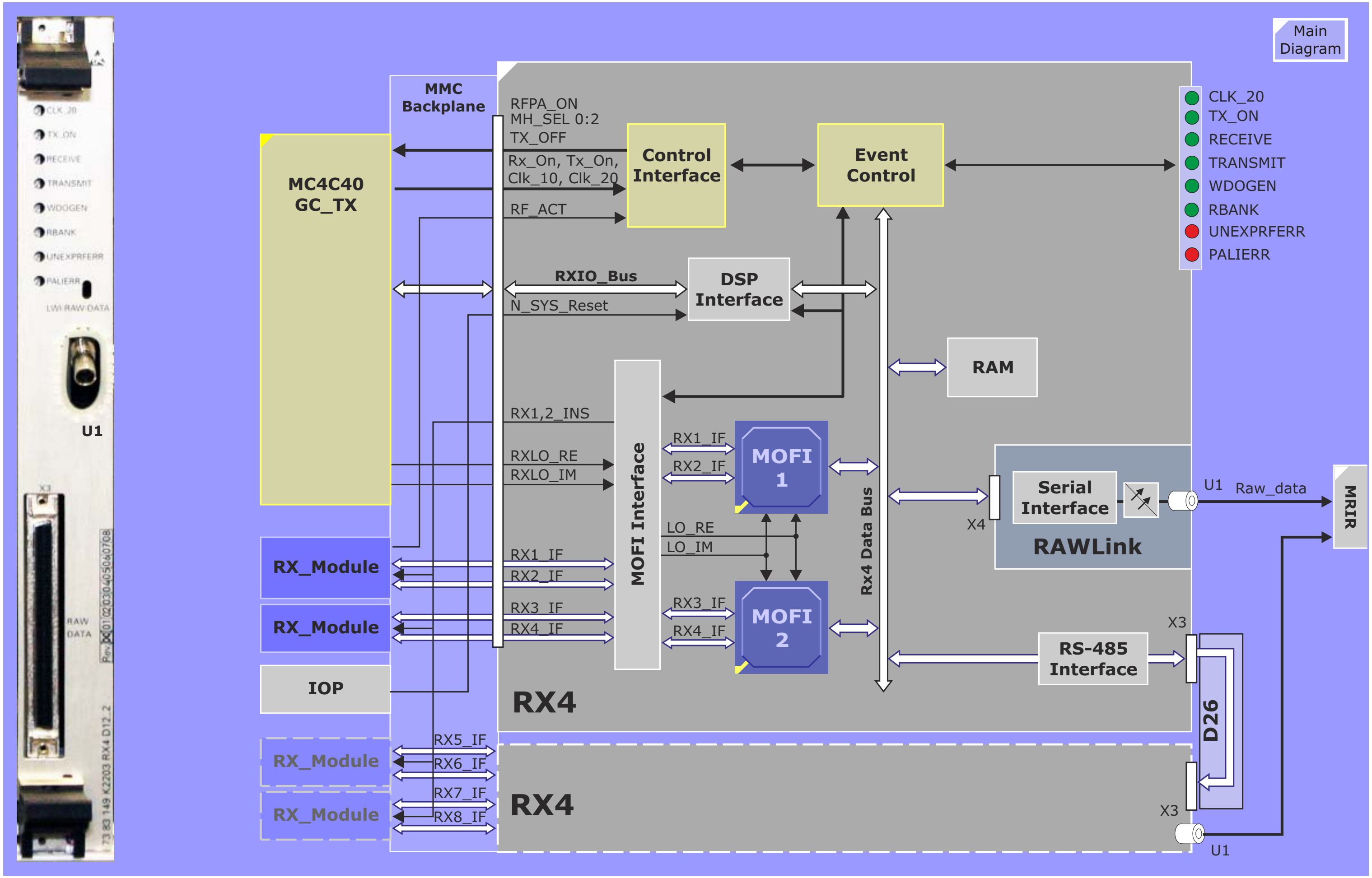
Output

A high-speed glass fiber optic cable delivers the processed MR signals (from this point on called raw data) to the Imager. If the Advanced IPA option is installed, the data output of the first RX4 board is sent to the second RX4 board over a connection board (D26) on the front side connector X3. The fiber optic link to the imager is taken from the second RX4 board.

LEDs

LED	Description
CLK_20	on when the 20 MHz clock (generated by the synthesizer of the *TX_Module* in the RFSU) is available, essential for the entire RF system and MMC
TX_ON	transmitting of RF power is enabled, this signal is necessary for PALI supervision
RECEIVE	data are being received from RFSU
TRANSMIT	data are being transferred to the Imager (MRIR) for image calculation
WDOGEN	the PALI watchdog is enabled
RBANK	Read Bank, internal signal necessary for sorting the data
UNEXPRFERR	there is RF from the RFPA without the TX_ON signal, this is an error condition.
PALIERR	PALI watchdog error. PALI did not react to an watchdog interrupt in less than 20μs. Software may have crashed.

Figure 48 *RX4 Block Diagram*



IOP

The IOP board has three main functions:

- Conversion of electrical control and CAN signals to fiber optic
- Voltage Monitoring
- Reset control

FOC outputs

Some control signals for the RFPA, TALES and BCCS are connected at the IOP fiber optic connectors U1 to U14. A number of the signals coming from GCTX and RX4 are not yet used.

CAN Bus Interface

The SLIO of the CAN bus (see next paragraph) mainly monitors the MMC power supply and the MPCU fan status. Besides the external CAN connections from MPCU and to RFCI, the TX_Module is connected via the backplane.

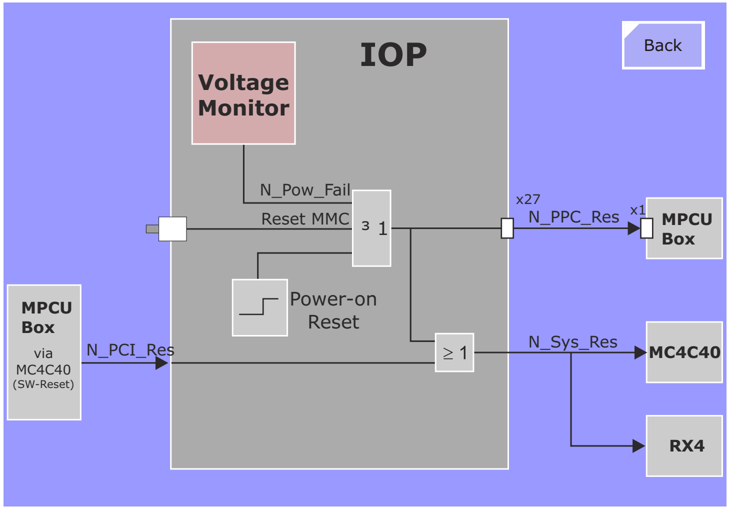
External Trigger Input

Is currently not supported by software.

Reset

The Power On Reset or the manual Reset via the push button will reset all MMC boards and MPCU via the backplane. The software can reset the MMC via MPCU, MC4C40 and IOP.

Figure 49 *IOP Reset Circuit*



Power Monitoring.

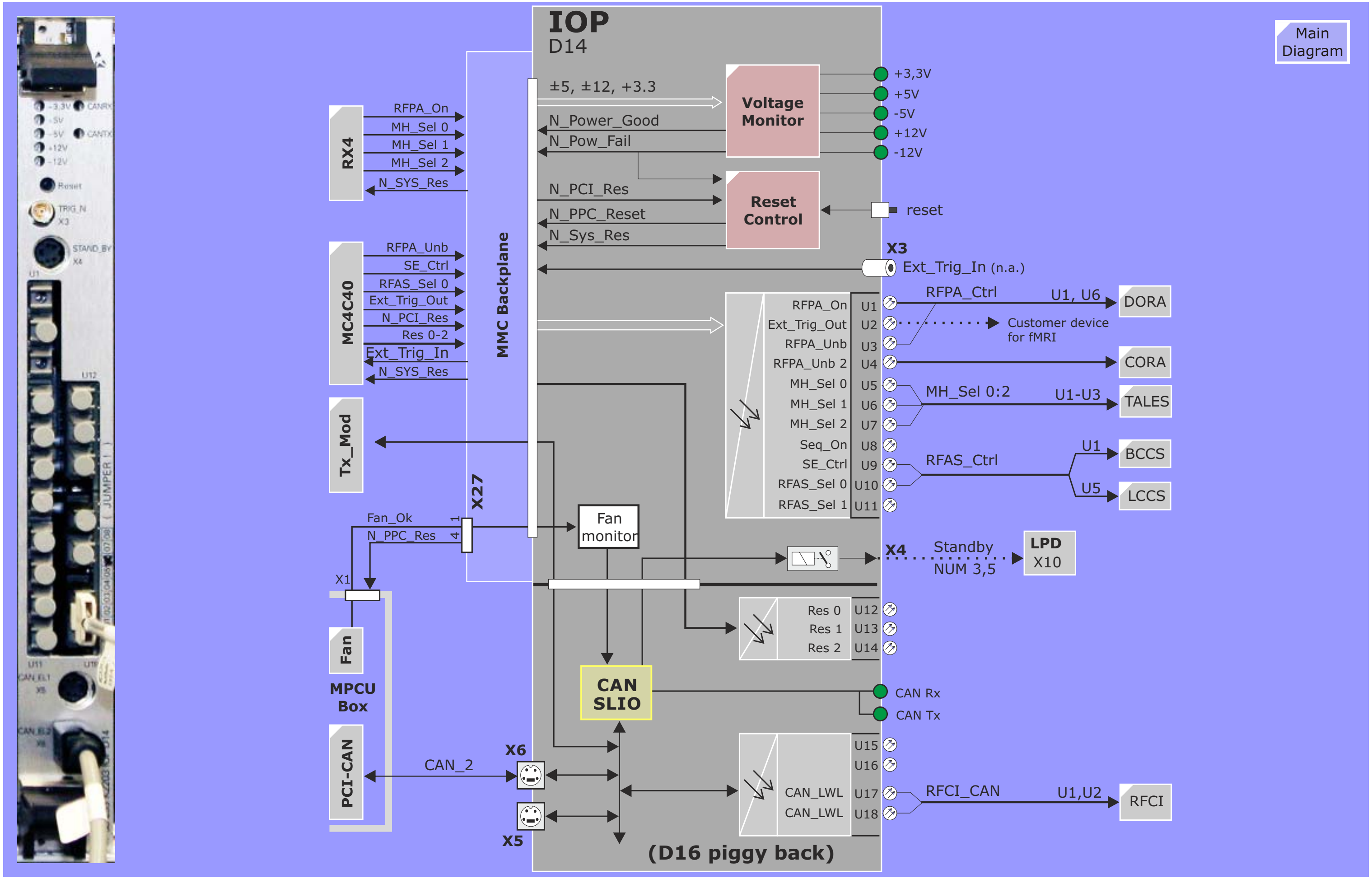
At the left top the LEDs of the MMC power supplies must always be on.

A monitoring system checks for the +/- 5% limits. Since there is no -5V supply, the jumper x13 (2-3) bypasses the check in order to enable the LED and avoid an error message. Any power failure would reset the MMC.

Standby

The **Standby** signal (X4) is used ONLY on NUMARIS 3.5 systems. For *syngo* systems a system standby is performed by the Host over a serial connection to the LPD over the Alarm Box. See HOST section for details.

Figure 50 *IOP Block Diagram*



Section

6

RF System

Introduction

RF Small Signal Unit - RFSU

The role of the RFSU has been reduced to that of a converter. The RF pulse modulation and MR signal demodulation are performed by DSPs of the GCTX and RX4 respectively. The Tx_Module must only convert the digitally modulated RF pulses from the GCTX and then mix the result up to the required system frequency. Conversely, the Rx_Module mixes the MR signal down to about 1MHz and A/D converts the 1MHz IF frequency for the RX4 which then does the filtering and demodulation digitally.

There are no adjustments required and there are no filters to clean and maintain.

The RFSU components are broad-band and are used for 0.2T (8.25MHz), 1.0T (40.45MHz) and 1.5T (63.6MHz) systems.

RF Power Amplifier - RFPA

A new double-resonant RFPA has been designed for the Harmony, Symphony product family. It is *fully solid-state* (no tube, no external power supplies, no discharging points and no manual adjustments), *extremely small,* is water cooled, can produce up to 15kW and can be used on both the 1 and 1.5 Tesla systems. The amplifier recognizes the input frequency automatically and sets the max power limit accordingly. Communication to the RFPA is achieved with a CAN SLIO component.

For 1.5T systems a multi-nuclei spectroscopy option is available which includes an additional 2kW solid state amplifier. We were in a hurry, so we borrowed the CORA from the Open system. At spectroscopy frequencies it produces more RF than we need.

RF Application System - RFAS

The functions are all there: T/R switch, pre-amps, impedance matching networks. For the most part, the previous circuits have been consolidated into a single unit to reduce the component count. The BTB has grown somewhat from three fixed tuning ranges of the Impact/Expert to 128 or 256 tuning positions. A calibration program, similar to the one used in the Vision, will be used to "map out" the tuning ranges so that the impedance matching process (patient tuning) can be automated and performed within seconds.

The SAMI is a frequency switch device included in the MKO spectroscopy option. It allows the connection of two RF power amplifiers operating in two different frequency bands to a common transmit coil.

Control for these components is exercised by CAN controllers or SLIOs. In its simplest form, a CAN SLIO is principally a programmable I/O chip with a CAN bus serving as the communication link to a CPU. A CAN controller has an integrated CPU. Both types are used in the RFAS.

Local Coils

Integrated Panoramic Array

A major technological advancement has been achieved in the Local Coil design concept which has been termed Integrated Panoramic Array. The Integrated Panoramic Array, or IPA, is the latest evolution in coil handling and CP array coil technology. The basic approach is to:

- reduce patient setup time by having some coils integrated into the patient table
- reduce the number of individual coils
- increase flexibility and optimize image quality by the possibility of combining almost every single coil with others for the intended application

The advantage of the Integrated Panoramic Array is to reduce patient setup times by leaving the head and spine array on the patient table for almost all examinations with the exception of female breast imaging. Therefore one has no longer a "Body Array Coil" or a "Neck Array Coil" as on Expert and Vision but the CP Spine array serves as the lower part of these coils; they are used in conjunction with the CP Neck Array or the CP Body Array, respectively. The user has to select the corresponding spine array elements in these cases. The image area is not just where the coil is located but it is variable.

Harmony and Symphony systems can be equipped with IPA (standard) or Advanced IPA (option). In addition, both systems are using analog linear signal combination of elements as well as array image combination. This will be explained in more detail in the Local Coil section below.

All array coil resonator elements employ circular polarization technology for optimized homogeneity and utilize built-in preamplifiers for unparalleled S/N performance. The coil casings use a specially formulated plastic with a high static load rating and excellent RF permeability to ensure good RF field homogeneity. All local coils are factory tuned and do not require a patient dependent impedance matching.

The RF System Menu

The RF System hardware components have been placed into the following blocks:

- RF System Block Diagrams for:
  - Harmony
  - Symphony
  - Symphony with Spectroscopy option
- RF Signal Unit (RFSU)
  - TX_Module
  - TX_Module for Spectroscopy
  - RX_Module
- RF Power Amplifier (RFPA)
  - 15kW RF Power Amplifier (DORA)
  - 2kW RF Power Amplifier (CORA)
- RF Application System (RFAS)
  - TAS for 42/63 MHz
  - SAMI for 63 MHz
  - BCCS for 42 MHz
  - BCCS for 63 MHz
  - LCCS for 42/63 MHz
  - TALES for 42/63 MHz
  - BTB for 42 MHz
  - BTB for 63 MHz
- Antenna Systems
  - Body Coil for 42 MHz
  - Body Coil for 63 MHz
  - Local Coils
- RF Cabin Interface (RFCI)

Figure 51 *Harmony RF System Block Diagram*

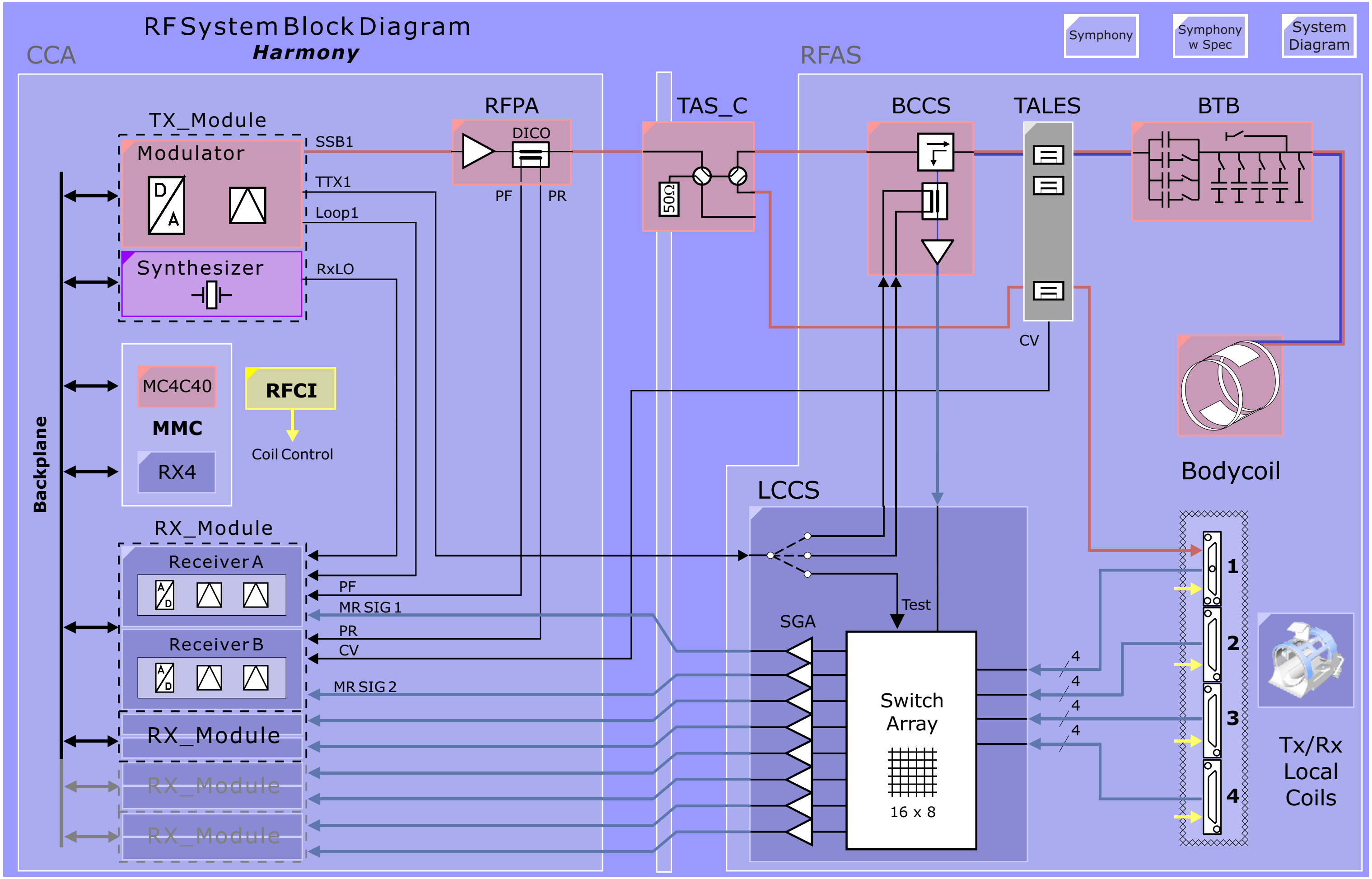


Figure 52 *Symphony/Sonata RF System Overview*

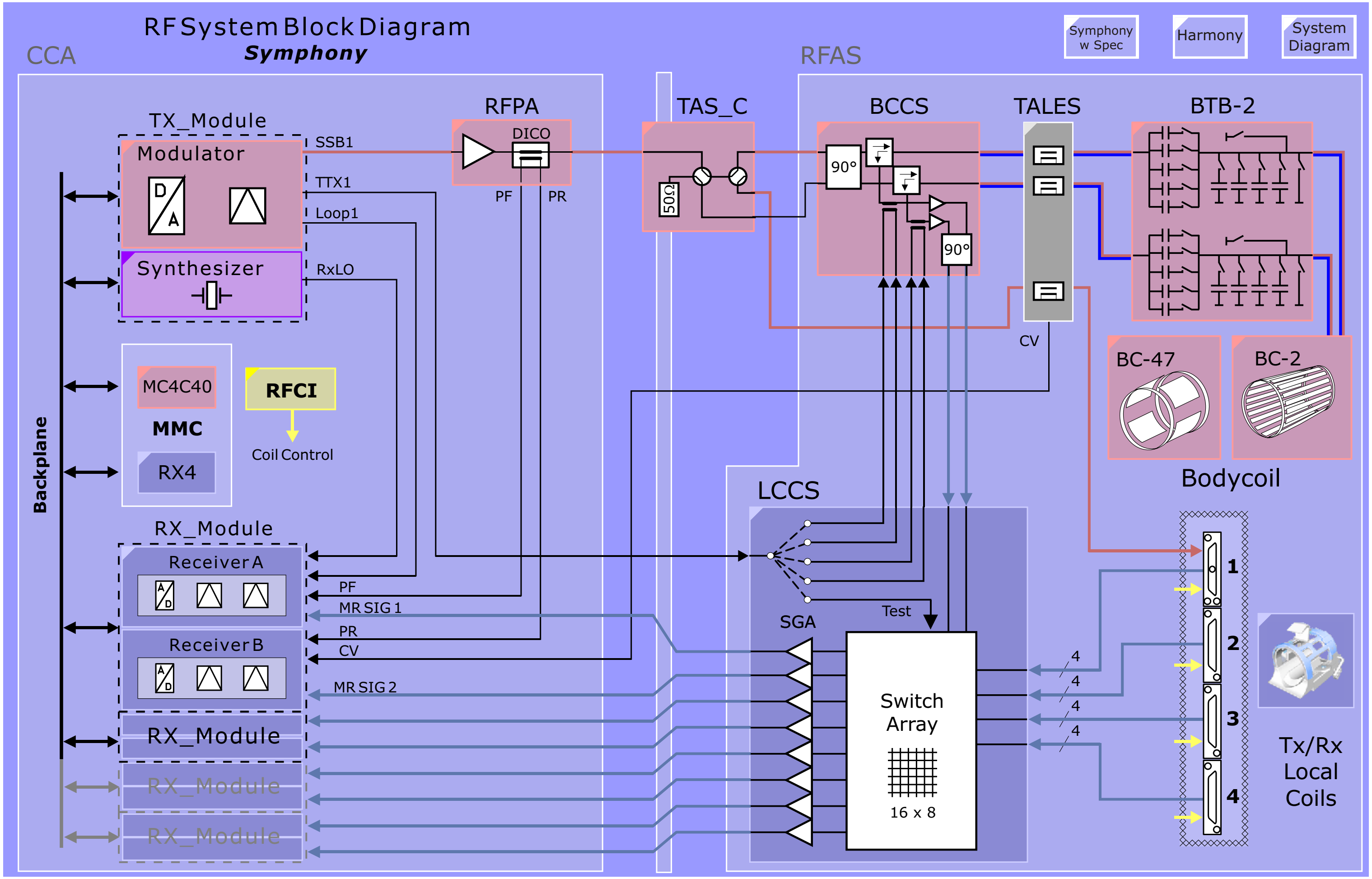
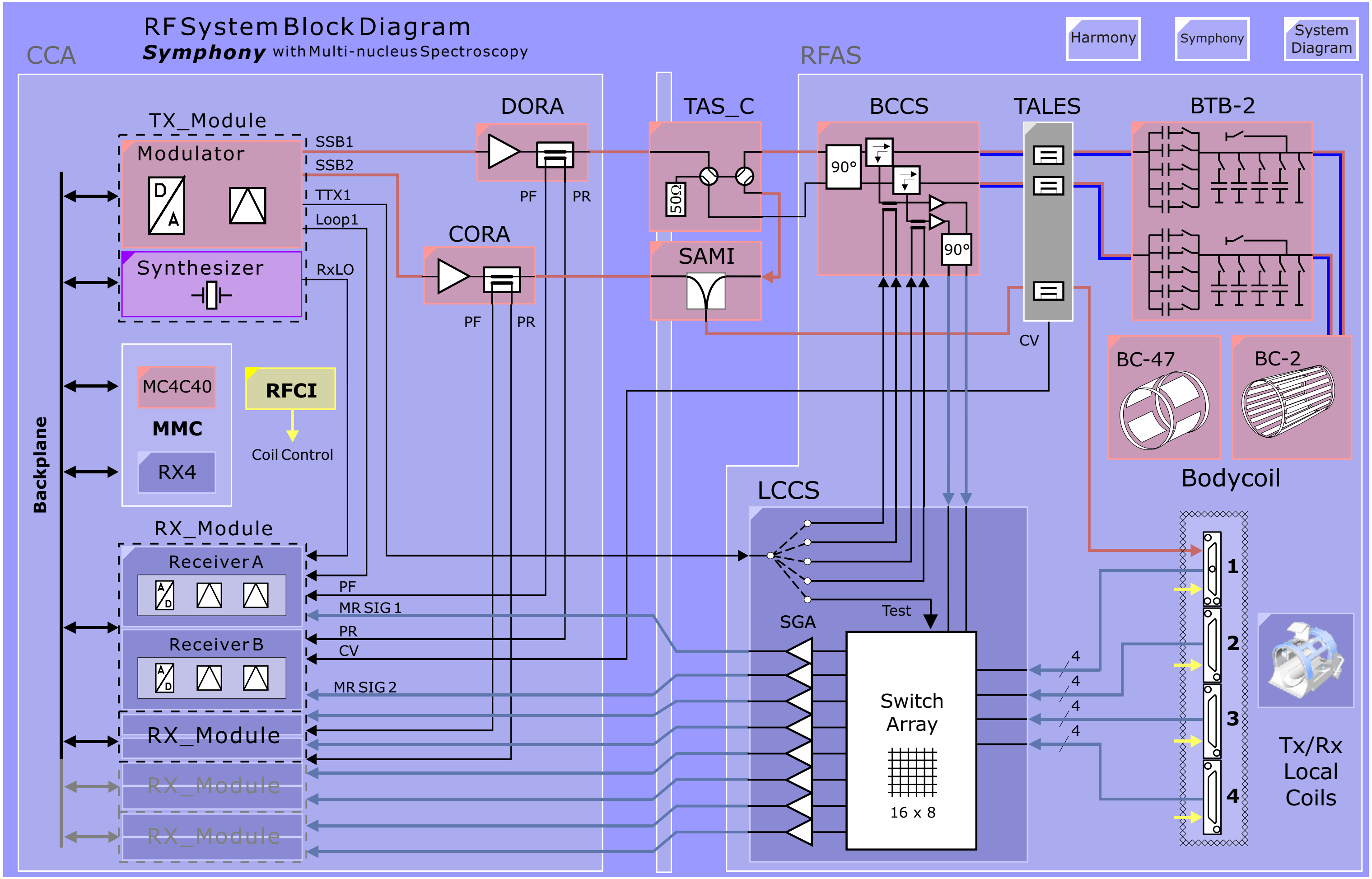


Figure 53 *Symphony/Sonata RF System Overview*



RF Signal Unit

Overview

The RF Signal Unit consists of just two components:

- TX_Module
- RX_Module

Both modules are produced by Siemens Albis, a Swiss company specializing in telecommunications hardware. One can see in the picture at right that each module is comprised of two cassettes which are assembled to form one plug in unit.

The **TX_Module** 42/63 MHz consists of a Synthesizer and a Modulator.

The **RX_Module** consists of two RF Receivers.

These components are field replaceable units. They require no maintenance and have no adjustments. Testing of these units is performed through the Service Software (SeSo) platform.

TX_Module 42/63 MHz

Overview

The TX_Module consists of a modulator cassette and a synthesizer cassette assembled together into a single module.

The Synthesizer generates the carrier signals for the mixers in the Modulator and Receivers as well as synchronization clocks required by the DSPs of the MC4C40 and RX4.

The Modulator, wrongly so called, is responsible for D/A conversion of the digital SSB RF pulses from the GCTX and mixing the results up to the required system frequency of 40.45 MHz or 63.6 MHz. Modulation is performed digitally by a DSP on the GCTX.

Figure 54 *TX_Module (left) and RX_Module*

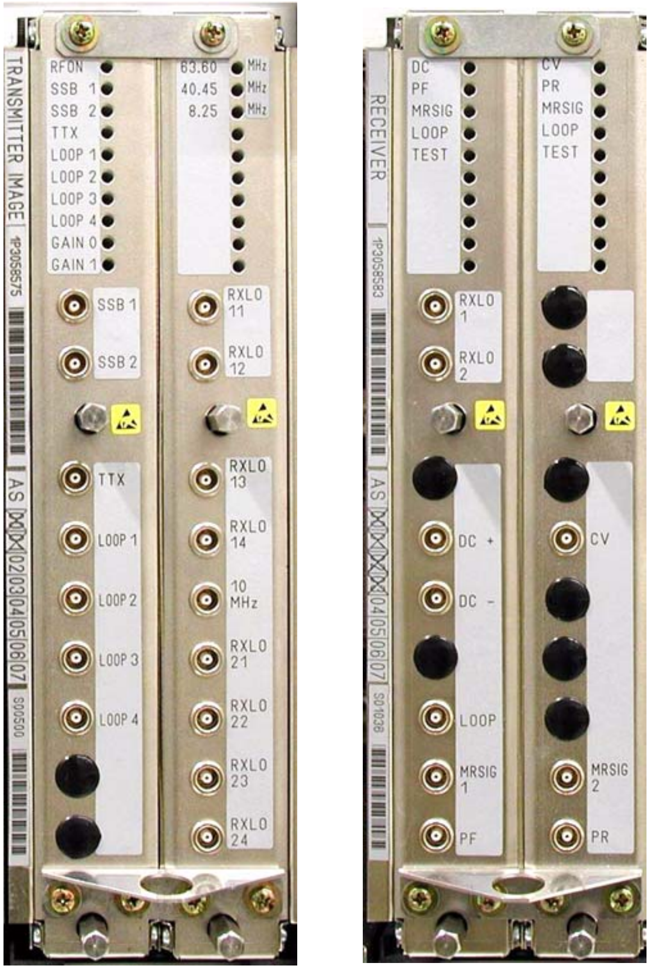
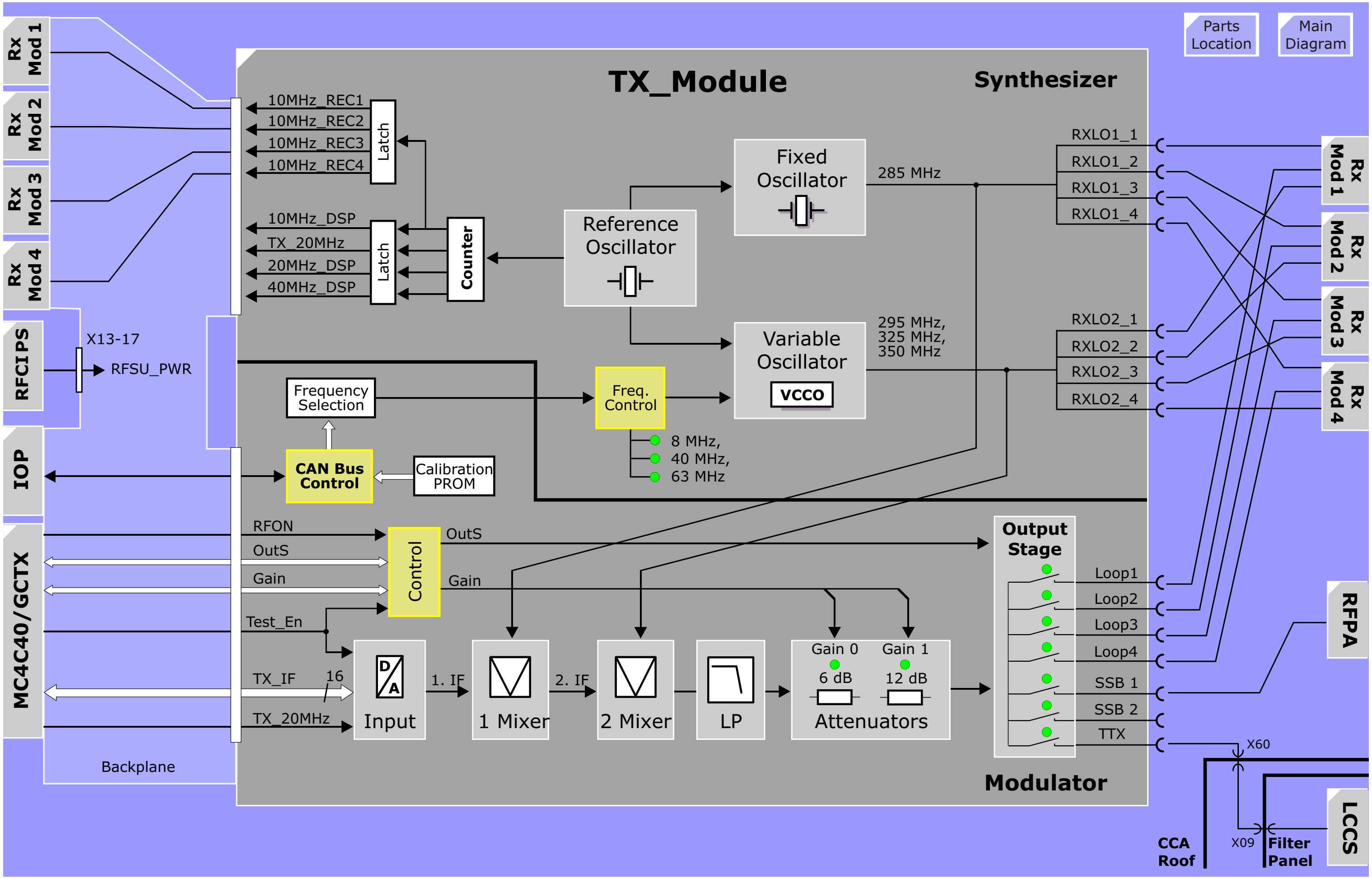


Figure 55 *TX_Module Overview*



Synthesizer

The synthesizer has the task of producing stable clocks used by the DSPs performing the modulation of RF excitation pulses and demodulation of received MR signals as well as provide clock signals used by other DSPs and control circuitry responsible for sequence control and gradient signal generation.

Inputs

The frequency range selection is determined by the variable oscillator which is set via the CAN SLIO in the Modulator half of the TX_Module.

Function

Control

The synthesizer's Variable Frequency Oscillator output frequency and phase are selected via commands from the MMC over the CAN unit in the modulator.

System Clock Generator

The various clocks used by the RFSU and MMC components are derived from a central 10 MHz reference clock. The reference frequency of 10 MHz is multiplied up to 80 MHz from which 40 MHz, 20 MHz and 10 MHz clocks are produced.

The clocks provide overall synchronization of the data measurement system. The DSP clocks are used to synchronize the digital RF modulation and demodulation by the GCTX and RX4 boards respectively and the generation of the gradient pulses. They are also used to synchronize data transfer in general.

Outputs

The output level for RXLO1_1:4 and RXLO2_1:4 is typically +7 dBm (500mV).

The 10MHz_REC 1:4 signals are used in the RX_Module by the A/D converters for the acquisition of the MR signals. They are routed to the RX_Module slots over the backplane.

LEDs

The software selected middle frequency of 63.6, 40.45, 8.25 MHz will be indicated by a front panel LED.

Testing

Testing of the synthesizer is performed under the Quality Assurance section of the service software. The test is conducted using a phantom and a SE sequence. The frequency and phase are varied during excitation and the received MR signals are evaluated accordingly. Please be aware that this test is also sensitive to any other system instabilities.

Modulator

Modulation of the SSB pulse is performed digitally by a DSP of the GCTX. The digital data of the SSB pulse is then converted by the Modulator to an analog signal and mixed up to the system frequency of 40 or 64 MHz.

Inputs

The TX DSP on the MC4C40 computes the center frequency, bandwidth, and pulse shape from the sequence parameters and produces the digital amplitudes which are then digitally SSB modulated on the GCTX module. The final pulse amplitudes, **TX_IF 1:16**, are clocked into the TX_Module over the backplane, synchronized to a 20 MHz clock. Additionally, the dynamic control signal RFON from the TICO is required for enabling the DAC.

Function

DAC

The digital amplitudes from the MC4C40 are converted to an analog signal with a 16-bit DAC. The RF digital amplitudes are varied to achieve the desired RF power at the transmit coil which is required for the flip angles in the sequence.

Mixer Stages

The D/A converted RF signal **1. IF** has a frequency of around 1 MHz. This has to be mixed up to the system frequency of 40,45 or 63,6 MHz. This is done by using two mixer stages instead of just one in order to achieve a very clean output spectrum, i.e., free of harmonic distortions and spurious. In the first stage the 1. IF signal is mixed with an LO frequency of 285 resulting in the sum and difference of 284 and 286 MHz. The difference signal is filtered out and the sum signal is applied to the next mixer stage (**2.IF**). In the second stage the 2. IF is mixed with the variable LO frequency of 325 or 350 MHz resulting in a sum of 611 or 636 MHz and the difference frequency of 40,45 or 63,6 MHz respectively. The following Lowpass filter filters out the sum signal and all harmonics and spurious resulting in a clean, harmonic-free signal.

Attenuators

The RFPA gain is fixed at 70dB for 40.45 MHz or 71.8 dB for 63.6 MHz. The RF transmit pulse amplitude is determined by the digital amplitudes and the output attenuators of the modulator. The DAC has 16 bits of resolution providing a large dynamic range eliminating the need for variable attenuators as in previous designs. Instead, two series attenuators with values of 6 dB and 12 dB form four power levels in which the 16 bit DAC resolution can be scaled to achieve any output amplitude. The power level is coil dependant. Level 3 is used for the Body Coil and range 0 or 1 for the CP Extremity coil depending on required transmit voltage. The range is selected by the Gain 0:1 control signals from the GCTX and displayed with GAIN 0:1 LEDs on the front panel.

The exact attenuator values are factory measured at all capable frequencies and stored in an on-board calibration PROM which will be read out by the MMC for correcting the RF transmit pulse amplitudes.

Outputs

The modulator has seven outputs placed in two groups - imaging/tuning and test - for increased isolation. Output selection is made via **OutS 0:2**. The dynamic control signal **RFON** is required for output and gives the SAR monitor a means to block the RF in the event of a SAR error. Unused outputs are terminated internally with 50 ohm.

LEDs

Power Level	LED: Gain 1..............0		Attenuator	Coil (Max Output)
3	off	off	0dB	Body Coil (3dBm)
2	off	**on**	6dB	-3 dBm
1	**on**	off	12dB	-9 dBm
0	**on**	**on**	18dB	-15 dBm

NOTE These LEDs are only active during the transmit phase of a sequence!!! They will be off otherwise.

Specifications

	Harmony	Symphony
Center Frequency	40.45 MHz	63.6 MHz
0.1dB Bandwidth	± 125 kHz	± 125 kHz
Gain Asymmetry	≤ 0.1 dB	≤ 0.1 dB

Testing

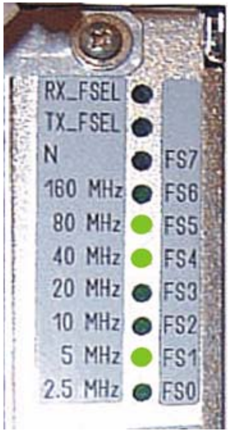
The data path between the GCTX and TX_Module can be tested via software. With the **Test_En** and **Test_Rd** signals the GCTX can write and then read back a data pattern into the Rd/Wr Buffer over the **TX_IF** bus (see Figure 55) thus verifying the TX_Module input buffer and the complete data path to the GCTX.

The Output Select and Gain control buffers can also be read back by the GCTX for testing purposes.

Synthesizer for Spectroscopy

The Spectroscopy option includes a new TX_Module with a Synthesizer part which is capable of a wider frequency spectrum for the various nuclei frequencies at 1.5 Tesla.

Figure 56 *TX_Module for Spectroscopy Option*



The LEDs of the Synthesizer half of the TX_Module are different to the normal TX_Module. The frequency is selected via a frequency cascade.

NOTE The TX_Modules for the Symphony/Sonata for Multi Nuclei option and for the Trio look alike but are not the same!

Function

The function of the spectroscopy Synthesizer is the same. the exceptions are described below.

Control

There are **two** Variable oscillators. The synthesizer's 2 Variable Frequency Oscillators output frequencies and phases are selected via commands from the MMC over the CAN unit in the modulator. A set of 7 select signals set the desired output frequency via a frequency cascade.

Variable Local Oscillator

The Variable Local Oscillator is generated by two identical PLLs, that can be selected by a fast switch. It covers two frequency ranges from 160MHz-275MHz and 295MHz-410MHz.

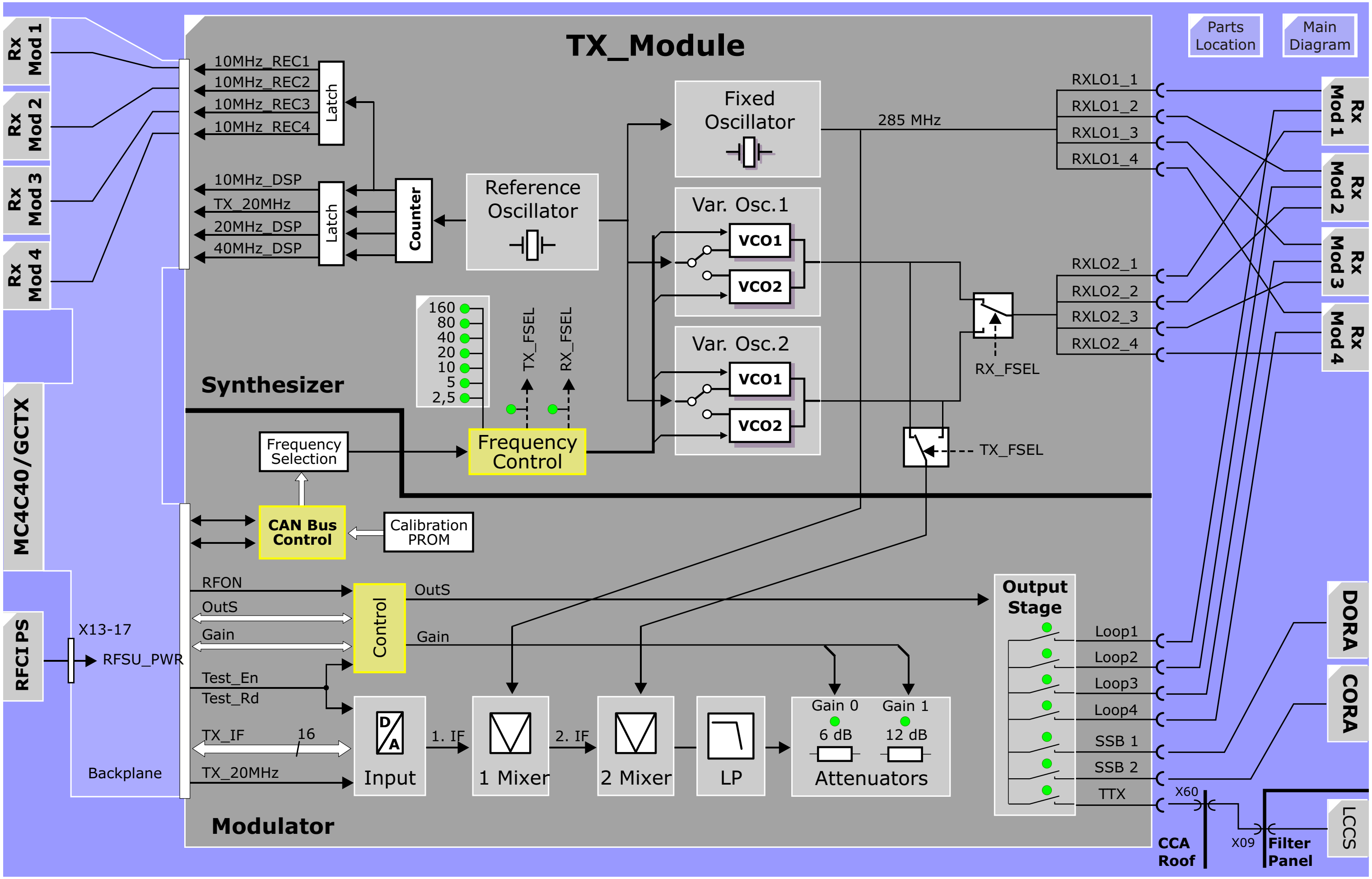
LEDs

The software selected middle frequency is indicated by LEDs. The LEDs indicate the actual status of the PLL. For example the frequency of 64 MHz will be indicated by 3 front panel LEDs 40MHz, 20MHz and 5MHz. In addition the status of the control line for fast switching of TX_FSEL and RX_FSEL is indicated.

1.5T systems (Symphony and Sonata) support the following additional nuclei/frequencies:

Element	Isotope	Gamma [MHz/T]	Freq at 1.494T	Bandwidth (in kHz)	total Bandwidth (kHz)
Hydrogen	^{1}H	42,575	63,6	250	700
Fluorine	^{19}F	40,053	59,833	50	288
Helium	^{3}He	32,433	48,450	200	552
Phosphorus	^{31}P	17,235	25,746	105	291
Lithium	^{7}Li	16,546	24,717	105	288
Xenon	^{129}Xe	11,776	17,591	35	125
Sodium	^{32}Na	11,261	16,822	70	193
Carbon	^{13}C	10,705	15,992	70	190
Oxygen	^{17}O	5,772	8,622	35	97

Figure 57 *TX_Module for Spectroscopy*



RX_Module

Overview

The RX_Module consists of two Receiver channels. Each cassette provides one high resolution RF receive channel. The RX_Module is in essence, as is the TX_Module, an analog to digital converter. The receiver reduces the MR signal frequency down to about 1.8MHz via two mixer stages, and digitizes the resultant at a fixed sampling rate of 10MHz. The increased sampling rate, together with the digital decimation filter of the RX_4, the effective ADC resolution is increased to between 21-24 bits depending on the pixel bandwidth. This translates into a dynamic range of about 130dB, an increase of 40dB over older systems. It is this effective increase in dynamic range that eliminates the need for a receiver adjustment.

The receivers also provide inputs for various other sources e.g., RFPA output forward and reflected waves, service loop inputs for testing purposes and an input for the TALES used for power monitoring.

The basic system configuration will contain two RX_Modules providing four receivers supporting up to four local coil elements. Additional two RX_Modules expand the receive capability by allowing four additional coil elements to actively receive without combining the MR signals.

Inputs

Inputs	Used for
MRSIG	MR echo from receive coils
PF, PR	loop test of RFPA and verification of TALES values.
Loop	loop testing of RFSU components
DC	Gradient service loop input
CV	forward and reflected transmit values (from TALES)

Differential Input (DC)

The differential input (DC+, DC-) with a high common mode rejection for direct measurement of DC signals from the Gradient System is filtered and fed through a de-coupling amplifier to a dedicated 12 bit ADC. The digital output is selected by test software via the output MUX. This input is used for gradient power amplifier testing and calibration.

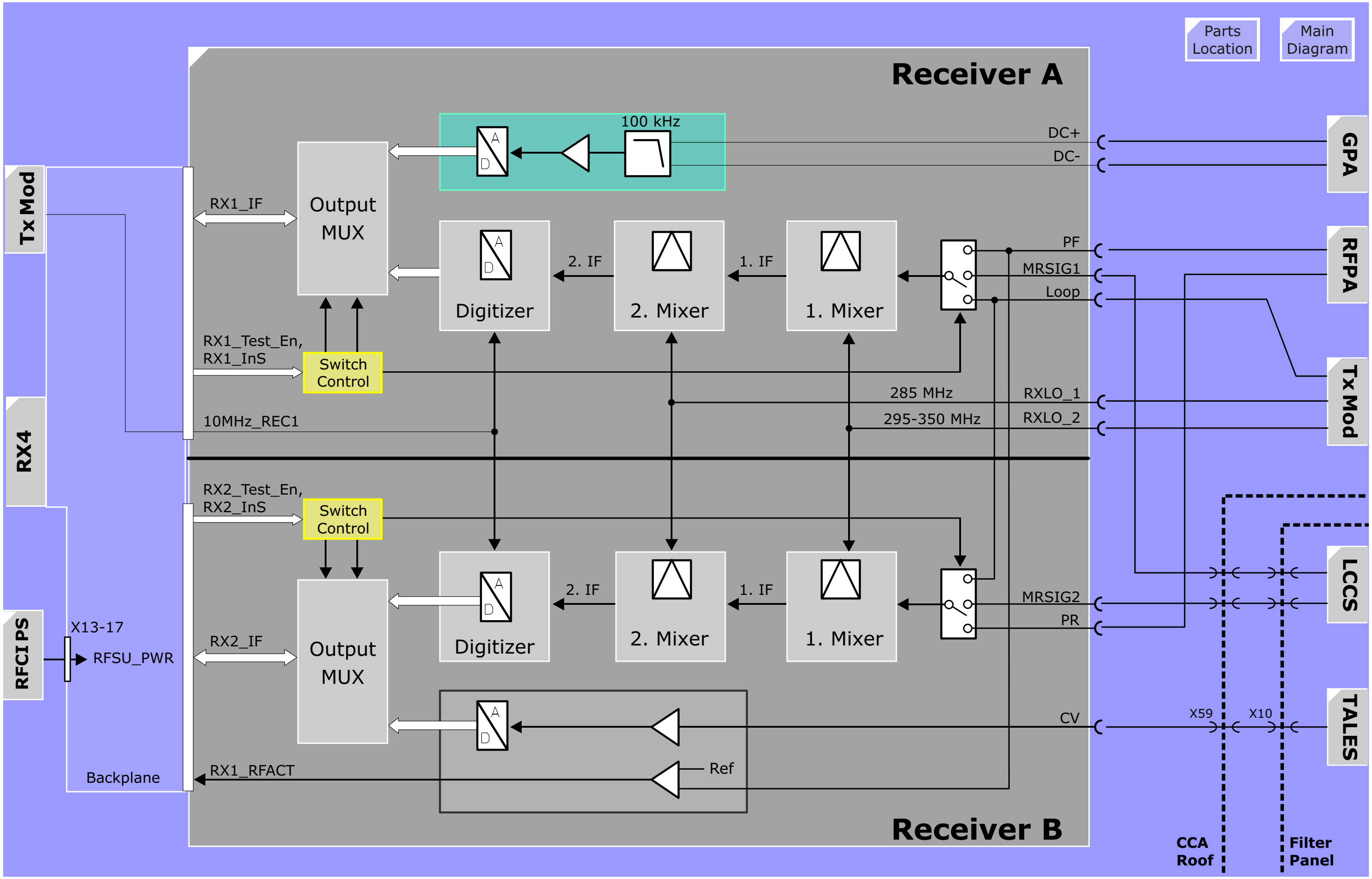
CV

The **CV** input originates from the TALES. The TALES is a precision RF voltmeter which measures the amount of RF power into the Body Coil or Local-TX coil (e.g. CP extremity coil). The TALES converts the power levels into DC voltages which are digitized by the RX_Module. The digital values are used for the SAR monitoring. As a means of verifying the validity of the CV signal, a 500mV DC reference is used.

RX1_RFACT

Over and above the normal SAR monitoring of the RF power by the MMC, a “RF present” detector has been incorporated into the receiver A of the RX_Module. When RF is present, the RX1_RFACT signal will be active. It is used to verify the presence or absence of RF at the output of the RFPA. A simple check is made: if the RFON signal is active (transmitting) and the RX1_RFACT signal also can be (should be) active. If the RFON signal is NOT active (we should not be transmitting), and the RX1_RFACT IS active (but we are transmitting).... oops. The software will take the appropriate action of disabling the RFPA.

Figure 58 *RX_Module Overview*



Function

Mixers Stages

The mixer stages of the receiver reduce the MR signal down to a frequency of around 1.8MHz in preparation for digitized by the ADC.

Digitizer

The 1.8MHz IF signal is digitized at a constant rate of 10 MHz, a rate much higher than the nyquist frequency. These additional sampled values can be processed in the digital domain (decimation filtering) in a way that results in additional signal dynamic. The actual increase in dynamic, however, is also dependant on the frequency bandwidth of the MR signal. The added dynamic is usually 2 to 6 bits.

Outputs

The digitized MR signals are sent to the RX4 board in the MMC over the backplane via opto-couplers which maintain the electrical isolation between the MMC and RFSU.

LEDs

Indicate the selected RF input.

Specifications

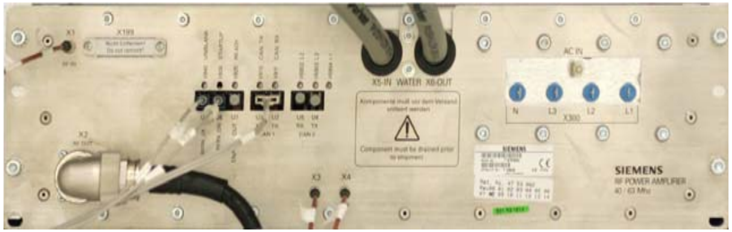
	1 Tesla	1.5 Tesla
Center Frequency	40.45 MHz	63.6 MHz
0.1dB Bandwidth	± 250 kHz	± 250 kHz
Max Input Level	± 1.2 dBm	± 1.2 dBm
Stability (5 min.)	≤ 0.01 dB	≤ 0.01 dB

Testing

Built-in test loops covering all of the RF System hardware components have been incorporated into the RF System to facilitate testing and trouble shooting. Test sequences specially designed to test functionality and performance - linearity and stability - can be configured and started from the service software platform. Results are displayed graphically and alpha-numerically with tolerances and performance data being given.

RF Power Amplifier DORA

Figure 59 *The DORA RFPA*



The DORA (DOuble Resonant RF Amplifier) is a solid-state power amplifier designed to operate at the frequencies of 40,45 and 63,6 MHz, or frequencies close to these bands (60MHz for 19F and 48 MHz for 3He).

Inputs

Signal	Description
AC_IN	The three-phase AC primary voltage of 93V (measured phase/neutral) for the RFPA is tapped off the main system transformer. The internal power supply generates all the required voltages from this.
RF_IN	Input of RF signal to be amplified. The nominal input level for the full output power level of 15kW is 0dBm.
RFPA_UNB	To reduce noise and loss of the MR echo during reception the amplifier is blanked during the receive cycle. When signal is active (light on), the amplifier is activated.
RFPA_ON	An enable signal allowing the SAR monitor to shut the amplifier down independently from the CAN bus controller. If the signal is missing (light out), the amplifier shuts down.
Enable_Out	A copy of the RFPA_On signal as output to an additional RF amplifier.
CAN1, 2	Fiber optic CAN bus communication interface.

NOTE CAN1 and CAN2 are labels on the RFPA, but don't confuse it with the CAN1 and CAN2 busses!

NOTE There are two RFPA types. All RFPA's with ODD NUMBERED REVISION LEVELS have CAN bus connectors reversed and REQUIRE ADAPTORS. Adapters are delivered with these RFPAs.

Function

Power Up

To bring the amplifier up to operating condition, the Power Supply requires two inputs: the internal **enable** signal from the CAN SLIO which will be set via a command sent from the MPCU over the CAN bus during the system start up, and the **RFPA_ON** signal from the IOP. When both signals are present, the **Start Up** LED will light and stay on until the power supply voltages have reached their proper levels. This takes only a couple of seconds. After a successful power up, the **Start Up** LED goes OUT and the **Ready** LED lights and to signify the amplifier is operational.

Bias Control

The bias control circuitry is responsible for blanking the amplifier (put the amplifiers into cut-off, i.e. disables it) when not in use and for regulating the bias of the driver amplifier and power stage amplifiers during the transmit cycle for a stable amplification. The **RFPA UNB** signal determines the blanked (disabled) or unblanked (active) state. An LED is supplied to display the state of this signal.

Amplifiers

The amplifier consists of a driver and a power stage amplifier. In total, the two amplifier stages are capable of producing a maximum power of 15kW at 63,6 MHZ or 10kW at 40,45 MHz.

Monitoring

The RFPA has several internal monitoring circuits monitoring several vital functions:

- over voltage or over current of the power supply
- excessive forward (5%) or reflected (approx 30%) output power levels (**RF Power**)
- unblank duty-cycle (6-8%)
- power stage temperature (50°C)

In the event of an internal failure or an operational limit of the amplifier has been exceeded the amplifier provides a status report via the CAN bus and the **READY** LED goes out.

Outputs

Signal	Description
Enable Out	Can be used to cascade the RFPA_ON enabling signal to an additional RFPA. Is not used on current system configurations.
PF, PR	Outputs of the internal Directional coupler (DICO) representing the forward and reflected output power levels of the RFPA. These signals are read in over the RX_Module and used to: • monitor the reflection factor • verify the power level measurements made by the TALES • provide test loop feedback to test RFPA
RF_OUT	This is hopefully where the amplified RF comes out.

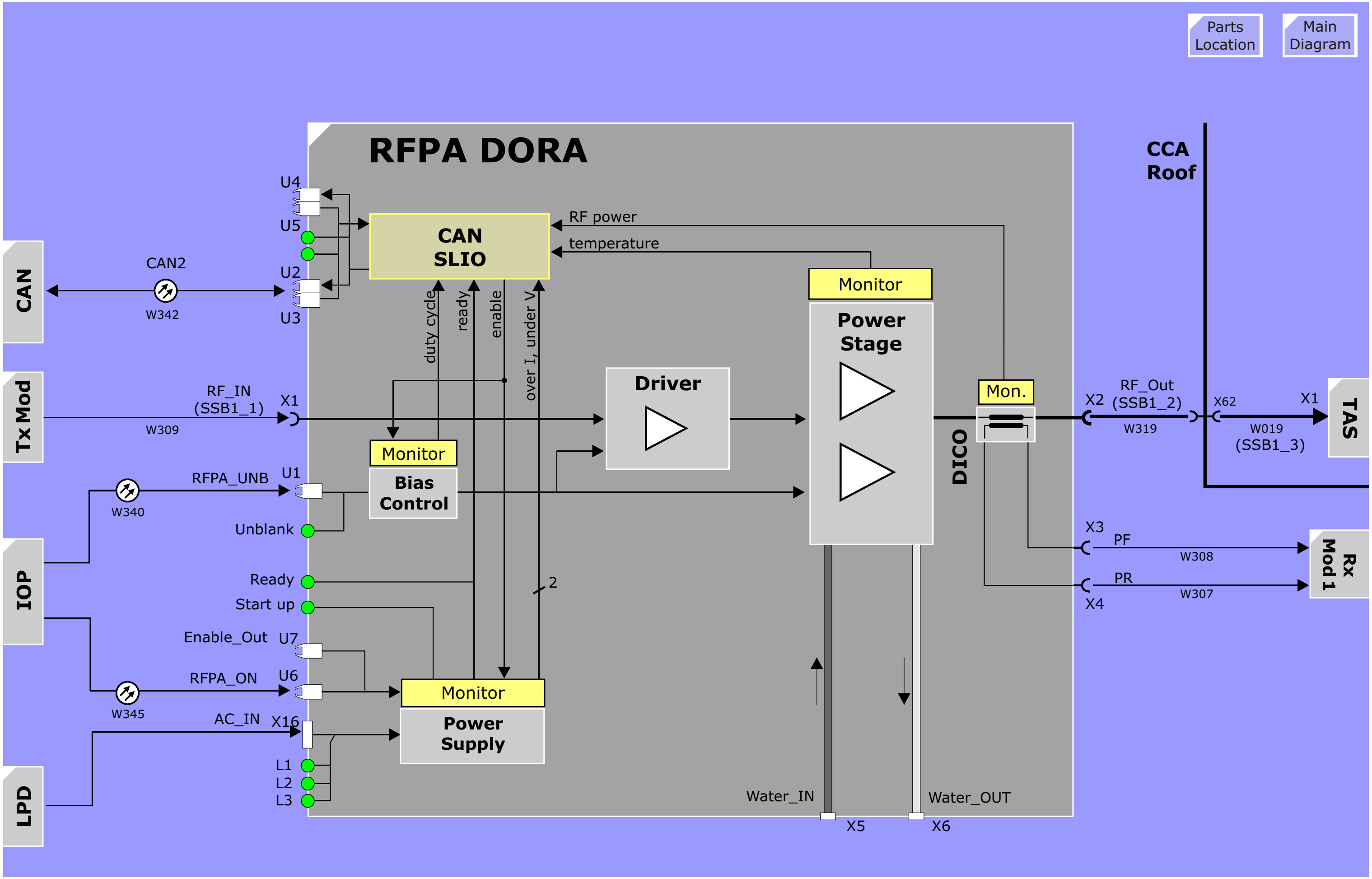
LEDs

LED	Description
UNBLK	Lights when being unblanked (i.e., when transmitting)
Start Up	Indicates the RFPA is being enabled via a command over the CAN. This LED is only temporary and will go out when the Ready condition is reached.
READY	Amplifier power supplies OK.
CAN TX, RX	Indicates activity on CAN bus.
L1-L3	AC Power supplied to unit.

Testing and Tune Up

Testing of the RFPA is performed through the service platform. A correction of the non-linearities in amplitude and phase will be compensated via calibration software in the Tune Up section.

Figure 60 *RF Power Amplifier Overview*



RF Power Amplifier (CORA)

Inputs

Signal	Description
AC_IN	The one-phase AC primary voltage of 230V. An internal power supply generates all required voltages.
RF_IN	Input of RF signal to be amplified. The nominal input level for the full output power level is 0dBm.
RFPA_UNB	To reduce noise and loss of the MR echo during reception the amplifier is blanked during the receive cycle. This is accomplished with the fiber optic signal RFPA_UNB. When signal is active (light on), the amplifier is activated.
PA_ON	An enable signal (light on) required to bring the amplifier to operating condition. The purpose of this signal is to be able to disable the amplifier, a requirement for the SAR monitoring.

Function

The function of the CORA (Cost Optimized RF Amplifier) is principally the same as the DORA. Differences are mentioned below.

Power Up

To bring the amplifier up to operation the **PA_ON** signal must be applied. When the amplifier is ready, it will set the **Ready** signal (light on). This signal is routed to the RFPA_On input of the DORA amplifier. This will tell the MPCU via CAN bus CORA is okay and MPCU will enable the DORA amplifier via CAN bus; the Start up - LED in DORA lights up. Finally the Start up - LED switches off and the Ready LED in the DORA lights up. Both RFPAs are ready now.

Monitoring

The CORA has integrated monitoring circuitry for various internal functions. If any of these monitored functions fails, the **Ready** condition is extinguished (LED goes out) and fiber optic signal to DORA RFPA is removed (light out).

Two LEDs represent temperature related errors **TS-ERR** and **HOT-Sink**.

In case of an error the Ready fiber optic signal to the RFPA_ON input of the DORA RFPA will be set low (light out) causing the DORA RFPA to shut down. This will be reported by the DORA’s CAN unit back to the MPCU. Currently, the MPCU will react by setting the PA_ON signal to CORA low and hence disable CORA and DORA.

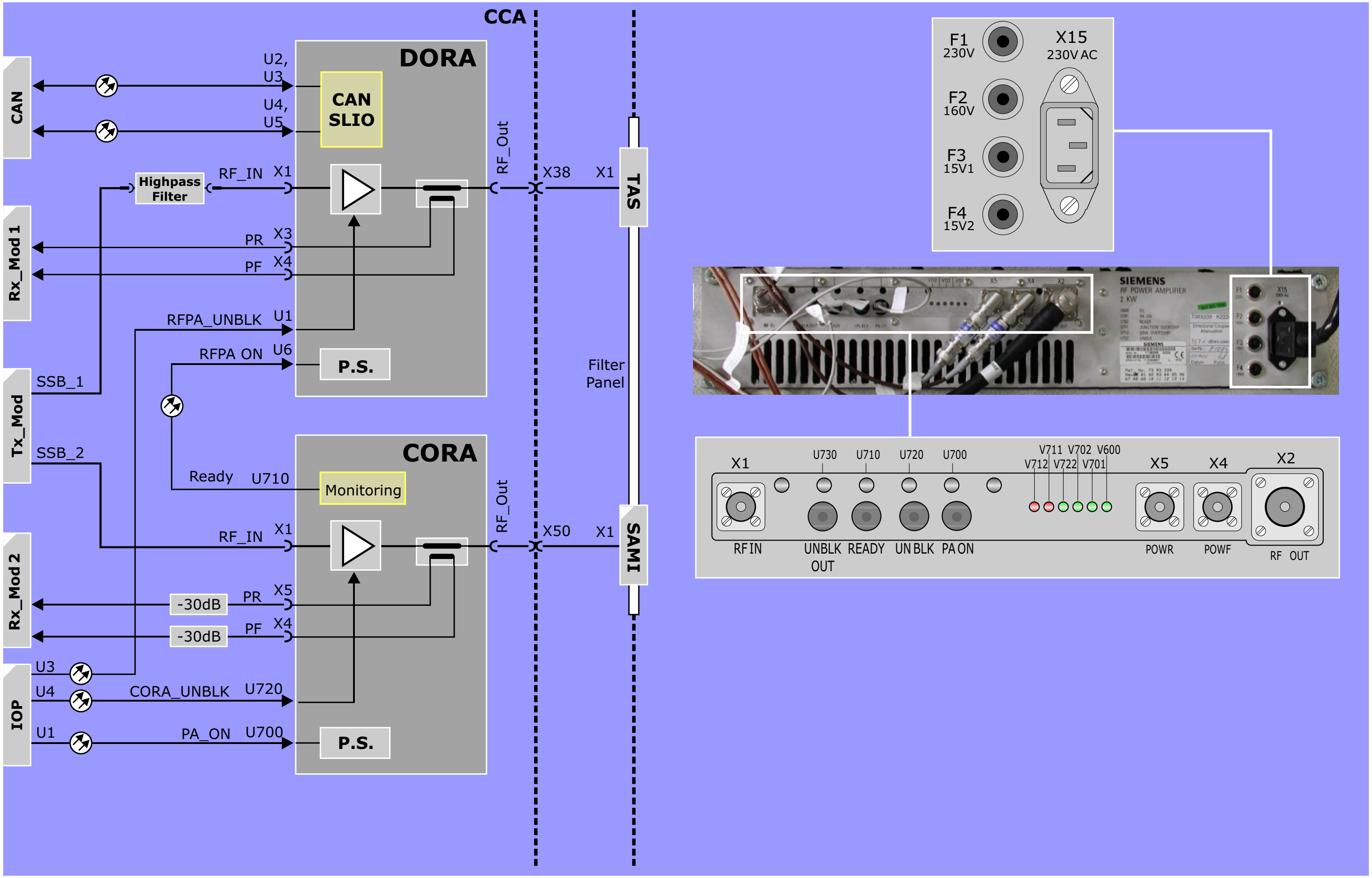
Outputs

Signal	Description
PF, PR	Directional coupler (DICO) outputs of the forward and reflected power levels of the RFPA output
RF_OUT	This is hopefully where the amplified RF comes out

LEDs

LED	Description
DC	indicats internal power supplies OK
PA_ON	indicates amplifier is being enabled (no RFSWD errors)
READY	Amplifier start-up completed and unit now ready
UNBLK	indicates unit being unblanked (i.e., when transmitting)
TS-ERR	Transistor junction temperature too high (**non-latched**)
HOT-TSINK	Heat sink temperature too high (**non-latched**)

Figure 61 *RFPA Configuration for Spectroscopy Option*



RFAS - RF Application System

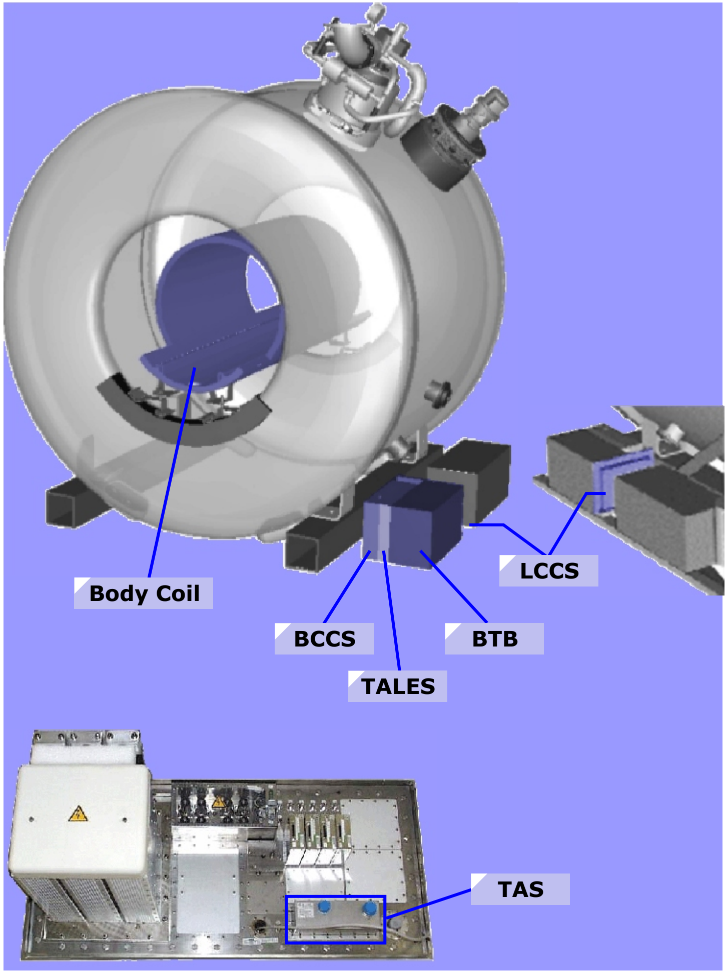
Overview

The RFAS is a group of components whose functions are to interface the RF front-end electronics to the individual transmit and receive coils. In such a system where a wide variety of antenna types are employed, the RFAS must provide several functional components to support the various coil requirements. In all, five components (or six if the spectroscopy option is installed) are involved. These are located at the **rear** of the magnet on the **right** side (the TAS, not shown, is located in the filter panel). If you can't find them there, then try the *other* rear right side of the magnet.

These are the components that make up the RFAS:

- Transmit Antenna Selector (TAS)
- SAMI
- Body Coil Channel Selector (BCCS) Harmony
- Body Coil Channel Selector (BCCS) Symphony
- Transmit Antenna LEvel Sensor (TALES)
- Body Tune Box (BTB) Harmony
- Body Tune Box (BTB) Symphony
- Local Coil Channel Selector (LCCS)

Figure 62 *RFAS Parts Location*



TAS - Transmit Antenna Selector

The TAS primary function is to route the RFPA output to the body coil or TX-local coil (e.g. CP Extremity). Secondly, it has a built-in 50 Ohm dummy load to terminate the 90° hybrid of the BCCS_63 (Symphony) and the RFPA when running test loops.

Function

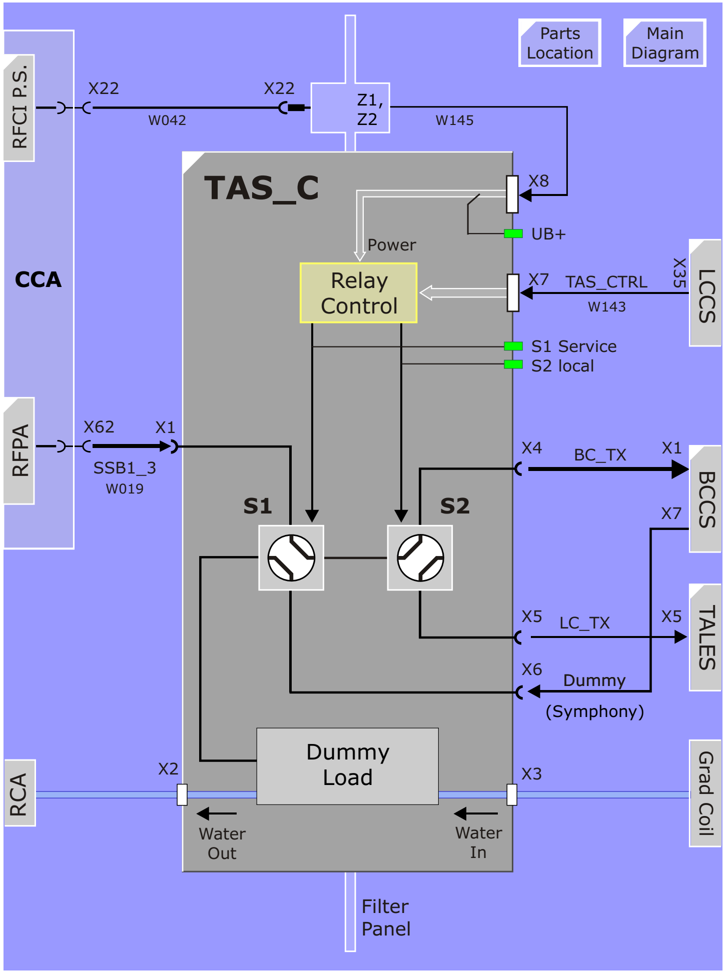
Relay Control

The switches are actuated with DC relays operating at 15 volts which is supplied by the RFCI power supply.The control signals for S1 and S2 are applied by the CAN controller of the LCCS. LEDs on both the LCCS and TAS provide the actual select status.

LEDs

LED	State	Description
UB	ON	power present (15V OK)
S1	ON	Service (RFPA out to Dummy)
	OFF	Imaging (RFPA out to S2)
S2	ON	TX to Local coil
	OFF	TX to Body coil

Figure 63 *Transmit Antenna Selector*



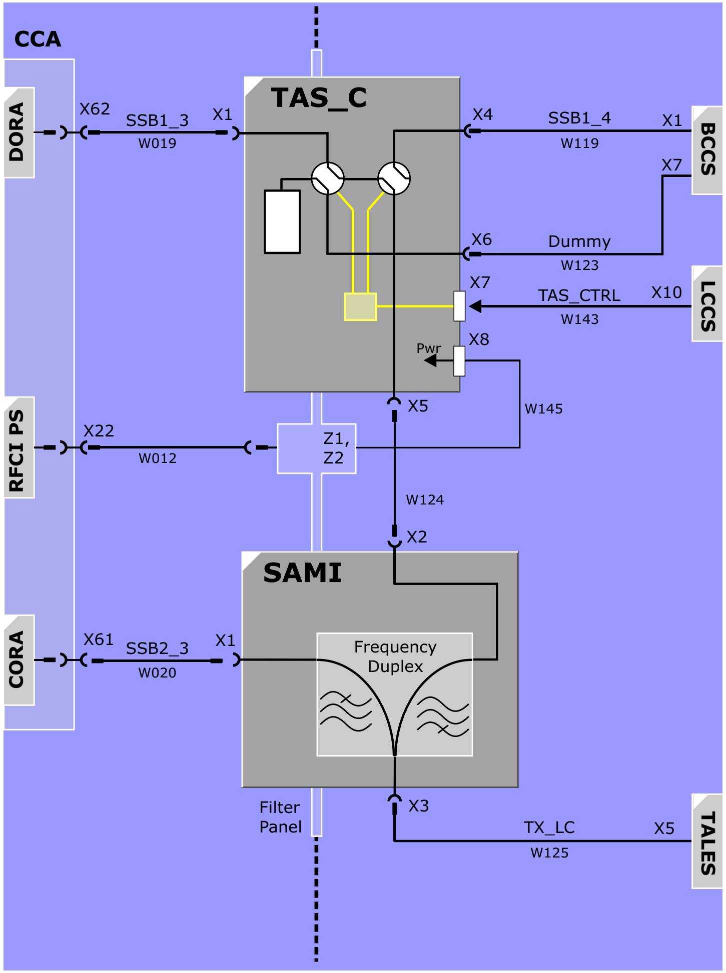
SAMI for 63MHz

For broad-band spectroscopy applications the Local Coil transmit path must be able to operate, in addition to the 1H frequency of 63,6 MHz, also over the various spectroscopy frequencies. At the higher spectral frequencies for the 3He and 19F nuclei, the standard DORA RFPA can be used. For the lower spectral frequencies of the 13C, 23Na, 129Xe, 7Li and 31P nuclei an CORA RFPA will be used. The Spectroscopy AMplifier Interface allows the use of both amplifiers over the same Local Coil transmit path without interfering with each other.

Function

The Frequency Switch prevents feeding the high frequency (63 MHz) output of the DORA RFPA from into the broad band RFPA and to prevent feeding the x-nucleus low-frequency RF pulse into the DORA RFPA.

Figure 64 *SAMI for 63MHz Systems*

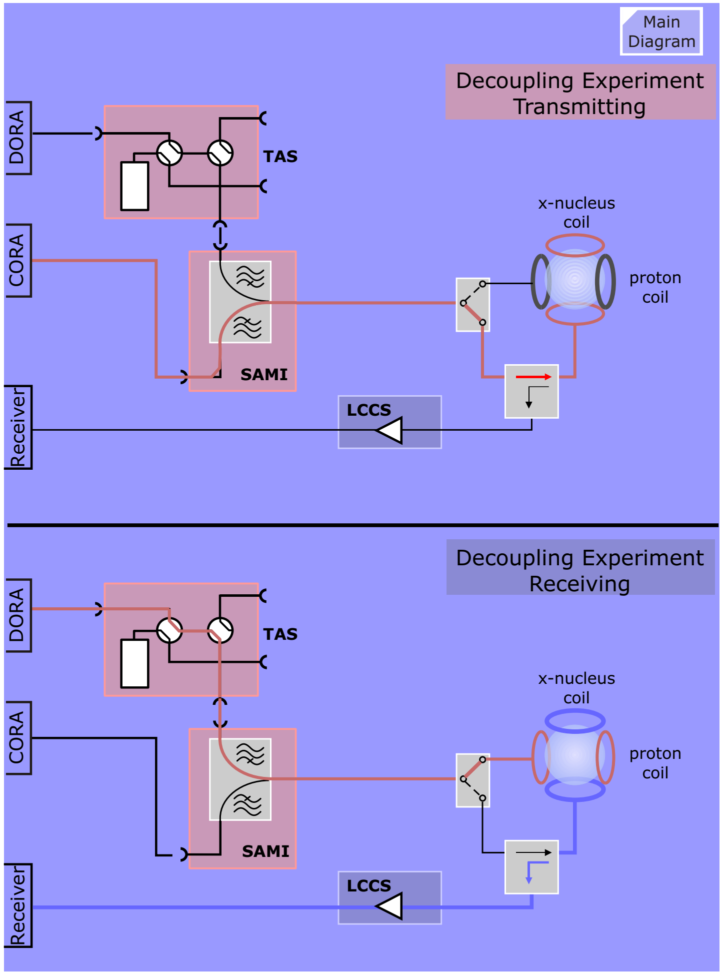


Application of SAMI

Decoupling experiment: during receiving the x-nucleus spectrum proton RF is transmitted: spectra become less complicated, fewer lines with better signal-to-noise result.

The **Decoupling** works as follows: A local Tx/Rx dual resonance coil is used. The broad band RFPA generates x-nucleus magnetization and during receiving the x-nucleus signal the DORA RFPA applies a proton frequency RF pulse to the local coil in order to decouple the spectra and hence getting a better signal to noise ratio.

Figure 65 *Decoupling Experiment*



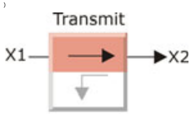
BCCS_42

The Body Coil Channel Selector for the Harmony provides the components to allow using the Body Coil as a transmit and a receive coil. The T/R switch connects the Body Coil to the transmit or receive path, a directional coupler provides a tuning and testing port and a pre-amplifier for the receive signal from the Body Coil.

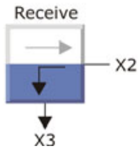
Function

Transmit / Receive Switch

The T/R switch is required when imaging with the Body Coil. It is actively switched to provide a linear feed-through characteristic.



When transmitting, the SE_CTRL is active and X1 is connected to X2.The path X1 to X3 is electrically isolated (very high Z). If the SE_CTRL signal is missing, the switch will still work, but non-linear however.



When receiving, the SE_CTRL is inactive (0V) and X2 is connected to X3. Path X1-X2 is electrically isolated.

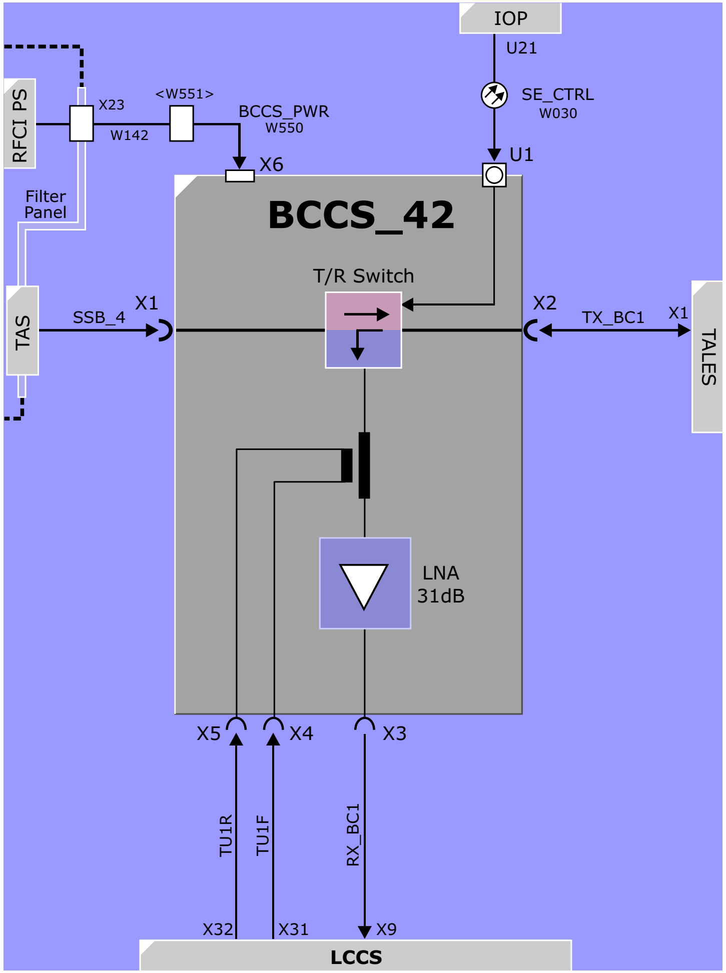
Pre-amplifier (LNA)

The narrow-band, low noise amplifier has a fixed gain of 31dB.

Directional Coupler

The directional coupler provides a port for tuning. A tune signal is applied alternatively to both the forward and reflected side of the directional coupler. The ratio of the amount of tune signal being coupled in the forward and reflected directions between the body coil and the amplifier is influenced by the impedance of the body coil. These signals are measured and compared to a matrix acquired during the tune_up and |r| values then determined.

Figure 66 *BCCS for the Harmony*



BCCS_63 - Body Coil Channel Selector

The BCCS_63 version is in fact a double BCCS_42 with a hybrid at the input and output. The hybrids are required to set up a 90° phase shift for the circular polarization of the body coil.

Function

The T/R switch, preamplifiers and directional couplers are described in the BCCS_42.

TX_Hybrid

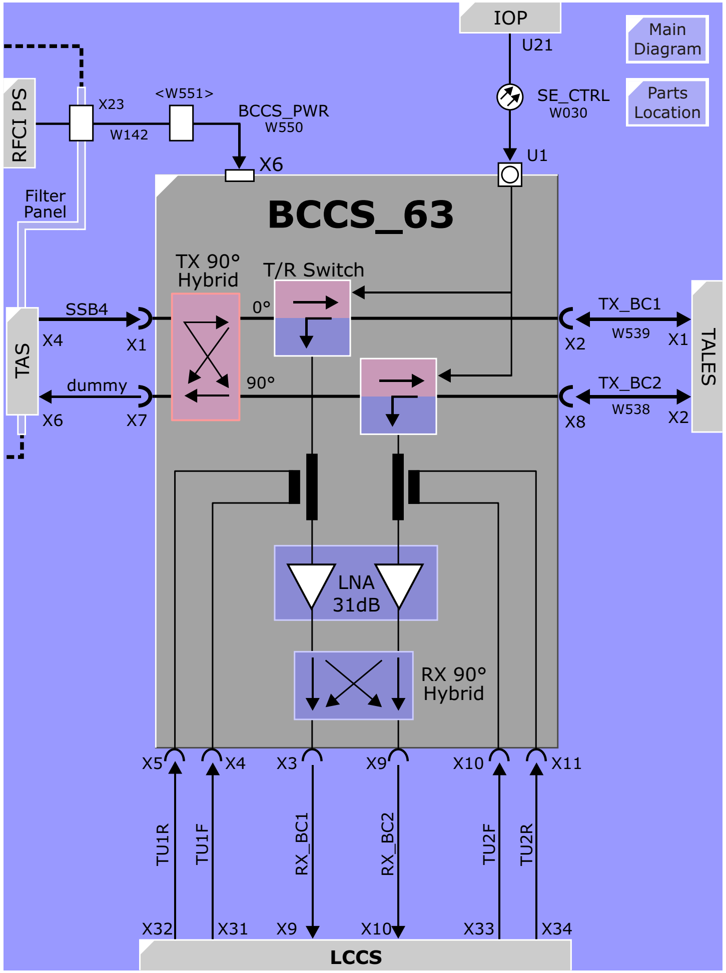
The Symphony/Sonata systems employ a circular polarized body coil which require two 90° phase-shifted signals. The TX_Hybrid is a passive-reactive device which provides the 90° phase shifting and splitting of the transmit signal required by the two CP Body Coil systems. Assuming both 0° and 90° Body Coil systems are equally loaded and have the same impedance the power draw will also be equal.

RX_Hybrid

A part of the MR signal is received (picked up) by each of the two Body Coil 0° and 90° resonator systems and needs to be combined to one coherent MR signal. This is achieved with the RX Hybrid. For imaging the output is taken from X3. The output at X9 will be terminated by the input of the LCCS Switch Matrix.

NOTE The output at X9 is used ONLY by the BC Tuning procedure in the Tune Up and BTB/BC service tests in the SeSo TestTools.

Figure 67 *BCCS Symphony*



TALES - Transmit Antenna Level Sensor

The TALES is an RMS RF voltmeter for measuring forward and reflected RF levels into the transmit coils. The measured values are used as the basis of the SAR calculations.

Function

Demodulation, Power Scaling

The RF signals picked up by the directional couplers are demodulated and filtered to produce a DC voltage proportional to the RF levels. The power scaling circuitry linearizes the measured values to correct the demodulator's non-linearities. See note below.

MUX

During sequence activity, the MMC reads out all six values in regular intervals, whether transmitting or not. The measured values are read out over a MUX which is addressed by the **MH_SEL** select signals. The resulting **CV** signal is output to the RX_Module where it is digitized and sent to the SAR DSP on the MC4C40 board. The SAR DSP will be informed by the RF_ACT signal (in Receiver) which CV values were mesured when the RF was active.

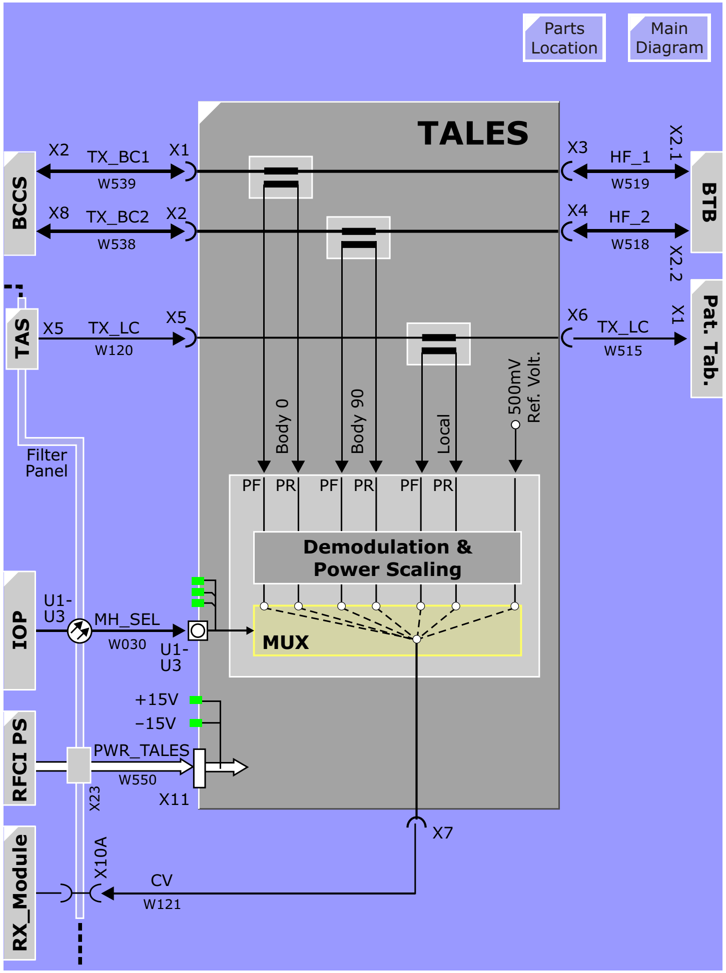
LEDs

The LEDs used for indicating the MH_Sel signals and the their respective port selections (e.g., Body 1 Fore, Body 1 Refl) will seem to light constantly. They are blinking, but faster than your eyes can percieve.

Outputs

In addition to the RF signal being measured, the TALES outputs a control dc **reference voltage** of 500mV. The reference is used to verify the accuracy of the signal coming from the TALES.

Figure 68 *TALES*



Accuracy of TALES

1. Directional coupler Body Coil

The voltage error is 2% at 700V, i.e. +-14V. This deviation is an absolute number. For example when measuring 40V, it is possible to get values from 26V to 54V.

2. Directional coupler Local Coil Transmit Path.

Here are two measurement ranges: **Low Range** from 0 to 100V, **Medium Range** from 100V to maximum voltage.

Voltage error for **Low Range**: 3% at 100V, i.e. +-3V.

Voltage error for **Medium Range**: 2% at 100V, i.e. +-2V.

The accuracies given above hold for proton frequency. For lower frequencies, for example 25MHz (31P) the voltage error is about 1% higher.

NOTE **Note:** The accuracy of the DICO (directional coupler in the RFPA) is better. But the DICO is calibrated by means of the TALES at a voltage of about 400V. Assuming, at 400V the TALES has an error of 14V, this corresponds to an error of 3.5%. In that case all measurement values of the DICO have an error of 3.5%. At a level of 10V this is 0.35V deviation.

Service Requirements

NOTE The TALES will be replaced every two years with a newly calibrated one. The old TALES will be sent to Erlangen for re-calibration.

BTB_42 - Body Tune Box Harmony

For the 63MHz version go here.

The BTB provides impedance matching between the 50 ohm of the RF transmit path and the body coil whose impedance varies with patient size and condition.

Those of you interested in a great explanation of the characteristics of the BTB should go here.

Relay Control

The control signals for the relays are generated by the internal CAN controller. LEDs are provided to indicate the selected capacitors.

Relay Power Supply

The switches are actuated with DC relays. The relay control voltage is generated by a +/-15 volts regulator on board that is supplied by the RFCI power supply.

Static Detune

When transmitting with a transmit-capable local coil, the body coil will be statically detuned by clamping the coil to ground, thus preventing and absorption of the transmit energy into the body coil.

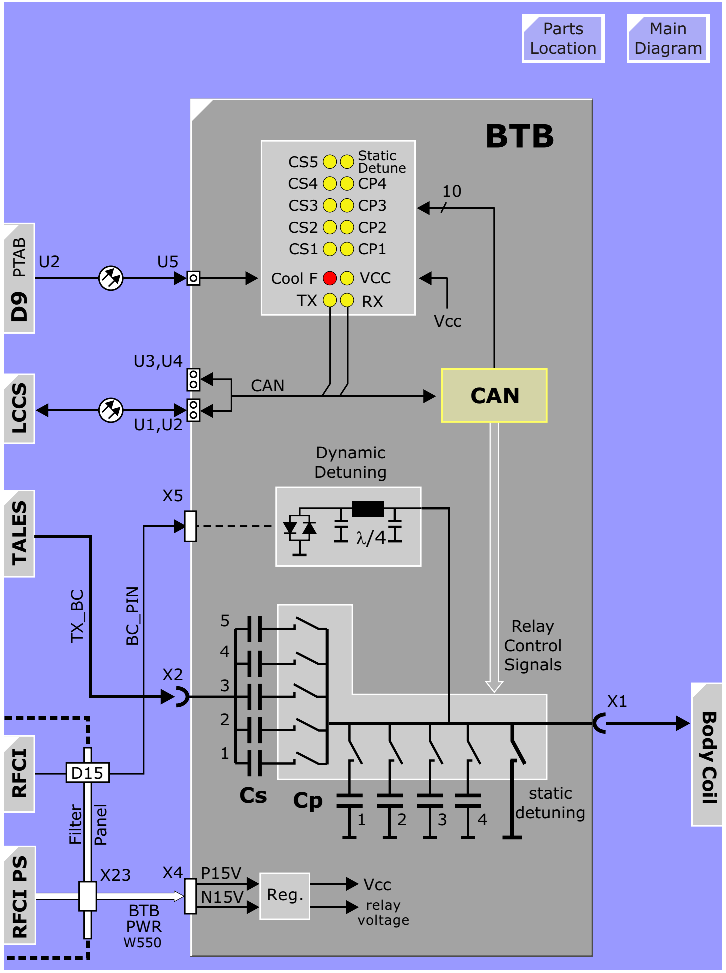
Dynamic Detuning

When used in conjunction with the receive local coils, the body coil will be tuned only when transmitting the excitation pulse. During reception it will be detuned to prevent absorbing any MR-signal. A fast switching PIN diode circuit on a λ/4 line is used for quick dynamic detuning.

LEDs

At the moment there are fourteen of them.

Figure 69 *BTB Harmony*



Body Tune Box Symphony

The BTB_63 provides impedance matching between the 50 ohm of the RF transmit path and the body coil whose impedance varies with patient size and shape.

Function

The BTB_63 is functionally identical to and performs in the same way as the BTB_42 version, but has two matching networks, called BTB1 and BTB2, one for 0° and one for 90°, to support the circular polarized body resonator.

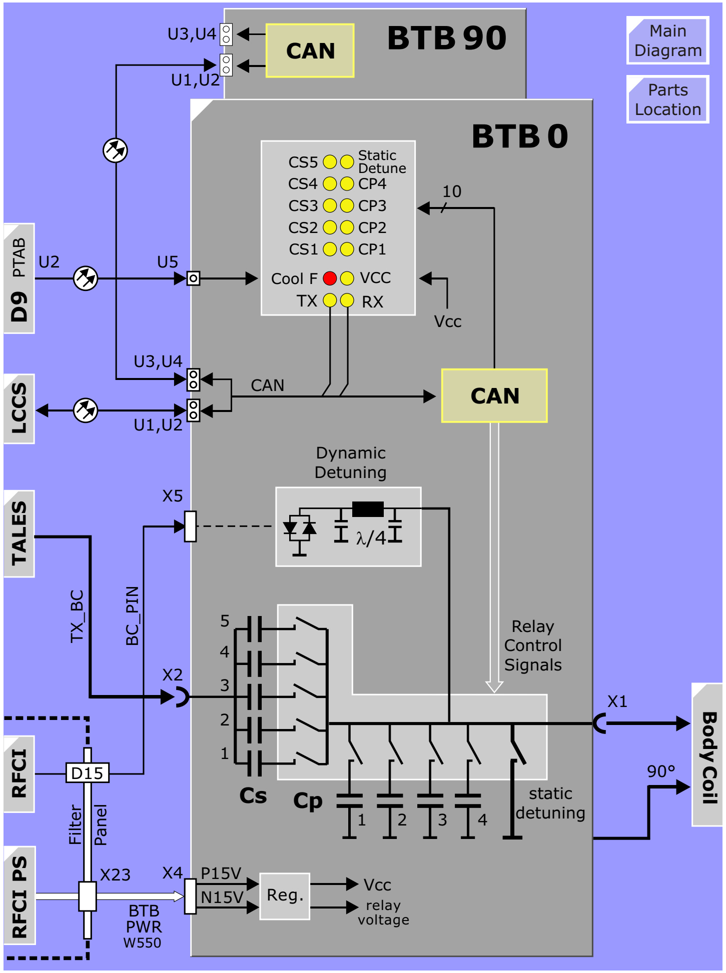
Although not visible in the diagram, BTB2 has the same BC_PIN, TX_BC and BTB_PWR connections as BTB1.

LEDs

There are twice as many as the Harmony version.

The BTB 90° on older systems may always show a Cooling Fault (Cool_F LED always on). That does not indicate a system deficiency since the monitoring is done at BTB 0° only. On newer BTBs the Cool_F at BTB 90° is jumpered out.

Figure 70 *BTB Symphony*



Body Coil

The Body coil is used for whole-body imaging and for transmission of the excitation pulses when using the specialized local coil antennas for reception. The fact that practically all clinical examinations are using the specialized local coils the role of the body coil is mainly to acquire scout scans and for use as transmit coil.

The Body Coil employs a Alderman-Grant design. With this design, it is not necessary to phase split the input signal. The coil is fed at a single point.

The body coil comes in three flavors:

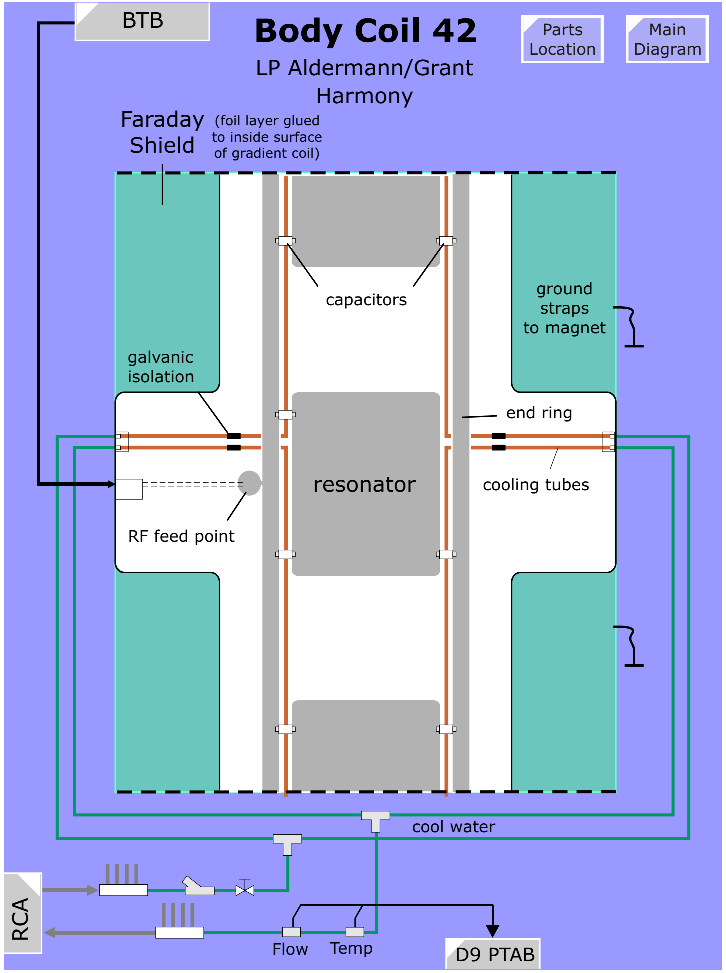
- Body Coil 42 - Harmony
- Body Coil 63 - Symphony
- Body Coil-2 - Symphony

Body Coil 42 - Harmony

The body coil for the Harmony is a linear polarized (LP) resonator assembly.

When the incoming RF signal is at the resonant frequency of the coil, the current flow through the coil resonator halves is inherently distributed between each resonator half over the end-rings. Due to high currents flowing over the capacitors, they are kept cold by the secondary cool water.

Figure 71 *Body Coil Harmony*



Body Coil Symphony

The body coil for the Symphony employes a circular polarized (CP) coil. A CP coil has been chosen for its higher efficiency which is necessary to keep the SAR limits within tolerances due to higher RF transmit pulse amplitudes required by 1.5T systems. It is primarily used for whole-body imaging and for transmission in conjunction with the specialized local coil antennas.

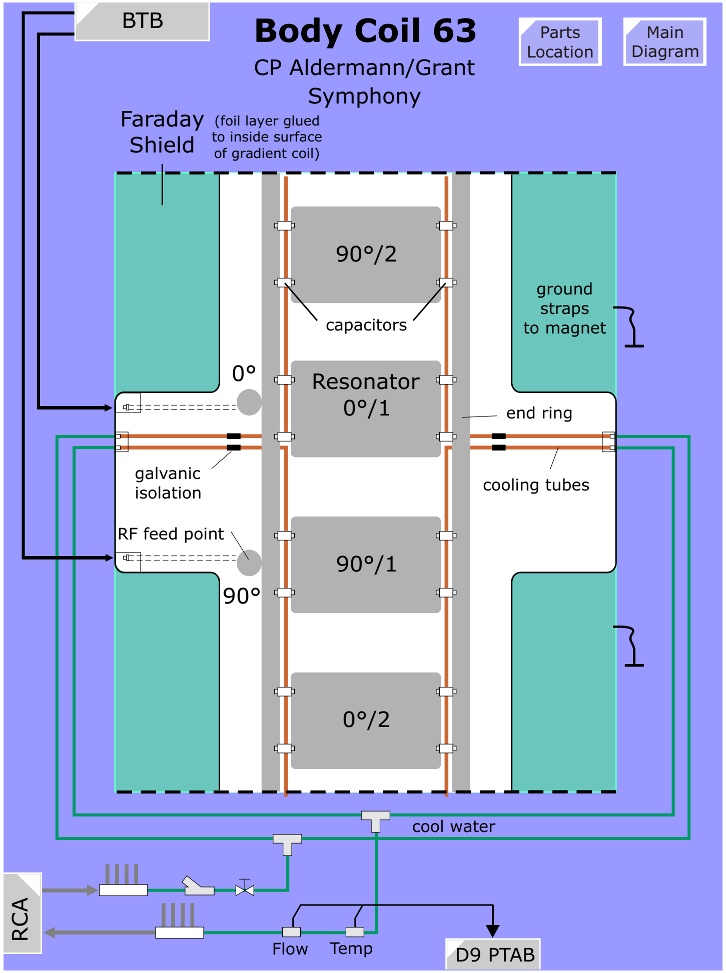
The Body Coil employs a Alderman-Grant design. The coil is fed at two points placed at 90° distances, and thus the reason for the 90° Hybrid phase shifter in the BCCS. This coil was in production up until August 2002, at which time it was replaced wiht the BC-2 coil (see below)

When the incoming RF signal is at the resonant frequency of the coil, the current flow through the coil resonator halves is inherently distributed between each resonator half over the end-rings. Very high voltages and currents exist over the capacitors and require cooling. The capacitors are placed on top of copper tubes carrying secondary cool water.

Since this Body Coil is a circular polarized coil, decoupling is an important issue. Strong coupling between the two resonator systems leads to high transmitter reference values and hence reduces the performance of the RF-transmit path. Decoupling is checked with SESO RF-Testtools, "BTB/BC Test" and can only be corrected mechanically by evenly mounting both sides.

One principal advantage of all Harmony/Symphony body coils compared with Impact/Expert and Vision is the fact, that the amplitude of the RF field outside the 500mm imaging values drops very quickly to low values. Hence ambiguity artifacts (also called "third arm artefact") are generally avoided in the Z direction.

Figure 72 *Body Coil Symphony*



Body Coil Cool-water Monitoring

The Harmony Body Coil and the first version of Symphony Body Coils (BC-47) use cooling water. The flow and temperature of this water is monitored on the D9 in the Patient Table Control Unit.

Flow

The cooling water flow is adjusted with the valve until the Flow sensor has reached an output of 70Hz. The D9 monitors this 70Hz ±3Hz. When the flow becomes too low (refer to RF-repair instructions for adjustment) LED V 83 (left) lights up.

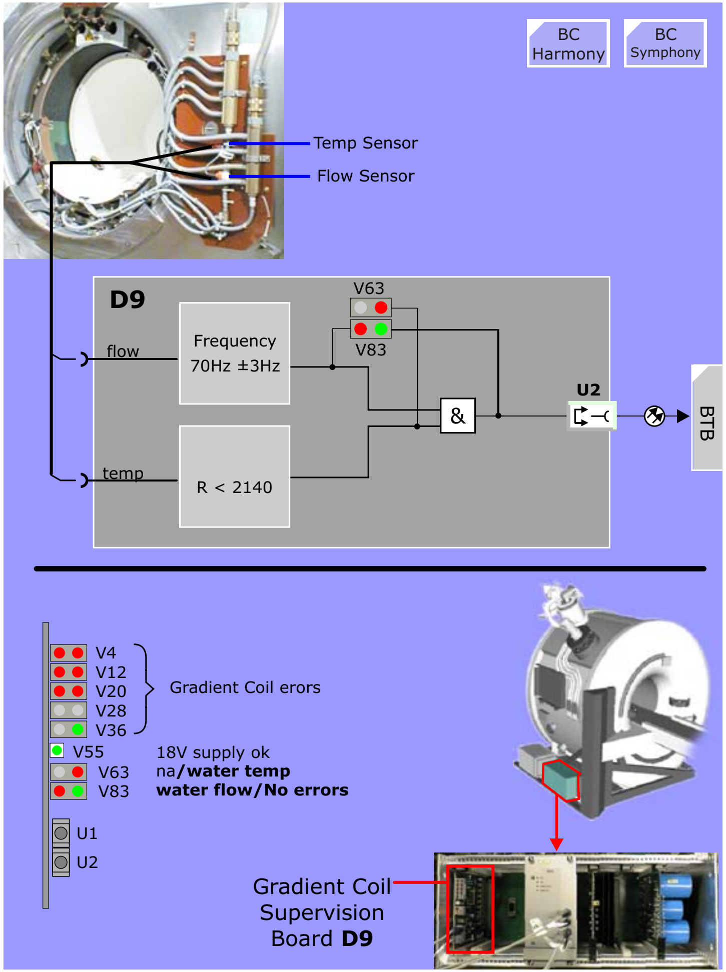
Temperature

Measurement of the cooling water temperature sensor resistance in the body resonator (return cooling water flow) can be done at connector X21/D9 between Pins 9 and 10.

Resistance values of the temperature sensor:

- R = 2000 Ohm at 20 °C
- R > 2140 Ohm -> over temperature
- R < 2000 Ohm -> OK

Figure 73 *Monitor for Body Coil chill water*



Body Coil-2 Symphony

In spring 2002 a new version for the 63 MHz Body Coil was introduced. The principal change is the capacitors being built in the end-rings. This design is called "high-pass birdcage". This coil requires a new Body Tune Box (BTB 047-2) which uses different capacitor values. The advantages compared to the first version 63 MHz body coil are:

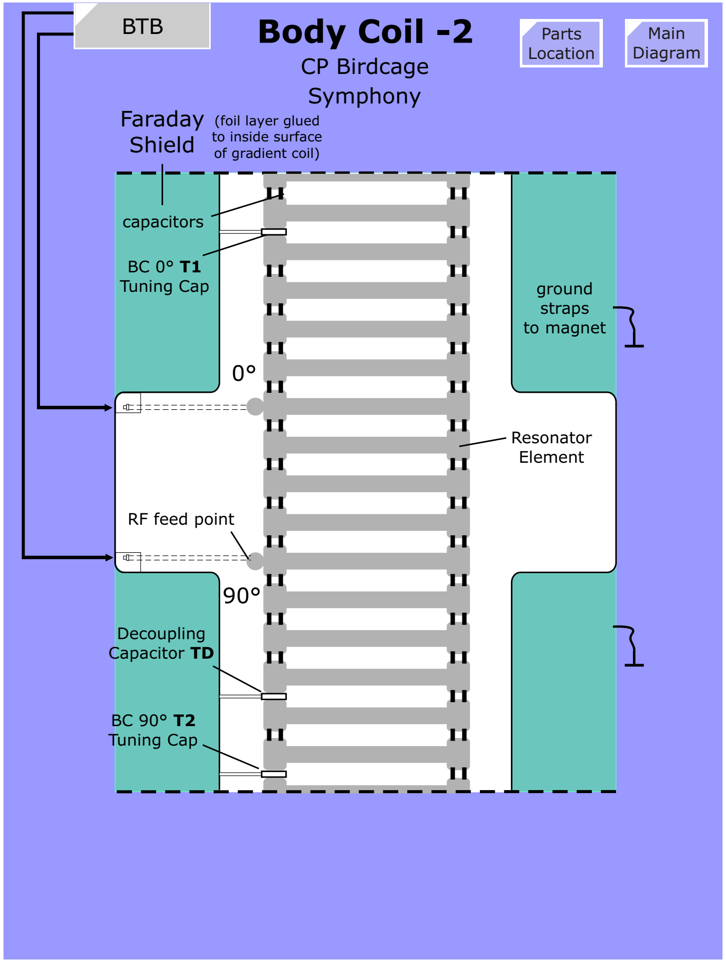
- much less RF losses in the coil now, the value being about 500 to 600W, in the first version the losses were about 1200W
- consequently the transmitter reference values (Adjust/Transmitter) will be much lower, typically values of about 300 V will be measured for loader and large spherical phantom (360V for the old Body Coil)
- better signal to noise compared with the old Body Coil
- no water cooling necessary

The following 63MHz Body Coils are delivered:

System	"Old BC"	"New BC"
Symphony	BC 047	BC 047-2
Sonata	BC Sonata	BC Sonata-2

NOTE You have to configure the 63 MHz Body Coil type in the Service Software.

Figure 74 *Body Coil-2 Symphonys*



LCCS - Local Coil Channel Selector

Overview

The LCCS is a multi-channelled pre-amplifier providing receive inputs for up to 16 local coil elements and 2 inputs for the Body Coil. All inputs can be routed to one of 8 output preamplifiers via a programmable switch matrix. This offers the maximum routing flexibility. The pattern of switch selection depends on coil and hardware configuration. See IPA for more info.

Switch Control

The switch matrix is configured by the sequence according to user's selection of connected coil elements. Although many selection possibilities are available, certain combinations should be adhered to in order to assure good IQ. See IPA for more info.

Amplifiers

Each of the output channels has its own amplifier stage switchable between 0dB (no gain) and +20dB. The gain level is set by the CAN or can be switched dynamically during the sequence via RFAS_SEL signal (for 3D sequences and HASTE sequences with thick slices).

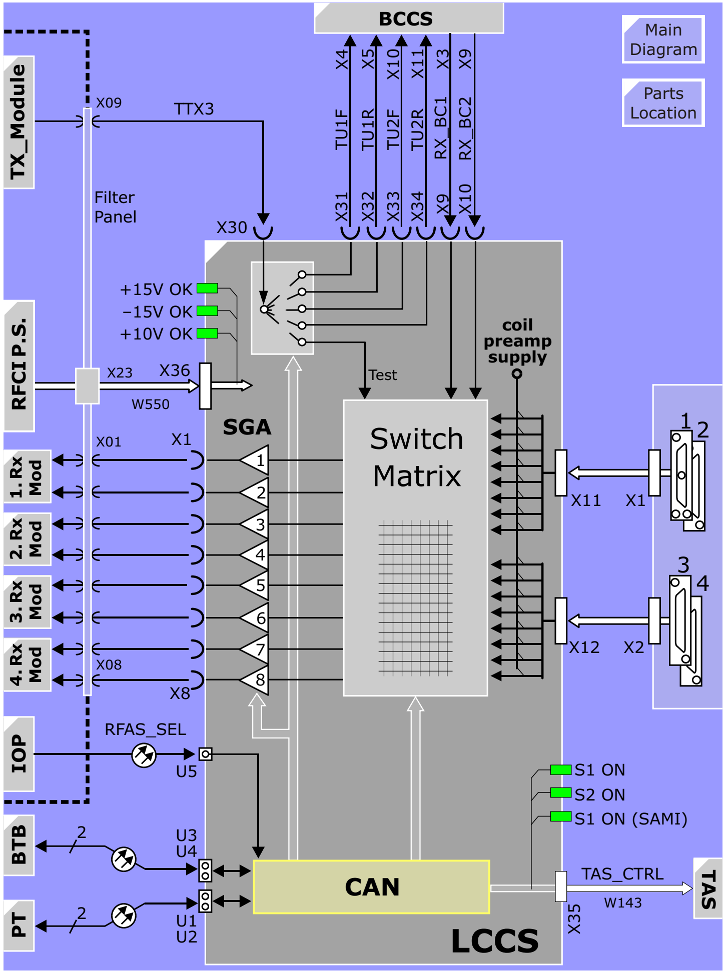
Coil Supply Voltage

Most local coils contain their own pre-amplifiers and are fed with a +10V supply over the connecting coax cables. This 10V coil preamp supply is generated in the LCCS from the +15V.

LEDs

To facilitate testing of the amplifiers several loops have been incorporated into the switch array. It is possible to switch the incoming tune signal output of the TX_Module to all of the eight amplifier stages but the first one for testing gain linearity and stability. Also, the service software is able to test all of the switches of the array.

Figure 75 LCCS



Local Coils

The Siemens **Integrated Panoramic Array™** concept for the MAGNETOM Harmony and the MAGNETOM Symphony is the latest evolution in coil handling and CP Array technology in the industry.

Integrated Panoramic Array

The modular concept of combining CP array coils into a sophisticated Panoramic coil increases productivity, expands anatomical coverage, and streamlines examinations – thus having a tremendous impact on clinical efficiency.

Increases Productivity

- The new coil design by Siemens Medical Engineering allows you to image the patient panoramically from head to thigh. Multiple CP array coils can be used simultaneously without having to reposition.
- To increase patient throughput and reduce set up time, the lower section of the CP head array coil and the whole CP spine array coil are integrated into the patient table. The coils remain on the table for almost all examinations except for knee (or other extremities) and breast imaging.
- These unique Siemens MAGNETOM features greatly accelerate patient set-up times, coil handling, patient throughput, and patient comfort.

Flexibility and Image Quality

- Up to four coils can be placed on the patient, largely eliminating patient and coil repositioning.
- This panoramic combination of individual coil elements into new configurations is the key to higher flexibility and excellent image quality.
- This increased flexibility for large field of view imaging is highly advantageous for head/neck imaging, whole spine examinations, body screening and peripheral MR Angiography.

The selection of local coil elements is achieved through the sequence change platform.

Trend setting developments in the CP Array coil arena have lead to the production of antenna assemblies which can cover a very wide area and at the same time provide excellent **signal quality**.

An array coil consists of several resonator elements in an array configuration the elements of which can be variably selected. Array antennas have the main advantage that through their combination larger FOVs can be achieved while maintaining the S/N advantages smaller antenna exhibit.

All array coils employ circular polarized antenna elements for good RF-field homogeneity and preamplifiers built into the coils thereby increasing the signal to noise separation for unparalleled S/N performance.

Figure 76 *The use of Array Coils in the Integrated Panoramic Array Concept*

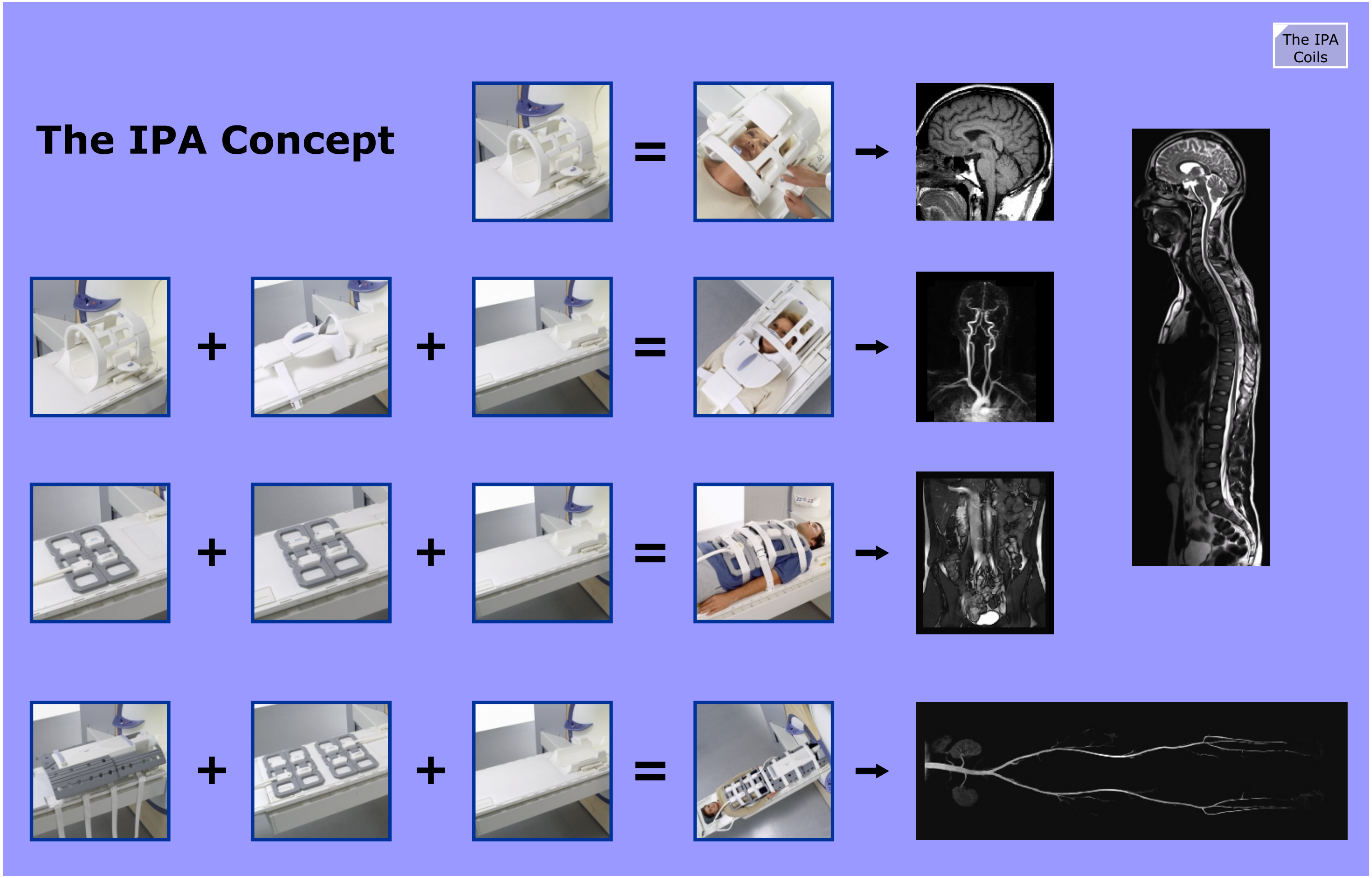
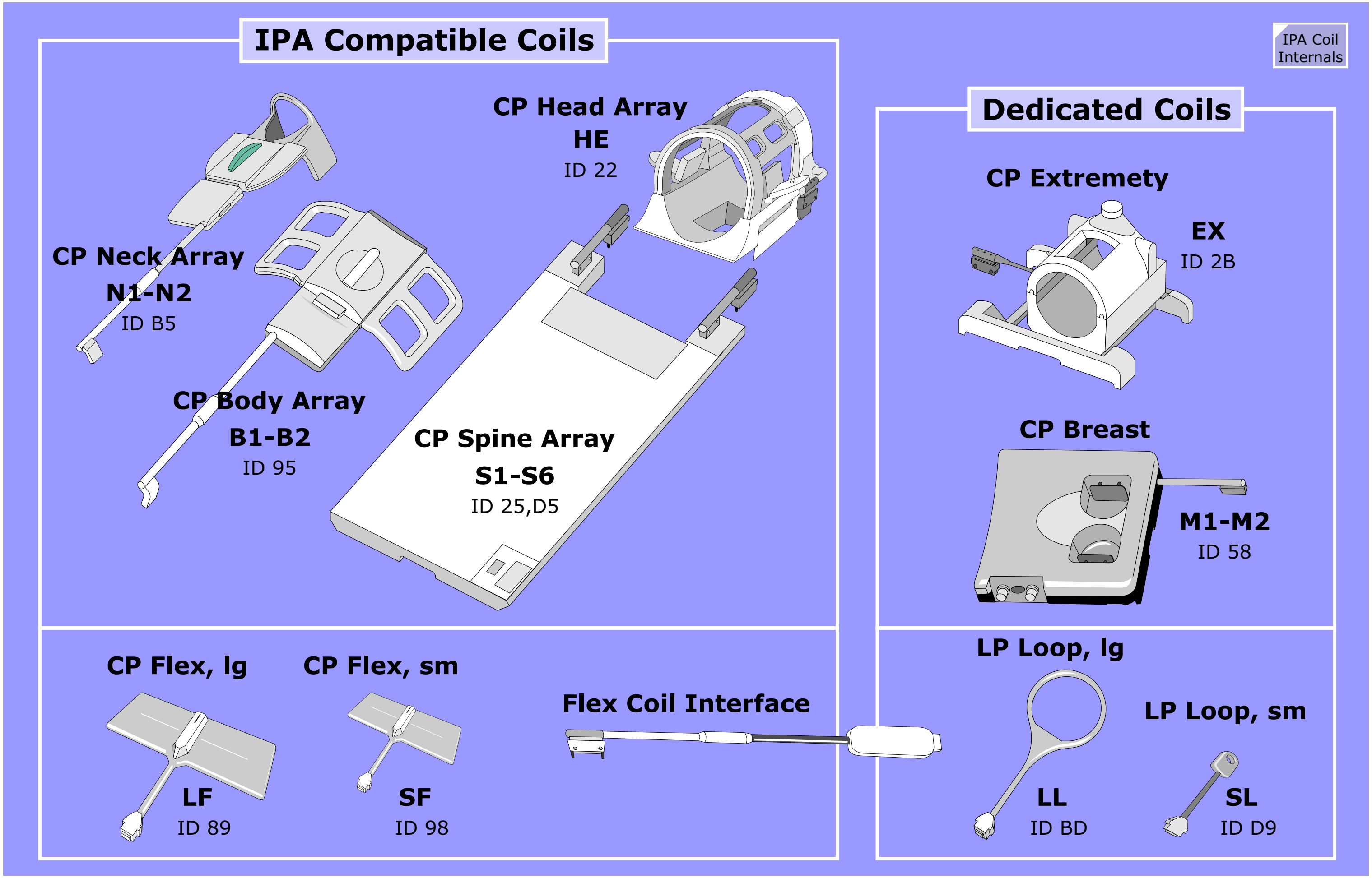


Figure 77 Local Coils



Coil Connectors

The RF coil connectors are integrated within the socket elements of the patient table. Each socket/plug element itself has four RF Receive pins (RX1, RX2, RX3, RX4) and various Control pins.

The transmit RF Pin (A1 TX/TX_GND) is only available in plug 1. In Magnetom Trio the transmit RF pin is reversed: male is on top. Therefore you cannot use the same Service Plug for tests, see below.

Involved in Coil Code Detection are Pins CODE1, CODE2 and CODE_GND. If no Coil is connected, you will measure approximately 9.07V between Pins 12 and 05 and between Pins 17 and 05. If the 10V generated on the RFCI motherboard is missing or faulty an error message will be displayed on the LCD Display and no measurement will be possible.

The output control signals LCx_PIN_1-4 (for local coil connector x=1, 2, 3, and 4) are applied to the local coils over the local coil connectors on the patient table, see the description "Dyscon Module A and B", this chapter.

Testing

The Service Plug can be used for continuity tests of signal lines to the local coil connectors. These test, described under RFCI Interactive Tests in the TSG, include:

- control signals LCx_PIN_1-4
- the 10V for the preamplifiers located in the surface coils
- coil code recognition lines

Internal Wiring of CP Array Coils

The diagram above shows the internal wiring of CP Spine Array Coil, CP Neck Array Coil and CP Body Array Coil and its connections to the coil connectors. The CP Body Array Extender is normally connected to connector 1.

Large FOV Adapter

The Large FOV Adapter shifts the CP Spine Array Coil by 36cm out of the bore allowing MR Angiography from kidneys to feet with one coil setup. You can connect the Body Extender Coil + the CP Body Array Coil (connected to the CP Spine Array) + the CP Spine Array + the 8 element Peripheral Angio Array-Coil (PAA).

The blue connector is only to connect the Body Array Extender, other coils (for example the Large Flex coil) are not supported.

When the Body Extender is connected, the CP Spine Array elements SP1 and SP2 are automatically disconnected and not selectable, see the switches on the right hand graphics.

Figure 78 *The Large FOV Adapter*

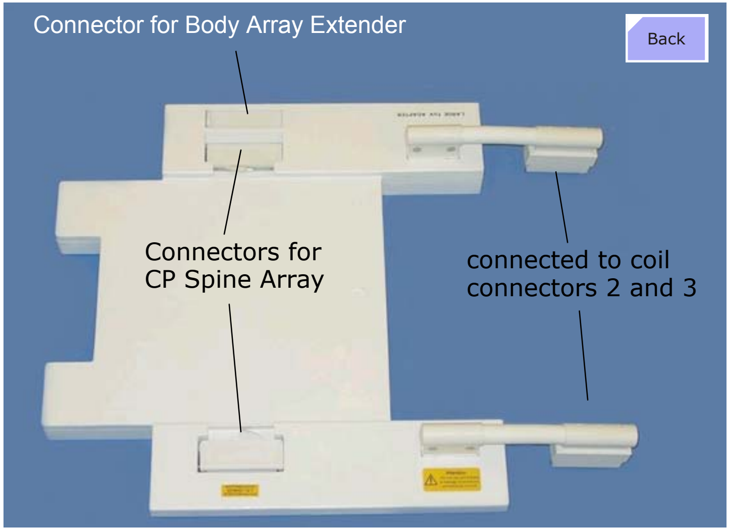
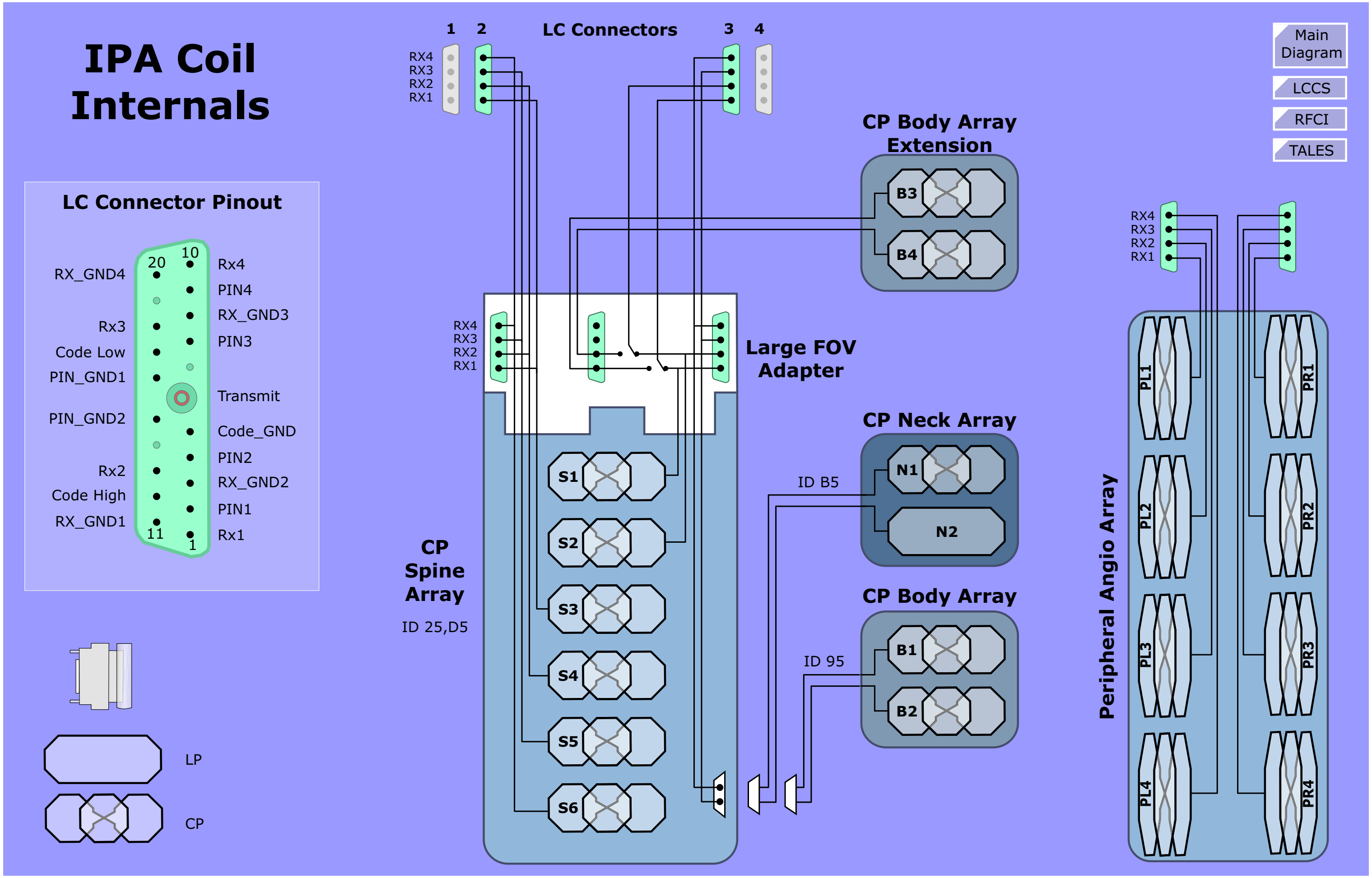


Figure 79 *Coil Connector*



RF Cabin Interface (RFCI)

Before you get confused, RF Cabin interface has nothing to do with the RF cabin (and thus the name). The RFCI is the control electronics for the dynamic detuning of the Body Coil and local coils. What you'll find in this chapter:

- Overview
- CAN Module Interface
- Motherboard
  - Voltage Monitoring
  - Coil Code Detection
  - Body Coil Detuning
- DYSCON Module A
  DYSCON Module B
- RFCI Power Supply

Overview

The RFCI's provides the tasks of generating the control signals for the PIN diodes in the local coils and the BTB for static and dynamic detuning of the Local coils and Body Coil and to recognize the coil code. On-board monitoring supervises all power supplies and detects if a coil change occurs during a sequence.

Figure 80 *RFCI Parts Location*

Block Diagram

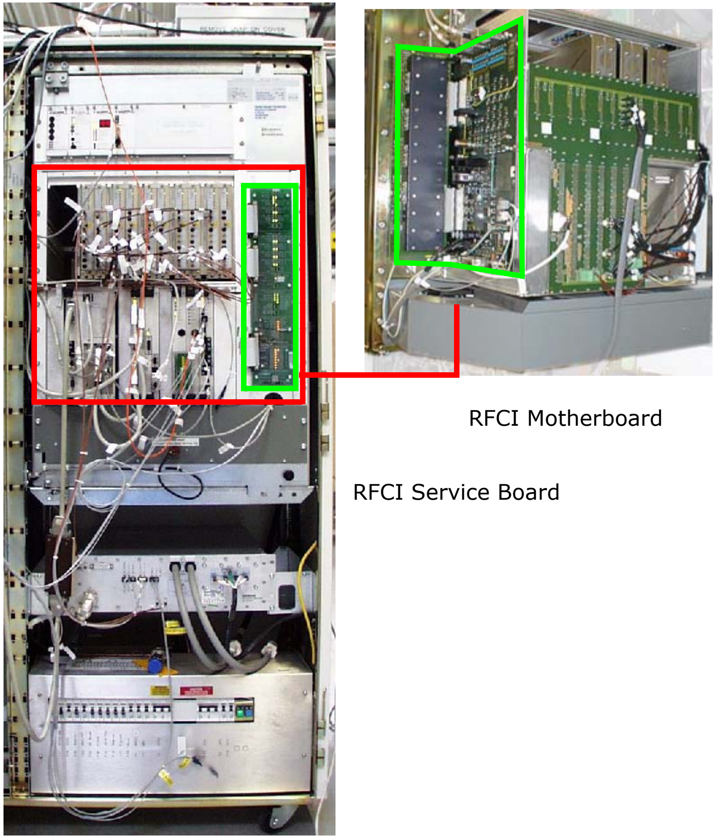
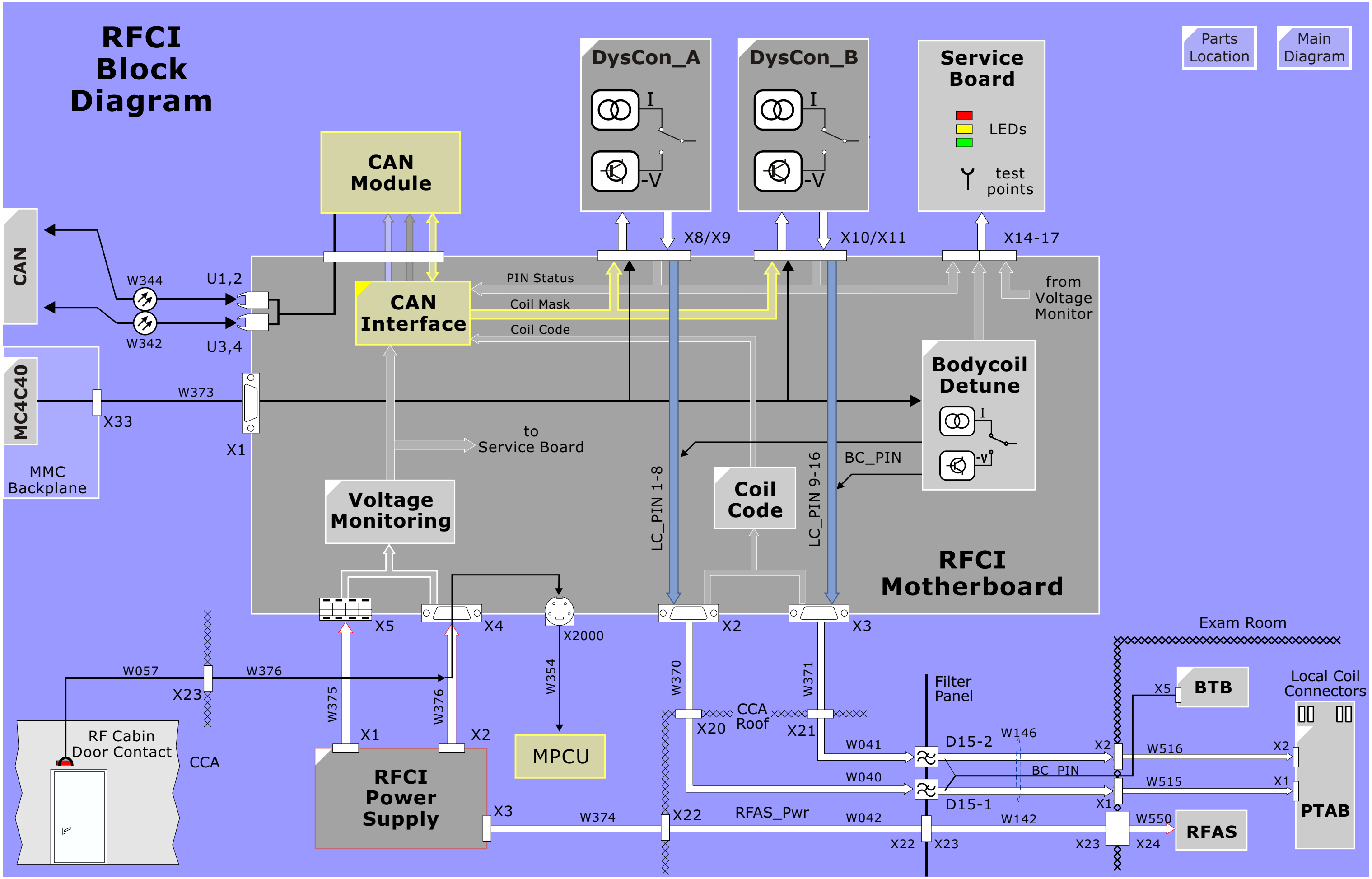


Figure 81 *RFCI Overview*



CAN Module

The CAN Module acts as the principle controller for the RFCI electronics. The CAN Module employs a CAN microcontroller whose functions are:

- CAN Bus communication
- Local Coil Code recognition
- Error Interrupt Handling
- Coil Detune Signal Masking

Unit Code

This CAN Module is also used in other components. The Unit code is an identifier byte used to identify the component in which the CAN Module is plugged.

N_INT

Two error interrupts, power_fail, coil_change, will cause an immediate stop of the sequence.

Status LED

The CPU will activate the FW LED on the service board if the CPU is in the firmware mode, the LW LED when in the loadware mode.

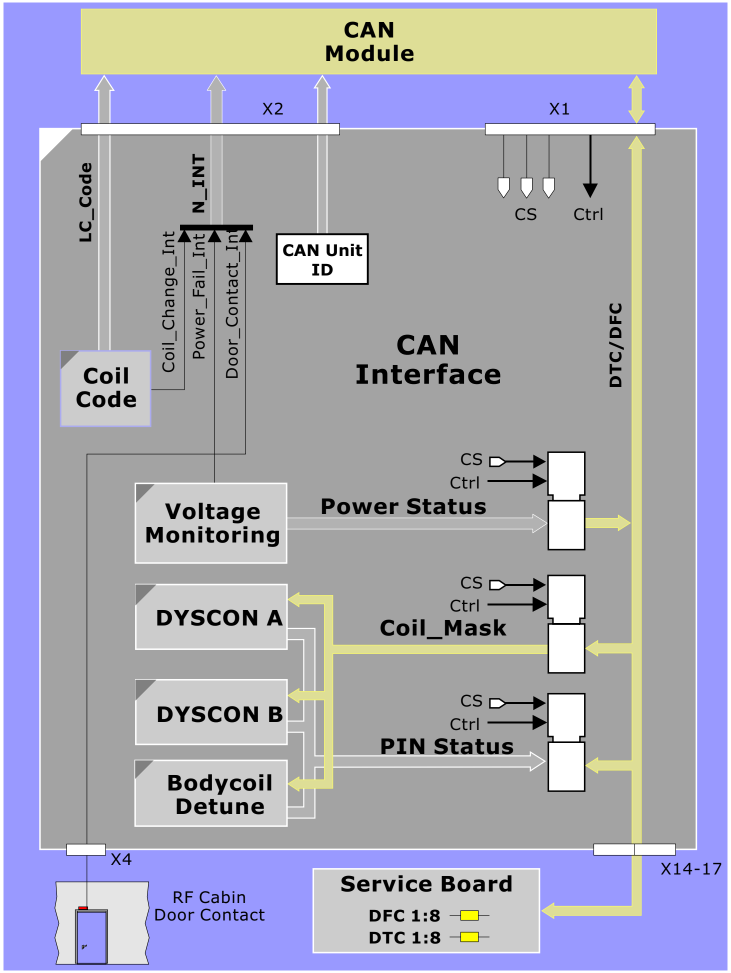
CAN Module Interface

The CAN Module interface provides hardware specific interface circuitry for the CAN Module which has been designed as a general control unit for use in any sub-system.

DTC/DFC

The Data To Can (DTC) and Data From Can (DFC) busses transfer the various status information and coil masks to and from the CAN Module. LEDs on the Service board show the digital state of each of these busses.

Figure 82 *CAN Module Interface*



Voltage Monitoring

To assure proper coil control and operation, all powers supply voltages are monitored.

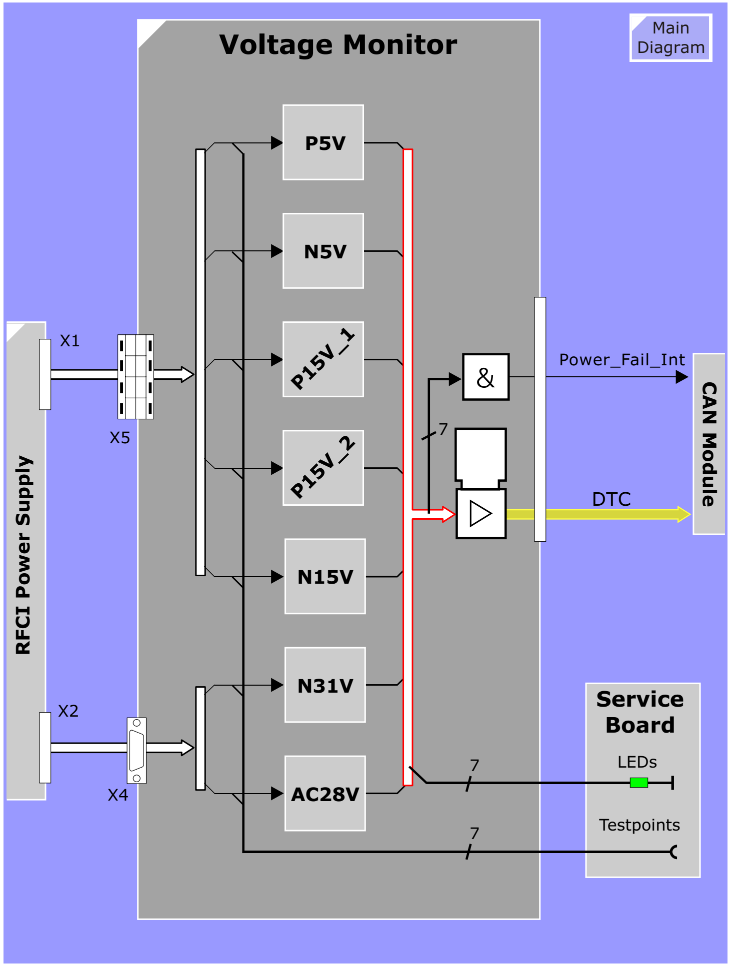
The Service Board indicates voltages are ok by green LEDs, the voltage levels can be measured at test points also (see TSG).

Power Supply Failure

All voltages are monitored for under voltage. A failing or weak voltage will generate an interrupt to the CAN Module. Subsequently the sequence will be stopped.

The failing voltage is also read in by the CAN Module and status sent to the NUMARIS error log.

Figure 83 *Voltage Monitoring*



Coil Code Detection

The coil section has two functions:

- Coil code detection
- Coil code change recognition

Coil Code Detection

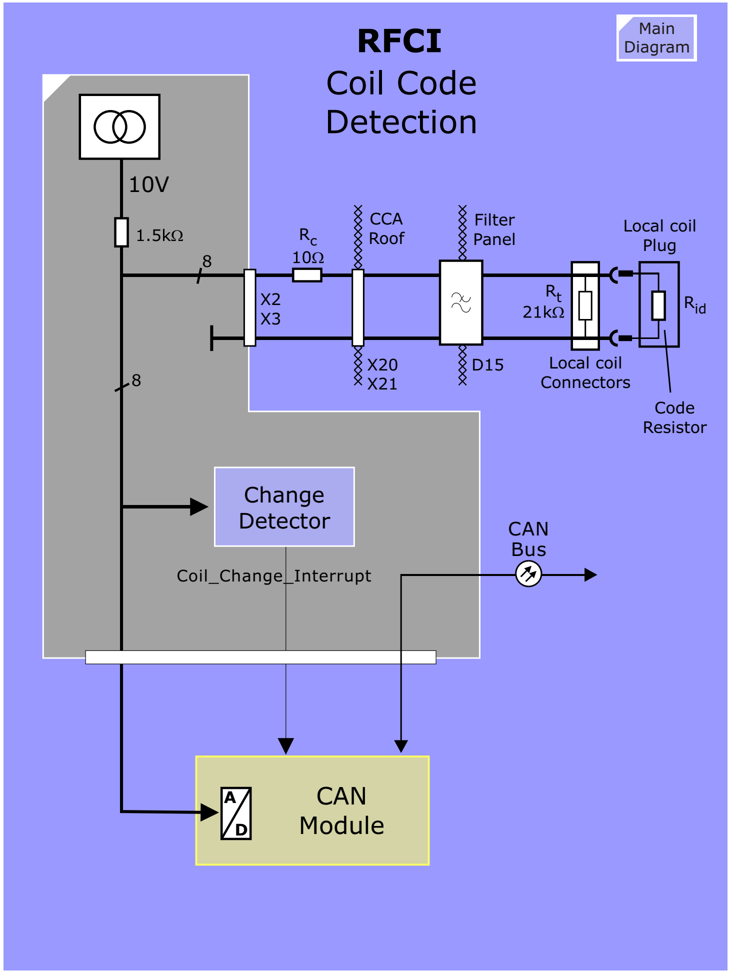
Recognition of connected local coils is realized using resistive coding. Every coil has two code resistors in each of its coil plugs. There are 13 defined resistive values for the code resistors whereby each resistive values corresponds to a hex code value from 1 and D (see the table on page 147). To read out the code resistor value a voltage divider is used. The voltage divider consists of a 10V source, a 1.5 kΩ resistor on the board and the code resistor in the coil plug together with the reisistance of the signal cables. The voltage between these two resistances is determined by the code resistor values (found in table on page 147).

When the CAN Module is not in sleep mode (during a running scan) ia connected coil will be recognized immediately and the coil code sent via the CAN Bus to the Host. The Host will read the name of the coil from the coil file and send it to the Patient Table display. If the conecting coil has an unrecognized code, the error message "coil file error" will be displayed.

Coil Code Change

During the sequence, the CAN Module will be in sleep mode and cannot read the coil change directly. In this case, a Change Dectector circuit will recognize the change and wake up the CAN Modue via an interrupt. This will cause the sequence to be stopped.

Figure 84 *Coil Code*



List of Coil Code Resistor Values

Coil code in Hex	Resistance in Ohm	Voltage at MUX	Notes
0	<63	0,314	Signifies a short = error!
1	132 - 162	0,938	
2	256 - 291	1,564	
3	402 - 444	2,19	
4	577 - 627	2,816	
5	790 - 853	3,442	
6	1056 - 1135	4,066	
7	1397 - 1500	4,692	
8	1849 - 1989	5,318	
9	2478 - 2680	5,944	
A	3412 - 3729	6,570	
B	4946 - 5511	7,196	
C	7930 - 9207	7,82	
D	16259 - 21493	8,446	
E	14643	9,072	signifies no coils connected
F	37572	9,698	signifies an open = error!

Listing of Coil Codes

Short Name	Coil Name	Coil Code
	CP Neck	11
HLS	Heart Liver (Spec.)	12
USI	USAI Head Spec	14
HE	CP Head Coil	22
SP1	CP Spine Array (Connector 1)	25
EX	Extremity Coil	2b
DL	Double Loop Array Left	2c
DR	Double Loop Array Right	2d
CA	MAI Carotid Array	31
UE	MAI Upper Extremity Flex	32
LE	MAI Lower Extremity Flex	33
WT	MAI Wrist Array	34
WR	Wrist	44
SH	CP Shoulder Array	4a
HB	MRIDC High Res.Head/Brain	4b
Shim	Shim Device	55
BR	CP Breast Array	58
PR	PAA Peripheral Angio Array (right side)	73
PL	PAA Peripheral Angio Array (left side)	83
LF	CP Flex Large	89
FL7	Flex Loop 7	8b
BO	CP Body Array (connected to Spine coil)	95
SF	CP Flex Small	98
EN	Endorectal	9d
FH1	FMRI Head Array	a4
FH2	FMRI Head Array	aa
LL	Flex Loop Large	bd
C1	Customer Coil 1 (RX - Coil)	c1
C2	Customer Coil 2 (Tx/Rx - Coil)	c2
SP2	CP Spine Array (connector 2)	d5
SL	Flex Loop Small	d9
BC	Integrated Body resonator	ee
Ser._P	Service Plug	TX mode = CC RX mode = CD
SF	CP Flex Small	98
EN	Endorectal	9d

Body Coil Detuning

The **dynamic detuning** of the Body Coil is accomplished with a λ/4 line tied to a PIN diode. The diode, when shorted (600mA), is transformed by the λ/4 line to a high impedance and the de-tuning circuit is without effect: the Body Coil will be tuned. When the diode is opened (-30V) the Body Coil will be detuned.

The dynamic control signal **IN_BC** is produced by the two dynamic signals **Tx_Rx**, **Dyn_Stat** and a digital memory consisting of 2 sets of 2 bits each. This allows the CAN module to freely program the diode state for tuning or detuning according to coil or coil combination. This concept offers the greatest flexibility and is easily programmable.

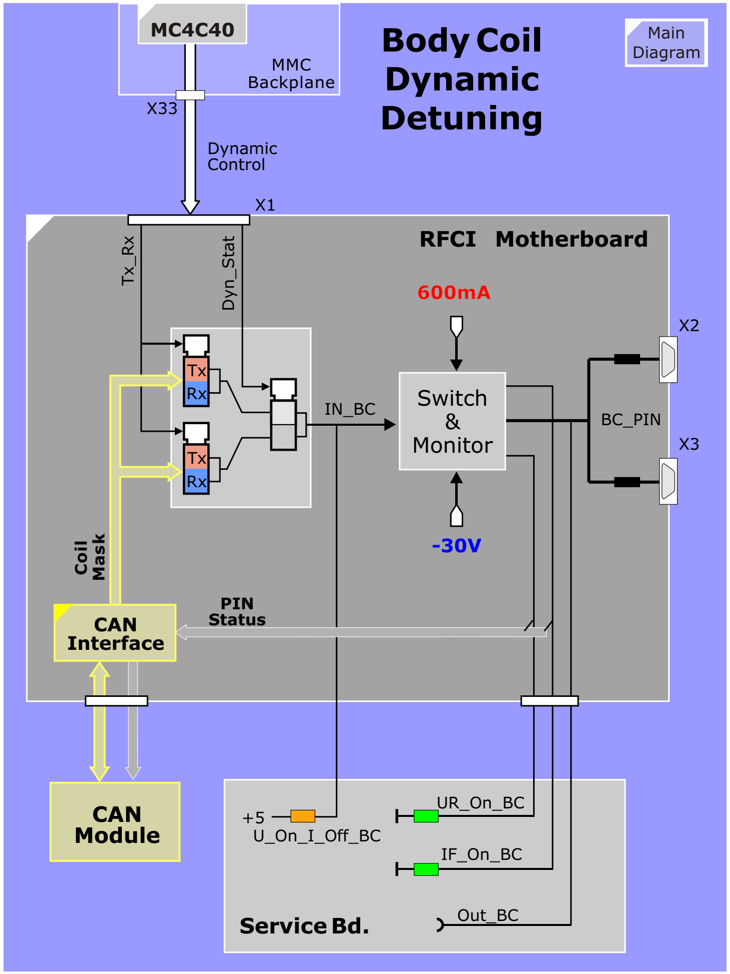
The level of the output signal **BC_PIN** is determined by **IN_BC** according to the table below. For the Symphony/Sonata systems with two BTBs the signal is doubled.

For trouble shooting purposes the status of the dynamic tuning/detuning signals is indicated by LEDs at the service board. Additionally, test points are available.

		LEDs			
IN_BC	**BC_PIN**	**U_On_I_Off_BC**	**UR_On**	**IF_On**	**Coil State**
H	600mA	OFF	OFF	ON	**tuned**
L	-30V	ON	ON	OFF	**detuned**

The monitor checks the output voltage and current levels. The **UR_On** and **IF_On** LEDs signify that the current or voltage level of the output signal are ok.

Figure 85 *Body Coil Dynamic Detuning Control*



DYSCON Module A

The dynamic detuning of the local coils is accomplished using PIN diodes which are strategically placed in the coil.

The dynamic control signal IN_LC is produced by the two dynamic signals Coil_Mask_Sel, LC_Ctrl and a digital mask from the CAN module. This allows the CAN module to freely program the diode state for tuning or detuning according to coil or coil combination. This concept offers the greatest coil programming flexibility.

The output signals LC1_PIN_1-4 (local coil connector 1) and LC2_PIN_1-4 (local coal connector 2) are applied to the local coil over the local coil connectors on the patient table.

For trouble shooting purpose the status of the dynamic tuning/ detuning signals is indicated by LEDs at the service board. Additionally, test points are available.

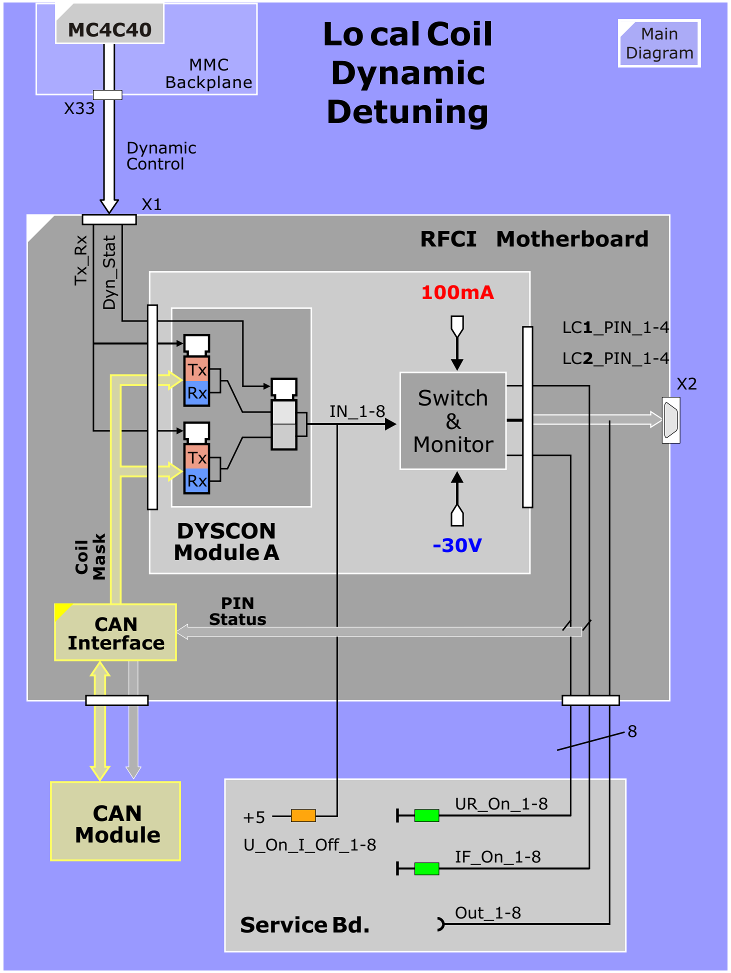
IN_LC	LC_PIN	LEDs			Coil
		U_On_I_Off_LC	UR_On	IF_On	
H	100mA	OFF	OFF	ON	detuned
L	-30V	ON	ON	OFF	tuned

As seen in the table above, the IN_LC signal controls the output of the PIN driver circuit. A monitor is also built in which checks the output voltage or current levels. The UR_On and IF_On LEDs signify that the current or voltage level of the output signal is ok.

NOTE The table above applies for ALL Siemens coils, but not necessarily for third party coils where the tune logic may be opposite.

NOTE The cable from the MMC delivering the dynamic control signals is NOT tested by the service software

Figure 86 *Local Coil Dynamic Detuning Control*



DYSCON Module B

The dynamic detuning of the local coils is accomplished using PIN diodes which are strategically placed in the coil electronics.

The dynamic control signal IN_LC is produced by the two dynamic signals Coil_Mask_Sel, LC_Ctrl and a digital mask from the CAN module. This allows the CAN module to freely program the diode state for tuning or detuning according to coil or coil combination. This concept offers the greatest programming flexibility.

The output signals LC3_PIN_1-4 (local coil connector 3) and LC4_PIN_1-4 (local coal connector 4) are applied to the local coil over the local coil connectors on the patient table.

For trouble shooting purpose the status of the dynamic tuning/ detuning signals is indicated by LEDs at the service board. Additionally, test points are available.

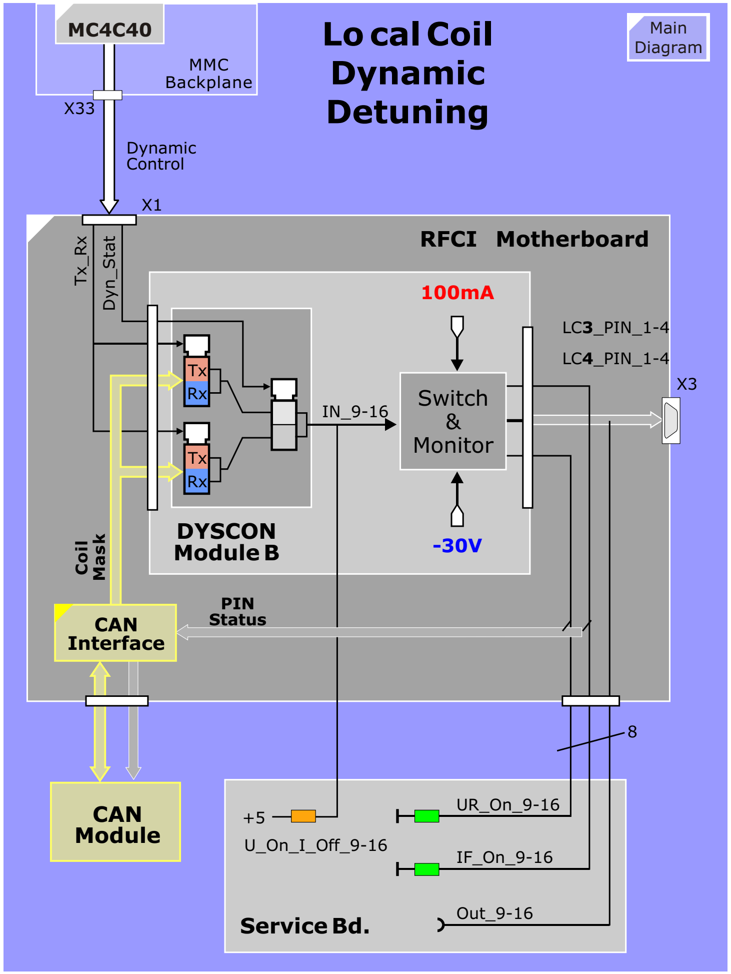
		LEDs			
IN_LC	LC_PIN	U_On_I_Off_LC	UR_On	IF_On	Coil
H	100mA	OFF	OFF	ON	detuned
L	-30V	ON	ON	OFF	tuned

As seen in the table above, the IN_LC signal controls the output of the PIN driver circuit. A monitor is also built in which checks the output voltage or current levels. The UR_On and IF_On LEDs signify that the current or voltage level of the output signal is ok.

NOTE The table above applies for ALL Siemens coils, but not necessarily for third party coils where the tune logic may be opposite.

NOTE The cable from the MMC delivering the dynamic control signals is NOT tested by the service software

Figure 87 *Local Coil Dynamic Detuning Control*



RFCI Power Supply

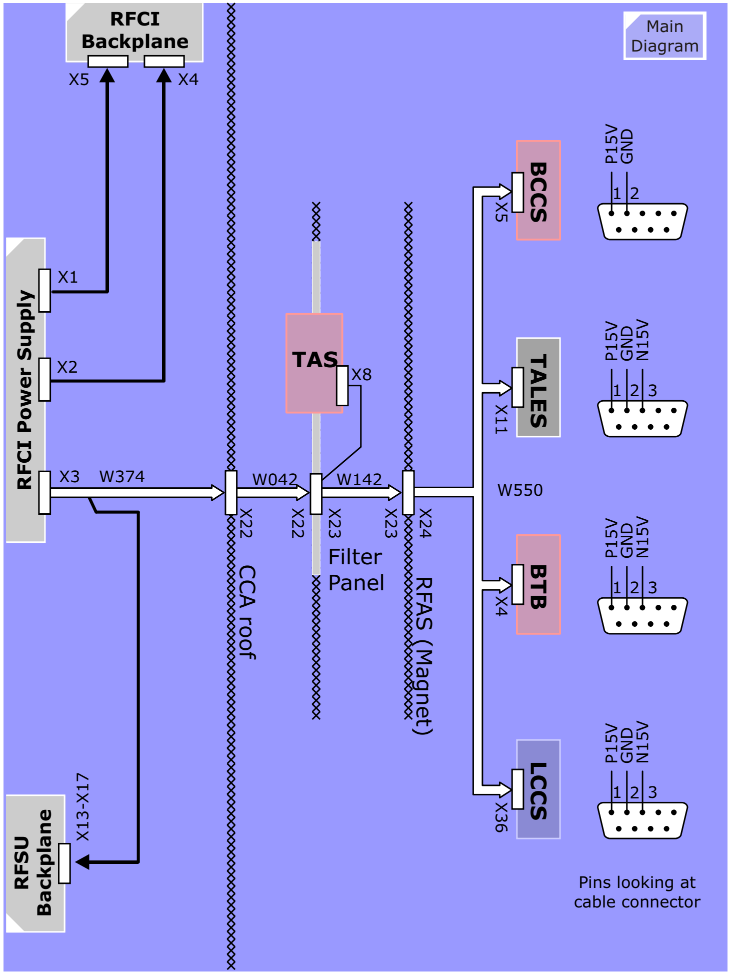
The power supply is an OEM component. It has built-in monitoring for protection against short circuit, over voltage and over temperature. And it's water cooled.

LEDs

LED	Description
I fold back	Indicates normal operation
I Constant	Indicates an error. Output current will be limited.
Overtemp 1	>50°C - Warning
Overtemp 2	>60°C - Fatal. Power supply shuts down
Output Voltage	One for each output voltage. Indicates only that a voltage is present. Level monitoring is done on the RFCI motherboard.
Output Error	Indicates that any error has occurred.
Crow bar Active	Indicates that an over voltage has occurred. It will stay on until the power supply has been turned off.

The Overtemp status signal going to the CAN Module on the RFCI will generate an error to the MMC.

Figure 88 *RFCI Power Supply*



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Section

7 Patient Handling

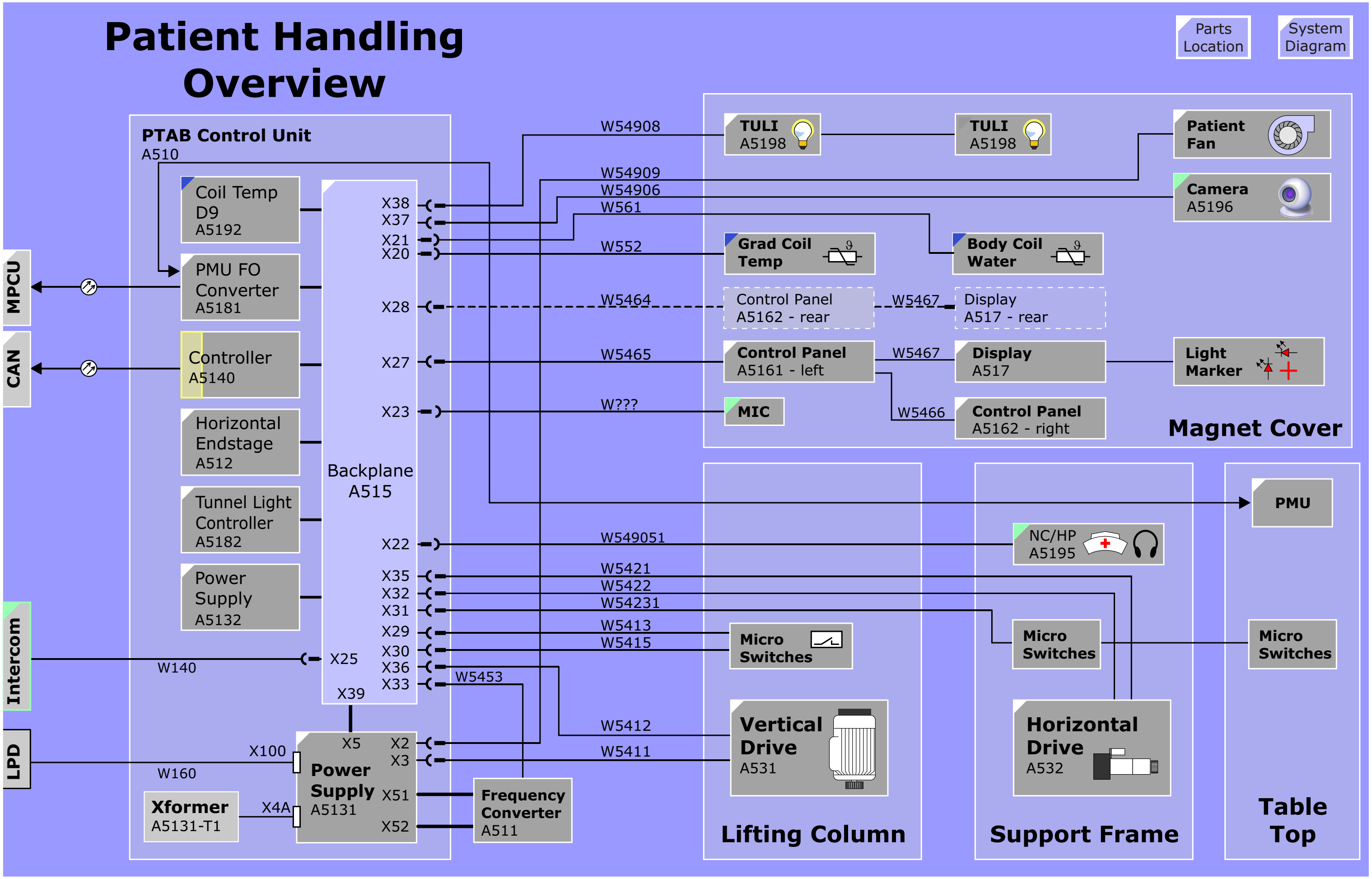
Introduction

The Patient Handling system includes all those components which in any way interact with the patient and consists of these functional blocks:

- Patient Table
- Trolley (optional)
- Intercom and Patient Video Monitor
- Physiological Measurement Unit (optional)

Figure 89 *Patient Handling Components*

Figure 90 *Patient Handling Overview Diagram*



Patient Table

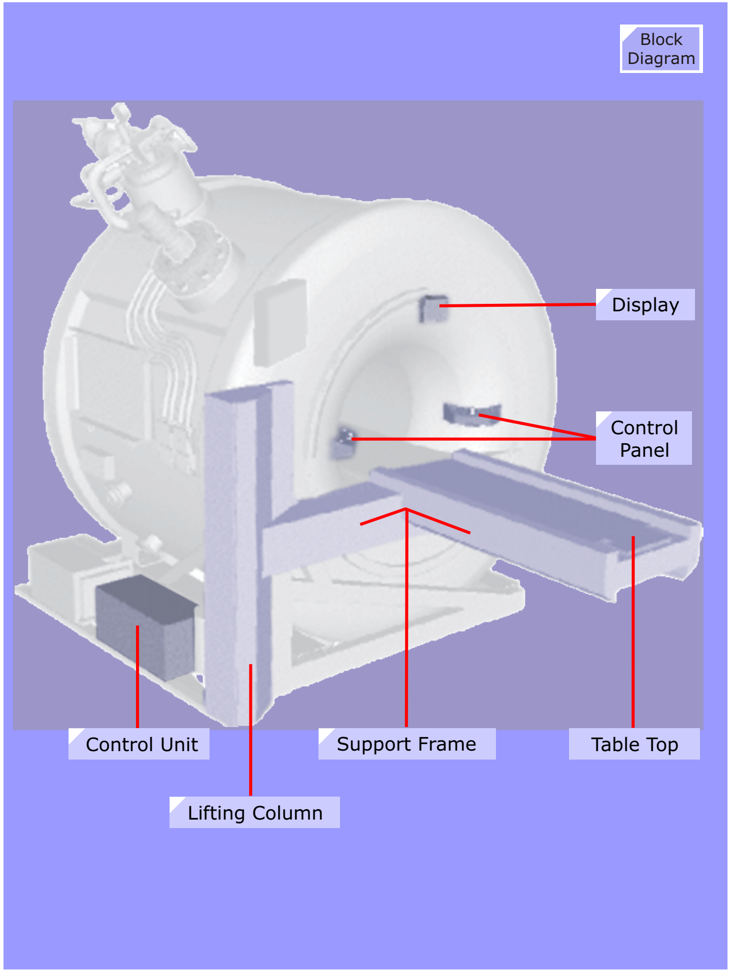
Integrated Panoramic Positioning (IPP) allows remote patient handling at the *syngo* Acquisition Workplace for optimized workflow efficiency.

In combination with the automatic table move and a scan range of up to 154 cm, IPP allows the planning of whole examination series and automates the dynamic acquisition process, e.g. peripheral angiography.

A free-floating table design provides better access to the patient by giving the attending staff free foot room. The table extends approx. 42 cm beyond the back of the magnet for additional patient access.

Specification	Value
Max. patient weight	200kg (440 lbs)
Max scan range	154 cm (5’11")
Horizontal table movement	212 cm (7’)

Figure 91 *Patient Table Parts Location*

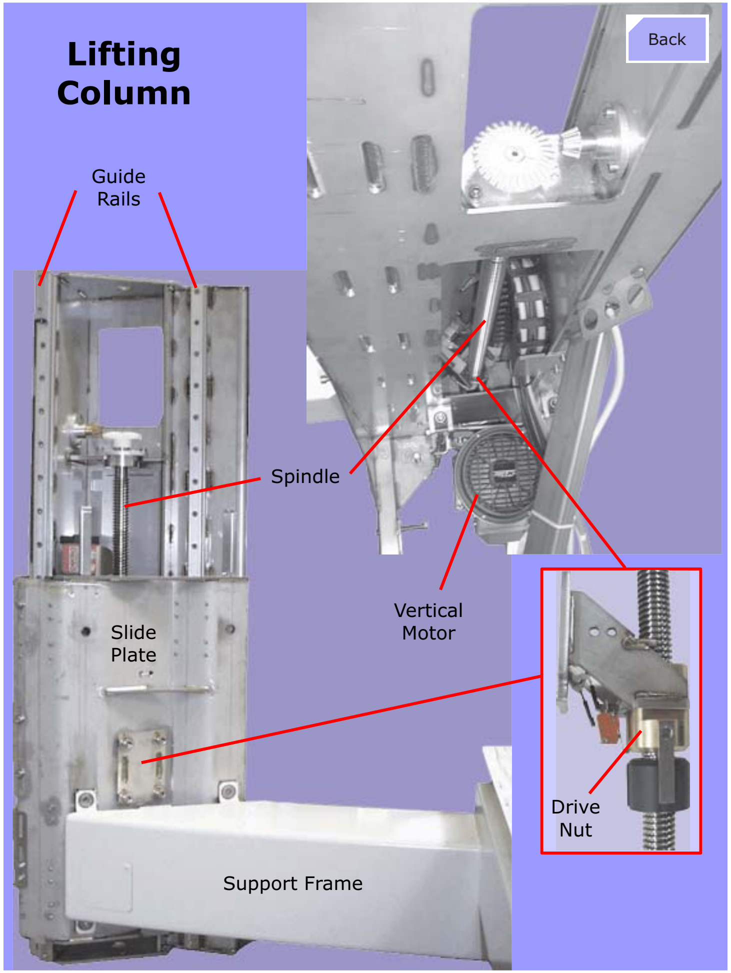


Lifting Column

The Patient Table lifting column is a mechanical assembly using a motor driven screw-spidle to move a drive nut in the vertical direction. The drive nut pushes up or lets down the attached support frame. The Support Frame is not mechanically fixed to the drive nut but is resting upon it through its own weight. The support Frame is sliding on guide rails attached on either side of the lifting column.

The lifting column is fixed to the magnet over an adjustable frame that is attached to the front and left side of the magnet frame. The Patient Table can be de-attached from this frame to facilitate installation in tight rooms conditions.

Figure 92 *Lifting Column*

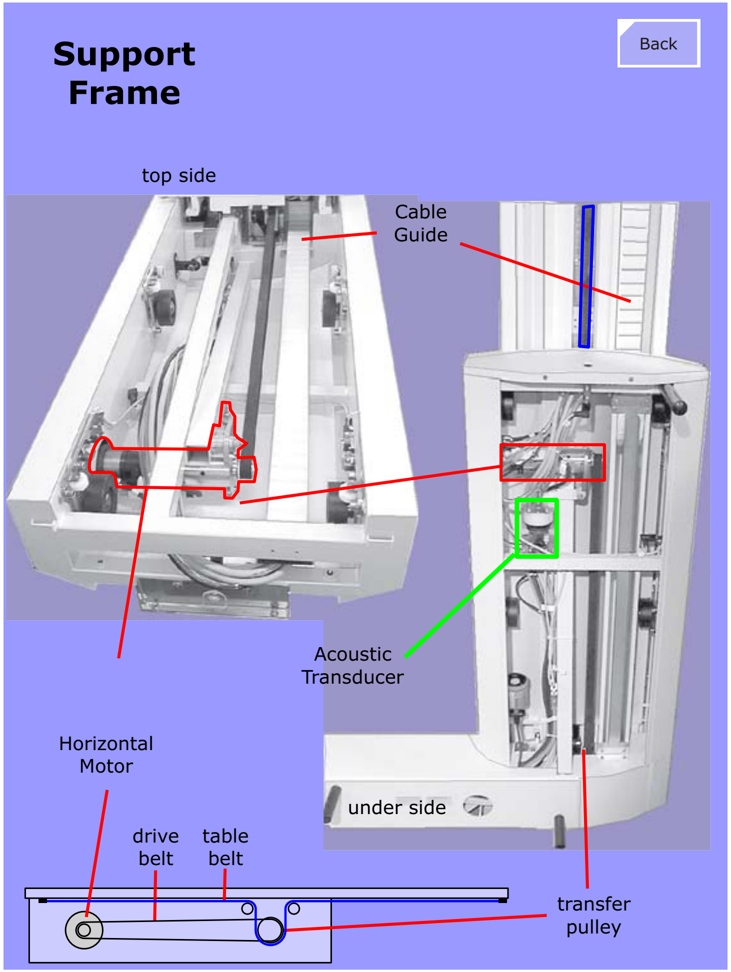


Support Frame

The support frame must support the weight of the patient (plus his clothes) and the Patient Table top. The components located within the support frame:

- Horizontal Drive motor.
- Acoustic Transducer (see Intercom section)
- cable guide
- feedback switches and sensors

Figure 93 *Support Frame*



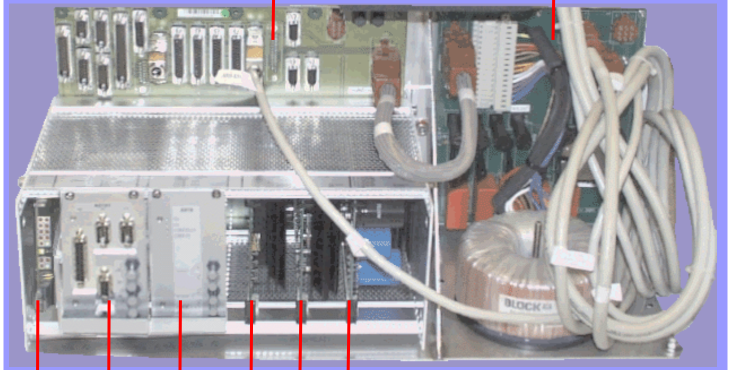
Control Unit

The Control Unit contains the electronic control components and consists of the Control Unit rack, Control Unit backplane, Power Supply A5131 and transformer.

These components are described in detail in further pages. Go to the Block Diagram for details.

Figure 94 *Control Unit Assembly*

Block Diagram
Parts Location



Vertical Drive

The Vertical Drive mechanics consists of a vertical screw spindle driven by a 3 phase AC motor

Function

The vertical motion is realized on a three phase asynchronous motor that drives a self-locking spindle assembly via toothbelt. The motor is controlled by a programmable Frequency Converter. The parameter set of the Converter is programmed/checked by the Main Controller. In case of parameter set differences the Main Controller is reloading the Converter automatically.

After replacement the Frequency Converter has to be initialized (see TSG, Mode M7, parameterizing the converter).

Vertical motion is enabled only if the table top is completely out of the magnet bore. This condition is detected by the SZ10 switch.The actual vertical position of the table is detected by the SY0 - SY5 switches.

The SY6 and the two (A/B) SZ15 switches composing an interlock interrupting the vertical motion in case of a mechanical blockage (i.e. something is below the table).

NOTE For detailed description of switches see TSG.

Interconnections

The power for the Patient Table is supplied by the Power Distribution system in the CCA (Fuse breaker F15) via connector X100 at the Power Supply backplane.

The Frequency Converter converter can be switched on or off with help of the INV_SUPPLY signal from the Main Controller and relay K4, located at the Power Supply Backplane.

By generating a three-phase AC voltage that can be changed in frequency and phase rotation, the Converter can control direction and speed of the Vertical Motor.

Control signals for the Frequency Converter are generated by the Main Controller and distributed via X33 Patient-Handling Control-Backplane to X501 at the Converter.

Figure 95 *Vertical Drive Control*

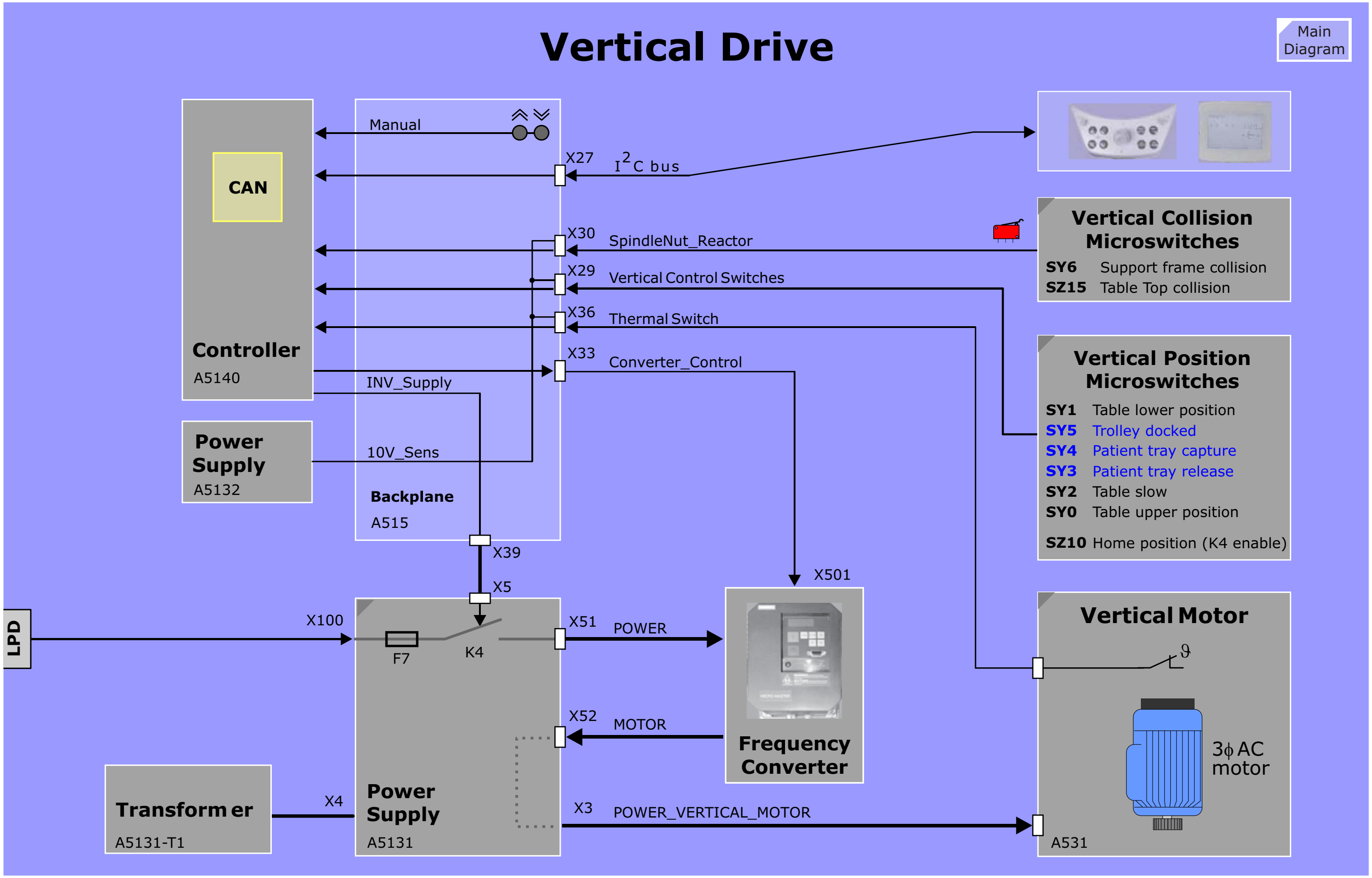
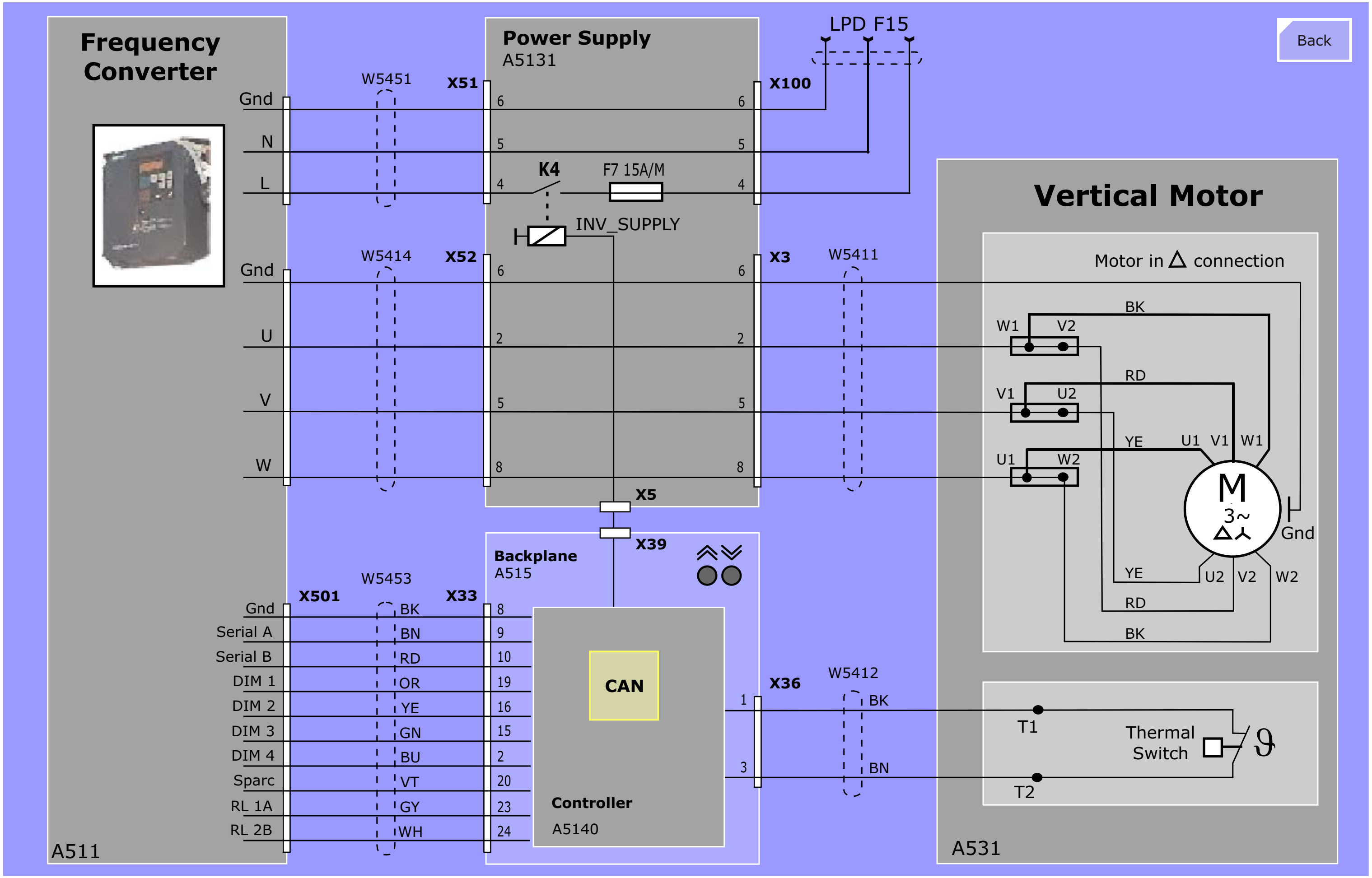


Figure 96 *Vertical Drive Interconnections*



Horizontal Drive Control

Function

Horizontal motion of the patient table is enabled if the table has reached the vertical top position (detected by toggle switch SY0). The Main Controller A5140 and the Horizontal Power Stage A512 are responsible for the horizontal movement control.

The Horizontal Drive motor is a DC type with integrated gearbox and electro-mechanical brake (brake impedance 100 Ohms, to be measured at X1, Pin 5/6). An incremental encoder is coupled to the motor detecting the relative motion of the table.

During horizontal positioning stops the table plate is locked actively by delta-position detection and motor force (the electro-mechanical brake is not active yet). After start of measurement, the Main Controller A5140 turns into a sleep mode and switches-off the horizontal motor. The electro-mechanical brake gets active now, the table plate is locked again (~20 VDC from A5140, applied via connector X1 Pin 5/6).

In vertical mode, the electro-mechanical brake is always active.

LEDs

LED	
POWER	38 V_AC - Power supply for end stage
SUPPLY	Power supply off during MR measurement
RUN	End stage is enabled and no error present
ERROR_C	Over current: Motor A532, cable W5421, end stage A512
ERROR_S	Over or under voltage or overtemperature of the end stage

Figure 97 *Horizontal Motor Interconnections*

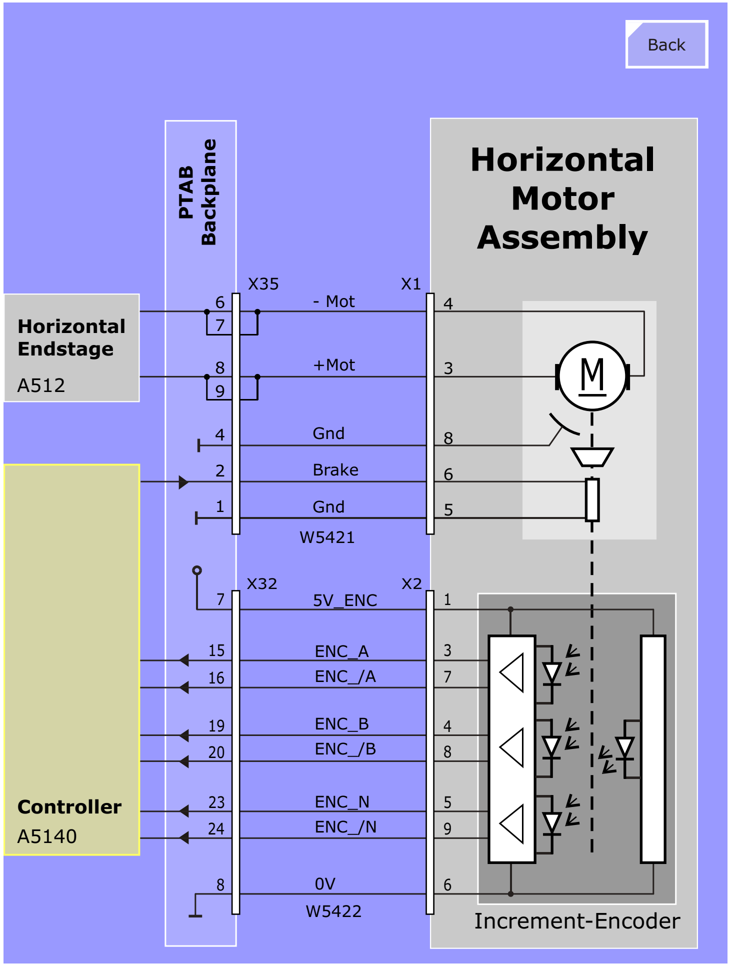


Figure 98 *Horizontal Drive Control*

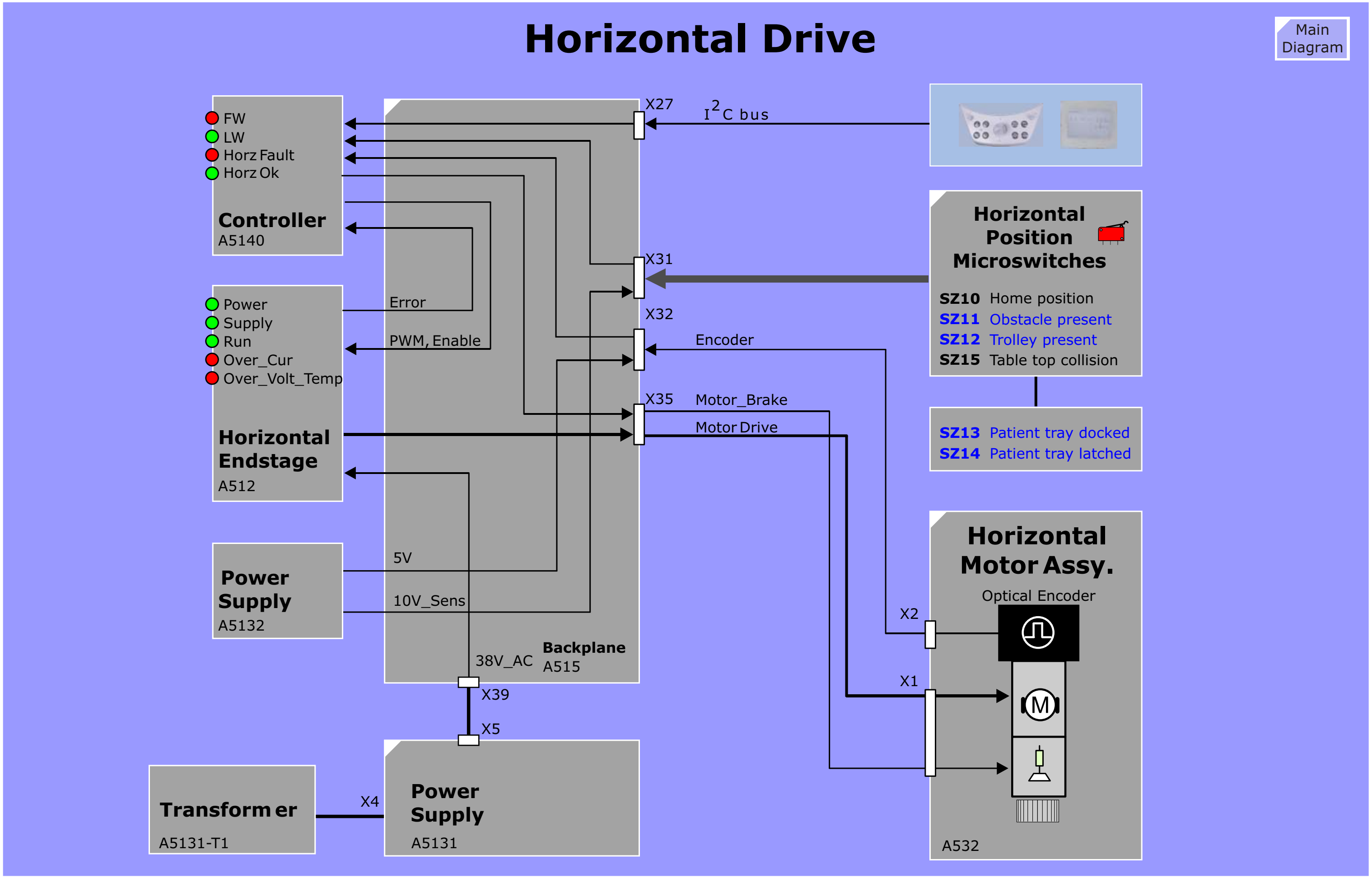
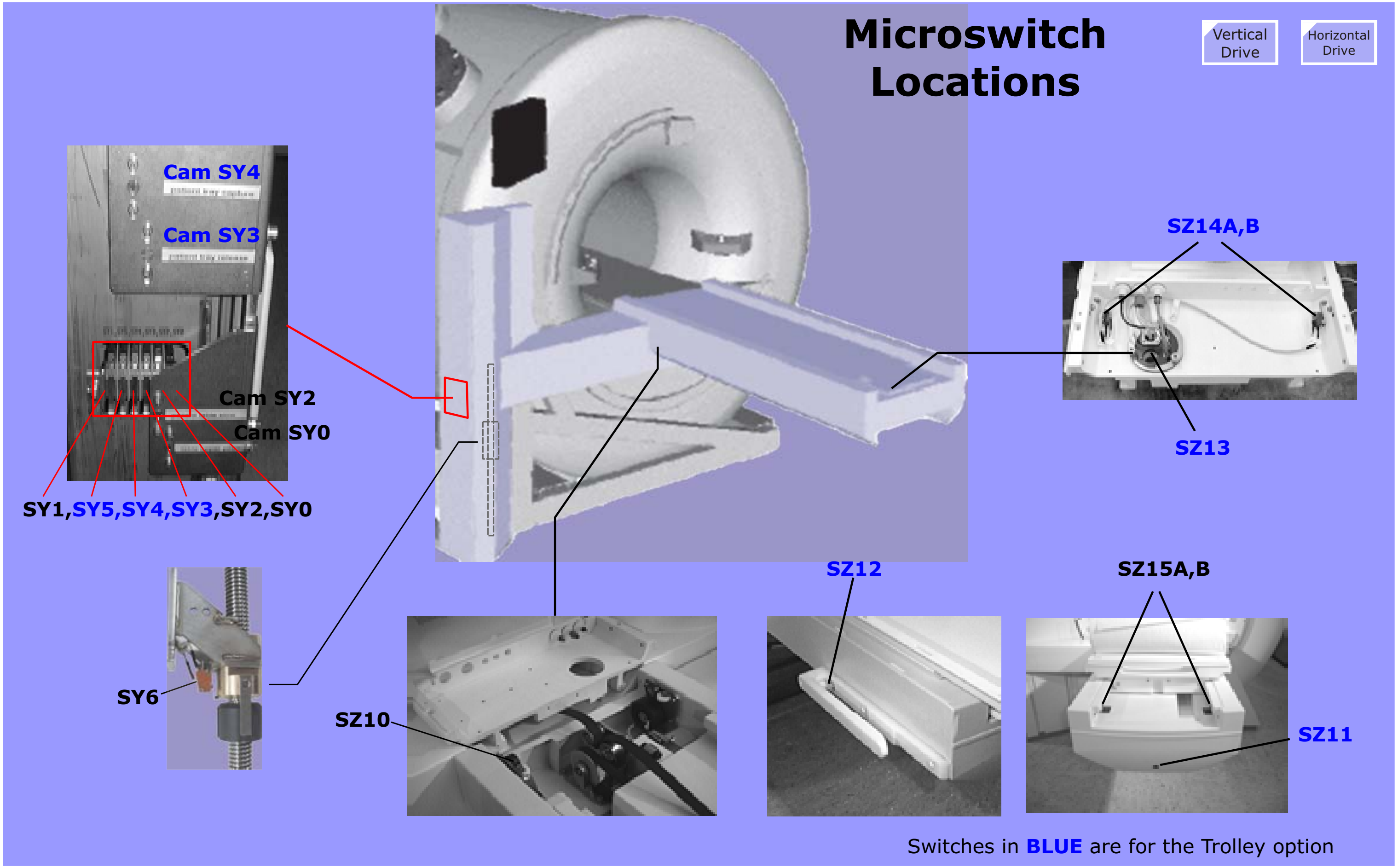


Figure 99 *Patient Table Microswitch locationsl*



Power Supply A5131

The Power Supply A5131, also called Power Supply backplane, provides the interconnections between the mains power from the Line Power Distribution, transformer, Frequency Converter and Patient Table Control Unit as well as provide relay circuitry for powering the Frequency converter on and off and control the Patient Fan speed.

Function

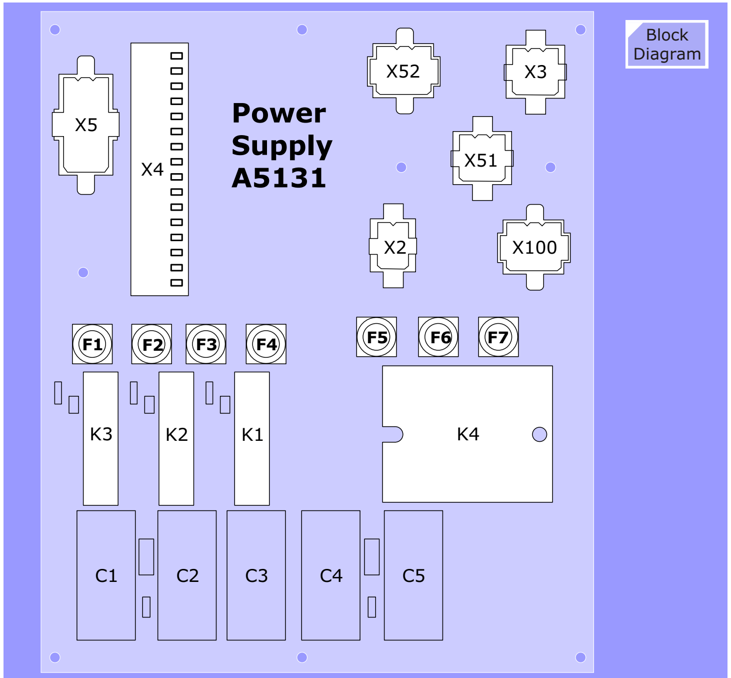
Transformer Interface

The A5131 provides the interconnects for the 230V AC mains from the LPD to the transformer. The transformer secondary voltages are fed back to the A5131 board and passed on to the Power Supply within the Control Unit. Some of these voltages are fused (see diagram below).

Fan Control

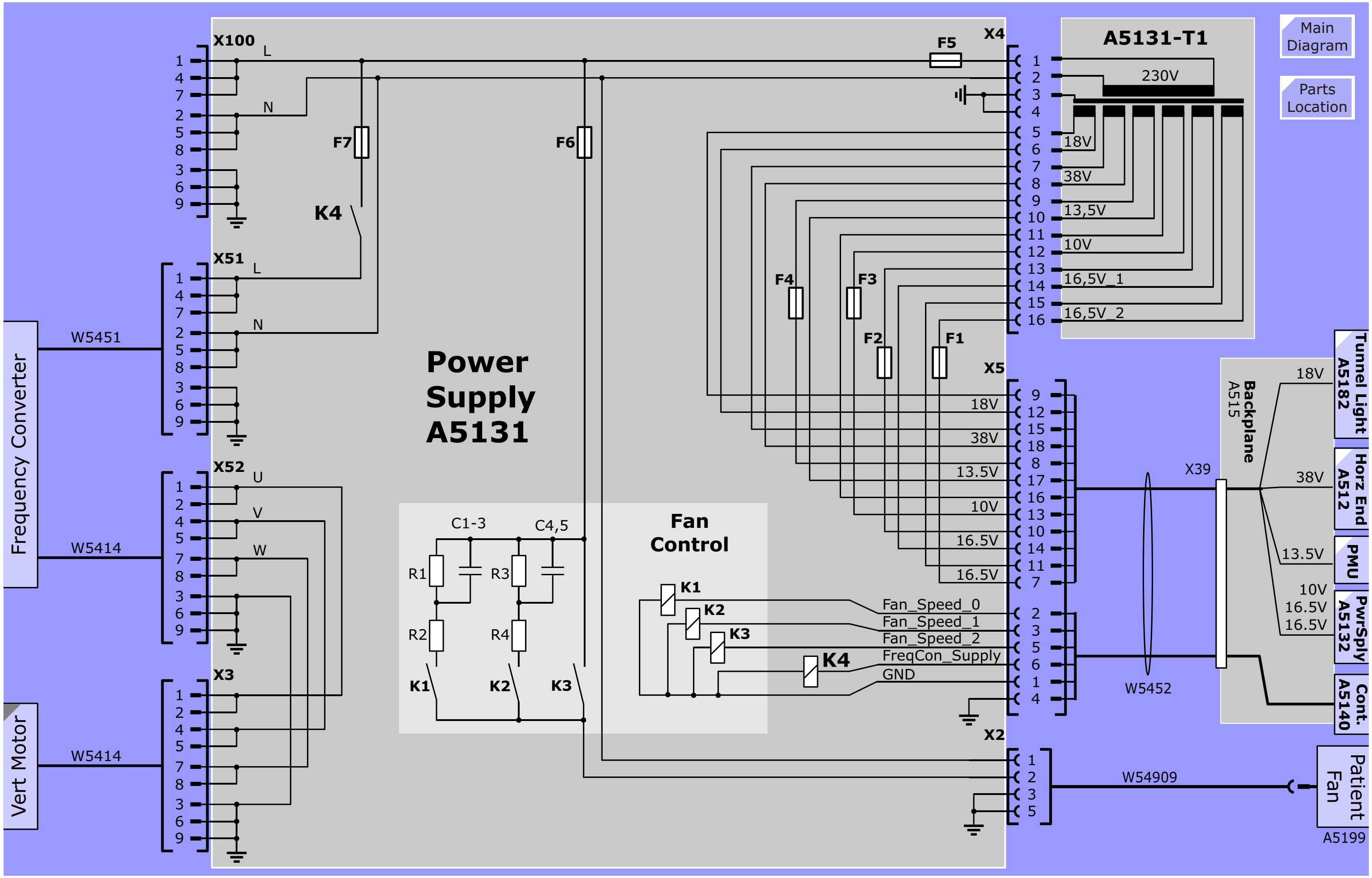
The fan speed is controlled by varying the applied voltage to the fan via relays K1-K3. The relays are activated by the Controller.

Figure 100 *Power Supply A5131 Layout*



X100 Mains input
X2 Patient Fan
X3 Vertical Motor
X4 transformer secondary
X5 connection to PTAB
X51 Freq Converter
X52 Freq Converter

Figure 101 *Power Supply A5131*



Power Supply A5132

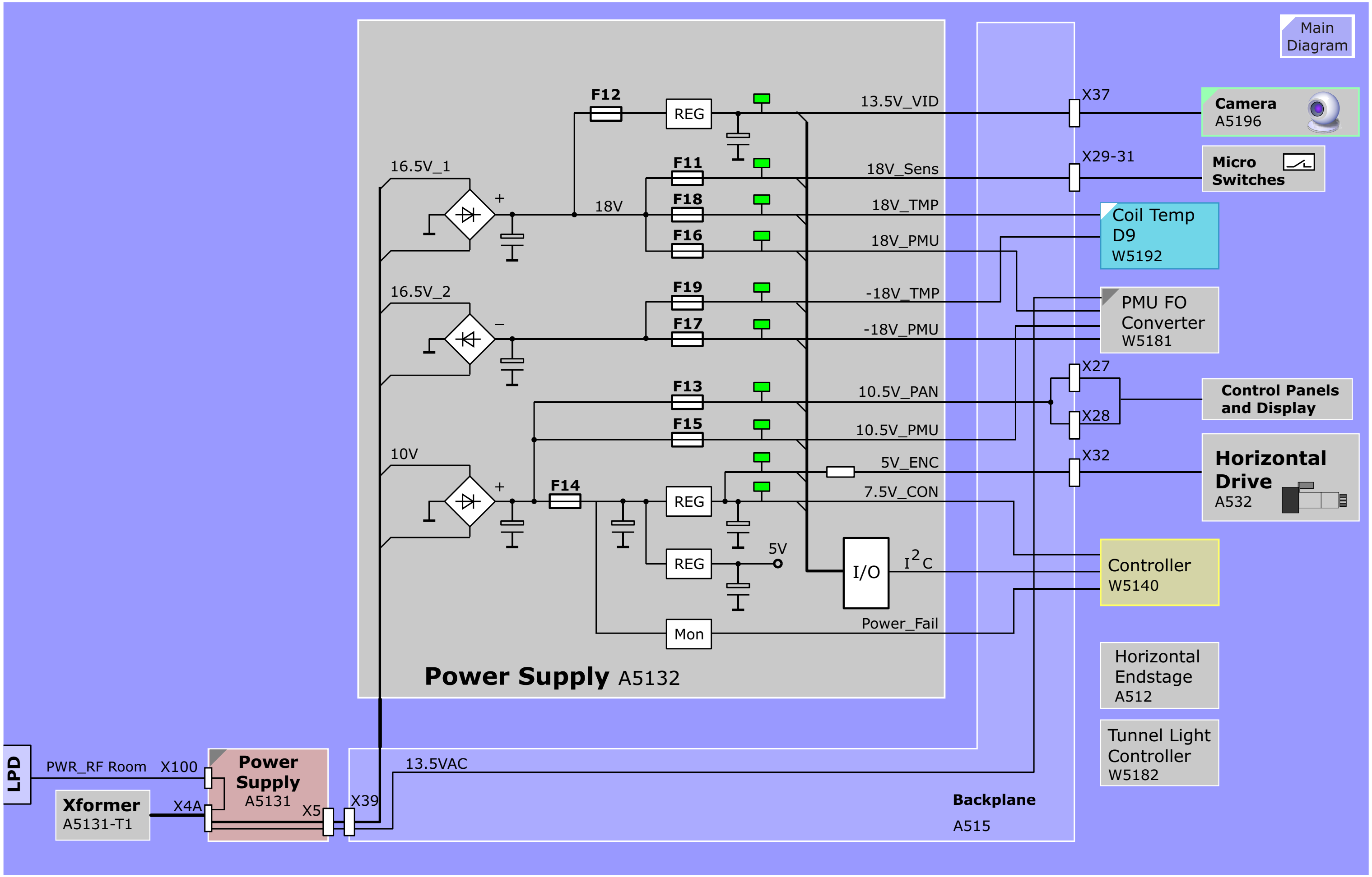
The Power Supply A5132, provides the various operating voltages required by the Control Unit, Intercom components on the magnet and PMU. See diagram below.

Function

Voltage monitoring

All voltages are monitored directly by the Controller. The controller can read the voltages in via the I2C bus. A seperate monitor is monitoring the +10V_DC voltage and will issue the Power_Fail error signal if this voltage fails.

Figure 102 *Power Supply A5132*



Tunnel Light

The Tunnel Light board provides the circuitry that enables the controller to vary the intensity of the tunnel lights.

Function

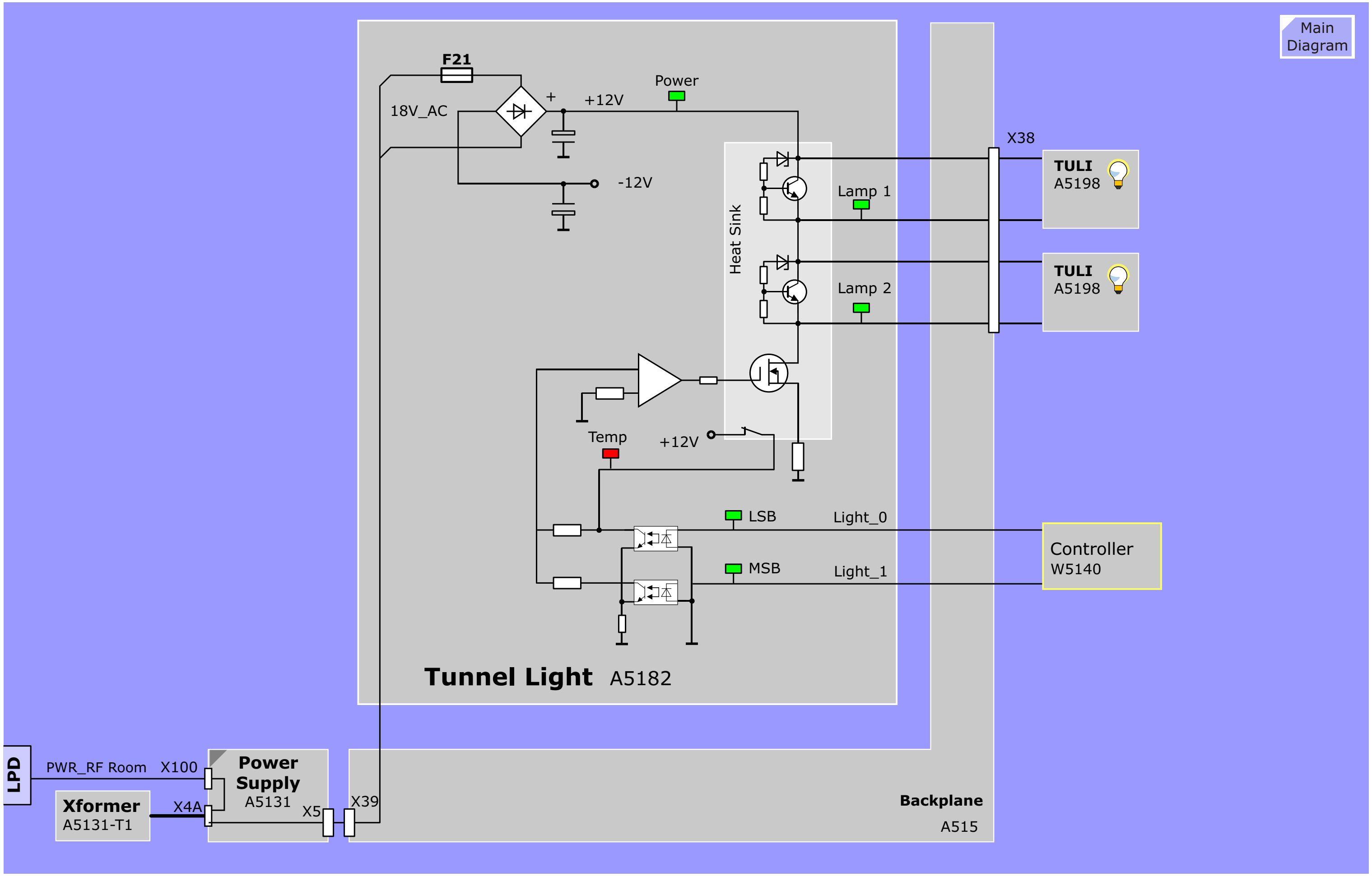
The tunnel lights are 12V halogen lamps. The voltage to the lamps is varied by the Controller with the signals Light_0 and Light_1:

Light Level	light_0	light_1
light off	0	0
LOW light level	1	0
MEDIUM light level	0	1
HIGH light level	1	1

Heat sink thermal switch

The driver elements are mounted on a heat sink the temperature of which is monitored by a thermal switch. If the heat sink temp opens the thermal switch, the 12V is removed from the OP-Amp and the lights are turned off. The TEMP LED will also light.

Figure 103 *Tunnel Light*



Control Panel and Display

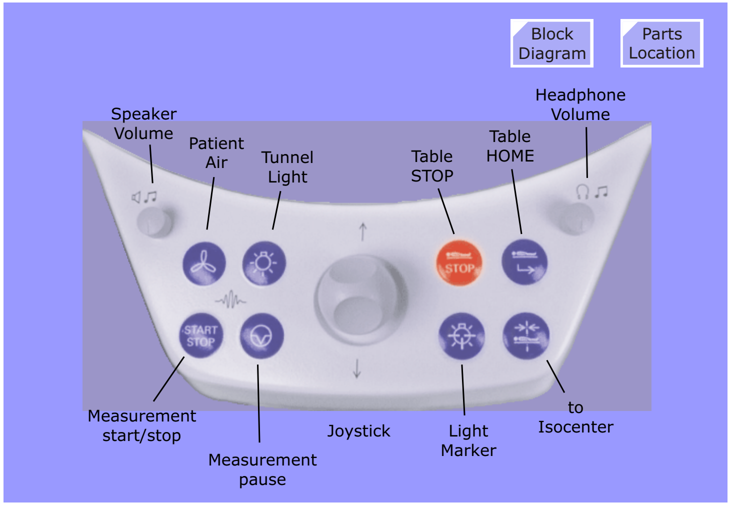
Interconnections

Up to three operating panels can be connected to the system. The number of connected Control Panels is recognized automatically during powerup of Patient Table electronics.

Front Panels PAN_0 and PAN_2 and Front Display DISPLAY_0 are connected in series to X27 at the Patient Handling Control Backplane. An optional Back Panel PAN_1 and DISPLAY_1 can be connected to X28.

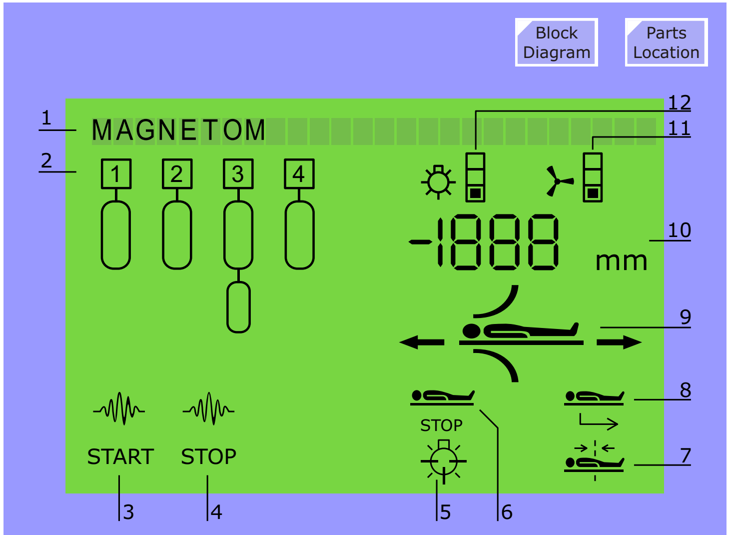
An Operating Panel can be removed from the HW-configuration (e.g. for troubleshooting). In that case the two free cable ends (connectors X1/female and X2/male) have to be connected together.

Figure 104 *Patient Handling Control Panel (front-left)*



NOTE Left buttons for patient convenience, right buttons for patient positioning.

Figure 105 *Patient Table Display*



1 Text field for displaying status messages
2 Coil socket assignments
3,4 Start and Stop
5 Status of laser light localizer
6 Table Stop
7 Center Position
8 Home Position
9 Current tabletop movement
10 Relative position of the tabletop
11 Tunnel ventilation
12 Tunnel lighting

Figure 106 *Operating Panel and Display interconnections*

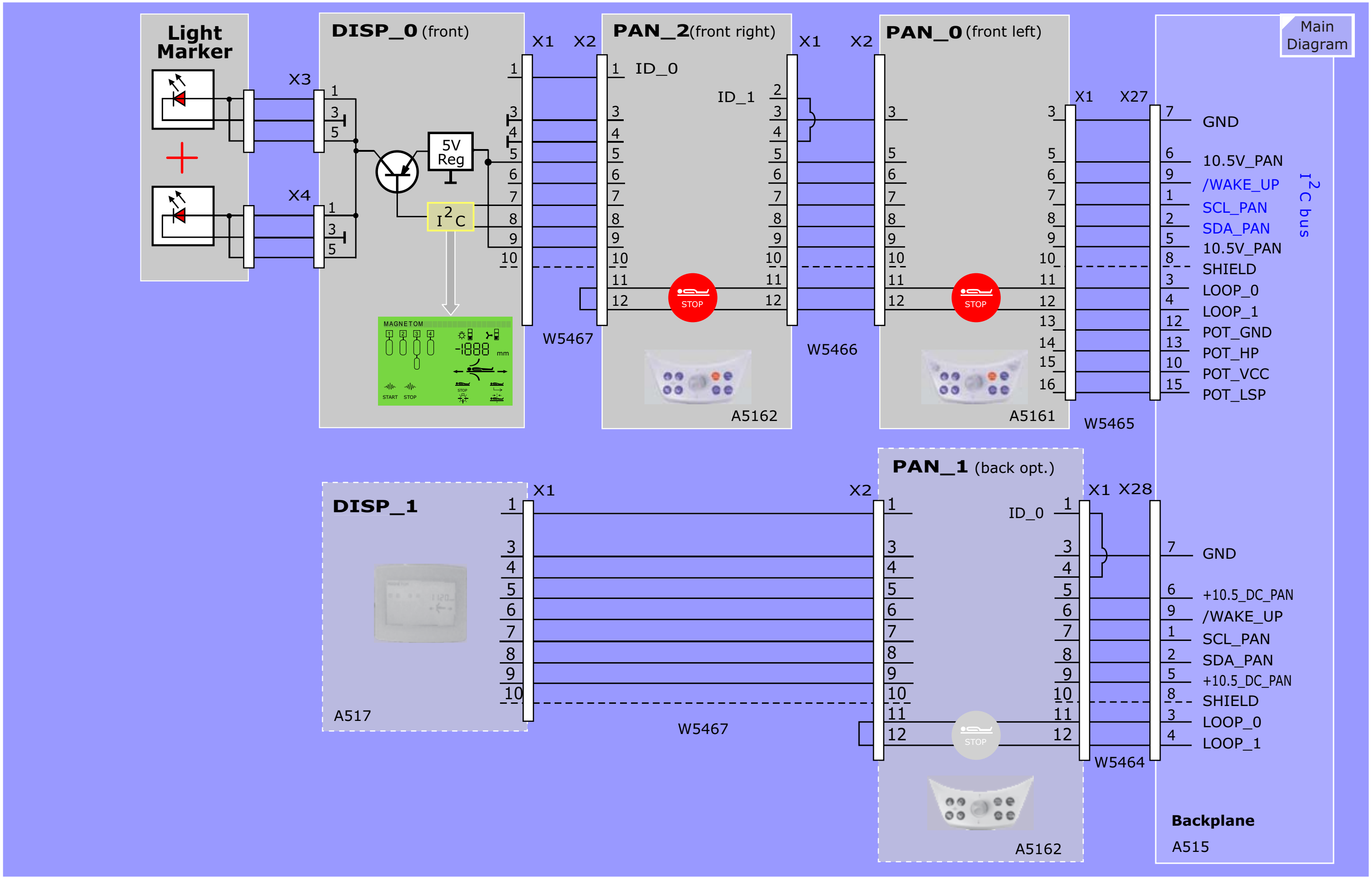
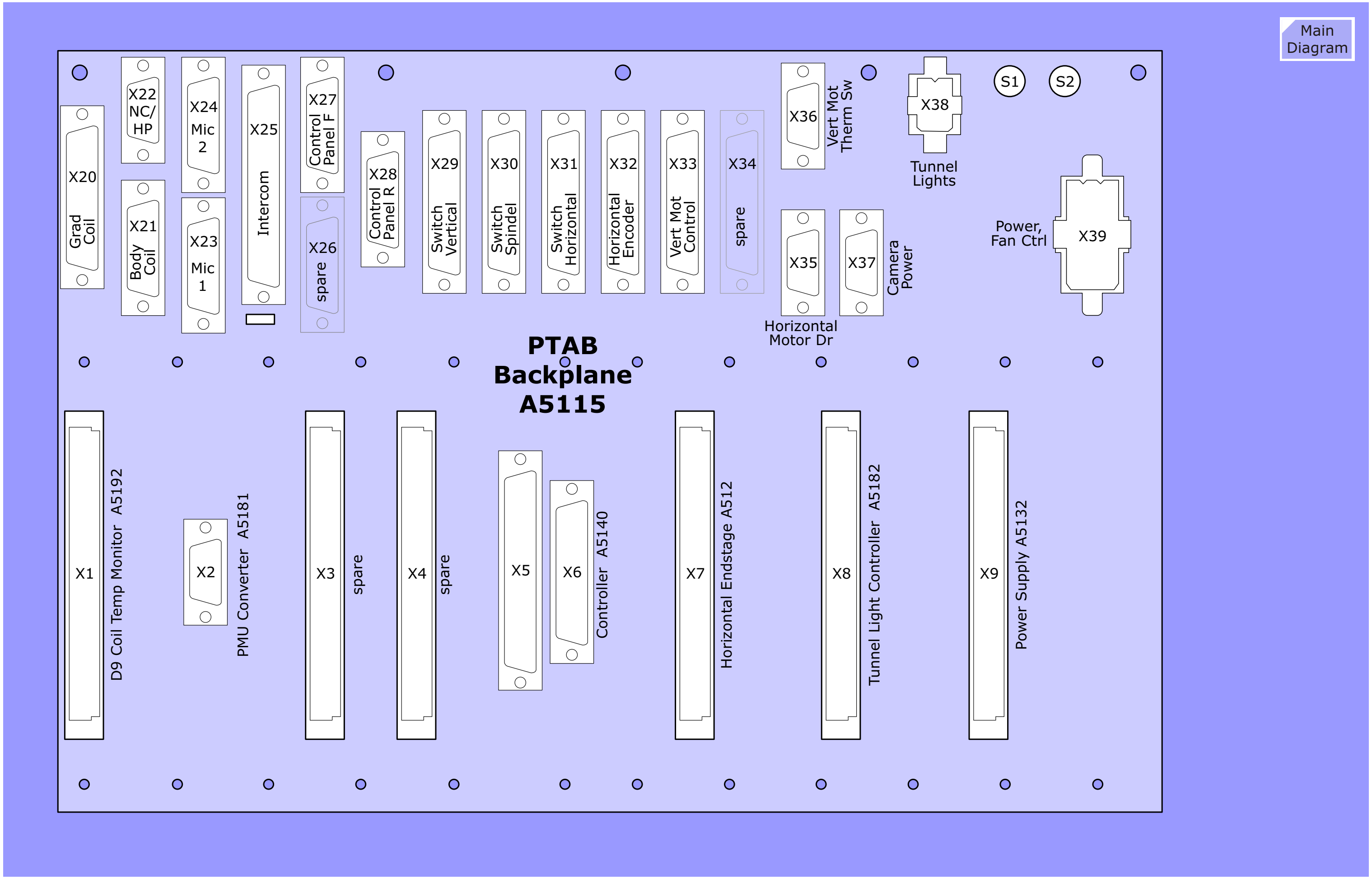


Figure 107 *Control Unit Backplane*



Intercom System

The Intercom system contains the following standard components:

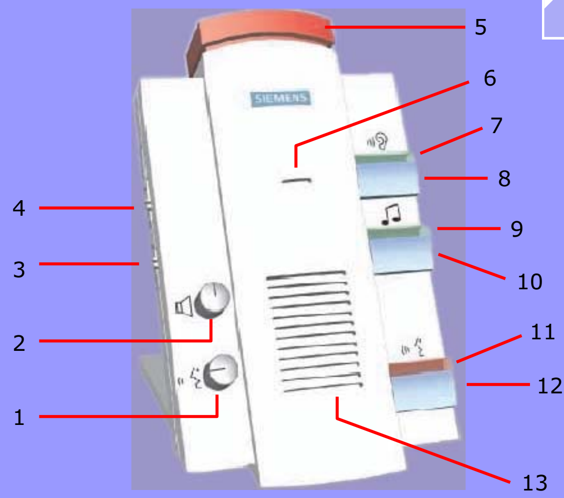
- **Central Unit Intercom K2201**. This box is mounted at rear of the operator's console. Here an audio source (CD player, MP3, etc) can be connected. R1 is a potentiometer to adjust the level of the speech from Control Room to RF cabin (Operator speakes) in order to avoid acoustic feedback when the door is opened, R2 is a potentiometer to adjust the level of speech from RF cabin to Control Room (Patient speaks, Noise of Coldhead, Gradient Noise) to avoid acoustic feedback when the door is opened.
- **Operating Unit Intercom.** Contains a red table stop button on top. Pressing this button once releases the table top, pressing twice will stop a running sequence. For the functions see graphics on the right.
- **Microphone board.** Located on front funnel cover.
- **Squeeze Bulb.** Pressing the bulb (also called hand ball) initiates an audible signal sounds on the Operating Unit Intercom. This is also called the Nurse Call.
- **Head Phone.** Via head phone the patient can hear operator's anouncements and music. The music is interrupted for anouncements.
- **Wall Loud Speaker**, mounted on magnet frame. Plays music and operator's anouncements.

Additional options are:

- **CCD Camera** at rear side of magnet. The image is transferred via fiber optic to the
- **Patient Monitor.** The monitor is adjustable, see description system manual.

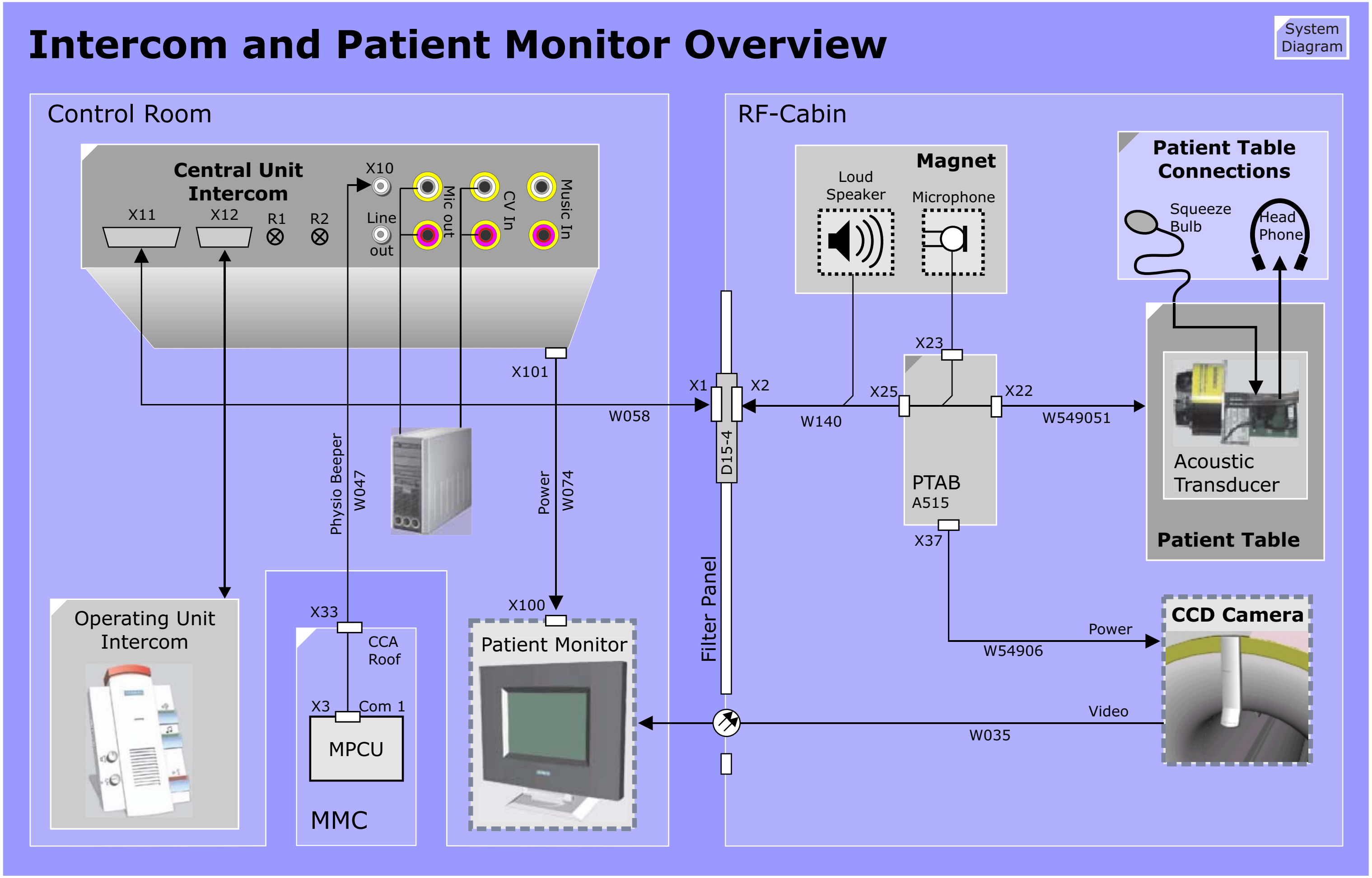
Figure 108 *Operating Unit Intercom.*

Back



1) Volume in RF cabin (speech only)
2) Volume console (listening)
3) CV/CBT switch (Computer Voice or background music)
4) Physio beeper volume trigger (4 levels)
5) Table stop/sequence stop button
6) Microphone
7) LED listening is on
8) Button listening on/off
9) LED music on
10) Button play music in cabin on/off
11) LED squeeze bulb active
12) Speak button and squeeze bulb reset button
13) Loudspeaker

Figure 109 *Intercom Overview Diagram*

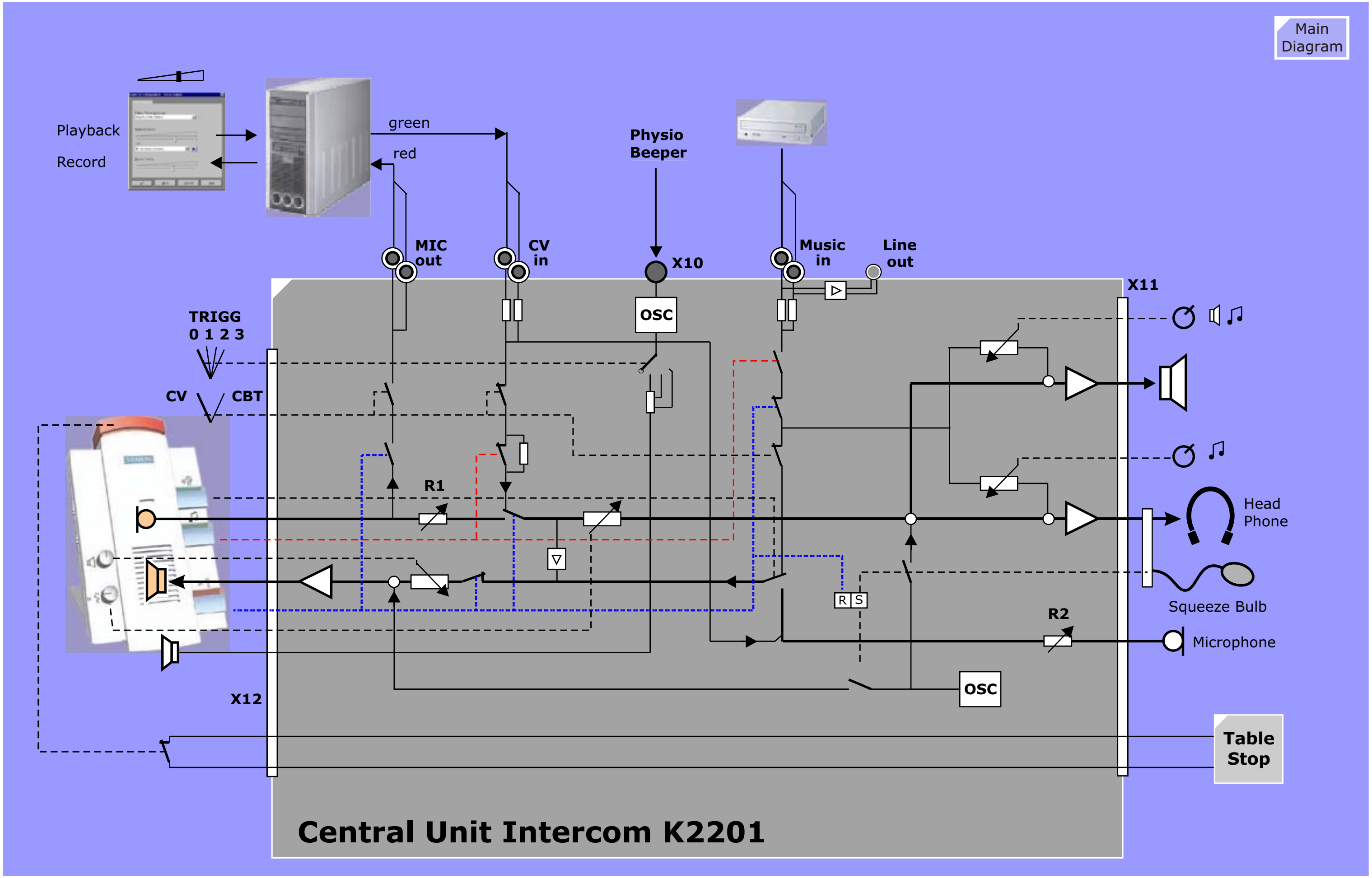


Central Unit Intercom 7.5.3

Figure 110 shows a simplified logic. The real logic is more complex and realized in ICs, so it is not possible to measure any "switch".

Anyway the drawing shows more details than described above or in the Operating Instruction.

Figure 110 *Central Unit Intercom Overview Diagram*



PMU

Overview

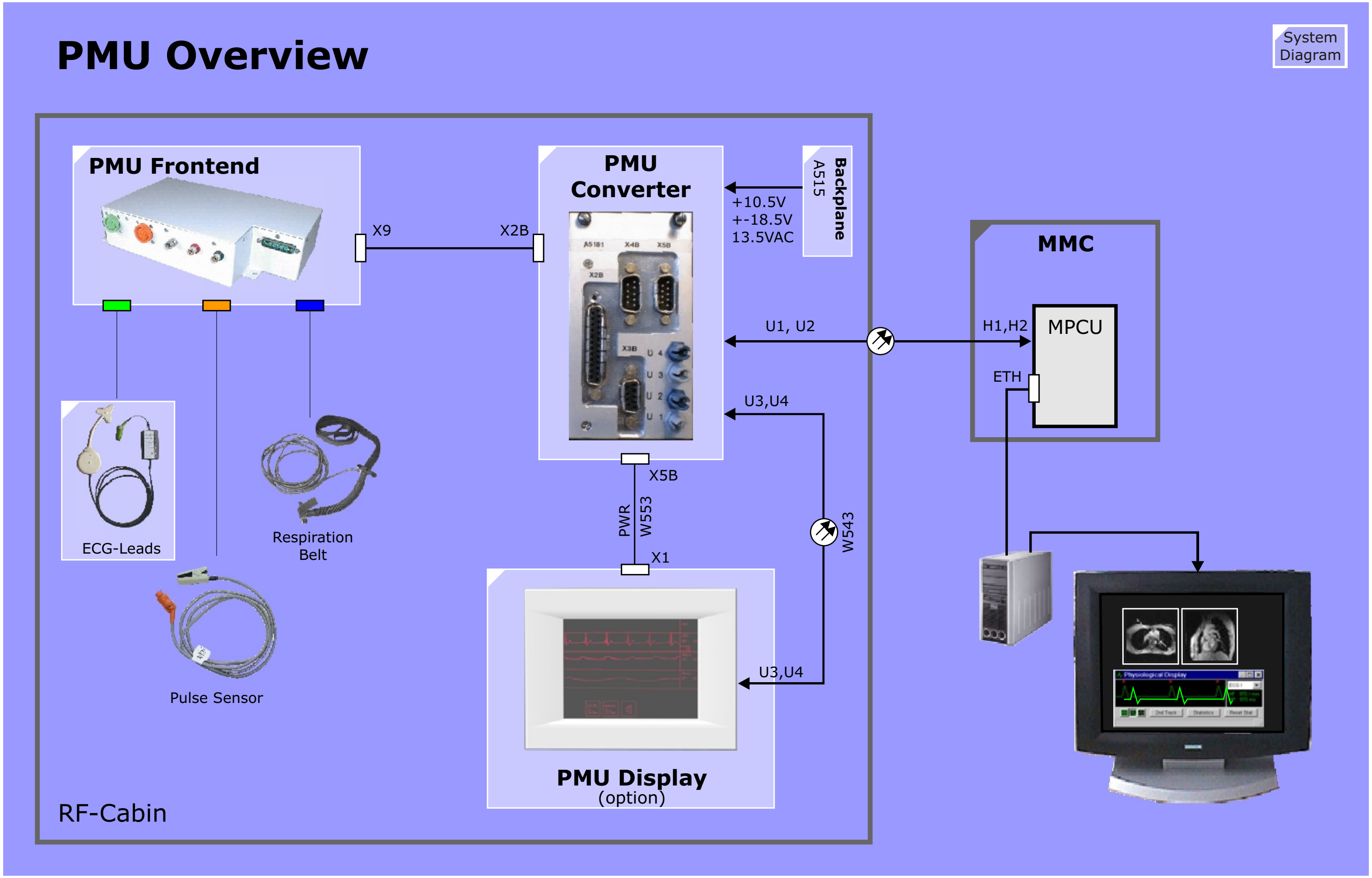
Physiological monitoring of the patients heart rate, ecg signals and respiratory condition can be used for either general purpose monitoring and/or sequence control. Some sequences require physiological triggering to allow imaging of organ dynamics, primarily of the heart, or to avoid movement artefacts caused by breathing. The PMU option provides hardware for EKG, pulse and respritory triggering.

The PMU option consists of these parts:

- PMU Frontend - a single unit including all amplifiers, acquisition and signal processing hardware)
- ECG, pulse and respiratory sensors
- PMU Converter - electrical-to-optical signal converters and interconnects
- In-room PMU display monitor (a separate option)

The PMU Frontend also provides an External Trigger Input for use with external PMU units and an External Trigger Output for fMRI.

Figure 111 PMU Overview



PMU Frontend

The ECG-signals measured at the patient are amplified and filtered in the PMU Frontend. The pulse-signals and the respiratory-signals are preprocessed in special analogue units. The galvanic isolation happens in the Frontend. After A/D-Conversion the signals are processed by a DSP. There is an EPROM in the PMU Frontend containing the Loadware. The current Frontend Loadware is downloaded from the MPCU upon startup, if necessary.

The 24-wire-connection between Frontend and Converter is for signals, communication and the 15Volts for the Frontend. In the case of the "Standard Patient Table" this is a one piece 6m long cable, in case of the removable patient table an intermediate connector (ODU-Connector) is required.

The communication links provide for the transmission of the physiological signals to the MPCU and the PMU Display, as well as for status messages and controls in the opposite direction (e.g. ECG lead selection).

The PMU Converter board A5181 performs RS485-optical conversion and vice versa. From the Converter the signal are fed:

- via FOC to the MPCU-Box. The PMU software on the MPCU provides for the computation needed for the transmission of the signals to the host, where they are displayed. In addition the signals - as selected by the USER - are processed in the PMU Software for the physiological control of the measurement.
- via FOC to the PMU-Display. The Graphic Controller in the PMU-Display calculates the graphics which are displayed by means of a TFT display.

Further functions of the PMU-Display are:

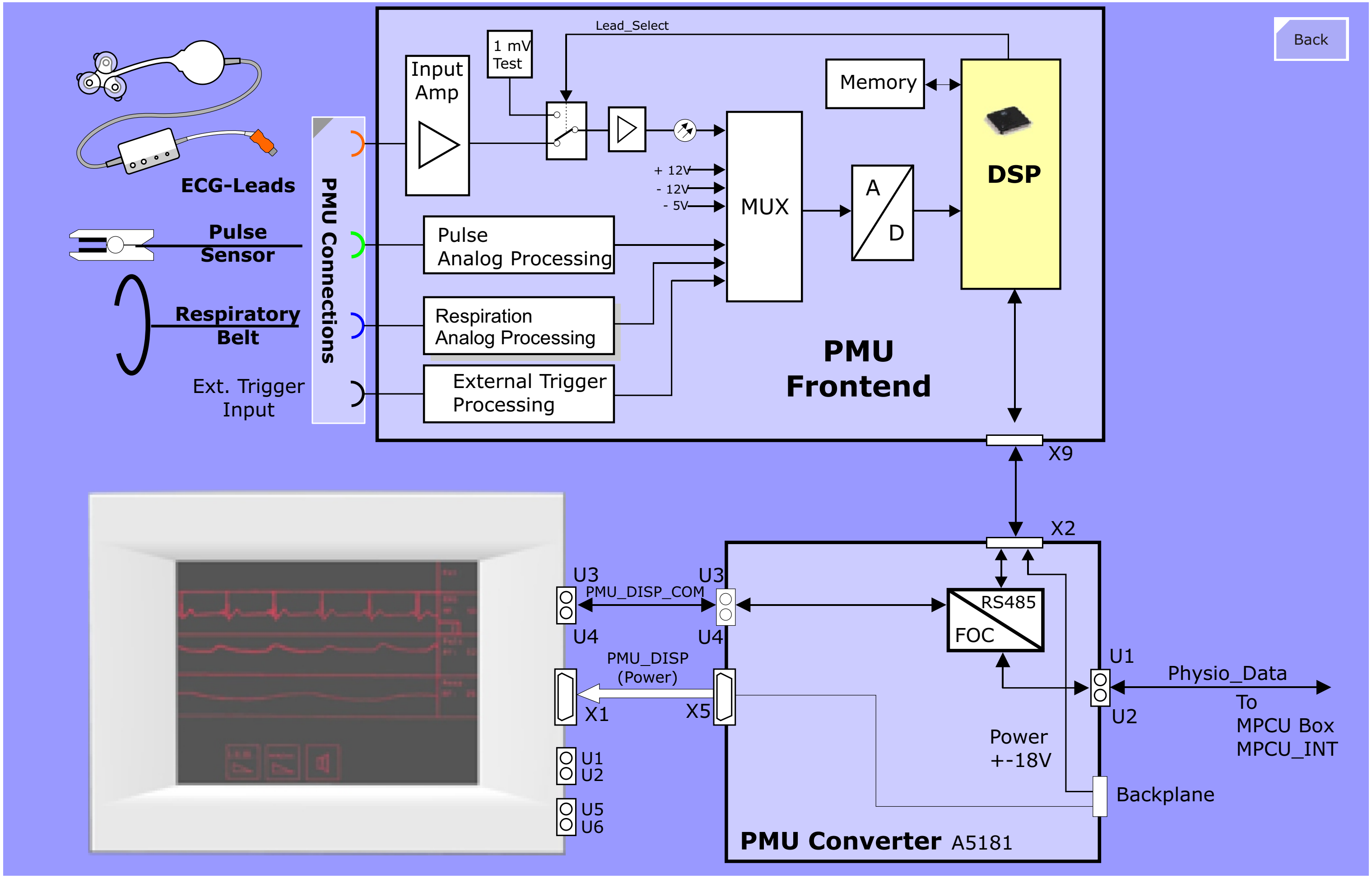
- Display of the curves and output of frequencies of ECG, pulse and respiration,
- selection of scroll speed,
- lead select facility. The different ECG-lead positions are: I, II, III, AVR, AVL, AVF or test signal,
- switching ON/OFF the beeper.

TESTS

- The PMU Frontend is equipped with a built-in test signal generator. It is used by the SeSo TestTools "Functional" test. This signal can also be used by sequences requiring an ecg trigger.
- There is a "hidden button" on the PMU display left of the control buttons for the User. A test signal can be activated by pressing the "hidden button" shortly and the "lead select" button afterwards. After the test press the "lead select" button again to switch to the default mode lead II.

NOTE The ECG-Test Phantom used in Vision and Impact/Expert cannot be used, because the connector does not fit

Figure 112 *PMU Block Diagram*



PMU Display (option)

The PMU Display is able to display the Physiological signal curves inside the examination room to facilitate the correct positioning of ecg electrodes to the patient. The PMU Display is mounted at the top-left side of the magnet front cover.

A 13.8 VAC supply voltage for the PMU Display is derived by connector X5 at the PDAU. The bidirectional Fiber-optic interface (FOC-connectors U3 and U4) supplies communication to the PDAU.

Four signals can be displayed:

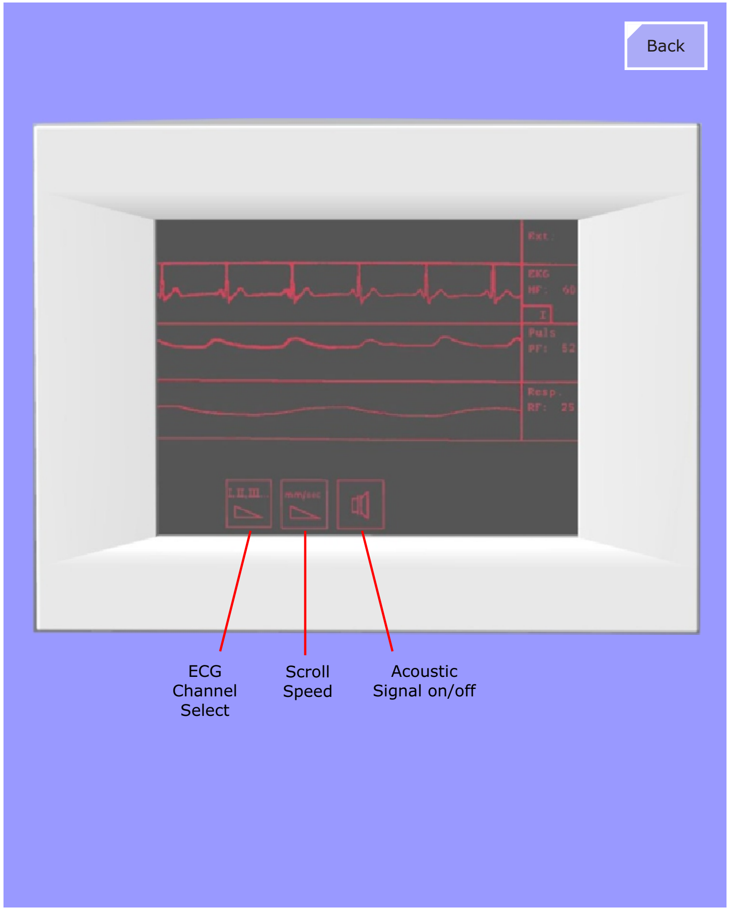
- ECG
- Pulse
- Respiratory signal
- External Triggering signal

ECG, Pulse and Respiratory signal are displayed simultaneously in real time curves, the External-Triggering signal is indicated as blinking "*" symbol for each low-to-high transition.

The PMU Display is equipped with three touchscreen buttons to adjust:

- ECG Channel Select (I or AVF for best R-wave signal)
- Scroll Speed (three speeds)
- Acoustic Signal (beeper activate/deactivate)

Figure 113 *Active ECG electrodes, Block Diagram*



Active ECG Electrodes

The acquisition and quality of the received ECG can be affected by interferences caused by gradient switching. The level of such interference depends on the amplitude and the slewrate of the applied gradient fields. These undesired interferences can cause additional peaks in the ECG and consequently incorrect triggering and poor image quality (blurring, ghosting). The use of active ECG electrodes can help overcome such effects.

Buttons:

Button 1: to switch on and off, the active ECG switches off automatically, if an electrode fault is detected for about 10 minutes.

Button 2: for lead selection. One push switches one ECG lead further. Only the leads I,II,III are available, compared to 6 leads with the standard cable.

ECG Amplifier Battery

The battery lifetime depends on the use. As a guideline, use of 15 hours per week results in about 2 years lifetime, a use of 60 hours per week in about 1 year lifetime.

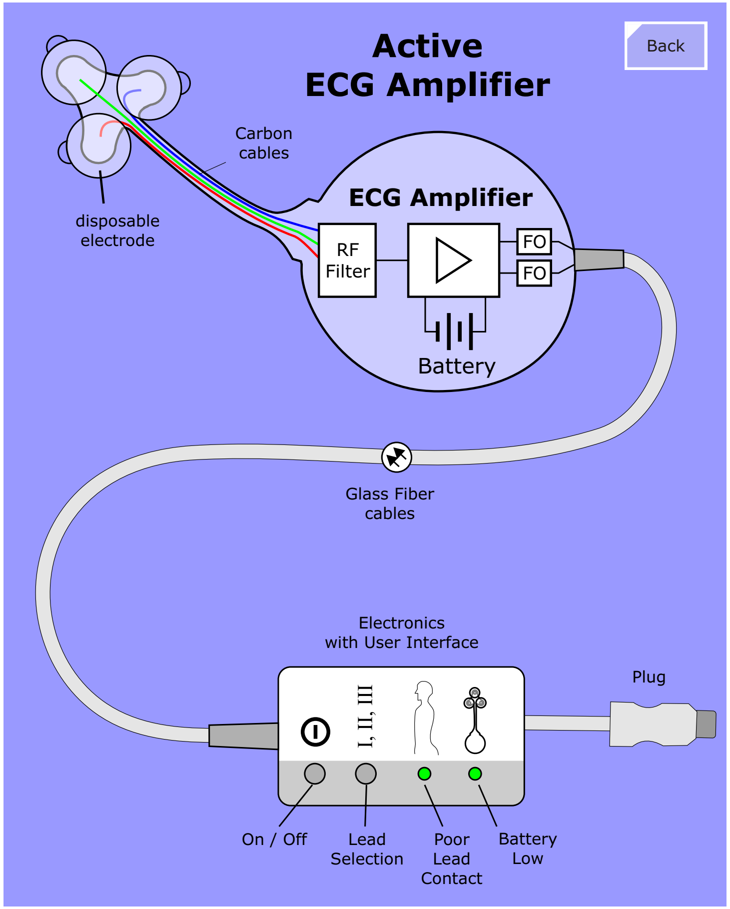
NOTE The Active ECG is an FRU. The battery can not be replaced, because the ECG amplifier box is sealed against liquids.

LEDs:

LED Electrode contact: lights up intermittently, if the electrodes have no proper contact to the patient's skin.

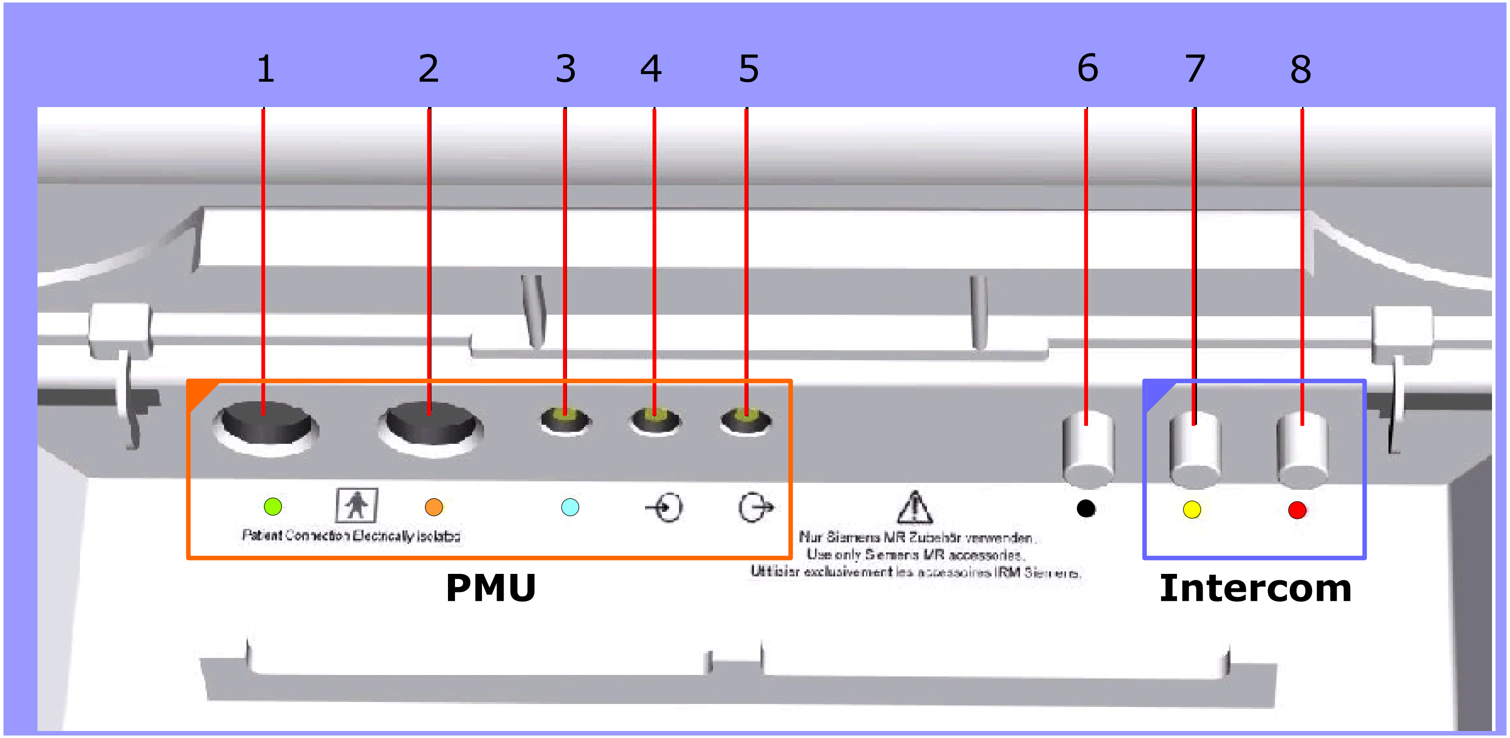
LED Battery of the amplifier: lights up intermittently when the battery is low (about 100 hours use left).

Figure 114 *Active ECG electrodes, Block Diagram*



PMU Sensor Conections

Figure 115 *Connections at the foot-end of Patient Table*



Pos. 1 ECG (green)

Pos. 2 Pulse Sensor (orange)

Pos. 3 Respiratory Belt (blue)

Pos. 4 Input for external trigger signal

Pos. 5 Output for external trigger signal (not supported)

Pos. 6 Vacuum pump (black)

Pos. 7 Headphones (yellow)

Pos. 8 Nurse Call / Squeeze bulb (red)

Section

8 Gradient System

The gradient systems of the MAGNETOM Harmony/Symphony/Sonata are based on switched mode current generators and actively shielded gradient coils. In order to achieve a compact design at the required high performance both components are water cooled.

The type of gradient system used depends on the MAGNETOM system, the NUMARIS software version (NUMARIS 3.5 or *syngo* MR) and the options bought by the customer. The following description covers the first generation of the Gradient System used in the MAGNETOM Harmony/Symphony (K2209) and the newer Cascade-gradient system (K2217) first used in the MAGNETOM Sonata, then in all new systems delivered from August 1999 onwards.

Introduction

This chapter serves as a general introduction to the hardware components of the gradient systems.

The majority of the Field Replaceable Units (FRU) in the gradient system are integrated into the left part of the twin cabinet called GPA.

Gradient System Performance

The performance and power of the gradient system is described by a variety of different parameters:

- Maximum gradient amplitude in $^{mT}/_{m}$
- Effective gradient amplitude $GRAD_{eff.} = GRAD_{max.} \cdot \sqrt{3}$
- Minimum rise time to maximum gradient amplitude
- Maximum voltage
- Slew rate SR

The SR value is a bottom-line performance quotient expressing the overall performance of both the gradient amplifier and gradient coil and is calculated from the maximum gradient field (coil) and the minimum time needed to reach this field (rise time as a factor of the applied voltage) by the formula:

$$SR = \frac{\text{max. gradient field}}{\text{min. rise time}}$$

For example, for a system with a 33mT/m coil and a GPA capable of ramp voltages needed for a 264 µs rise time, the SR would be:

$$SR = \frac{33\,\frac{mT}{m}}{0.264\text{ ms}} = 125\,\frac{\frac{mT}{m}}{ms} = 125\,\frac{\frac{T}{m}}{s}$$

Gradient Configurations

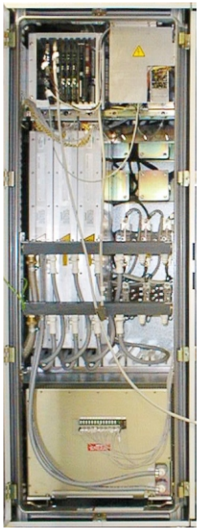
The table below gives an overview of the components, options and configurations and their performance values for the various gradient systems implemented in the MAGNETOM systems.

System	Gradient Coil	FoV	Gradient Amplifier	G_{max} in mT/m	G_{eff} in mT/m	I_{grad}	V_{max}	Option	Risetime	SR
Harmony Symphony	AS39S (old) AS39SR	500	K2209 see note	20	35	300 A	400 V	Turbo	800 µs	25
							800 V	Ultra	400 µs	50
			K2217 Cascade Light	20	35	300 A	500 V	Turbo	800 µs	25
							1000 V	Ultra	400 µs	50
Symphony only	AS39T	500	K2217 Cascade Quantum	20	35	300 A	1000 V	Ultra	400 µs	50
				30	52	380 A	1500 V	Sprint	400 µs	75
				30	52	380 A	2000 V	Quantum	300 µs	125 (Z) 100 (X,Y)
Sonata	AS85	400	K2217 Cascade Quantum	40	69	500 A	2000 V	Quantum	200 µs	200

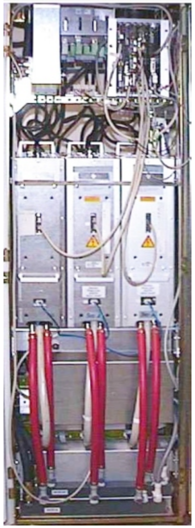
*** Notes**:

- There are two different K2209 Power Stages. The Papillon400 is a new gradient Power Stage replacing the older ones which are no longer in stock. This new power stage uses IGBT switching elements instead of FETs increasing its reliability. The newer power stages are pin-compatible and can be used with older versions. For the Ultra option, however, both power stages used in one axis must be of the same type.
- The Cascade Gradient Amplifier K2217 replaces the K2209 Gradient Amplifier. As can be seen in the table, this amplifier comes in two variants: Cascade light and Cascade Quantum.
- The Cascade Quantum Gradient Amplifier hardware is standard for all MAGNETOM Symphony and Sonata systems. The performance level options are realized in software (licensing and configuration).
- The AS39SR coil replaces the AS39S, but has the same specifications and performance characteristics.

Click below to jump to the Functional Description



K2209



K2217

Gradient System K2209

Introduction

The K2209 Gradient Amplifier is the first amplifier designed for the MAGNETOM Harmony and Symphony systems. It can be called the "classic" gradient system.

Specifications

	Turbo	Ultra Option
Max Gradient Field	20 mT/m	20mT/m
Max Current	300 A	300 A
Max Voltage	400 V	800 V
Min Rise Time	800 µs	400 µs
Slew Rate in T/m/s	25	50

Figure 116 *Gradient System K2209*

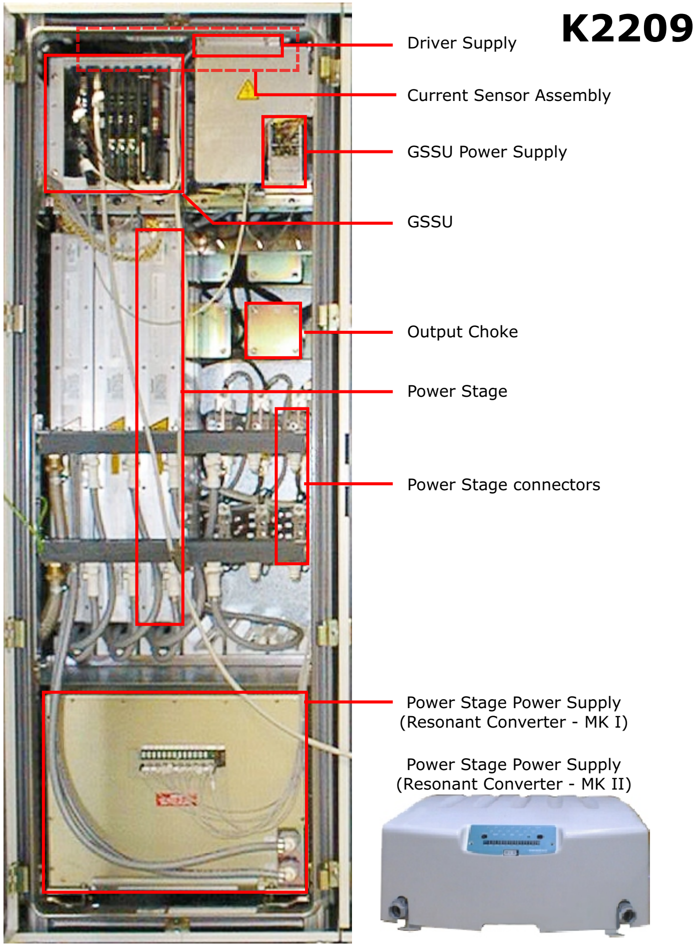
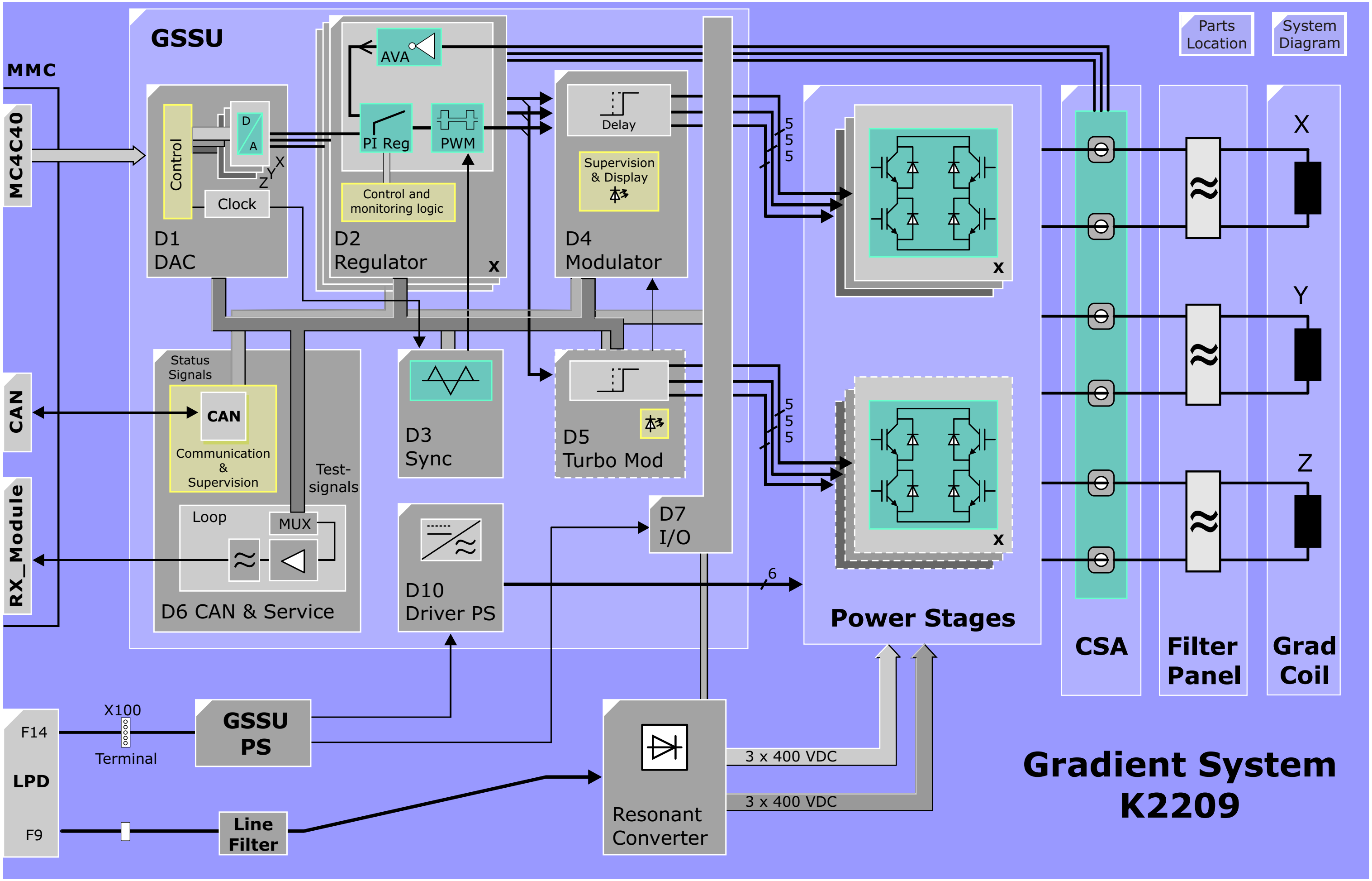


Figure 117 *K2209 Block Diagram*



Gradient Small Signal Unit (GSSU)

General

The GSSU is generating the pulse width modulated control signals for the switches in the Power Stage based on the digital data coming from the MMC. It's additional task is the overall supervision of the GPA.

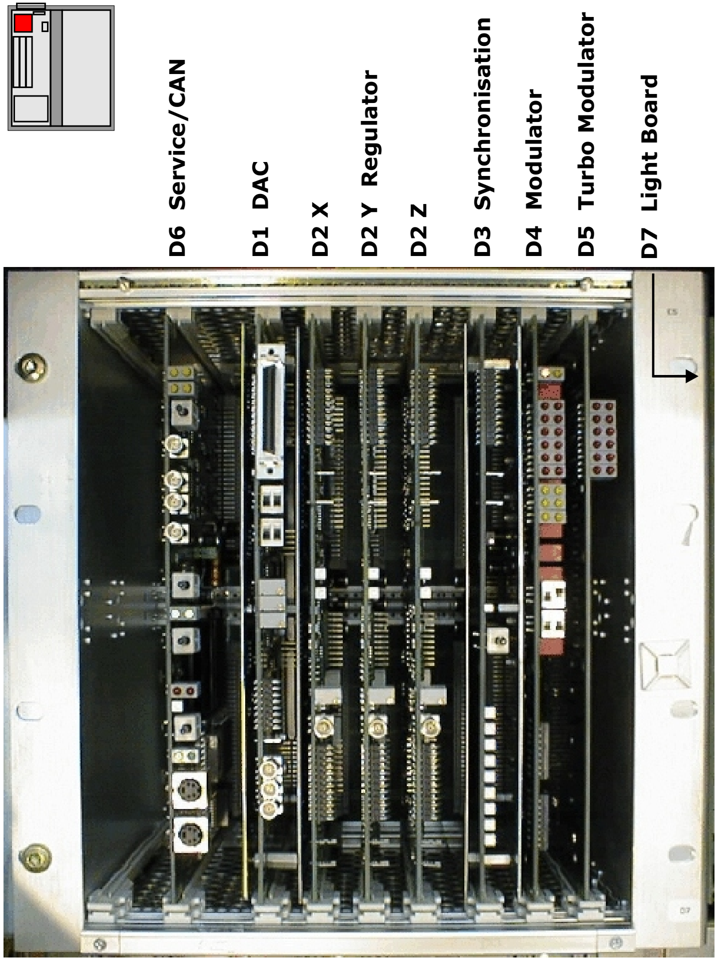
The PCBs or FRUs (field replaceable units) of the GSSU connected to the one common backplane (D8) are listed below:

- Service/CAN D6
- DAC D1
- Regulator D2 (3x)
- Synchronization D3
- Modulator D4
- Turbo-Modulator D5 (only Ultra-gradient option)
- Light board D7

Backplane D8

The backplane is the interconnection between the components of the GSSU and the rest of the gradient system. It is part of the GSSU-rack mounted to the cabinet frame by four quick release screws.

Figure 118 *GSSU Layout*



D1 DAC

The DAC-board contains the DA convertors for all three X, Y, and Z channels. The digital data supplied by the MMC contains the gradient pulse amplitude data including the eddy current pre-distortion and the gradient delay.

Control Logic

The 18 bit digital data **GRAD_D** and data latch signals **G_SEL_** from the MC4C40/GCTX to the Control Logic section are simply buffered and passed through to the DACs as long as the Test signal is not active.

If the **Test** signal is activated (MC4C40 test is SeSo) the data in the X DAC latch (**SU_GRAD**) is read back to the MC4C40. This is intended to check the data bus cable connection.

Nominal Value

The output voltage of the DACs (DAC_ X, Y and Z) is ± 10 V. It corresponds to ± 300 A driven trough the gradient coils.

Switches **S1** and **S2** are used to zero the DAC outputs to allow the adjusting the DAC offsets with potentiometers R153(X), R177(Y) and R200(Z). See TSG for the procedure.

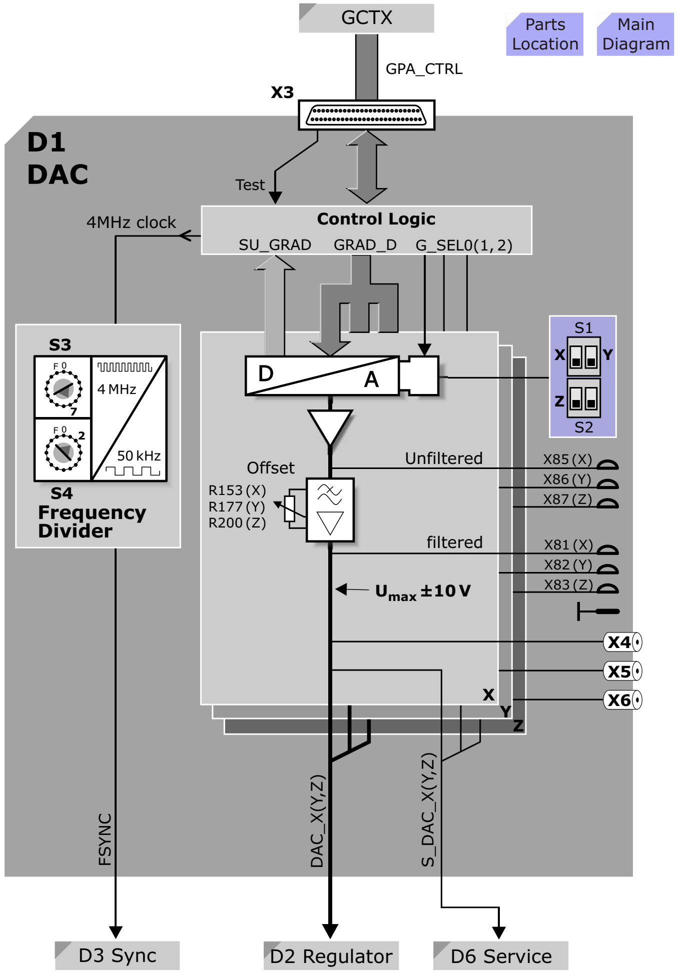
Monitoring

The DAC output signals (DAC_X, Y, Z) are available for monitoring via QLA-connectors (X4, X5 and X6) and test pins (X81-83; GND on X9) on the front panel of the board and they are also connected to the loop multiplexer on the Service/CAN board (S_DAC_X, Y, Z) for the automated gradient test loops in SeSo.

Frequency Divider

Also a 4 MHz clock signal is connected to the board via the same connector. This clock signal is divided by a frequency divider to selectable 25 kHz, 50 kHz (FSYNC) or 100 kHz. The division ratio can be adjusted by two rotary switches S3 and S4.

Figure 119 *D1 DAC Block Diagram*



Switches

Switches **S1** and **S2** are used to zero the DAC inputs and outputs for the offset adjustment. When set to the upper position, the Offset potentiometers **R153**(X), **R177**(Y) and **R200**(Z) are adjusted for **0V ±50µV** at connectors **X4**(X), **X5**(Y), **X6**(Z).

Switch	Description	Position
S1 left	DAC_X enable/disable	down/up
S1 right	DAC_Y enable/disable	down/up
S2 left	DAC_Z enable/disable	down/up
S2 right	not used	not used

Potentiometers

The static DAC offset can be adjusted with potentiometers. During the adjustment the corresponding DAC has to be disabled by using DIP-switches S1 and S2. See above.

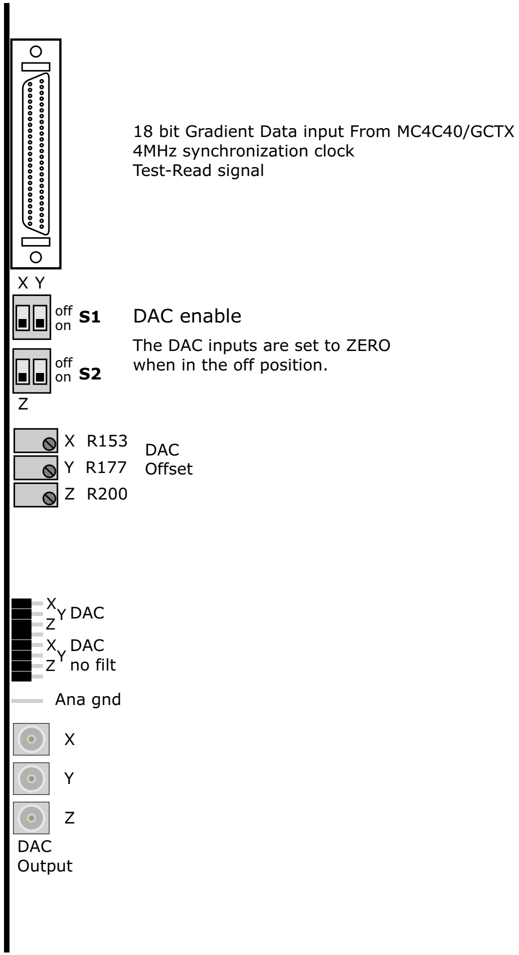
Potentiometer	Description
R 153	DAC_X offset adjustment
R 177	DAC_Y offset adjustment
R 200	DAC_Z offset adjustment

Test Points

The DAC output signals (DAC_X, Y, Z) are available for monitoring via QLA-connectors (X4, X5 and X6) and test pins **X81-X88** with ground connection **X9** at the front panel of the board.

Figure 120 *D1 Front View*

Block Diagram



D2 Regulator

General

The Regulator Board D2 is divided into three functional blocks:

- Actual Value Amplifier,
- P-I Regulator
- Modulator

Actual Value Amplifier

The current sensors in the Current Sensor Assembly deliver a compensation current proportional to the current driven through the gradient coil. This compensation current is converted into a corresponding voltage level by the Actual Value Amplifier. The output is scaled for **40A/V**

The offset of the amplifier is adjusted by potentiometer R191. See TSG for the procedure.

Soft-start

In order to prevent pre-magnetization of the current sensors the actual value amplifier is soft-started via photo-resistors to the sensors.

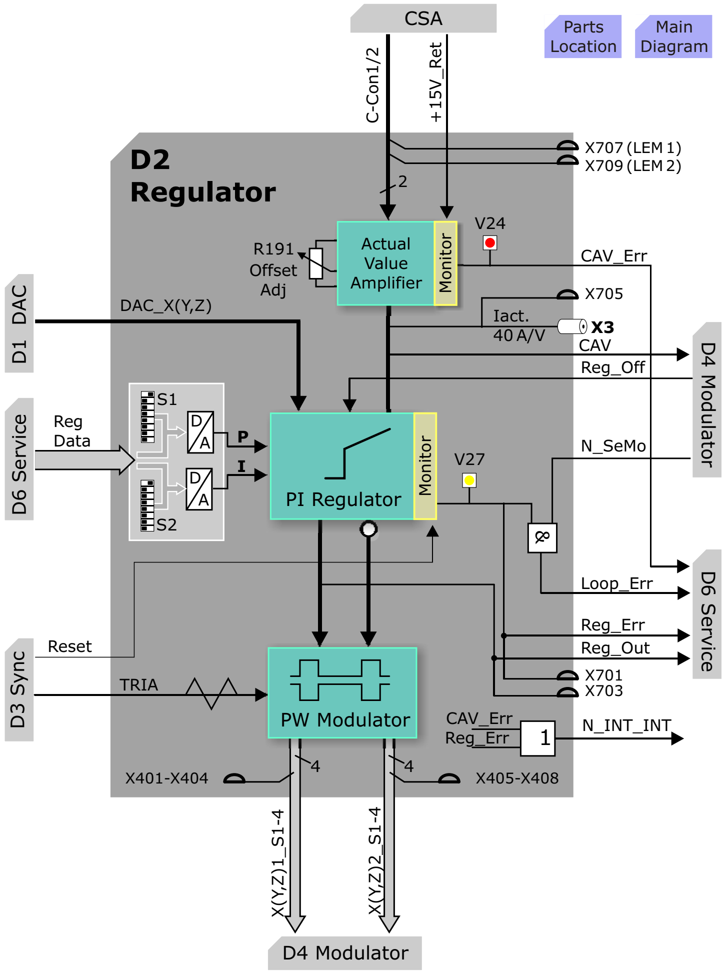
Monitoring

The AVA contains a soft-start circuit to run up the current sensors slowly. It also checks if the amplifier is connected to the sensors and the cable connection to the sensors is checked by an interlock signal (P15_RET). Also, the difference between the compensation currents of the two sensors is checked that it is not greater than 30A. If any of the above conditions are not met, a **CAV_ERR** is generated and LED **V24** is lit.

Loop Testing

The analog signal CAV is connected to the loop multiplexer on the Service/CAN board for automatic testing.

Figure 121 *D2 Regulator Block Diagram*



P-I Regulator

The closed loop regulation is based on a digitally controlled PI-regulator with 7 bit resolution for both the Proportional (P) and the Integral (I) part. The initial setting of the regulator after power on is determined by DIP-switches (**S1**: P-part; **S2**: I-part. The nominal setting, a result of the regulator adjustment procedure, is activated by software.

The regulator compares the actual current value to the nominal current value from the DAC and the resulting regulator difference is amplified by the regulator according to its PI-characteristic.

Monitoring

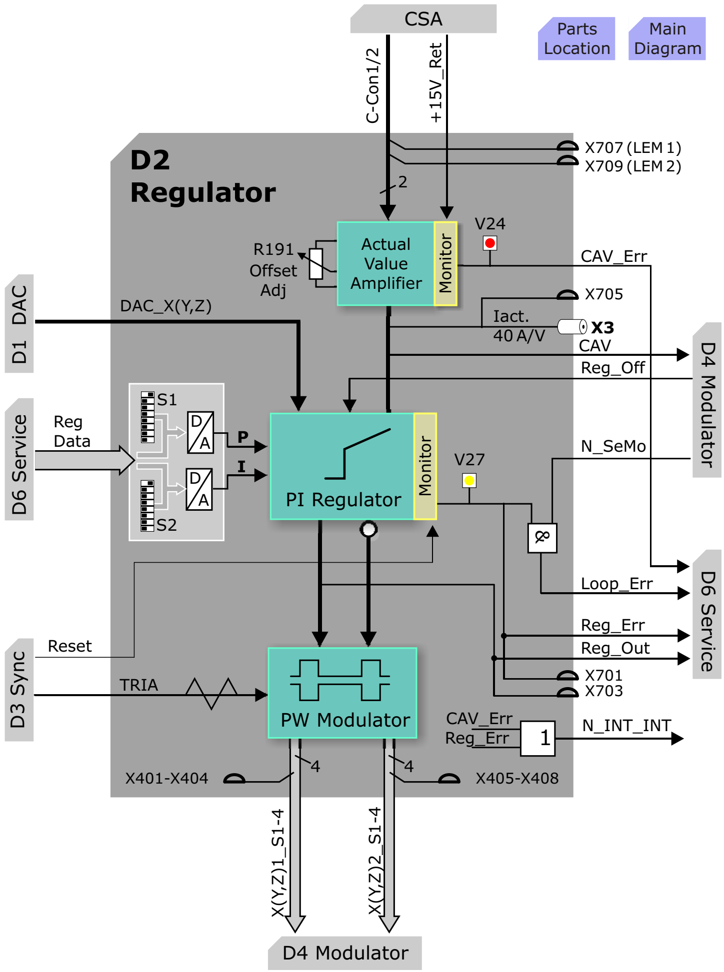
The difference between the nominal value from the DAC and the actual value and the output signal of the regulator (**Reg_Ou**t) are monitored by an error logic. If any of these signals are out of range a yellow LED (**V27**) on the front panel is on. This state (**Loop_Err**) is detected by the Service/CAN board (D6).
The analog outputs of the Regulator are connected to the loop multiplexer on the D6 Service/CAN board for testing, although this signal can not be selected in the SeSo test tools.

The **Reg_Off** signal generated by the supervision logic on the D4 Modulator disables the regulator by forcing its output to zero.

Pulse Width Modulator

The pulse width **Modulator** compares the output signal of the regulator to the TRIAngle voltage coming from the D3 Synchronization and generates two pairs of four switch control signals 1S1-4 and 2S1-4 for up to two Power Stages.

Figure 122 *D2 Regulator Block Diagram*



Switches

Switches S1 and S2 are used to set the default P and I regulator values and are always set to S1.1 ON, S1.2-8 OFF .

LEDs

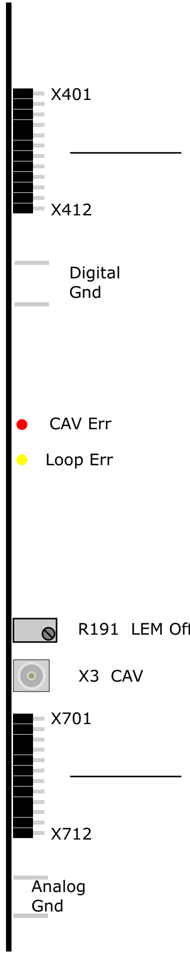
LED	Description
CAV Err	Indicates a current difference of more than 32A between the current measured by the two current sensors.
Loop Err	Indicates the difference between the Actual and Nominal inputs to the regulator or the regulator output level is too high. There is an OPEN loop!

Testpoints

Testpoints X401-X412 and X701-X712 are for internal (factory) use only and not intended for field service procedures.

Figure 123 *D2 Front View*

Block Diagram



X401	Bridge 1 - Switch 1
X402	Bridge 1 - Switch 2
X403	Bridge 1 - Switch 3
X404	Bridge 1 - Switch 4
X405	Bridge 2 - Switch 1
X406	Bridge 2 - Switch 2
X407	Bridge 2 - Switch 3
X408	Bridge 2 - Switch 4
X409	not used
X410	not used
X411	not used
X412	not used

X701	Regulator error signal
X702	no connection
X703	Regulator output
X704	no connection
X705	CAV (40V/A)
X706	no connection
X707	Current sensor 1
X708	no connection
X709	Current sensor 2
X710	no connection
X711	no connection
X712	Input : control test

D3 Synchronization

General

The D3 consists of two functional blocks:

- Triangle Generation and Synchronization
- Supervision.

The D3 generates a triangular signal required for the pulse width modulators on the three regulator boards. It also generates a rectangular synchronization signal for the driver supply circuit in the GSSU power supply. The above mentioned signals are phase locked to the system frequency.
In addition to the signal generation a multipurpose supervision is implemented on the board.

Triangle Generation and Synchronization

Triangular Signal

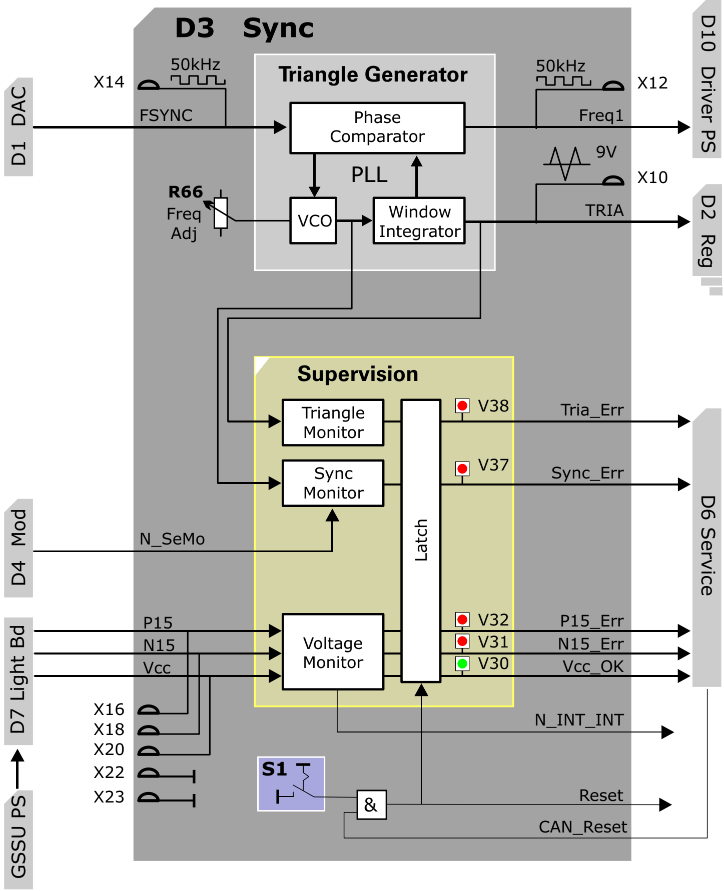
The triangular signal **TRIA** is generated by a free running Voltage Controlled Oscillator (VCO). The frequency of the signal is 50 kHz (phase locked to the system frequency) the amplitude is ± 9 V.
This signal is available for monitoring on the front panel test pin **X10**. The free running frequency of the VCO is adjustable by the potentiometer **R66**.
The synchronization signal (**FSYNC**) can be monitored on the front panel test pin **X14**.

Driver Supply Clock

A rectangular 50 kHz synchronization signal (**FREQ1**) is generated for the Driver Supply Board (D10 in the GSSU power supply).

Figure 124 *D3 Synchronization Block Diagram*

Parts Location
Main Diagram



Supervision

Triangular Signal

The frequency, amplitude and offset of the triangle signal is monitored. If any of the monitored parameters are out of tolerance LED **V38** will be turned on and the **TRIA_ERR** signal sent to the D6 Service/CAN board.

Synchronization

If the PLL is not locked it is detected by an error logic. In this case LED **V37** will come on and the **SYNC_ERR** sent to the D6 Service/CAN board. This error will be over-ridden if the Service Mode switch (**N_SeMo** signal from D4) is activated.

Supply Voltage Supervision

The supply voltages of the GSSU (+5 V and ±15 V) are monitored on this board. The state of the supervision logic is indicated by LEDs on the front panel. Two red LEDs (**V31,V32**) are indicating if the ±15 V is out of tolerance and one green LED (**V30** normally on) indicates if the +5 V supply is O.K. The state of the supervision logic is detected by the Service/CAN board. These voltages can be measured over test pins **X16, X18** and **X20**.

General Functions

Reset

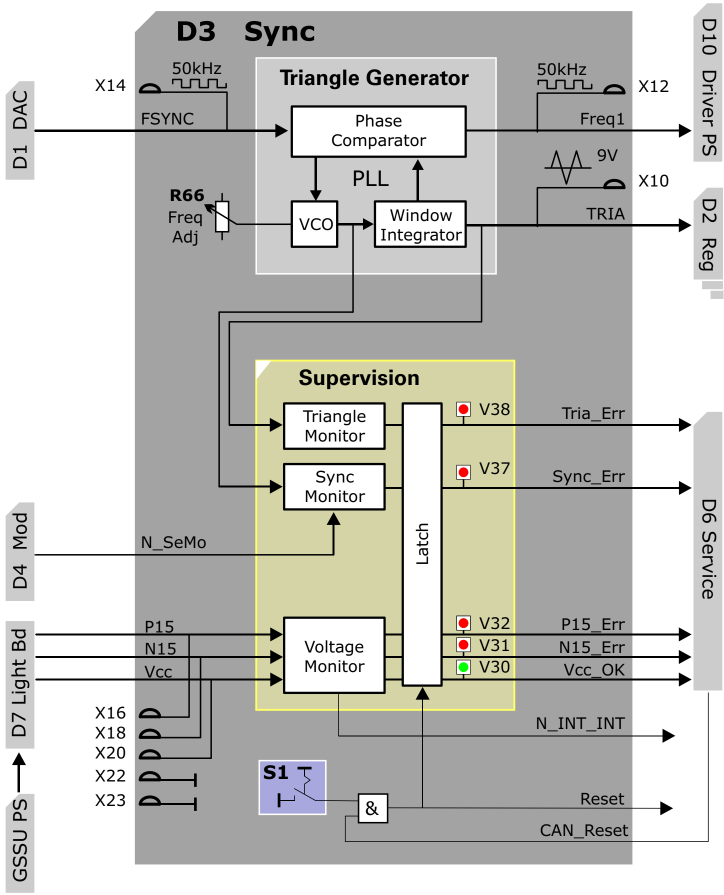
All error signals are latched. Push button **S1** on the front panel can be used to reset the error latches on the board. This reset function can also be activated by the Service/CAN board.

Internal Interrupt

All error states will activate the internal interrupt line (**N_INT_INT**) except Sync-Error in Service-Mode (see D4 Modulator description).

Figure 125 *D3 Synchronization Block Diagram*

Parts Location
Main Diagram



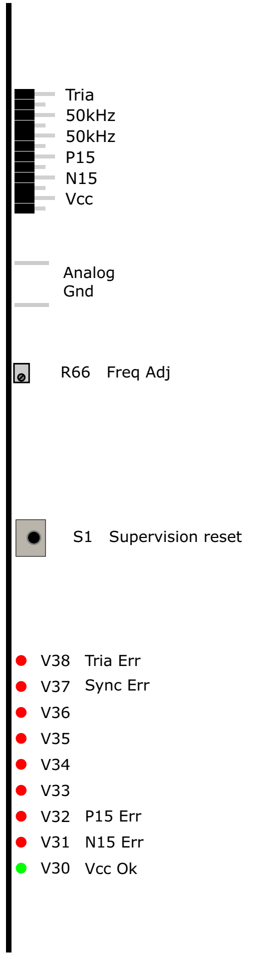
LEDs

LED	Description
Tria Err	On when Triangle signal not ok: amplitude and/or offset
Sync Err	On when the Sync signal is either missing (clock from MC4C40) or if the PLL cannot lock on (internal eror)
P15 Err	P15 power not ok when ON
N15 Err	N15 power not ok when ON
P5 Err	P5 power not ok when OFF

Potentiometer

R66 can be used to adjust the frequency of the Triangle signal. Se Replacement of Parts for details.

Figure 126 *D3 Front View*



Block Diagram

D4 Modulator

General

Two main functions are implemented on the Modulator Board D4:

- Dead-time delay generation
- Supervision of Power Stages (the first set)

The Modulator board is based on a programmable logic that contains the digital part of the pulse width modulator and the monitoring circuit of time critical power stage related signals.
The switch control signals from the three regulators passing through the board with an added dead-time delay and a double security disable function.
The supervision monitors the supply voltage and temperature of the power stages and limits the current generated by them to prevent damage of the hardware.

Modulator Function

Dead-time Delay

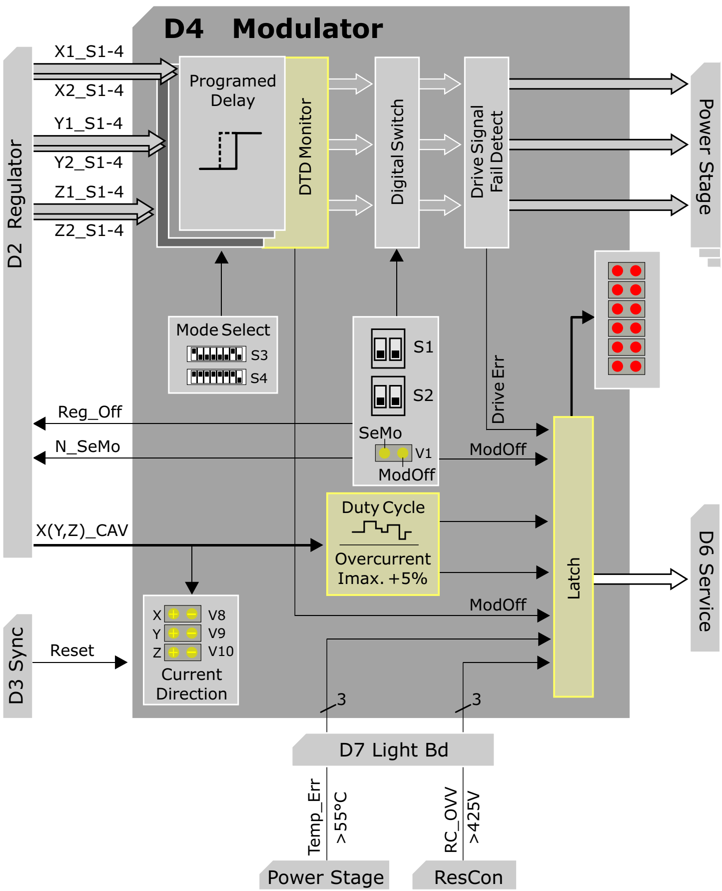
In order to prevent cross-current in the power stage bridge a delay is added to the switch-on signals. This delay (dead-time delay), generated by the programmable logic is selectable by a DIP-switch (sw1 and sw2 of S3).
The logic is also responsible to prevent firing the two switches in the same half of the bridge at the same time (cross-current). A second safety level based on a separate circuit is responsible to detect the cross-current state in case the prevention logic fails (DRIVE_ERR).

Current Indicator

The actual current signal (SU_CAV) from the regulator board reaches the current indicator detecting the current direction and correspondingly driving the yellow twin-LEDs V8, V9 and V10. The current indicator also generates signals for the supervision logic used to disable driver signals of the respective counter direction.

Figure 127 *D4 Modulator Block Diagram*

Parts Location
Main Diagram



Supervision and Control

The supervision and control based on the programmable logic allows a very high flexibility. The internal 32 MHz clock (MOD_CLK) of the supervision is generated on board and it is phase locked to the 4 MHz system clock.
The control section is responsible for generating the switch control signals if it is enabled by the supervision section. It is also generating signals supporting the Turbo Modulator (D5).
In the supervision section the power dissipation of the power stages is simulated by an electrical circuit. Additionally the supply voltage, the temperature of the power stages and the current generated by the those is monitored.

Status Signals

The following error states are detected and partially displayed by the supervision logic:

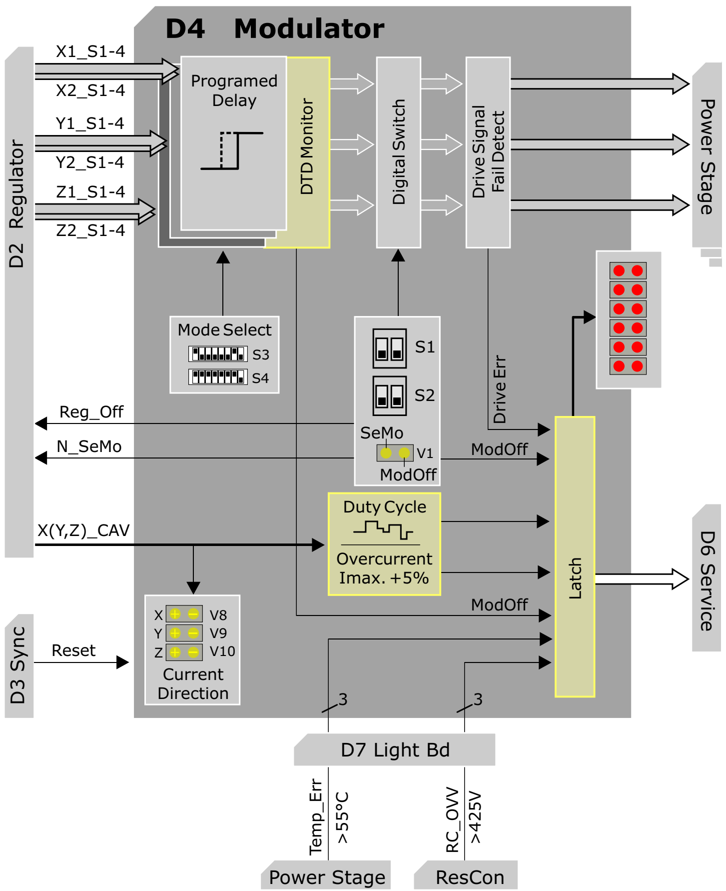
- Over-current (OVERCUR_ERR)
- Duty cycle error (DC_ERR)
- Temperature error (TEMP_ERR)
- Over-voltage (OVV_ERR)
- Driver error (DRIVE_ERR)

Beside the error states also the internal interrupt line (INT_INTERN) and front panel switches (S1_left, S2_right, S2_left) can disable the generation of switch control signals. These states are detected by the supervision logic, too.

Error Memory

The above mentioned seven states are stored locally on the modulator board for all three axis. The Service/CAN board is able to read these states sequentially using the ERR_X, ERR_Y, ERR_Z data and ERR_SEL1, ERR_SEL2, ERR_SEL3 control lines.

Figure 128 *D4 Modulator Block Diagram*



Service Mode

The supervision logic can be switched into service mode (SEMO) by a front panel switch (S1_right) and remotely from the Service/CAN board. The service mode is indicated by a yellow LED (V1 left) on the front panel and it is also detected by the Service/CAN board.

Internal Switches

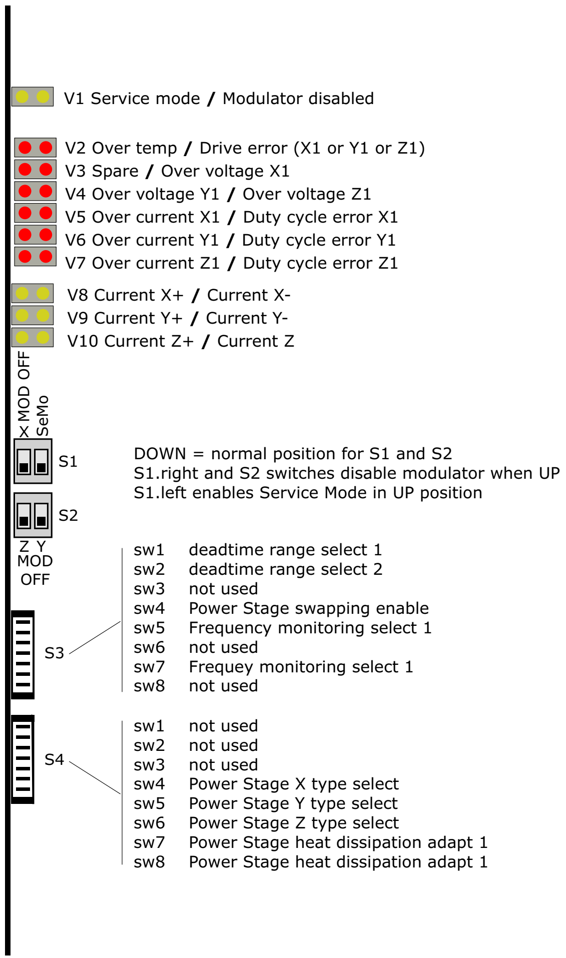
The setting of the two DIP-switches (S3 and S4) on the board determines the function of the supervision and control. The following two tables are a brief explanation of these functions.

Switches

Switch	Function	Explanation
S3		
sw1	dead time 1	Dead time range select switches
sw2	dead time 2	
sw3	mode 1	not used
sw4	mode 3	Power Stage swapping enable
sw5	fqa	Frequency monitoring select switch (with fqb)
sw6	mode 2	not used
sw7	fqb	see sw5
sw8	mode 4	not used
S4		
sw1	modsemo x	Local service mode select switches (not used)
sw2	modsemo y	
sw3	modsemo z	
sw4	sunuse x	Power Stage switch device type select switches
sw5	sunuse y	
sw6	sunuse z	
sw7	swloss 1	Power Stage switch heat dissipation adapt switches
sw8	swloss 2	

Figure 129 *D4 Front View*

Block Diagram



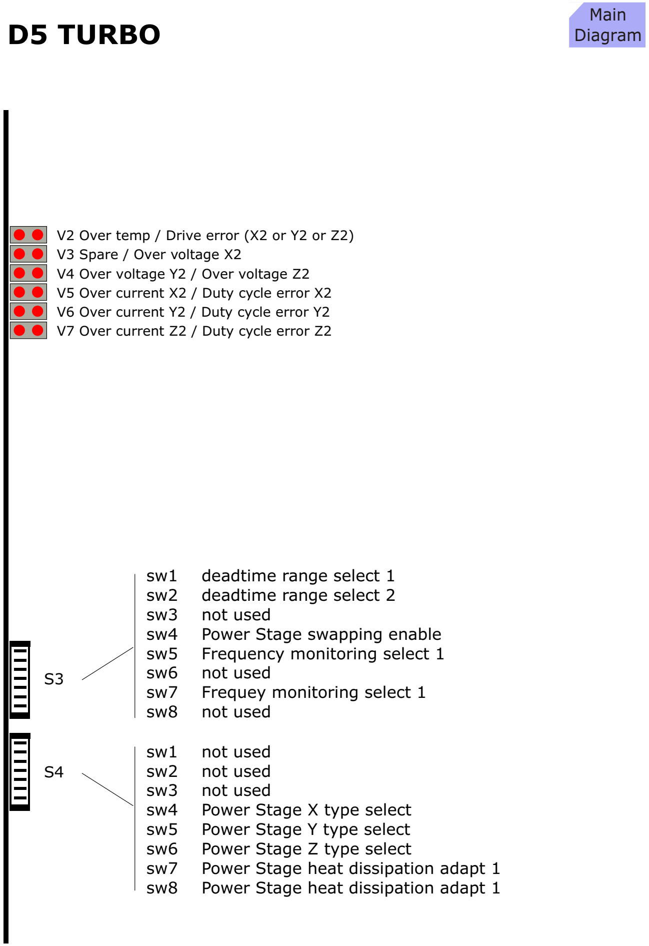
D5 Turbo Modulator

Function

The D5 Turbo Modulator is a slave of the D4 Modulator and is responsible for performing the same functionality for the Power Stages of the ULTRA option.

NOTE The Turbo-Modulator D5 is only necessary for gradient systems with the Ultra-option. It is not installed for the Turbo option!

Figure 130 *D5 Turbo-Modulator*



D6 Service/CAN

General

The Service/CAN Board consists of two functional blocks:

- the communication and supervision (CAN-Module) and
- the analog signal multiplexer.

Function

The CAN-Module provides communication with the MMC. Control data for the GPA is transferred from the MMC and GPA-status information to the MMC.

The Loop section allows feedback of analog signals from the GPA to be captured by an ADC in the RF Receiver for evaluation by test and tune_up procedures. This section is used only by the SeSo TestTools for the GPA. The Ext_In is not supported by software.

CAN-Module

Communication

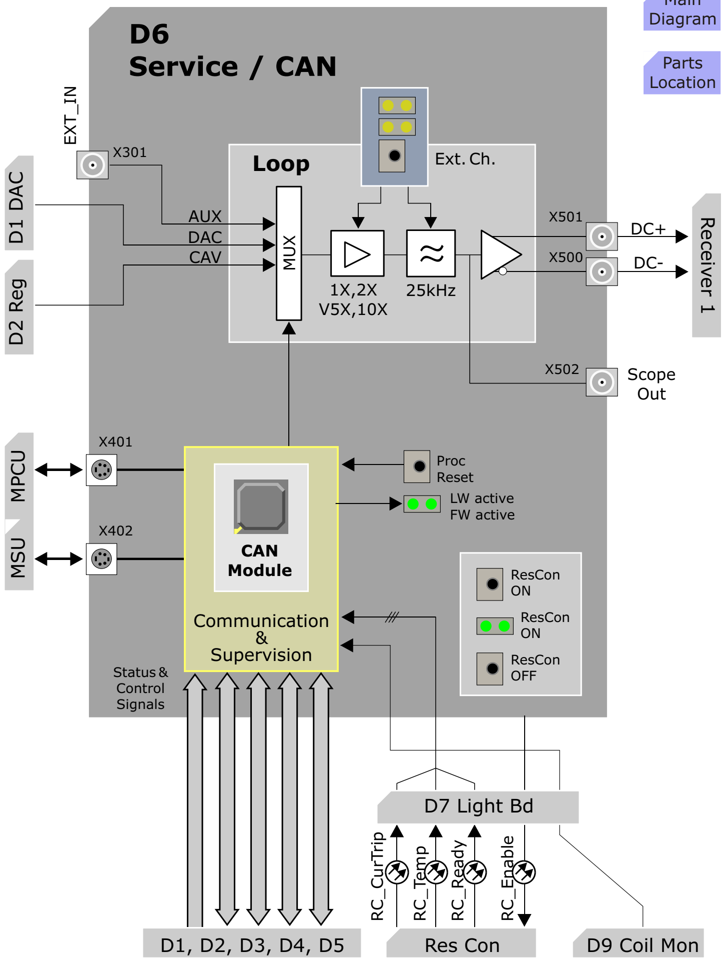
The firmware (FW) stored in a PROM allows the processor to run a self-test to communicate on the CAN-Bus and to load the application software. The application software (LW) for the processor is stored in an EEPROM. According to this program the processor is able to receive control data from the MMC and transfer status information from the GPA to the MMC.

Reset

The front panel push button S404 is used to reset and re-start (warm-start) the Processor.

Figure 131 *D6 Service/CAN*

Main Diagram
Parts Location



ResCon Control

ResCon on/off

The front panel push button S402 is used to switch on the Resonant Converter; button S403 is used to switch it to standby. The switched on state is indicated with the green twin-LED V407.

Power -Fail message

The + 5 V supply of the CAN-Module is buffered at + 15 V level. If the + 15 V fails it is detected by an error logic and the buffered + 5 V is still available long enough to send a Power-Fail message to the CAN-System. This message is processed with the highest priority. The Power-Fail state is indicated by the red LED V607.

Analog Signal Multiplexer

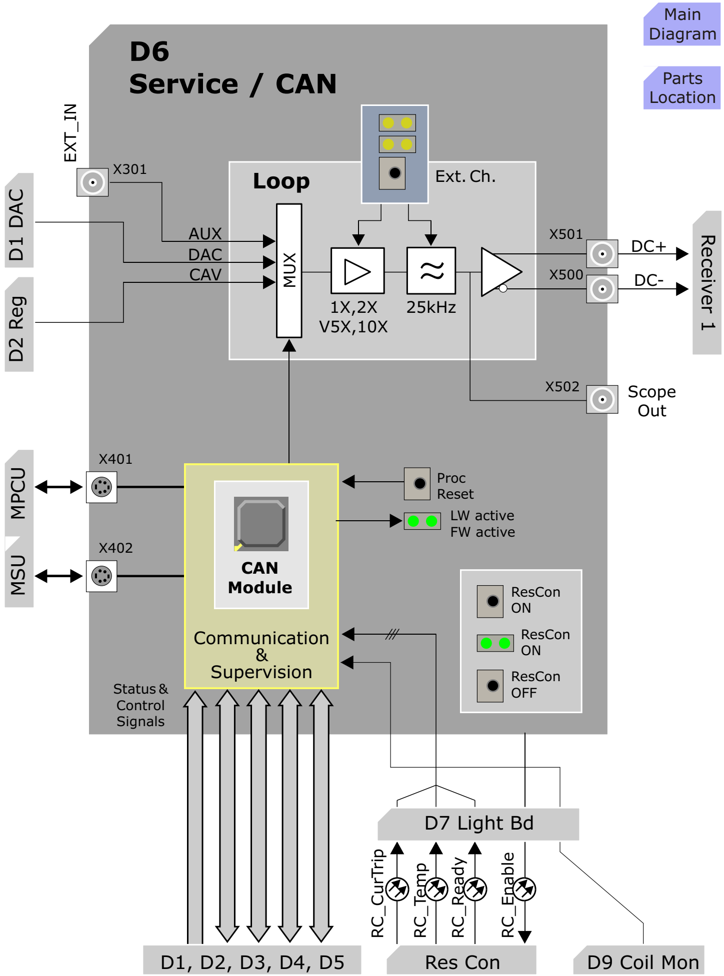
Loop multiplexer

The DAC-output signal (S_DAC), the actual value signal (S_CAV), the regulator difference (S_REGERR) and regulator output signal (S_REGOUT) from all three channels are fed via the backplane to the signal multiplexer. The selected signal is amplified by the selected gain (1, 2, 5 or 10) and additionally a low-pass filter (25 kHz) can be activated. The differential output signal DC+ (X500) and DC-(X501) are input to the RX_Module of the RF-system where it is digitized.

The settings described above can also be used for an external signal connected to the multiplexer via the front panel connector X301. In this case the setting is done by the front panel push button S401.

(For detailed explanation of this function see Troubleshooting Guide.)

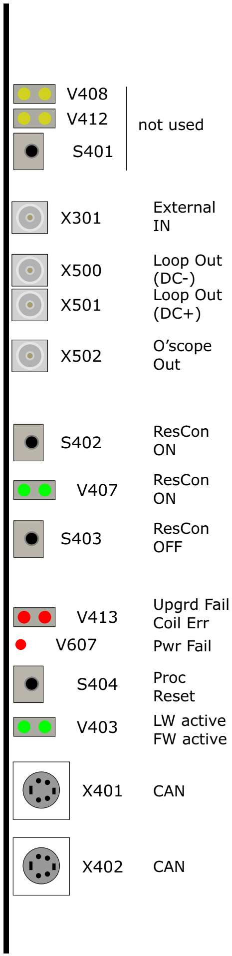
Figure 132 *D6 Service/CAN*



LEDs
See diagram.

Figure 133 *D6 Service/CAN*

Main Diagram



D7 Light Board

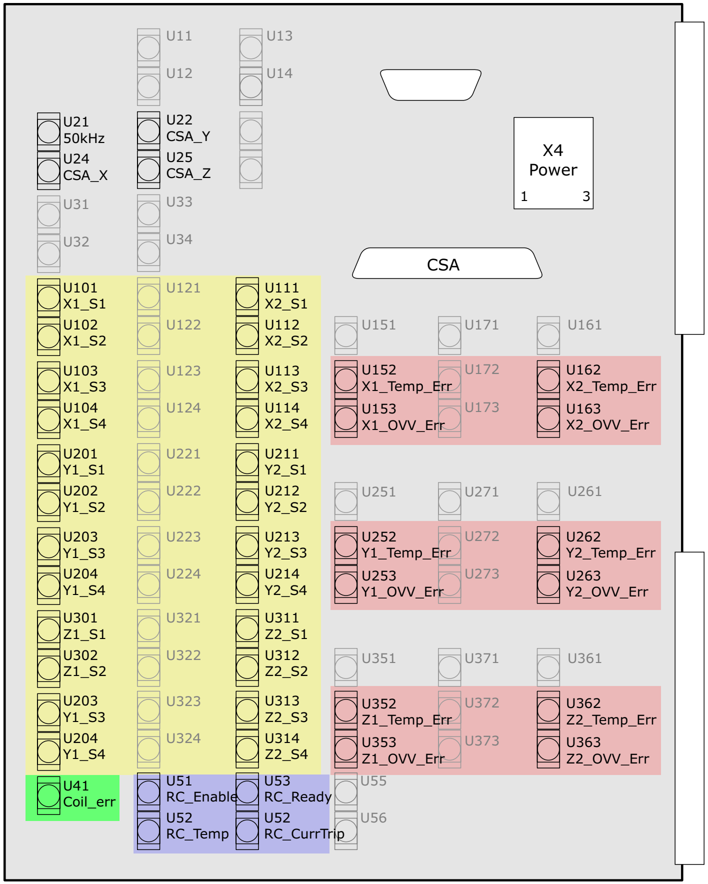
General

The optical transmitters and receivers between the GSSU and the rest of the Gradient System are implemented on the Light Board D7. Additionally the Power Supply connection and the Current Sensor Assembly connection is done via this board.

Function

- Electrical to optical and optical to electrical signal conversion for the Power Stages.
- Control signal decoupling for the ResCon.
- Clock signal (50 kHz) decoupling for the Driver Supply board D10.
- Gradient coil temperature monitoring signal decoupling.

Figure 134 *D7 Layout*



GSSU Power Supply

DC Supplies

Stabilized DC-voltages for the GSSU. Voltages and tolerances with disabled modulator:

Voltage	Tolerance	Measure Point	Current
5.075 V	±25 mV	D17 X63	3 A
15.050 V	±25 mV	D17 X61	2 A
−15.050 V	±25 mV	D17 X65	2 A

D10 Driver Power Supply

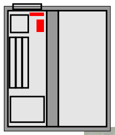
Slot 2 of the GSSU power supply provides a DC output to the D10 Driver power supply which converts this to AC @25 kHz, I_{max} = 10 A which is distributed to the drivers in the Final Stages.

For the DC/AC conversion the Driver Supply requires an external clock (jumper X4 1-2) delivered from the D7 over a fiber optic cable.

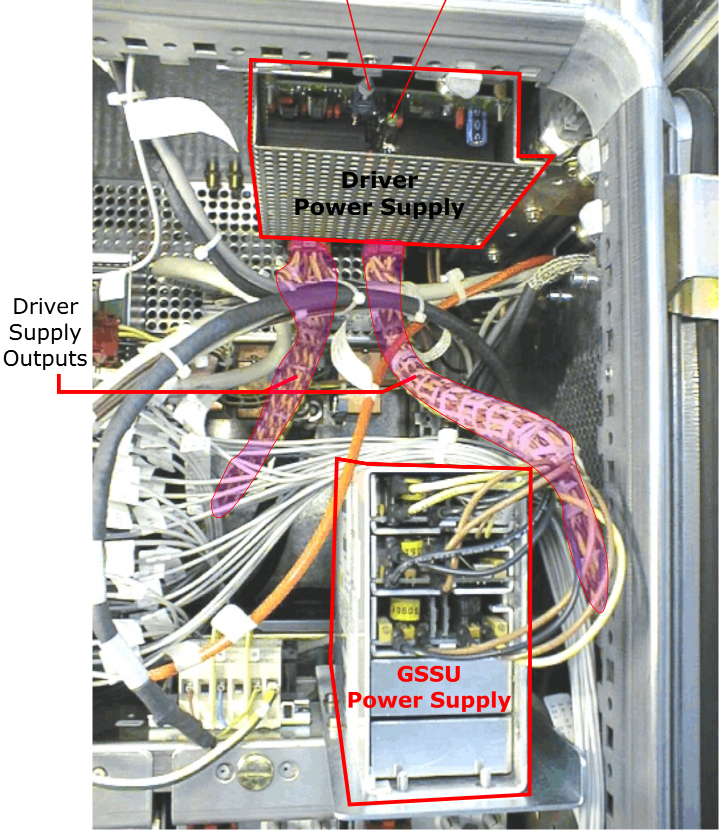
An LED indicates if the output voltage is available

NOTE The stabilized DC-voltages are adjustable by potentiometers on the respective power supply. Refer to the TSG and ROP for the adjustment procedures

Figure 135 *GSSU Power Supply*



Main Diagram



Power Stage

Each of the water-cooled Power Stages contains a 4-switch bridge supplying maximum 400V and capable of a maximum output current of 300 A pulsed, 160 A constant current.

The number of Power Stages is as follows:

- Standard Power Stage = Turbo option 3 power stage modules
- Upgrade Power Stage = Ultra option 6 power stage modules

Function

The **400V DC supply voltage** from the Resonant Converter is delivered to the Power Stages over tongue connectors at the rear.

The **PWM drive signals** from the Modulator are delivered over fiber optic cables and connected at the top-side of the Power Stage modules.

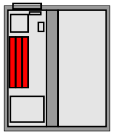
WARNING Swapping the fiber optic drive signals will result in blowing up the Power Stage. This is very loud and can cause heart failure and/or hearing impairment! Therefore, don’t do it.

The **16V AC, 25 kHz power supply** for the internal Drivers are delivered over yellow and brown twisted pair cables and connected also at the top of the Power Stage modules.

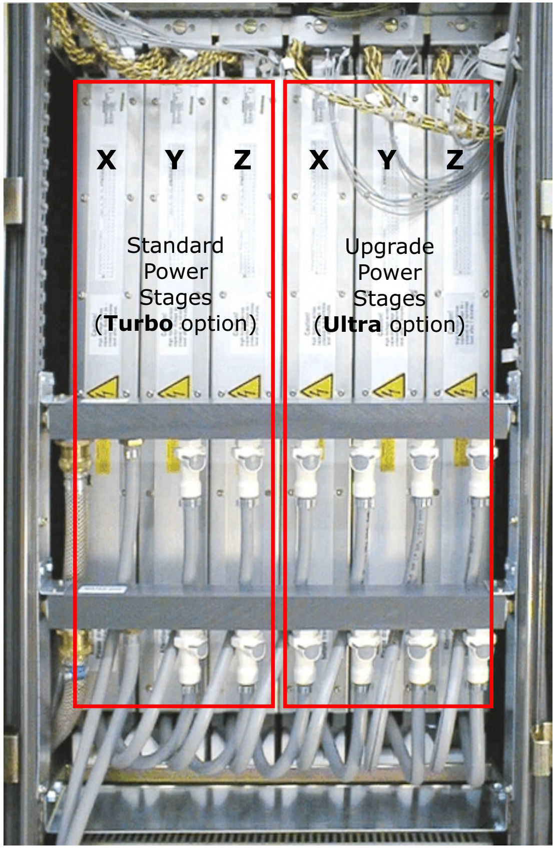
The power stages have internal heat-sink **temperature monitoring** circuits. Their output is one fiber optic cable at the top of the Power Stage module

NOTE The Power Stage is an FRU!

Figure 136 *Power Stage*



Main Diagram



Resonant Converter

General

The Resonant Converter (ResCon) supplies the Power Stages with stabilized DC-voltage. Each Power Stage has its own supply to minimize the coupling between the axis. It is water cooled.

Specifications

The adjustable output voltage is set to DC400 V and the ResCon is capable to handle at least 25 kW output power for an unlimited time.

There are two variants of the ResCon:

- MK I
- MK II

Refer to the spare parts catalog on ordering replacements.

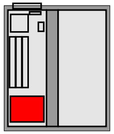
Supervision

The monitoring of the ResCon controls the power-on sequence and it detects the over temperature and overload state (total power) of the unit and provides a voltage and current supervision for every output channel (6x).

LEDs

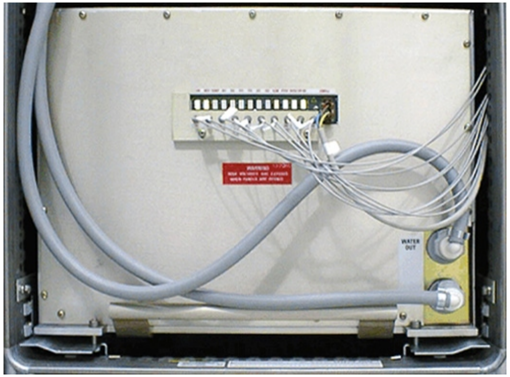
See **TSG** for a detailed description of status LED’s and FOC-links.

Figure 137 *Resonant Converter*



Main Diagram

Resonant Converter Mk I



Resonant Converter Mk II

Current Sensor Assembly

Overview

The Current Sensor Assembly (E7) consists of three pairs of current sensors each for one gradient axes. In addition, filter capacitors are installed in this unit.

Sensor Variants

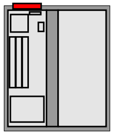
There are two types of current sensors asseblies in use:

- containing sensors from **LEM** only, (NO LONGER AVALABLE)
- containing sensors from **LEM** and **Danfysik**

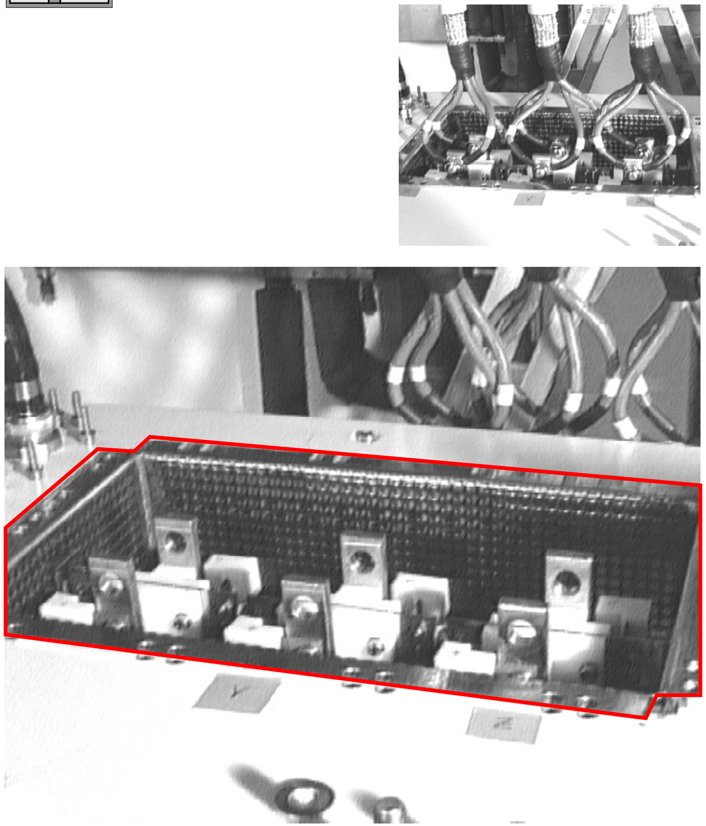
The Danfysik sensors require a synchronization signal supplied by the D7 and delivered over a fiber optic cable.

NOTE The CSA with Danfysik sensors also require a different D2 Regulator board. Heed notes in the spare parts catalog when ordering!

Figure 138 *Current Sensor Assembly*



Main Diagram



Gradient System K2217

Introduction

The Quantum gradient system K2217 is the standard gradient system in the MAGNETOM Sonata. Additionally, it is available as an option for the MAGNETOM Symphony and the MAGNETOM Harmony.
(For more details refer to "The Different Gradient Systems" in this description.)

The Quantum gradient system in the MAGNETOM Sonata features the following performance:

	Sprint	Quantum
Max. Gradient Field	30 mT/m	30mT/m
Max. Current	380 A	380 A
Max. Voltage	1500 V	2000 V
Min. Rise Time	400 µs	300 µs
Slew Rate in T/m/s	75 (X,Y,Z)	100 (X,Y) 125 (Z)

Figure 139 Gradient System K2217

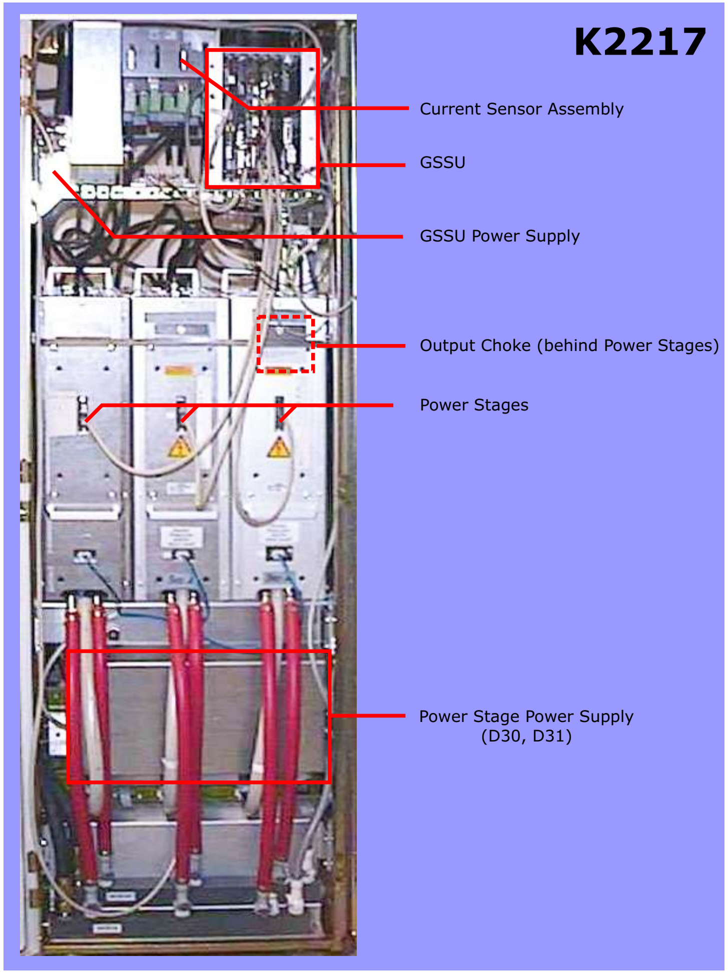
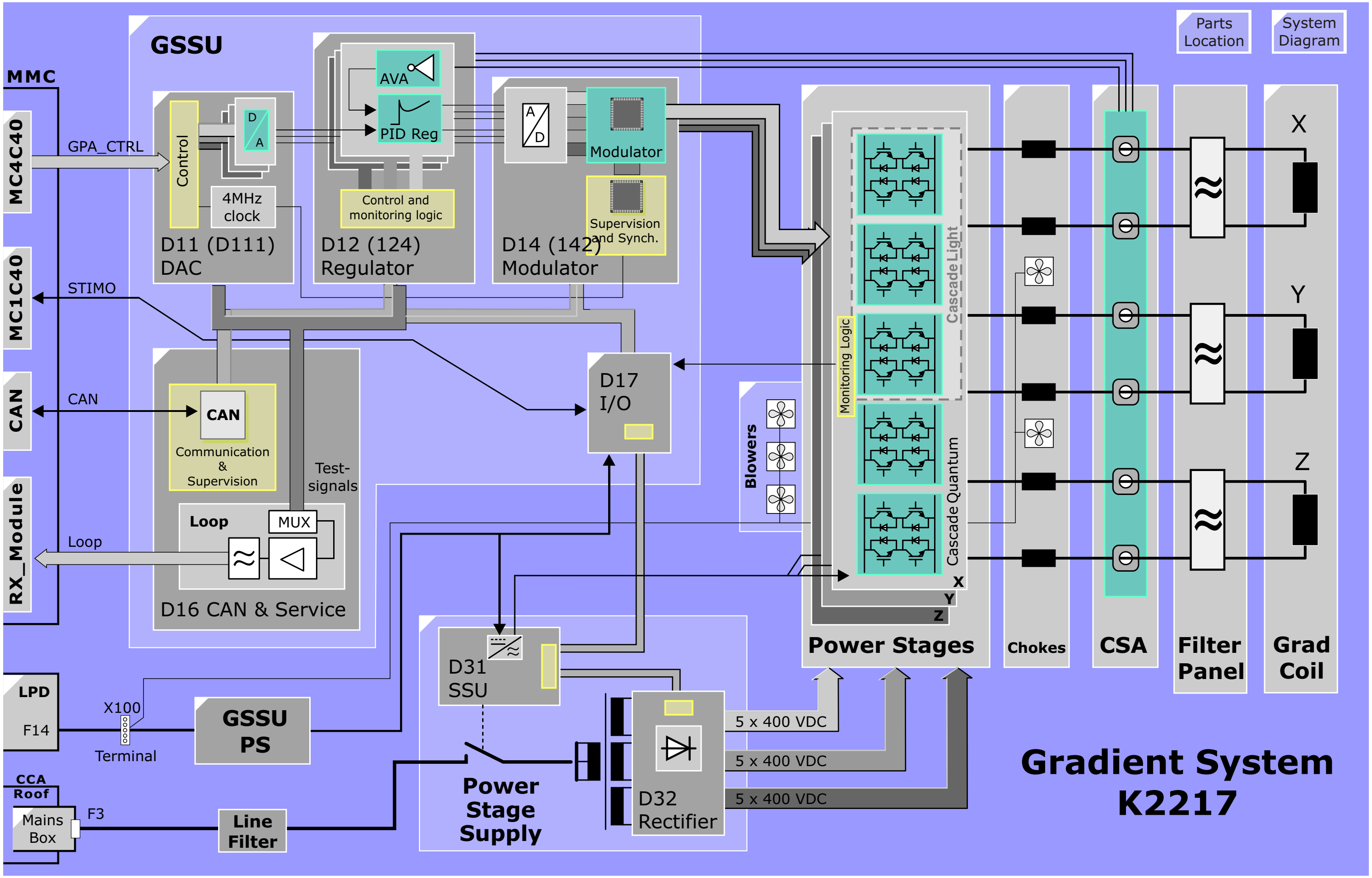


Figure 140 *K2217 Block Diagram*



Gradient Small Signal Unit (GSSU)

General

The GSSU is generating the pulse width modulated control signals for the switches in the Power Stage based on the digital data coming from the MMC. It's additional task is the overall supervision of the GPA.

The PCBs or FRUs (field replaceable units) of the GSSU connected to the one common backplane (D18) are listed below:

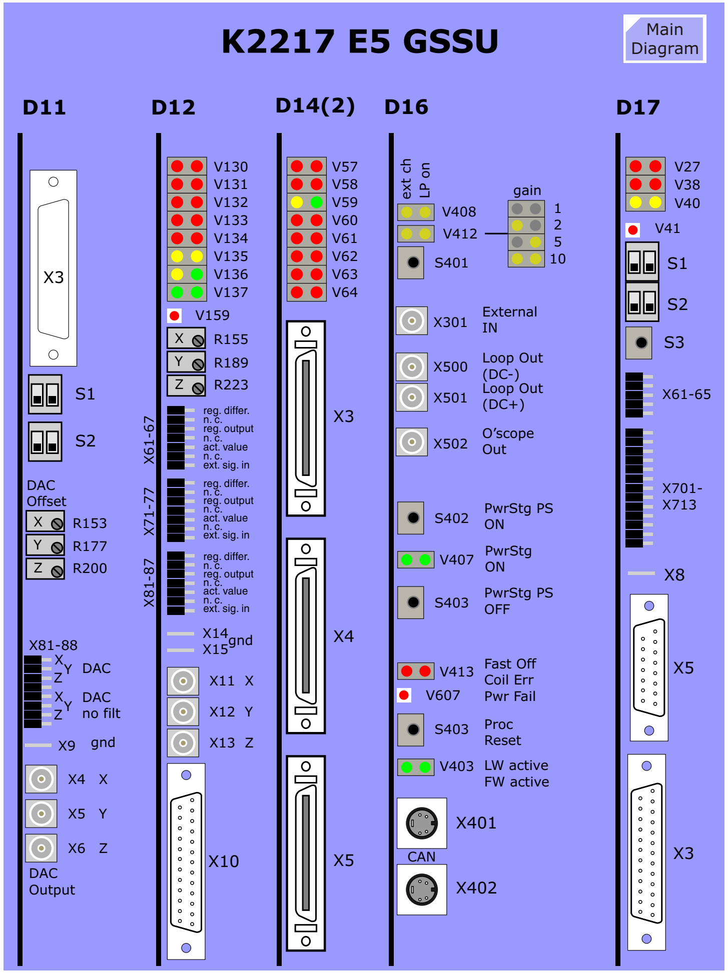
- DAC D11 (or D111 with Light)
- Regulator D12 (or D124 with Light)
- Modulator D14 (or D142 with Light)
- Service/CAN D16
- I/O-Board D17

D18 Backplane

Function

The backplane is the interconnection between the components of the GSSU. It is part of the GSSU-rack mounted to the cabinet frame by four quick release screws.

Figure 141 *GSSU Front View*



D11 (D111) DAC

The DAC-Board exists in two versions:

Board	System	Differences
D11 K2217	Quantum or Sonata	selected DACs
D111 K2217	Light	non-selected DACs

The DAC board can be divided into these functional blocks:

- Control Logic
- DACs
- Frequency Divider

Control Logic

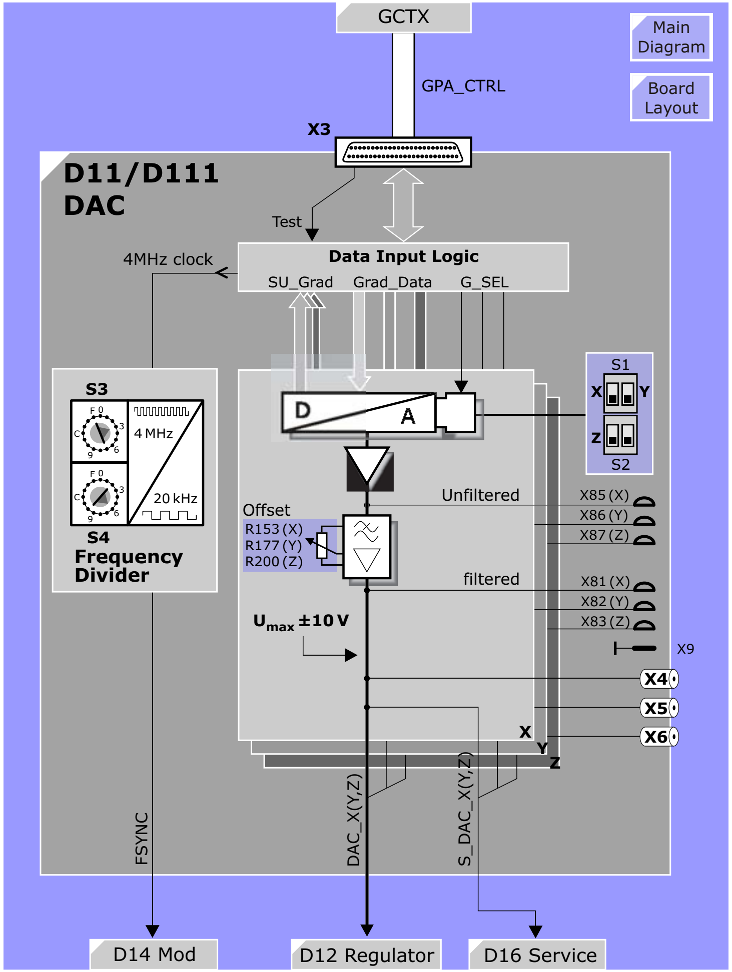
The gradient amplitude data (including the eddy current pre-distortion and the gradient delay) coming from the MMC is connected to a 50 pin female front connector (X3). The Control Logic contains circuitry to galvanically isolate from the MMC, *data change recognition* and a data feedback for test looping.

The 18 bit amplitude data (**Grad_Data**) for the three gradient axes is transferred multiplexed and controlled by the select signals G_SEL. The **G_SEL** select signals are used to clock in new gradient data values. Unchanged values are not clocked in, thereby increasing the signal stability.

Loopback Test

An input data loop-back test between the MC4C40/GCTX and the D11 can be run from the Service Software. When the **Test** signal is activates (SeSo), the 18 bit data of the X channel (**SU_Grad**) and the control lines can be looped back to the GCTX to verify the data cables and connections.

Figure 142 *DAC-Board D11/D111*



DAC

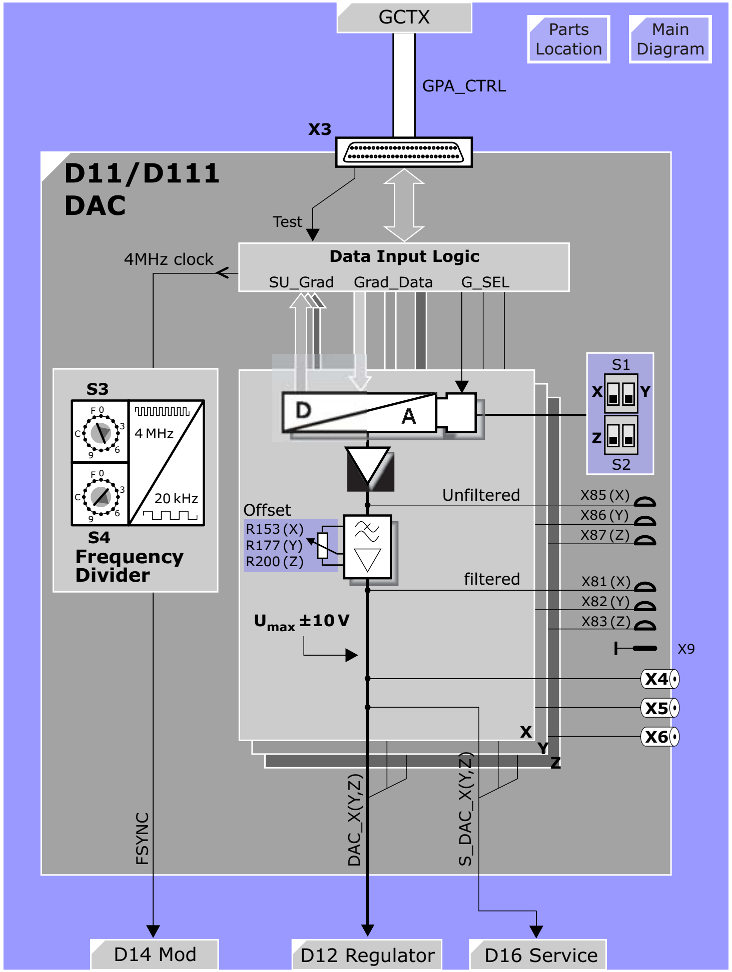
The 18 bit amplitude data is transferred to the three DACs controlled by the select signals. The DACs have an output voltage of ± 10 V which corresponds to the maximum current allowed by the gradient option.

The main DAC output is fed over the GSSU backplane to the D12 Regulator board as the nominal gradient amplitude. A copy is also sent to the D16 CAN & Service board where it is used as a feed back to the measurement control for loop testing and for the tune up procedures found in the Service Software.

Frequency Divider

A 4 MHz clock signal delivered from the GCTX is used to synchronize the GPA to the rest of the MR system. This clock signal is divided by the frequency divider according to the settings of the two rotary switches **S3** and **S4**. They are set to 3 and 6 respectively for an output frequency of 20kHz.

Figure 143 *DAC-Board D11/D111*



Switches

Switches **S1** and **S2** are used to zero the DAC inputs and outputs for the offset adjustment. When set to the upper position, the Offset potentiometers **R153**(X), **R177**(Y) and **R200**(Z) are adjusted for **0V ±50µV** at connectors **X4**(X), **X5**(Y), **X6**(Z).

Switch	Description	Position
S1 left	DAC_X enable/disable	down/up
S1 right	DAC_Y enable/disable	down/up
S2 left	DAC_Z enable/disable	down/up
S2 right	not used	not used

Potentiometers

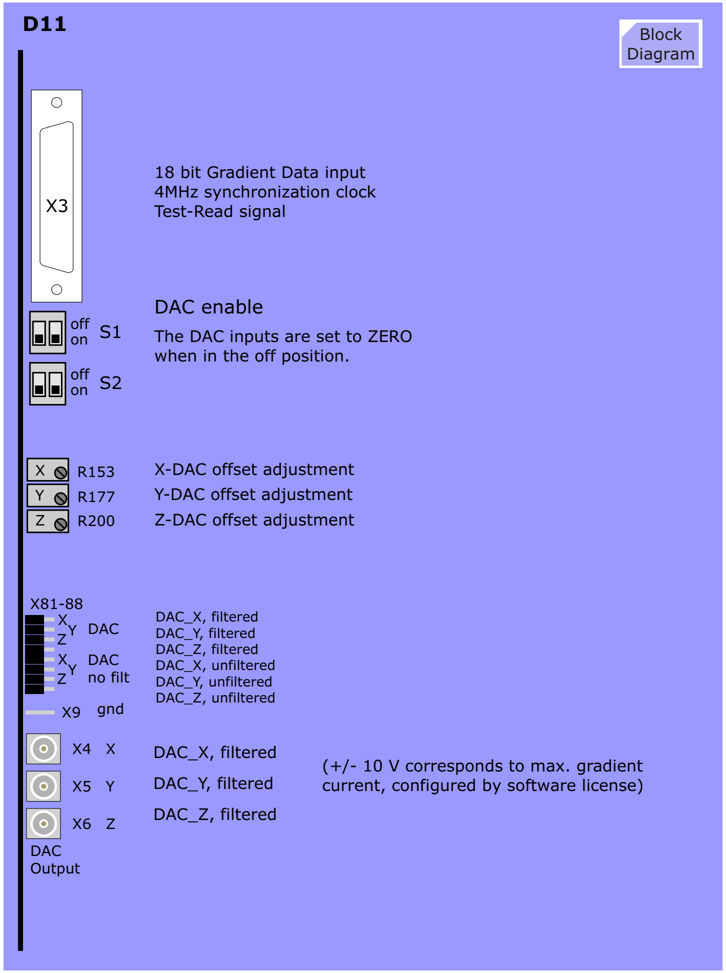
The static DAC offset can be adjusted with potentiometers to the specification of **0V ±50µV**. During the adjustment the corresponding DAC has to be disabled by using DIP-switches S1 and S2. See above.

Potentiometer	Description
R 153	DAC_X offset adjustment
R 177	DAC_Y offset adjustment
R 200	DAC_Z offset adjustment

Test Points

The DAC output signals (DAC_X, Y, Z) are available for monitoring via QLA-connectors (X4, X5 and X6) and test pins **X81-X88** with ground connection **X9** at the front panel of the board.

Figure 144 *D11 Front View*



D12 (D124) Regulator Board

General

The Regulator-Board exists in three versions:

Regulator	Gradient System	Differences
D12 K2217	Quantum or Sonata	
D124 K2217	Light	regulator parameters
D123 K2217	Allegra	

The Regulator board includes the regulator function for all three axes. It contains the following functional blocks:

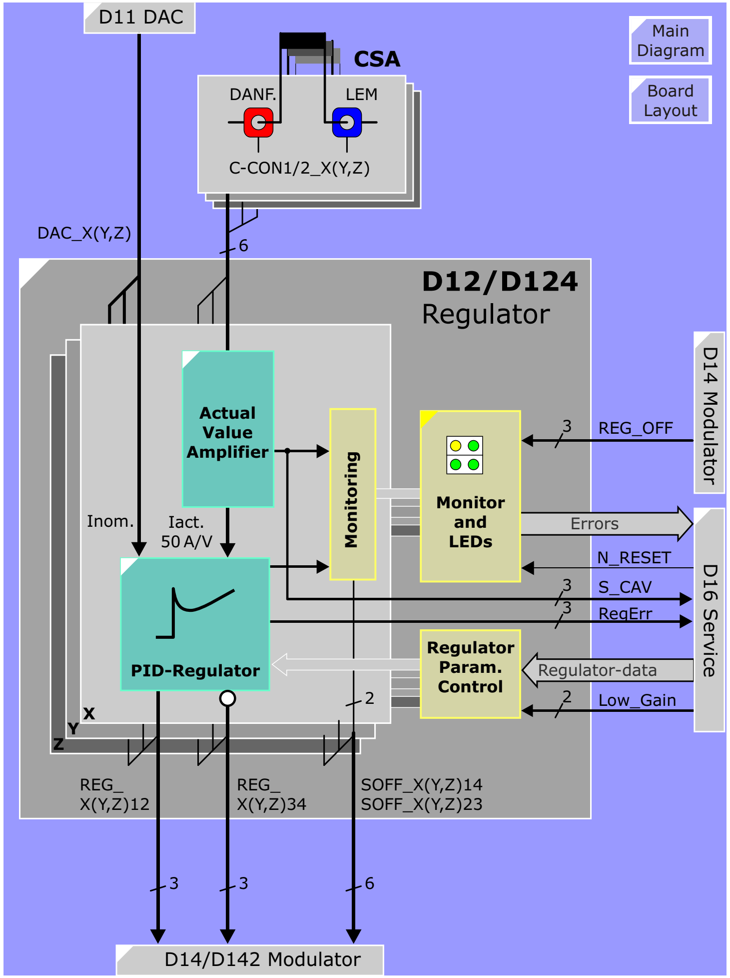
- Actual Value Amplifier (3x)
- PID Regulator (3x)
- Supervision Logic (3x)
- Monitoring and LEDs
- Regulator Control

Function

The current sensors in the Current Sensor Assembly (CSA) deliver a compensation current proportional to the current driven through the gradient coil. This compensation current is converted into a corresponding voltage level by the actual value amplifier.

In the PID-regulator the actual current value is compared to the nominal current value DAC_X(Y, Z) from the DAC-board. The resulting regulator difference is amplified by the regulator according to its characteristic. The output signals REG_X(Y, Z)12 and REG_X(Y, Z)34 of the regulator are connected to the input of the modulator board D14 (or D142).

Figure 145 *Regulator Board D12/D124*



Actual Value Amplifier (AVA)

Amplifier

The measurement current (C-CON1) from the DANFYSIK current transformer is amplified so that 50 A coil current corresponds to 1 V. The signal from the LEM current transformer (C-CON2) is used for monitoring only.

The **offset** of the amplifiers can be adjusted with potis R155(X), R189(Y) and R223(Z). To adjust, the modulators are first disabled using switches S1 and S2 located on the D17. The potis are then adjusted for 0V ±50µV as measured at X11(X), X12(Y) and X13(Z).

Monitoring

To assure that the Danfysik current sensor is working a second current sensor (the LEM) is used for comparison. The two sensor outputs are checked for differences greater than 32 A.

The power supply (±15 V and ANA_GND) for the current transformers on the current sensor assembly (CSA) is coming from the D12/124-board and the +15 V is looped back as P15_RET-signal. This signal is combined with the ±15 V voltage monitoring. If one of the above monitoring signals is faulty, the respective current error (CAV_ERR) is sent to the D16-board. This error signal also lights a red LED on the front panel of D12/124 and initiates the interrupt internal signal (N_INT_INT) that blocks all three axes.

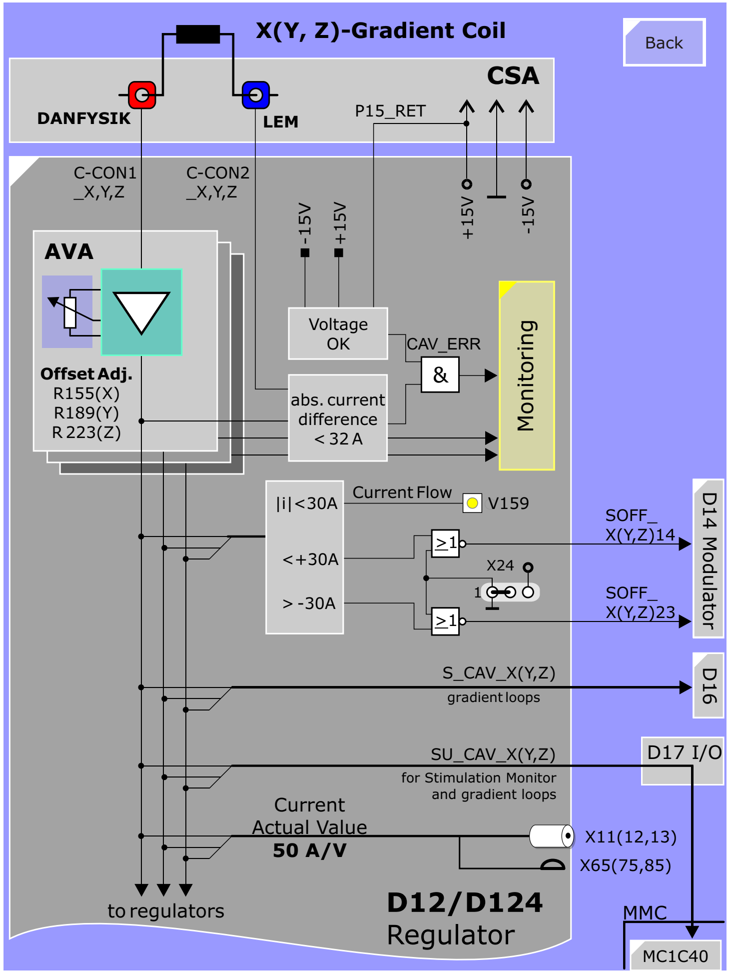
Loop Test

The analog signal S_CAV is connected to the loop multiplexer on the Service/CAN board for automatic testing.

Stimulation Monitoring

The actual current value is an important signal for the software stimulation monitor on the MC1C40 in the Modular Measurement Control (MMC). Therefore, the signals SU_CAV and SU_GND are sent to this board via the D17 I/O-board.

Figure 146 *Actual Value Amplifier*



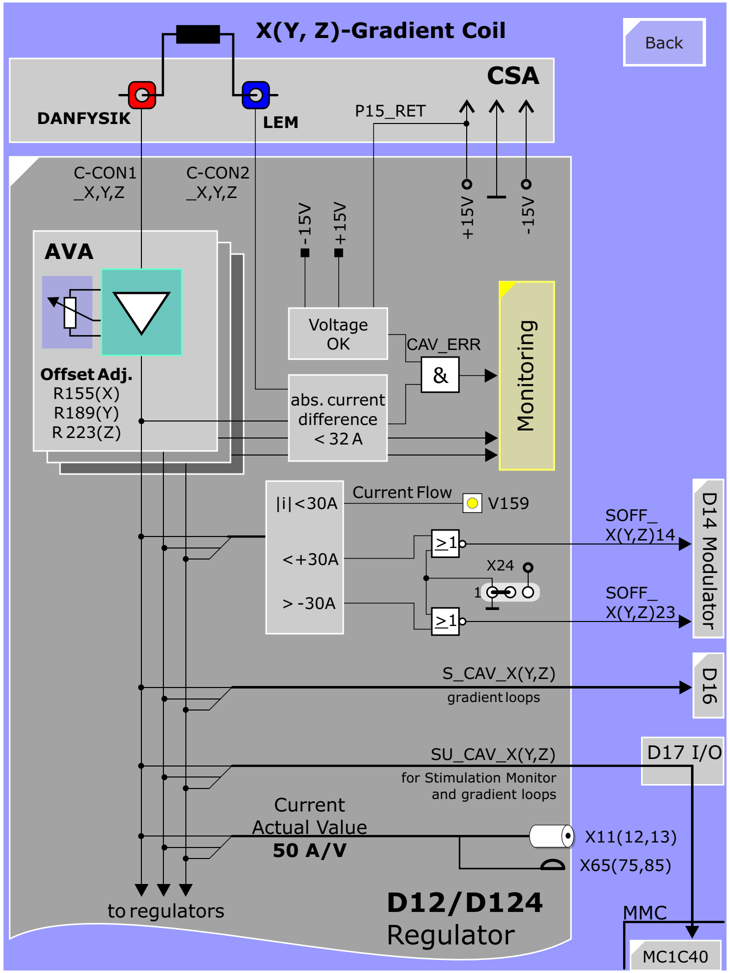
Opposite Current Direction Disabled

The actual value is monitored for a threshold of approximate 30 A absolute. A green LED indicates if a current of more than +30 A or less than −30 A is running. Furthermore, the other two outputs of this comparator logic (< 30 A and > −30 A) are used to disable the transistors of the opposite current direction.

The signal SOFF_X(Y, Z)23 disables switches 2 and 3 when current flows in a positive direction (> 30A) and the signal SOFF_X(Y, Z)14 disables switches 1 and 2 when current flows in a negative direction (< -30A). Both signals are sent to the modulator board D14 or D142, respectively.

Jumper X24 determines wether this disable logic for the opposite current direction is used (1-2) or not (2-3).

Figure 147 *Actual Value Amplifier*



PID-Regulator

Current Setting

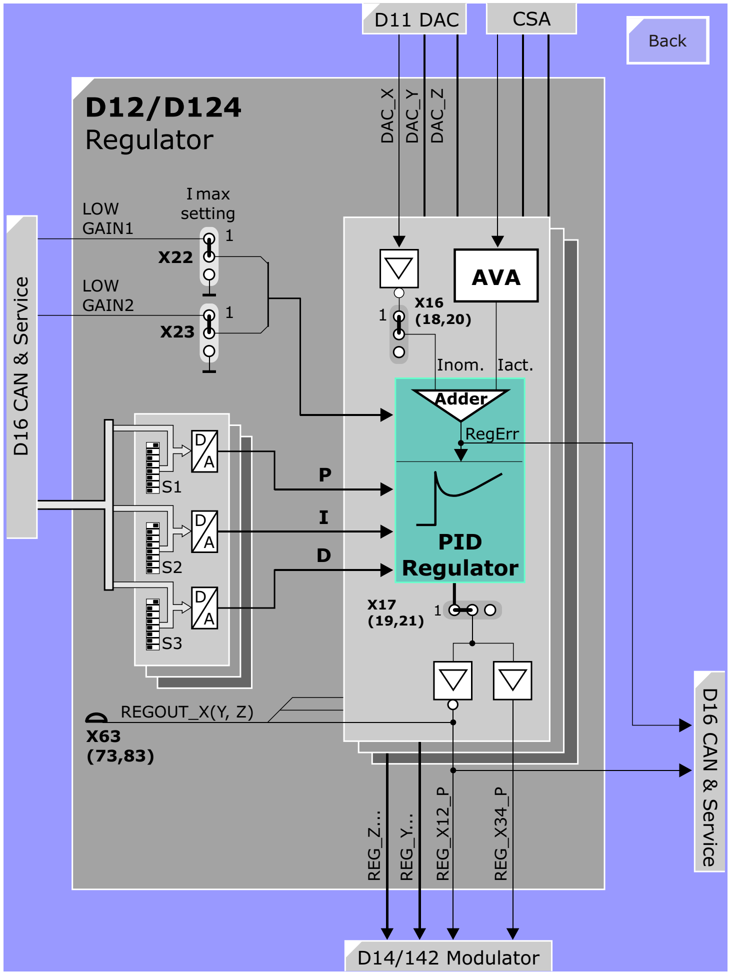
The nominal value sent from the DAC-board is inverted and via jumper X16(18, 20) added to the actual value coming from the AVA. The two jumpers **X22** and **X23** are used to set the maximum current value for the gradient system in use:

X23	X22	$I_{max.}$	Gradient System	Software
1-2	1-2	software controlled	what ever	NUMARIS/4
non	non	300 A	Light	
non	2-3	380 A	Sym/Har with Quantum option	NUMARIS/3.5
2-3	non	435 A	not used	
2-3	2-3	500 A	Sonata	

Regulator Unit

The closed loop regulation is based on a digitally controlled PID-regulator with 7bit resolution for the proportional (P), the integral (I) and the differential (D) part.
The initial setting of the regulator after power on is determined by DIP-switches S1 to S9 (see table on the left). During the scanner boot process the regulator values from the tune-up will be loaded by software.
The regulator output signal passes jumper X17(19, 21) and is sent to the D14-board (inverted and non-inverted). The inverted regulator output signal can be measured on test point X63(73, 83).
Test point X67(77, 87) can be used to supply an external nominal value (via X16) or an external regulator signal (via X17).

Figure 148 *PID-Regulator*



Monitoring

Errors and LEDs

The following table shows the monitoring on the D12/D124-board:

Error	Monitored Signal	LEDS	INT_INT
CAV-error	current sensors; 15 V return from CSA; ±15 V supply	3 red	YES
Over-current	actual current value	3 red	YES
Duty Cycle	actual current value	3 red	YES
Ohmloss	actual current value	1 red	NO
Loop-error	regulator difference regulator output signal	3 yellow	YES, if not in service mode

The interrupt internal (**N_INT_INT**) always blocks all three axes for safety reasons.
An Ohmloss-error additionally results in a Duty Cycle error for identification of the faulty axis.

Reset

All of the above mentioned errors are latched in error flip-flops and sent to the D16-board. The error flip-flops are reset by software via the D16-board or manually using button S3 on D17-board.

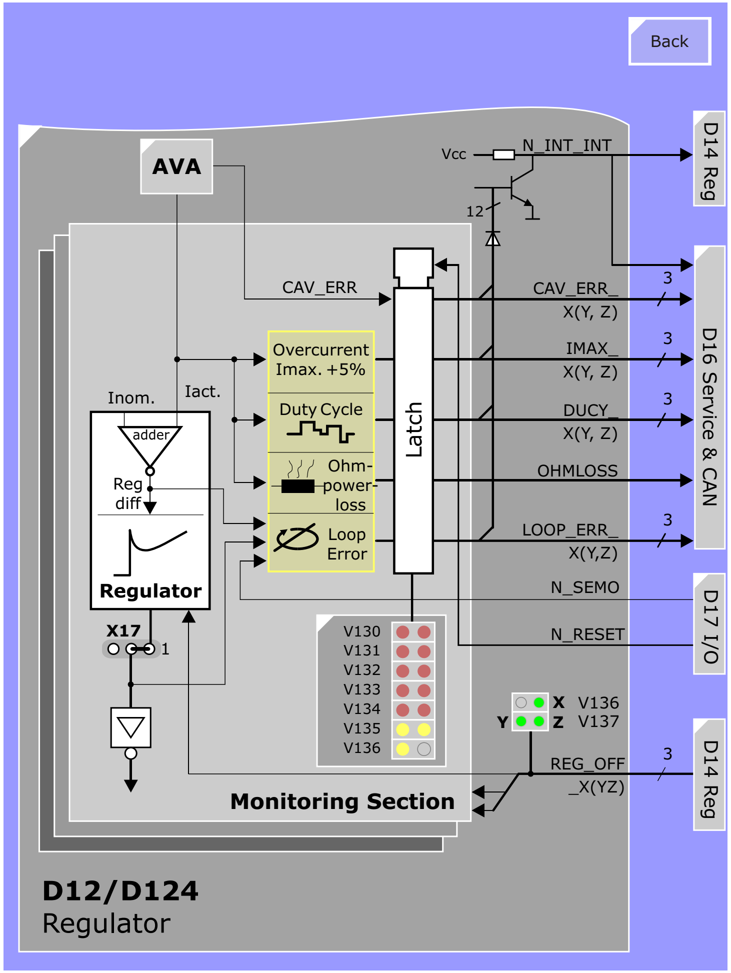
Service Mode

The service mode can be activated either by software via the D16-board or by switch S2 on the D17-board. The service mode is indicated by a yellow LED on the D17-board. If service mode is activated, the loop error does not result in an interrupt internal. Nevertheless, the yellow LEDs will light and the error is forwarded to the D16-board.

Regulator Enable

The modulator board D14 or D142 sends the signal REG_OFF to enable and disable the regulator board. The enable status is shown by three green LEDs on the front panel of the regulator board.

Figure 149 *Monitoring on Regulator Board*



LEDs

LED	Description
V136 left	current flow (one or more axes) > |87 A|
V136 right	regulator X enabled
V137 left	regulator Y enabled
V137 right	regulator Z enabled

Test Points

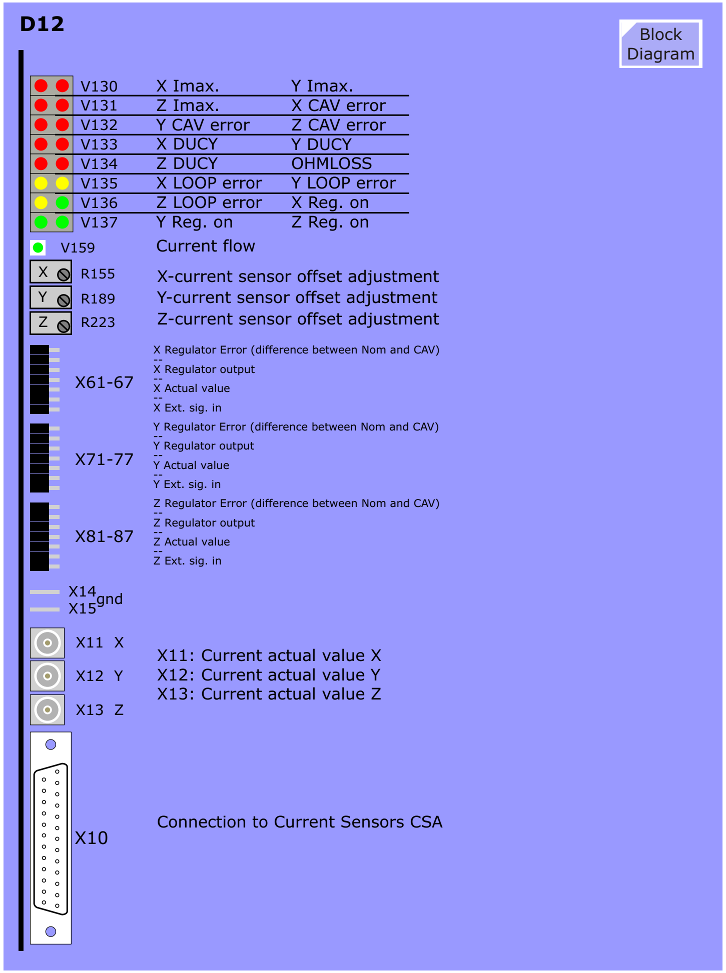
These test points are generally intended for factory use.

Potentiometers

Potentiometer	Description
R155	X-current sensor offset adjustment
R189	Y-current sensor offset adjustment
R223	Z-current sensor offset adjustment

The adjust is made in much the same way as the DAC offset adjustment. For this adjustment, the Modulators must be disabled with switches S1 and S2 on the D17 I/O board to assure there is no current being generated by the amplifier. The pots are adjusted for 0V ±50µV at connectors X11(X), X12(Y), X13(Z).

Figure 150 *D12 Front View*



D14 (D142) Modulator Board

General

The Modulator board exists in two versions dependent on the type of Power Stage used:

Gradient System	# of Cascades	Modulator
Quantum	5	D14 K2217
Light	3	D142 K2217

The Modulator Board has two main functions:

- generation of the pulse-width-modulated switching signals
- supervision of power stage related signals

Pulse-Width-Modulator

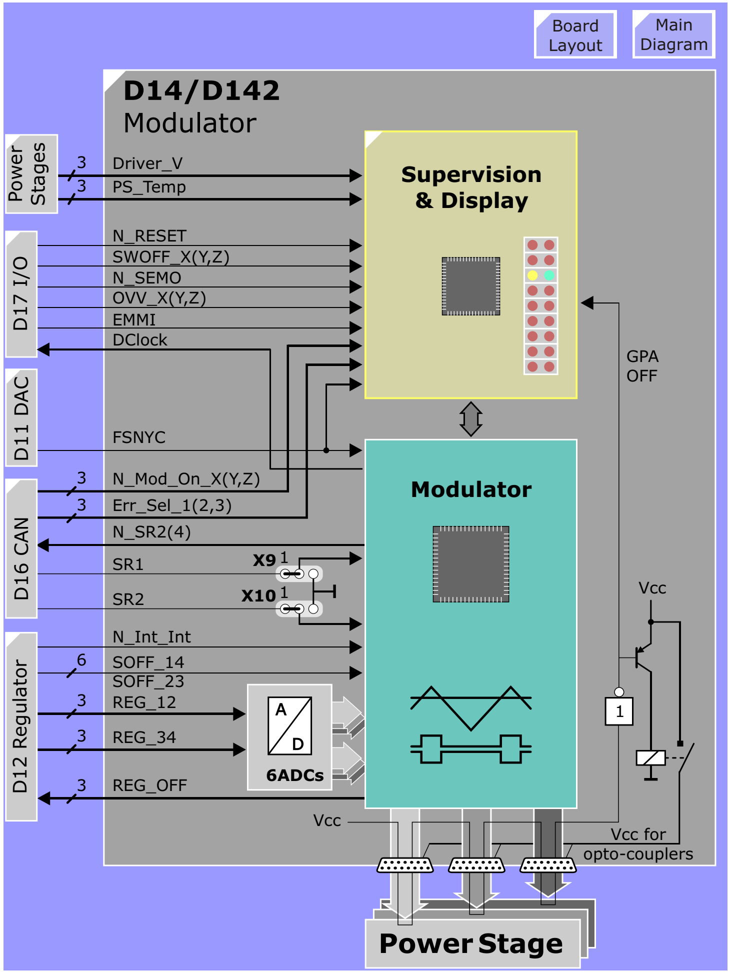
Overview

The D14(2)-board is based on a programmable logic that contains the pulse-width-modulator and the monitoring circuit of time critical power stage related signals.

Modulator

The regulator output signals **REG_12 (34)** are analog-to-digital converted and fed to the modulator chip where the pulse-width-modulation takes place. The resulting switch signals (active low) are sent over 3 cables to each Power Stage module. These cables are monitored by looping **VCC** through a bridge between pin 1 and pin 50 inside the plug on each of the Power Stage modules. If this signal (**GPA_On**) is missing, the VCC-supply for the opto-transmitters on the driver boards in the power stages is switched off by means of a relay.
The signals **SOFF_14(23)** inform the modulator chip whether the opposite current direction should be enabled or disabled.
The **N_INT_INT** signal from the Regulator informs the modulator chip about a severe problem of the regulator. Consequently, the modulator disables the regulator of the desired axis by means of signal **REG_OFF**_X(Y, Z).

Figure 151 *Modulator Board D14 or D142*



Jumpers

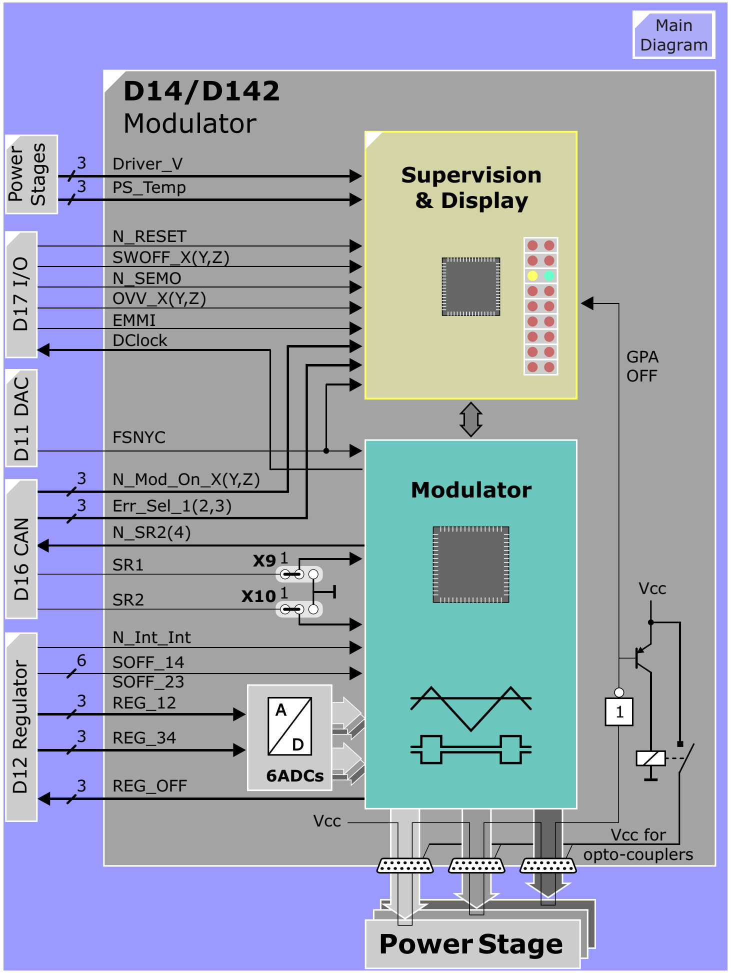
The Quantum gradient system is available in different versions dependent on the MAGNETOM system it is used for. The max. power stage output voltage is selectable by software or jumper X9 and X10:

X9	X10	Max Output Voltage	System	Software
1–2 (SR2ADJ)	1–2 (SR4ADJ)	software controlled	all	NUMARIS/4
2–3	2–3	2000 V	S/H Quantum; Sonata	NUMARIS/3.5
none	2–3	1000 V	S/H Ultra	
don’t care	none	500 V	S/H Turbo	

The actual setting of the Power Stage voltage is fed back to the D16 Service/CAN. If the Power Stage voltage is set to >1000 V, the gradient system requires a stimulation monitor. Systems running with NUMARIS/3.5 use a hardware stimulation monitor (D13 or D131) and systems with NUMARIS/4 use a software stimulation monitor:

Option	NUMARIS/3.5	NUMARIS/4
Sprint	not existing	software controlled stimulation monitor
Quantum	D13 K2217	
Sonata	D131 K2217	

Figure 152 *Modulator Board D14 or D142*



Supervision, Display and Synchronization Logic

Monitoring

The supervision monitors the supply voltages (±15 V and +5 V), clock and synchronization. Furthermore, monitoring signals from other boards can disable the modulator:

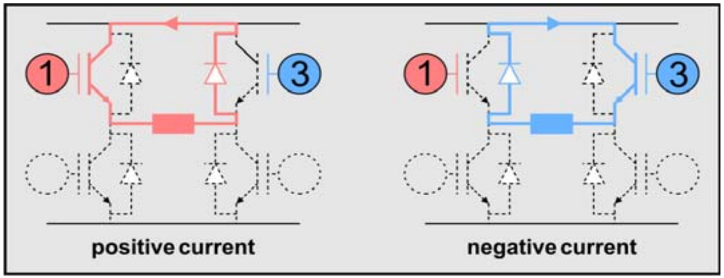
- Software Stimulation Monitor from MMC;
- Power stage voltage monitoring from D17;
- Manual modulator off by switches S1 and S2 on D17;
- Modulator enable/disable from the processor board D16.

If the modulator detects an error, the D16 is informed by signal **ERR_X(Y, Z)**.

Switch Off Modes

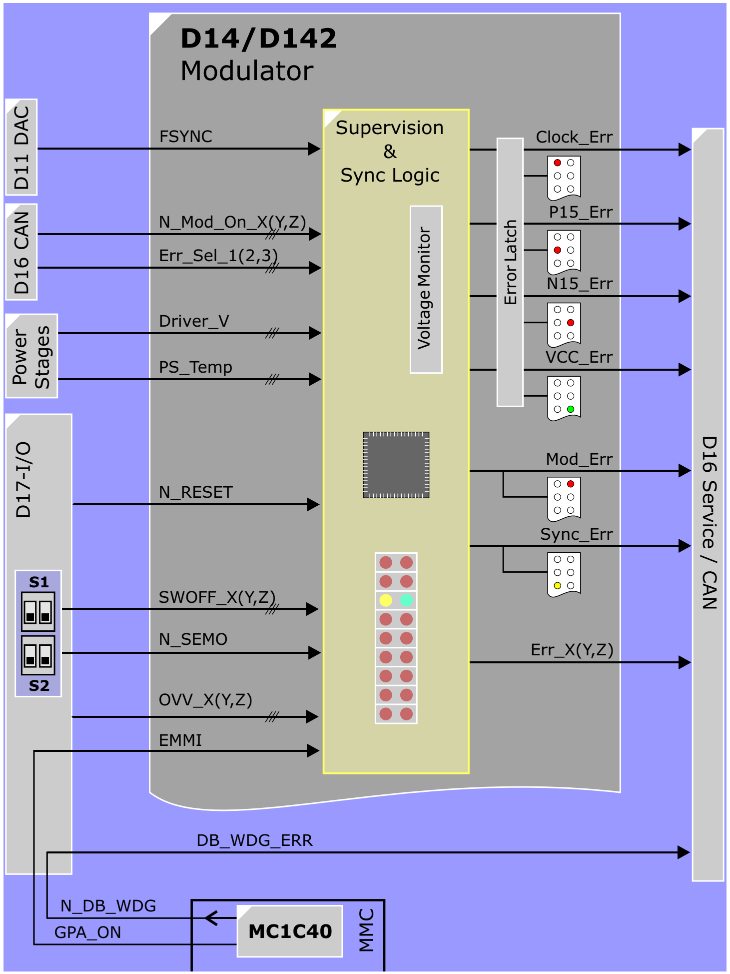
In case of an error, the modulator can disable a running gradient current in two ways:

- The "hard" way; all switches of the power stage are blocked.
- The "soft" way; the modulator enables switch 1 and 3 of the power stage permanently (upper freewheeling).



This "soft off" is necessary to prevent stimulation during switch off. Only the processor controlled modulator disable using signal N_MOD_ON_X(Y, Z) and the manual off (SWOFF...) by the switches on D17-board result in a hard switch off of the gradient current.

Figure 153 *Supervision on D14/D142*



LEDs

The errors indicated by the LEDs here on the D14 are coming form monitoring circuits both on and off the baord (see the block diagrams for details).

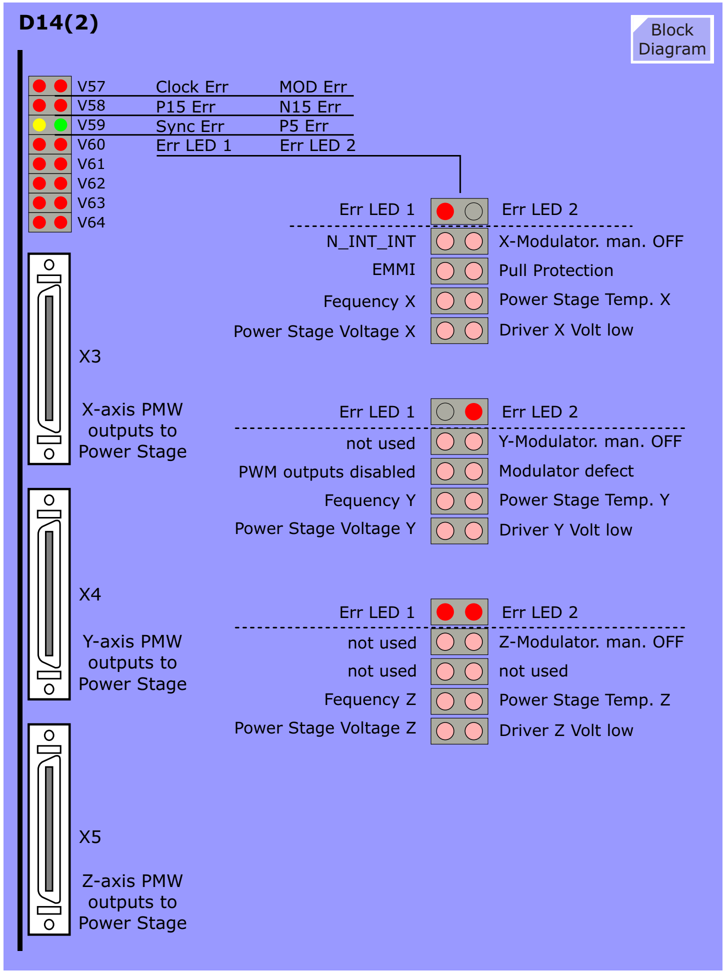
LEDs **V57-V59** indicated errors corresponding to the error names. LEDs **V60-V64** are multiplexed to display up to 24 errors using only 10 LEDs. And this is how this works:

The pattern of the two LEDs of V60 dictates which errors will be displayed by LEDs V61 through V64 according to the diagram at right.

Error Reset
All errors can be reset by the processor D16 (N_CAN_RESET) or manually by switch S3 on I/O-board D17.

Service Mode
The supervision logic can be switched into a service mode (N_SEMO) by front panel switch S2_right on D17-board. When in the service mode masking out (disabing) certain errors to allow troubleshooting. and remotely from the D16-board (N_CAN_RESET). The service mode is indicated by a yellow LED on the front panel of the I/O-board D17.

Figure 154 *Error Display D14(2)*



D16 Service/CAN Board

General

The Service/CAN Board consists of two functional blocks:

- Communication and Supervision (CAN-Module)
- Analog signal multiplexer.

Function

The CAN-Module is responsible for the data transfer between the MMC and the GPA. Control data for the GPA is transferred from the MMC and GPA-status information to the MMC.

The analog signal multiplexer is used to connect signals generated in the GPA to the ADC in the data acquisition system. This link allows the software evaluation of the selected signals.

CAN-Module

Communication

The firmware (FW) stored in a PROM allows the processor to run a self-test to communicate on the CAN-Bus and to load the application software. The application software (LW) for the processor is stored in an EEPROM. According to this program the processor is able to receive control data from the MMC and transfer status information from the GPA to the MMC.

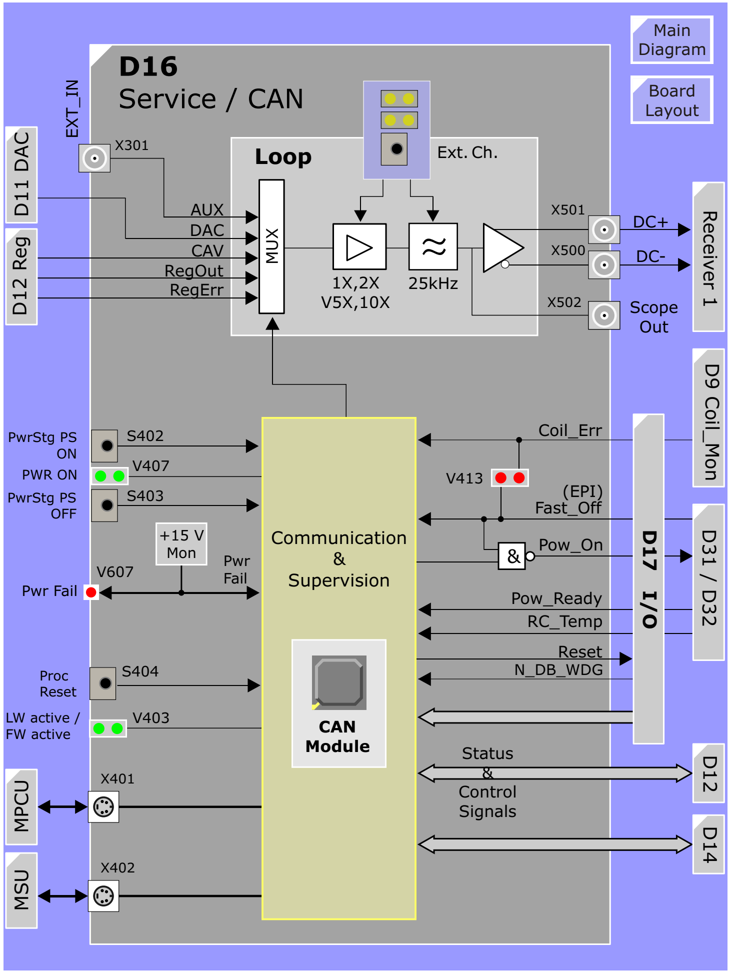
Front Panel Control

The state of the front panel switches is checked via input ports and the front panel LEDs are driven by output ports of the CAN-Module.

Reset

The front panel push button **S404** is used to reset and re-start (warm start) the Processor.

Figure 155 *D16 Service/CAN Board*



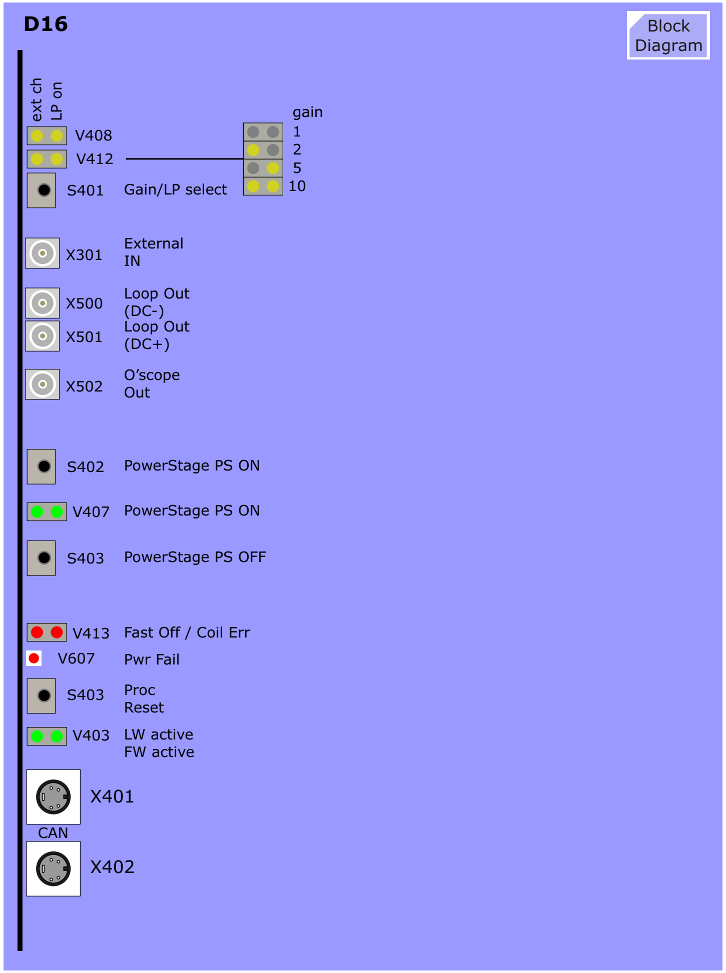
Power Stage Power Supply ON / OFF
Switches **S402** is used to switch ON the Power Stage Power Supply switch **S403** to switch it OFF. LED **V407** indicates ON/OFF state and lights when ON.

Power Fail Message
The + 5 V supply of the CAN-Module is produced from the + 15 V and buffered. If the + 15 V fails it is detected by an error logic and the buffered + 5 V will be available long enough to send a Power-Fail message to the CAN-System. This message is processed with the highest priority. The Power-Fail state is indicated by the red LED **V607**.

Analog signal multiplexer

Loop Multiplexer
The DAC-output signal (S_DAC), the actual value signal (S_CAV), the regulator difference (S_REGERR) and regulator output signal (S_REGOUT) from all three channels are fed via the backplane to the signal multiplexer. The selected signal is amplified by the selected gain (1, 2, 5 or 10) and additionally a low-pass filter (25 kHz) can be activated. The differential output signal DC+ (**X500**) and DC- (**X501**) are input to the RX_Module of the RF-system where it is digitized.
The settings described above can also be used for an external signal connected to the multiplexer via the front panel connector **X301**. In this case the setting is done by the front panel push button **S401**.
(For a detailed explanation of this function see Troubleshooting Guide.)

***Figure 156** D16 Service/CAN Boar Layoutd*



D17 I/O Board

General

The D17 I/O board acts as an interface for the control and monitoring signals to and from the D14 Modulator and D16 Service CAN boards and the components external to the GSSU (refer to Figure 158).

It also provides several service switches (see below) and the distribution of the GSSU power supply voltages to the GSSU boards over the backplane.

STIMO Interface

For systems running syngo MR and having the Sprint, Quantum or Sonata gradient options the MC1C40 stimulation monitor is used. The MC1C40 outputs two signals, **EMMI** and **DB_Wdg_Err**, to the D14 and D16 over the D17 where their states are indicated. They can also be bypassed with jumper X9.

Power Control

See the Power Stage power supply section for details.

Service Switches

S1 and S2 provide a means o disable the modulators for service purposes. They are used, for example, to set the current sensor offsets (see D12 AVA description)

Synchronization for DANFYSIK-Unit

The signal DCLOCK from the modulator is sent to the DANFYSIK-current sensors in the Current Sensor Assembly (CSA) to synchronize these units to the system clock.

Power Supply Distribution

The GSSU power supply voltages are fed to the GSSU boards over the backplane via the D17. These voltages can be measured at the front over X61, X63 and X65 (see D17 board layout diagram).

Figure 157 *D17 I/O Boar Layoutd*

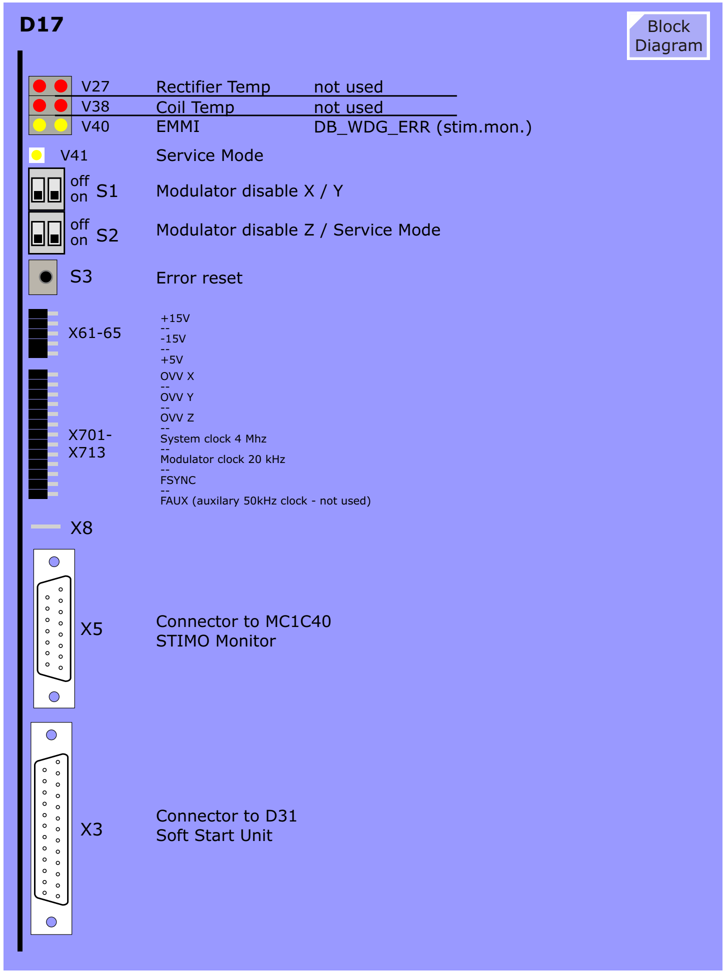
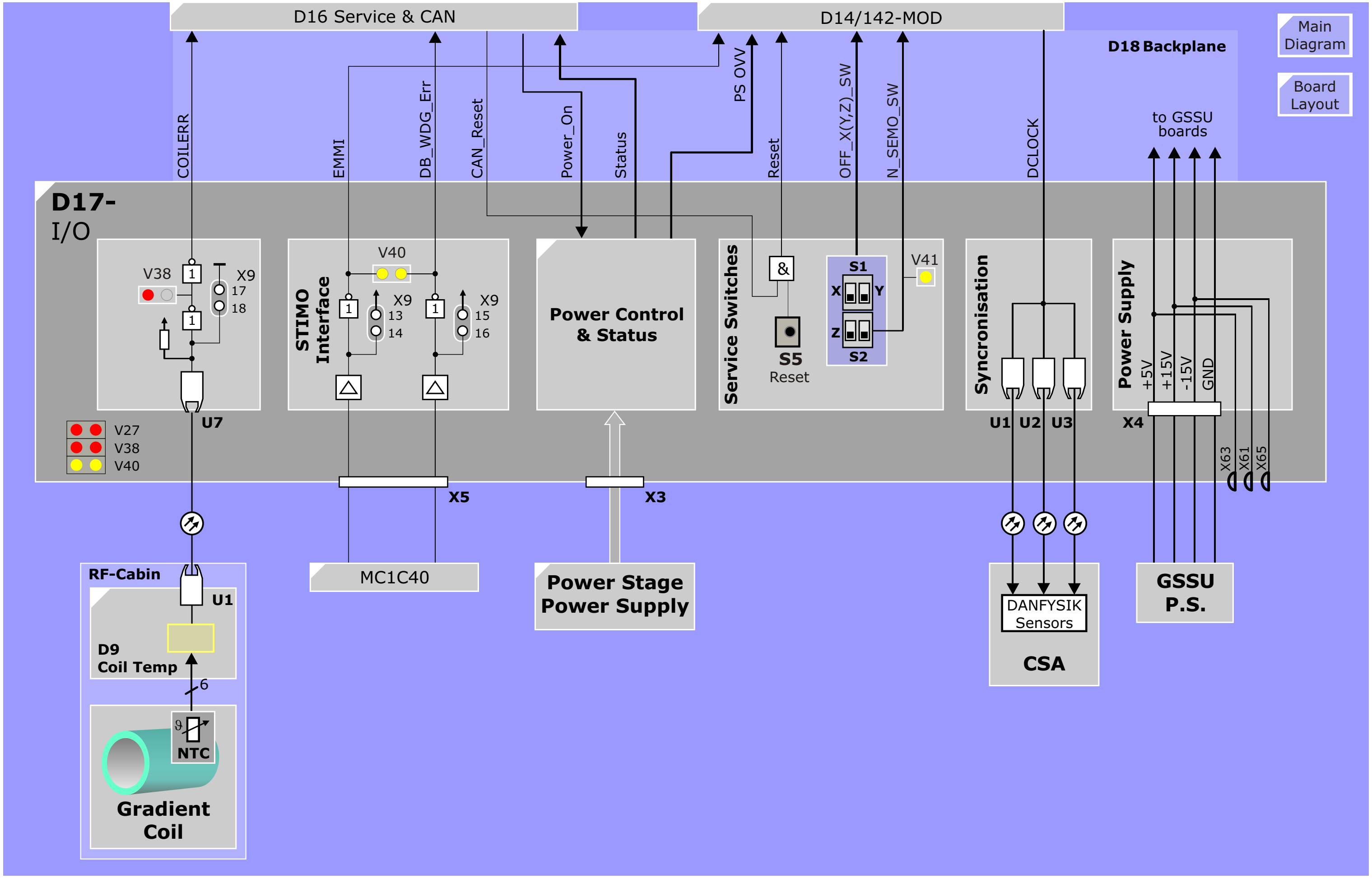


Figure 158 *I/O-Board D17*



GSSU Power Supply

DC-Supply for GSSU

Stabilized DC-voltages for the GSSU. Voltages and tolerances with disabled modulator:

Voltage	Tolerance	Measure Point	Current
5.075 V	±25 mV	D17 X63	3 A
15.050 V	±25 mV	D17 X61	2 A
−15.050 V	±25 mV	D17 X65	2 A

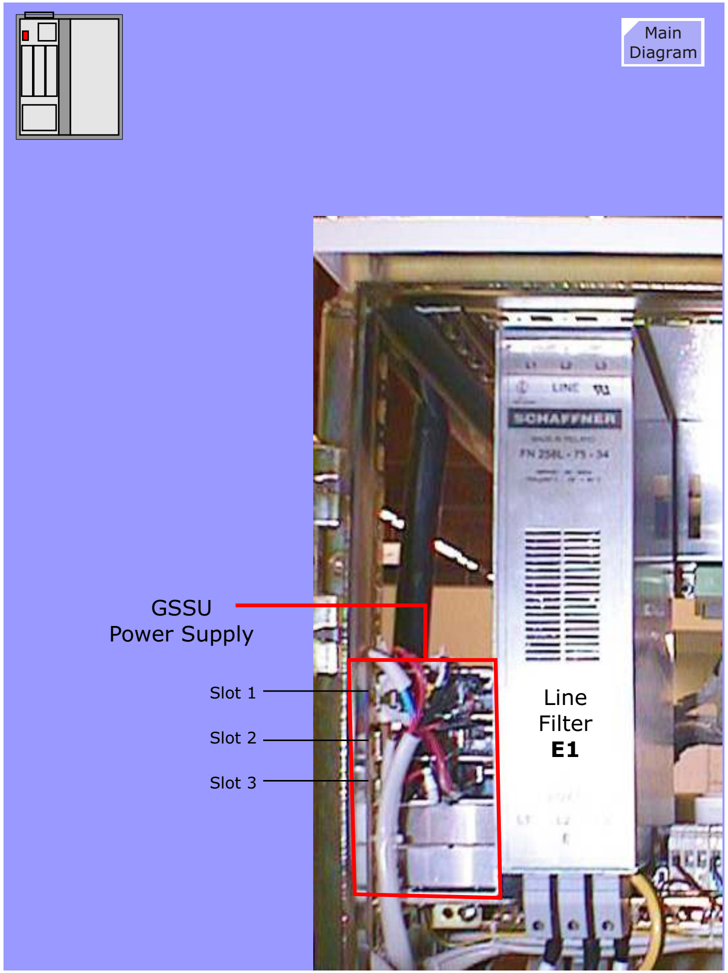
DC-Supply for the Driver Supply

Stabilized DC-voltage for the driver supply in the power stage modules.

Voltage	Tolerance	Measure Point	Current
16.500 V	±100 mV	direct on the power supply	12 A

NOTE The stabilized DC-voltages are adjustable by potentiometers on the respective power supply.

Figure 159 *GSSU Power Supply*



Power Stage Power Supply

Mains Supply

The mains (L1, L2, L3) are supplied by the 80 A circuit breaker F3 located at the top of the CCA cabinet. Line filter E1 provides decoupling from the customer mains. Three yellow LED-pairs on the D31 Soft Start Unit indicate that circuit breaker F3 is closed and there is lines voltage available.

WARNING Circuit breaker F3 MUST be switched on **BEFORE** powering up the GSSU. Otherwise severe damage to the Power Stage power supply and/or Power Stages is possible!

Power Up/Down

The Power Stage power supply is enabled/disabled with the **N_Power_On** signal and is performed either automatically via a command from NUMARIS over the CAN bus or manually with switches **S402** and **S403** on the Service/CAN D16.

When set the **N_Power_On** signal activates relay K1 which feeds the mains voltage to transformer T1 over current-limiting resistors to allow a slow charge of the buffer capacitors in the Power Stages. After a 2 second delay contactor K2 will be activated to connect the mains voltage directly to the transformer.

The current over the damping resistors is monitored and when too high, the respective relay (K100 for L1, K200/L2 and K300/L3) is activated resulting in a extreme power input (**EPI**) error message to the D16 (this signal changes to **Fast_Off** which lights the left side of LED V413 and informs the CPU.) At the same time the N_Power_On signal to K1 is removed deactivating both K1 and K2.

Power-ON Acknowledge

When contactor K2 is activated a normally-open contact of K2 is used to switch on relay **K3** on the D31 via a cable interlock over the D32 Rectifier board. A normally open contact of K3 informs the processor (**N_Power_Ready**). Service jumper X9 on D17 allows to bypass this acknowledge signal.

Transformer T1

The primary of Transformer T1 must be tapped for the proper incoming mains voltage. Its secondary provides fifteen 285V AC outputs each fused with 25A fuses. The secondary voltage is sent to the D32 Rectifier board for rectification to 400V DC.

Rectifier Temperature Monitoring

The 15 rectifier units on the D32-board are mounted by a water-cooled heat sink whose temperature is monitored via a PTC-resistor. A monitoring circuit on the D31 checks the rectifier heat sink temperature, if it exceeds 55°C relay **K4** is activated. In case of an error, relay K4 is deactivated and D17 sends an error signal to the D16-board. The left side of LED V27 on the D17 indicates this error.

Power Stage Voltage Monitoring

Each power stage module needs 5 times 400V DC for the five cascades. The D32 Rectifier board has 15 rectifiers for the three axes with over-voltage monitors. The three error signals pass the soft start unit D31 and block the modulator via the I/O-Board.

Figure 160 *Power Stage Power Supply*

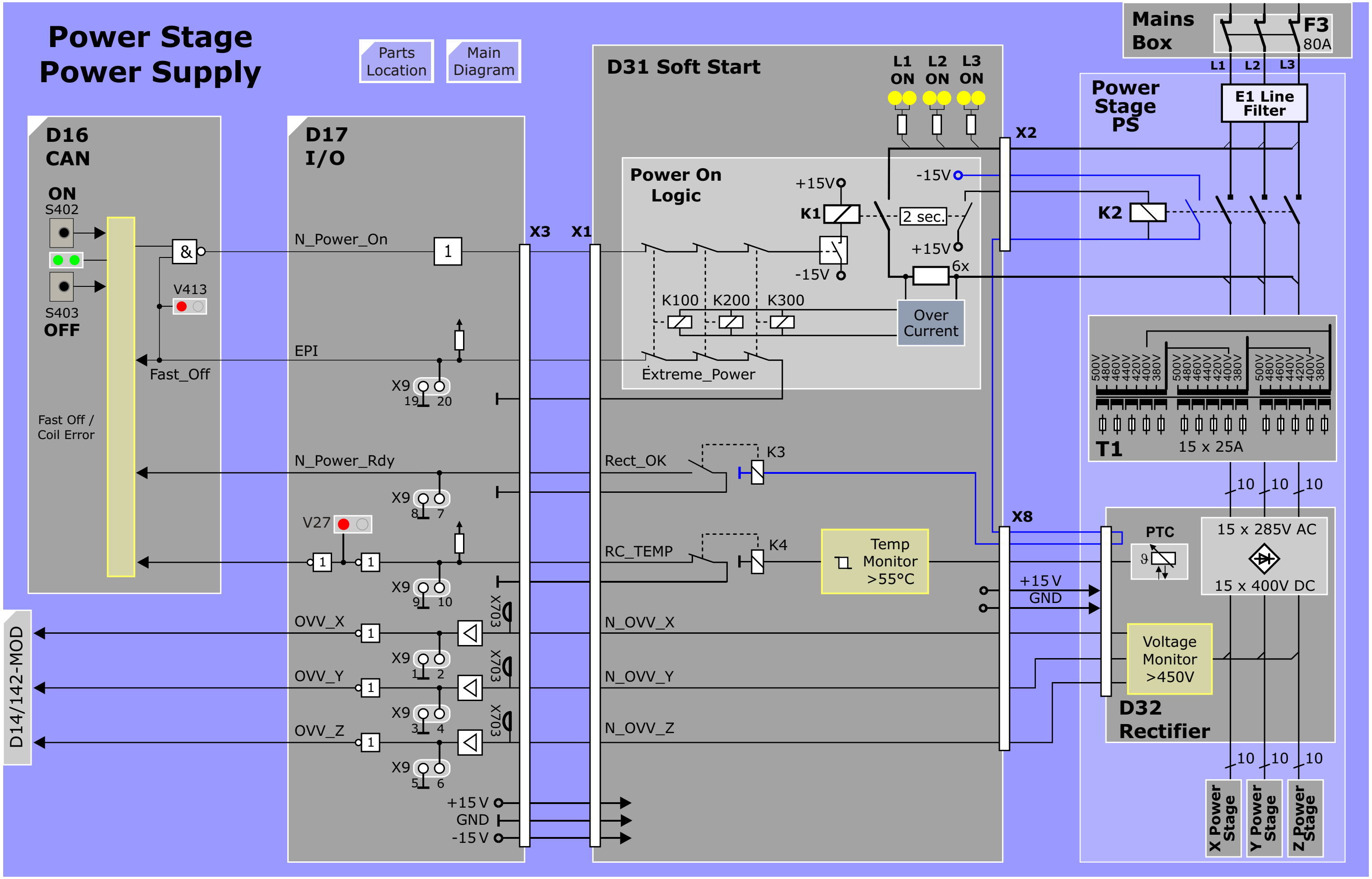
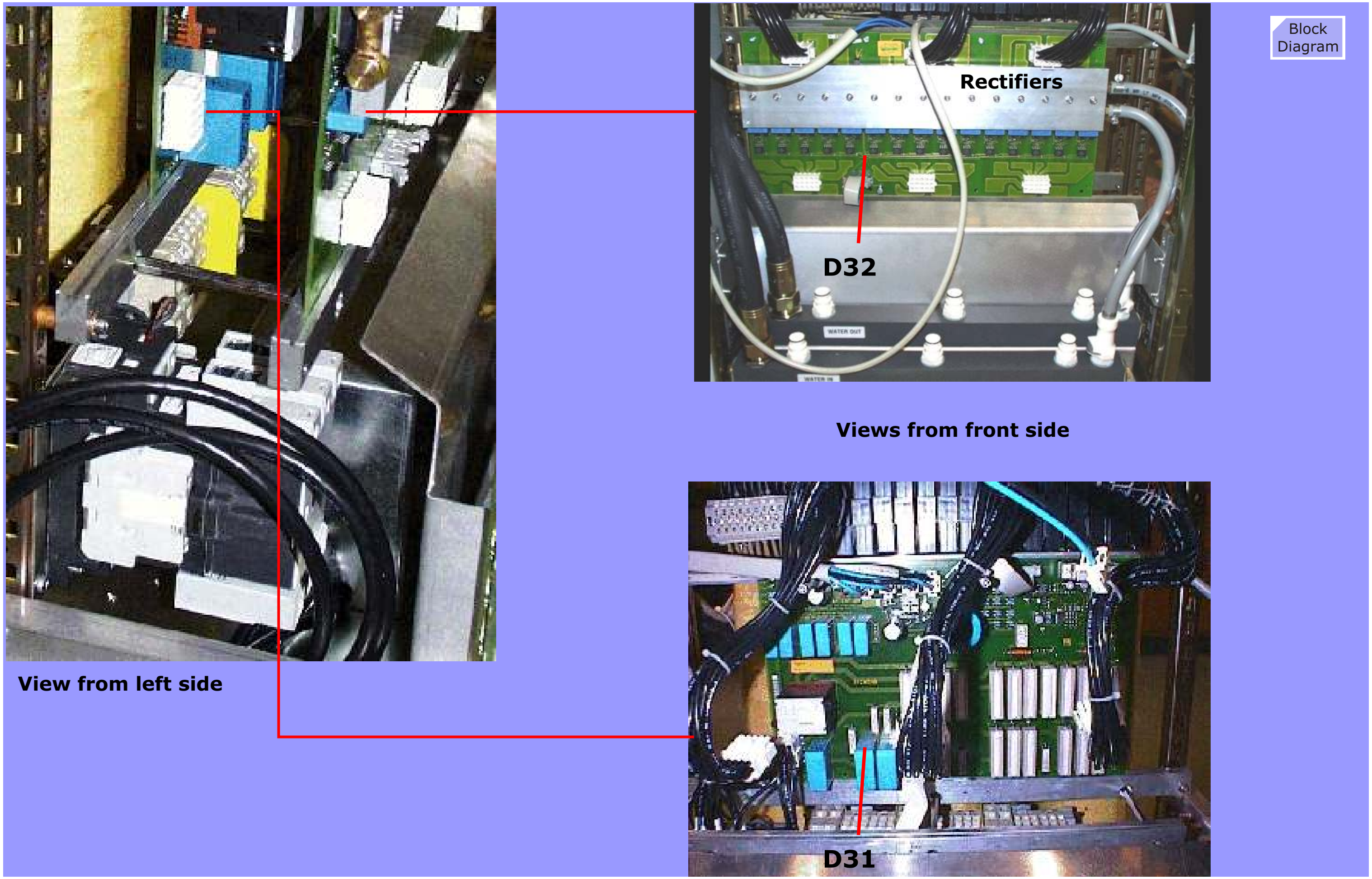


Figure 161 *Power Stage Power Supply*



Power Stage

Overview

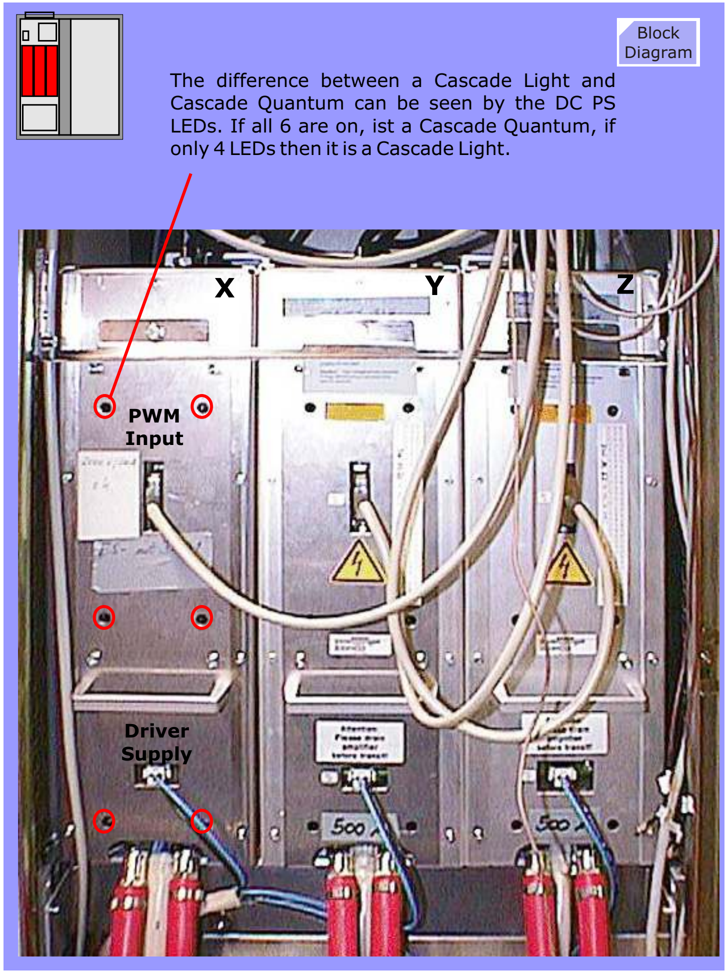
Each of the three power stage modules consists of the following:

- 5 water cooled bridges (cascades) with 4 IGBT-switches each (3 water cooled bridges in case of Light).
- 5 times DC400 V power stage supply.
- Integrated driver supply, capacity block and filter units.
- Integrated opto-transmitter and opto-receiver for drive- and monitoring-signals.
- Maximum output 500 A and 2000 V

CAUTION The Power Stage capacitors require approximately 20 minutes to fully discharge after switch-off.

NOTE This time can be shortened to under 1 minute by enabling the Service Switch **S4 right** on the D17 before turning power Stage power supply off using **S403** on the D16 board.

Figure 162 *Power Stage Modules*



LEDs

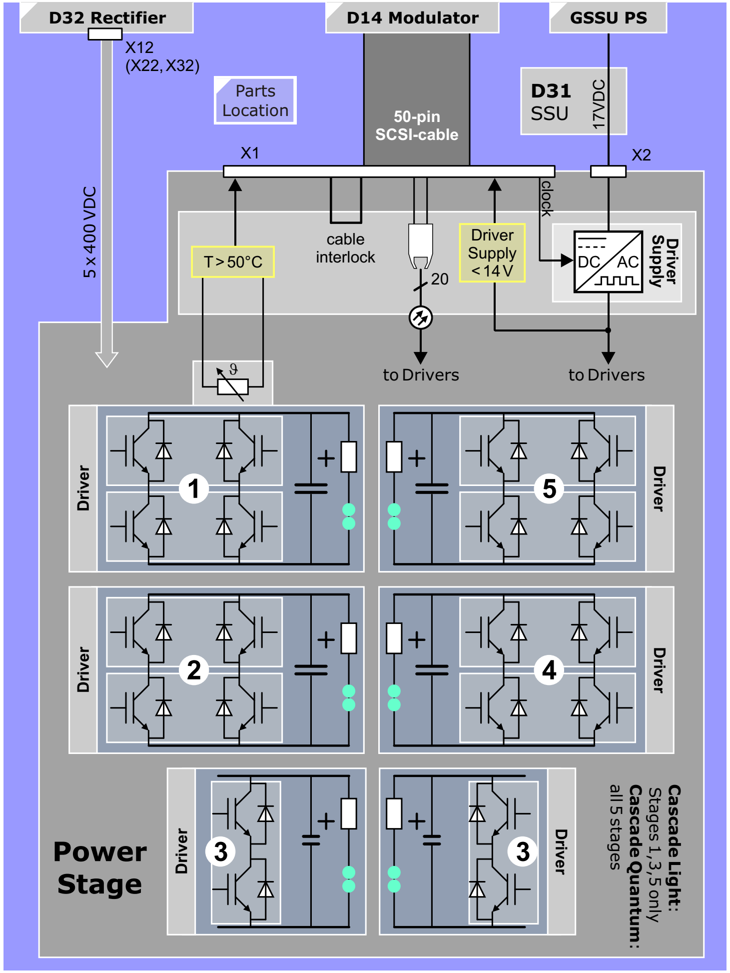
Each of the Power Stage modules consists of 5 cascades (bridges) (or 3 for the Cascade Light version) each with its own DC400 V power supply. The presence of the supply voltage is indicated by green double LED sets which are visible through holes on the front side of the Power Stages.

NOTE Power Stages for the Symphony, Sonata and Quantum systems having revision level 07 (as of August 2006) these LEDs have been removed! But the holes are still there :-)

Cable Connections

- 5 times DC400 V from the rectifier board D32
- 16.5V DC supply from the GSSU-power supply unit routed via the soft start unit D31.
- 50-pin SCSI-cable for the drive signals coming from the modulator board D14(2).

Figure 163 *Power Stage Module*



Output Filter Chokes

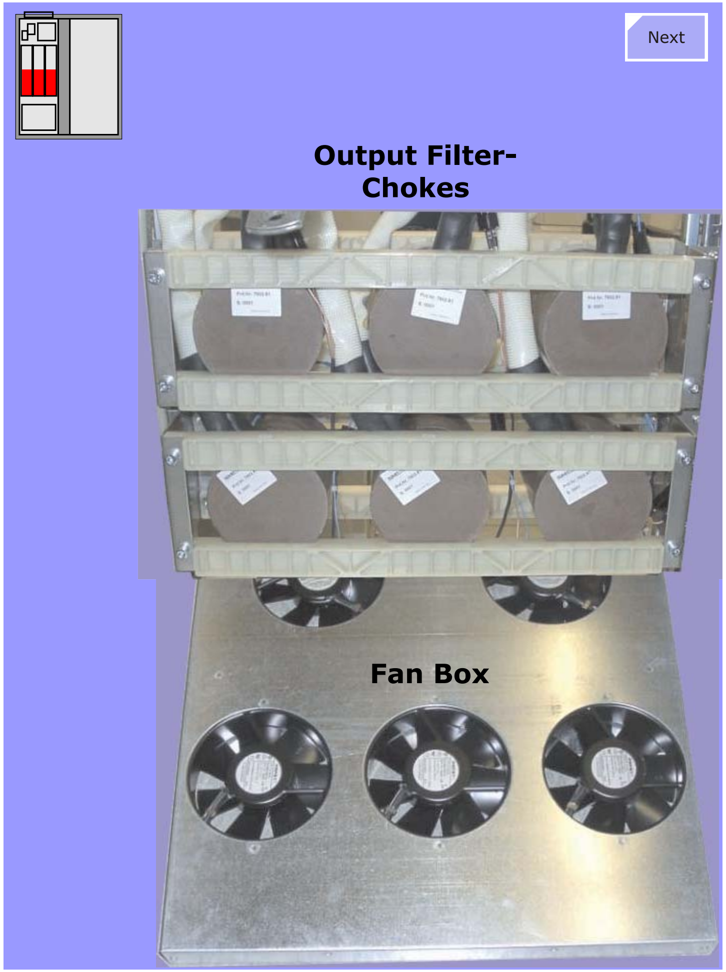
Six output filters consisting of series chokes and parallel capacitances filter out the 60/100 kHz switching ripple of the gradient current. The chokes are located behind the Power Stages. The capacitors are mounted on the under-side of the Current Sensor Assembly.

Fan Box

For cooling the Power Stages electronics and Output Filter-chokes, five 230 VAC fans are mounted on an assembly as seen in the diagram at right.

NOTE	Each fan is a Field Replaceable Unit (FRU)!

Figure 164 *Output Filter-Chokes and Fan Box*



Current Sensor Assembly

Overview

The Current Sensor Assembly consists of three DANFYSIK current sensors which supply the precise current value (actual value) for the regulation and three LEM current sensors for monitoring purposes.

The capacitors for the output filtering consisting of the output chokes, are installed in this unit.

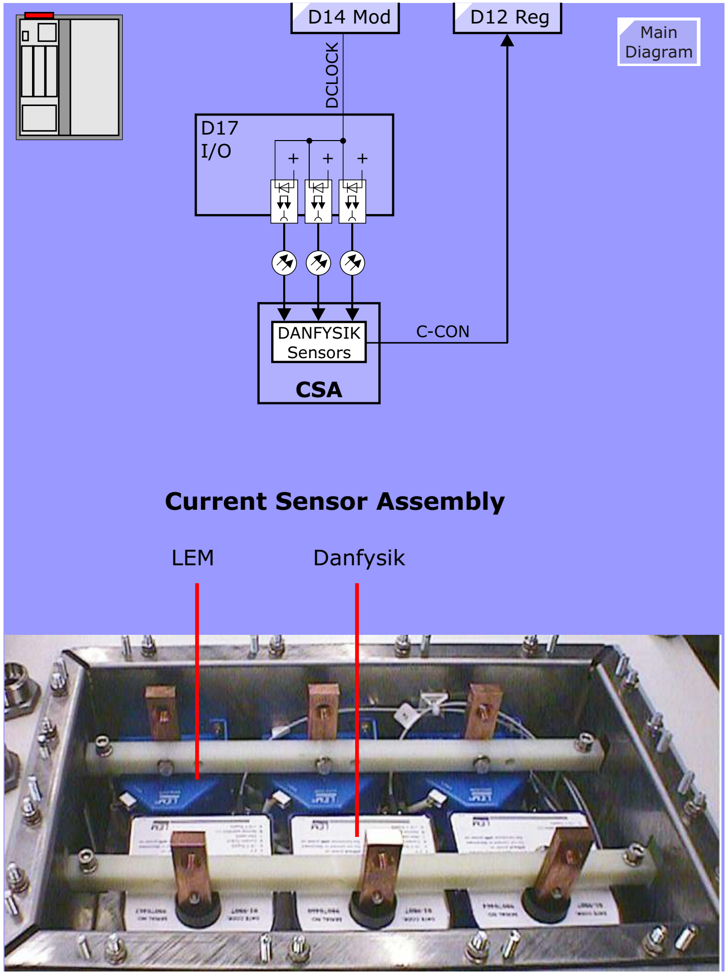


Figure 165 *Current Sensor Assembly*

Gradient Coil Assembly

The Gradient Coil is actively shielded and water cooled. The electrical parameters of the coil allow a ramping time of 40 µs/mT/m with the standard Power Stage (Turbo-Gradient) or 20 µs/mT/m with the Upgrade (Ultra-Gradient). The coil is able to handle the current that is required for a 20 mT/m gradient field.

Coil Supervision

The temperature of the Gradient Coil Assembly is monitored by the Coil Supervision Board D9 located in the Patient Table Control frame. The GPA-control (Service/CAN D6) detects the state of the monitoring via a fibre optic link (Light Board D7).

D9 Coil Temp Monitoring

The temperature of the gradient coil is measured with six thermal resistors (NTC) located in the coil. The monitoring circuit is on the Coil Supervision Board D9 where a defective NTC can be by-passed by a DIP-switch.

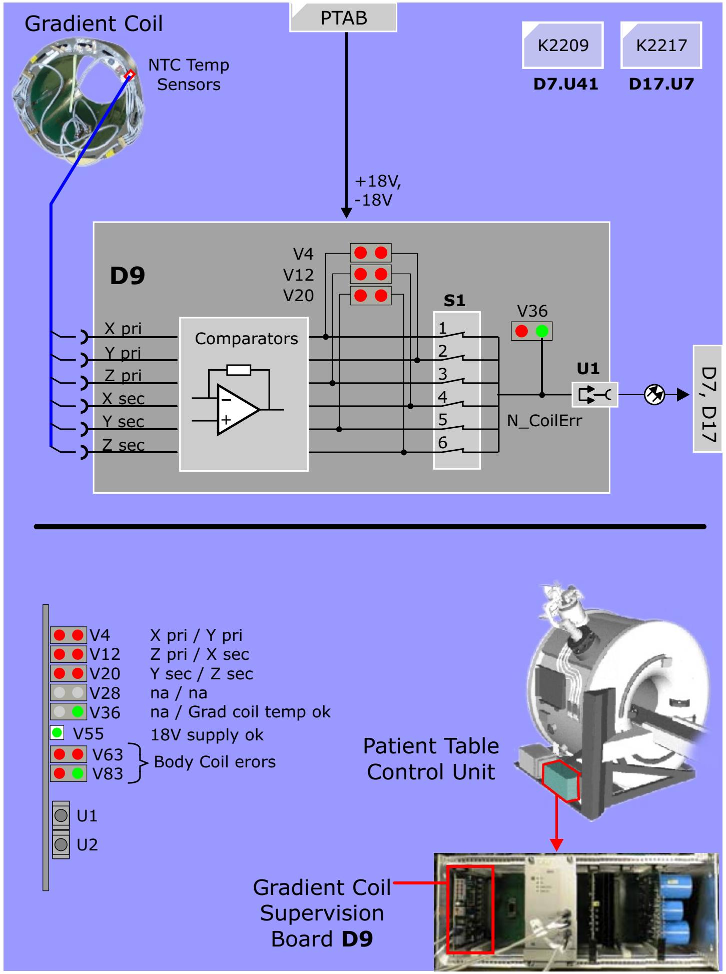
NOTE For safety reasons there must be at least four working NTCs in the gradient coil.

In addition the Body resonator temperature (water cooled) is monitored by the board D9. If the temperature is within range the green LEDs Grad. temp. ok (V36/A2) and Body Coil ok (V83/A2) are on. (For detailed description of the Coil Supervision Board D9 see TSG.)

Service Jumper X9

By means of this jumper each of the errors passing the D17-Board can be jumpered out. In such a case the desired error is not monitored anymore!

Figure 166 Coil Supervision Board D9



Section

9 Magnet System

Introduction

The Magnet Systems of the 1 Tesla Harmony and the 1.5 Tesla Symphony/Sonata systems consist of the following components:

- **OR24 Magnet** - a 1 Tesla superconductive magnet for the Harmony systems.
- **OR70 Magnet** - a 1.5 Tesla superconductive magnet for the Symphony and Sonata systems.
- **Magnet Supervision - MSUP**
  Monitoring system for magnet and refrigerator functions.
- **Alarm Box**
  Indicator for warning and alarm messages of the MSUP including the system power up-, stand by-, power down- and magnet stop buttons.
- **Magnet Refrigerator System - MREF**
  Compressor and cold head unit for cooling the cryo shields of the magnets.
- **Advanced Shim**
  An optional electrical shim unit for localized shimming of imaging volumes.
- **Magnet Power Supply - MPS**
  A mobile power supply for energizing (ramping up) and de-energizing (ramping down) the magnet. The MPS, including the ramp cables and accessories, is a service tool and not part of the system delivery.

Figure 167 *OR24, OR70 Magnets*

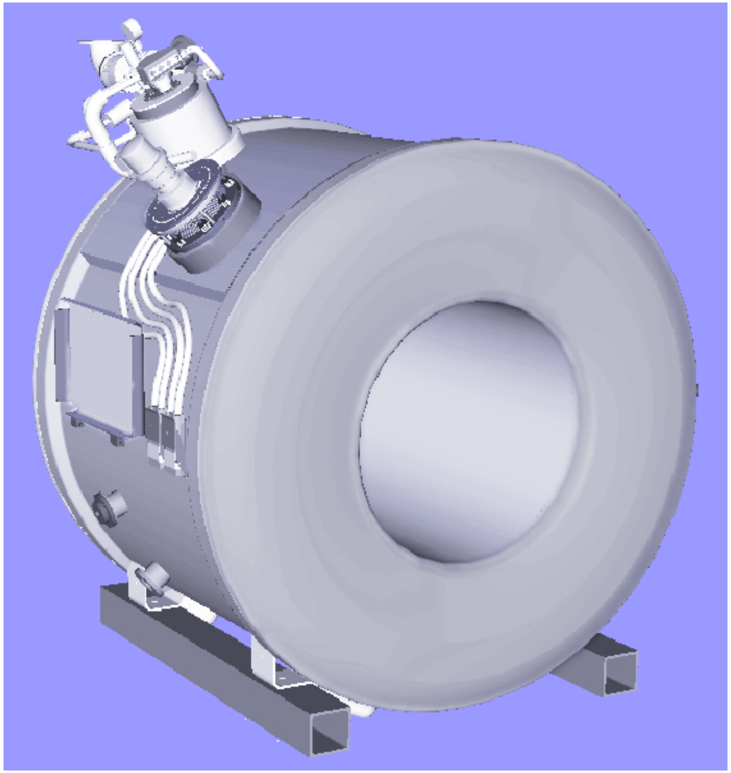
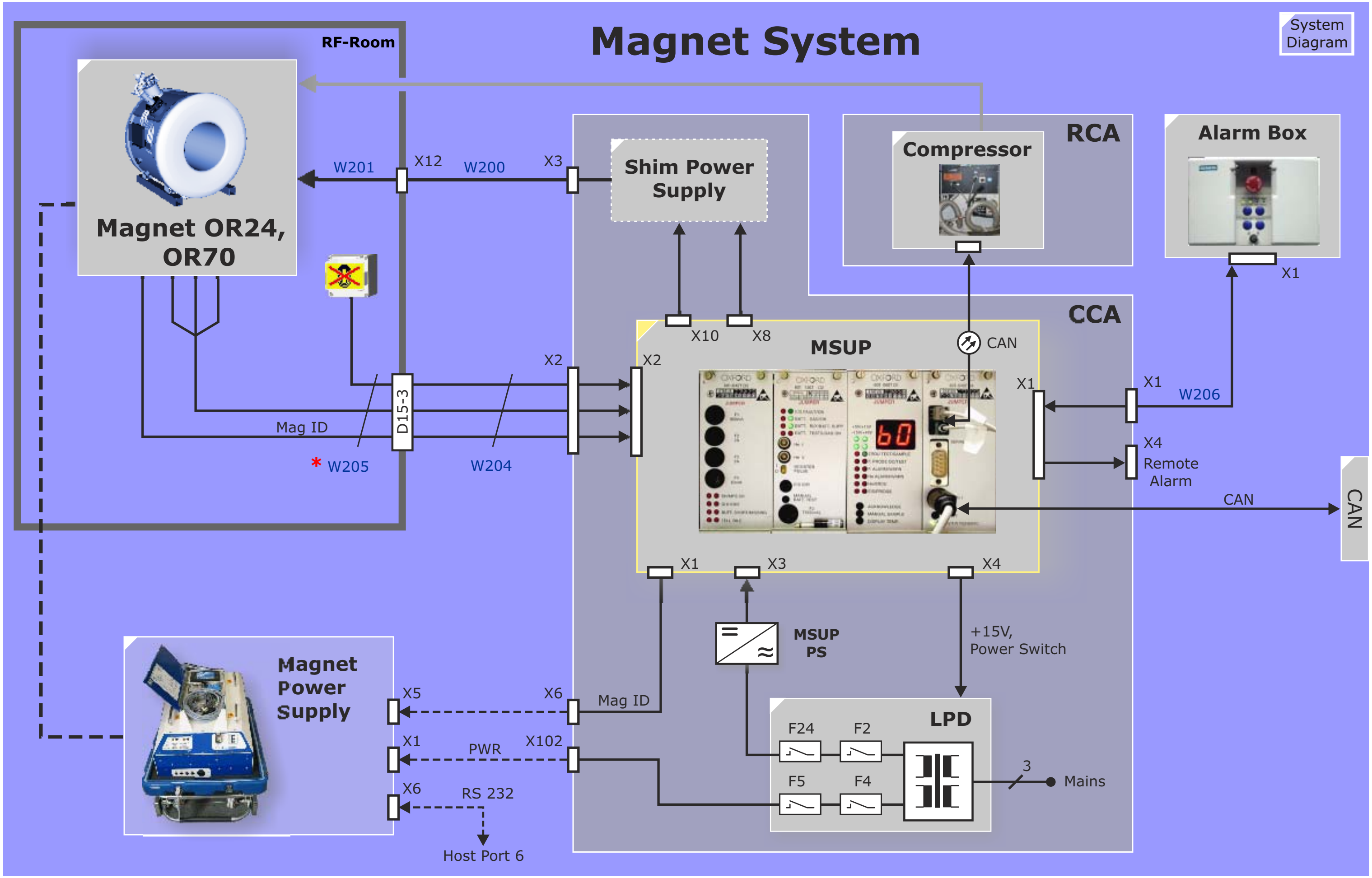


Figure 168 *Magnet System Overview*



OR24/OR70 Magnets

Overview

The OR24 and OR70 are high-field actively shielded super-conductive magnets designed for the generation of a high homogeneous magnetic field, B_0. The mechanical outer design, venting system and the electrical connections and interfaces are identical for both magnets. These magnets have gone through several revisions and improvements. The documentation will focus on the latest version.

The magnets employ active shield coils to reduce the fringe fields The magnet primary and active shield coils are sealed into a welded stain-less steel vessel and filled with liquid helium. This vessel, called a cryostat, magnet or helium vessel, is assembled into an outer stainless steel shell which is placed under vacuum to eliminate convected heat from entering the magnet vessel as a means to reduce helium boil-off. Between the outer shell and the magnet vessel are two aluminium radiation shields whose purpose is to reduce the influx of radiated energies which would cause an increased helium boil-off. As additional measures to reduce helium boil-off, the magnet is suspended within its shell by fiberglass suspension elements as a means to reduce the influx of convected energies.

Cryogenic and electrical access to the is provided by an integrated turret.

The radiation shields are cooled by a two stage Gifford-McMahon refrigerator (see also Refrigerator System) to buffer the outer temperatures of ~300K from the internal magnet vessel temperature of 4K (liquid) to 6K (gas).

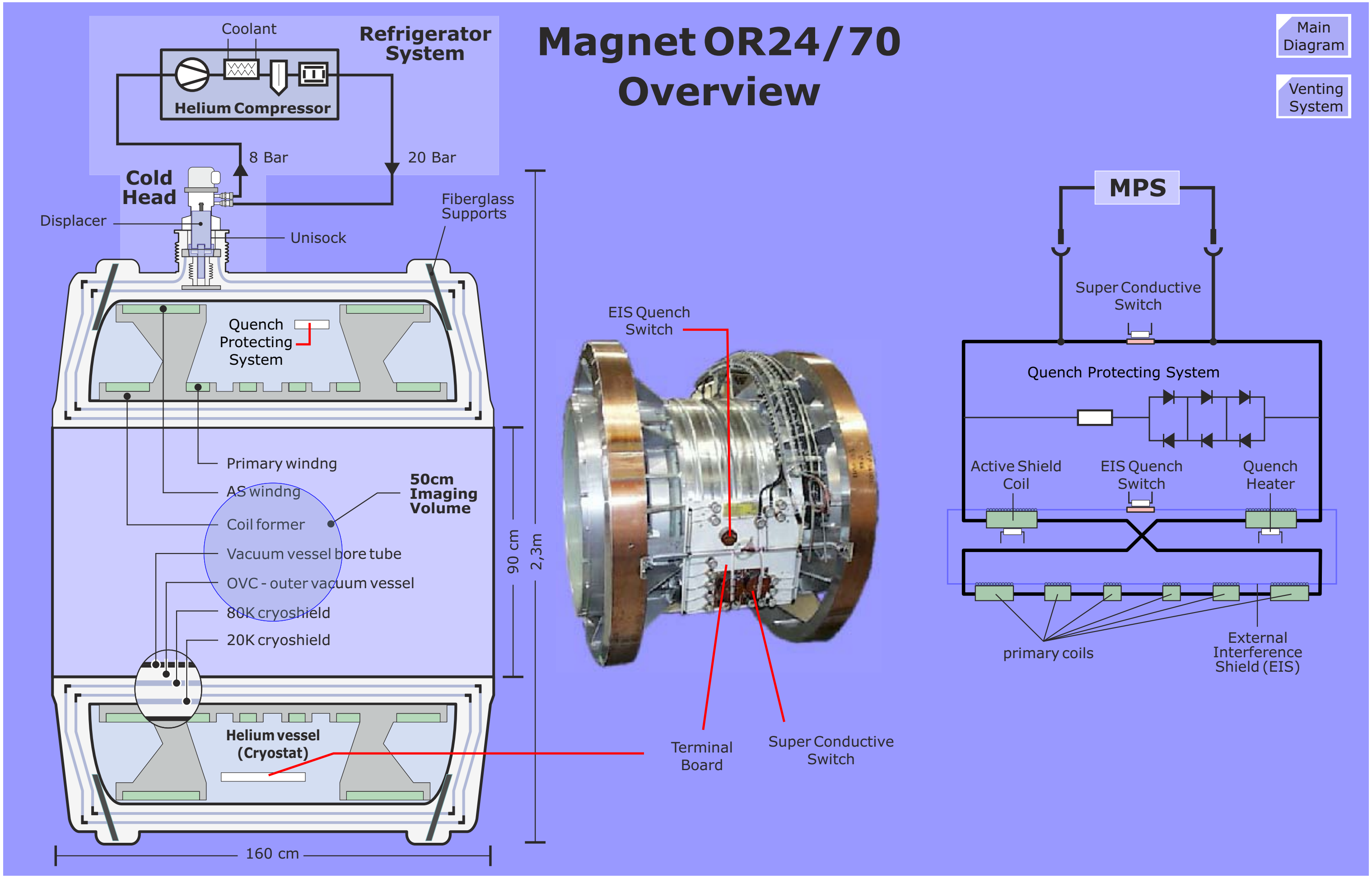
The Magnet consists of two main components:

- **Electrical Circuits** - coil windings, temp sensors, LHe sensor
- **Venting System** - bleed and bypass valves, and plumbing for the boil-off and quench gasses

Specifications

	OR 24 Harmony	OR70 Symphony	OR41 OR42 Impact	OR35 Vision	OR64 Trio
Field strength	0.95 T	1.494 T	0.95 T	1.50 T	2.895 T
0.5mT fringe field distance	Z 3.9 m X,Y 2.3 m	Z 4.0 m X,Y 2.5 m	Z 3.9 m X,Y 2.3 m	Z 4.7m X,Y 2.6 m	Z 6.0m X,Y 3.9m
Max. field decay	37 kHz /Year	55 kHz /Year	37 kHz /Year	55kHz /Year	108kHz /Year
Magnet weight (100% He, no coils or table)	3350 kg	3900 kg	5000 kg	8650 kg	9970 kg
Magnet length	1.6 m		1.89 m	2.284 m	2.15 m
Magnet height (with turret)	2.33 m		2.37 m	2.508 m	2.53 m
Magnet outer diameter	1.97 m		1.99 m	2.15 m	2.15 m
Magnet bore	0.9 m		same	0.905 m	0.92 m
Imaging Volume	500mm DSV	same	same	same	same
Ramping time to nominal field	30 min	45 min	30 min	40 min	120 min
Typical Current	380A	585A	355A	690A	487A
max. Voltage during ramping	9.9 V		same	same	same
Stabilization after ramping	<=5ppm/hr after 10 min		<=5ppm/ hr after 10 min	<=5ppm/ hr after 20 min	<=5ppm/ hr after 15 min
100% Helium level	1806 l	1710 l	1320 l / 1233 l	1630 l	1319 l
30% Helium level (min level)	454 l	436 l	380 l / 317 l	300 l	715 l
maximum He boil off	0.075 l /h		not specified	0.10 l/h	not specified
typical He boil off	0.050 l /h		0.1 l /h	0.10 l/h	not specified

Figure 169 *Magnet Supervision Block Diagram*



Venting System

Hazards and Risks

Given the extreme temperatures and the phenomenal liquid-to-gas expansion ratio of liquid helium, it should almost go without saying that the hazards which exist with a vessel containing up to 1600 liters of liquid helium are not exactly small. The main dangers confronting both man and machine are frostbite, asphyxiation and explosion of the magnet due to an over-pressure situation (ice-blocked magnet vessel).

With these dangers vividly before our minds-eye we want to spend some time here to discuss the components of the magnet venting system which provides features for both safety and serviceability.

WARNING This information is provided for general informational purposes only. Work on the magnet should not be performed by inexperienced personnel.

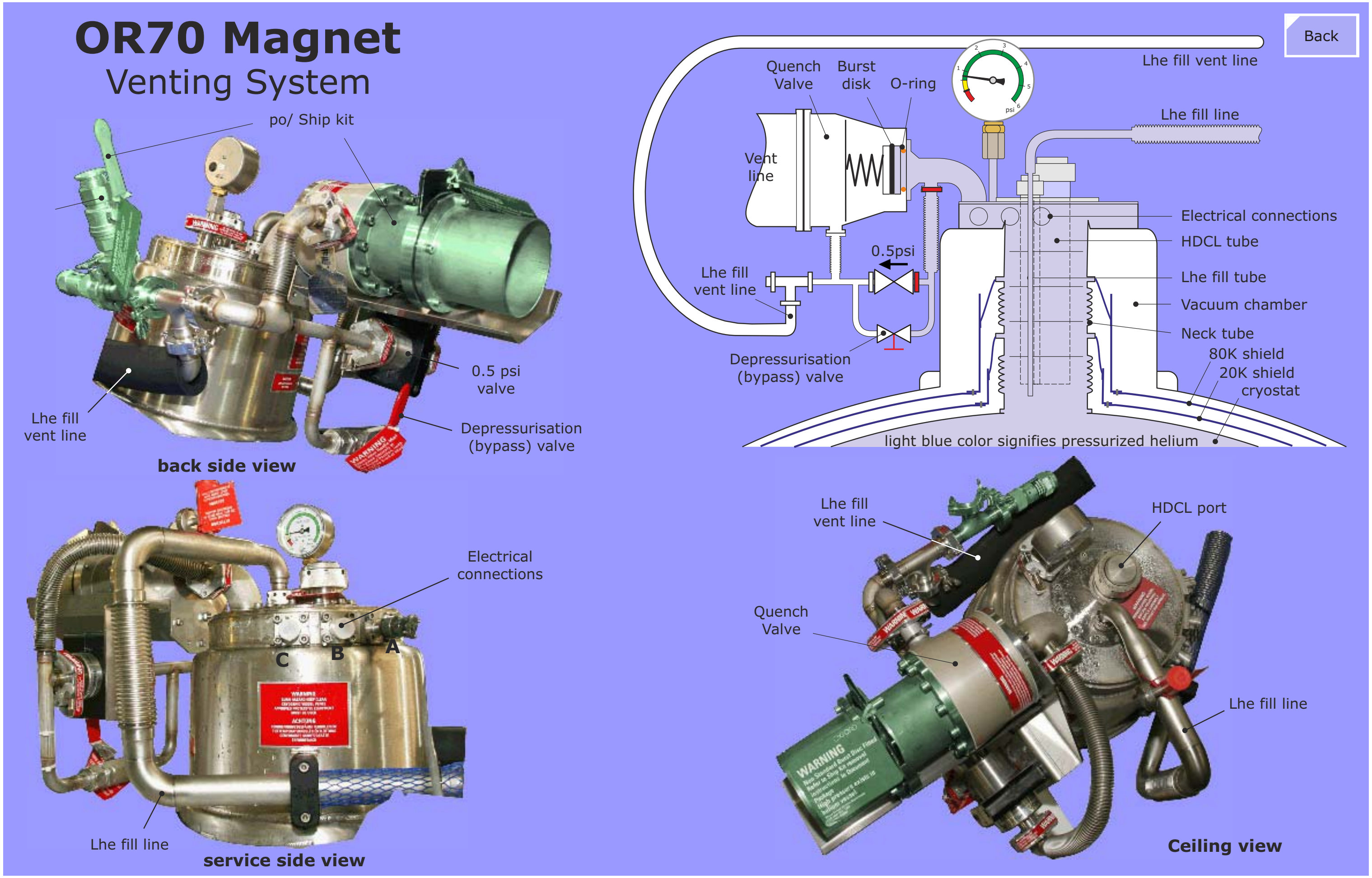
Introduction

The components of the venting system provide these functions:

- Allow an unimpeded exiting of the normal helium boil-off gasses or quench gasses developed in the magnet safely out to atmosphere.
- Prevent air from entering the magnet - ingression of air will cause an ice blockage which could lead to a total blockage of the magnet and eventually bring the magnet to explode.
- Provide a service entry point into the magnet.
- Provide indication of pressure within the magnet.

Component	Description
Service turret	The service turret is the interface into the He vessel for cryo and gas transfer as well as for all electrical connections
Cold Head	The cold head is a part of the refrigerator system, keeping the He boil off at a low level
He syphon	The vacuum pipe is a part of the He filling device
Ramp connectors	The quench valve and overpressure valve is connected there to allow safe venting of boiled-off He gas to quench pipe
Quench pipe	Piping system for directing boil-off and quench gasses to the outside of the building
Quench valve	Valve that opens in case of a quench to release the large amount of boiled-off LHe safely into atmosphere. It contains a spring loaded valve that opens at a pressure of 15 psi. In addition a graphite bursting disc will break if a pressure of 20 psi is exceeded.
Overpressure valve	The overpressure non return valve keeps the magnet He-pressure at 2/3 psi
Depressurization (Bypass) valve	Opens a connection from the He vessel into the quench pipe to depressurize the vessel in case of service actions
Pressure gauge	Indicates the pressure inside the He vessel
Current lead cooling pipe	Used for cooling the current lead during ramping (will be installed only during ramping)

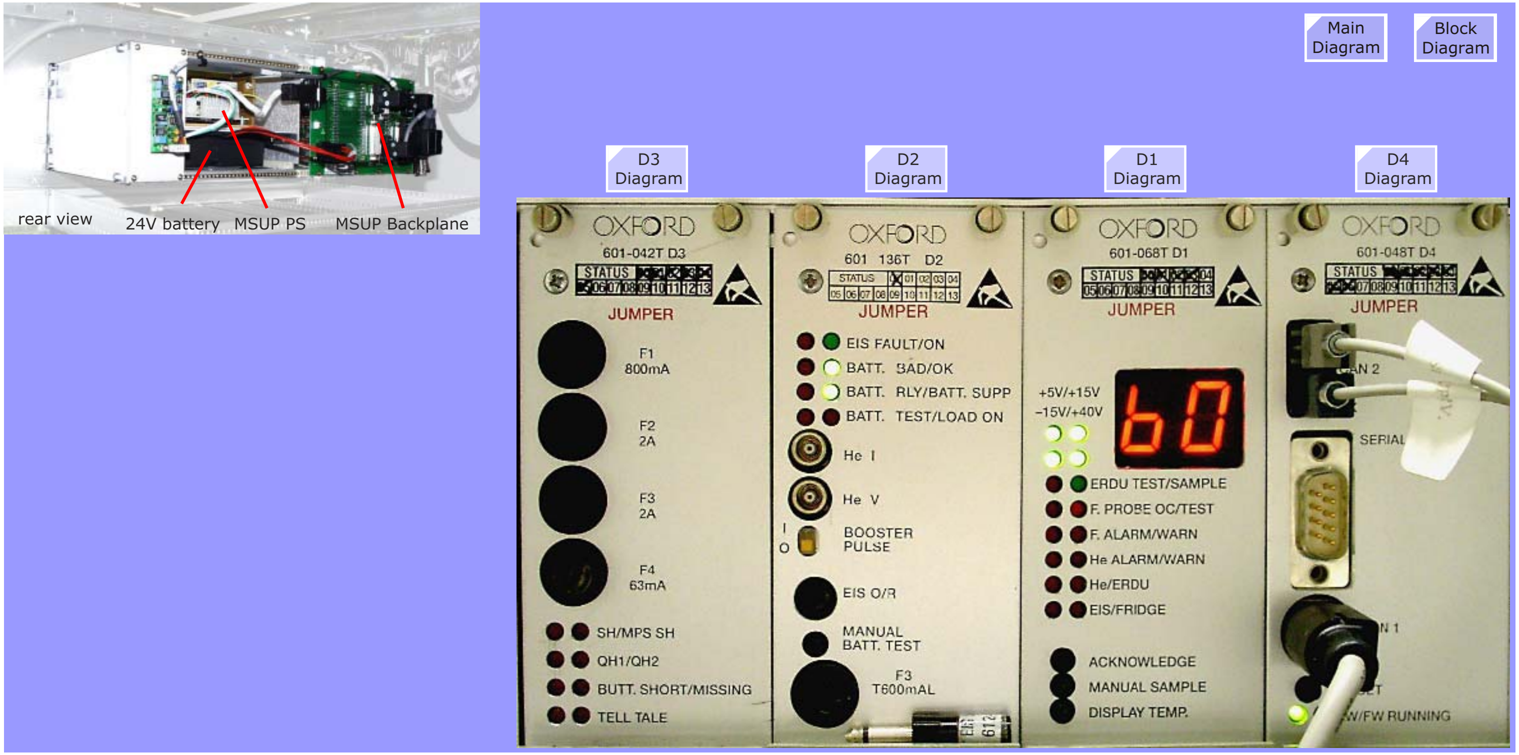
Figure 170 *Magnet Venting System OR24, OR70*



Magnet Supervision

The Magnet Supervision **MSUP** is responsible for supervising various magnet vital statistics and providing an emergency run down function. The MSUP is designed for use with both the OR24 and OR70 magnets as well as with the older OR41/OR42 and OR35 magnets.

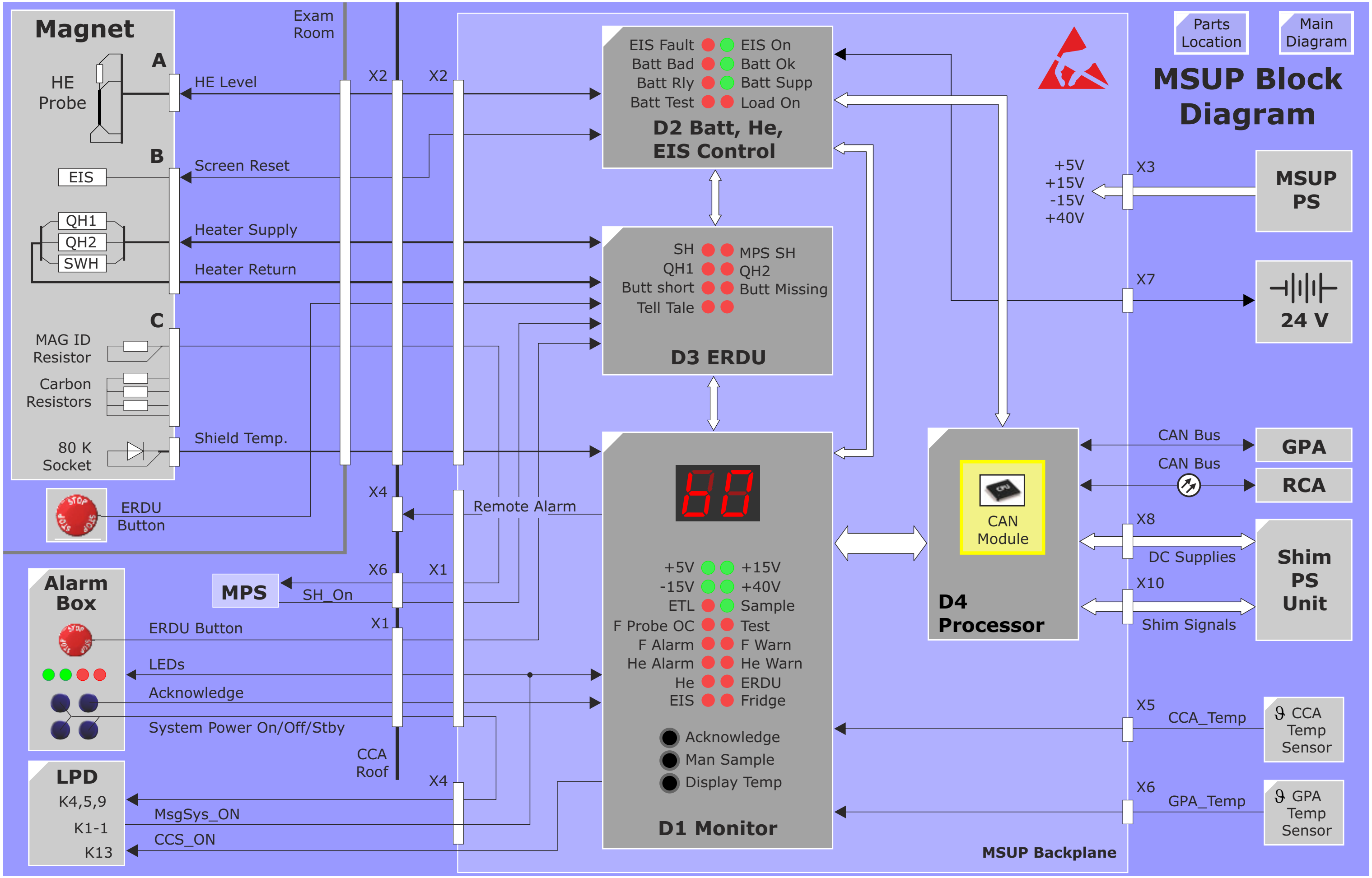
Figure 171 *MSUP*



The MSUP includes the following features:

- Emergency run down unit (ERDU)
- Switch and quench heater supervision
- Helium level metering
- Shield temperature monitoring (Cold Head efficiency)
- Cabinet temperature monitoring
- Screen coil reset
- Error / Status reporting

Figure 172 *Magnet Supervision Block Diagram*



D1 Monitor

The D1 provides the following functions:

- error / status logic - error and status message latching and LED indicators, numeric display (for He level), signal level conversion (analog to TTL)
- magnet shield temperature monitoring
- voltage monitoring
- cabinet temperature sensor interface

Error & Status Logic

The error/status signal outputs of the monitoring circuits on the D2 and D3 boards have analog voltage levels and need to be converted to TTL levels before being sent to the D4 Processor. Error signals are latched, warning and status signals are indicating only with LEDs and messages to the customer. Latched errors set off an audible alarm on the Alarm Box via the **Buzz** signal. The audible alarm can be silenced with the Acknowledge button on the Alarm Box and with **S1** on the D1 board. S1 additionally resets the error latch.

Remote Alarm

An error also activates relay **K1** connected to Plug X4 on CCA Roof is driven together with the buzzer activating an external alarm. Since relay contacts are used galvanic separation is provided from any external connection.

Switch Heater Error Masking

When ramping the magnet the **HTR_OK** error must be disabled. It is masked out with the **SH_I_On** signal.

Helium Level Display

The two digit display on the D1 is for display of the helium level. The value is driven by the D4 Processor. LEDs: there are two LEDs. The HE-WARN LED is turned on when the He-level is below the warning level. The level is set under SESO, Magnet and Cooling. The HE_ALARM LED is turned on to indicate that the minimum level ($\leq$30% LHe-level) is reached. The alarm box Helium alarm and buzzer are also activated.

ERDU Test Load Connected

Connecting the ERDU lest load (ETL) activates the buzzer and lights the ERDU TEST LED. This serves as a reminder to the CSE not to forget to remove the test load, since the ETL disables the ERDU buttons making an emergency quench impossible.

Voltage Monitoring

The +5 V, +15 V, –15 V and +40V voltages are continuously monitored. If any of these voltages drop too low the **PWR_FAIL** or **40V_FAIL** signal will inform the D4 Processor.

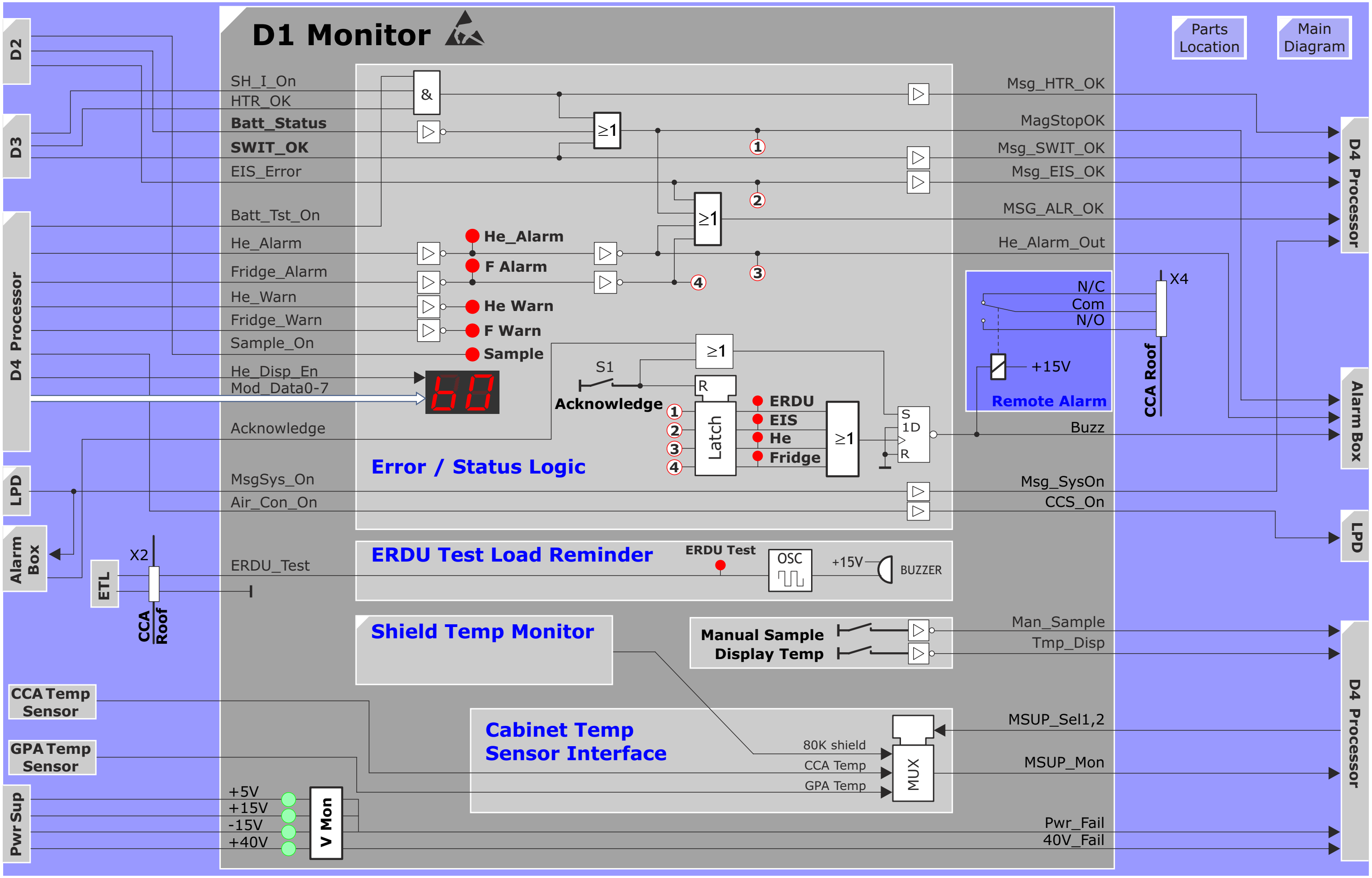
LEDs

LED	Description
He	Helium level alarm - helium level has dropped below 30%
ERDU	an error with either the 24V battery, ERDU buttons or heaters
EIS	EIS circuit or probe fault
FRIDGE	magnet shield temperature above 80 K. Either Cold Head is not running or has become too old

Buttons

Button	Function
Acknowledge	to acknowledge He, EIS and ERDU errors. Turns off the corresponding LEDs
Manual Sample	initiates a helium level and shield temperature measurement and an EIS reset
Display Temp	for displaying the cabinet temperature

Figure 173 D1 Monitor Block Diagram



Shield Temperature Measurement.

The function of the refrigerator system is indirectly monitored via the shield temperature of the magnet. If the refrigerator fails or becomes inefficient (Cold Head wears out) the shield will heat up. This condition needs to be recognized for two reasons: firstly, warm cryo-shields will increase the helium boil-off and, secondly, will lead to higher Eddy currents causing poor fat saturation adversely effecting image quality. and a warning or alarm will be issued after the next shield test.

NOTE In case of a complete refrigerator failure the helium boil-off rises to 3% or more per day after just 24 hours after the failure!

The shield temp measurement is initiated by the D4 Processor with the Fridge_On signal. When active, +5V is applied to a constant current source which runs a current through the temperature sensor diode located on the 80 K Unisock. The diode resistance is proportional to the temperature. The voltage drop across the diode is read into the D4 Processor's ADC and evaluated by software. The measurement takes approximately 10 seconds.

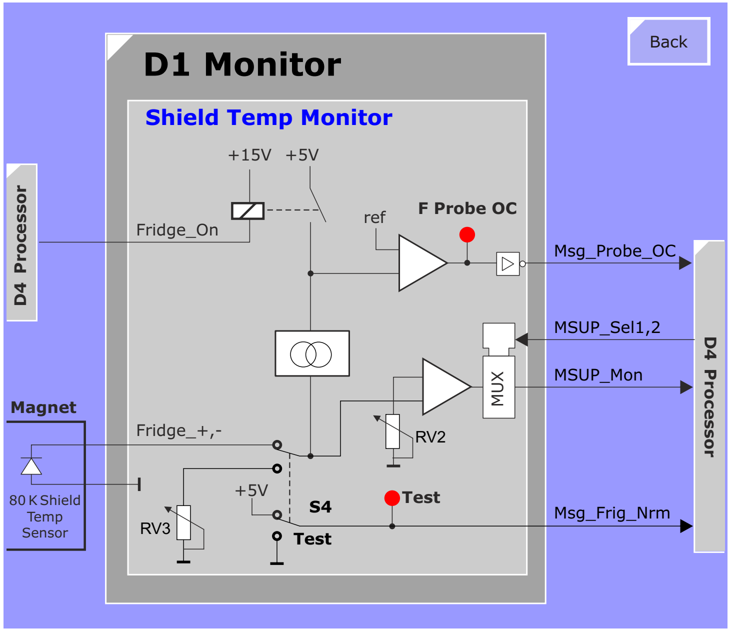
If the temperature exceeds 75 K the **Fridge_Warn** signal is activated turning on an LED on the D1 and the customer will be informed with a warning message in a pop-up window under NUMARIS customer.

If the shield temperature is above 80 Kelvin the **Fridge_Alarm** signal is activated setting the F. ALARM LED on and causing an annoying audible alarm.

If the diode is defective or the connection to the diode interrupted, a probe fault will be generated which turns on the F. PROBE O/C LED during the sampling period, and also causes a fridge alarm.

There are three additional shield temperature sensor diodes (two of them mounted on the 20K shield) which can be measured manually. See the TSG.

Figure 174 *Shield Temp Monitor*



D2 Battery, He and EIS Control

ERDU Battery Control

A sealed rechargeable 24V battery in the MSUP keeps the magnet stop (ERDU) in operation in case of line failures. The battery voltage is checked every 24 hours by the D4 processor via signal BAT_TST_ON. The voltage is compared to a reference value and sent back to the processor D4 as BATOK. From this the D4 drives an OK or fail LED on the D2 and transmits the signal MAGSTOP OK to the alarm box. If during test the voltage of the battery falls below 20V the BAT. BAD LED at the D2 board gets activated.

Battery Charging

The battery is trickle charged. The voltage for charging is fixed. The battery has to be replaced every 3 years during preventive maintenance.

USTOP and UBATT

In case of Magnet Stop (see D3 ERDU) the regulated 24V voltage USTOP is used for driving current to the quench heaters. In case of power failure, the battery will provide the 24V for USTOP.

Screen Reset

This module periodically quenches out any circulating current in the screen coil of the magnet (by opening it) and it prevents any current from being induced into the screen coil during ramping the magnet. The function is activated by:

- EIS_ON. This signal comes from processor D4 once every 24 hours, at a time when the system is not used. The time point is set under SESO, Magnet and Cooling. The output is turned on for 15 seconds.
- SH_I_ON from D3 ERDU. The interface to this signal causes an output to be delivered for as long as the magnet switch heaters are on, plus approximately 10 minutes extra for settling.

Connector EIS O/R: You can also permanently open (heat) the screen coil (for example in order to increase the pressure in the magnet) by shortening the connector **"EIS O/R"**.

Screen Reset Monitoring

If a false output or no supply when requested is detected, the message EIS_Error is transmitted to board D1 Monitor.

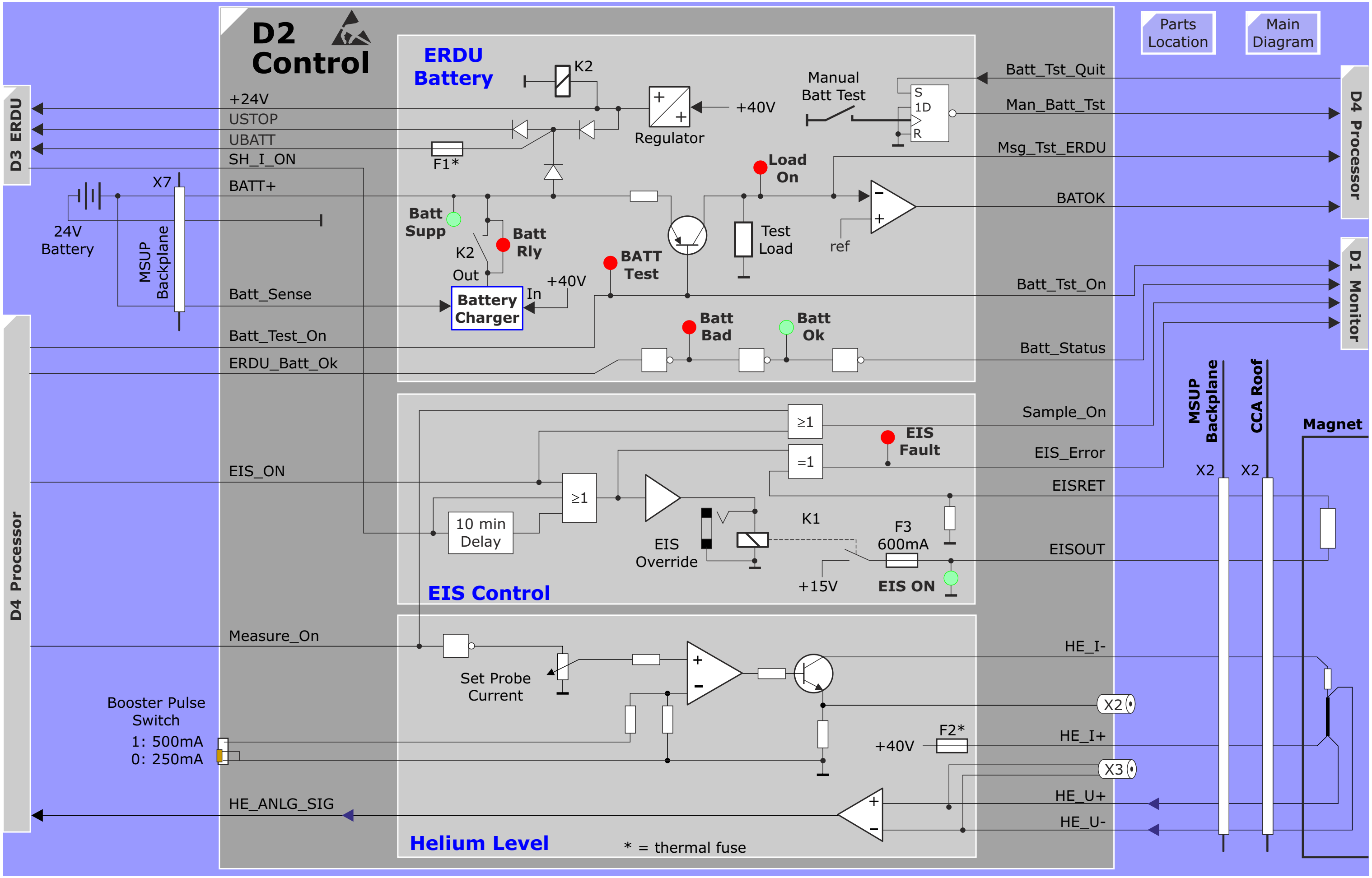
Helium Level Measurement.

The He level is measured by a superconductive wire located in the He-tank. During the measurement a constant current is driven through the wire (He measuring probe), which heats it up. This makes the wire normally conductive where it is not covered by liquid helium. The measured resistance of the normally conductive part of the probe is thus a measure for the He level. The constant current for the He measuring probe is 250 mA, that can be adjusted with the Poti RV1. The processor in D4 starts the measurement in 24 hour intervals with the signal MEASURE_ON. The voltage drop generated by the constant current in the probe is measured via a differential amplifier and is transmitted to the processor D4.

Manual measurement: in addition it is possible to measure with a DVM connected to X3 (He V, see TSG) and X2 (He I, see TSG) directly the He level. The measurement has to be started by pressing the button "Manual Sample" on the D1 Monitor.

Switch Booster Pulse: For some magnets, especially when not under field, a bigger current for the He measurement is required. If you activate the switch "**Booster Pulse**", a current of **500mA** will be driven to the He-probe during the He measurement.

Figure 175 *MSUP D2 Block Diagram*



D3 ERDU

ERDU (Magnet Stop)

The ERDU has the capability to quickly run down the main magnetic field in an emergency. This is done by heating the superconductor by means of the quench heaters and switch heaters. The current for these heaters is supplied by the ERDU when a quench button is activated. When pressed, relay K1 activates and applies voltage to the Quench Heaters (QH1, QH2) and the Switch Heater (SH). The voltage is supplied by the He, EIS & Battery D2.

When one of the magnet stop switches is depressed relay K1 is activated applying 24 VDC from either USTOP or UBatt (supplied by D2 Control board) to quench heaters 1 and 2 and switch heater 1 via the D81 ERDU Power board. The voltage is normally supplied by the TEDI board D80 (USTOP_Net). In case of voltage failure the 24 Volt battery supplies the voltage via a de-coupling diode (USTOP).

The heating current flows to ground via diodes/resistors which are part of the heater monitoring circuit.

Tell Tale Indicator

When K1 energizes (a quench button has been pressed) fuse F4 (63mA) will blow and the Tell Tale LED will light.

Heater Monitoring

Via a protective resistor the voltage UHTR permanently drives a small current (about 10 mA) through QH1, QH2 and the Switch heater. The voltage drop is compared to reference values in two comparators. It is thus possible to detect breaks in the lines to the heaters or short-circuits. Each heater can activate a LED on board to indicate a heater fault. The actual alarm signal is summarized from all messages and transmitted to the D1 Monitor.

Switch Monitoring

The two ERDU remote buttons driving relay K1 on D3 each have one resistor. Via this resistor, the relay and a resistor against ground a permanent current is driven. A voltage drop compared to reference values in comparators indicates whether both switches are connected. In case of a fault, a LED will light and a message SWITOK is transmitted to the monitor board D1.

DIP-Switch SW2

Allows to connect a third (optional) ERDU button to the MSUP.

- 2 ERDU buttons (standard): SW2A closed, SW2B open
- 3 ERDU buttons (optional): SW2A open, SW2B open

Switch Heater

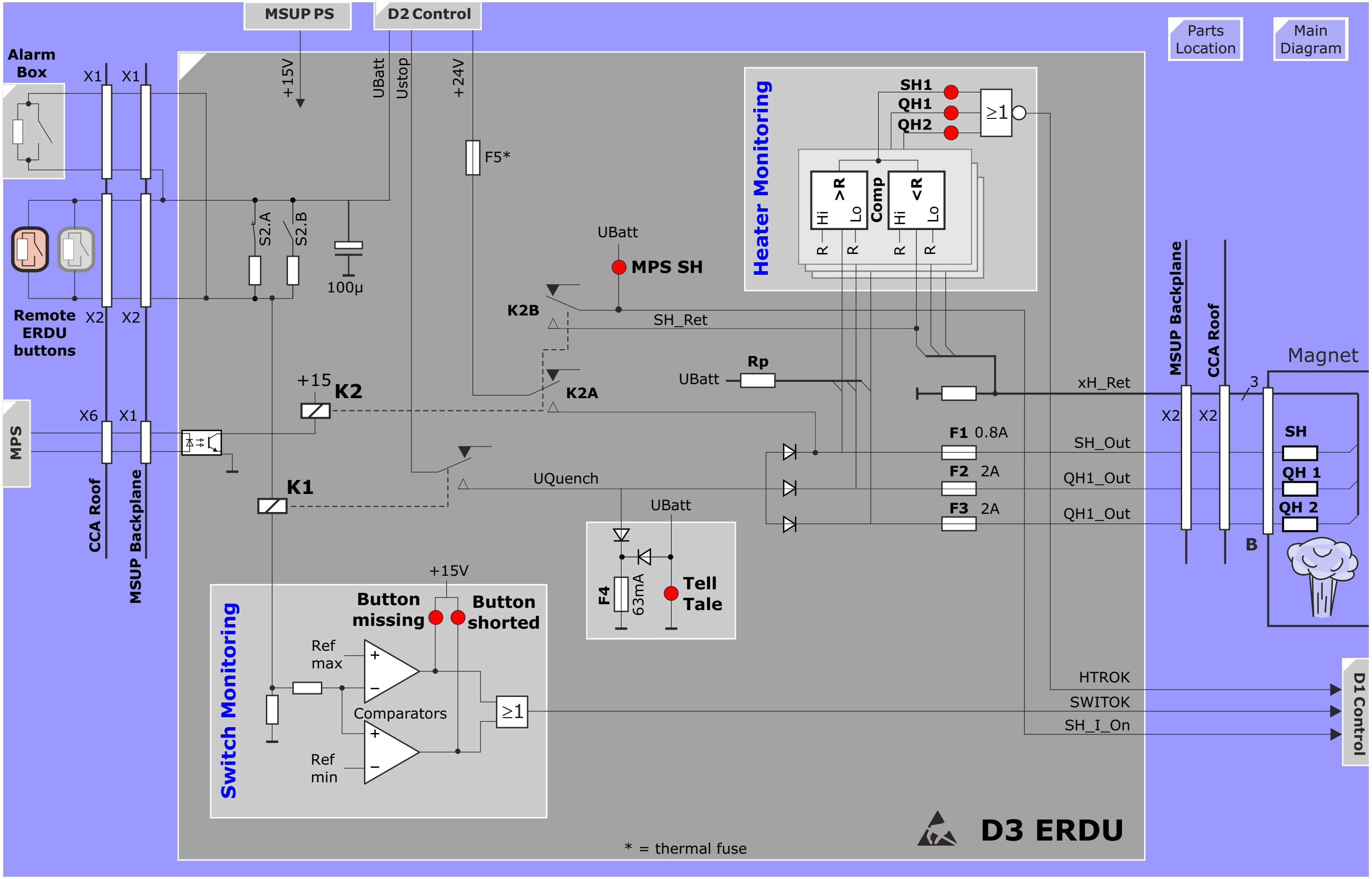
The Switch Heater is switched on when the signal +5V_MPS is energized from the MPS.

ERDU Test Load (ETL)

The ERDU Test Load is a **service tool** used to test the monitoring functions of the MSUP. Push buttons on the test load simulate faults such as heater load resistance too low or too high. These simulated faults have to be recognized by the D3 ERDU switch monitoring. When the ERDU buttons are activated, LEDs on the test load light up to indicate correct operation.

For a detailed description how to use the ERDU TEST LOAD see the Safety Related Tests documentation.

__Figure 176__ D3 ERDU Block Diagram



D4 Processor

Function:

The D4 is the communication interface between the magnet monitoring system and the MMC. The Monitoring functions are:

- ERDU: battery, heater and buttons
- Helium level, Screen reset, cryo-shield temperature, CCA/GPA air temperature

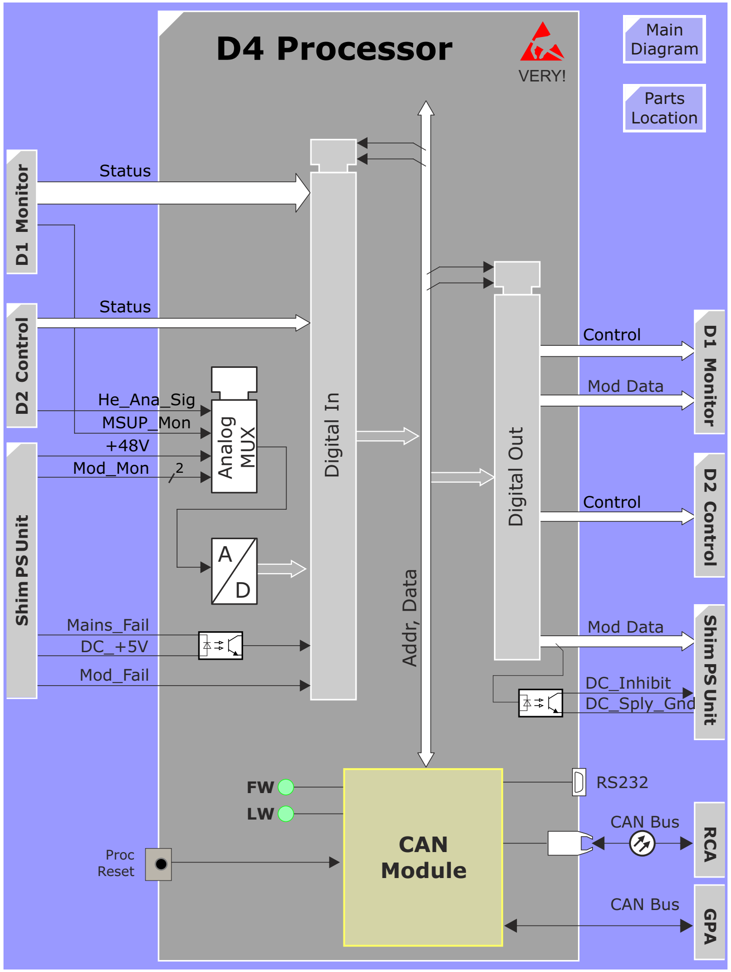
In addition, the processor board controls the Advanced Shim.

Actual Magnet Values

Date	He-Level [%]	Shield Temp. [K]	Screen Reset
3. 7. 6	78.6%	53K	OK
3. 7. 6	78.7%	55K	OK
2. 7. 6	78.8%	55K	OK
1. 7. 6	78.9%	55K	OK
30. 6. 6	78.8%	53K	OK
30. 6. 6	78.8%	53K	OK
30. 6. 6	78.9%	55K	OK
29. 6. 6	78.9%	53K	OK
29. 6. 6	79.0%	55K	OK
28. 6. 6	79.0%	53K	OK

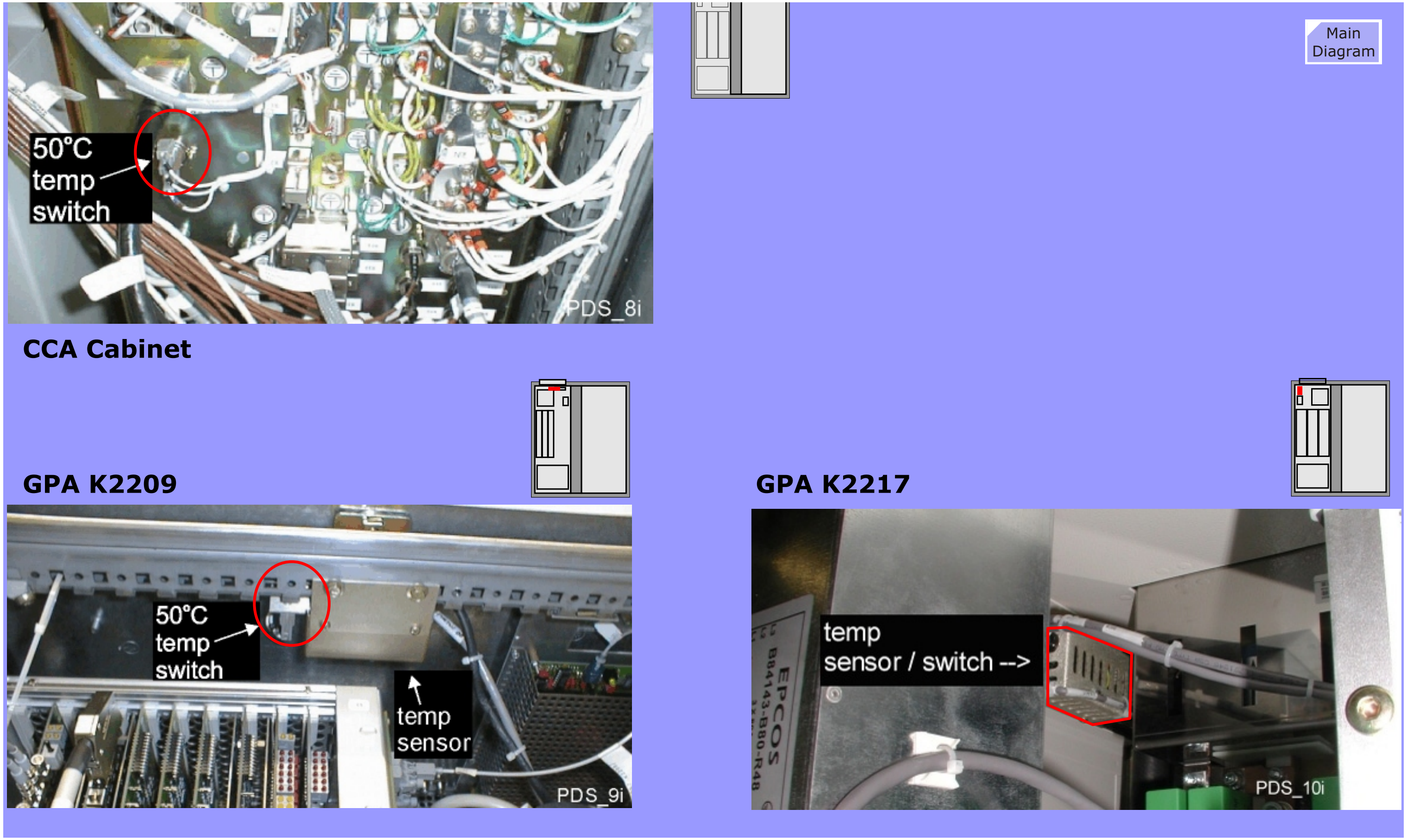
Magnet	Helium Level Meter	min (0%)	102.2	Ohm	max (100%)	2.0	Ohm
	Time Setting	Daily at	02:00	o'Clock			

Figure 177 *D4 Processor Block Diagram*



Cabinet Temp Sensors

Figure 178 *CCA Cabinet Temperature Sensors*



Alarm Box

Overview

The Alarm Box is a user interface for:

- Power switching of the MR-system
- Magnet Supervision Indicators
- ERDU

System Power Switching

The System On, System Off and System Standby buttons are hard-wired to relays K4, K5 and K9 in the Power Distributor respectively. In addition, the Host can also activate the system **Standby** or **On** relays via software under the *System / Control / MR Scanner* menu.

Figure 179 *Alarm Box*

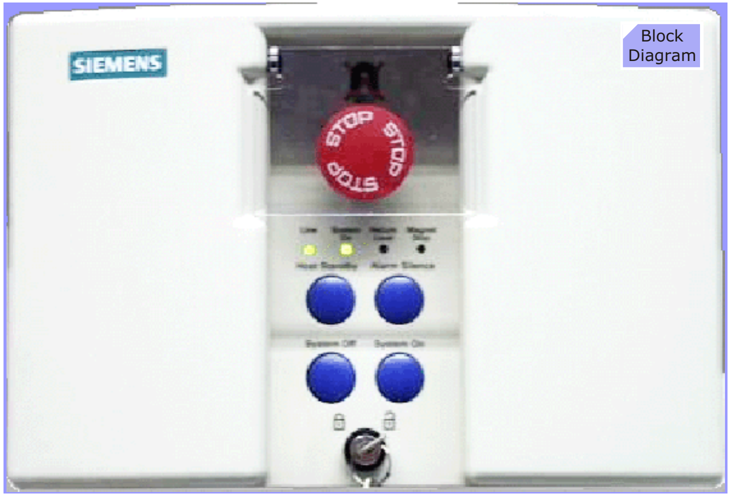
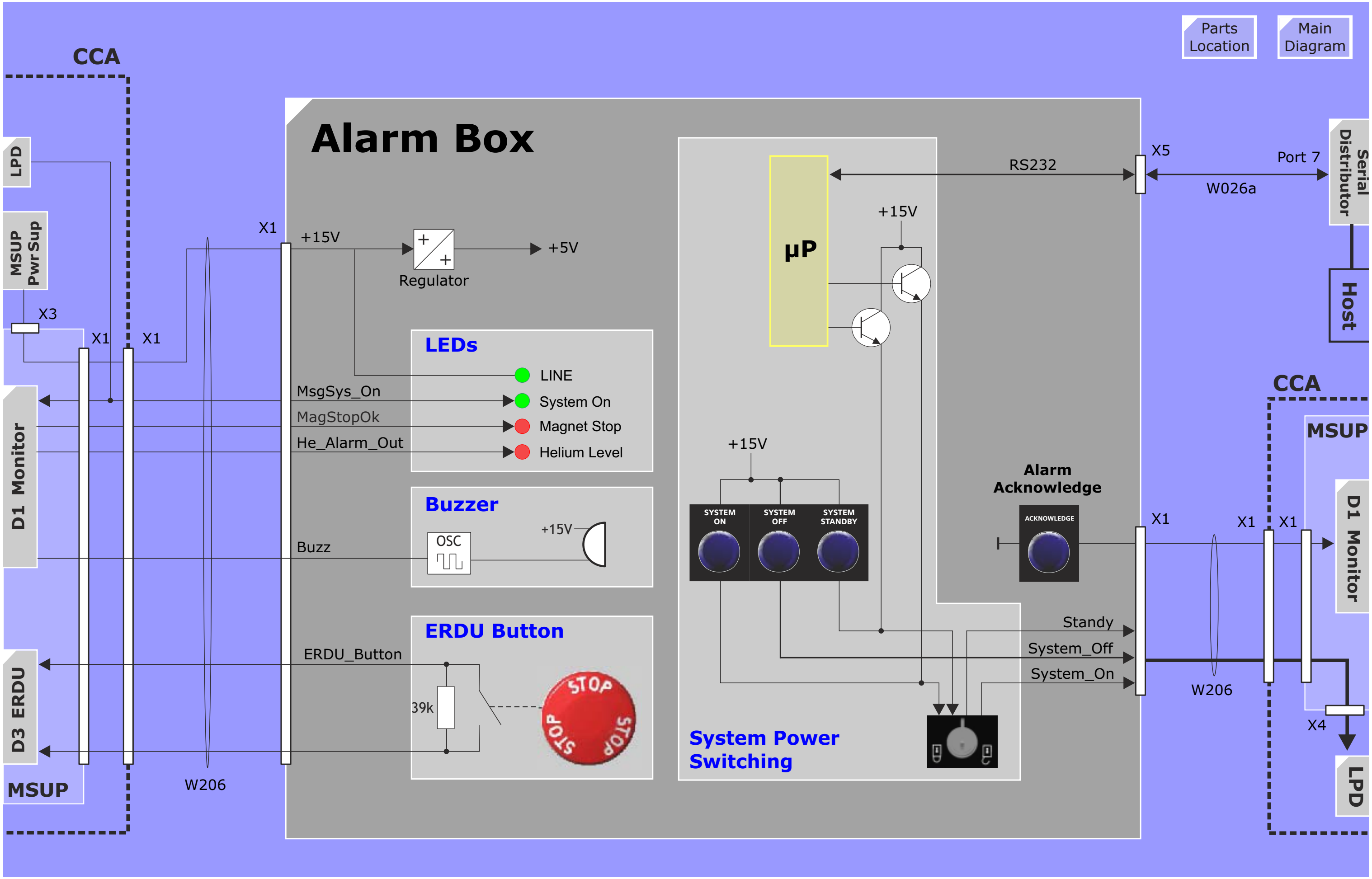


Figure 180 *Alarm Box Block Diagram*



Refrigerator System

Overview

The cryogenic refrigeration system consists of these components:

- Compressor Unit,
- Cold Head,
- the flex lines between compressor unit and cold head.

The first refrigeration system used in the Harmony/Symphony systems was the Coolpak 6000 from LEYBOLD. In 2001 it was replaced by the APD HC-10.

NOTE As of Jan. 2007 the **Sumitomo F-70** replaces all previous compressor types. Displacers for the older compressors are still available, however, if replacement of the compressor is required, you must order the Sumitomo F70. See SPC for details.

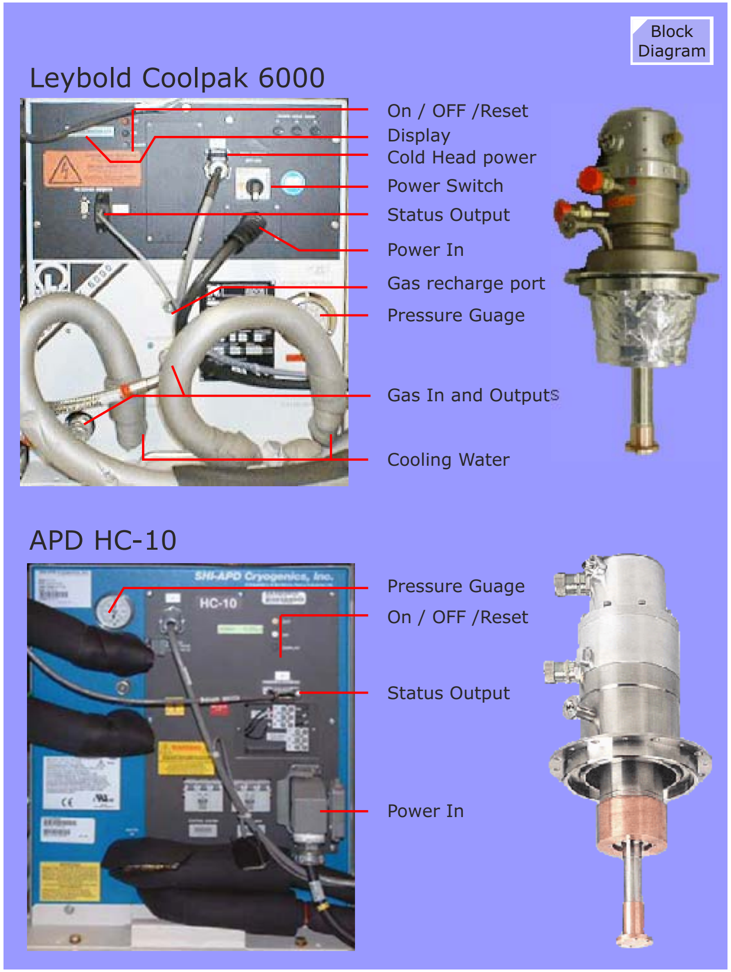
The function and operation of all these compressors is practically the same.

Concept

Heat is being developed in the cryo shields, mainly caused by induced **radiation.** This heat, if not removed, would lead to an increased liquid helium boil-off rate. Helium costs, at today's rate, around €10 a liter.

The Refrigerator System, consisting of the compressor and cold head, works on exactly the same principle as refrigerator appliances you find in any home. The compressor provides the high pressure gas. The cold head provides the expansion chambers to produce the heat extraction from the cryo-shields to which it is connected. The cold head has two stages. Both the design and the materials used within the cold head allow it to reach temperature of 20K and 80K

Figure 181 *Compressor Unit Layout*



Compressor Unit

Function

The oil-lubricated compressor compresses the helium gas from approx. 7 bar (P_l) to approx. 21 bar (P_h). This causes the helium to become very hot. For this reason it is passed through a helium/ water heat exchanger located immediately downstream from the compressor. The gas temperature is lowered to about the temperature of the coolant. The major portion of the oil vapor mixed with the helium in the compressor is condensed to droplets within the oil separators and returned to the compressor by means of capillary pipes.

The helium gas exiting the second oil separator, however, still contains small quantities of oil vapor. A charcoal-based adsorber removes the rest of the oil. The adsorber must be replaced every 18000 hours (two years) since it will become saturated with oil over time.

CAUTION	If any oil were to escape the compressor, it would contaminate the flex lines and also enter the cold head. There it would freeze and damage the cold head. Therefore: REPLACE THE ADSORBER REGULARLY!

To cool the compressor, its oil is pumped through a heat exchanger. Oil circulation is maintained by the pressure difference between low and high-pressure helium. The temperature of the cooling water should be 5...25°C at a flow rate of 5...10 l/min.

Safety Devices

The compressor unit has the following safety equipment:

- Pressure switch **PSL**: shuts down the compressor unit if the pressure difference between low-pressure helium and atmospheric pressure drops below 2.0 bar.
- Thermal Switch **TSH1**: shuts down the compressor, if the temperature of the helium behind (downstream of) the heat exchanger exceeds 38° C.
- Thermal Switch **TSH2**: shuts down the compressor if the compressor-exit helium temperature exceeds 100° C.
- Thermal switch in compressor capsule.
- Motor protection switch. It protects the compressor unit against excessive main currents.
- Fuses for cold head
- Fuses for control voltage (internal)
- Bypass valve. This will open a shunt between the high- and low-pressure helium circuits if the pressure difference exceeds 17 bar.
- Safety valve. This will open the helium circuit to the atmosphere if the high-pressure segment exceeds 29 bar.
- Microprocessor for checking the phase rotation.

If the compressor is shutdown due to a failure, the compressor must be MANUALLY re-started. Please refer to the delivered OEM documentation for detailed information on error status and handling.

Figure 182 *Compressor Status Feedback*

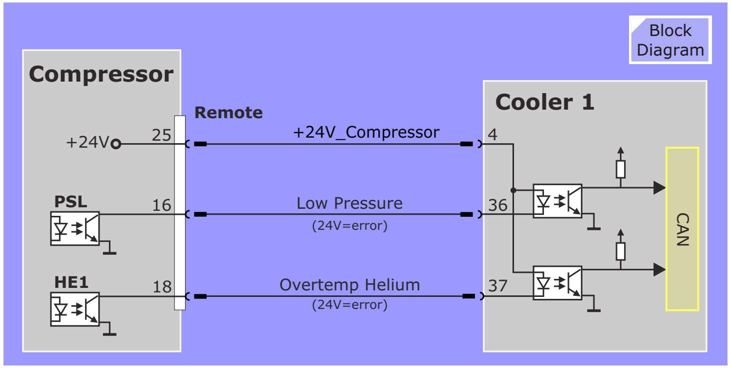
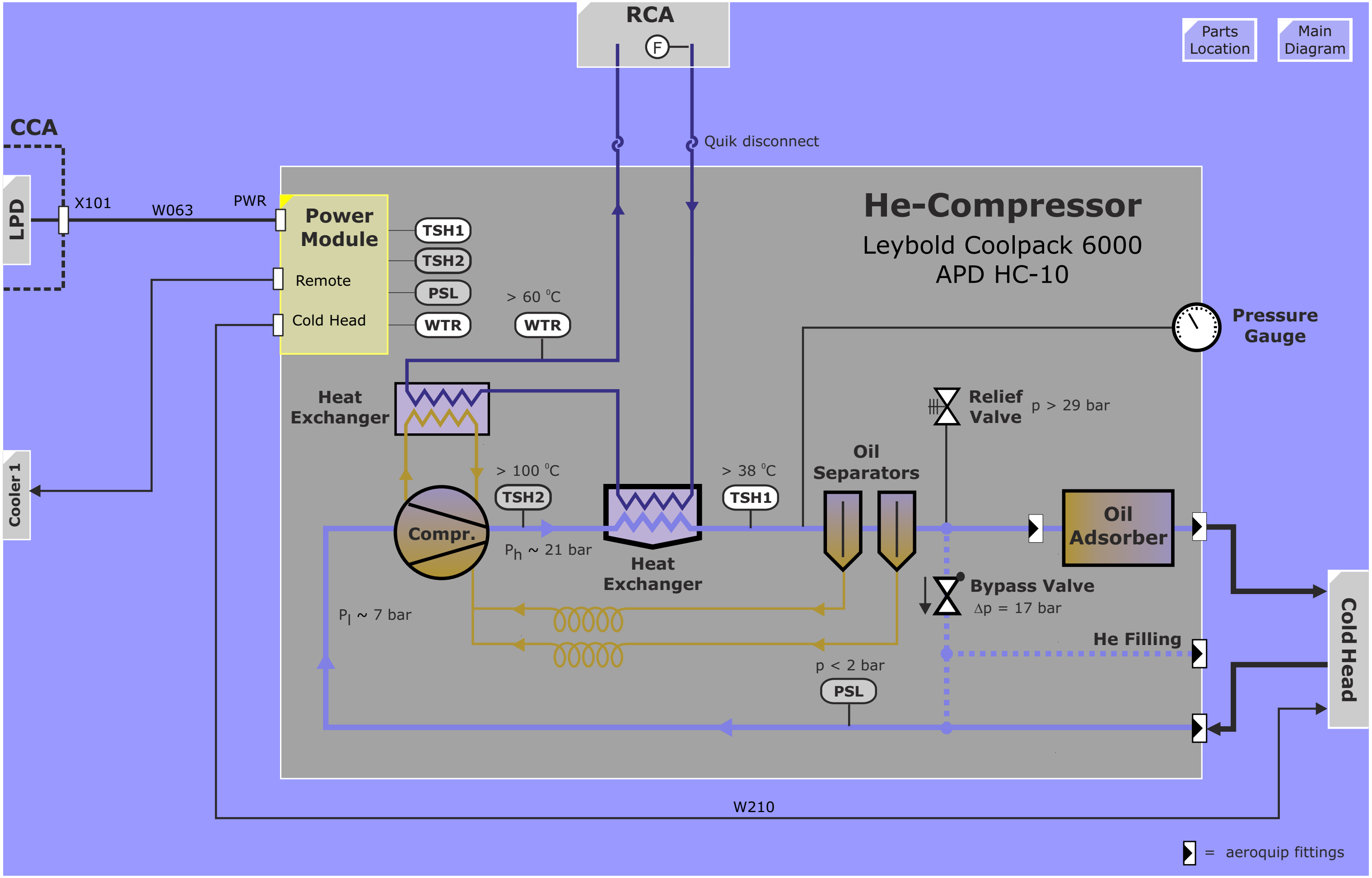


Figure 183 *COOLPAK 6000 Compressor Unit*



Advanced Shim

Introduction

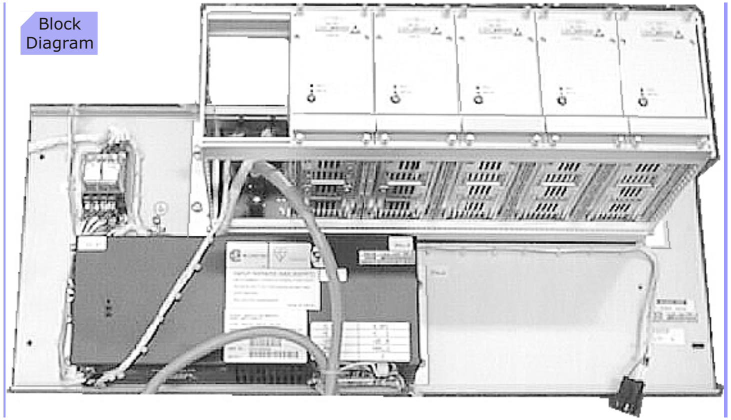
Passive shimming with iron plates reduces low and high order inhomogeneities caused by magnet tolerances and ferromagnetic objects in the environment. Objects to be measured (patients as well as phantoms) may generate local field inhomogeneities that have to be compensated for applications sensitive to inferior shim e.g. Spectroscopy, Fatsat, EPI (Echo Planar Imaging), TGSE (Turbo Gradient Spin Echo).

Correction of the first-order (linear) inhomogeneities is achieved by applying DC offset currents to the gradient coils. The higher second-order terms are corrected with a 5-channel shim coil embedded within the Gradient coil. The currents for these coils is produced by the Shim Power Supply Unit, which is described here. The Advanced Shim option is optional for the Harmony and Symphony systems, standard for the Allegra and Sonata systems.

Layout and Function

The Shim Power Supply Unit (Shim PSU) consists of 5 identical shim amplifier modules housed in a rack having a common backplane and a +48V DC power supply both of which are mounted onto a metal frame which conveniently attaches to the back of the CCA cabinet. A shim filter box is also part of the option and is mounted on the RF Filter Panel for feed of the shim currents to the shim coils.

Figure 184 *Shim Power Supply Layout*



Inputs

Power

The +48V DC power supply requires 220V AC provided by the LPD over circuit breaker F12.

The shim amplifier modules get their +5, +15 and -15 V supplies from the MSUP over K12 which will be activated as soon as the 230V from the LPD is available.

Shim Control

The communication is passed through the Magnet Supervision via the CAN bus connection and a hardware inter-face. The 5 demand currents are loaded via Host - MPCU - Can Bus - D4 Processor in MSUP - Data Bus into the 5 power modules. The second order field inhomogeneities have been measured via the tune-up step Phantom Shim. The compensation currents for the shim coils are calculated by the host.

The amplifier bus (cable from X9 MSUP to X80 Shim Motherboard, Shim Signals) comprises:

- 8 bit data bus MOD_DATA0 - MOD_DATA7. The data bus is used to load the 12 bit channel demand currents to the amplifiers current DAC and to determine the amplifier control
- ADDSEL line to select the channel writing of demand data and amplifier operation
- 4 bit control bus with a. MOD_COM used for controlling amplifier inhibit, fault, polarity conditions, b. DAC_DATA used for loading DAC current demand, c. MOD_FAIL used to indicate an amplifier fault, d. H/L BYTE is used for loading demand current in two halves.

Amplifiers output currents and voltage monitoring is selected onto the MOD_MON-lines. Typically, for output current an input reading of 0 - 4.02V corresponds to 0 - 5A. For troubleshooting you can measure the current by measuring the voltage at QLA outputs on the shim channel amplifiers.

The power modules require 48V DC for current generation (generated by a 48V DC Supply, which belongs to the Advanced Shim) and +5V, +-15V for logics and ADCs. The +5V, +-15V voltages are generated by the auxiliary power supply in the magnet monitoring section. The Output currents from the power modules run via (connector X3 at CCA Roof)-(connector X12 at SHIM Filter)-(connector X1 at magnet rear side) to the shim coils.

The functionality of the Advanced shim is tested during tune-up procedure "Phantom Shim".

Technical Specification

Specification	Value
Main Input Voltage	230VAC+-10%
Frequency	50/60 Hz +-1Hz
Power Consumption	<800W
Max current	5A per channel
Load resistance	1 to 6 Ohms
Inductance	0 to 30mH
Capacitance	100nF

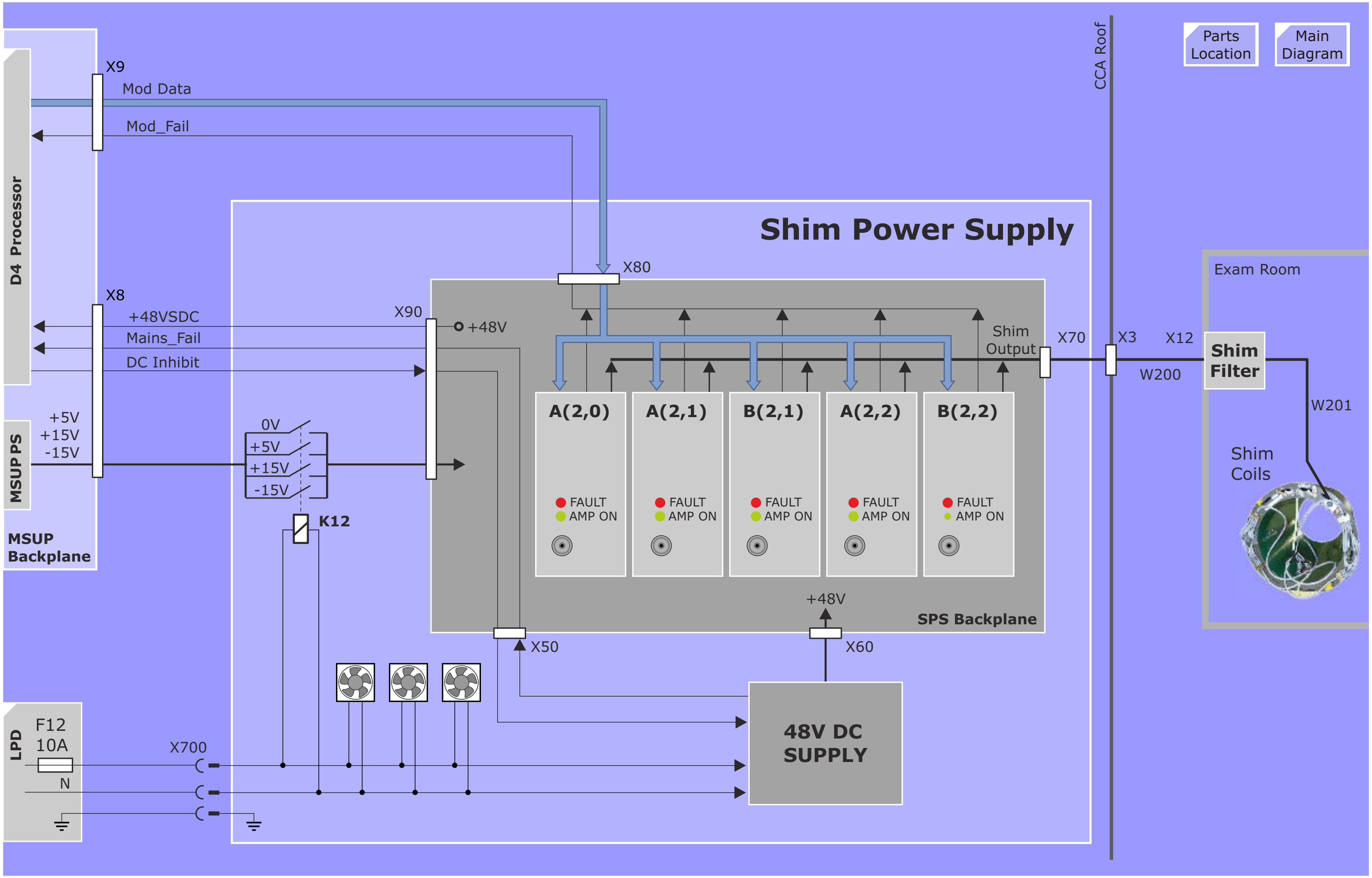
Protection and Monitoring

- The DC supply current is fused at 5A/48V on each of the shim amplifier boards,
- the amplifier output current is limited by a hardware current trip adjusted to 5.5A,
- the 48V DC is monitored,
- Main line voltage failures are monitored by signal MAINS_FAIL,
- The max power is 100W/channel and hence 500W in total for all channels. If that is exceeded, the 48V DC Supply will be shut down via signal DC_Inhibit. With 5A demand, the coil + filter + cable resistance must stay below 4 ohms in total (i.e. I**2*R must be <100W),
- Via the MOD_MON-lines currents and voltages are monitored. Output load disconnection or incorrect voltage reading by MSUP D4 processor board will cause faults such as: Advanced Shim error: Amplifier fault of shim channel X (over voltage),

LEDs

LED	Description
AMP ON	The green LED shows a digital signal sent from the processor to the module. This signal switches on the power electronics that regulates the output current. To turn on the outputs, first the 48V PSU is enabled, then the processor waits a short time for 48V to rise, then modules are turned ON.
FAULT	amplifier fault

Figure 185 *Advanced SHIM*



Magnet Power Supply 3600

Overview

The Magnet Power Supply Unit (MPS 3600) is an air-cooled, moveable service tool to energize the superconductive magnet coil to the required nominal field strength of 1.495T.

The MPS functions are controlled by a serial connection to the host computer or to a Service-PC. Additional cable connections have to be established for connecting the MPS control section to the Magnet Supervision for switch heater control, to the Mains Box for line power and the connections to the magnet.

The MPS consists of:

- MPS 3600
- Line power connection cable (X1 MPS to X102 CCA roof)
- Ramp cables (contained in separate crate)
- RS 232 connection cable (X6 MPS to host serial port #6)
- CCA control cable (X5 to X6 CCA roof)
- Operation Manual (under lid)

Pos. 1 CB1, Mains circuit breaker

Pos. 2 X1, MPS power connection

Pos. 3 Fuses (F1 - F4)

Pos. 4 X6, RS 232 connection to Host or Service PC

Pos. 5 X5 Control cable to MSUP

Pos. 6 Ramp cables

Figure 186 *MPS 3600)*

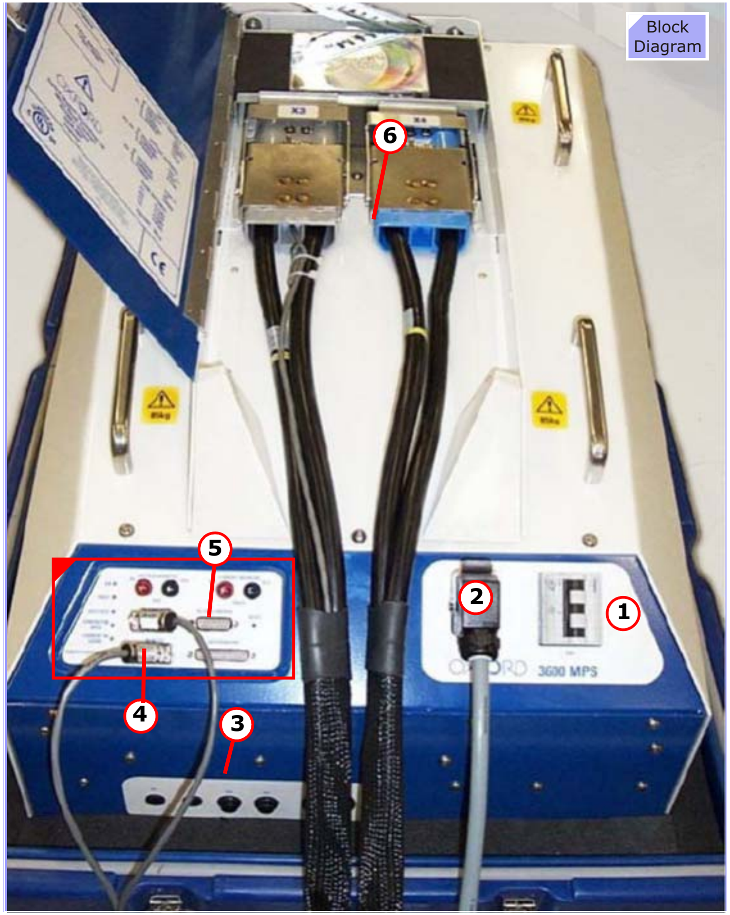
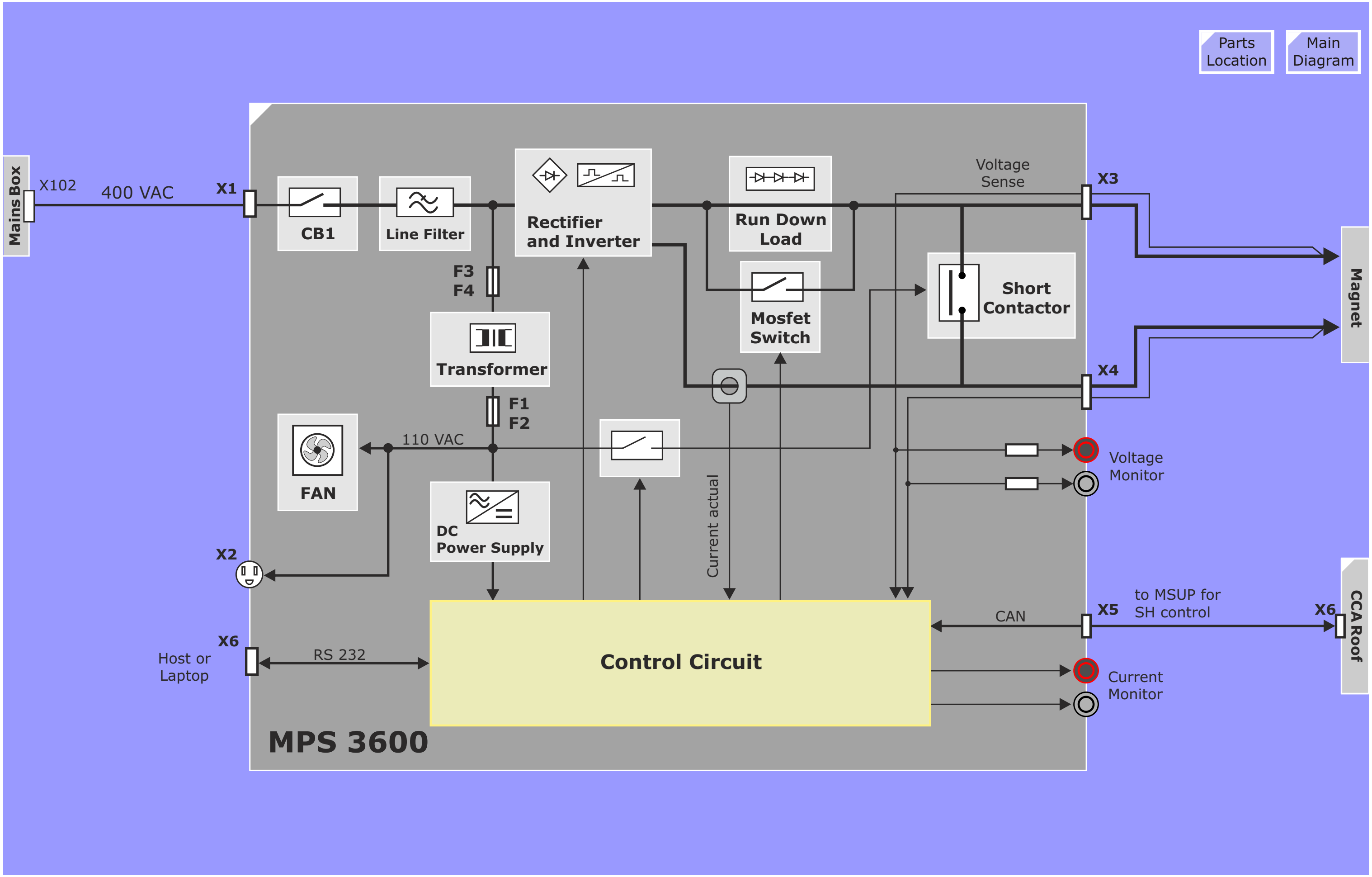


Figure 187 *MPS 3600 Block Diagram*



Function

Power Supplies

The 400 VAC 50/60 Hz 3-phase line voltage is fed via circuit breaker CB1, which routes the input supply to a line filter. Two outputs are taken from there, one via fuses F3 and F4 to a transformer that reduces the voltage to 110 VAC, the second one to the rectifier and inverter.

The power for the four cooling fans as well as for MPS control circuits is protected by fuse F1 and F2. That 110 VAC voltage is used as energizing voltage for the main output shorting-contactor and is also routed to connector X2 at the MPS front. It can be used to connect a low power-consumption device like a Service-PC.

Ramping Power Supply

The 3-phase input is rectified and fed to a full bridge inverter running at about 20 kHz. The modulated square wave is fed to a 20 to 1 step down transformer that supplies a schottky rectifier and a LC filter. The MPS output current (max. 725 A @ 12 V) is fed via a Run Down Load (RDL) to the MPS output connector X3 and X4. The RDL can be shorted by a MOSFET switch.

Actual output current is measured by a current transducer that is connected to the Control Circuit. For service, it can be directly measured at connectors X11 and X12.

Run Down Load, MOSFET switch

To run down the magnet, the MPS is equipped with a Run Down Load (RDL) which consists of heavy-duty diodes embedded in heat sinks and cooled by fans.

To activate the RDL, the MPS opens a MOSFET switch that is connected in parallel. The energy that is stored in the magnet coil is converted into heat inside the RDL and the magnet current is ramped down. During ramping up, the MOSFET switch is closed to directly feed the MPS current to the magnet coil.

Shorting Contactor

The MPS is equipped with a shorting contactor that is normally closed. During different ramp phases, the control circuit will open the contactor if required.

The shorting contactor is opened if:

- The MPS ramps up the magnet
- The MPS ramps down the magnet

The shorting contactor is closed if:

- The MPS is switched off
- During ramping, a line power failure occurs
  (Magnet current will safely run down via the ramp cables and closed shorting contactor)
- MPS performs internal 700 A self-test prior to ramping
- Magnet is captured at beginning of the ramping-down cycle

Control Circuit

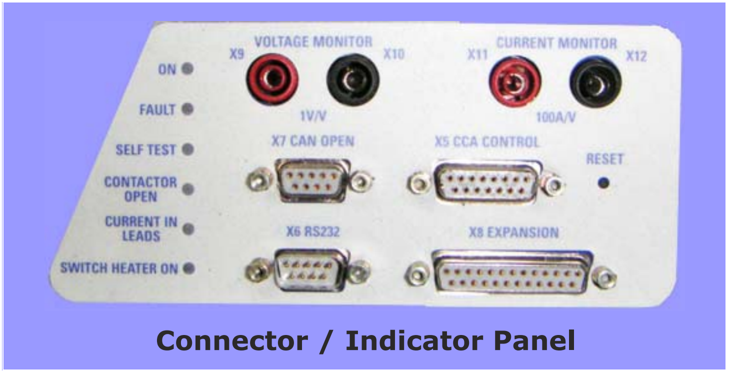
The control circuit monitors and controls the operation of the MPS. Ramping commands are initiated by the operator using the host or a Service-PC, to be connected to the MPS. The algorithm for ramping is stored in a table at the control circuit. Required magnet specific data are loaded from the Magnet Supervision (MSUP) via CAN-bus connection (X7). During ramping, the MPS activates the switch heater by sending CAN commands to the MSUP.

If the CAN-bus is disconnected, the MPS is inhibited and the fault displayed at the host or Service-PC.

Current and Voltage Monitoring

The MPS output current and voltage is monitored by the Control Circuit. It is reported via RS 232 to the host or Service-PC. Additionally, a DVM can be connected to the voltage monitor terminals (X9, X10) or current monitor terminals (X11, X12) on the MPS connector panel.

Figure 188 *MPS 3600 Connector / Indicator Panel)z*



Service Routines

Connecting Host or Service-PC

To start ramping, the host or Service-PC has to be connected to X6 of the MPS (RS 232 interface). To establish communication, Hyper Terminal software is required and must be set-up.

Hyper Terminal Setup

Hyper Terminal has to be configured and started before the MPS is switched on.

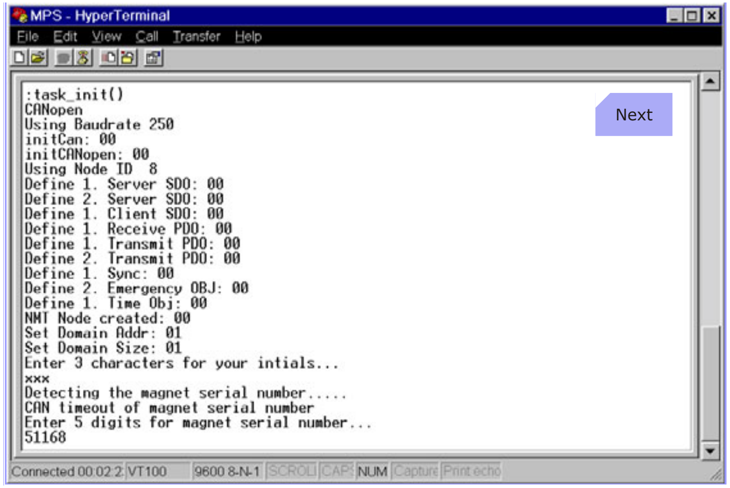
- Select appropriate Com Port (port 8 if cable is connected to Serial Distributor port 6)
- Baud rate = 9600
- Data bits = 8
- Parity = none
- Stop bits = 1
- Flow Control = None.
- File, properties, settings tab, Emulation = VT100

Welcome Screen

A welcome screen is displayed, 3 digits representing the operator (i.e. your initials) have to be entered and accepted with <ENTER>. In a next step, the magnet serial number has to be entered.

The MPS will now retrieve the magnet type, last set current, He-level, date and time from the MSUP. Once the data has been obtained, the MPSU will initiate a self test to ensure its correct functioning.

Figure 189 *MPS 3600 Logon)*



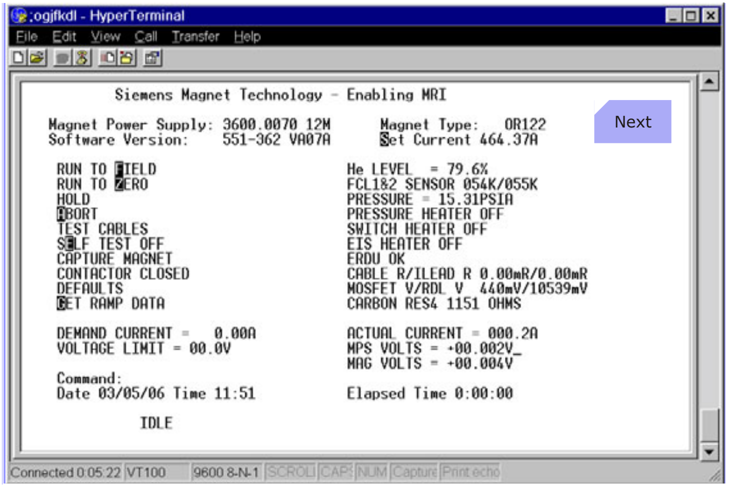
Self Test

The MPS is run up to 700 A with shorting contactor closed, the voltage drop across the MOSFET switch is monitored. When the current reaches its peak, the voltage is recorded (V1). Next, the MOSFET switch is opened and the current runs through the Run Down Load. Before the current starts to decay, the voltage drop across the RDL is recorded at Peak Current (V2).

Operational Screen

On completion of the self test the Operational Screen is displayed:

Figure 190 *Operational Screen)*



Ramping Commands

Command	Key combination
To set magnet current:	S <space> 513.04 <Return> (Example)
To Run To Field:	F <Return>
To Run To Zero:	Z <Return>
To abort current action:	A <Return>
To refresh the screen:	O <Return>
To go to the Error Screen:	G <Return> Ramping parameters

Errors

If a fault occurs, an error screen is displayed automatically. To access it manually, a command can be entered (see table above). The Magnet Peak Current is recorded on this screen, so if the magnet should quench, the current level at that time is stored.

Run to Field

Prerequisites for Ramp to Field:

- The MPS must be connected properly.
- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including cable tests).
- Following a helium fill, wait at least 2 hours before ramping.
- Helium level to be >40%.

MPS system Test:

- Perform Helium sample
- Check if magnet has not been filled with LHe the last 2 hours
- Check if current leads are not opened or shorted
- Check if FCL temperature > 28 K
- Determine actual FCL temperature
- Switch on pressure heater, wait one minute
- Check if magnet pressure < 17 psiA
- Check if magnet pressure increases
- Check if FCL temperature has dropped by 10 K (time-out 30 min.)
- Vent Delay, wait 5 min.
- Turn off pressure heater
- Turn on EIS heater

Cable Test:

- Inverter and MOSFET switch is turned on, shorting contactor open
- The current in the cables is ramped to 100 A
- MPS checks if actual current is 100 A +/- 1 A and measures voltage drop across the FCL
- Current in the cables is ramped down to 0 A

- Calculation of Turret Lead and Cable lead resistance

Ramp to Field:

- Switch heater is turned on
- Check if magnet is running down already. If so, the magnet is captured and ramped down safely
- Shorting contactor is opened
- MPS demand current is set to entered current value + 3 A
- As soon as 10 A are reached, the switch heater is switched off
- 10 A before reaching nominal current, switch heater is switched on again
- Check if entered current value + 3A is reached (over fielding)
- Wait 15 seconds
- Ramp to entered current value
- Turn off switch heater, wait 2 minutes
- Ramp down cables to 0 A, wait 2 minutes
- Turn EIS off

Ready!

Run to Zero

Prerequisites for ramp to zero:

- The MPS and cables must be connected properly
- Wait at least 30 minutes after completing any previous ramp or aborted ramp (including cable tests).
- After helium fill at field, wait at least 2 hours before ramping.
- Helium level to be >40%.

MPS checks prior to ramping down:

Same as checks prior to ramping up!

Cable test:

Same as for ramping up!

Ramp to zero:

- Switch heater is turned on
- Magnet capture is performed
- Shorting contactor is opened
- MPS demand current is set to 0 A, wait 10 seconds
- Switch heater is turned off
- When magnet voltage has fallen to < 1 V switch heater is turned on again
- Check if actual current is 0 A, turn off switch heater, wait 2 minutes
- Turn EIS heater off

Ready!

Bumping the Magnet

Bumping the magnet involves making a small change to the magnetic field strength either up or down. The procedure is essentially the same as ramping the Magnet.

Prerequisites for bumping:

- The MPS and cables must be connected properly
- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including cable tests).
- Following a helium fill, wait at least 2 hours before ramping.
- Helium level to be >40%.

Saving MPS logfile

A logfile of the last ramp is recorded by the MPS and has to be downloaded and sent to the factory at the following e-Mail address: **magnet.hsc.med@siemens.com**

The following commands have to be entered:

- G <Return>, to enter the Error screen

From Hyperterminal menu select:

- Transfer

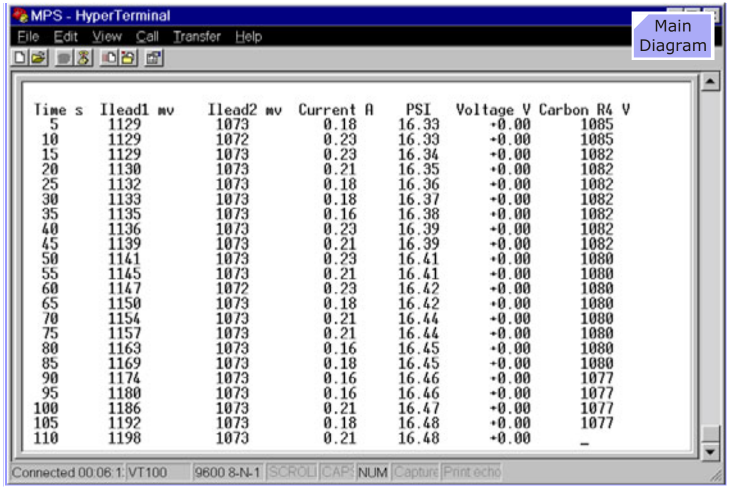
- Capture Text
- Browse
- Select the target directory and filename: e.g. 3600 Ramp Data\OR105_SerNr_RampUp_Date_Time
- Start
- L <Return> (a data screen will appear)

From Hyperterminal menu select:

- Transfer
- Capture Text
- Stop

To return to the Operational Screen press <Return>.

Figure 191 *MPS Ramp Data Dump)*



Time s	Ilead1 mv	Ilead2 mv	Current A	PSI	Voltage V	Carbon R4 V
5	1129	1073	0.18	16.33	+0.00	1085
10	1129	1072	0.23	16.33	+0.00	1085
15	1129	1073	0.23	16.34	+0.00	1082
20	1130	1073	0.21	16.35	+0.00	1082
25	1132	1073	0.18	16.36	+0.00	1082
30	1133	1073	0.18	16.37	+0.00	1082
35	1135	1073	0.16	16.38	+0.00	1082
40	1136	1073	0.23	16.39	+0.00	1082
45	1139	1073	0.21	16.39	+0.00	1082
50	1141	1073	0.23	16.41	+0.00	1080
55	1145	1073	0.21	16.41	+0.00	1080
60	1147	1072	0.23	16.42	+0.00	1080
65	1150	1073	0.18	16.42	+0.00	1080
70	1154	1073	0.21	16.44	+0.00	1080
75	1157	1073	0.21	16.44	+0.00	1080
80	1163	1073	0.16	16.45	+0.00	1080
85	1169	1073	0.18	16.45	+0.00	1080
90	1174	1073	0.16	16.46	+0.00	1077
95	1180	1073	0.16	16.46	+0.00	1077
100	1186	1073	0.21	16.47	+0.00	1077
105	1192	1073	0.18	16.48	+0.00	1077
110	1198	1073	0.21	16.48	+0.00	

Section

10 Cooling System

Introduction

The Harmony and Symphony/Sonata MR systems employ a number of high-power components that require cooling to maintain a constant operating temperature and stable performance. The Cooling System is responsible for providing temperature-stabilized cool water to these components and to provide monitoring of the cooling water flows and temperature.

The Cooling System consists of two separate water circuits, a primary and a secondary.

Primary Cool Water

An external water chiller or customer in-house chilled water provides primary cooling water of 9°C used to cool the following components:

- RCA Water Cooler (secondary cooling for system components)
- RCA Coolpak (Cold head compressor)
- Water-CCS Option

The RCA Water Cooler contains a heat exchanger which uses the primary chilled water to maintain a secondary cooling water circuit at 20°C.

The RCA Coolpak (cold head compressor) uses the primary water for cooling the compressor.

The water-CCS option uses primary water for internal climatic cooling of the CCA and GPA cabinet components. This option eliminates the need for expensive and space-requirement-intensive external air conditioners and thus benefitting the customers pocketbook.

Secondary

The secondary water circuit of the Water Cooler is used for the following components:

- Electronics Suite :
  RFPA
  RFCI power supply
  GPA (resonant converter, rectifiers and power stages)
- Gradient Coil (alternatively MPS)
- RF Body Coil
- RF Dummy load (TAS)

Double Cooler

The RCA cabinet is designed to accept a second chiller unit necessary for Sonata, Trio and Allegra. The distribution of the secondary water can be seen in the next 3 block diagrams.

Flow Regulation

The flow reducers in the CCA and GPA cabinets assure that the proper amount of water flow requirements for the components in each cabinet are maintained.

Figure 192 *Cooling System Overview - Single Chiller*

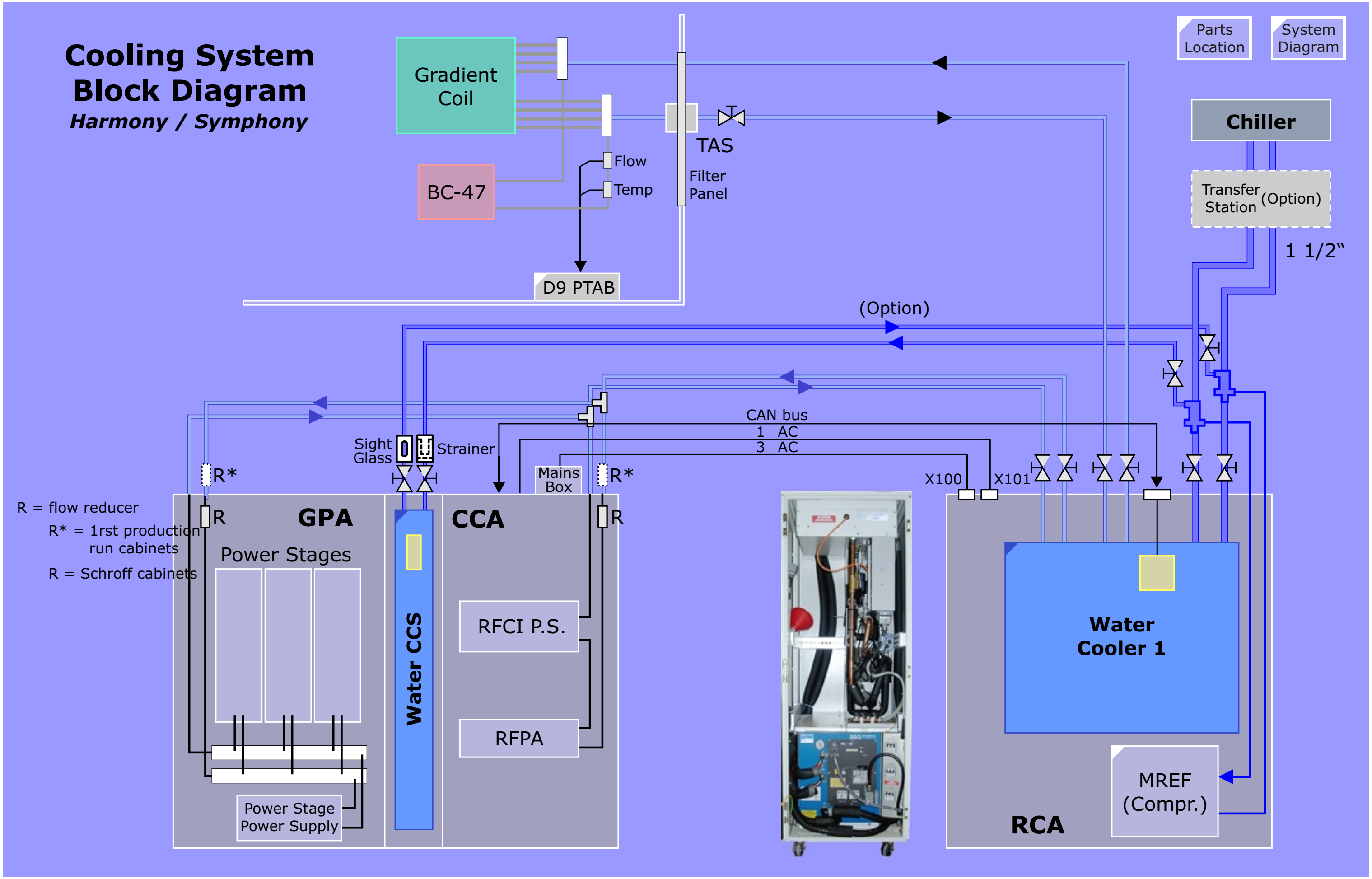
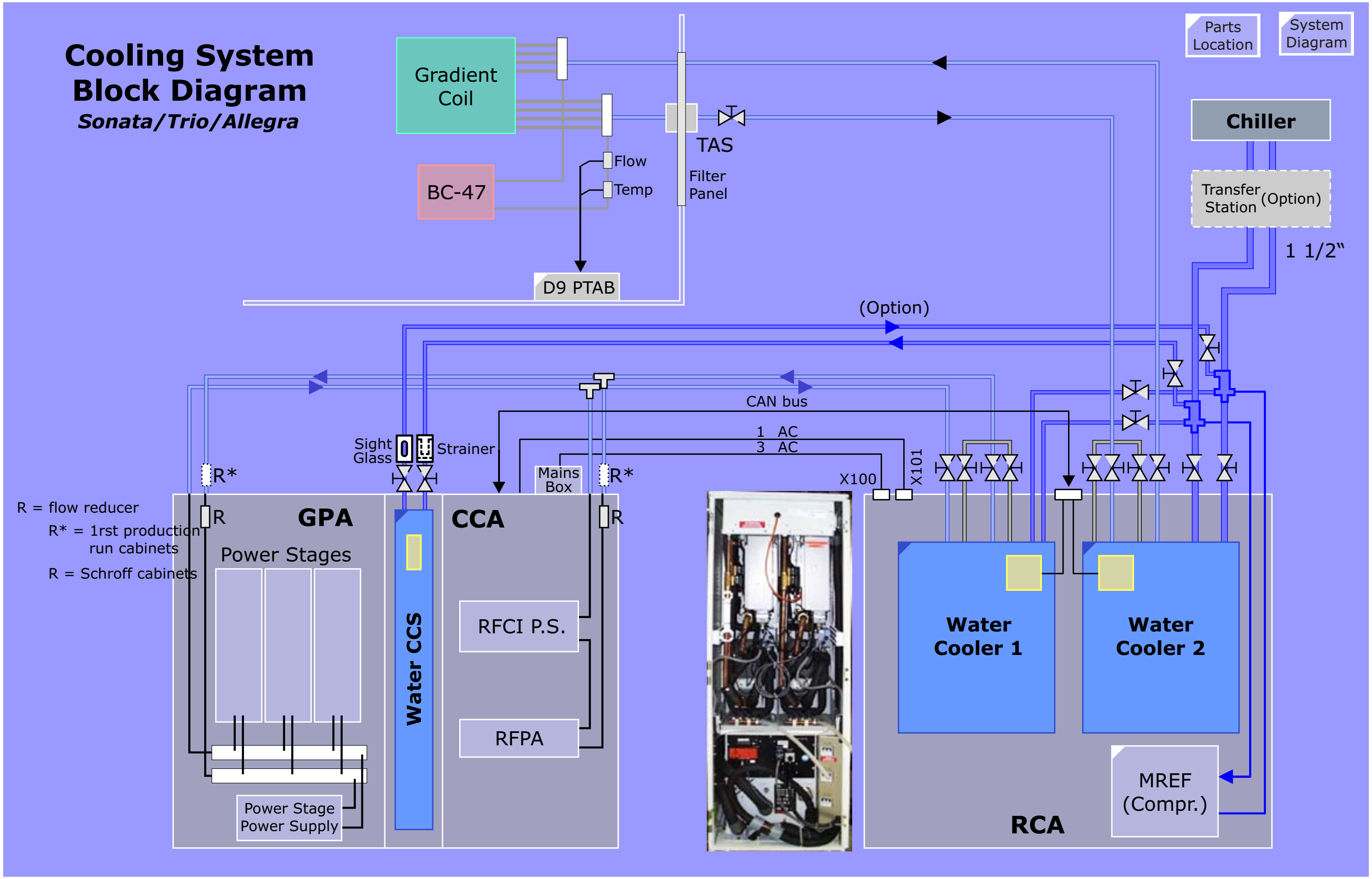


Figure 193 *Cooling System Overview - Double Chiller*



Water Cooler

Overview

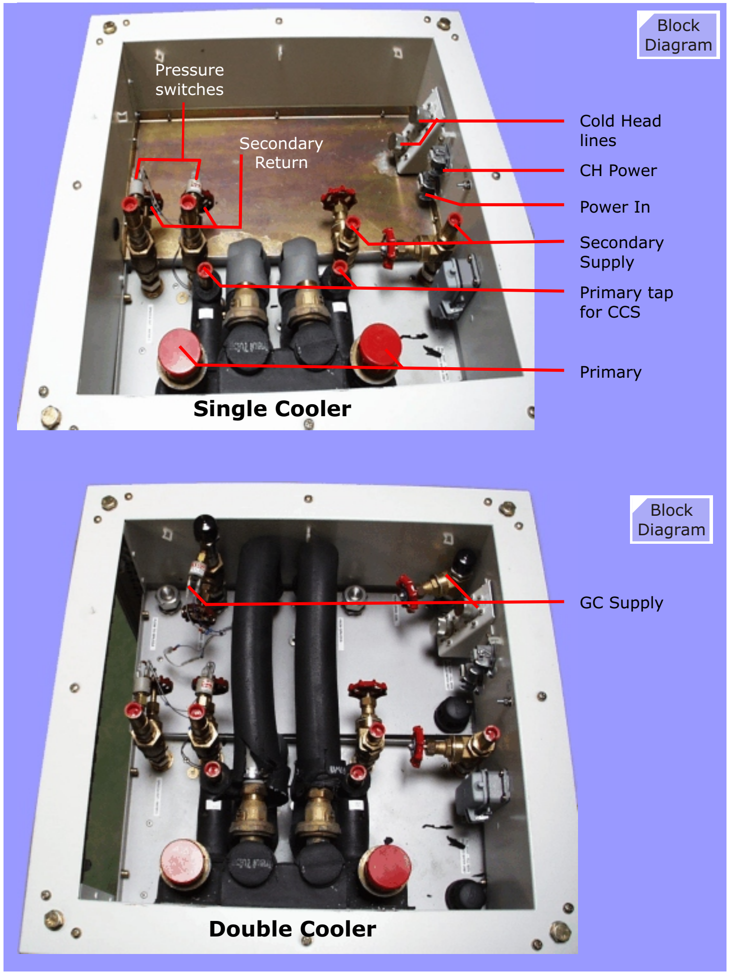
The RCA water cooler contains a water/water heat exchanger which uses the primary chilled water to maintain the secondary cooling water circuit at 20°C.

The water/water plate heat exchanger in the RCA is responsible for the pressure difference of max. 2.5 bar of forward and backward running primary water.

The secondary water circuit of the Water Cooler is used on the following components organized in two circuits:

- Electronics Suite:
  - RFPA,
  - RFCI power supply,
  - GPA.
- Coil:
  - Gradient Coil,
  - RF Body Coil,
  - RF Dummy load (TAS).

Figure 194 *Water Cooler Connections*



Control Electronics

Monitoring is done by the Cooler Control Electronics. The controller is an IAC (intelligent controller) manufactured by Satchwell and is linked to a CAN-Module. According to its function the processor in the CAN-Module receives status and error information from the Satchwell controller and will transfer this information from the RCA to the MMC. It also receives control data from the MMC, for example for switching ON the secondary water circuit via the IAC or switching it OFF during standby mode.

The IAC has the following INPUTS and OUTPUTS:

Analog INPUTS:

- Electronic Suite Waterflow,
- Gradient Coil Waterflow,
- Primary Waterflow,
- Water Overpressure Electronic Suite,
- Water Overpressure Gradient Coil.

The waterflow is monitored electrically by measuring the magnetic variation in a coil caused by the rotation of a piece made of ferromagnetic material placed in the water flow. This means, the flow measurement will also work in polluted water.

Resistive INPUTS:

- Primary water temperature,
- Secondary water temperature.

Digital INPUTS:

- Pump Fault (for detection of overcurrent),
- 3 INPUTS indicating the waterlevel (FULL, LOW, VERY LOW) of the seconday water reservoir (refilling of the secondary water is necessary, when the yellow light glows constantly)
- Low pressure and Over temperature signals from the coldhead (these are status lines only),
- Local control. This is a manual switch labeled AUTO/MAN. Its normal position for remote control of the pump is AUTO. It is set to MAN during startup, when filling the secondary water reservoir the first time.

Analog OUTPUTS:

- Actuator. The temperature of the secondary water is controlled and adjusted to the "temperature set point" (normally 20 degree, set under SESO, Magnet and Cooling, Initialization) by changing the primary waterflow. The primary waterflow rate through the Heat exchanger is controlled by the IAC via motor setting of the primary bypass valve. This is necessary in order to protect the pump of the primary water chiller, when the system is switched off. The flowrate of the primary water will be dependent on the primary water temperature.

The IAC will switch off the pump for the secondary water circuit, if:

- system switched to "standby",
- water level in secondary water tank too low,
- water pressure too high,
- pump overload.

In the case the pump was switched off, due to overpressure for example, you can switch it on again (after check of water connections and hoses) by setting the system to "standby" and then "ON" again.

Setting the flow rates of the secondary water through Electronic Suite and Gradient coil is done by manually adjusting the valves and monitoring the flow rates in the SESO under "MAGNET & COOLING", "STATUS". This procedure as well as the setting of the temperature measurement points is described in the "Technical Documents", Register Start-up.

Figure 195 *RCA Water Cooler Overview*

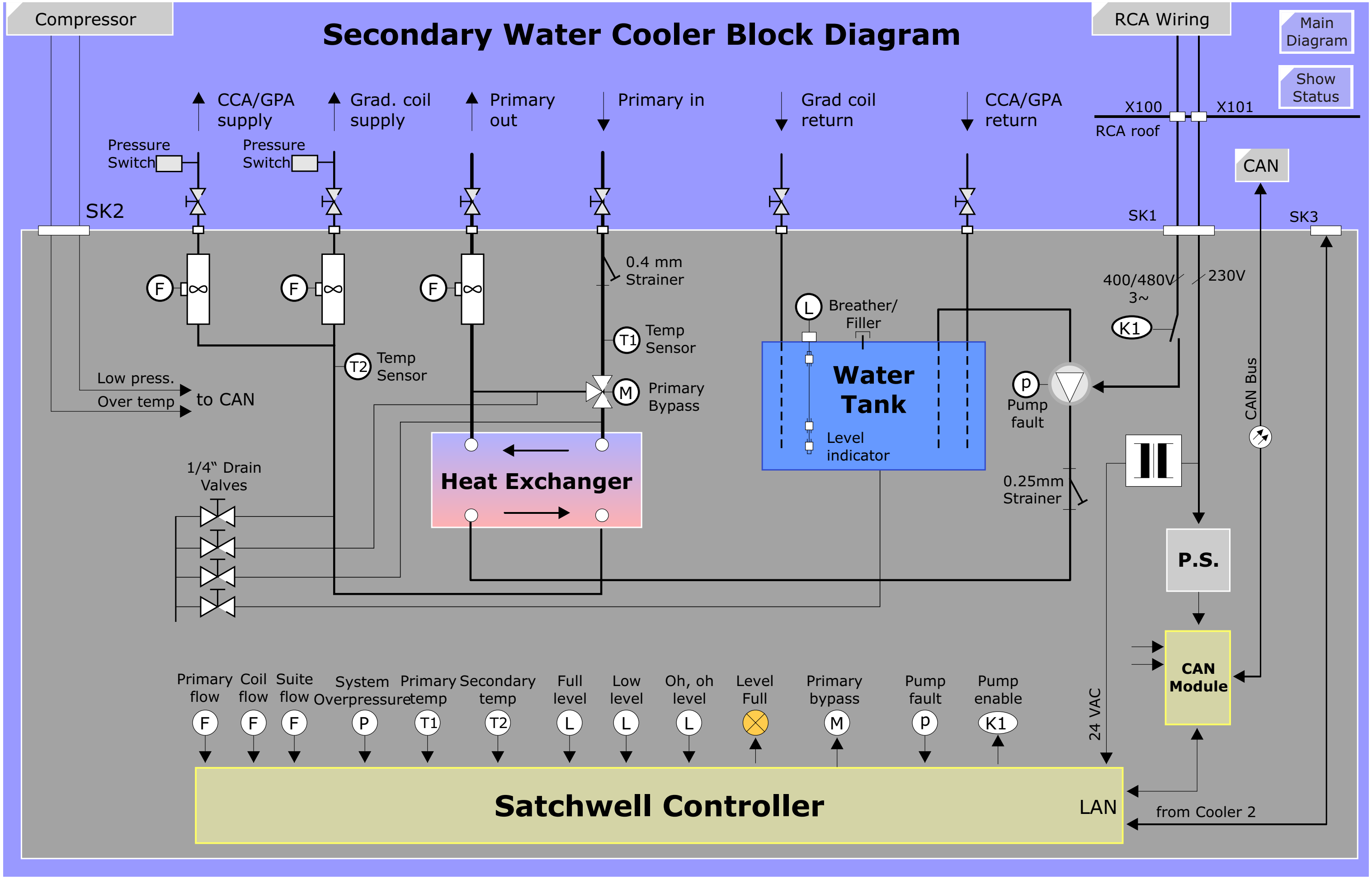
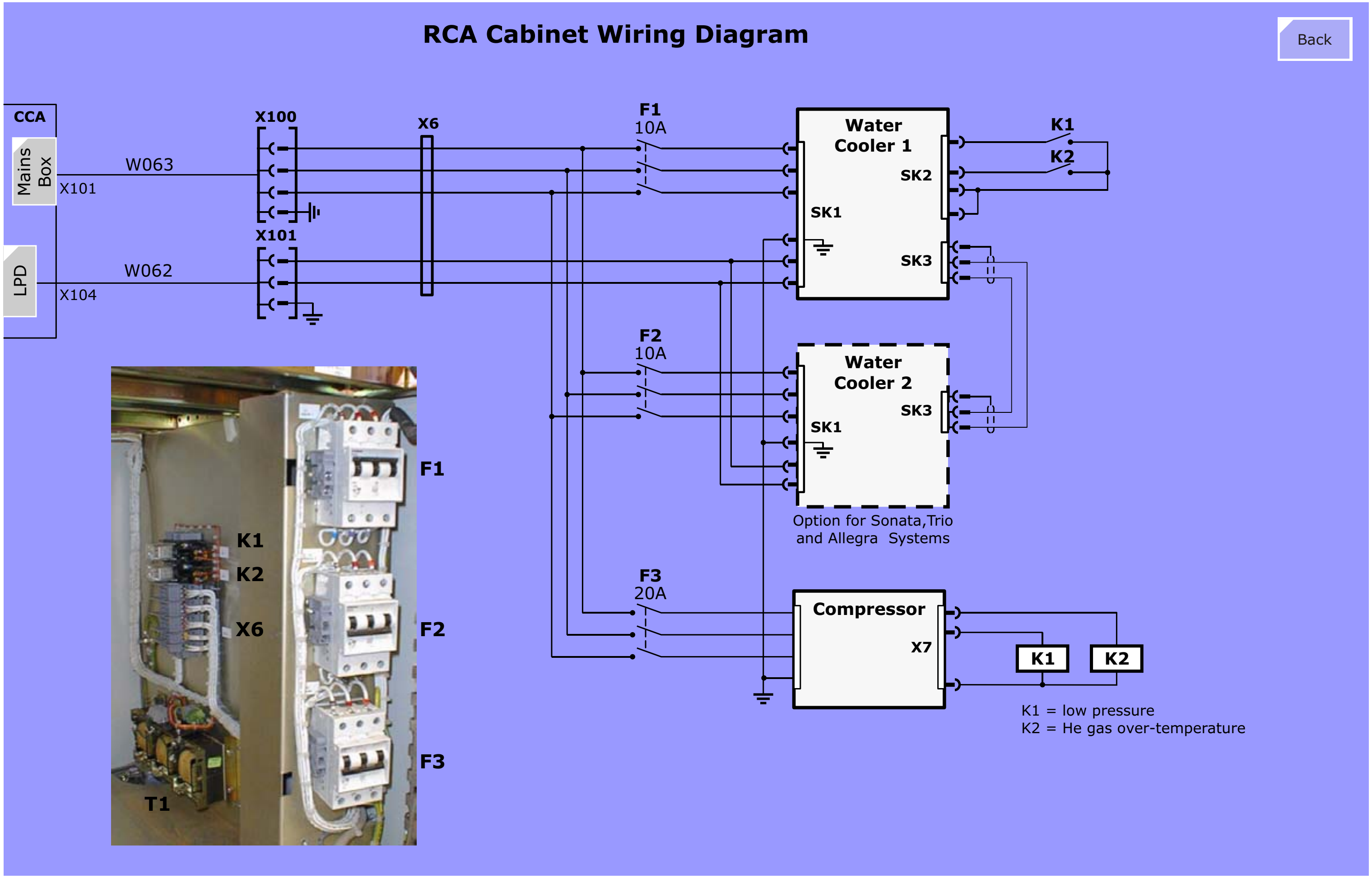


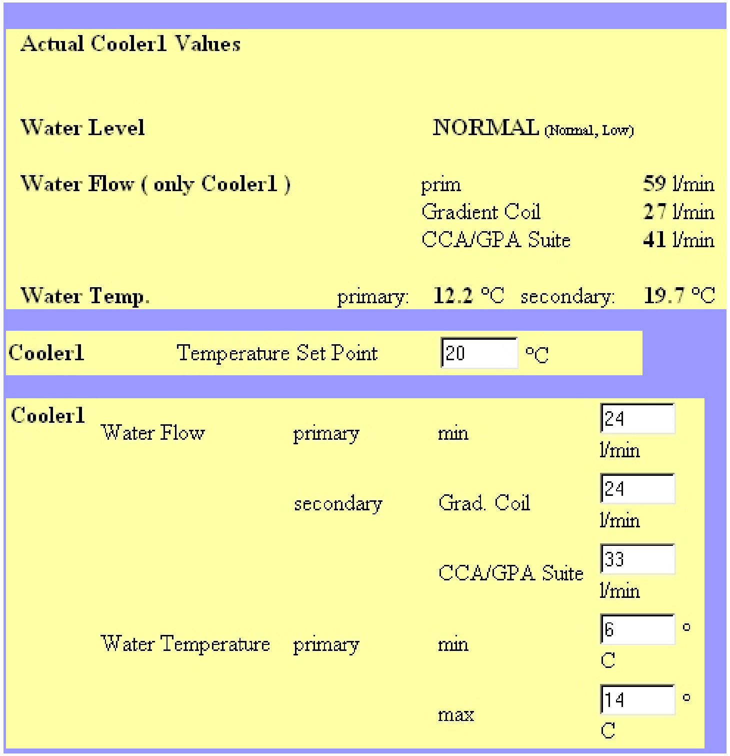
Figure 196 *RCA Power Wiring*



Cooler Status

The Cooling System status masks (viewed under SeSo > Magnet & Cooling) are shown below:

Figure 197 *Cooler Status Masks*



Cabinet Cooling - CCS

Overview

The CCS incorporates two functional blocks:

- CCA/GPA cabinet climatisation
- Monitoring

Cabinet Climatisation

CCS Water version

The blowers located in the Control-Blower Module draws in warmed air from the GPA and CCA cabinets and forces it over a heat exchanger which is supplied by the 6-12°C primary cooling water. The cooled air enters the CCA and GPA cabinets through cut-outs located cut out at the bottom of these cabinets. The CCA has an additional cut-out at its mid-section. Air circulates over the CCA and GPA electronics and drawn into the blowers again through cut-outs at the top of the CCS completing the air circulation path.

CCS Air version

If the technical room is air conditioned, the CCS Air version can be used. Cooled air is taken in from ventilation holes in the CCA cabinet door and circulated through the CCA and GPA cabinets in the same way as mentioned above.

Regulation

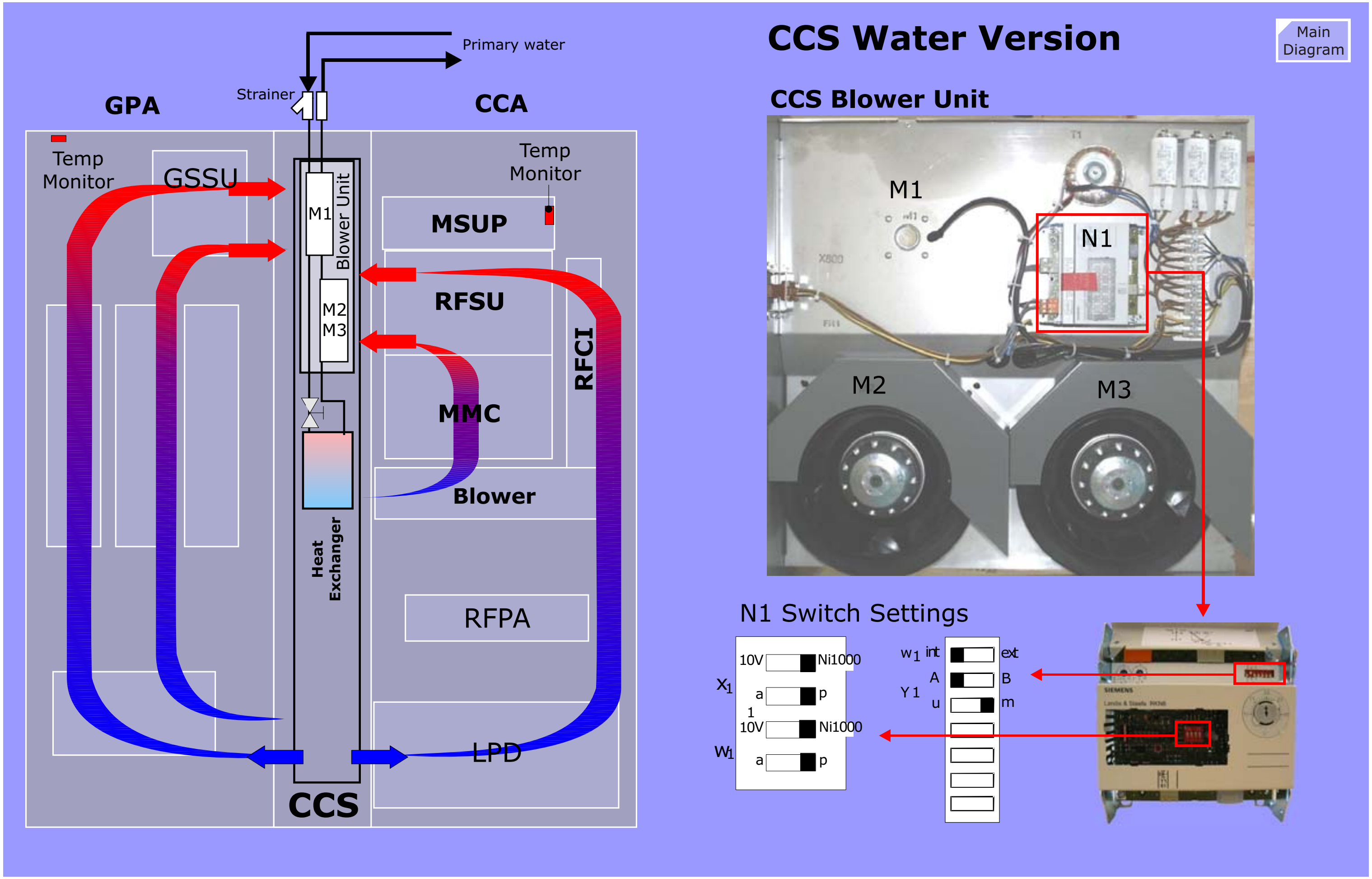
Regulation of the air temperature is achieved with a 3-way valve (**actuator**) controlled by a PI regulator located on Control Unit N1. Temperature sensor **R1**, a PT100 sensor located at the air outlet, is used as actual value.

Monitoring

The Cabinet temperature monitoring is performed by temperature sensors located at the top of the GPA abinet and at the backside of the MSUP electronics rack in the CCA cabinet. The MSUP is also responsible for the monitoring.

Figure 198 *CCS Versions*

Figure 199 *Cabinet Cooling Overview*



Section

11 Power Distribution (LPD)

Wow!! Hallo, it's been a l o n g time since someone's been *here*... probably made a wrong turn on the way to the RF section., eh? That usually what happens... hey, hey... wait! PLEASE DONT LEAVE!!! If you stay and keep me company for a while, i'll give tomorrows lottery number! Really.

Introduction

The LPD has the monumental task of distributing the lines power to the various components withing the system. Overview schematics of the LPD are found below. Please also refer to the Diagrams document for details.

Mains Box

There are two versions of the LPD and Mains box, both are shown below. The electrical connections for both versions are displayed in the schematics on the following pages.

Figure 200 *Line Power Distributor Mains Box*

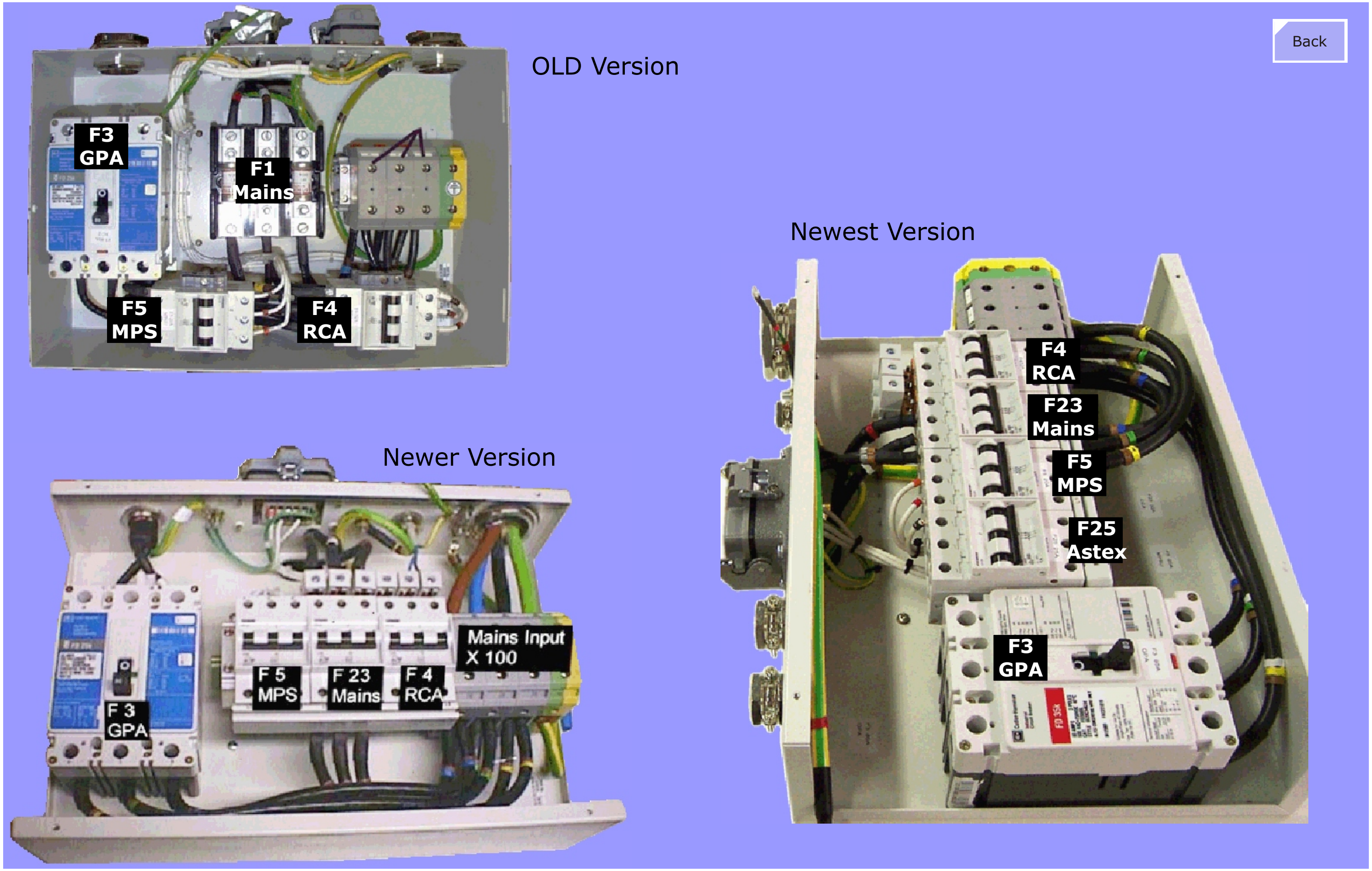


Figure 201 *LPD Schematic Page 1 (**OLD** version with K2209 GPA)*

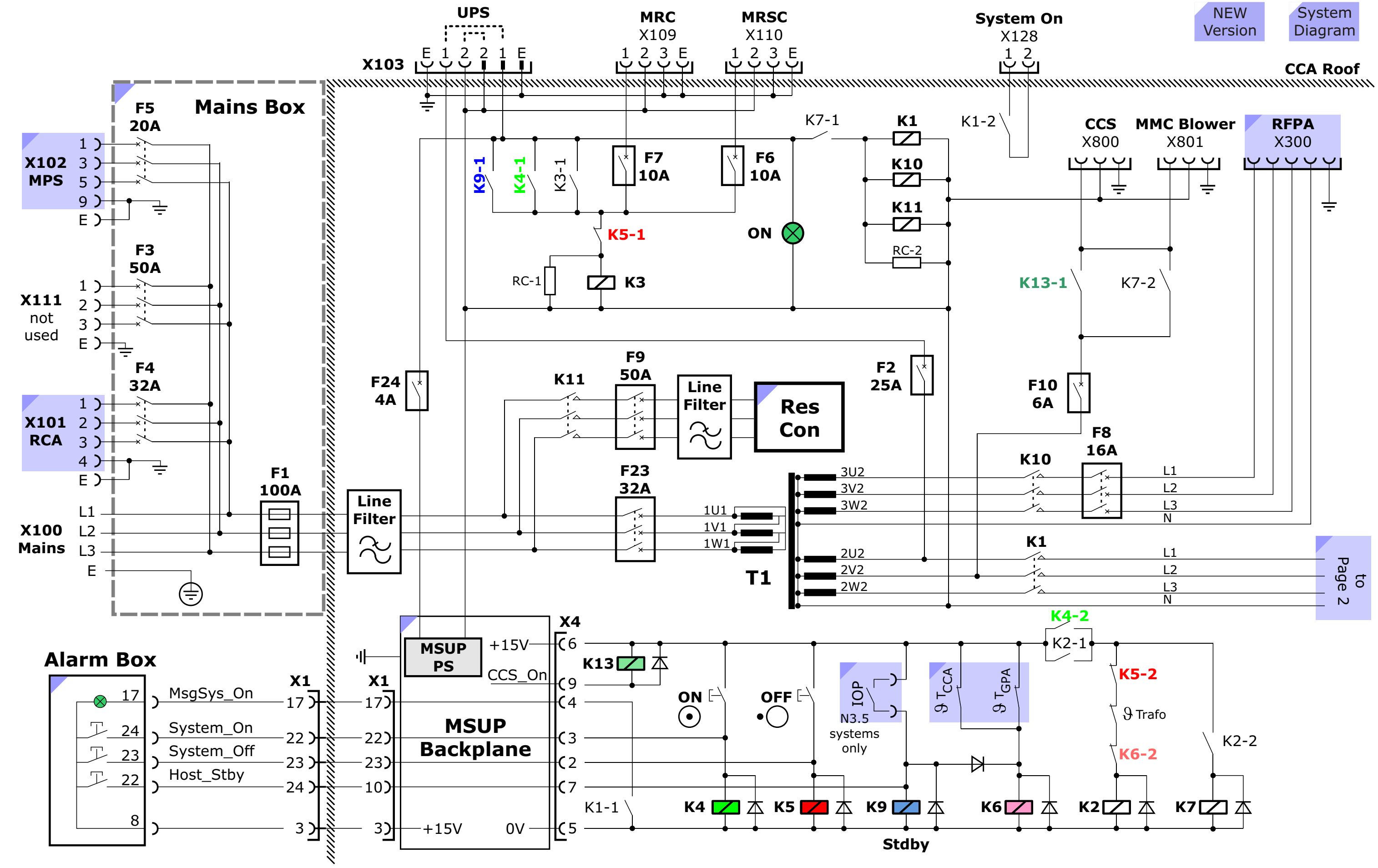
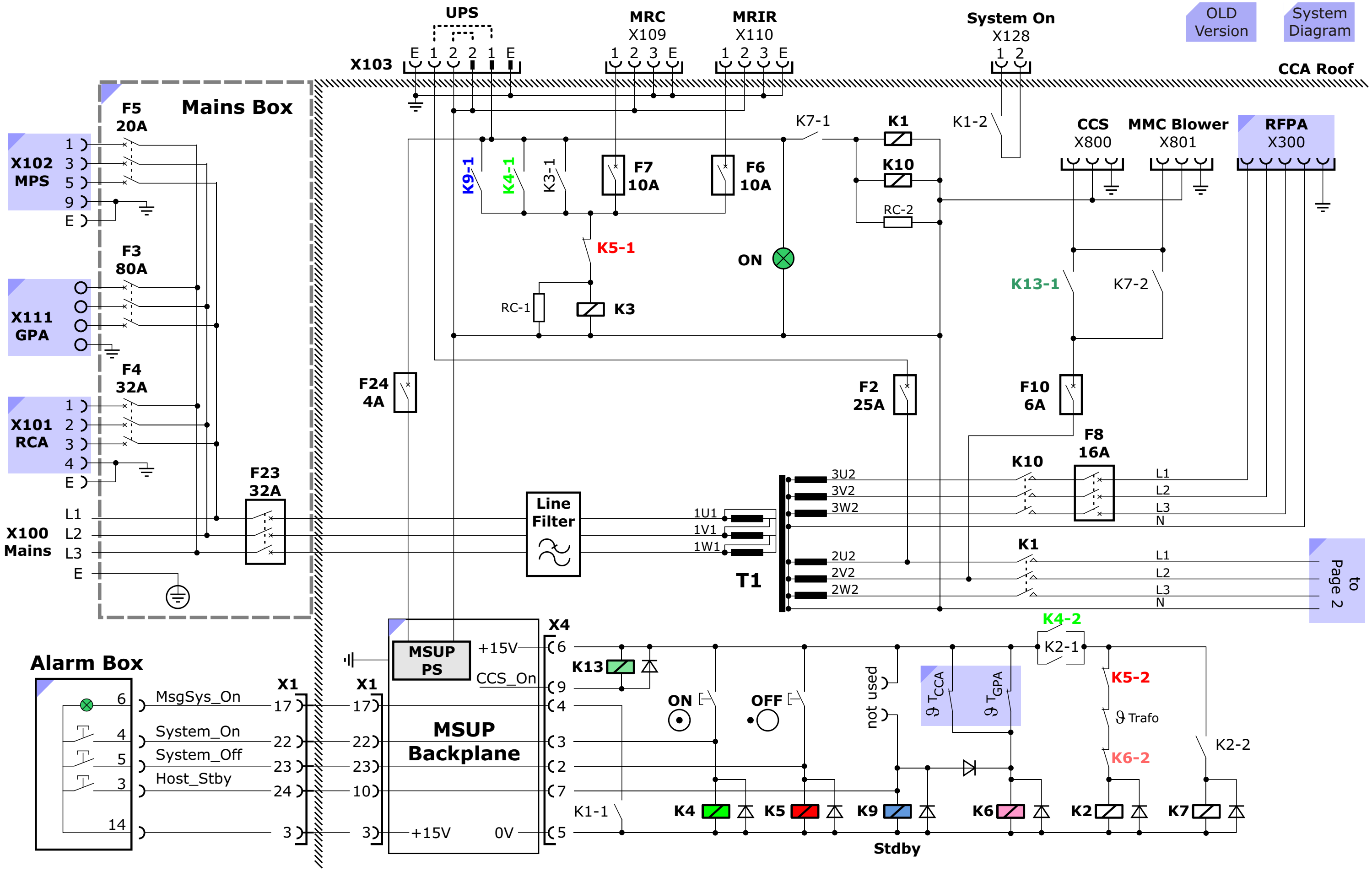


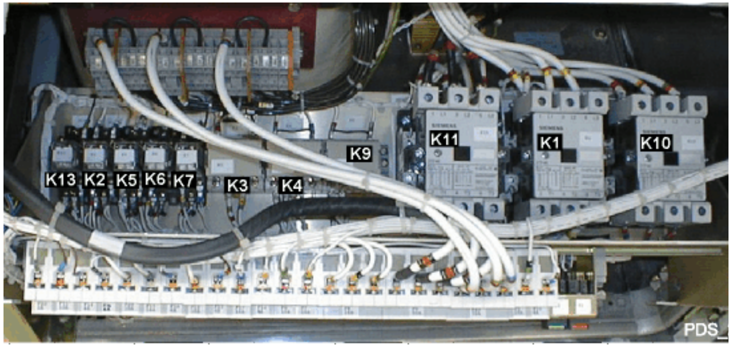
Figure 202 *LPD Schematic Page 1 (**NEW** version with K2217 GPA*



Relay Control

Relay Assembly

Figure 203 *Inner view to PDS*



System On

Manually

Pressing either the "System on" button on the Alarm Box or the GREEN button S1 in the LPD K4 is activated temporarily closing contact K4-1 and K4-2. Contact K4-1 activates K3 which pulls in self-holding contact K3-1 and supplies the MRC and Imager with power. Contact K4-2 activates K2. When K2 activates the self-holding contact K2-1 keeps itself on and contact K2-2 activates K7. Contact K7-1 activates relays K1, K10 (and K11 on oler systems) which supplies power to the rest of the scanner components.

Via Software

"System On" can also be activated the software under the Options > System Control menu on the Scanner task card : System Standby. When activated, the Host sends a command to a μP in the Alarm Box over a serial link and the μP activates the System On signal to the LPD.

System Off

Pressing either the "System off" button on the Alarm Box or the BLUE button S2 in the LPD activates K5 which opens contacts K5-1 and K5-2 deactivating relays K3 (MRC, MRIR power) and K2 (K7, K1, K10 and K11 removing power from the scanner components).

Host Standby

Manually

Pressing the "Host Standby" button on the Alarm Box activates K9 and K6. K9 will enable K3 (self-locking) which powers MRC and MRIR (MRSC). If the scanner was ON, K6 will disable K2 and so K7, K10, K11 and K1 removing power from the rest of the scanner components. If the scanner was OFF then K6 has no function.

Via Software

This feature works the same as for System On described at the left, however, the Standby signal is activated.

Overtemperature Protection

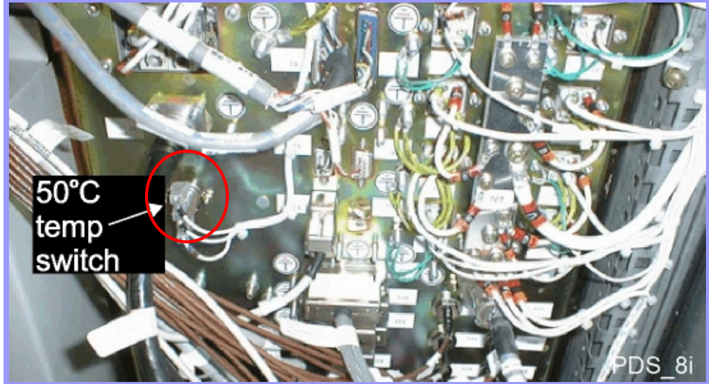
In case the temperature in either the CCA or GPA cabinets exceeds 50°C K6 will be activated and the system is switched to Standby. If the system is already switched off, but for any reason the overtemperature switches get activated, a diode between K6 and K9 will prevent the system from switching to Standby.

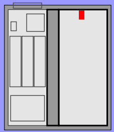
CCA and GPA 50°C Overtemp Switches

Figure 204 *Cabinet Overtemperature Switches*

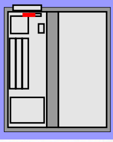
OLD Version

NEW Version

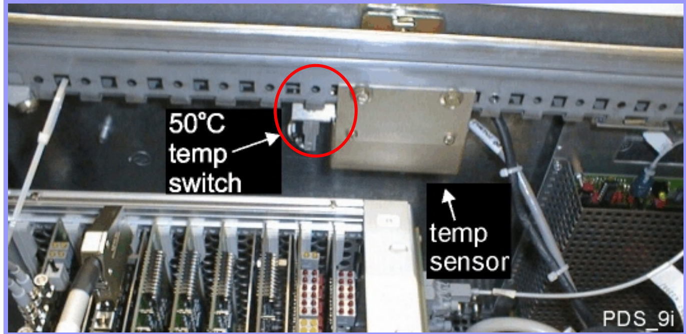


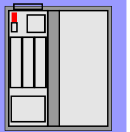


CCA Cabinet



GPA K2209





GPA K2217

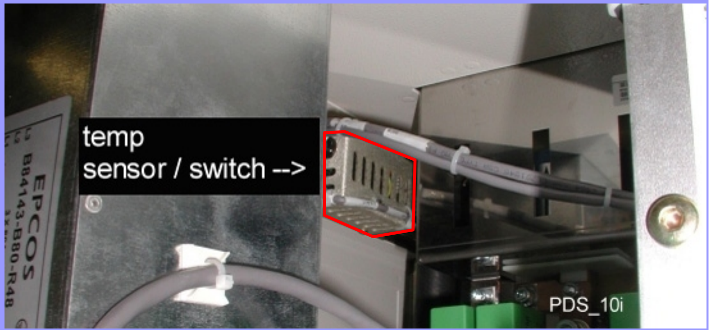
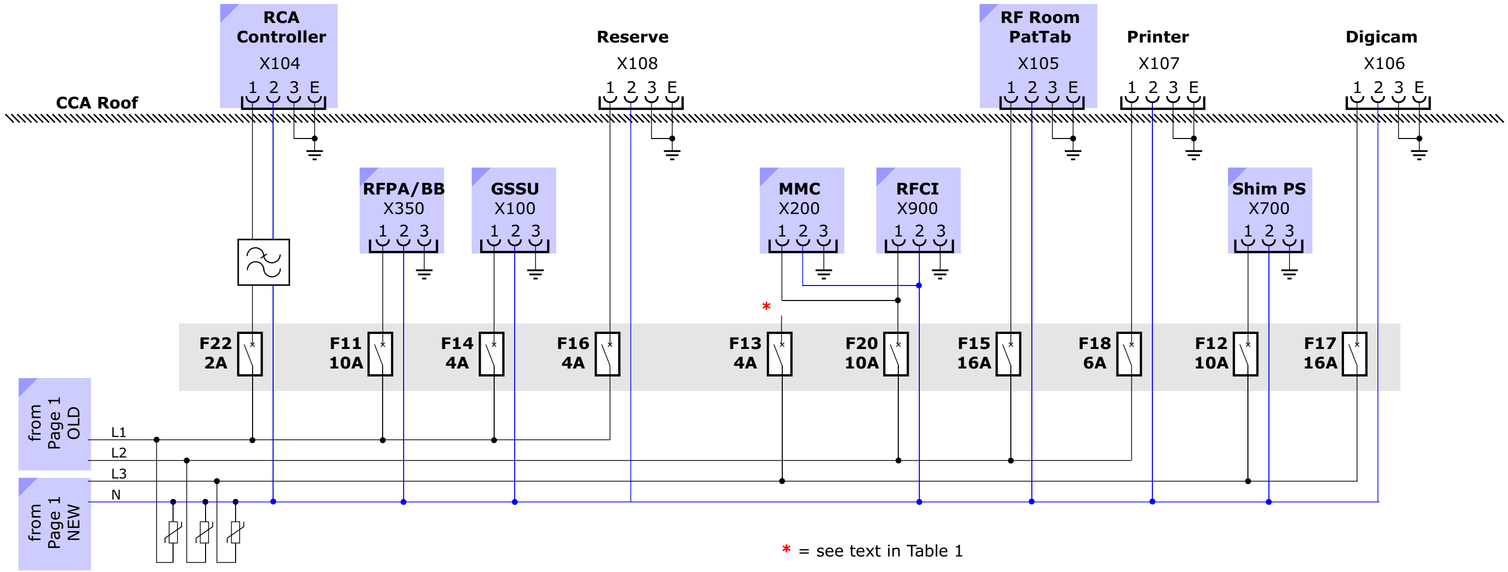
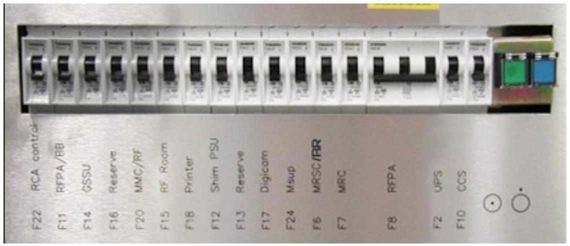


Figure 205 *LPD Schematic Page 2*



Old version

New version



Circuit Breakers

NOTE The diagrams above show the relay contacts in the "System Off" state.

LPD Circuit Breakers	
F22	RCA control electronics
F11	RFPA / BB (CORA amplifier for Spectroscopy)
F14	GSSU, driver and multiple DC supply, K2209 Gradient only: resonant converter control
F16	spare
F20	D14 motherboard,TX-modul D1, RX-modul D12 (TALES, LCCS, BCCS, BTB), **in newer systems**: additionally MMC power supply, MPCU-box, backplane with MC4C40, RX4, IOP
F15	PTAB, patient camera, patient light, fan, localizer-light, vacuum pump
F18	Printer option
F12	Shim control crate, DC supply, fans
F13	MMC power supply, MPCU-box, backplane with MC4C40, RX4, IOP, **in newer systems**: spare
F17	Digital camera (optional)
F24	MSUP, also supply for relay control!
F6	MRSC / BR (image processor)
F7	MRC with LCD monitor, intercom
F8	RFPA (160 volts, 3-phase)
F9	K2209 Gradient only: resonant converter, **in newer systems**: removed
F2	Main fuse for relay control and MSUP,in case no UPS is installed. With UPS, this will take over relay control and MSUP supply.
F10	X800 to CCS, X801 to air distributor (fan) below MMC frame

Table 1 *Circuit Breaker Listing*

LPD Circuit Breakers	
Circuit Breakers in Mains Box (on CCA roof)	
F1	Mains fuses (OLD systems), **in newer systems**: replaced with F23
F23	Main fuse for transformer T1 (NEW systems)
F3	K2217 GPA Final Stage supply (Light, Quantum, Sonata)
F4	RCA
F5	MPS (service tool)

Table 1 *Circuit Breaker Listing*

CAUTION F8 should not be used to turn power off and on to the RFPA, or damage to the RFPA may result. The complete system must be turned off to remove power to the RFPA.

Section

12 Tune-Up / QA

Main Diagram   System Diagram

Introduction

The Tune-Up is a collection of procedures used to calibrate, compensate and/or determine hardware variables needed to optimize the scanners performance in its environment. The Tune Up (TU) will be performed when an MR system is installed and when certain components have been exchanged during repairs. It is also recommended to performed a TU at regular intervals (at least yearly) to maintain maximal system performance since some system components (particularly DAC and ADC offsets) will drift with time. The replacement of parts wil usually require one or more Tune-Up procedures to be performed as well. This information is found in the Replacement of Parts procedures.

The QA offers a set of procedures which are intended to verify that the scanner is operating within specifications and as well as procedures that whose main intention is to help determine the cause of poor system performance.

This description will include physical background information of the individual Tune-Up and QA procedures. The more simplified and automatized the service-software has become, the more important it is to have a good understanding of these procedures and what they are doing.

Tune Up Procedures

The procedures fall into three major parts:

- **RF-related** Tune-Up Procedures
  - Tuning Calibration
  - Body Coil Tuning
  - RF Characteristic
- **Gradient-related** Tune-Up Procedures
  - Regulator Adjust
  - Table Adjustment
  - Phantom Shim
  - Cross Term Compensation (CTC)
  - Eddy Current Compensation (ECC)
  - Coil Power Losses (CPL)
  - Gradient Delay
  - Gradient Sensitivity
- **Receive Path-related** Tune-Up Procedures
  - Receive Path Calibration (to be done with the Head Coil)

Quality Assurance Procedures

The description for the QA procedures begins on page 320

Modes

Each of the Tune-Up steps can be performed in one of two modes:

- **Normal mode**: this is the mode that has to be used to achieve a status of "Done".
- **Expert Mode**: As the name implies, this mode is intended for experts only. In this mode, some procedures offer the possibility to change or make individual selections of sub-steps of a procedure. For example, a single gradient axis could be measured alone or certain parameters can be selected, such as the terminations for the tuning calibration. The determined or measured values will also be saved in this mode!

Procedure Status

If a procedure is successful, it will be set to **Done**.

A status of **To Do** will be set if it is a fresh installation or when a procedure having interdependencies was performed retrospectively, For example, if a regulator adjustment is performed, all gradient-related procedures will be set to To Do, even if they were set to Done previously.

Also, if a Tune-Up procedure has changed its status then the procedure(s) in the Quality Assurance menu for the corresponding coil are changed to **To Do.**

The ERROR status is set if a procedure has stopped or aborted due to an error or time-out.

Reports

A report is generated each time one or a set of procedures have been performed and are found under "Reports", "Session History".

Active Coil Menu

The Active Coil menu shows the name of the currently selected coil that will be used by the Tune-up procedures. Since the body coil is always connected, it will always be the default selection when the Tune-up platform is opened.

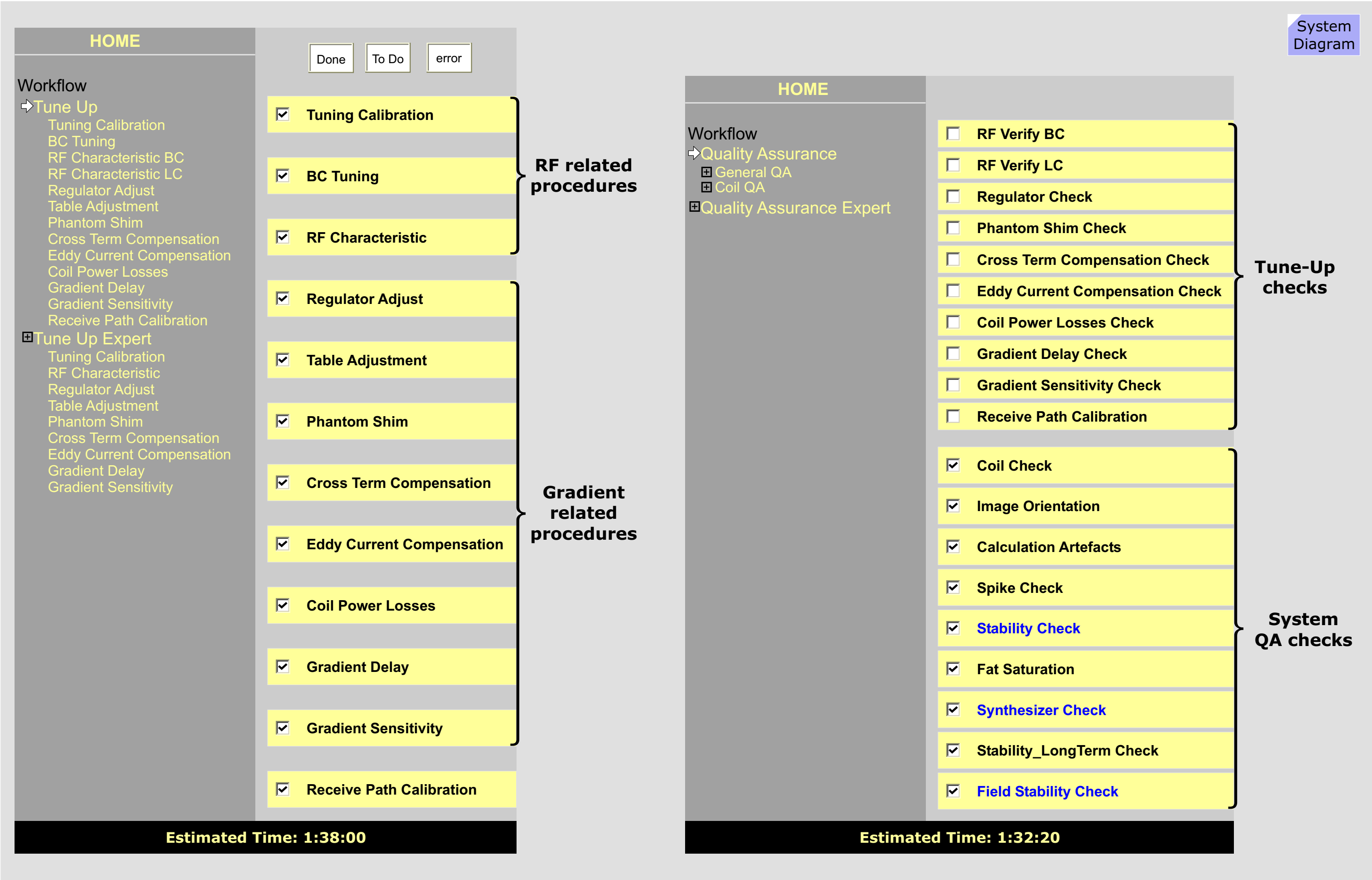
Dependencies

All selected procedures are performed from top to bottom. The individual steps follow a specific order and should not be changed due to the dependencies between those prior and those that follow.

If a Tune-Up procedure has changed its status then the procedure(s) in the Quality Assurance menu for the corresponding coil are changed to To Do.

NOTE The Tune-Up procedures measure and **SAVE** system parameters. Therefore the Tune-Up procedures should NOT be used for troubleshooting or if the system is not in proper working order. The **Quality Assurance** procedures only measure and **VERIFY** system specifications without saving any system parameter.

Figure 206 *Tune Up and QA Procedural Steps*



RF Related Tune-Up

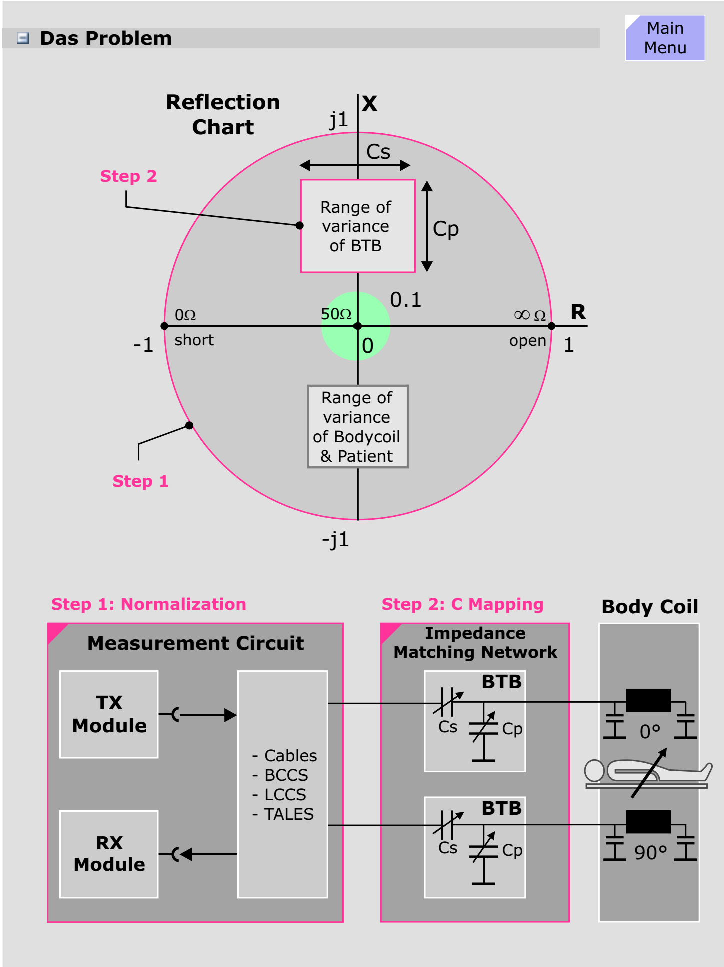
Tuning Calibration

The Body Coil's center frequency (electrical length) and loading conditions are constantly changing with every new patient or patient position thus shifting its impedance away from the nominal RF System impedance of 50 ohms resulting in an increasing power reflection. The Harmony and Symphony have been designed with an impedance (Z) matching network called the Body Tune Box (**BTB**). The BTB consists of a serial and a parallel capacitor array that can be set to compensate the effects of the patient and bring the impedance of the Body Coil to the nominal 50 ohm impedance, or close enough so that the reflected power is 10% or less (green circle in center of reflection chart). The task of measuring and analyzing the BC impedance (*reflection factors*) and finding the appropriate BTB capacitor setting to compensate is performed by the tuning algorithm in the Adjust/Tune procedure.

Before the Adjust/Tune procedure can perform the task of matching the impedance it must first calibrate the Tune Measurement Circuit which is used to measure the reflections and also the actual effective range of the BTB. This calibration is accomplished in two separate steps:

- *Measurement Circuit Calibration* - normalizing the *impedance characteristics* of the components in the Tune Measurement Circuit including the TX_Module, RX_Module, all connecting cables and the internal components of the BCCS, LCCS and TALES.
- *BTB Calibration* - *mapping out* the range of variance of the series (Cs) and parallel (Cp) capacitances in the BTB.

Figure 207 *Simplified BC Tuning Circuit*



Tune Measurement Circuit Calibration

Measurement

For this measurement, the TALES output will be terminated with an open, a short and a 50 ohm terminations which covers the complete impedance resitive range. Under these different impedance conditions the forward and reflected values are measured by applying a rectangular RF pulse (tuncal sequence) to the Ur and Uf side of the directional couplers in the BCCS. For each termination condition, the frequency is varied through the complete operating range. For Harmony the frequency range is 40.35 to 40.55 MHz with a frequency step of 10 kHz (21 measurements). For Symphony/Sonata the frequency range is 63.45 to 63.75 MHz with a frequency step of 10 kHz (31 measurements).

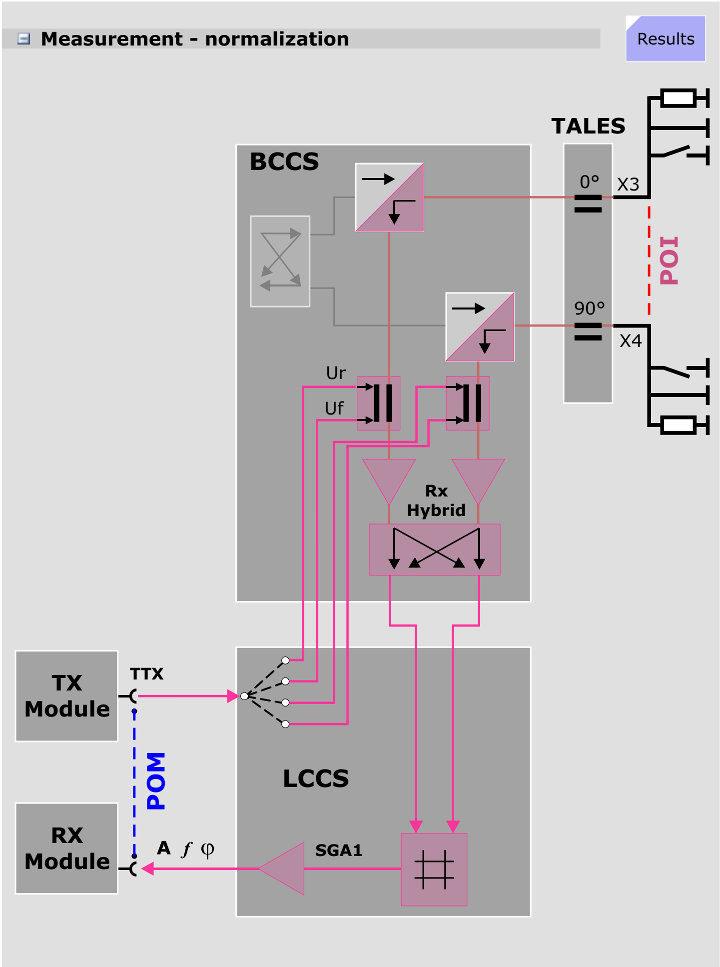
The Symphony/Sonata system has a CP Body Coil consisting of two coil systems so an additional measurement is necessary since there is also a coupling of the two Body Coil systems over the Rx Hybrid. This coupling has the result that impedances of the 0° system couple through to the 90° system effecting its impedance and vice versa, so it must be measured and compensated as well. This is done by measuring the 90° path when sending with the 0° and vice versa. The coupling factors are expressed as H-parameters.

After this measurement the Tune Measurement Circuit will be normalized and the point of measurement (POM) (i.e., the point where the system knows the exact amplitude, frequency and phase of the signal being sent or received) will have been transferred to the Point of Interest (POI) and the system is now able to "see" the impedance conditions at the POI.

Expert Mode

Individual terminations can be selected in the Expert mode. When would you want to do this? It depends.

Figure 208 *Normalization of Tune Circuit*



Results

The graphic displays the results of the H parameter (Rx hybrid coupling). Each colored line represents one of the four measured H parameters with each line displaying the 21 (harmony) or 31 (Symphony/Sonata) measured frequencies (tick marks).
It is difficult to interpret this graphic since the lines will always be round. You may notice, however, that the blue line begins, for example, at real part 0 and imaginary part of approximately 0.75 and runs to r=0.6/i=0.2. All other lines begin and end at different points. The graphic may be of help if compared with curves measured at a time when the system was functioning properly. Otherwise: forget about it.

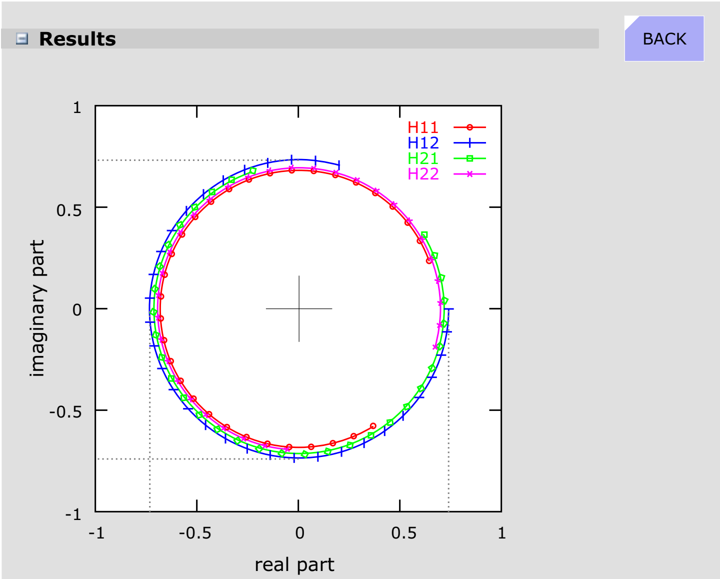
Before the BTB Calibration is performed (the next measurement, see below), the tuning calibration is first verified. This is accomplished with two measurements. The first measures the forward and reflected values on the 0° system with a 50 ohm termination. After applying the compensation derived from the tuning calibration measurement both real and imaginary parts should be zero (= 0% reflection). The second verification measurement measures the 90° system with a short which should give a result of real = -1, imaginary = 0 (i.e., 100% reflection). The results are displayed in the two tables.

If the results are good, it will continue with the BTB Calibration.

Testing the Tuning Calibration

There is a service test in the TestTools under RF system / BTB Interactive that also checks the tuning calibration. This test uses, however, an open termination instead of a short (as used in the measurement described above) which should result with a value of real=1, imag=0 if the last saved tuning calibration is still valid (i.e., nothing in the tuning circuit has changed).

Figure 209 *Results of Normalizations*



TALES Reflection Verification 0-system (50 ohm)

	Value	Low Spec	High Spec
TALES_0_Reflect_Real	0.00	0.00	0.02
TALES_0_Reflect_Imag	0.00	0.00	0.02

TALES Reflection Verification 90-system (short)

	Value	Low Spec	High Spec
TALES_90_Reflect_Real	-1.00	-1.02	-0.98
TALES_90_Reflect_Imag	0.00	0.00	0.02

BTB Calibration

After successfully normalizing the tuning circuit, the capacitor array of the BTB will be "mapped out", i.e., its range of variance determined. The tuning calibration must have been successful since the impedance and coupling data are used.

In the circle in the diagram at right represents the reflection factor for any given load (R) and frequency (X) value. The outer perimeter signifies 100% reflection (r=|1|), the center point 0% reflection (r=0 i.e., 50 ohms). This chart has been overly simplified for explanatory purposes. It can be seen, that the Cp capacitance varies along the X axis, whereby the Cs capacitances vary along the R axis. The amount of variance along either axis depends on the *number of* and the *values of* the individual capacitors. For example, if the BTB has 4 switchable Cs capacitors, call them C1 through C4, and C1=1pF, C2=2pF, C3=4pF, C4=8pF, then there are 16 different capacitance values ranging from 1-15pF. There are three different versions of BTB:

- Äst < 5 : 4 Cs capacitors, only three selectable (= 8 values)
- Äst > 5 : 4 Cs capacitors, all selectable (=16 values)
- BTB-2 : 5 Cs capacitors, all selectable (= 32 values)

(Äst = revision level) All BTB versions have 4 selectable Cp capacitors yielding 16 values.

Results

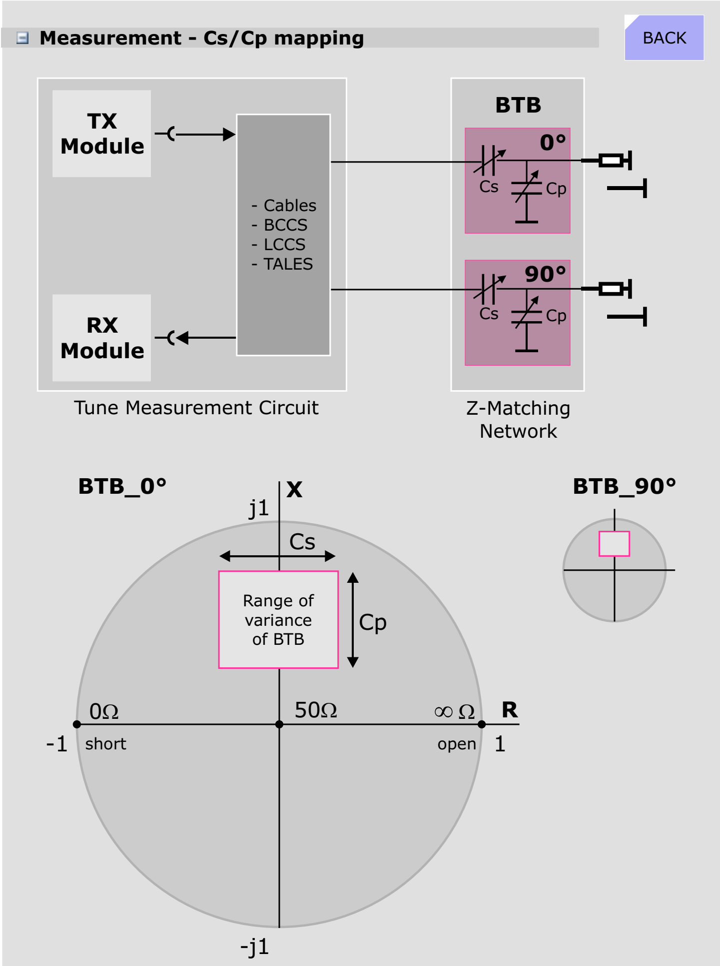
The results are not displayed but are stored in the tuncal.dat file under C:\MedCom_047\MriSiteData\Measurement if the program finished successfully.

The output in the report is as follows:

```
Performing BTB Calibration measurement for Open/50 ohm setup
Performing BTB Calibration measurement for Short/Open setup
Performing BTB Calibration measurement for 50 ohm/Short setup

Results are in specification. Have a nice day.
```

Figure 210 *Mapping out the BTB capacitances*



Body Coil Tuning

This procedure is used to measure and, if necessary, adjust the central *resonant frequency* of each of the two Body Coil systems as well as minimize the *coupling* between them.

This procedure is only possible for systems with the bird-cage Body Coil (BC47-2).

Measurement

The measurements are performed with fixed Cs/Cp values and over an 800 kHz frequency range. The resonant frequency is found at that point where the reflection of the BC has a minimum. The decoupling is given by the frequency where the transmission has its maximum value. The transmission is determined at that frequency where the transmission (coupling) has its maximum value.This should be lower than -16 dB. A -16dB coupling translates into a 2.5% power loss! Therefore, the lower the coupling, the less losses.

Evaluation

Resonant Frequency

The procedure measures the frequency response of the Body Coil over an 800 kHz frequency range. The evaluation determines:

- if the center frequency of each Body Coil system is between 63.45-63.75 MHz
- that the center frequencies of both coil systems are tuned to within 100 kHz of each other
- that the reflection factor is not more than 30% ($r \leq 0.3$)

Transmission (coupling)

The transmission through the Body Coil, that is, the amount of coupling between the two Body Coil systems, describes a mutual inductance between the Body Coil systems due to coil construction. The transmission is measured by sending a signal into the 0°-system and measuring what comes out of the 90°-system. A

Adjust Procedure

This is a manual adjustment requiring the rear tunnel cover to be removed. On the Body Coil are three 50-turn trim capacitors (refer to Figure 211 below), two for adjusting the frequency of each Body Coil system and one for minimizing the coupling between the two.

Tuning

The trim capacitors **T1** and **T2** vary the resonance frequency of the system 1 and 2 respectively. These 50-turn capacitors vary the frequency by approximately **5 kHz per turn**. Turning the cap **CW LOWERS** the frequency, **CCW INCREASES** the frequency.

Decoupling

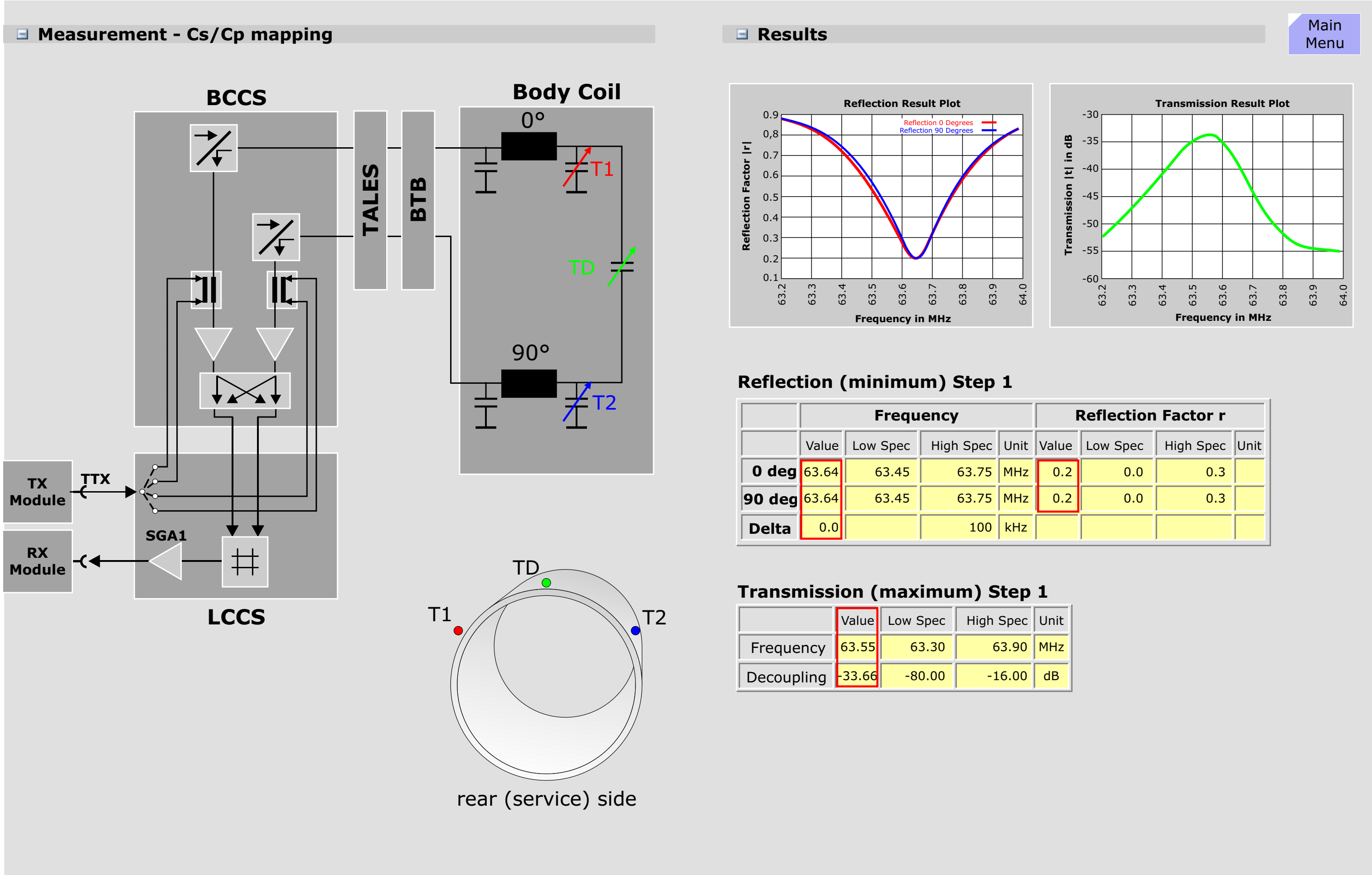
TD is also a 50-turn capacitor and effects the de-coupling between the two BC systems. It is not possible to know which direction to take at first, but there is only one minima.

NOTE There is a interdependency between all three adjustments! Refer to the TSG for a detailed description of this procedure.

Expert Mode

There is no Expert Mode for this procedure.

Figure 211 *Body Coil Resonant Frequency and De-coupling Adjustments*



RF Characteristic

The calibration of the RF transmitting system is important for obtaining optimum image quality. The RF pulse amplitude defines not only the flip angle, but the pulse lobe amplitude relationship also defines RF pulse profile and hence the slice profile.

Effects of amplitude distortions

Amplitude distortions of the RF envelopes caused by the non-linearity of the amplifier may produce distortions of the slice profile and slice thickness with the following consequences:

- reduced signal to noise caused by crosstalk (in multi-slice sequences)
- reduced contrast due to the resulting partial volume effect

The principle of amplitude corrections is shown in the next graphics

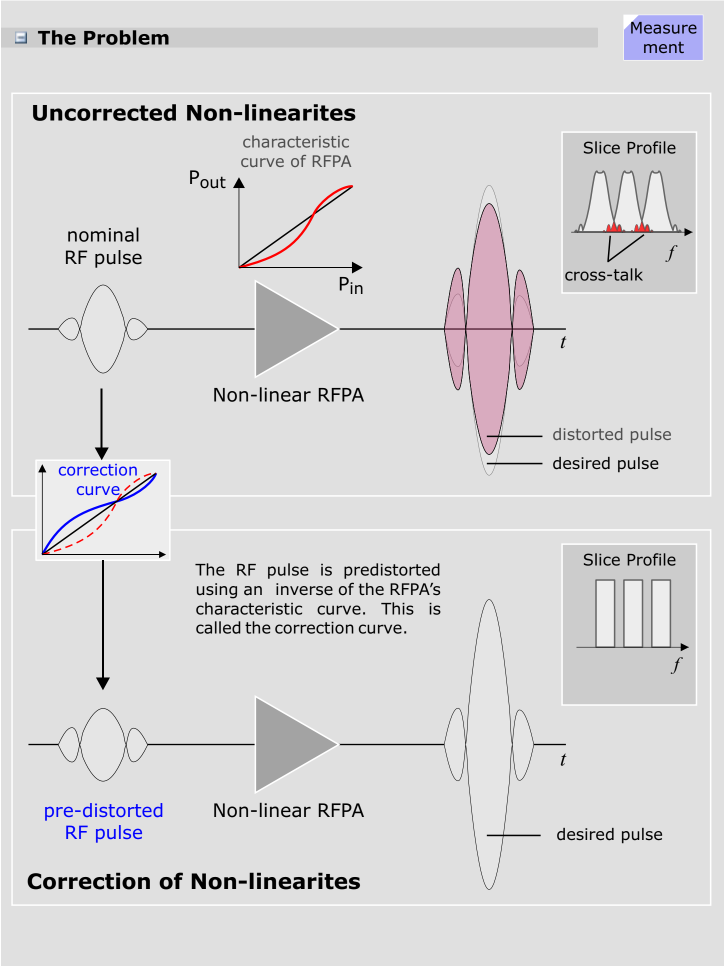
Effects of phase distortions

A phase distortion would lead to a shift or drift in the frequency, which would result in false slice positions hence the phase distortion must also be corrected.

Method of correction

The distortion caused by the non-linearity of the RF amplifier is corrected by measuring the output of the RF transmission in terms of the amplitude and phase based on a known input to the RF amplifier for the entire operating range of the RF amplifier, thereby defining the input/output characteristic of the amplifier. The corresponding inverse function of the amplitude and phase characteristics are used for pre-distortion of the input signal in order to achieve a linear output characteristic and to optimize the RF excitation.

Figure 212 *Pre-distortion of RF-pulses*

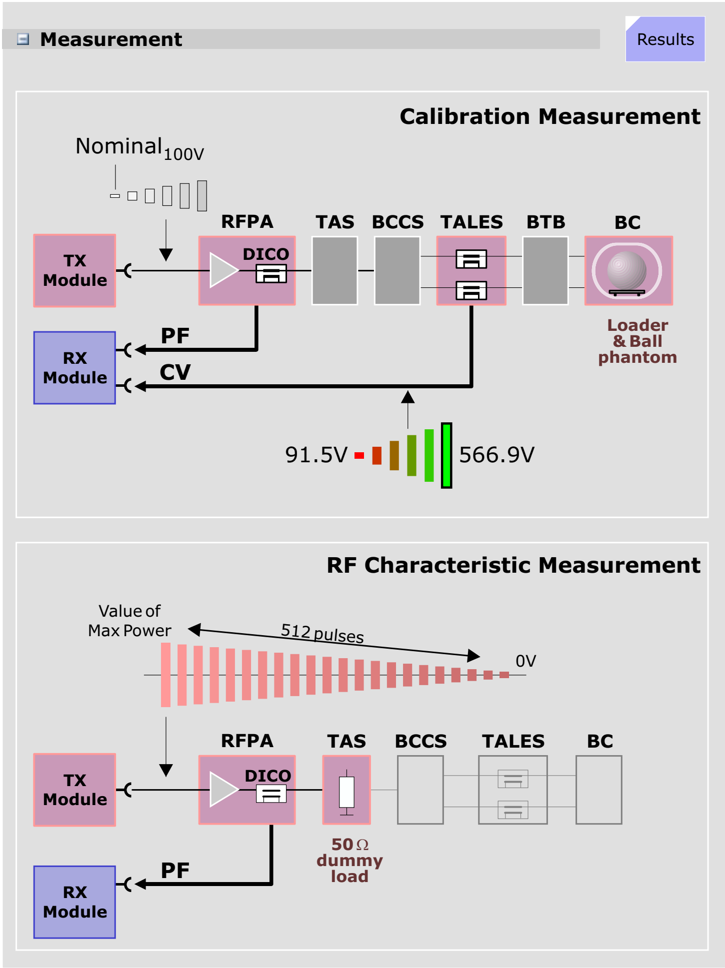


Measurement

The RF-characteristic-measurement is performed with the following steps:

1. **Calibration Voltage Search:** Starting with a nominal value of 100 Volts, the output is increased until a value of 566 Volts (400V for Harmony) is measured at the TALES.
2. **RF Calibration:** The characteristic measurement will be made using the DICO, so it must first be calibrated. Using the results of the first step, the DICO value will also be measured giving a relation between the TALES and DICO. This value is used as a correction factor for the DICO values.
3. **RF Max Power Search:** The system now determines the input voltage to the RFPA required to achieve an output power at the coil of 12.2kW (8.5kW for Harmony), the power range over which the RFPA will be calibrated. It begins with a nominal value and increments the input until the 12.2 kW has been reached.
4. **RF Characteristic measurement:** For the following measurement the RF is sent to the 50Ω dummy in the TAS. 256 pulses with different amplitudes are applied. The first pulse starts with the maximum voltage, then the voltage drops by 0.25 dB for each pulse. In this way the complete characteristic is measured.
5. **Save the RF Characteristic** result in the *rfchara.dat* file.
6. **Verify**: The final step is to perform a verification measurement. The inverse of the measured characteristic curve is now used to pre-distort the RF pulses. The result should be a linear output transfer characteristic of the RFPA.

Figure 213 *RF Characteristic Measurement*



Results

Plot of Measure Characteristic

The first set of graphics show the measured amplitude and phase characteristic curves each measured with 512 measure points. with a normalized RF input.

Plot of Calculated Characteristics

The second set of graphics displays a calculated version of the measured data mentioned above. The measured data is used to generate an equidistant RF Characteristic correction data base on input amplitude, slope and phase necessary for the RF Characteristic data file. The output data is first made by dividing the maximum voltage into 1024 equidistant points. The corresponding input and phase data is then calculated by interpolating with the actual measured data. Non-unique amplitude data is avoided by simply ignoring this data when generating the input data. Finally, the output data and phase data is converted to slope and phase values.

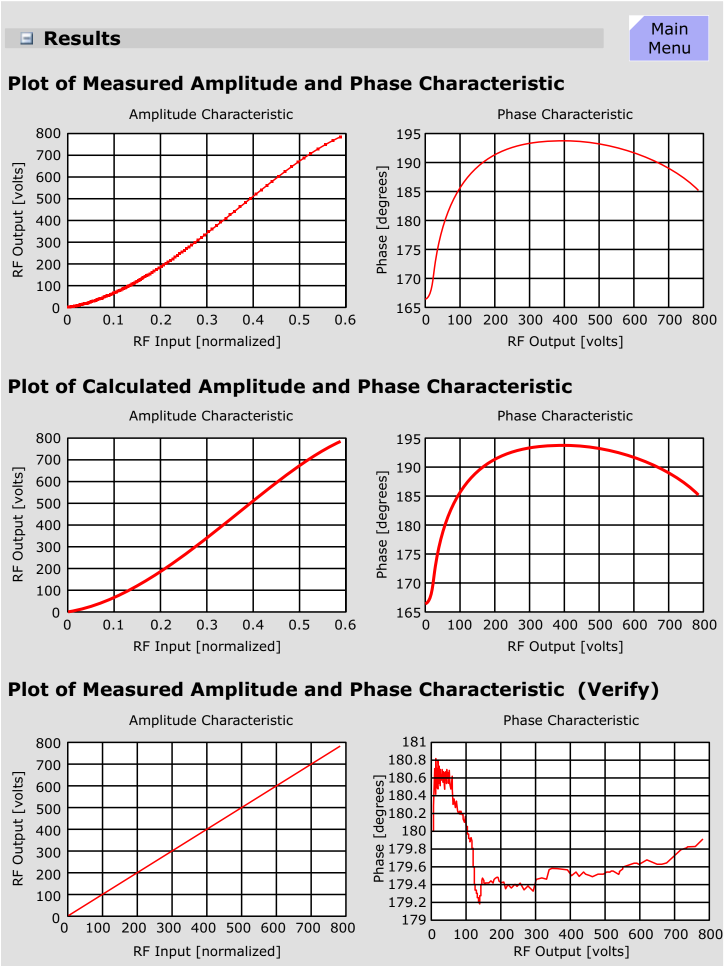
Plot of Verification Measurement

The last set of graphics display the evaluation results of the verification measurement. The verification measurement is made by using the inverted characteristic curve to pre-distort the RF pulses. For a correct compensation, the result should be a linear input to output transfer characteristic. The evaluation is made by calculating the deviation of the measured data to an ideal linear function which is derived by simply creating a straight line between the first data point (0V) and the last data point (maximum voltage). A check for monotony is also made, assuring that the amplifier at no time went into saturation.

Expert Mode

The expert mode allows the user to specify the maximum voltage. The voltage input must be between the range 50% to 100% of MaxPowerAllowed.

Figure 214 *RF Chara Results*



RF Characteristic for Local Coil Path

- Non-spectroscopy systems:

The Local coil characteristic can be measured in **Expert** Mode interactively by connecting the dummy load to the LC output of the TALES. This measurement is not mandatory because the RF Characteristic BC is applied also for local TX-coils.

- Spectroscopy systems:

The RF-Characteristic must be measured using all RF-amplifiers and all available nuclei (H and all X-nuclei) in the Local coil (LC) path. An interaction of the service technician is necessary:

Measurement

The dummy load in the TAS_C must be connected to the LC-output of the TALES (i.e. disconnect output cable from TALES X6, disconnect input cable at BCCS X7 and connect it with TALES output X6) and the service plug must be connected to patient table coil plug 1.

The RF characteristic is measured as follows:

1. **Calibration Voltage Search:** Starting with 100 Volts output (this is just a nominal value). The TALES readings are used to increase the output until 250 Volts output are measured.
2. **RF Calibration:** The DICO has to be calibrated since it is used for the final measurement of the characteristic. The DICO value is measured based on the resulting input voltage from step 1. The DICO is calibrated using the known relationship between TALES and DICO.
3. **RF Max Power Search:** By carefully increasing the output voltage, the max. power of 1900 W, respectively 2440 W (depending on nucleus) is searched.
4. **RF Characteristic measurement:** For the following measurement, the RF is sent to the 50Ω dummy in the TAS. 256 pulses with different amplitudes are applied. The first pulse starts with the maximum voltage, decreasing by 0.25 dB for each subsequent pulse. This ensures that the complete characteristic is measured.
5. The **RF Characteristic** is saved in the *rfchara.dat* file.
6. The final step is the **verify measurement**. This includes the calculation of amplitude and phase deviation and checking if the results are within specification. The results and graphics can be viewed in the service software under Tune-Up, Report, Tune-Up Results.

Results

Same as for RF Characteristic procedure for Body Coil.

Expert Mode

In Expert Mode, the function operates similar to standard mode except that the following options are available:

1. You can click **Save** to save the **RF Characteristic** after measurement.
2. You have an option to modify the "Maximum allowed RF power".
3. Selection of RF-amplifier, coil path and nucleus.
4. Option to switch off the frequency adjustment
5. Option to disable switching of TAS to dummy load.

Also, Expert Mode does not include RF-verification at the end of the measurement.

Main Menu

Gradient Related Tune-Up

Regulator Adjust

The regulators in the newest very-high-power Gradient Amplifiers (500A and above at 2000V) have a **proportional**, P, (= gain), **integral**, I, (= time) and **differential**, D, regulation characteristics, thus each regulator consists of a P, I and D regulator circuit in series. The purpose of this procedure is to find the optimal P, I and D regulator adjustment values for an optimized regulation characteristic.

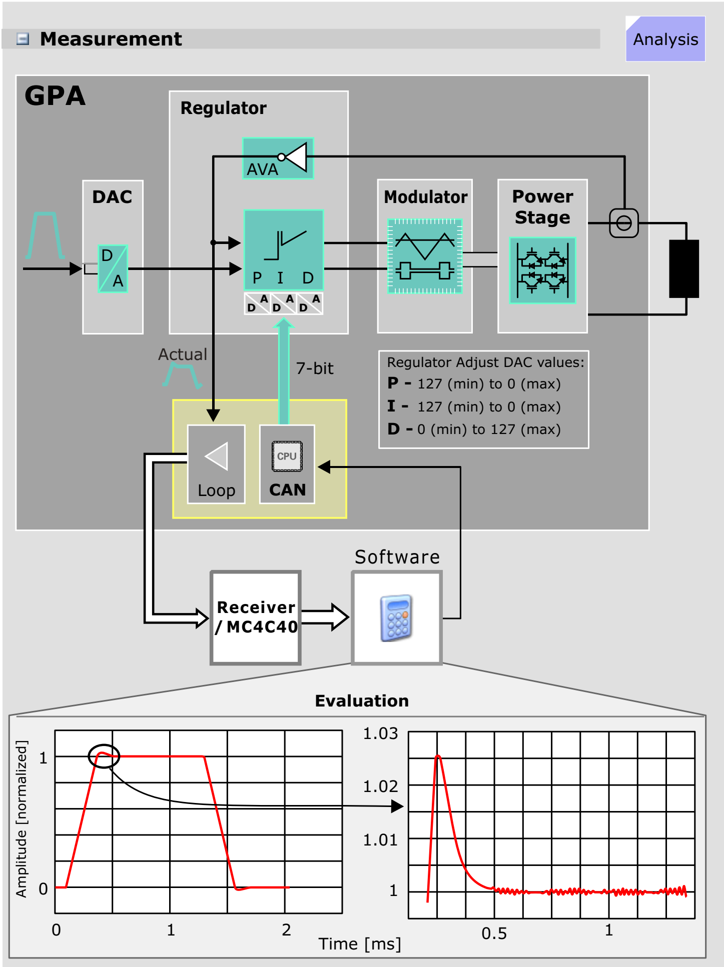
Measurement

The regulators are tuned one after the other in the order PID. Each regulator begins with a configured start value or in Expert Mode with the user given start value. The regulator value then is incriminated by one single bit and a measurement is done. The measurement results are evaluated and a new regulator value is calculated. This step is repeated until the regulator is optimized. Finally, a configured *safety factor* is added to the regulator.

The actual value gradient waveform is sent over the gradient loop path from the D16 Service Can board in the GPA to the RX_Module in the RFSU and acquired by the ADC, with 256 sample points. The GradReg program then performs the following:

- checks whether the data is in the real or imaginary part
- displays the measured pulse: the first graphic shows the complete gradient pulse, the second graphic displays the overshoot of the gradient pulse
- performs an evaluation (described below)

Figure 215 *Regulator Adjustment*



Evaluation

The pulse is evaluated for three criteria:

- **Overshoot** amplitude to the nominal pulse amplitude (in %)
- **Decay time** for overshoot to decay to 15% of its amplitude
- **Smoothness** is the maximum positive deviation of the gradient pulse top portion.

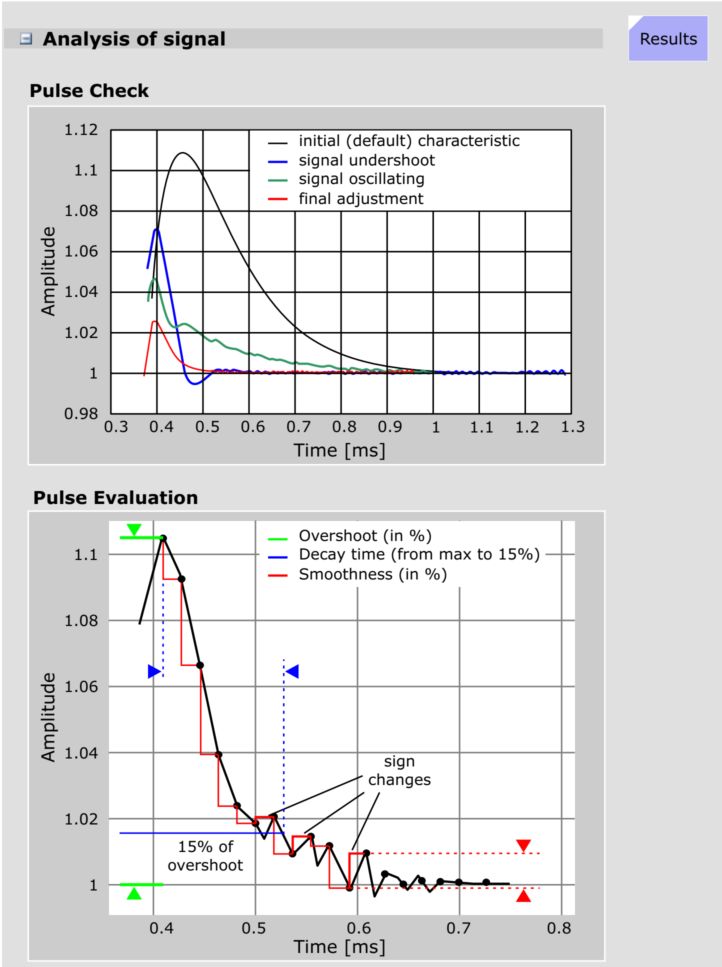
The smoothness of the overshoot is determined in this way: Starting from the maximum data point of the overshoot, the difference to the next successive data point is determined. This is repeated for all data points of the overshoot. The smoothness factor is the maximum difference determined by the above process divided by the nominal pulse amplitude an expressed in %.

Pulse Check

A check if the regulator is being over-driven or the pulse signal oscillates is also made, indicated by 2 or more sign changes in the data points of the overshoot.

If there are points with an amplitude below the nominal amplitude an undershoot after an existing overshoot is detected. That means over-driven regulator or oscillating signal.

Figure 216 *Regulator Signal Evaluation*



Results

The final regulator adjustment values and achieved values are displayed in tabular form.

Figure 217 *Evaluation Results (final results)*

Results

Main Menu

	Value	Low Spec	High Spec	Unit
P	34			
I	50			
D	42			
Overshoot	2.572	1.500	3.500	%
Decay Time	0.103	0.000	0.150	msec
Smoothness	0.097	0.000	0.300	%

Regulator results of X gradient are in specification

After all regulators (PID) are tuned it is checked if the signal is in specification. If so then the regulator values are set permanently. If an error occurs or the program is aborted the old regulator values (saved before Tune-Up starts) are restored for the GPA.

Expert Mode

If the factory default values are too tight and cause the regulator to oscillate and result in an aborted measurement, the default values can be changed in the Expert Mode menu.

Figure 218 *Evaluation Results (final results)*

RegAdj Expert Mode

Orientation	X gradient Y gradient Z gradient	
Proportional	25	127 (min) to 0 (max)
Integral	38	127 (min) to 0 (max)
Differential	58	0 (min) to 127 (max)

Table Adjustment

The goal of the Table Adjustment is to determine:

- center positioning of the phantom in the magnet iso-center
- sufficient field homogeneity
- distance between light marker and magnet iso-center

The Phantom shim, CTC and ECC procedures require a centered phantom for accurate results. The Table Adjustment determines the phantom position with respect to the center of the gradient coils (iso-center). From this, the distance from the light marker to the magnetic iso-center is determined.

Measurement

The measurement uses a double echo sequence. The first echo is acquired without applying a readout gradient resulting in an FID the length of which is a rough check of the fundamental imaging volume homogeneity.

During the second echo a readout gradient is applied resulting in a signal spectrum of the phantom. The center of the phantom's signal spectrum is compared to the center of the scan volume frequency window to calculate the phantom position in readout direction. The sequence measures the three axis in this way. Figure 219 shows as example only the X axis.

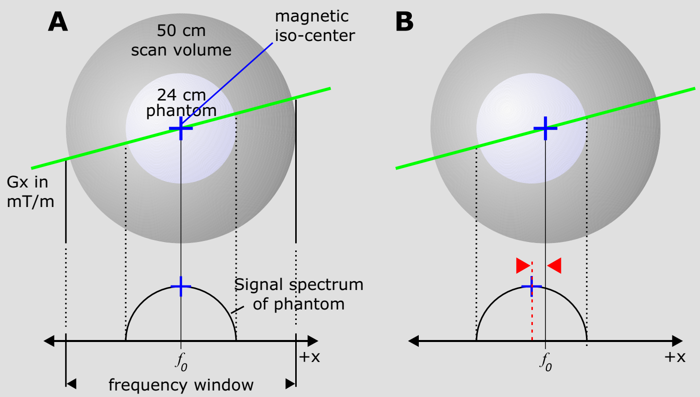
Results

The results of the Z axis measurement is used to determine the table position and will be corrected automatically and stored as the new light marker distance. All three phantom position values are displayed in a table, shown in the table in the graphic. Corrections necessary in the X direction are made by re-positioning the phantom. Corrections in the Y axis may be made by adjusting the table height with SY0 (described in the TSG).

Figure 219 *Table Adjustment Results*

Measurement

Main Menu



With a gradient applied, a linear frequency deviation over distance results providing a Hz/mm relationship.

A If the phantom is centered within the scan volume, the center of its signal spectrum will correlate with the center of the frequency window defined by the applied gradient. The magnetic center of the gradient coil defines the iso-center of the scan volume.

B If the phantom is off-center, its signal spectrum will not be centered to the frequency window. The deviation from center in mm can be derived from the correlation of frequency per mm value known from the gradient amplitude in mT/m.

Result

Phantom Position

	Value	Low Spec	High Spec
X Position	-3	-5	5
Y Position	4	-10	10
Z Position	3	-2	2

Phantom is not centered in z-direction!
Moving patient table by -2 mm.
Correcting patient table distance by 2 mm to 899 mm.

Phantom Shim

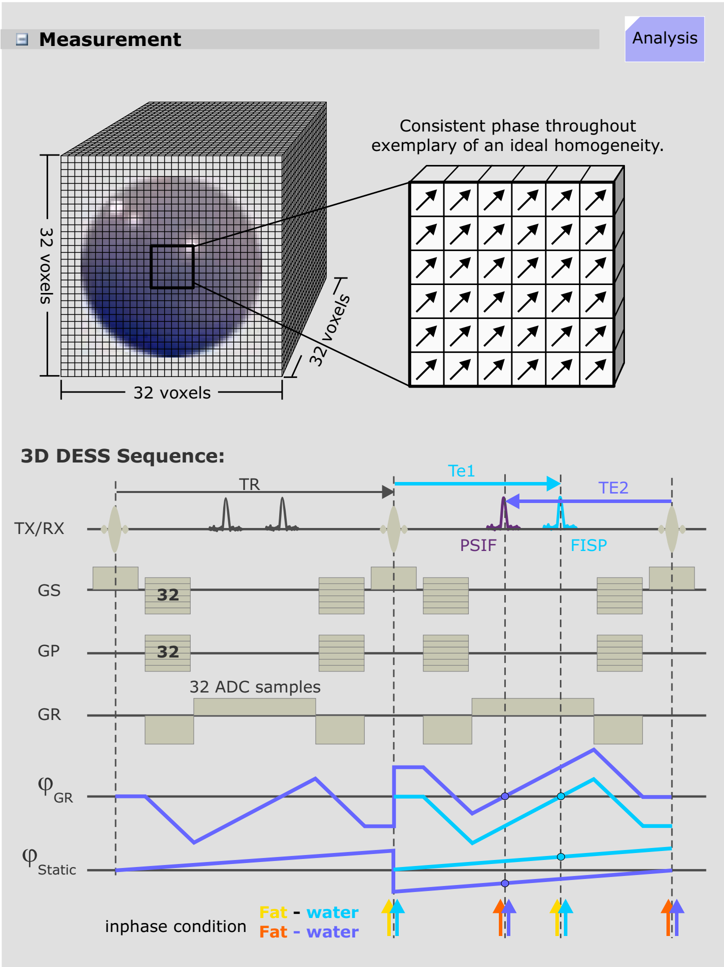
Phantom Shim measures magnet field inhomogeneities and calculates the gradient offset and shim coil currents (option) necessary for optimization. It is performed at this point in the Tune Up set as preparation for the following procedures using the ball phantom for measurement. The correction currents determined here also serve as default values for the pre-sequence adjust shim procedure used for patient imaging.

Background

The phase of an MR signal is directly proportional to the magnetic field. Phantom Shim makes a 3D-measurement, a so-called "Field Map" of the measurement volume using a 32 x 32 x 32 voxel matrix covering a 35 x 35 x 35 cm volume. The resulting voxel volume is about 0.5ml and the voxels are cubes, so the resolution is isotropic. An advantage with this method is in having actual field values for each of the voxels and allows the user to define a specific volume within the measurement volume that is to be optimized. This is called local volume shimming. Also, the evaluation algorithm detects voxels with little or no signal and will not use the data for correction improving the optimization.

To avoid hardware anomalies (e.g., gradient delays or array coil phase errors) it is better to do the measurement with two echo times and examine the phase difference of both measurements. To save time the two echoes can be measured in one sequence using a 3D-DESS-sequence (Double Echo Steady State). The measurement is done twice, for the second measurement all gradients are reversed to eliminate eddy current effects.

Figure 220 *The 3D - Shim Sequence*



Measurement

Shim System Check

First, an operational check of the shim hardware is made. A nominal shim current is applied to each shim channel followed by a shim measurement and the actual field change effect is compared to the expected, predicted field change effects. In this way wrong cabling, wrong polarity, open or short coils can be detected.

Field Map

The sequence is a 3D-DESS-sequence, i.e. two echoes, a FISP and a PSIF echo are generated. The echo times - 4.7ms - are so chosen, that fat and water are in phase, hence, the sequence is prepared for patient use as well.

As seen in Figure 220 the $Phase_{GR}$ line represents the phase rotation due to the readout gradient and the $Phase_{static}$ is the phase rotation due to the static field inhomogeneities which is the factor we are looking to determine.

Chemical shift (Fat - Water) = 3.35ppm. We operate with 1.5T:

3.35 * 63.6 MHz / 10**6 = 213Hz.

1/213Hz = 4.7ms.

Evaluation

For 1.5T systems the echo time difference of FISP and PSIF signals is 9.4ms. This has the shortcoming, that the **true absolute** off resonance-frequencies cannot be analyzed: Due to the Nyquist-theorem the highest frequency, which can be analyzed correctly is only

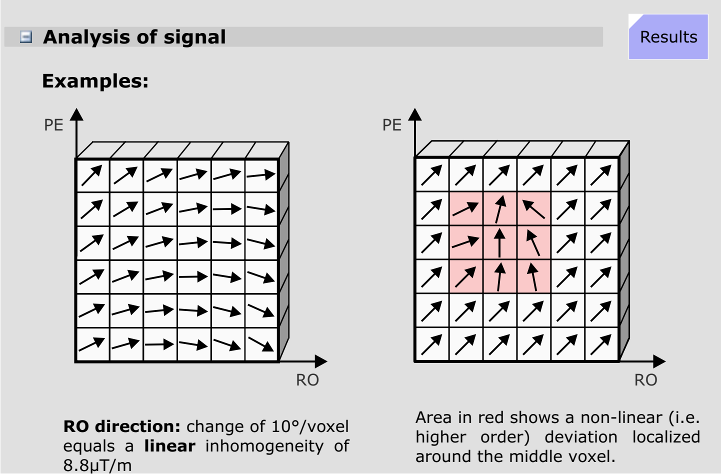
$$1/9.4ms = \pm 53Hz$$

However, for the check and optimization of the B_o-field it is sufficient to compare the **field differences** between neighboring voxels. The field difference is obtained by comparison of the phases of the magnetization vectors. With an echo time difference of 9.4ms a shift of 10 degrees from voxel to voxel in one direction results in a linear homogeneity of about 9µT/m in that direction.

The complete data set is evaluated by a differential shim equation. The equation minimizes the difference between:

- the field generated by the three gradient coils and the 5 shim coils
- the measured magnetic field inhomogeneities.

Figure 221 *Phase rotation of magnetization vectors in the voxels*



Numerically, the problem is a least squares fit leading to a system of many linear equations with 8 unknowns: the 3 gradient offset currents and the 5 shim currents. In short: the comparison of the B_o-fields in two voxels - performed by analyzing the phases - results in one equation. For all voxels 98304 calculations are required. However, a threshold is set for the signal to noise: voxels with no or little signal-noise ratio will be rejected, so the number of equations is typically reduced to about 20000.

Results

The first table displays the measured field terms for each of the 8 terms that can be corrected. The 3 first order field terms and the 5 second order field terms that can be optimized are:

Spherical Harmonic	Field	Correction Device
First Order Terms		
A(1,0)	Gz	Z gradient coil
A(1,1)	Gx	X gradient coil
B(1,1)	Gy	Y gradient coil
Second Order Terms		
A(2,0)	$z^2 - (x^2+y^2)/2$	Shim Coil
A(2,1)	xz	Shim Coil
B(2,1)	yz	Shim Coil
A(2,2)	$(x^2-y^2)/2$	Shim Coil
B(2,2)	xy	Shim Coil

Pos. 1 The following two tables give the correction currents for the gradient and shim coils. The gradient offset current are expressed in mT/M and the shim coil currents in mA.

Expert Mode

No extra functionality is provided by the Expert Mode for either Tune-Up or QA

Figure 222 *Phantom Shim Results*

Results

Main Menu

Field Terms

	Value	Low Spec	High Spec	Unit
Bpp	2.02			ppm
Brms	0.23			ppm
A10	0.67	-0.70	0.70	ppm
A11	0.89	-0.70	0.70	ppm
B11	0.06	-0.70	0.70	ppm
A20	-0.05	-0.70	0.70	ppm
A21	0.04	-0.70	0.70	ppm
B21	0.03	-0.70	0.70	ppm
A22	-0.00	-0.70	0.70	ppm
B22	-0.01	-0.70	0.70	ppm

TU Gradient Offsets Table

	Old		New		Change	
	Value	Unit	Value	Unit	Value	Unit
Gx	0.0011	mT/m	-0.0035	mT/m	-0.0045	mT/m
Gy	-0.0041	mT/m	-0.0044	mT/m	-0.0003	mT/m
Gz	0.0030	mT/m	-0.0011	mT/m	-0.0041	mT/m

QA Shim Currents Table

	Old		New		Change	
	Value	Unit	Value	Unit	Value	Unit
A20	-10	mA	-2	mA	8	mA
A21	43	mA	29	mA	-14	mA
B21	-29	mA	-140	mA	-11	mA
A22	-455	mA	-454	mA	1	mA
B22	-13	mA	-9	mA	4	mA

Cross Term Compensation (CTC)

An applied gradient field in any axis produces not only eddy currents along that axis, but also small amounts of eddy currents are coupled over the conductive magnet components onto the other two axis', the so-called cross terms. The CTC (cross term compensation) procedure measures the eddy currents created by the cross terms. Compensated is made by pre-distortion of the gradient waveform. Cross term amplitudes can be up to 0.2% of the main applied gradient amplitude. Unlike the main eddy currents which are complicated and need to be modeled with 5 time constants, the Cross Term current decay exponentially in time in general with only one time constant.

Measurement

The procedure measures the decay of the eddy currents caused by the cross terms. Amplitudes and time constants of the eddy current contributions are then calculated. The compensation is straightforward: gradient fields of equal and opposite amplitude and time constants to the measured cross terms are applied perpendicular to the main applied gradient field. The result is a much cleaner main gradient with very small residual cross terms improving the final image quality.

Result

The amplitude and time constants if the compensations are displayed graphically and in tabular form. Only the Z axis is shown.

Expert Mode

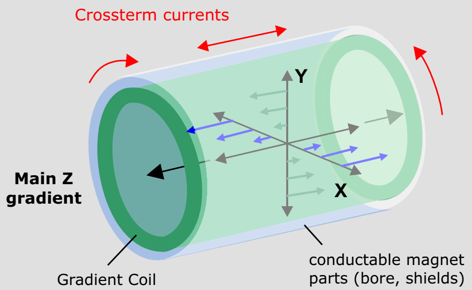
The CTC can be made individually for each axis.

Orientation
- X gradient
- Y gradient
- Z gradient

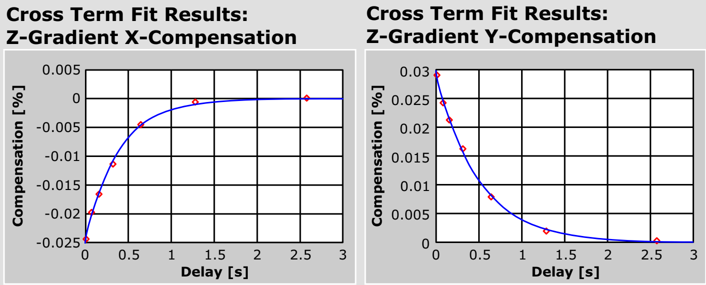
Figure 223 *Cross term Currents*

The Problem

Main Menu



Results



Cross Term Fit Results for Z-Gradient

	Amplitude		Time Constant	
	Value	Unit	Value	Unit
X Compensation	-0.0246	%	0.388	s
Y Compensation	0.0293	%	0.505	s

Eddy Current Compensation (ECC)

The dynamic gradient fields produce currents (called eddy currents) in all the surrounding conductive structures, mainly the Body Coil, magnet bore and the cryo-shields. The eddy currents produce in turn magnetic fields which oppose and distort the applied gradient fields. Eddy currents in the warm components (Body Coil and Magnet bore) have relatively short decay times, whereby eddy currents developed in the cold cryo-shields can have relatively long time constants.

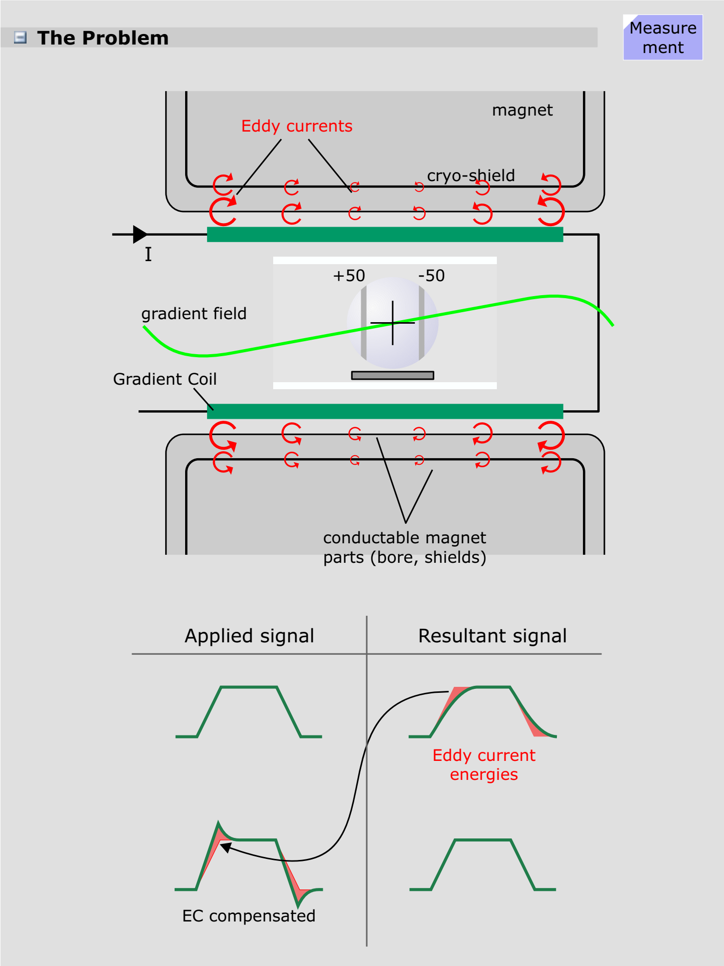
The eddy currents have to be accurately quantified and characterized by their time constants and amplitudes to achieve a good compensation. The compensation is made by adding the reciprocal of the measured eddy currents to the gradient pulse thus neutralizing their effects. For accurate compensation 5 time constants are used.

Spatial Dependency of Eddy Currents

Experience has shown, that for smaller slice shifts it is sufficient to express the eddy fields into a 0^{th} and a 1^{st} order term.

Term	Description
0th	This term arises from an ***asymmetry*** of the gradient coil with respect to the magnet bore and cryo shields. This term is space independent and present in the complete imaging volume and adds to the nominal B_o-Field. The amplitude (x) is given in the unit µTm/mT. The time constant of the most significant 0th term component is about 500ms, although there are shorter time constant components as well.
1st	This is the most important term as it has the same symmetry as the gradient field itself. The amplitude is given in % of the applied gradient pulse.
2nd or above	There are also high ordered eddy currents present, they are usually small and negligible as long as the slice shifts and off center zooms are not too large. No compensation for the high order eddy currents can be made.

Figure 224 *Eddy Current Measurement*



Measurement

First, the phantom center position is checked. The same preparation measurement is used as for Phantom shim, however, the small spherical phantom is used. If the phantom is not correctly positioned the measurement stops with an error.

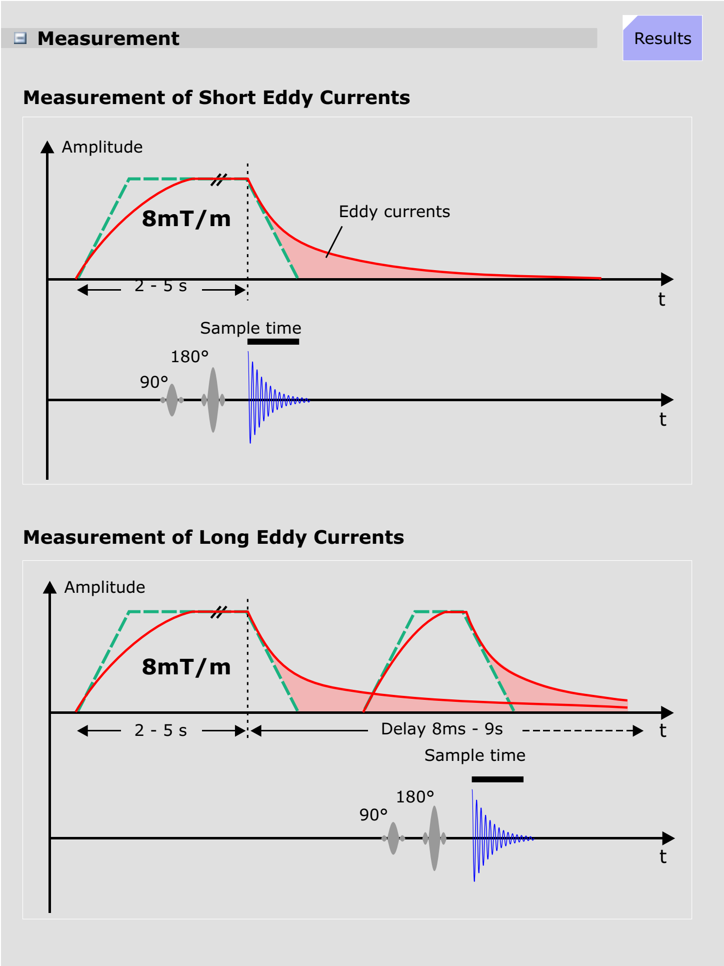
The sequence for the measurements generates 23 spin echoes with delay times between 0.4 ms to 9000 ms of the applied gradient.

The measurement is made at the moderate slice shifts of ±5 cm to prevent contributions from higher order terms.

```
V1: Training Center          File: ecct_prep.dat
--------------------------------------------------
2006-12-14 10:22:17  Sequence: eddyprep
Gradient: Echoes:   12           Points:  256 * 30us

Slice Grad  Shift  Frequ  Ampl1  SN  Points  Pos  Ampl2  SN
            [mm]   [Hz]                      [mm]
-------------------------------------------------------------
  1     X    -50      4  12199  354   254      2  25943  402
  2     X     50     -2  11018  393   254      0  24383  486
  3     Y    -50     -7  11084  315   254     -3  26051  550
  4     Y     50      8  11999  291   254     -2  27071  546
  5     Z    -50     -4  10365  347   254      3  23582  516
  6     Z     50      5  11917  298   254      2  25538  460
-------------------------------------------------------------
Phantom Position: X:-2.7mm Y:-0.8mm Z:-2.7mm
Tolerances:          5.0mm    5.0mm    3.0mm
```

Figure 225 *Eddy Current Measurement*



Results

The program will then calculate:

Gradient Compensation (1st order term)

```
=========== Gradient Compensation for X ==========
  Current Param.         Change      Suggested Param.
  Time[s] Amp[%]    Time[s] Amp[%]     Time[s] Amp[%]
--------------------------------------------------
 1.982741   0.25|-0.068626   0.03| 1.914115   0.28
 0.409179   7.49|-0.000460  -0.01| 0.408719   7.48
 0.263029  -1.45| 0.005060  -0.01| 0.268090  -1.46
 0.012902   0.41| 0.000000  -0.00| 0.012902   0.41
 0.002000  -0.05| 0.000000   0.01| 0.002000  -0.05
--------------------------------------------------
  Overshoot =    6.64 %       Fit Quality = 0.103916
```

In this example the overshoot tells us that the initial gradient amplitude has to be 1.13% higher than the normal value without eddy current compensation. For the Fit Quality the RMS-value is given.

B_0 Compensation (asymmetry)

```
============== B0 Compensation for X ============
  Current Param.         Change      Suggested Param.
  Time[s] Amp[%]    Time[s] Amp[%]     Time[s] Amp[%]
--------------------------------------------------
 0.509204   0.09|-0.014494  -0.00| 0.494710   0.09
 0.007394   0.05| 0.000255  -0.00| 0.007650   0.05
 0.000499  -0.14| 0.000017   0.03| 0.000516  -0.11
--------------------------------------------------
  Overshoot =    0.09 %       Fit Quality = 0.069140
```

The first plot of the uncompensated eddy fields shows a smooth curve. The next measurement will establish already a large improvement.

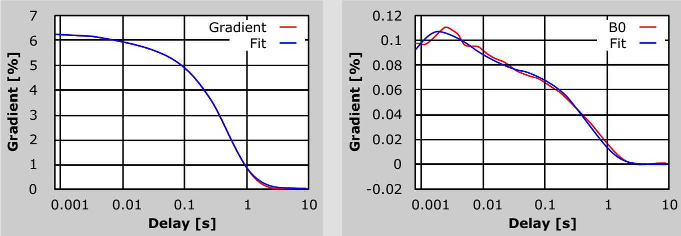
The final iteration, usually after 2 or 3 measurements, shows an oscillating pattern indicating the eddy-fields have become smaller than the measurement system is able to resolve.

Figure 226 *ECC Measurement Results and Specifications*

Results

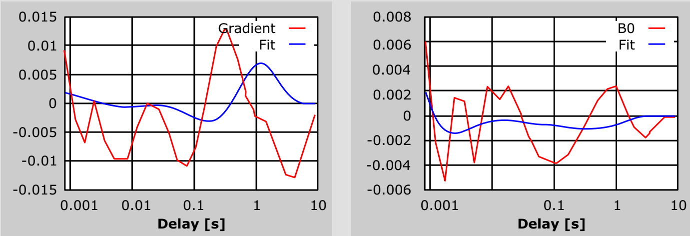
Main Menu

Results of First Measurement



X	Gradient			B0		
Range	Value	Spec	Unit	Value	Spec	Unit
<10ms	6.222	0.0 - 0.1	%	0.110	0.0 - 0.1	uTm/mT
>10ms	5.894	0.0 - 0.1	%	0.087	0.0 - 0.1	uTm/mT
2-100ms	1.049	0.0 - 0.04	%	0.041	0.0 - 0.03	uTm/mT

Results of Final Measurement



X	Gradient			B0		
Range	Value	Spec	Unit	Value	Spec	Unit
<10ms	0.009	0.0 - 0.1	%	0.006	0.0 - 0.1	uTm/mT
>10ms	0.013	0.0 - 0.1	%	0.004	0.0 - 0.1	uTm/mT
2-100ms	0.011	0.0 - 0.04	%	0.006	0.0 - 0.03	uTm/mT

Coil Power Losses (CPL)

The purpose of this procedure is to measure the amount of power loss of the Body Coil. This value is used by the RF-Safety Watchdog (RFSWD) monitor to achieve a more accurate SAR measurement.

Measurement

A complete inline adjustment with a non-volume-selective frequency adjustment is performed, followed by a transmitter adjustment. The voltage from the Adjust/Transmitter result is used to calculate the reference power:

Reference Power from TALES = (Voltage from Adj/Tra)^2 / 50 Ω.

The small spherical phantom is necessary because of the reduced losses in the phantom and the dielectric resonances.

A subsequent measurement is made only to verify that the small (17cm diameter) spherical phantom is used and that is it centered.

Evaluation

The measured reference power from the TALES is checked. Low Spec = 250 W, High Spec = 630 W. If an oil-filled 170 mm phantom were to be used the low specification might not be reached or if the body loader phantom is in the magnet the high specification might be exceeded.

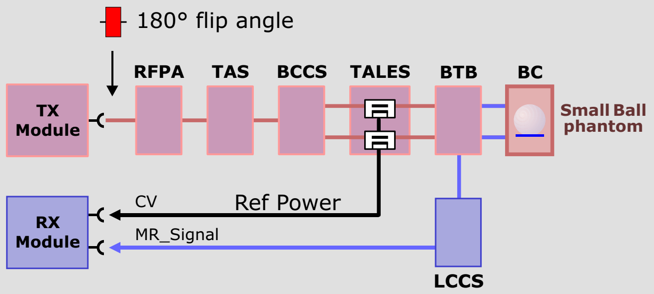
Results

The value for the Coil Power Losses is calculated: CPL = Reference Power from TALES * (1 + σ), where σ is a correction factor taking into account electrical conductivity of the phantom fluid and dielectric resonance effects. The factor σ is greater than 1 so the Coil power loss value is larger than the measured reference power.

Figure 227 *Coil Power Loss Measurement*

Measurement

Main Menu



Result

Phantom Center

	Value	Low Spec	High Spec	Unit
Phantom Center X	-2.67	-20.00	20.00	mm
Phantom Center Y	2.81	-20.00	20.00	mm

Phantom Diameter

	Value	Low Spec	High Spec	Unit
Phantom Diameter X	169.97	150.00	190.00	mm
Phantom Diameter Y	170.01	150.00	190.00	mm

Ref Power from TALES

Name	Value	Low Spec	High Spec	Unit
Ref Power from TALES	422.98	250.00	630.00	W

Coil Power Losses result 507.57 W saved

Gradient Delay

The gradient amplifiers have a delayed response time (t_{nom} to t_{act}), due mainly to the coil inductance and is slightly different for each of the three gradient axis. The Gradient Delay measurement determines the delay time for each gradient axis. A precise timing of gradient pulses and RF is very important for good image quality.

Measurement

The measurement uses a Spin-Echo-sequence without phase encoding, hence only one line is measured. The gradient to be tested acts as the readout gradient and is switched on once between the 90°- and the 180°-pulse and then again during the readout time of the spin echo.

The first gradient pulse causes a dephase of the magnetization vector in the direction of the gradient axis. The amount of dephase is proportional to the time-amplitude integral of the gradient pulse. Subsequently, a 180° refocusing pulse is applied normally resulting in a spin echo at TE/2 time later. However, due to the dephasing of the vector by the first gradient pulse, the echo will only land in the center of the ADC window (which is also centered exactly around the TE time point) when an equal rephasing gradient has been applied at the proper time so that the rephasing is finished at the center of the ADC cycle.

For a non-corrected gradient delay the two gradient pulses are time-delayed. This does not influence the dephasing effect of the first gradient pulse, however the delayed second gradient pulse causes a rephasing delay and thus a time-delayed echo signal. From the echo signal delay and the gradient amplitude the system can calculate the required correction time.

Deviation Check

Each gradient is measured at three different gradient strengths - 1.5, 4 and 8 mT/m - giving three gradient delays for each gradient. For an ideal gradient system, all delays should be the same.

The **Max. Diff. Deviation** is the maximum difference of the delay times with different gradient strengths, i.e. high gradient pulses may be different from low amplitude pulses. If the tolerance is exceeded, this indicates some kind of nonlinearity in the gradient system.

Results

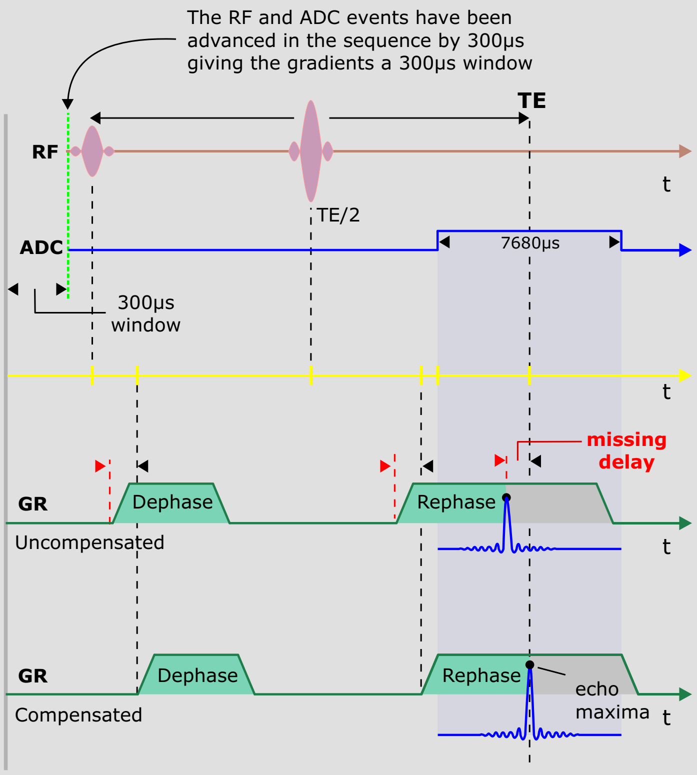
The new delays and maximum deviation values are displayed in two separate tables. If the procedure was successful, the values will be saved.

Expert Mode

No extra functionality is offered in the expert mode.

Figure 228 *Gradient Delay Measurement and Results*

Measurement



Result

Main Menu

New Gradient Delay Values

	Value	Low Spec	High Spec	Unit
X Gradient Delay	16.000	0.000	300.000	μs
Y Gradient Delay	14.000	0.000	300.000	μs
Z Gradient Delay	12.000	0.000	300.000	μs

Max Diff Deviation

	Value	Low Spec	High Spec	Unit
Max diff deviation X	2.000	0.000	5.000	μs
Max diff deviation Y	0.000	0.000	5.000	μs
Max diff deviation Z	1.000	0.000	5.000	μs

New Gradient Delay values saved

Gradient Sensitivity

The purpose of this measurement is to establish the actual LSB weighting (sensitivity) of the gradient amplifier so that accurate gradient field strengths can be calculated assuring accurate FoV (image size).

Nominal Sensitivity

The nominal sensitivity is simply the maximum gradient strength of the Gradient System (amplifier and coil) divided by the digital resolution of the DAC. In the example below, the gradient system has a maximum 30 mT/m field strength and an 18-bit DAC, whereby 17 bits is the dynamic range since 1 bit is used for polarity (it is a ±10V DAC). This gives a nominal sensitivity of 0.229 µT/m (value of the LSB). The first Gradient Sensitivity measurement will use this value. As can be seen in the first image results, the phantom has the shape of an oval.

Actual Sensitivity

The actual sensitivity, however, will depend on the characteristics of the DAC, the amplifier and the coil. In order to determine the actual gradient sensitivity an **object of known diameter** (the large 24 cm spherical phantom) is measured using a 500 FoV. and the resultant image evaluated and the actual phantom diameter determined. The actual sensitivity is the old value time the correction factor which is determined by dividing the nominal phantom diameter with the actual measured image diameter.

Table Distance

During this procedure the patient table distance - the distance from the light localizer to the iso-center - is also measured (again, it was first measured some steps ago) and, if necessary, automatically corrected. If a correction is necessary the measurement is repeated.

Measurement

The gradsens sequence is a standard Gradient Echo (FLASH) sequence with a TR = 50 msec, TE = 10 msec. The sequence is run three times, one for each orientation. For the sagittal orientation the usual orientation of the readout and phase encoding gradients are swapped.

Evaluation

The 24cm phantom is measured and checked initially for size and center.

Results

Afterwards the three values for the gradient sensitivity will be stored in the status file.

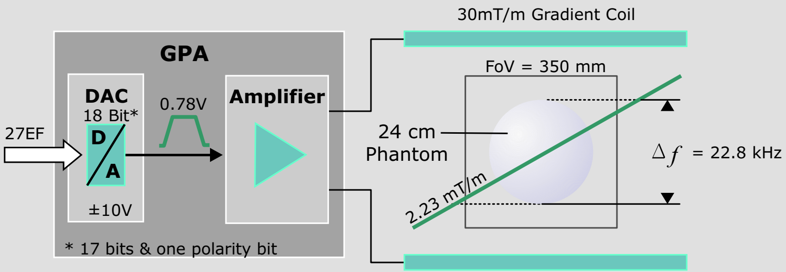
In addition, the Grad Sens program is used to determine the distance from the light marker to the gradient coil center.

Expert Mode

In **Expert** Mode, the Grad Sens procedure operates similar except that the results are not written into the database.

Figure 229 Gradient Sensitivity Measurement and Results

Measurement



Sequence

PBW = 130 Hz
MA = 256
FoV = 350 mm

$$GR = \frac{PBW * MA * 1/FoV}{42.577}$$

GR = 2.23 mT/m

Analysis

Default Sensitivity

$$= \frac{30mT/m}{2^{17}}$$

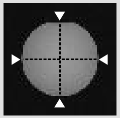
$$= 0.229\mu T/m \ (LSB)$$

Actual Sensitivity

$$\text{New Value} = \text{Old value} \bullet \frac{\Delta f \text{ Actual}}{\Delta f \text{ Nominal}}$$

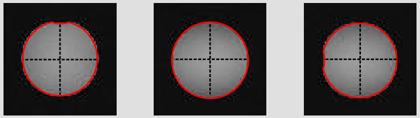
$$(2.23mT/m * 0.24m) * 42.577MHz/T = \Delta f \textit{ nominal in kHz}$$

Image



Result

Main Menu



Phantom Position

	PhaseEnc		ReadOut		Average	
	Value	Unit	Value	Unit	Value	Unit
X Position	-0.5	mm	-0.5	mm	-0.5	mm
Y Position	+3.5	mm	+3.6	mm	+3.6	mm
Z Position	+3.3	mm	+3.5	mm	+3.4	mm

Phantom Diameter

	PhaseEnc			ReadOut			Average		
	Value	Spec	Unit	Value	Spec	Unit	Value	Spec	Unit
Diameter X	240.4	238 - 242	mm	239.9	238 - 242	mm	240.1	238 - 242	mm
Diameter Y	240.0	238 - 242	mm	240.3	238 - 242	mm	240.1	238 - 242	mm
Diameter Z	241.3	238 - 242	mm	239.5	238 - 242	mm	240.4	238 - 242	mm

Gradient Sensitivities

	Current		New	
	Value	Unit	Value	Unit
Sensitivity X	0.000229	mT/m	0.000265	mT/m
Sensitivity Y	0.000229	mT/m	0.000276	mT/m
Sensitivity Z	0.000229	mT/m	0.000272	mT/m

Saving new gradient sensitivities to GPA.
Moving patient table by -3 mm.
Correcting patient table distance by 3 mm to 899 mm.

Receive Path Calibration

The receiver system was designed to support the Integrated Panoramic Array technology which allows the combined use of multiple coils simultaneously. There are 16 receive lines from the 4 table connectors to the switch matrix on the LCCS. The **switch matrix** can switch any of the 16 inputs to any one of the 8 outputs. Each output goes through a **switched-gain amplifier** and over **receive cables** of varied lengths (between 7 and 20 meters) to the receive modules containing the **ADCs**. Since any coil element can be switched through the LCCS matrix to any ADC in any of the receive modules it is necessary to calibrate every receive path so that the same image brightness for a given coil element will be achieved independent of the receive path used. The factors that will influence the MR signal from a coil, in amplitude or phase, are highlighted above in bold letters.

This procedure replaces the former Coil Tune-Up procedure (aka, image brightness adjustment) that had to be performed for EVERY installed Local Coil (AdR: which was very time consuming).

Measurement

The Receive path calibration is performed with the **CP Head Array** coil and the large bottle phantom.

The **image brightness** and the signal to noise ratio (S/N) is determined first. The **receiver scale factor** of the first receiver channel is determined by evaluating the image brightness. The scale factors of all other receiver channels are adjusted by evaluating the raw data amplitudes. The scale factors are necessary to correct the differences of the amplification of each receiver. In addition a complex factor is calculated for each receive path, by measuring the relation of the raw data signals for receiver high gain and low gain.

The **ReceiverChannelCorrection factor** compensates the different sensibilities of the individual ADC's. This means that you will get the same brightness for a given phantom and coil element, independent of the ADC you are using (with preamplifier switched to high gain).

The **ReceiverGainCalibration factor** characterizes the preamplifier in the LCCS belonging to an ADC. The preamplifier can be switched to high and low gain. The absolute value of the ReceiverGainCalibration factor describes the ratio of the magnitudes of the high and low gain signal.

The **phase factor** describes the phase difference of the high and low gain signal.

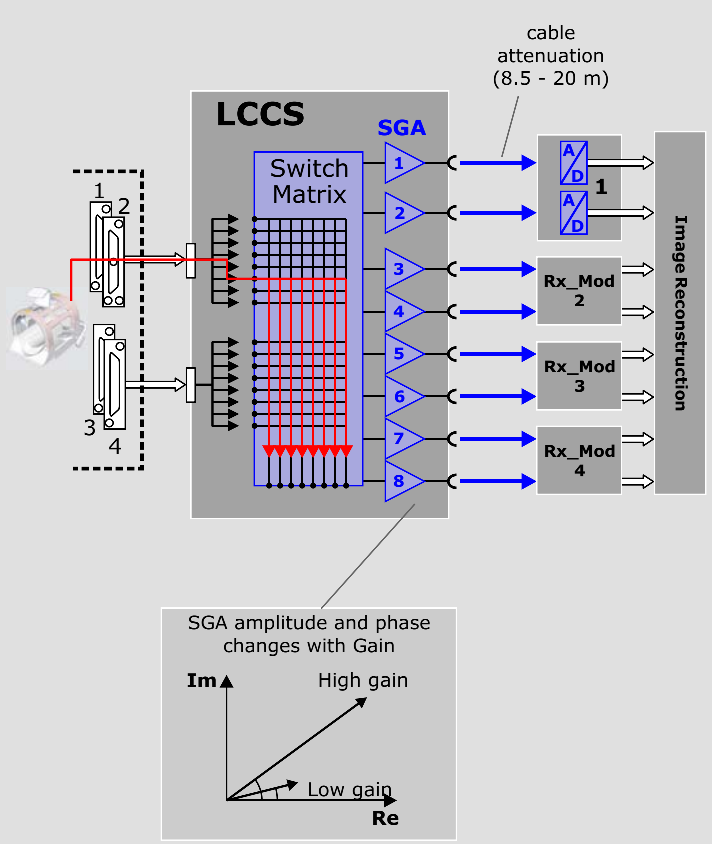
Expert Mode

In Expert Mode, the Receive Path Calibration (only for the head coil) operates similar except that the results are not written into the database. For all other coils the standard procedure Image brightness is available.

Figure 230 *Receive Path Calibration Measurement and Results*

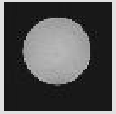
Main Menu

Measurement



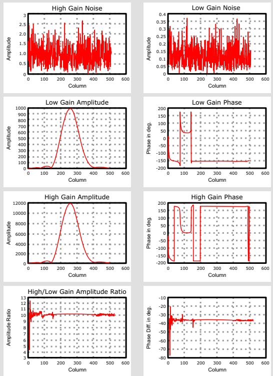
Result

Protocol CP_Head_Array_Tra



	Value	Spec
Protocol	CP_Head_Array_Tra	
Brightness (Ref.:1500.00)	1502.62	
S/N	65.37	65.00
RxChannelCorrection Factor Old	1.03	0.50 - 1.50
RxChannelCorrection Factor New	1.03	0.50 - 1.50

Raw Data from Receiver Channel #1



RxChannelCorrection

	Old		New	
Rx Channel	Value	Spec	Value	Spec
#1	1.03	0.50 - 1.50	1.03	0.50 - 1.50
#2	1.05	0.50 - 1.50	1.03	0.50 - 1.50
#3	1.06	0.50 - 1.50	1.06	0.50 - 1.50
#4	1.06	0.50 - 1.50	1.06	0.50 - 1.50
#5	1.04	0.50 - 1.50	1.05	0.50 - 1.50
#6	1.08	0.50 - 1.50	1.11	0.50 - 1.50
#7	1.05	0.50 - 1.50	1.10	0.50 - 1.50
#8	1.16	0.50 - 1.50	1.20	0.50 - 1.50

ReceiverGainCalibration factor

	Old Values		New Values					
Receive Channel	Abs	Phase	Abs	Spec	Unit	Phase	Spec	Unit
#1	10.24	-35.25	10.26	9.00 - 11.00		-35.80	-180.00 - 180.00	deg
#2	10.34	-35.74	10.35	9.00 - 11.00		-35.67	-180.00 - 180.00	deg
#3	10.19	-34.75	10.19	9.00 - 11.00		-35.39	-180.00 - 180.00	deg
#4	10.30	-35.90	10.29	9.00 - 11.00		-36.61	-180.00 - 180.00	deg
#5	10.37	-35.93	10.36	9.00 - 11.00		-36.51	-180.00 - 180.00	deg
#6	10.30	-35.87	10.29	9.00 - 11.00		-36.31	-180.00 - 180.00	deg
#7	10.33	-35.52	10.31	9.00 - 11.00		-36.00	-180.00 - 180.00	deg
#8	10.42	-35.61	10.40	9.00 - 11.00		-36.17	-180.00 - 180.00	deg

Quality Assurance

The QA procedures are a set of procedures which are used to verify system performance and include all the procedures found in the Tune-Up as well as additional procedures for testing overall system performance. Only these will be described since those procedures found in Tune-Up are identical, with the exception that not values are saved:

- If you are out of specification after a Quality Assurance step, you must perform the corresponding Tune-Up step.
- Every Tune-Up and Quality Assurance procedure can be run in either Normal or Expert mode with one exception: the "field stability" procedure under Quality Assurance (very useful during shim procedure or trouble shooting stability problems) can only be run under Expert mode.

Go back to MAIN MENU

Coil Check

This procedure replaces the former SN DIP used for the coil-QA and determines:

- Signal to noise ratio S/N
- Image brightness
- Image uniformity (inhomogeneity)

for the Body coil and all LC-coils except the Head-coil.

One or more protocols have been defined for each coil. After each protocol is measured, the routine will analyze the resulting image or images for S/N and in some cases the image intensity profile and image size. For transmit-capable coils the specific absorption ratio (SAR) is checked as well.

In standard mode and for coils where only one phantom position is necessary, the SN_DIP procedure performs all required SN_DIP protocols without user intervention. If more than one protocol is required, the procedure measures and evaluates each image from the protocol before continuing with the next protocol. For coils that require more than one phantom position to perform all the required SN_DIP protocols, the user will be guided on screen through the various phantom positions.

Expert Mode

Expert Mode lets the user select the protocols to be performed. A list of available protocols is displayed. Any combination of protocols can be selected. When selecting Start from the QA platform, the selected protocols will be performed.

Image Orientation

The 6 electrical connections of the Gradient Coil are checked here. If any of the amplifier or coil connections are swapped between two (or more) axis or any of the polarities of any axis is swapped, the orientation of the image will not be correct. This procedure will determine if the connections are correct and if not, which ones are or could be wrong.

The check for correct gradient coil connections is performed in two steps:

- First, for swaps between the three axis (x to y, y to z, etc.).
- Second, for correct polarity of each axis.

The Image Orientation procedure measures the position of the loaded phantom and the spherical phantom in order to determine the correct gradient orientation and polarity.

Measurement

With the first image the correct connection of the gradient cables (x/y/z axis) are tested; with the second image the correct polarity of the gradient currents is checked.

Because the image of the loader is not symmetrical with respect to interchanging of gradient axes, an exchange of gradient axes (i.e. wrong cable connections) can be detected.

Incorrect gradient polarity can be detected by identifying the unsymmetrical cap of the spherical phantom.

In case of incorrect gradient axis orientation, polarity or positioning of the phantom, the procedure is finished with "error".

Evaluation

The two measured images are automatically evaluated following the measurement.

Axis Swap

Because the image of the loader is not symmetrical with respect to interchanging of gradient axes, an exchange of gradient axes (i.e. wrong cable connections) can be detected.

The program searches for the outer edges of the loader phantom from all sides and compares it to the expected geometry. If correct, it will continue on to the polarity swap measurement.

If, however, a swapped axis is identified, the program terminates with the report of which axis are swapped. A reversed polarity check will not be made since this would only make sense when no axis-swap is present. after fixing the swapped axis problem, the program must be made again.

Polarity Swap

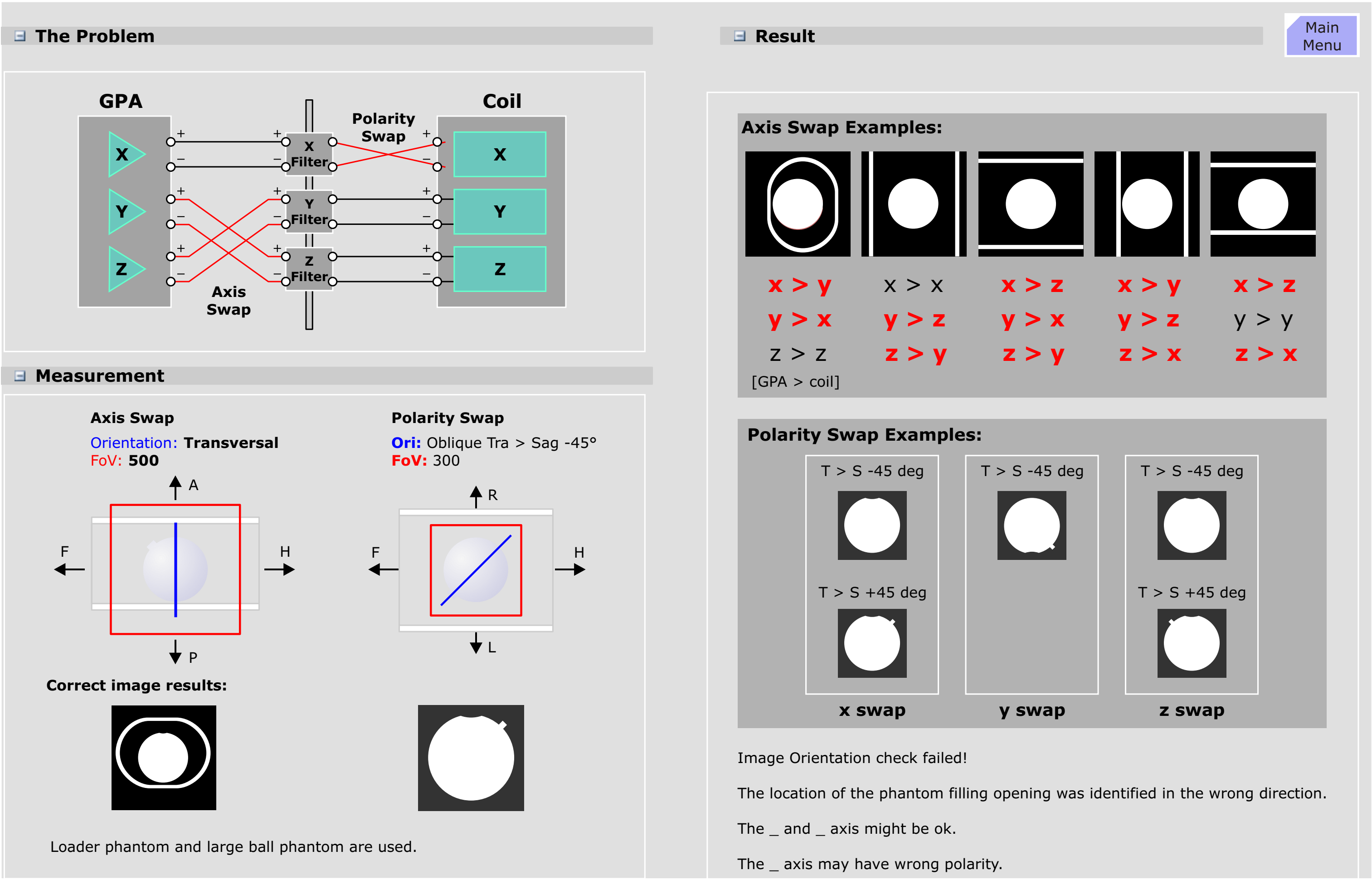
Incorrect gradient polarity can be detected by identifying the position of the unsymmetrical fill-cap of the spherical phantom. If the fill-cap is not in the expected position the program will make further measurements and reversing the gradient polarities until the fill-cap is found. When the fill-cap has been found the program terminates and reports which gradient axis or axis have been swapped.

The images and error messages can be displayed by clicking Reports and selecting the corresponding procedure in the Session History Log or by selecting the corresponding log file under QA.

Expert Mode

The only difference in Expert Mode is that no data are written into the database.

Figure 231 *Image Orientation*



Calculation Artefacts

The Calculation Artefacts (CalcAr) QA procedure is used to quantify the artefact magnitudes (i.e. signal intensities found *outside* the image). Artifacts in MR images can be produced by a variety of mechanisms. For example, gradient deficiencies (instability, non-linearity, 50/60Hz ripple) or RF instabilities will produce periodic blurring or ghosting in the direction of the phase encoding.

Sequence

Calcar is a double-echo Spin Echo sequence run in all three slice orientations at the iso-center (slice shift = 0). The transverse orientation is performed first, followed by the sagittal measurement and the coronal measurement. (Note: for the SAG slice orientation, the Z-gradient is used as phase encoding gradient.)

Evaluation

For a good description on the use and interpretation of the CalcAr results refer to the Tune-Up section of the TSG under QA Troubleshooting Tables.

Expert Mode

Expert Mode lets the user select the slice shift 0, 50 mm, and the three slice orientations separately.

The Evaluate option allows to evaluate any image in the actual segment for artifacts.

Figure 232 *CALCAR*

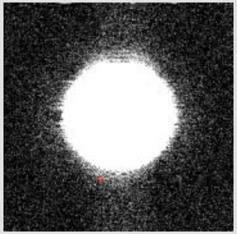
Measurement

Main Menu

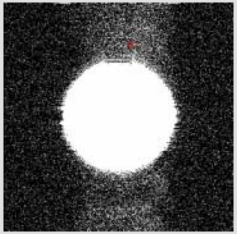
sequence calcar
Transversal
FoV 500.0*500.0 (width*heigth)
256*256 (Lines*Columns)
TR 600.00
TE 35.00, 105.00

Results

Transversal 1. Echo



Transversal 2. Echo



Artifacts for Transversal

	Value	Low Spec	High Spec	Unit
slice shift 1	0.000			mm
Transversal_Artifact_Echo1	0.780	0.000	2.000	%
slice shift 2	0.000			mm
Transversal_Artifact_Echo2	1.034	0.000	3.000	%

Spike Check

The Spike Check looks for occurrences of "spikes" or RF interferences.

Spikes are electrostatic discharges picked up by the RF coils during the measurement which are usually caused by loose cables or components anywhere in the RF room. These discharges are very short in time and therefore seen as "spikes" in the raw data set. How the spikes affect the image depend on where the spikes are in the rawdata set and how big their amplitudes are. Typically, spikes result in a moire-looking pattern that can occur in any direction, very often diagonally.

RF interferences are, generally speaking, RF energies at or near the system frequency that are being picked up by the receive coils. Normally such interferences do not originate from the system but from electronic components inside the RF cabin capable of producing RF frequency noise. Rf interferences can also be coming from sources outside the RF cabin if the integrity of the cabin has in some way been compromised (e.g., bad door gaskets or leaky filters on the Filter Panel, or cables introduced into the room without appropriate RF filters.)

Testing Method

Spike Measurement

The spike sequence starts with an initial series of strong gradient pulses of both positive and negative polarities on all three gradients intended to cause a buildup of static electrical energy caused by any moving components (cables, covers, etc). The ADC is then enabled and again a pattern of gradient pulses with amplitudes ranging from max positive to max negative are applied. Any static electrical discharges caused by moving or loose parts will picked up by the receiving coil (usually the body coil) and ADC and stored in the raw data. The evaluation looks for intensities in the raw data set exceeding a threshold (usually noise) and counts both the number and intensities of the spikes and displays them numerically and graphically. The electrical discharges will be seen as "spikes" in the raw data. The position of the spikes in the raw data set relates to the gradient that was on when the spike occurred which may give an indication of the direction in which the cause of the spike may be found.

RF Interference

This is basically a simple noise measurement. With such a measurement the receiver is opened wide and the ADC is turned on, anything and everything measurable at or around the system frequency is quantified. Ideally, nothing but noise will be measured. The evaluation of this measurement does not look at the raw data but the fourier transformed "image". In case RF-interferences have been picked up during the measurement the fourier transform will result in dots or streaks in the image data. The same algorithm for spike searching is used in this evaluation as well.

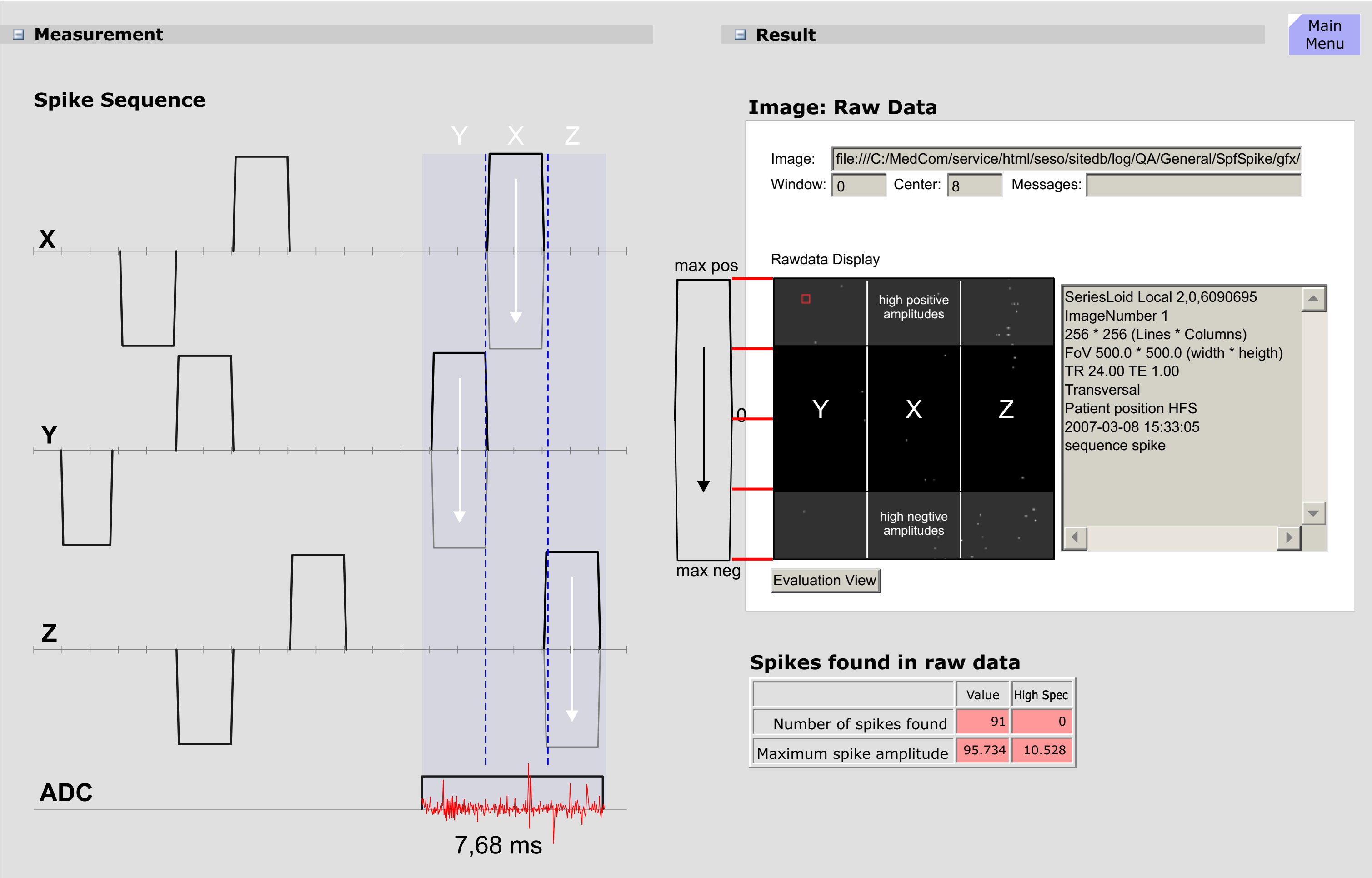
Advanced Spike Check

TRUFI and HASTE sequences and sequences used for EPI or BOLD-Imaging may generate spikes which cannot be verified by the normal "Spike Check". In these cases the "Advanced Spike" can be used for spike trouble shooting. The "Advanced Spike" should be repeated at least three times to be assured the system is free of spikes. But be aware, if there weren’t any spike-causing problems present before using this measurement, there may be afterwards!

Results

The measured raw data and RF-interference images will be loaded and displayed.

Figure 233 *Spike Check*

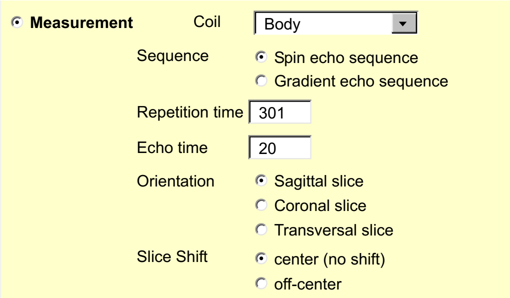


Stability Check

The process of aquiring the large number of MR signals necessary for an MR image over a relatively long acquisition time, usually in the order of seveal minutes, puts a high demand on the field generating sub-systems (RF, gradients, magnet and shim) to have good **stability** and **reproducibility**. Instabilities in any of these sub-systems will result in smearing (ghosting) artefacts. The Stability test, together with the other QA tests, may help determine what is responsible for instability artefacts.

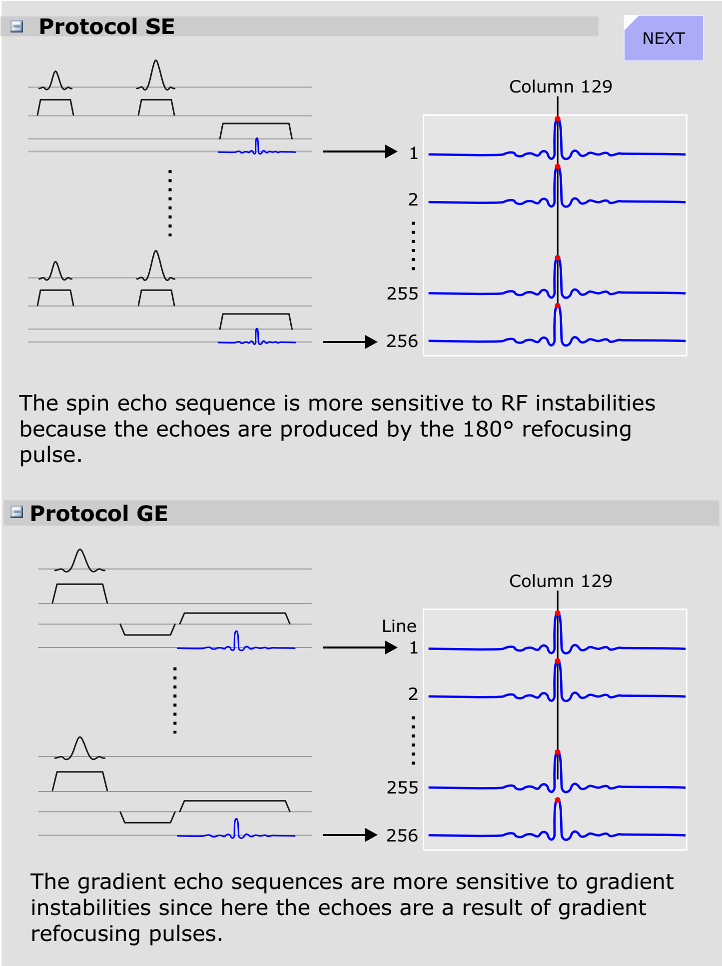
Measurement

The stability test consists of a series of measurements using both spin echo and gradient echo sequences. These sequences are **without phase encoding** so all echoes should be identical. All three orientations are measured so that each gradient is performed as a read-out gradient. In addition, measurements are made at center and off-center (+50 mm) positions.

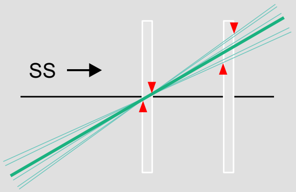


The spin echo sequence is more sensitive to RF instabilities because the echoes are produced by the 180° refocusing pulse. The gradient echo sequences are more sensitive to gradient instabilities since here the echoes are a result of gradient refocusing pulses.

Figure 234 *Stability Sequence*



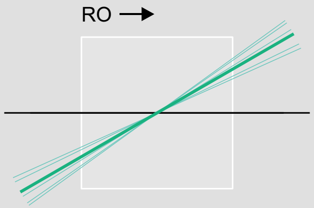
Slice Shift ◉ center (no shift) ◉ off-center (+50mm) NEXT



The field deviations caused by an instable gradient are greater off-center than at center. Therefore gradient instabilities will effect off-center slices more than center slices. The center slice is sensitive to B_0 field changes as well.

The off-center measurements are also more sensitive to gradient instabilities then the center slices due to the enhanced efects of the gradient fields the further away from center the measurement is made.

Orientation ◉ Sagittal slice RO = Y (Y swapped for RO)
○ Coronal slice RO = Z
○ Transversal slice RO = X



An instable gradient axis will cause greater deviations when used as a read out gradient.

The orientation determines the role of each gradient axis. For example, lets assume there is an instability in the X gradient. The values for the sagittal orientaion (X = slice selection) may be in spec while the spec for the transversal orientation (X = read out) could be out of spec.

Evaluation

Spin Echo

The spin echo sequence is sensitive to instabilities in the range of a few Hz up to 100 Hz and so sensitive to 50/60 Hz line hum.

Gradient Echo

The gradient echo/flash sequence is sensitive from DC up to a few Hz which may be caused by slow external field interferences, field instabilities or mechanical vibrations.

Off-center

The measurement result of the center slice is sensitive to instabilities in the B0 Field, whereas the 50 mm offset slice is more sensitive to gradient instabilities.

Amplitude Stability

The amplitude value in the column of maximum signal is displayed in the graphical output.:

The phase shift of a spin echo is determined by the integral over the magnetic field from 90° to 180° pulse minus the integral from 180° pulse to echo:

For a constant field or a slowly varying field, both integrals cancel each other. However, for an oscillating field with a period of TE (20 ms = 1/50 Hz) this expression becomes maximum.

For the gradient echo sequence, the phase stability is evaluated by fitting a straight line to the phase data. Subtracting the value of the linear fit from the phase data results in the phase stability changes. The slope of the straight line indicates the linear phase drift.

Expert Mode

Expert Mode lets the user select either a Spin echo or Gradient echo sequence, modify the default parameters (slice shift of 0 and +50 mm, orientation and TR and TE values.

Figure 235 *Stability Check Evaluation*

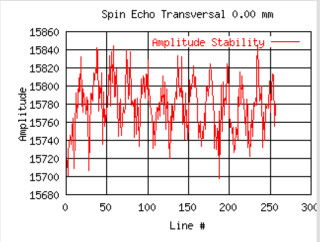
Results

Main Menu

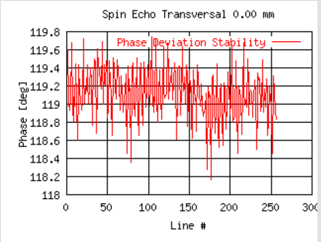
Spin Echo Sequence: Transversal Slice 0 mm

	Value	Low Spec	High Spec	Unit
Amplitude	0.931	0.000	2.000	%
Phase	1.585	0.000	4.000	deg

Amplitude Stability



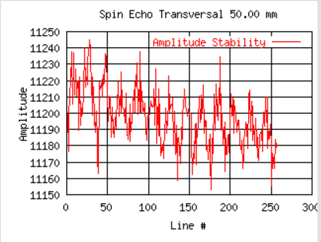
Phase Deviation Stability



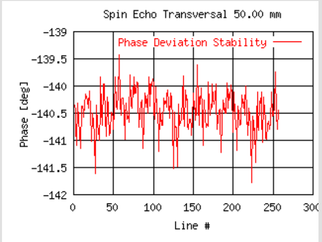
Spin Echo Sequence: Transversal Slice 50 mm

	Value	Low Spec	High Spec	Unit
Amplitude	0.931	0.000	2.000	%
Phase	1.585	0.000	4.000	deg

Amplitude Stability



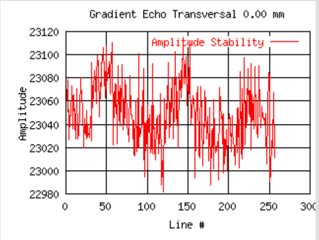
Phase Deviation Stability



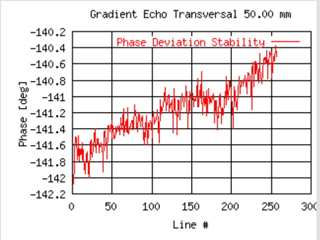
Gradient Echo Sequence: Transversal Slice 0 mm

	Value	Low Spec	High Spec	Unit
Amplitude	0.577	0.000	2.000	%
Phase	0.783	0.000	4.000	deg
Phase Drift	0.700	0.000	4.000	deg

Amplitude Stability



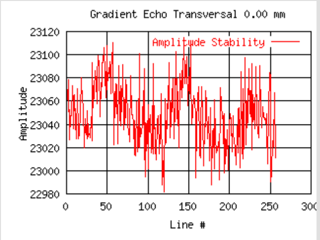
Phase Deviation Stability



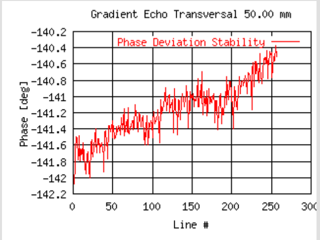
Gradient Echo Sequence: Transversal Slice 50 mm

	Value	Low Spec	High Spec	Unit
Amplitude	0.453	0.000	2.000	%
Phase	0.862	0.000	4.000	deg
Phase Drift	1.064	0.000	4.000	deg

Amplitude Stability



Phase Deviation Stability



Fat Saturation

Fat saturation is a widely used feature to suppress fat signal in MR images. There are various methods for doing this, the most common however is the *spectral method*. This method applies an RF pulse at the fat frequency (3.4 ppm lower than water frequency) before each Fouier line deflecting the fat signal thus greatly reducing its attribution to the final MR signal. This method requires a very good magnet homogeneity (1-2 ppm peak-peak) and a well defined RF pulse to assure adequate saturation of fat throughout the FoV and also to prevent the water from being saturated as well. The RF flip angle of the Fat Sat pulse should be about 90 degrees for optimal results.

Evaluation of Measured Images

First, the normal image is analyzed for position and diameter. Then, a region of interest (80% of the phantom diameter) for both images is defined and the ratio of both signals is calculated point by point. To reduce noise at every point, a 3x3 average over neighbor points is performed. Point by point saturation ratios are calculated, resulting in a mean and a maximum value. These two values are compared to specification.

Results

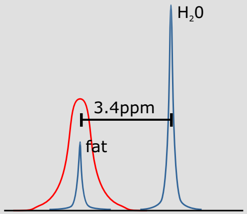
The images measured are loaded and displayed. Image windowing is done by clicking and dragging the mouse or by direct input of image window and center values.

As a rule, for a mean saturation ratio of 15%, the peak-peak homogeneity of the field within the imaging volume has to better than 1ppm.

Figure 236 *FAT Saturation*

Measurement

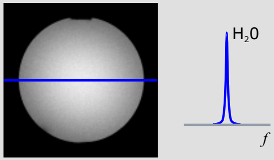
Main Menu



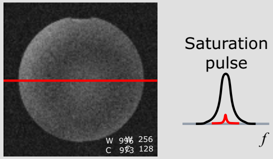
Normally, the saturation pulse is at the fat frequency. This measurement sets the saturation pulse frequency to water. If the magnet homogeneity is good (good shim, good eddy current compensation, good RF pulse selectivity) the result should be the same as for fat.

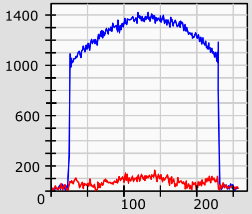
Result

Reference Image



Saturated Image





Saturation Ratio

	Mean				Max			
	Value	Low Spec	High Spec	Unit	Value	Low Spec	High Spec	Unit
Transversal	5.9	0.0	15.0	%	9.6	0.0	30.0	%

Synthesizer Check

The Synthesizer Check procedure is an important test to verify the proper operation of the synthesizer. It is used to validate the frequency and phase quality of the synthesizer with respect to the accuracy of the slice position.

Principle of Synthesizer Test

The Avanto system contains a maximum of 32 receiver channels. Four receiver channels are connected to 1 double (Numeric Controlled Oscillator) NCO so that we have in sum 16 RX-NCO's. Double NCO means that we have two groups of NCO's, so that it is possible to switch the group of 4 receiver channels either to NCO-group 1 or to NCO-group 2.

For transmit we have one double NCO. As well the transmit path can be switched to NCO-group 1 or NCO-group 2.

During simultaneous sending and receiving, the phase and frequency of the TX-NCO and RX-NCO within one group is the same. In the following QA-procedure asynchronous behavior of the TX- and RX-NCO is needed, that means one NCO is stepping phase and frequency while the other is fix. For that reason we always are using different groups of NCO's when sending and receiving simultaneously, i.e. TX NCO is fix and RX-NCO is stepping.

In a first step the verification of the frequency and phase operation of the RF reference signal is performed by using the MR signal itself (NCO-sequence). After the demodulation the received MR signal, the reference signal is compared to the MR signal in terms of frequency and phase. In one test, the reference signal is varied in frequency over the whole range of the MR system. A second test modifies the phase in fixed intervals. In this way the whole operating range of the oscillator can be tested. The test is performed for one RX-NCO from group 1 and one RX-NCO from group 2. So the two RX-NCO's are validated and can be used for further tests of the other NCO's.

In a second step we perform loop-measurements (nco_loop-sequence) between the just tested RX-NCO's and the TX-NCO's. By doing so, we approve the correct function of the TX-NCO's.

In the further steps all other RX-NCO's are tested against the approved TX-NCO's by testing separately the group 1 RX-NCO's and group 2 RX-NCO's.

Expert Mode

The only difference of Expert Mode is that no data is written into the database.

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Main Menu

Stability_LongTerm Check

A new application of MR-Imaging is to make activities in the brain visible. For that purpose the patient will be exposed to a periodical optical, acoustical or other stimulation. During the exposition, a series of MR-images will be measured. Afterwards the images will be evaluated for changes of the signal intensity. The regions of the brain activated by the stimulation can be localized by a very small variation of signal intensity. These variations can easily be hidden by various instabilities of the system and/or the environment. Therefore the instability of the MR-system and interferences from environment must be below a specified level.

The Stability_LongTerm Check determines the sum of all instabilities while doing a phantom measurement. Typical causes of instabilities are:

- Local B0 field variation caused by temperature variation in the shim-unit generate frequency instabilities. The result is a virtual movement of the measured object during the scans.
- B1 field variations by instable RF produce image intensity variations.
- Gradient fields generate frequency and phase variations, causing also virtual movement of the object.
- Mechanical vibrations, e.g. by cold head, produce signal intensity variation by the movement of the measured object relatively to the magnetic iso-center.
- Movement of phantom fluid cause phase errors.

Measurement

NOTE Between phantom positioning and measurement start wait at least 15 min, since even small movements of phantom liquid will affect measurement results.

The measurement is done with the 8 Channel Head Matrix coil and the large bottle phantom (7.3 l).

A series of 512 measurements (EPI_FID sequence) with 11 slices will be performed and repeated once again.

The first measurement is done with a flip angle of 30 degree (evaluation of stability of B1 field, i.e. instabilities in the Tx- chain), the second with a flip angle of 90 degree on 1 and 1.5T systems and

70 degree flip angle on 3T systems (minimization of unstable RF, detection of B0 and gradient like interferences).

Evaluation of Measured Images

The following graphical and numerical evaluations will be performed:

- Mean values of a 15x15 ROI related to the mean value of all measurements as function of image number
- Standard Deviation of a 15x15 ROI mean values (linear and logarithmic plot)
- Center position in x and y direction of phantom, dependent of image number (Expert mode only)
- Measurement protocol values
- Statistics of Mean values (for ROI size 15x15 and 25x25):
  - Mean Image Intensity
  - Variation peak-peak
  - relative Variation peak-peak
  - Standard Deviation
  - total relative linear Drift
  - Standard Deviation from linear Fit
- Image quality parameters such as SNR and Ghosting
- Statistics of Phantom Movement (Expert mode only)

Normally these evaluations will not be displayed.

Results

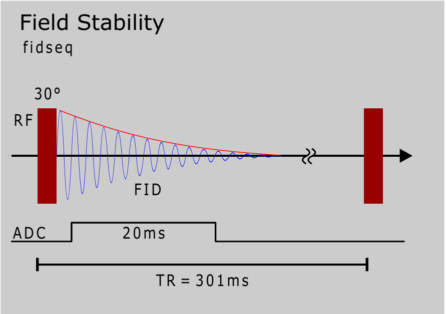
The numerical results and graphics can be viewed in the service software under Reports.

Field Stability

Principle

This procedure is intended to measure field changes caused primarily by the magnet. To eliminate influences by the gradients and RF the FID is measured and evaluated.

The phase of an MR-FID (Free Induction Decay) is given by:



$$\varphi(t) = \gamma \int_0^t B(t)dt \approx \gamma B \cdot t$$

γ is the gyro magnetic ratio of the proton
$B(t)$ is the time dependent magnetic field
t = 20ms measured every 301 ms.

To replace the integral, it is assumed that the field B(t) does not change much during sampling of the FID. Time 0 begins at the end of the RF-pulse. So the field can be calculated directly from the phase at time t. However, because of hardware imperfections it is better to take the slope of the phase over time which also gives the field.

A totel of 512 FIDs are measured over a 154 second period (repetition time of 301 ms). From every FID the phase over time is calculated and a linear fit is done, the slope giving the frequency.

The time range of the fit should be long to get a high accuracy but is usually limited by the length of the FID. A good compromise is 20ms, which averages over 50Hz line hum which is usually of no interest in this measurement.

Evaluation of Field Stability

The result of the field measurement is the field (=frequency) over the measurement time. As a measure of quality the linear drift over 10 min (=typical measurement time of an imaging sequence) and the maximum deviation from this linear drift are taken. When the measurement time is different from 10min, the drift is corrected for 10min by multiplying with the ratio (10min / actual measurement time).

The drift over 10min should not exceed the pixel bandwidth of low bandwidth sequence (e.g. 20Hz). The second one is important for gradient echo sequences to avoid smearing. To be able to use long echo times it should be typically less than 2Hz.

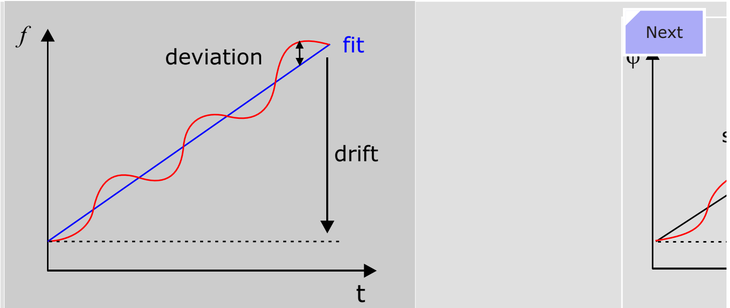


Figure 237 *Field Stability Check Evaluation*

Frequency Check

	Value	Low Spec	High Spec	Unit
Frequency Drift	0.39	-4.00	4.00	Hz/10min
Maximum Deviation	0.08	0.00	2.00	Hz

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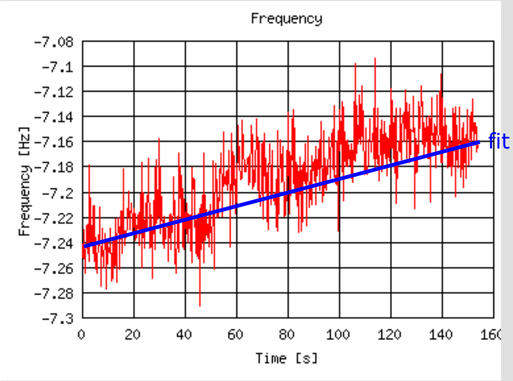

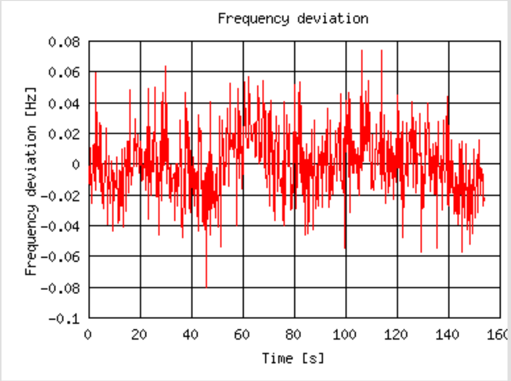

Section

13 Changes

This chapter provides a listing of the changes to the previous version

Introduction

Page 3 -Removed warning stating this document is for training only.

Gradient System

Page 236 - Removed text stating fans and choke temperatures are monitored. They are not.

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